

AI Workflow Documentation

Supplementary Appendix to “Skin in the Chain: Locked Supply and the Cross-Section of Cryptocurrency

Moazzam Khoja – AFA 2027 Special Session of Papers Written with Generative AI
Workflows

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Overview

This document is the complete contemporaneous record of the generative-AI workflow used to produce the paper, per the AFA 2027 Call for Papers. The project began on June 10, 2026 with an initial prompt to Claude (Anthropic); all subsequent work proceeded through a two-tier workflow: interactive strategy-and-writing sessions (Claude in Cowork) in which the human author set direction, reviewed output, and made all substantive decisions, and autonomous build sessions (Claude Code) launched with written kickoff prompts and reviewed after completion. The project proceeded in four phases. Phase A, theory and literature formulation (June 10–19, 2026; sessions 001–005): the theoretical framework – convenience yield, monetary search, seigniorage, the coin/token distinction, and the conviction measure λ – was developed in interactive conversation, alongside the literature review and first drafts of the introduction, theory, and hypotheses. Phase B, data construction (June 22 – July 30; sessions 006–041): on-chain panels built in autonomous Claude Code sessions launched from written kickoff prompts. Phase C, empirical testing (August; sessions 042–045): pre-specified analysis, results review, and mechanism testing. Phase D, writing and revision (August): results written up, pruned, and iterated under author direction.

Contents, ordered so the record reads chronologically: Part 1, all 47 contemporaneous session logs, beginning with the theory-formulation sessions; Part 2, the human time log; Part 3, all 33 verbatim kickoff prompts – the literal initial prompts given to autonomous AI sessions (these begin at the data phase; the theory phase was interactive and is documented in the session logs); Part 4, the data-decisions log (127 numbered entries covering the empirical phases; theory-phase decisions are recorded in the session logs and time log). Complete machine-readable session transcripts are archived in the project repository (06_documentation/ai_transcripts/).

Part 1: Session Logs (contemporaneous, chronological)

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session__001__2026-06-10.md

AI Conversation Log — Session 001

Date: June 10, 2026

Model: Claude Sonnet (Anthropic) via Claude Cowork

Human: Moazzam Khoja (khoja.research@gmail.com)

Summary

Initial project scoping session. Human shared the AFA 2027 call for submissions and a 2-page write-up of the QTM-based cryptocurrency valuation framework. Discussed workflow strategy (Cowork for research/writing, Claude Code for empirical testing), set up project folder structure, and created GitHub repository.

Key Outputs

- Project workflow plan
- GitHub repo: https://github.com/moazzamkhoja/AFA_2027_QTM_Crypto
- README.md
- time_log.md template
- workflow_notes.md template
- Full folder structure (01_literature through 06_documentation)

Human Inputs This Session

- Shared AFA 2027 call for submissions (email text)
- Shared QTM Valuation Framework PDF (2-page write-up)
- Decided on GitHub as storage location (public repo)
- Provided GitHub handle: moazzamkhoja
- Decided on C:/AFA_2027_QTM_Crypto as local working directory

Human Time This Session

~25 minutes total (see time_log.md)

Full conversation transcript to be appended here or exported from Claude Cowork and saved as a separate file in this folder.

/newpage

session__002__2026-06-10_theory.md

AI Conversation Log — Session 002

Date: June 10, 2026 **Model:** Claude Sonnet (Anthropic) via Claude Cowork **Human:** Moazzam Khoja

Topic: Theoretical Foundations of the QTM Framework

Summary

Extended theoretical discussion establishing the academic grounding for the QTM-based crypto valuation framework. Covered: convenience yield, monetary search theory, seigniorage, coins vs. tokens distinction, governance participation as MOAT signal, TVL disambiguation, and the coins-as-currencies / tokens-as-currencies framing.

Key Theoretical Decisions Made

1. Primary Theoretical Frame: Convenience Yield + Monetary Search

- **Convenience yield** (Sockin-Xiong 2020) is the primary rational motive for holding
- **Monetary search theory** (Kiyotaki-Wright 1993) grounds the coordination equilibrium
- **Seigniorage** (Cong-He NBER w33640) complements as the income mechanism
- These three are complementary, not competing explanations

2. Rational Expectations Confirmed

- The lambda $\rightarrow$ MOAT signal is primarily rational, not behavioral
- Behavioral amplifiers (loss aversion, lock-in) acknowledged but secondary
- Cong-Li-Wang (2021) adoption S-curve is the key rational dynamics model

3. Coins vs. Tokens — Fundamental Distinction

Coins (ETH, SOL): - PQ = total on-chain transaction volume in the native coin - lambda = staked / total circulating supply (clean on-chain measure) - Signal = monetary coordination + convenience yield + seigniorage - Cong & He (NBER w33640) directly covers this

Tokens (UNI, AAVE): - PQ = protocol throughput (DEX volume, TVL-as-activity) = “nominal GDP” of the protocol economy - lambda = governance staked + voting participation / circulating supply - Signal = MOAT (economic conviction, voice vs. exit, switching costs) - Edmans-Manso (2011) “voice vs. exit” is the key framework - Each token is the currency of its protocol economy — NOT an equity claim

4. TVL Disambiguation (Critical)

Two distinct uses of TVL in the framework: - **TVL as lambda (for coins):** Staked coins = M_Locked (supply measure) - **TVL as PQ (for tokens):** Protocol economic activity = nominal GDP analog These must never be conflated.

5. Rejected: Claim Coefficient (theta)

User correctly rejected theta (fraction of protocol revenue flowing to token holders) as overcomplicating the framework. Rationale: tokens are currencies, not equities. Whether revenue flows to holders is immaterial — the dollar has value because the US economy is large, not because GDP flows to dollar holders. Cong-Li-Wang’s approach (present value of future platform utility) is the correct framing.

6. Governance Token lambda — Composite Measure

For governance tokens, lambda should be a composite: - lambda_passive = tokens staked but not voting / total supply (capital commitment only) - lambda_active = voting participation rate x staked / total supply (conviction signal) - Active governance participation is a costly signal (Spence 1973): gas fees, time, reputational risk

Key Literature Identified

Paper	Relevance
Cong, Li & Wang (2021) RFS	Dynamic token adoption and valuation — primary theory reference
Sockin & Xiong (2020) NBER w26816	Convenience yield pricing of crypto
Cong & He (2024) NBER w33640	Staking tokenomics — coins theory
Kiyotaki & Wright (1993) AER	Monetary search theory — coordination equilibrium
Edmans & Manso (2011) RFS	Voice vs. exit — governance token theory
Hirschman (1970)	Exit, Voice, Loyalty — conceptual foundation for token governance
Spence (1973) QJE	Signaling theory — active governance as costly signal
Jiang, Krishnamurthy & Lustig (2024) NBER	Exorbitant privilege / reserve currency analog
Krishnamurthy & Vissing-Jorgensen (2012) JPE	Convenience yield in Treasury market
Nagel (2016) QJE	Liquidity premium of near-money assets
Liu & Tsyvinski (2021) RFS	Risks and returns of cryptocurrency
DAO Governance ACM (2024)	Voting participation -> token returns (+4.7%)

Paper Outline Agreed

1. **Introduction** — QTM framing, λ/V_{Free} , coins vs. tokens, findings, related lit, roadmap
2. **Theoretical Framework** — formal derivations, propositions for coins vs. tokens
3. **Hypotheses** — H1 ($\lambda \rightarrow$ returns for coins), H2 (governance $\lambda \rightarrow$ returns for tokens), H3 (quadrant $\rightarrow$ cross-sectional returns)
4. **Data and Empirical Design**
5. **Results**
6. **Robustness**
7. **Conclusion**

Literature embedded in introduction (JF/RFS convention), with “Related Literature” paragraph near end of introduction.

Human Time This Session

~60 minutes — theoretical discussion, framework refinement, outline decisions Activity type: Decision + Review

/newpage

session_003_2026-06-12.md

AI Conversation Log — Session 003

Date: June 12, 2026 **Model:** Claude Sonnet (Anthropic) via Claude Cowork **Human:** Moazzam Khoja

Topic: Introduction Revision (8 Issues) + Empirical Design Brainstorm

Introduction Revisions Made

Issue 1 — Cong-Li-Wang citation conflict

Removed Cong-Li-Wang from the opening DCF criticism. They are now presented as supporting the QTM view, not as exemplars of what we are critiquing.

Issue 2 — Convenience yield explanation

Rewritten: convenience yield = non-pecuniary benefit of network membership (access to platform services, liquidity, future transactional capability). SoV premium = PV of expected future convenience yields. Explicitly tied to MC_SoV.

Issue 3 — SoV and MoE definitions

Added a dedicated paragraph defining both components before the lambda equations: - MoE = value of current transactional function, priced by free velocity - SoV = premium paid today by forgoing current exchange utility in anticipation of future value appreciation from network growth, adoption, utility, and innovation

Issue 4 — “Speculative conviction” removed

Replaced throughout with “fundamental conviction” — grounded in network effects, utility expansion, competitive positioning, and innovation trajectory.

Issue 5 — Seigniorage channel

Rewritten: in PoS networks, newly minted tokens go to stakers (monetary seigniorage). Staking reveals belief that seigniorage income + appreciation > opportunity cost of locking = revealed preference for positive NPV.

Issue 6 — Governance token staking

Clarified: staked governance tokens are illiquid (cannot be sold while locked). Voluntary illiquidity cost = costly Spence signal of conviction. Distinct from coin staking which earns block rewards.

Issue 7 — Quadrant labels corrected

- AVOID = High NVT/G + Low SoV/MoE (expensive + weak conviction) — primary short signal
- OVERVALUED = High NVT/G + High SoV/MoE (expensive but believers locked in) — priced for perfection
- STAR/MOAT = Low NVT/G + High SoV/MoE — primary long signal
- GARP = Low NVT/G + Low SoV/MoE — utility-driven value

Issue 8 — Minor fixes

- Email corrected to moazzam.khoja@utsa.edu
- Fisher (1911) added as citation for $MV=PQ$
- Duplicate sentence removed

Structural Changes to main.tex

1. **Equations moved to Appendix A** — introduction is now prose-only
2. **“They are currencies” removed** — replaced with “exhibit currency-like behavior”
3. **Appendix A added** — full derivation:
 - $V_{\text{free}} = PQ / M(1-\lambda)$
 - $MC_{\text{MoE}} = M(1-\lambda)$

- $MC_SoV = M\lambda$
 - $SoV/MoE = \lambda/(1-\lambda)$? key result, no velocity estimation needed
4. **Related literature condensed** to one paragraph; three-part contribution statement added
 5. **Abstract updated** to reflect $SoV/MoE = \lambda/(1-\lambda)$ result

Key Theoretical Result Confirmed

$SoV/MoE = \lambda/(1-\lambda)$

This is a monotone function of λ alone. No velocity estimation required. The ratio is directly observable from on-chain data. This is the central empirical variable.

Empirical Design Brainstorm

λ Measurement — Coins

Measure	Definition	Source	Role
λ ?	Staked supply / circulating supply	Glassnode, Staking Rewards	Primary
λ ?	Supply unmoved ≥ 1 year / circulating (HODL waves)	Glassnode	Robustness
λ ?	TVL / circulating supply	DeFiLlama	Robustness only

λ Measurement — Tokens

Measure	Definition	Source	Role
λ ?	Governance staked / circulating supply	DeFiLlama, on-chain	Primary
λ ?	λ x voting participation rate	Snapshot, Tally	Enhancement

NVT/G — Levelized PQ approach

- Project TXN forward 10 years at trailing 3-year CAGR
- Terminal value at $g_term = 3\%$, discounted at CAPM r_e
- $PQ^* = PV$ of projected TXN stream (levelized)
- $NVT_fundamental = MC / PQ^*$
- Simple version ($MC / \text{current TXN x growth rate}$) tested as robustness

Asset Universe

- **Coins:** L1 PoS chains with native staking (ETH, SOL, AVAX, BNB, ATOM, ADA, etc.)
 - Exclude L2s (security derived from ETH, not independent monetary coordination)
- **Tokens:** DeFi governance tokens (UNI, AAVE, COMP, CRV, GMX, LDO, MKR, etc.)
 - Include regardless of chain
- Survivorship bias: include delisted assets from CoinGecko/DeFiLlama
- Minimum threshold: ≥ 6 months data, $\geq \$10M$ average market cap

Cross-Sectional Classification

- Coins and tokens analyzed **separately**
- High/low defined by **within-group median** at each rebalancing date (monthly)

- Token protocol-type fixed effects as control (DEX, lending, liquid staking, infrastructure)
- Within-group tests as supplemental robustness

Portfolio Construction

- **Primary:** Long Stars / Short Avoids, equal-weighted, monthly rebalancing
 - **Factor model:** regress on 3 factors — market (BTC), size (log MC), momentum (3-month)
 - **Alpha** = unexplained return after factor adjustment
 - **Sharpe ratios** reported for Stars-only, Avoids-only, long-short, benchmark
 - Separate results for coins sample and tokens sample
-
-

Empirical Design Decisions — Continued

NVT Clarification

- $PQ^* = \text{PMT equivalent of discounted TXN stream (Excel PMT logic)}$
- $PQ^* = \text{PV(TXN stream, 10yr + terminal value)} / \text{annuity_factor}(r_e, n)$
- Primary measure: $\text{NVT_fundamental} = \text{MC} / PQ^*$ — growth already embedded, no further G adjustment
- Simple NVT/G (current TXN ? trailing growth rate) used as robustness only

Asset Categories Finalized

Coins (L1 independent consensus): ETH, SOL, AVAX, BNB, ATOM, ADA, HYPE, others

L2 tokens (separate category): ARB, OP, STRK — rollup governance, not independent consensus

Token groups: - DEXes (UNI, CRV, BAL, RAY) - Lending (AAVE, COMP, MKR) - Liquid staking (LDO, RPL) - Perpetuals/Derivatives (GMX, DYDX, SNX) - Prediction markets (REP, GNO — small, merge if needed) - Infrastructure/Oracles (LINK, GRT)

Blockchain Agnostic Decision

Primary analysis: blockchain-agnostic with chain fixed effects (ETH ecosystem, SOL ecosystem, own-chain)

Robustness: within-chain tests

Rationale: protocol economy conviction signal should hold regardless of underlying chain

HYPE Classification

HYPE = Coin. Rule: native asset of independent consensus mechanism = coin. HYPE staked on HyperEVM's own HyperBFT consensus. GMX on Arbitrum = token.

Robustness check: reclassify HYPE as token, test sensitivity.

Hypothesis Structure Finalized

Key theoretical insight

lambda is the primary theoretical signal (convenience yield + moat). NVT/fundamental is a **conditioning variable**, not an independent predictor. The interaction of the two is the main empirical claim.

H1a — Coins

Higher $\lambda/(1-\lambda)$ positively predicts future returns for L1 coins (convenience yield + coordination mechanism).

H1b — Governance tokens

Higher governance staking $\lambda/(1-\lambda)$ positively predicts future returns for DeFi tokens. Voting-weighted λ has incremental power (Spence costly signal test).

H2 — Interaction (key insight)

Return predictability of $\lambda/(1-\lambda)$ is strongest when NVT/fundamental is low — the conviction signal is most valuable when not yet priced in. Tests as interaction term in Fama-MacBeth.

H3 — Portfolio

Long Stars (high $\lambda/(1-\lambda)$, low NVT/f) / Short Avoid (low $\lambda/(1-\lambda)$, high NVT/f). Factor-adjusted alpha. Sharpe vs. benchmark. Separate for coins and tokens.

Literature Added

- Pagnotta & Buraschi (2022 JF) — equilibrium valuation of Bitcoin
- Biais et al. (2019 RES) — blockchain folk theorem
- Easley, O'Hara & Basu (2019 JFE) — mining to markets (transaction fees)
- Liu, Tsyvinski & Wu (2022 JF) — common risk factors in crypto
- Makarov & Schoar (2020 JFE) — trading and arbitrage in crypto
- Howell, Niessner & Yermack (2020 RFS) — ICO financing
- Chod & Lyandres (2021 MS) — ICO theory
- Lagos & Wright (2005 JPE) — unified monetary theory framework
- Schilling & Uhlig (2019 JME) — Bitcoin economics

Paper Status

- Section 1 (Introduction): COMPLETE (revised)
- Section 2 (Theory): placeholder
- Section 3 (Hypotheses): COMPLETE
- Sections 4-7: placeholder
- Appendix A: COMPLETE

Pending Next Steps

1. Section 2 — Theoretical Framework (formal propositions)
2. Section 4 — Data and Variable Construction
3. Claude Code — empirical pipeline

Human Time This Session

~105 minutes total — introduction review (8 issues), revised draft review, empirical design decisions, hypothesis refinement
Activity type: Review + Decision

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AI Conversation Log — Session 004

Date: June 19, 2026 **Model:** Claude Sonnet (Anthropic) via Claude Cowork **Human:** Moazzam Khoja

Topic: Section 2 (Theoretical Framework) — Approach and Structure

Decision 1: Light Formalization, Not a Full Structural Model

Two options were considered for Section 2:

- **Option A — Full structural model.** Build a new agent-optimization/equilibrium model from scratch (e.g., search-theoretic, multi-period) deriving our predictions formally, with full propositions and proofs. High rigor, but multi-week effort and high risk of redundancy with existing models (Kiyotaki-Wright, Sockin-Xiong).
- **Option B — Applied theory with light formalization.** Motivate the empirical signal using existing theory, with brief stylized one-period models that yield clean, citable propositions mapping directly onto the hypotheses. Standard approach in empirical asset-pricing papers built around a new signal (e.g., Liu-Tsyvinski-Wu).

Decision: Option B. Rationale: the paper's contribution is the empirical decomposition and signal, not a new structural model; the AFA special session rewards a documented, efficient AI-assisted research process over a maximally rigorous theory section; the H2 interaction (signal strongest when unpriced) is a price-discovery story that is easier to motivate informally than to derive from a fully specified equilibrium model.

Decision 2: Merge Theory and Hypotheses into One Section

Human asked whether Section 2 should be structured as hypothesis development rather than a separate Theory section followed by a separate Hypotheses section. Agreed: under the light-formalization approach, keeping Theory and Hypotheses as two sections is redundant — the mechanism paragraphs in the old Hypotheses section already restated the same economic logic that a separate theory section would contain. Restructured into a single **Section 2: “Theoretical Framework and Hypothesis Development,”** with each hypothesis immediately following the proposition that motivates it. The old standalone Section 3 (Hypotheses) was removed; all subsequent sections renumber automatically via LaTeX auto-numbering (Data is now Section 3, Results Section 4, Robustness Section 5, Conclusion Section 6).

Section 2 Content Drafted

Four subsections, each a short stylized one-period model + proposition + hypothesis:

1. **2.1 The Staking Decision for Coins.** Agents choose to stake based on seigniorage s_t and private convenience yield $c_{i,t+1}$ vs. heterogeneous liquidity cost φ_i . Aggregate $\lambda_t = F(s_t + \bar{b}_{t+1})$. Private information in $\bar{b}_{t+1}$ is only partly priced (Grossman-Stiglitz noisy aggregation), so returns are increasing in λ_t . -> **Proposition 1 -> Hypothesis 1a.**
2. **2.2 Costly Signaling for Governance Tokens.** Agents have private binary beliefs q_i in q_H, q_L about protocol quality. Staking cost κ creates a single-crossing/separating equilibrium (Spence) where only high-belief agents stake, so λ_t reveals the share of holders with favorable private beliefs. -> **Proposition 2 -> Hypothesis 1b**, including the voting-weighted extension (additional monitoring cost ψ_i selects an even higher-quality subset of stakers).
3. **2.3 Price Discovery and the Conditioning Role of NVT/Fundamental.** Defines δ_t = fraction of the conviction signal already absorbed into price; shows NVT/fundamental is increasing in

$/\delta_t$, so marginal predictive power of $/\lambda_t$ is decreasing in NVT/fundamental. -> **Proposition 3 -> Hypothesis 2**. Explicitly states NVT/fundamental carries no independent return prediction of its own.

4. **2.4 The Quadrant Portfolio**. Combines Propositions 1–3 into Star/Avoid definitions -> **Hypothesis 3** (long-short alpha test), unchanged from prior session’s H3 language.

New citation added: **Grossman and Stiglitz (1980)**, *American Economic Review* — informationally efficient markets / noisy price aggregation, used in both 2.1 and 2.3 to justify why private information in $/\lambda_t$ is only partially priced.

Structural Changes to main.tex

1. Replaced placeholder Section 2 + standalone Section 3 (Hypotheses) with combined Section 2: “Theoretical Framework and Hypothesis Development.”
2. Updated introduction roadmap paragraph: removed reference to `sec:hyp`, now states Section 2 “develops the theoretical framework and states the paper’s testable hypotheses.”
3. Added `grossman1980impossibility` bibliography entry (alphabetically after Fisher, before Hirschman).
4. Verified no leftover `sec:hyp` references; all new internal labels (`sec:theory_coins`, `sec:theory_tokens`, `sec:theory_nvt`, `sec:theory_portfolio`) resolve correctly.

Paper Status

- Section 1 (Introduction): COMPLETE — will need one more revision pass once empirical results are available (currently has placeholder [X]%, [Y] Sharpe ratio).
- Section 2 (Theoretical Framework and Hypothesis Development): COMPLETE (combined theory + hypotheses, light formalization).
- Section 3 (Data and Variable Construction): placeholder.
- Section 4 (Results): placeholder.
- Section 5 (Robustness): placeholder.
- Section 6 (Conclusion): placeholder.
- Appendix A: COMPLETE.

Pending Next Steps

1. Section 3 — Data and Variable Construction: formalize the multi-channel $/\lambda$ index (staking/locking, holding period, voting engagement) decided in this session, plus NVT/fundamental construction, asset universe, and classification rules from Session 003.
2. Claude Code — empirical pipeline (separate workflow, not in Cowork).
3. Final revision pass on Introduction once empirical results exist.

Topic: The $F(\cdot)$ Aggregation Critique and Governance-Token Staking Mechanics

Decision 3: Reject the Closed-Form Aggregation, Then Properly Fix It

Human read the drafted Section 2.1 and rejected the original aggregation step $/\lambda_t = F(s_t + /barc_{t+1})$: substituting the cross-sectional *average* convenience yield $/barc_{t+1}$ into the CDF of $/varphi_i$ inside F does not follow from the individual threshold condition $s_t + c_{i,t+1}/geq/varphi_i$ unless $c_{i,t+1}$ is common to all agents — which would eliminate the private-information premise the whole signal relies on. Human asked for the underlying “trick” to be explained.

Explained the rigorous fix: a continuum of agents with a common unobserved future state $/\theta_{t+1}$, private noisy signals $x_i = /theta_{t+1} + /varepsilon_{i,t}$, and idiosyncratic liquidity costs $/varphi_i$, with a law-of-large-numbers argument washing out idiosyncratic noise in the aggregate. This was first proposed as background only (“why your intuition is sound”) with a recommendation *not* to add it to the paper, since it conflicts

with the Option B (light formalization) scope. Human’s interim instruction was to keep it simple and flag the result as an assumption to be tested empirically — implemented as “Proposition 1 (Assumption).”

On reflection, human asked for two further fixes in the same pass:

1. **Small fix** — generalize Section 2.2’s governance-token cost structure. The model’s $/barq = (/ell + /kappa)/Delta$ already nested the no-lock case ($/ell = 0$), but the prose didn’t say so. Added a paragraph making explicit that vote-escrow protocols (Curve veCRV, Balancer veBAL) have $/ell > 0$ (a real illiquidity cost), while snapshot/balance-voting protocols with no lock (Uniswap, Compound) have $/ell = 0$, leaving $/kappa$ (monitoring/ engagement cost) as the only force sustaining the separating equilibrium. Same threshold logic, different observable action.
2. **Big fix** — replaced the unjustified $F(s_t + /barc_{t+1})$ and the bare “Proposition 1 (Assumption)” hand-wave with the actual continuum-of-agents derivation: $c_{i,t+1} = /theta_{t+1} + /varepsilon_i$ (common state + private noise), threshold $/eta_i /equiv /varphi_i - /varepsilon_i /leq s_t + /theta_{t+1}$, and SLLN aggregation giving $/lambda_t = H(s_t + /theta_{t+1})$ for a fixed increasing H — a genuinely *derived* result, not an assumption. The one remaining assumption is narrowly scoped: that recovering $/theta_{t+1}$ from $/lambda_t$ requires costly information extraction (Grossman-Stiglitz 1980), so price only partially reflects it at t . Proposition 1 is now stated as a clean proposition again (label “(Assumption)” removed), with the partial-revelation step called out explicitly as the model’s one substantive assumption.

Decision 4: $/lambda_t$ as a Measurement Index, Not a Single Formula

Human’s question — “why would someone stake a governance token, what does staking even mean, and should holding period be a $/lambda$ measure if people don’t vote?” — was researched via web search. Findings: governance “staking” spans three distinct mechanisms. (1) Vote-escrow locking (Curve veCRV, Balancer veBAL, Aerodrome ve(3,3)): tokens locked up to several years for voting power plus a real fee/yield share — genuine illiquidity cost. (2) Yield-bearing staking with no lock requirement (GMX): stakers earn a direct share of protocol fees; governance rights often attach to holding rather than staking — closer to the coin model’s seigniorage channel than to costly signaling. (3) Pure snapshot/balance voting, no staking at all (Uniswap, Compound, Aave’s base layer): voting power equals token balance, fully liquid, gasless off-chain voting. Category (3) means the existing $/lambda$ definition (governance staked / circulating) is literally undefined for several planned-universe tokens (UNI, COMP, AAVE).

Human’s resolution: $/lambda_t$ -as-conviction is the right concept, but the *empirical measurement* should be an index over three observable channels — staking/locking, holding period, and voting engagement — with the specific construction (weights, thresholds, which channels apply to which protocol) decided later, once we are looking at the actual data, rather than fixed now in the theory section. This is now stated directly in Section 2.2 (see Decision 3, small fix) as a forward pointer to Section~/{ref{sec:data}}, rather than as a fixed formula in the theoretical framework. No empirical design choice was finalized in this session; only the principle that $/lambda$ is multi-channel was confirmed.

Structural Changes to main.tex (this pass)

1. Rewrote Section 2.1 in full: replaced $/lambda_t = F(s_t + /barc_{t+1})$ and “Proposition 1 (Assumption)” with the continuum-of-agents/private-signal/LLN derivation and a clean “Proposition 1.”
2. Added a paragraph to Section 2.2 generalizing the cost structure across lock and no-lock protocols, and introducing $/lambda_t$ as a multi-channel empirical index (staking, holding period, voting engagement), construction deferred to Section 3.
3. No other sections changed; hypothesis numbering (H1a, H1b, H2, H3) and proposition numbering (1–3) unchanged.

Topic: Unifying the Coin and Governance-Token Models

Decision 5: Merge Sections 2.1 and 2.2 into a Single “Locking Decision” Model

Human pushed back on the two-model structure (separate Section 2.1 for coins, 2.2 for governance tokens), arguing both are self-regulated digital economies in which λ signals the same underlying object — aggregate holder conviction about future growth, network effects, and product/service innovation — and that splitting them into two models was an artificial, over-complicating distinction. Agreed, with one addition: the original 2.2 (discrete-type Spence signaling, $\lambda_t = P(q_i/geq/barq)$ over a static type distribution) had no time-varying common state, so it never actually explained why λ_t should predict *future* returns the way the coin model’s θ_{t+1} does — unifying the two models fixes this gap for free, not just a simplification.

Human’s clarification: the difference between coins and tokens is in how λ is *measured* empirically, not in what it theoretically represents. Confirmed and implemented: both asset classes now share one continuum-of-agents/threshold-crossing derivation, with the coin/token distinction entering only through (a) the current benefit term, generalized from seigniorage s_t to $b_t/geq0$ (seigniorage for coins; fee/yield share, possibly zero, for governance tokens), and (b) the idiosyncratic cost, generalized to $\varphi_i = \ell + \kappa_i$ for tokens (lock-induced illiquidity cost + engagement cost) versus a pure liquidity cost for coins.

Separately, human proposed simplifying the common/idiosyncratic decomposition $c_{i,t+1} = \theta_{t+1} + \varphi_i$ to a time-series random walk $c_{t+1} = c_t + \varphi_t$. Pushed back: the $\theta_{t+1} + \varphi_i$ term is a *cross-sectional* decomposition (common state shared by all agents at a fixed date, idiosyncratic noise across agents) that the SLLN aggregation step requires to produce $\lambda_t = H(b_t + \theta_{t+1})$; a time-series random walk describes one quantity’s evolution over time with no cross-section of agents and cannot substitute for it without breaking the result. Human accepted the correction.

Decision 6: Drop the “L1” Restriction; Coin/Token Is a Functional, Not Architectural, Distinction

Human asked why Hypothesis 1a was restricted to “L1 native coins” given that L2 native assets are already part of the planned asset universe (Session 2026-06-12 decision). Agreed this was an unforced restriction left over from the original draft: nothing in the locking-decision mechanism depends on whether a chain is an L1 or an L2. The relevant cut is functional — whether locking the asset earns a security-staking benefit $b_t > 0$ (coin, Hypothesis 1a) or not (governance token, Hypothesis 1b) — not which layer it settles to. Removed “L1” throughout Section 2.1; Hypothesis 1a now reads “for native coins with a seigniorage- or security-staking-eligible design” with an explicit note that the mechanism is architecture-agnostic (L1 or L2), and classification of specific assets into the coin vs. token bucket is deferred to Section 3, consistent with how λ ’s measurement channels were already being deferred (Decision 4).

Decision 7: Make the Cross-Sectional Asset Index Explicit in Proposition 1

Human noted λ_t is implicitly asset-specific and asked whether it should be written with an explicit index, e.g. $\lambda(i, t)$. Clarified: i is already the agent index used throughout the Section 2.1 derivation, so an asset index needs a different letter (j) to avoid collision. Confirmed it is standard and fine to suppress the asset index through the agent-level derivation itself (the threshold-crossing logic is identical for any asset held fixed; asset heterogeneity enters only through θ_{t+1} and b_t), but Proposition 1 as previously written ($\text{textCov}(r_{t+1}, \lambda_t) > 0$) was ambiguous about whether the covariance was a time-series or cross-sectional object — and the hypotheses (H1a/H1b) are explicitly cross-sectional. Restated Proposition 1 with an explicit asset index j : $\text{textCov}_j(r_{j,t+1}, \lambda_{j,t}) > 0$, covariance taken across the cross-section of assets at a given date. Flagged (not yet built into the text) that this cross-sectional comparison is only informative about conviction if cost/benefit structure (H, b_t) is roughly comparable across the assets being compared — a concern the empirical hypotheses already partly address via category/size/momentum controls.

Structural Changes to main.tex (this pass)

1. Merged Section 2.1 (“The Staking Decision for Coins”) and Section 2.2 (“Costly Signaling for Governance Tokens”) into a single Section 2.1, “The Locking Decision” (`sec:theory_locking`, replacing `sec:theory_coins/sec:theory_tokens`). Generalized s_t/tob_t and recast the token-side cost as $/varphi_i = /ell + /kappa_i$ inside the same continuum-of-agents/SLLN derivation used for coins.
2. Removed “L1” qualifiers; Hypothesis 1a restated as architecture-agnostic (coin vs. token is now a functional distinction based on b_t , classified in Section 3).
3. Restated Proposition 1 with explicit asset index j as a cross-sectional claim, covering both coins and governance tokens (replacing the separate coin-only Proposition 1 and token-only Proposition 2).
4. Renumbered: old Proposition 3 (NVT cross-partial) is now Proposition 2; old Section 2.3 (NVT) is now 2.2, old Section 2.4 (Quadrant Portfolio) is now 2.3 (automatic via LaTeX subsection numbering). Updated cross-references (`sec:theory_coins/sec:theory_tokens` $\rightarrow$ `sec:theory_locking`) and “Propositions 1 through 3” $\rightarrow$ “Propositions 1 and 2” in the portfolio subsection, reworded to describe the seigniorage- and engagement-cost-driven mechanisms as both nested within the unified Proposition 1. Hypothesis numbering (H1a, H1b, H2, H3) unchanged throughout.
5. Recompiled: 18 pages, clean (no undefined references or citations).

Topic: Renaming the NVT/Fundamental Variable

Decision 8: Rename “NVT/Fundamental” to “Growth-Levelized NVT”

Human flagged that “NVT/fundamental” is a poorly constructed name — the slash makes it read as “NVT divided by fundamental,” but “fundamental” was never separately defined as its own variable; the metric is just traditional NVT (Network Value to Transactions) with the denominator (TXN) replaced by a forward-looking, levelized projection PQ^* . Asked for a better name.

Proposed three adjective-prefixed alternatives: “Levelized NVT” (matches existing appendix language describing PQ^* as a “levelized annual equivalent”), “Normalized NVT” (echoes “normalized earnings,” more legible to a general finance audience), and “Permanent NVT” (Friedman permanent-income framing). Recommended “Levelized NVT.”

Human asked whether prepending “growth-” to either candidate (“growth-normalized NVT” or “growth-levelized NVT”) would be clearer, since the construction explicitly bakes in growth assumptions (g, g_{infty}) before levelizing. Agreed “growth-” sharpens the name by flagging that the denominator is a forward-looking projection, not a trailing average — and recommended “growth-levelized” over “growth-normalized,” since “levelized” precisely names the PMT/annuity-equivalent step already used in the appendix, whereas “normalized” is mechanically vague and risks conflation with the PEG ratio’s much simpler “divide by growth rate” operation.

Decision: rename “NVT/fundamental” to “Growth-Levelized NVT” throughout, with notation $/textNVT_f/to/textNVT_{GL}$.

Structural Changes to main.tex (this pass)

1. Replaced all occurrences of “NVT/fundamental” (Abstract, Introduction, Section 2.2 heading and body, Hypothesis 2, Quadrant Portfolio subsection/Hypothesis 3, Appendix A) with “Growth-Levelized NVT.”
2. Updated notation: $/textNVT_{f,t}/to/textNVT_{GL,t}$ and $/textNVT_f/to/textNVT_{GL}$ (Section 2.2, Proposition 2, Appendix A).
3. Recompiled: 18 pages, clean (no undefined references or citations); one minor (~8pt) overfull-hbox cosmetic warning on the new subsection heading, not addressed.

Human Time This Session (running)

~90 minutes so far — reviewed Option A vs. B tradeoff, approved Option B, raised and approved the Theory/Hypotheses merge question, identified the flawed $F(/cdot)$ aggregation step and requested the

rigorous fix, explored governance-token staking mechanics, decided */lambda* should be a multi-channel measurement index with construction deferred to the data stage, challenged and approved the coin/token model unification, corrected an over-simplification of the common/idiosyncratic decomposition, identified the unjustified L1 restriction, identified the asset-index ambiguity in Proposition 1, and identified and resolved the “NVT/fundamental” naming problem (renamed to “Growth-Levelized NVT”).

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AI Conversation Log — Session 005

Date: June 22, 2026 **Model:** Claude Sonnet (Anthropic) via Claude Cowork **Human:** Moazzam Khoja

Topic: Section 3 (Data and Variable Construction) — Planning Approach

Human proposed that, rather than drafting Section 3 prose directly, the AI should first write a standalone data specification document for a separate Claude Code session to execute (build the empirical pipeline). Rationale, in human’s words: real crypto data may not be available in the ideal form, and “we may have to improvise” — the spec should be written and the pipeline run before Section 3 is drafted, so the prose reflects what data actually exists rather than an aspirational design.

AI agreed this sequencing reduces rework risk and asked four clarifying questions before drafting, covering asset-universe scope, sample period, factor-model choice, and lambda-index construction.

Decision 9: Asset Universe — Point-in-Time Rank Screen, Not a Fixed-Dollar Floor

Human’s framing: need a minimum-capital criterion to exclude “penny” tokens without creating survivorship bias by excluding failed tokens; proposed building the candidate list via CoinGecko/DeFiLlama/Artemis Analytics, then pulling per-asset transactional and voting data via Etherscan.

AI proposed refining the minimum-capital criterion: a **fixed nominal-dollar floor is methodologically unsound** here because total crypto market cap grew by orders of magnitude from 2015–2026, so any fixed threshold would be miscalibrated at one end of the sample or the other. Adopted instead a **point-in-time rank/percentile screen** (default: top 250 by market cap each month, adjustable pending coverage data) with the rule that **once an asset qualifies, it remains in the panel for the rest of the sample regardless of subsequent decline or failure** — addressing the penny-token and survivorship-bias concerns simultaneously. Flagged the common implementation trap of ranking by *today’s* market cap and backfilling (which silently reintroduces survivorship bias by excluding anything already delisted from current listings) and required point-in-time historical rankings instead. Also generalized “Etherscan” to chain-appropriate explorers (Solscan, Snowtrace, Arbiscan, etc.) plus off-chain governance sources (Snapshot, Tally, Boardroom, DeepDAO) for protocols that vote off-chain — pure block-explorer reads would miss those votes entirely.

Decision 10: Sample Period — August 2015 Onward (Not January 2021)

Human pushed back on AI’s earlier default suggestion of starting the sample around January 2021, asking “why just 2021?”, and proposed starting from the post-Ethereum “token economy” era beginning 2015.

Decision: sample starts August 2015 (the month after Ethereum’s mainnet launch, July 30, 2015), common across the panel, monthly frequency, running to the most recently completed month. AI’s original 2021 suggestion was about balanced-panel feasibility for governance-token-dependent tests, not a claim that earlier data is unusable; the spec now requires both the longer 2015+ history and an explicit year-by-year coverage report so it’s visible where governance/voting-dependent tests (H1b, H2, H3) actually become

testable versus where only the coin-only test (H1a) has sufficient density in the early years. Flagged a concrete worked example of the kind of landmine this surfaces: Ethereum itself was proof-of-work (no staking mechanism) until the Beacon Chain deposit contract (~Dec 2020) and the Merge to proof-of-stake (~Sept 2022) — so even ETH, a 2015-era asset, has no staking-based lambda channel for most of the proposed sample window. To be verified against live sources, not assumed.

Decision 11: Factor Model — Both CAPM and Liu-Tsyvinski-Wu, as Mutual Robustness

Confirmed: build **both** a Liu-Tsyvinski-Wu-style 3-factor crypto model (market, size, momentum, per their 2022 *Journal of Finance* paper) **and** a CAPM beta with custom size/momentum controls constructed directly from this paper’s own universe, and report both as robustness checks against each other for the H1/H2/H3 risk-adjusted return tests, rather than picking one.

Decision 12: lambda-Index Construction — Equal-Weighted Average of Standardized Channels

Confirmed: `lambda_t` is the equal-weighted average of the standardized (z-scored) values of whichever of the three channels (staking/locking ratio, holding duration, voting engagement) are observable for a given asset-month. Missing channels are excluded from the average for that asset-month rather than imputed with zero or a cross-sectional mean; the count of contributing channels is recorded as a diagnostic.

Outcome: Data Specification Document Written

Drafted `04_code/DATA_SPECIFICATION.md` — a self-contained brief for a separate Claude Code session, covering: the theoretical purpose of each variable (tying back to lambda, the SoV/MoE ratio, and Growth-Levelized NVT as defined in `05_paper/main.tex`), the universe-construction rule from Decision 9, the sample period and known landmines from Decision 10, lambda-channel definitions and sourcing per channel, Growth-Levelized NVT input definitions (MC, PQ, trailing CAGR g , CAPM r_e , terminal growth g_{inf} , horizon n), the dual factor-model requirement from Decision 11, and a phased deliverable plan (Universe $\rightarrow$ lambda channels $\rightarrow$ NVT_GL inputs $\rightarrow$ returns/factors $\rightarrow$ assembly) with a coverage/feasibility report required at the end of each phase before proceeding to the next.

Also created `04_code/DATA_DECISIONS_LOG.md` — an append-only template for the Claude Code pipeline session to record every deviation from the spec (proxy substitutions, classification judgment calls, threshold adjustments, sources that didn’t pan out), explicitly operationalizing the human’s “we may have to improvise” expectation rather than letting such deviations go undocumented. This log is intended to feed both the eventual Section 3 prose and the AFA-required AI-conversation audit trail.

No hypothesis testing or Section 3 prose has been written yet. Per the spec, that work is gated on the Claude Code pipeline session producing a Phase 0 coverage report.

Status Carried Forward (Parked, Not Addressed This Session)

- **Bibliography sanity check** (raised in the prior session): two `/bibitem` entries in `main.tex` (`dao2024governance`, `daoreview2025`) lack real author names and have generic-sounding venue names, suggesting fabricated/placeholder citations. Human explicitly asked to park this until returned to; not addressed in this session.

Human Time This Session (running)

~25 minutes — reviewed AI’s proposed sequencing (spec-first, write Section 3 after data is collected), answered four clarifying questions on universe scope, sample period, factor model, and lambda-index construction, approved the resulting data specification document.

/newpage

session_006_2026-06-22_phase0_pipeline.md

AI Conversation Log — Session 006 (Phase 0 Empirical Pipeline)

Date: June 22, 2026 **Model:** Claude Opus 4.8 (Anthropic), exact model id `claude-opus-4-8` **Interface:** Claude Code CLI (agentic coding session, distinct from the Cowork planning sessions 001–005) **Working directory:** `C:/AFA_2027_QTM_Crypto` **Human:** Moazzam Khoja

AFA documentation note. This file is the structured companion record for the Phase 0 pipeline session. The *primary* AFA record is the verbatim terminal transcript of the Claude Code session; this log faithfully summarizes the initial prompt, the AI’s reasoning, every judgment call, and the human’s interventions, but the raw transcript should also be preserved alongside it.

Initial prompt (verbatim intent)

The session was opened with the standing kickoff prompt in `04_code/CLAUDE_CODE_KICKOFF_PROMPT.md` and the instruction:

“Read `04_code/CLAUDE_CODE_KICKOFF_PROMPT.md` and proceed with the project. When you reach 60% of the context space, create three documents: (1) Session status report, (2) Transition document for next session, (3) Next session’s prompt. Create an implementation plan before writing the code and have an agreement on it.”

The kickoff scopes the session to **Phase 0 only** (asset universe), per `DATA_SPECIFICATION.md` Section 7: build a reproducible, point-in-time, monthly, survivorship-bias-free ranked universe; a coin-vs-token classification table with evidence; and a coverage report. Stop at the end of Phase 0 for review.

Activity log (chronological)

1. Read-in and context

Read `DATA_SPECIFICATION.md` (full), `DATA_DECISIONS_LOG.md`, and `main.tex` Section 2 (Locking Decision model) + Appendix A (Growth-Levelized NVT derivation). Confirmed environment: Python 3.13, pandas/numpy/requests/openpyxl present.

2. Source feasibility audit (live, not assumed) — key findings

- **CoinGecko public API: historical data capped at the past 365 days** (error 10012); full history requires a paid plan. -> Cannot serve point-in-time rankings back to 2015 on the free tier. Documented as a deviation from the spec’s CoinGecko-first framing.
- **CoinMarketCap data-api/v3/cryptocurrency/listings/historical:** free, no key, **daily** point-in-time ranked snapshots back to 2013, includes delisted/dead coins, returns market cap / price / volume / supply / tags, limit ≥ 1000 . -> Adopted as the **primary ranking backbone** (survivorship-bias-free per Section 2.1).
- **DeFiLlama api.llama.fi/protocols:** free, works -> used for governance-token classification evidence (protocol category).
- **Artemis (api.artemisxyz.com):** 410 Gone / DNS failure on public hosts -> no accessible free API; deferred to a later phase (needs authenticated/paid access).

3. Implementation plan — presented and agreed

Presented a 5-part plan (fetch snapshots -> build universe -> classify -> coverage report -> decisions log). Two genuine decisions raised to the human via structured question: - **Data source:** human chose to **proceed with the free CMC historical endpoint** (logged deviation) rather than supply a paid CoinGecko key. - **Stablecoins:** human chose to **exclude stablecoins entirely from the universe** (drop at construction so they never enter downstream phases). AI noted this correctly keeps LUNA (the coin) while dropping UST (the stablecoin).

4. Build — scripts written to 04_code/

- `fetch_cmc_snapshots.py` — pulls top-1000 CMC historical listing for the last calendar day of each month, 2015-08 -> 2026-05 (130 months), caches raw JSON to `03_data/raw/cmc_snapshots/`. Ran clean: 130/130 fetched, 0 failed; early months served ~500–600 assets, recent months 1000.
- `build_universe.py` — applies Section 2.1 (top-250 entry, permanent retention, carry-forward of last observed price for in-panel-but-unobserved months), ranking inclusive of stablecoins but dropping them from the output panel. Output: 1,939 qualifying non-stablecoin assets; live top-250 cross-section ~215–247/month (the shortfall vs. 250 = excluded stablecoins occupying real rank slots); growing `carried_forward` tail = faithfully retained dead/fallen coins. Rank sensitivity (N=200/250/300) emitted.
- `classify_assets.py` — first pass tag-driven coin/token/stablecoin/other classifier with evidence strings + ambiguity flags + ETH time-varying staking onset (Beacon genesis 2020-12-01, Merge 2022-09-15). **First run surfaced classification bugs under review/fix** (see below).

5. Classification issues found in first pass (being fixed)

- Substring `stablecoin` match over-caught `stablecoin-protocol` (governance tokens of stablecoin issuers) and `stablecoin-algorithmically-stabilized` (LUNC = the Terra *coin*). Fix: exact-tag match, consistent with `build_universe`.
- L1 coins carrying `defi` ecosystem tags but no clean consensus tag (ATOM, AVAX, FTM) fell through to `token`; XRP/DOT fell to `other/token`. Fix: promote `layer-1` tag to a high-priority COIN signal; restrict DeFiLlama promotion to genuine governance-token categories; add a small curated native-coin override (documented judgment call).

6. Human intervention (mid-session)

Human paused work to (a) confirm the AFA record-keeping requirements and ask the AI to log conversations/activity, and (b) confirm the GitHub repo (`moazzamkhoja/AFA_2027_QTM_Crypto`, `main`) and the standing authorization to commit and push at the end of each session. AI created this log file, confirmed the repo and documentation scaffolding, and recorded the model/config distinction for this session.

7. Phase 0 completed

Classification corrected (all spot-checked majors verified: coins BTC/ETH/SOL/ADA/AVAX/DOT/ATOM/XRP/XLM/etc.; tokens UNI/AAVE/COMP/CRV/MKR/LDO/etc.; LINK left ‘other’). Coverage report generated (`03_data/PHASE0_COVERAGE_REPORT.md`): live cross-section ~290->630 assets/month; observed governance tokens 19 (2015) -> 233 (2025); H1b/H2/H3 testable only from ~2020–2021. Nine entries appended to `DATA_DECISIONS_LOG.md`. Three end-of-session documents written: `PHASE0_SESSION_STATUS.md`, `PHASE0_TO_PHASE1_TRANSITION.md`, `CLAUDE_CODE_PHASE1_KICKOFF_PROMPT.md`. Raw 76MB cache gitignored (regenerable). Committed and pushed to `origin/main`.

Status at last update

Phase 0 complete. Stopped for review before Phase 1, per the kickoff instruction.

/newpage

session_007_2026-06-22_phase0_review.md

Session 007 — Phase 0 Review (Cowork / Claude)

Date: 2026-06-22 **Interface:** Claude (Cowork mode), reviewing output produced by a separate Claude Code session (Session 006). **Scope:** Review of Phase 0 deliverables; resolution of open questions before Phase 1.

1. What this session reviewed

Read in full: 03_data/PHASE0_COVERAGE_REPORT.md, 06_documentation/ PHASE0_SESSION_STATUS.md, 04_code/DATA_DECISIONS_LOG.md (Entries 1-9), and re-checked 05_paper/main.tex Section 2 (Locking Decision, Price Discovery/H2, Quadrant Portfolio/H3) and the still-placeholder Data/Results/Robustness sections.

2. Findings on Section 2/3 framing (part a of the review request)

- No changes needed to `main.tex` Section 2 right now. It is written as a timeless cross-sectional theory section and correctly defers all empirical specifics (sample period, universe construction, thresholds) to Section 3 (“Data and Variable Construction”), which remains an unwritten placeholder.
- When Section 3 is written, it must state explicitly that H1a (coins) is testable across most of the 2015-08+ sample, while H1b/H2/H3 (governance-token- and voting-dependent) only become meaningfully testable from roughly 2020-2021 onward, per the coverage report’s year-by-year governance-token counts. Earlier H1b/H2/H3 estimates should be flagged out-of-support, not silently pooled with the well-supported window.

3. Findings on stablecoin exclusion / classification defensibility (part b)

- The stablecoin exclusion (ranking-inclusive, output-exclusive, exact-tag detection, LUNA/UST treated as the clean illustrative case) is defensible and consistent with precedent (e.g., Liu-Tsyvinski-Wu’s own stablecoin exclusion).
- The classification engine (tag-rules + DeFiLlama + curated native-coin override, `ambiguous_flag=True` on non-clean cases) is a reasonable first pass. The ‘other’ bucket (874 distinct names, smaller by observed asset-months) needed a bounded manual confirmation pass before any of those assets enter the tests — see Section 5 below for how this was scoped.

4. Four original open questions — resolved

Asked via structured question, answered by the user:

1. **Carry-forward death returns:** split first (presumed-failed vs. temporarily-out), decide the return-treatment formula only after seeing the split. (Decisions Log Entry 10.)
2. **Classification review scope:** raised initially with an unclear “thin” framing; user asked for clarification. Resolved in Section 5 below.
3. **Artemis / paid CoinGecko access:** deferred until Phase 2 actually needs it. (Decisions Log Entry 14.)
4. **Universe size N=250:** confirmed, no change. (Decisions Log Entry 15.)

5. Clarification: what “thin” meant, and the resolved review scope

The user asked whether the classification-review scope question was about a market-cap threshold, noting the universe already uses a top-250 rank screen. Clarified: “thin” referred to a different, independent dimension —

the count of *months* an asset is actually observed in the panel (persistence), not its rank or market cap. Every asset in the universe already passed the top-250-at-least-once rule; the persistence filter is a second-stage triage on top of that, used only to decide which ‘other’/ambiguous names are worth a manual evidence review (a one- or two-month pump-and-dump blip contributes negligible weight to any regression and isn’t worth reviewing by hand). Resolved: review ‘other’/ambiguous names with ≥ 12 observed asset-months. (Decisions Log Entry 11.)

6. New questions raised by the user this session

(i) **Meme coins / NFTs.** The user asked whether meme coins and NFT-like tokens should be excluded from the universe, on grounds that they don’t support genuine economic output (no PQ basis, no real lambda/conviction signal) — and whether they’re already excluded. Finding: meme coins (DOGE, SHIB, PEPE, etc.) are explicitly named in the Phase 0 coverage report as already landing in the ‘other’ classification bucket (no staking mechanism, no governance mechanism), so they’re already excluded from H1a/H1b/H3 by construction; they still count toward the top-250 ranking, the same treatment as stablecoins. NFTs are presumed absent entirely, since the CMC fungible-token listings endpoint used in Phase 0 is a different product from CMC’s NFT tracking — flagged for explicit confirmation rather than assumption. Resolved: keep current handling, no Phase 0 change; confirm NFT absence in the Phase 0 follow-up. (Decisions Log Entry 12.)

(ii) **Quadrant Portfolio (2x2 matrix) construction mechanics.** The user asked how the Star/Avoid quadrant categorization (H3) will actually be built. Clarified that this is a different categorization from coin/token classification: the quadrant axes are two continuous variables, lambda/(1-lambda) and Growth-Levelized NVT, each split at a cross-sectional median into high/low, crossed into four cells. The open design question was whether the medians should be computed pooled across coins+tokens or separately within each class (since H3 reports results separately for coins and governance tokens, and the two classes’ staking/governance norms may differ structurally for reasons unrelated to conviction). Resolved: compute medians within each asset class separately, not pooled. (Decisions Log Entry 13.)

7. Decisions log updated

Appended Entries 10-15 to `04_code/DATA_DECISIONS_LOG.md` covering all six resolutions above (death-return split-first approach, classification review threshold, meme-coin/ NFT handling, quadrant median-split methodology, Artemis deferral, $N=250$ confirmation).

8. Next step

Wrote `04_code/CLAUDE_CODE_KICKOFF_PROMPT_PHASE0_FOLLOWUP.md` — a short, scoped follow-up session (not Phase 1 itself) to: (1) implement the carry-forward presumed-failed/temporarily-out split with diagnostic counts, and (2) run the ≥ 12 -month classification confirmation pass, explicitly checking meme-coin and NFT handling along the way. This session is to stop and report before Phase 1 (lambda channels) begins, consistent with the project’s phased-gate discipline.

No Phase 1 work was started or authorized in this session.

9. Addendum — sector/economic-function classification (raised before committing)

Before committing Sections 1-8’s changes, the user paused with two questions.

(i) **“Class” was being used too narrowly.** The user clarified that Entry 13’s coin/token “class” (used for the H3 quadrant median splits) was not what they meant by “class” in general — they want a separate, narrower economic-function/sector variable that “defines the tokens or coins economy” (e.g., L2 vs. L1, Lending Protocol vs. Perpetual, DEX vs. Perpetuals, staking pools vs. lending protocols), with the specific taxonomy/comparisons to use deferred to later, but the need for the field itself flagged now. A quick check of `classification_table.csv` confirmed the raw ingredients already exist as classification evidence: 574/2010

assets carry a `defillama_categories` value (154 Dexs, 78 Yield, 59 Derivatives, 53 Lending, 34 Canonical Bridge, 20 Liquid Staking, 17 CDP, etc.), and 134/2010 carry a `layer-1` CMC tag with 43 carrying `layer-2`. Resolved: added spec section 2.6 (sector/economic-function classification, independent of the coin/token cut) and Decisions Log Entry 16 — capture a `sector` field from DeFiLlama categories + CMC tags now; defer deciding which sector-level comparisons to actually test until Phase 1-4 data exists.

(ii) **Whether to revise the next Claude Code prompt.** Resolved by revising `04_code/CLAUDE_CODE_KICKOFF_PROMPT_PHASE0_FOLLOWUP.md` directly (rather than waiting for a separate Phase 1 prompt, since that file is the next session and “our current discussion” needed to land there) — added a third deliverable: populate the `sector` field per spec 2.6/Decisions Log Entry 16 and report coverage by `asset_class`, with an explicit instruction not to decide sector-level comparisons in that session. The carry-forward split and classification-confirmation deliverables (1-2) are unchanged.

No Phase 1 (lambda channel) work was started or authorized. This addendum’s changes are included in the same not-yet-committed working tree as Sections 1-8.

/newpage

`session_008_2026-06-22_phase0b_followup.md`

AI Conversation Log — Session 008 (Phase 0B Follow-Up)

Date: June 22, 2026 **Model:** Claude Opus 4.8 (Anthropic), exact model id `claude-opus-4-8` **Interface:** Claude Code CLI (agentic coding session) **Working directory:** `C:/AFA_2027_QTM_Crypto` **Human:** Moazzam Khoja

AFA documentation note. This file is the structured companion record for the Phase 0B follow-up session. The *primary* AFA record is the verbatim terminal transcript of the Claude Code session; this log faithfully summarizes the initial prompt, the AI’s reasoning, every judgment call, and the human’s interventions, but the raw transcript should be preserved alongside it.

Initial prompt (verbatim intent)

The session was opened with:

“Read `CLAUDE_CODE_KICKOFF_PROMPT_PHASE0_FOLLOWUP.md` in `C:/AFA_2027_QTM_Crypto/04_code` and proceed from there”

That kickoff scopes a **focused follow-up to Phase 0 (not Phase 1)** with three deliverables implementing Decisions-Log Entries 10, 11, 16 (and confirming Entry 12): (1) split carried-forward asset-months into `presumed_failed/temporarily_out` with diagnostic counts; (2) a classification confirmation pass over `other/ambiguous` names with ≥ 12 observed asset-months; (3) populate an independent `sector` field per spec 2.6. Stop and report before Phase 1.

Activity log (chronological)

1. Read-in and context

Read in full: `DATA_SPECIFICATION.md` (incl. new ?2.6), `DATA_DECISIONS_LOG.md` (Entries 1–16), `PHASE0_COVERAGE_REPORT.md`, `PHASE0_SESSION_STATUS.md`, and the existing `classify_assets.py` (to reuse its rule logic). Verified data shapes: panel 156,838 rows / 1,939 assets / 130 months; classification table 2,010 rows; 89,535 carried_forward asset-months; 630 `other/ambiguous` candidates at the ≥ 12 -month gate. Python 3.13 / pandas 3.0 present. (Hit and worked around a Windows cp1252 console encoding issue by writing analysis to UTF-8 files.)

2. Deliverable 1 — carry-forward split (carry_forward_split.py)

Rule (vectorised, equivalent to gap analysis): for asset j with last observed month L_j , a carried forward month at t is `temporarily_out` iff $t < L_j$, else `presumed_failed`. Added column `carry_forward_subtype`; left `status` untouched. Results: **81,167 presumed_failed (90.7%) / 8,368 temporarily_out (9.3%)**; 2,107 closed gaps (min 1 / median 1 / mean 4.0 / max 47 months); 1,399 `presumed_failed` assets with terminal-gap median 64 mo. **Right-censoring flagged explicitly**: 98 `presumed_failed` assets have a ≤ 6 -month trailing gap (started after 2025-11-30) and 166 a ≤ 12 -month gap — indistinguishable from a still-mid-gap `temporarily_out` case as of 2026-05-31. **No death-return formula chosen** (deferred per Entry 10). Logged as Entry 17.

3. Deliverable 2 — classification confirmation pass (classify_confirmation_pass.py)

Reviewed all 630 candidates; per-asset proposals written to `classification_confirmation_review.csv`. The set is overwhelmingly genuine utility/ sector tokens (AI, gaming, payments, DePIN, storage, identity, exchange, meme) that correctly stay `other`. Applied **16 conservative, individually-verifiable flips**: 15 `other->coin` (KSM, POL, DYM, KUJI, XYM, IOST, STEEM, ARDR, QKC, VLX, WICC, NEBL, UOS, CENNZ, WTC — native PoS/DPoS chains whose CMC tags carried platform/ecosystem labels or a non-standard consensus tag, or whose DeFiLlama category mislabelled the base chain), and 1 `other->token` (STG / Stargate, veSTG vote-escrow fee-share). A further **16 genuinely ambiguous names were left other with an explicit note** rather than forced (L2 gas/governance tokens OP/MNT/MANTA/IMX; LST/staking-infra RPL/ANKR/SSV/STRD; weak/edge chains EWT/GBYTE/FCT; NFT-market BLUR/LOOKS/ME; PNK; PTS). Original label preserved in `asset_class_original`; reason in `confirmation_basis`. Class counts: coin 618->633, token 447->448, other 874->858. Logged as Entry 18.

4. Deliverable 2 (cont.) — meme/NFT confirmation

Scanned the full universe. **Meme**: 84 meme-tagged names -> 58 other / 21 token / 5 coin. The 5 coins (DOGE, MONA, ?) are genuinely mineable/PoS coins — `coin` is functionally correct, not a mis-flag. Several meme *tokens* (SHIB, FLOKI, BabyDoge, ELON, MEW, SNEK?) were promoted to `token` via an attached DEX/farm DeFiLlama category — **flagged for a human decision, left unchanged** per Entry 12. All memes among the 630 reviewed `other` candidates correctly stay `other`. **NFT**: 112 NFT-tagged names are all *fungible* tokens of NFT-ecosystem projects; **no non-fungible collection exists** in the panel (CMC fungible-listings source lists none) — closes the Entry-12 verification item. Logged as Entry 19.

5. Deliverable 3 — sector field (build_sector_classification.py)

Added independent `sector` column = DeFiLlama categories (as-is, multi-value) ? a curated CMC sector-tag whitelist (Layer-1/Layer-2/Oracle/Privacy/DePIN/Meme/etc.), with governance-axis and ecosystem/portfolio/listing tags excluded; blank where neither fires. **Coverage 1,113/1,939 (57.4%)**: coin 54.5% (leans on CMC L1/L2 tags), token 89.7% (leans on DeFiLlama), other 42.7%; residual 42.6% — the lean spec ?2.6 predicted. **No sector-level comparison proposed** (deferred per Entry 16). Caveat logged: the DeFiLlama join is by ticker symbol, so common tickers over-attribute (BTC inherits SoFi/Reserve Currency/Canonical Bridge); CMC-tag parts are clean. Logged as Entry 20.

6. Report + decisions log

Wrote `03_data/PHASE0B_FOLLOWUP_REPORT.md` (all three deliverables, in the coverage-report format). Appended Decisions-Log **Entries 17–20**. Removed scratch files; verified all outputs (panel subtype column, three new classification columns, 630-row review file).

7. Human intervention (end of session)

Human authorized: update the project memory (Phase 0B complete, Phase 1 next), revise the Phase 1 kickoff prompt to reflect Phase 0B state, and **commit + push everything (logs + prompt history) per the project cadence**. AI updated `PHASE0_TO_PHASE1_TRANSITION.md` and

CLAUDE_CODE_PHASE1_KICKOFF_PROMPT.md for the new repo state (new columns, updated class counts, 20 decisions-log entries, next session = 009), wrote this log + the time-log entry, and committed/pushed to origin/main.

Files produced / modified this session

- **Scripts:** 04_code/carry_forward_split.py, classify_confirmation_pass.py, build_sector_classification.py
- **Data:** universe_panel.csv/.parquet (+carry_forward_subtype); classification_table.csv (+sector, asset_class_original, confirmation_basis; 16 flips); classification_confirmation_review.csv; three_*.txt diagnostics.
- **Docs/logs:** PHASE0B_FOLLOWUP_REPORT.md; DATA_DECISIONS_LOG.md Entries 17–20; this log; time_log.md; updated transition brief + Phase 1 kickoff prompt.

Status at last update

Phase 0B follow-up complete. Stopped for review before Phase 1 (lambda channels), per the kickoff instruction. Items still needing a human decision are listed in PHASE0B_FOLLOWUP_REPORT.md ?4 (death-return formula; meme-token demotion question; gray-zone 16; sector DeFiLlama symbol-collision de-noising).

/newpage

session_009_2026-06-23_phase1_lambda.md

AI Conversation Log — Session 009 (Phase 1: lambda Channels)

Date: June 23, 2026 **Model:** Claude Opus 4.8 (Anthropic), exact model id claude-opus-4-8 **Interface:** Claude Code CLI (agentic coding session) **Working directory:** C:/AFA_2027_QTM_Crypto **Human:** Moazzam Khoja

AFA documentation note. This file is the structured companion record for the Phase 1 session. The *primary* AFA record is the verbatim terminal transcript; this log faithfully summarizes the prompt, the AI’s reasoning, every judgment call, the human’s interventions, and the verified-source decisions, but the raw transcript should be preserved alongside it.

Initial prompt (verbatim intent)

“Read CLAUDE_CODE_PHASE1_KICKOFF_PROMPT.md in C:/AFA_2027_QTM_Crypto/04_code and then proceed from there”

That kickoff scopes **Phase 1 only** (spec ?3/?7): build the three lambda conviction channels (staking/locking, holding duration, voting engagement) per observed asset-month on the frozen Phase 0 universe, z-score each within its monthly cross-section, equal-weight-average the available channels into lambda, and produce a coverage report — without imputing missing channels and after verifying every source’s live free access first.

Human intervention that shaped the session

After source verification began, the human **redirected the data sourcing:** build the lambda numbers from the *canonical chain source* (“etherscan or other chain main data source”) rather than an aggregator, “to ensure we are getting information from the real source.” This became the defining constraint of the session: DeFiLlama was demoted to a metadata registry (address book), and all lambda numbers were taken from on-chain event logs (Etherscan V2) and Snapshot’s own API. The human then supplied a free Etherscan V2 API key (“AFA Paper”), which was stored gitignored at 04_code/.api_keys.json (confirmed not committed).

Activity log (chronological)

1. Read-in and context

Read in full: DATA_SPECIFICATION.md (?1, ?3, ?7), DATA_DECISIONS_LOG.md (Entries 1–20), PHASE0_TO_PHASE1_TRANSITION. (incl. ?0 addendum), PHASE0B_FOLLOWUP_REPORT.md, PHASE0_SESSION_STATUS.md, and main.tex ?2 (the Locking Decision model). Verified data shapes and that only ETH carries a `staking_start`. Python 3.13 / pandas 3.0.

2. Source verification (live) — Entry 21

Probed every candidate source before building. Working/keyless: DeFiLlama, Snapshot GraphQL, CMC historical (reused). Paywalled now: beaconcha.in (401), Boardroom/Tally (401). **Key findings:** Etherscan V1 is dead and V2 needs a key; **no free RPC serves archive state**, and Etherscan free `eth_call` at a past block silently returns *latest* — so all historical Channel-1 data must come from **event logs**, not state reads.

3. Asset->on-chain-identity map — Entry 22

`phase1_build_identity_map.py` joins the universe to DeFiLlama’s registry on `cmcId` to get each asset’s token contract + Snapshot space. DeFiLlama’s `cmcId` is sparse (241/1,939 matched, 35 Snapshot spaces) — this thinness drove the curated extensions below.

4. Channel 1 — staking/locking (canonical on-chain) — Entries 23, 26

- **ETH native staking:** reconstructed cumulative staked ETH from the Beacon deposit contract’s `DepositEvent` logs (8-byte LE gwei `amount` parsed per event), respecting `staking_start=2020-12-01`. Validated (2.17M @ 2020-12, 5.2M @ 2021-05). Full ~2.5M-event run is long; left resumable (self-checkpointing monthly). Caveat: cumulative-deposited overstates net stake post-Shapella.
- **EVM vote-escrow/staking tokens:** locked supply = cumulative Transfer in ? out of the escrow, reconstructed from base-token logs for 6 curated escrows (veCRV, vlCVX, veFXS, xSUSHI, stkAAVE, veYFI). **All 6 validated to <3% of live on-chain balanceOf** (CRV 854M vs 855M live; YFI exact). veBAL (BPT lock) and SNX (C-ratio) excluded with reasons.

5. Channel 2 — holding duration: NOT BUILT — Entry 24

Coin-age across the panel needs full chain history or a paid API; per the spec’s operating principle the gap is flagged rather than proxied. Schema column kept (always NaN).

6. Channel 3 — voting (canonical, Snapshot) — Entry 25

`phase1_channel3_voting.py` pages all closed proposals per space and computes monthly token-weighted turnout = `mean(scores_total)/circulating`. Space map = DeFiLlama governanceID (35) ? 27 curated DAOs by explicit `cmc_id` (avoiding the UNI=4113-vs-7083 symbol-collision trap). A token-weight validity guard nulled 1p1v spaces (snxgov, enzyme, ilvgov). Result: 55 DAOs, 51 with valid turnout, 1,178 valid asset-months.

7. lambda assembly — Entry 27

`phase1_assemble_lambda.py`: z-score each channel within each monthly cross-section (requiring ≥ 2 assets so a std exists), equal-weight the available standardized channels, no imputation, record `n_channels`. Result: **1,308 asset-months, 51 assets, 2020-08->2026-05; 253 two-channel (the 6 vote-escrow tokens), rest one-channel**. Noted that `lambda_z` is a standardized ranking, not a [0,1] level — Phase 4 must map to `lambda/(1?lambda)`.

8. Coverage report + logs

Wrote `03_data/PHASE1_COVERAGE_REPORT.md`, appended Decisions-Log Entries 21–27, this session log, and updated `time_log.md`. Committed + pushed at session end.

Judgment calls made this session (all logged in the Decisions Log)

1. Canonical chain/Snapshot sources for lambda numbers; DeFiLlama demoted to a registry (human-directed).
2. Event-log reconstruction (not state reads) forced by the no-free-archive finding.
3. ETH staking from the deposit contract instead of the now-paywalled beaconcha.in.
4. Channel 2 left as a documented gap rather than a weak proxy.
5. Curated, explicit-cmc_id space map + token-weight validity guard for voting.
6. Curated 6-escrow lock set (clean base-token holds only); veBAL/SNX excluded.
7. =2-asset standardizability rule for the monthly z-score.

Open items carried forward

- Finish the full ETH staking series (resume the script); decide post-Shapella netting.
- Channel 2 (coin-age) is the highest-value missing channel — decide on a paid source.
- Widen Channel 1 via CMC `detail.platforms[]` + more curated escrows / non-EVM explorers.
- Phase 4 must convert `lambda_z` to a level for the SoV/MoE `lambda/(1?lambda)` map.

Activity Types: Prompt ? Direct ? Review ? Decision (per prior sessions' legend).

/newpage

session_010_2026-06-24_phase1_closeout_phase2_pq.md

AI Conversation Log — Session 010 (Phase 1 close-out + Phase 2 PQ scoping)

Date: June 24, 2026 **Model:** Claude Opus 4.8 (Anthropic), exact model id `claude-opus-4-8` **Interface:** Claude Code CLI (agentic coding session) **Working directory:** `C:/AFA_2027_QTM_Crypto` **Human:** Moazzam Khoja

AFA documentation note. This file is the structured companion record for the session. The *primary* AFA record is the verbatim terminal transcript; this log faithfully summarizes the prompt, the AI's reasoning, every judgment call, the human's interventions, and the verified-source decisions, but the raw transcript should be preserved alongside it.

Initial prompt (verbatim intent)

The kickoff (`04_code/CLAUDE_CODE_PHASE2_KICKOFF_PROMPT.md`) scoped **two parts in one session**: Part A — Phase 1 finishing touches (finish the ETH staking series; explore a free Channel-2 BTC coin-age workaround before concluding infeasible; confirm $N=250$; revisit the 16 gray-zone classifications with real lock data; fix the coverage report's internal asset-month/asset-count inconsistency; optionally enrich the identity map). Part B — build Phase 2 NVT_GL (MC / PQ/*) on the frozen Phase 0 universe, verifying every new source's live free access first, then a Phase 2 coverage report.

The session was instructed to read three documents in full first (`PHASE1_COVERAGE_REPORT.md`; Decisions Log Entries 21–28; spec ?4 + ?7), which it did.

Part A — completed (all items)

1. **ETH staking series — FINISHED.** Re-ran `phase1_channel1_eth_staking.py`; the monthly checkpoint already held all 66 months (2020-12 -> 2026-05) from a prior resumable run, so the script

rebuild the CSV cleanly with no new API calls. Cumulative deposits 2.17M -> 86.16M ETH (ratio 0.019 -> 0.714). Method kept as-is (no post-Shapella netting — new scope, per Entry 28). Validated against the known 2.17M-at-2020-12 anchor.

2. **Channel 2 BTC coin-age — re-checked live, still a gap.** Probed free explorers: mempool.space / blockstream (Esplora) expose no aggregate coin-age; blockchair /stats carries `cdd_24h + hodling_addresses` but only as a current 24h snapshot (historical CDD chart not on a free endpoint — 404/401 + bot-protection); **bitcoin-data.com / bgeometrics** has a genuine free, keyless CDD/dormancy/ancient-supply API but the free tier serves only a trailing ~4 years (2022-06->present) and is hard-limited to **10 requests/hour** (`RATE_LIMIT_HOUR_EXCEEDED`) — it cannot reach the pre-2020 depth that is the whole point. No free, panel-usable source; no paid source procured (deferred). Channel 2 stays a documented gap.
3. **N=250 — confirmed, no change.** Coin-side gap is a source problem, not a universe-size problem; stated on record.
4. **16 gray-zone names — revisited, all stay other.** Only the 6 curated escrows have lock data (none of the 16). OP/MNT/RPL/SSV have a Snapshot voting space (so they appear in lambda as **other**, voting-only) but no security lock; the other 12 have neither. Zero reclassifications; `classification_table.csv` unchanged. Per-name reasons logged.
5. **Report inconsistency — reconciled.** After the ETH resume the correct final figures are **1,374 asset-months / 52 distinct assets** (the old ?5 “1,326/52” vs ?8/Entry 27 “1,308/51” mismatch resolved; +66 = ETH’s now-complete series). Re-ran `phase1_assemble_lambda.py`; updated every section of `PHASE1_COVERAGE_REPORT.md`.
6. **(Optional) CMC platforms[] identity-map enrichment — deferred** (explicitly optional, not required before Part B).

All of the above logged as **Decisions Log Entry 29**.

Part B — paused at the PQ-definition decision (human-directed)

The AI verified Phase 2’s PQ sources live before building: **DeFiLlama fees** (protocol-level `/summary/fees/{slug}` and chain-level `/overview/fees/{chain}`) are free, keyless, and historical (Uniswap fees to 2018-11; Ethereum chain fees to 2018-03; 339 chains, 2,306 fee protocols). DeFiLlama slug coverage among observed assets is sparse (241/1,939), so PQ density will mirror lambda’s DeFi-token/major-L1 lean.

Before building, the AI surfaced a **methodological divergence** between the human’s stated instinct and the spec/theory + free-data feasibility:

- **Human intervention.** The human stated their understanding that PQ = economic activity, proposing **TVL for tokens** and **on-chain transaction volume (tracked from Etherscan via the contract address) for coins**, and asked whether that matched the AI’s strategy.
- **AI response.** Not exactly, on two specific points:
 - **Tokens/TVL:** PQ in this paper is a *throughput flow* ($M?V = P?Q$; Appendix A levelizes projected throughput), whereas **TVL is a stock**. MC/TVL is a real DeFi multiple but not NVT in the paper’s sense; the spec ?4.1 asks for fees / DEX volume / active users (flows). Recommended fees/DEX-volume as PQ, with TVL captured as a side column for a robustness MC/TVL cut.
 - **Coins/transfer volume:** agreed it is the spec ideal, but it hits the same free-archive
 - * rate-limit wall the human already approved in Phase 1 — native-coin value transfers are not logs (need archive), and whole-token ERC-20 transfer summation is the un-targeted full-set version of what Channel 1 did for 6 escrows (Entries 21/24). Recommended DeFiLlama **chain fees** as the free historical on-chain-throughput proxy, optionally with a flagship (ETH/BTC) true-volume cross-check.
- **Human decision.** Rather than resolve the two PQ-definition calls inline, the human asked for a **status report capturing the questions** to discuss in **Cowork** (where the full project context lives), and to **commit** so the logs/history are preserved for integrity. The AI accordingly **paused the Part B build** at the decision point, wrote `06_documentation/PHASE2_PQ_DECISION_STATUS.md` (Part A status + the two PQ decisions with trade-offs and recommendations), and committed.

No NVT_GL code was written this session — the build is intentionally gated on the two PQ decisions. No Phase 3 work was started.

Files changed / added this session

- 03_data/phase1/channel1_eth_staking.csv — completed 66-month series (rebuilt from checkpoint).
- 03_data/phase1/lambda_panel.csv, _lambda_channel_diagnostics.csv — re-assembled (1,374/52).
- 03_data/PHASE1_COVERAGE_REPORT.md — reconciled to final numbers; Channel-2 audit; N=250; gray-zone.
- 04_code/DATA_DECISIONS_LOG.md — **Entry 29** (Part A close-out).
- 06_documentation/PHASE2_PQ_DECISION_STATUS.md — **new** decision memo for Cowork.
- 06_documentation/ai_conversations/session_010_*.md — this log.
- 06_documentation/time_log.md — session entry.
- 03_data/raw/defillama/phase2/ — gitignored fee-source verification caches.

Verified-source ledger (this session)

Source	Probed for	Result
Etherscan V2 deposit-contract logs	ETH staking (resume)	? series complete (cached)
mempool.space / blockstream Explora	BTC coin-age	? no aggregate coin-age metric
blockchair /stats	BTC coin-age	?? cdd_24h current-snapshot only; no free history
bitcoin-data.com / bgeometrics /v1/cdd	BTC coin-age	?? free but trailing ~4y + 10 req/hr cap -> not panel-usable
DeFiLlama /summary/fees/{slug}	token PQ	? free/keyless/historical
DeFiLlama /overview/fees/{chain}	coin PQ	? free/keyless/historical (339 chains)

/newpage

session_011_2026-06-24_pq_theory.md

AI Conversation Log — Session 011

Date: June 24, 2026 **Model:** Claude (Anthropic) via Claude Cowork **Human:** Moazzam Khoja

Topic: Resolving the PQ Definition for NVT_GL (Phase 2)

Summary

Follow-on discussion to Claude Code session 010’s PHASE2_PQ_DECISION_STATUS.md, which paused Phase 2 at a PQ-definition fork (Decision 1: token PQ — fees/DEX-volume vs. TVL; Decision 2: coin PQ — chain fees vs. true transfer volume) and recommended fees for both. This session worked through the human’s pushback on that recommendation, landed on a corrected theoretical definition of PQ, and ended in a decision to pilot-test Etherscan feasibility for the corrected definition rather than resolve it by extrapolation.

Key Discussion Points

1. Industry-aggregator research (initial pass)

Searched how Artemis Analytics, Token Terminal, and DeFiLlama define “economic activity.” All three use flow metrics (fees, revenue, volume/GMV) and keep TVL as a separate, parallel stock metric — initial read: supports fees-over-TVL for Decision 1. (This read was later refined — see #2.)

2. The fees-as-tax-revenue correction (the turning point)

The human rejected fees on a sharper theoretical ground: economic activity is the *value* generated by a transaction; fees are the *cost* of facilitating it — structurally identical to a government’s tax revenue not being a measure of GDP. Re-deriving PQ from the paper’s own $M?V = P?Q$ identity confirms this: $P?Q$ is nominal GDP — the value of goods/services exchanged, not the toll collected on the exchange. A protocol’s fee rate is a discretionary, governance-set parameter (fee-switch votes, fee-tier choices) — exactly like a country’s tax rate — so fees move for policy reasons unrelated to real usage. Re-reading DeFiLlama’s own language (“fees... show how much the protocol is *facilitating* in economic activity”) literally actually supports this correction rather than the earlier fees-positive read: a toll booth facilitates billions in freight without its toll revenue measuring that freight’s value.

3. What TVL actually represents

Walked through TVL concretely by protocol type: AMM inventory (LP-supplied tradeable assets), lending pool (loanable funds), liquid-staking AUM. In every case TVL is the **capital stock** that enables activity — the K in a production function — not the output Y. This formalizes the human’s own “TVL organizes economic activity” intuition correctly: capital enables production, it isn’t the production. A high-TVL pool with no trading produces zero realized economic activity that period.

4. The corrected PQ definition

PQ = **transacted value** — the dollar value of what actually moved through the contract: swap/DEX volume for AMMs, loan-origination/borrow flow for lending, notional volume for derivatives — sector-appropriate via the project’s existing **sector** field (Entry 16), rather than one universal proxy forced across heterogeneous protocol types. This is also, not coincidentally, what the *original* NVT ratio (Willy Woo’s Bitcoin metric, this paper’s direct namesake) used on the coin side: on-chain transaction *value*, never fees. TVL and fees are both retained as secondary diagnostics (capital-stock control; cost-of-intermediation/take rate), plus a new Volume/TVL turnover diagnostic — a protocol-level restatement of $M?V=PQ$ with TVL as M. Logged as Decisions Log Entry 30.

5. Etherscan feasibility question

The human asked whether Etherscan’s existing Transfer-log mechanism (already proven in Channel 1) could directly compute true transfer volume: pull all Transfer events for a token, multiply by daily price. Mechanically yes — but Channel 1’s escrow approach is fast *because* it filters on one sparse counterparty address; pulling *all* transfers of an actively-traded token removes that filter and will hit the `getLogs` 1,000-log cap far more often, forcing much finer block-range bisection. Entry 24 already found full-supply Transfer enumeration “orders of magnitude beyond” the free key for a different purpose (coin-age); this session could not re-verify that with real numbers because the Cowork sandbox has no network route to `api.etherscan.io` (confirmed directly — even a plain request to google.com fails the same way, so it’s a blanket sandbox egress restriction, not Etherscan-specific). There’s also an independent methodological concern: one swap can fire multiple Transfer events via routing, and exchange-internal transfers add non-economic noise, so raw summation would need netting even if fully extractable.

6. Decision

Rather than resolve scale by extrapolation, run a small bounded pilot in the next Claude Code session (which has live Etherscan access): two tokens (UNI, AAVE), a 30-day window, measure real call

counts/timing, dollarize, and cross-check against DeFiLlama's reported DEX volume to quantify the noise multiplier — see 04_code/CLAUDE_CODE_PHASE2_PQ_PILOT_PROMPT.md. Decision 1 and Decision 2 in PHASE2_PQ_DECISION_STATUS.md stay open pending this pilot's findings.

Documentation updated this session

- 04_code/DATA_DECISIONS_LOG.md — Entry 30 (PQ-definition theoretical resolution)
- 04_code/CLAUDE_CODE_PHASE2_PQ_PILOT_PROMPT.md — new pilot prompt (next: session 012)
- 06_documentation/time_log.md — new row

/newpage

session_012_2026-06-24_pq_pilot.md

AI Conversation Log — Session 012

Date: June 24, 2026 **Model:** Claude (Anthropic) via Claude Code (Opus 4.8) **Human:** Moazzam Khoja
Working dir: C:/AFA_2027_QTM_Crypto

Topic: Phase 2 PQ — Etherscan Transfer-Volume Feasibility Pilot

Summary

Executed the bounded diagnostic pilot defined in 04_code/CLAUDE_CODE_PHASE2_PQ_PILOT_PROMPT.md (written by Cowork session 011, which had no network route to Etherscan). Goal: get *real numbers* on whether raw-Transfer-log transacted-value PQ can be built at panel scale on the free Etherscan key, or whether DeFiLlama's reported volume should be the working PQ source. Pulled all **Transfer** events (no counterparty filter) for UNI and AAVE over May 2026, dollarized, cross-checked against DeFiLlama, and extrapolated. Outcome: cost is *not* the binding constraint — **validity** is. Recommendation: Option (B), DeFiLlama reported volume. Full write-up in 03_data/PHASE2_PQ_PILOT_REPORT.md; decision in Decisions Log Entry 31.

Key Discussion Points

1. Setup correction — the “two UNIs” problem

The prompt named “UNI (Uniswap governance token)”. `universe_panel.csv` has **two** UNI rows: `cmc_id 4113` (a dead 2019 namesake, last observed 2020-12, price ~\$1.43 — not Uniswap) and `cmc_id 7083` (the real Uniswap UNI, observed through 2026-05). Used 7083. AAVE = 7278 (matches the Channel-1 escrow table). Both verified `status='observed'` for the window. Window = May 2026 (most recent complete month; panel `max month_end = 2026-05-31`).

2. Mechanism and a quick probe

Reused the `phase1_channel1_evm_locks.py` `getLogs` recursive block-bisection on the 1,000-log cap, with the `topic1/topic2` address filter dropped (all transfers, not escrow-only). A one-day probe (UNI, 2026-05-15) returned 3,530 transfers in just **9 calls / 8.7 s** — immediately showing the recent-window regime is cheap, contra a naive read of Entry 24 (which was about full multi-year enumeration, a different regime). Processed day-by-day for clean per-day measurement + resumable checkpoints + a per-token call/time budget guard.

3. Measured cost

- UNI: 133,350 transfers, **381 getLogs calls**, max depth 6, 345.6 s (0.91 s/call).
- AAVE: 116,910 transfers, **309 getLogs calls**, max depth 5, 305.1 s (0.99 s/call).
- ~345 calls/token-month, ~11 calls/day, ~0.9 s/call — trivial against the free 5 req/s & 100k/day caps. Refines Entry 24: the wall is full *multi-year* history, not a recent window.

4. The decisive finding — wrong quantity, not just cost

- **UNI** token-transfer volume = **\$0.79B** (May) vs DeFiLlama **Uniswap DEX swap volume = \$36.75B** -> swap volume **46.6x larger**, daily correlation only **0.30**. Summing a governance token's own Transfer events does not measure the protocol's swap throughput (swaps move paired assets through pool contracts, not the gov token).
- **AAVE** raw sum came out **physically impossible** (\$8.2x10?? vs a 15.4M-token supply): **6** transfers carried sentinel-scale values, one of **10?? tokens = 6.5x10??x supply**. Cleaned (outliers > supply removed) = **\$2.75B**, still unrelated to Aave lending throughput (loans are in USDC/ETH/etc.). Extreme form of the spec ?6 wash/internal-churn caveat.
- The correct on-chain swap-volume measure would require enumerating each protocol's pool **Swap** events — i.e. re-implementing DeFiLlama's adapter layer per protocol — out of scope on a free key.

5. Extrapolation

At ~345 calls/token-month, 0.9 s/call, 100k/day cap: 1 token full history ~23.5k calls (~6 h); 1 recent month x 127 DeFi-slug tokens ~44k calls (~11 h); **full history x 127 tokens ~1.75M calls (~17.5 days)**; x 241 sluggish assets ~3.3M (~33 days). Recent window or single flagship = feasible; full-panel multi-year history = infeasible as a routine build.

6. Decision

Option (B): DeFiLlama reported sector-appropriate volume as the working PQ source; TVL + fees as side diagnostics (Entry 30); raw Etherscan logs as occasional spot-check only. Ruled out (A) [panel-scale raw] on validity + cost; ruled out pure (C) [flagship raw as a *source*] because it still measures the wrong object — keep flagship raw only as a spot-check. Coin-side PQ (ETH/BTC native transfers) was not in this pilot and faces the harder archive-state wall (Entries 21/24).

Deliverables this session

- 03_data/PHASE2_PQ_PILOT_REPORT.md — full report (cost, dollarization, cross-check, extrapolation, rec).
- 04_code/phase2_pq_pilot.py, 04_code/phase2_pq_pilot_diag.py — pilot + noise diagnostic.
- 03_data/raw/phase2_pilot/* — cached numbers (gitignored; no large raw-log dumps committed).
- Decisions Log **Entry 31**; this log; `time_log.md` updated.

Open / handoff

- Phase 2 build remains **paused pending human review** of the (B) recommendation. Do not build `phase2_pq.py` on raw Etherscan volume. Once (B) is confirmed, build PQ on DeFiLlama volume, sector-routed, with TVL/fees side columns.

/newpage

session_013_2026-06-24_phase2_build.md

AI Conversation Log — Session 013

Date: June 24, 2026 **Model:** Claude (Anthropic) via Claude Code (Opus 4.8) **Human:** Moazzam Khoja
Working dir: C:/AFA_2027_QTM_Crypto

Topic: Phase 2 build — NVT_GL (PQ for tokens + covered coins, then g/r_e/PQ*/NVT_GL)

Summary

Resumed Part B of CLAUDE_CODE_PHASE2_KICKOFF_PROMPT.md (the build) on the now-resolved PQ definition (Entries 30–32). Built **token PQ** = DeFiLlama sector-routed reported volume (Part A), ran the **Step-B1 live coin-source verification**, and built **coin PQ** for the covered rungs (Part B). Then assembled the full **NVT_GL** machinery (MC, PQ0, g, beta, r_e, PQ/*) per spec ?4.1. B1 hit its designated STOP point — **Artemis REST API is paid-only**, leaving 81 material coins uncovered — so on the human’s decision those were deferred to a new **Phase 2b** and the session proceeded with the covered panel. Result: **NVT_GL for 1,821 asset-months, 59 assets**. Full numbers in 03_data/PHASE2_COVERAGE_REPORT.md; decisions in Log Entries 33–34.

Key Discussion Points

0. Finding the repo / unpushed work

Session opened with working dir C:/Users/zdd251 (not the repo). The repo at C:/AFA_2027_QTM_Crypto (C: root, outside the home tree) was located after the human confirmed the path. It had **uncommitted Part-A work** in the tree (modified DATA_DECISIONS_LOG.md with Entries 30–32, modified time_log.md, untracked CLAUDE_CODE_PHASE2_PQ_BUILD_PROMPT.md) — built on, not clobbered.

1. Part A — token PQ (DeFiLlama sector-routed volume), phase2_pq_tokens.py

Routed each of the 127 slugged tokens by DeFiLlama category to the matching free volume dimension. **16 tokens** got a monthly volume PQ (11 DEX, 4 aggregator [flagged double-counting], 1 options). The other 111 are flagged NaN with reasons: 93 have no transacted-value object (yield/farm/lending/ gaming/infra tokens — correct), 8 slug-absent/ambiguous (SunSwap version-split, VELO/SXP symbol collisions — left NaN, not guessed), and **10 perps tokens** whose DeFiLlama derivatives volume dimension is **now paid-gated (HTTP 402)** — a fresh landmine. Fee->volume backout fired for **0** protocols (no protocol-month met the strict single-stable-rate test; all candidates are multi-tier/ variable/reserve-factor fees, excluded by Entry 32). This is expected — most governance tokens are not DEXs and have no swap flow.

2. Step B1 — coin source verification (the human-review pause point)

Three live checks per the build prompt: - **Artemis:** old API dead (410/DNS); current product has **settlement_volume** as a standalone metric (right object) BUT the **free “Lite” tier is Terminal/Sheets-only — no REST API**; Pro (\$300/mo) doesn’t list REST access either. **Rung 2 closed for a reproducible pipeline.** - **DeFiLlama chain DEX** (/overview/dexs/{chain}, 134 chains): with an explicit materiality threshold (30-day DEX volume ? mcap >= 0.01; BTC 9x10?? fails, ETH 0.143/SOL 1.04 pass) only **49 coins (40 material)** qualify for Rung 1. - **blockchain.com** Estimated Transaction Value confirmed for **BTC** (Rung 3, change-excluded, 2010->present).

Net: 81 material coins (XRP, DOGE, LTC, BCH, XMR, ZEC, DASH, ATOM, DOT, MATIC, ?) left with no free transacted-value source. Per the prompt’s explicit instruction, this Artemis-paid / material-gap

case is a STOP-and-report point — not something to resolve by guessing a toll proxy.

3. Human decision -> Phase 2b

Reported the B1 finding + full rung table. **Human's call: "start on what you have and create a Phase 2b to find solutions for the 81 coins."** So: built the covered coin PQ (49 R1 + BTC), flagged the 81 as `GAP:artemis_paid_only`, and wrote `CLAUDE_CODE_PHASE2B_KICKOFF_PROMPT.md` (XRPL APIs for XRP; bitinfocharts/blockchair sent-in-USD for UTXO coins; Artemis-paid option; **XMR permanently impossible** — RingCT hides amounts). Rung 4 (chain fees) auto-applied to zero coins.

4. NVT_GL assembly, `phase2_nvt_gl.py`

PQ0 = trailing-12m sum of monthly PQ (annual throughput). g = trailing-3y CAGR of PQ0 ($\geq 1y$ fallback flagged, capped +/-). β vs **BTC** (the cap-weighted index is numerically unusable — penny-token returns $\rightarrow \infty$). $r_e = rf + \beta \cdot MRP$ with **$rf=4\%$, $MRP=30\%$ as documented robustness parameters** (the realized BTC premium $\sim 114\%/yr$ is not a forward required return), floored at 0.05. $g_{inf}=3\%$, $n=10$. $PQ/*$ = spec ?4.1 levelized annuity; $NVT_GL = MC/PQ/*$. **1,821 asset-months, 59 assets (46 coins, 13 tokens), 2016-08->2026-05**; 0 pathologies.

Key caveat surfaced: the g -cap binds on **43.4%** of rows and $PQ/*$ scales with $(1+g)^n$, so NVT_GL spans many orders of magnitude driven by g . It is therefore reliable as a **cross-sectional rank/conditioning variable** (exactly how H2/H3 use it via median splits), not a cardinal level. g_{cap} and n are the first sensitivity knobs. All intermediates (β , g , $PQ0$, r_e) emitted so $rf/MRP/g_{inf}/n/k$ vary without rebuild (spec ?4.2).

5. Diagnostics, `phase2_pq_diagnostics.py`

TVL (capital-stock control) + Volume/TVL turnover per Entry 30/31: 2,534 asset-months, median turnover 1.15x — a protocol/chain restatement of $M?V=PQ$ with $TVL \sim M$.

Deliverables this session

- Code: `phase2_pq_tokens.py`, `phase2_pq_coins.py`, `phase2_nvt_gl.py`, `phase2_pq_diagnostics.py`
- Data: `03_data/phase2/{pq_tokens,pq_coins,nvt_gl_panel,pq_diagnostics}.csv`, `03_data/phase2_coin_rung_`
- Docs: `03_data/PHASE2_COIN_PQ_VERIFICATION_B1.md`, `03_data/PHASE2_COVERAGE_REPORT.md`, `06_documentation/CLAUDE_CODE_PHASE2B_KICKOFF_PROMPT.md`
- Log: Decisions Log Entries 33–34; this file; `time_log.md`.

Open items

- **Phase 2b:** source PQ for the 81 GAP-R2 coins.
- **Sensitivity (deferred, spec ?5):** vary $g_{cap}/n/g_{inf}/rf/MRP$.
- **Do not start Phase 3 without review.**

/newpage

`session_014_2026-06-25_phase2b_coins.md`

AI Conversation Log — Session 014

Date: June 25, 2026 **Model:** Claude (Anthropic) via Claude Code (Opus 4.8) **Human:** Moazzam Khoja
Working dir: `C:/AFA_2027_QTM_Crypto`

Topic: Phase 2b — source native settlement-value PQ for the 81 deferred GAP-R2 coins

Summary

Ran 06_documentation/CLAUDE_CODE_PHASE2B_KICKOFF_PROMPT.md: source *native settlement value* (on-chain payment/transfer value in USD — the coin-side analogue of Bitcoin’s NVT denominator; NOT fees, DEX volume, or TVL) for the 81 material coins Phase 2 deferred (Entry 34, Artemis paid-only). Verified every source free/keyless and live before building; flagged rather than guessed; no raw multi-year block iteration. **04_code/.api_keys.json holds only an Etherscan key — no Artemis — so Rung 2 stayed closed.**

Outcome: 8 of 81 coins sourced via bitinfocharts native “Sent in USD” (DOGE, LTC, BCH, DASH, ETC, BTG full-history through 2026-06; BSV stale->2021-08, ZEC stale->2022-05). **73 remain PQ=NaN, all with live-verified reasons** — XRP and XMR are permanent gaps, the other 71 have no free ready-made native series. Re-ran the pipeline: **NVT_GL 1,821 -> 2,526 asset-months, 59 -> 67 assets (54 coins, 13 tokens)**. Decisions Log **Entry 35**; coverage report headline + ?1b updated.

Key Discussion Points

1. bitinfocharts “Sent in USD” — the covered 8, and the BTC-default landmine

bitinfocharts exposes a free, keyless, daily, long-history “Sent in USD” series for **13 tickers only** [btc eth xrp zec doge ltc xmr bch dash etc bsv vtc btg]. The GAP-R2 overlap is DOGE, LTC, BCH, DASH, ETC, BTG (current) + BSV, ZEC (stale).

Landmine caught: the {coin-name}-sentinUSD alias form and any *unrecognised* ticker silently serve **Bitcoin’s** series — verified directly (bch/bsv/btg/nano/peercoin/komodo/? via the alias all returned an identical \$12.2B/day, 2010-> series). The build therefore uses only the canonical **sentinUSD-{ticker}** form and **asserts each covered series’ latest value != BTC’s** (contamination guard in **phase2b_pq_coins.py**). XRP’s page exists but is **empty** (XRP is not a UTXO chain; no “sent in USD” is computed for it).

Honesty flags (carried in pq_coins.csv notes): “Sent in USD” = total on-chain *output* value, so for UTXO coins it is **change-INFLATED** (the opposite of BTC’s blockchain.com change-*excluded* Estimated-Tx-Value series). Consequence: these coins’ NVT_GL *levels* are depressed (e.g. LTC median NVT_GL ~ 5x10??), which reinforces the existing “NVT_GL is a cross-sectional rank, not a cardinal level” caveat (?2a). **ETC** is account-model (no UTXO change). **ZEC** captures only the transparent pool — shielded (zk) tx amounts are cryptographically hidden — and is stale post-2022-05; **BSV** stale post-2021-08. Daily values are summed -> monthly, matching **phase2_pq_coins.py**’s BTC handling (PQ is a flow).

2. XRP (highest-value GAP coin) — no free keyless XRPL volume series

Checked every source named in the kickoff, live: data.ripple.com (Ripple Data API v2, which served **payment_volume**) -> **403/dead**; api.xrpscan.com -> account endpoints work but **no** historical volume endpoint (docs + live probes 404); xrplmeta (s1.xrplmeta.org) -> token-metadata / clio node, not volume; api.xrpldata.com -> XRPL **NFT** API; bithomp -> 403 (key); data.xrplf.org -> nginx default / 404. Raw full-history XRPL ledger iteration (~21.6k ledgers/day) is the forbidden call-volume wall. -> **XRP stays PQ=NaN, documented (no_free_xrpl_volume_series)**.

3. XMR — permanent gap (as the kickoff anticipated)

RingCT cryptographically hides transaction amounts -> native transacted value is unobservable on any source. Marked permanent NaN (**xmr_ringct_unobservable**); never proxied.

4. Next tier (ATOM/Cosmos, KAS, DOT/KSM, FIL, THETA, XTZ, VET, IOTA, NEO, ?) — no free series

Probed live: Cosmos public LCD (`cosmos-rest.publicnode.com`) and `api.kaspa.org` return only **current state** (supply/network); Filfox returns **base-fee** (a toll, not value); Mintscan (`apis.mintscan.io`) and Subscan require **API keys**; blockchair has **no** free historical charts API (`/charts/...` -> 404), only current `/stats`. No free, keyless, ready-made historical USD transacted-value series -> these 71 stay PQ=NaN (`no_free_native_series_p2b`). Raw multi-year native iteration forbidden (Entry 31/32).

Artifacts

- New: `04_code/phase2b_pq_coins.py` (idempotent post-process on `pq_coins.csv`; run AFTER `phase2_pq_coins.py`, BEFORE `phase2_nvt_gl.py`; raw HTML cached gitignored under `03_data/raw/bitinfocharts/`).
- Updated: `03_data/phase2/{pq_coins,nvt_gl_panel,pq_diagnostics}.csv` (schemas unchanged).
- Updated: `03_data/PHASE2_COVERAGE_REPORT.md` (headline, by-year table, ?1b, ?4); `04_code/DATA_DECISIONS_LOG.md` (Entry 35); `06_documentation/time_log.md`.

Open / deferred

- If Artemis Settlement Volume or a Subscan/Mintscan/Glassnode key is later procured, Rung 2 reopens for most of the 71 uncovered coins at once (Settlement Volume only, never the Total-Economic-Activity composite). XMR remains impossible regardless.
- The 8 change-inflated UTXO series should be sensitivity-checked if a coins-only NVT_GL *level* (not rank) ever drives a result.
- **Do not start Phase 3 without review.**

/newpage

`session_015_2026-06-25_dune_pilot.md`

Session 015 — Dune Analytics Feasibility Pilot (token-side PQ gap)

Date: 2026-06-25 **Agent:** Claude Code / Opus 4.8 (local, normal network access) **Prompt:** `04_code/CLAUDE_CODE_DUNE_PILOT_PROMPT.md` **Scope:** Bounded diagnostic pilot — 3 tokens x trailing-30-day. **Not** a Phase 2c build. **Deliverable:** `03_data/DUNE_PILOT_REPORT.md` (+ this log, Decisions Log Entry 36, `time_log`).

Why this session existed

Token-side PQ has a structural gap: DeFiLlama has no transacted-value series for Lending / Liquid Staking and paywalls Derivatives (HTTP 402). Three test tokens — AAVE (Lending), LDO (Liquid Staking), GNS (Derivatives) — are all `pq_usd` NaN in `pq_tokens.csv`. The Cowork session that wrote `dune_pilot_test.py` had no network route to `api.dune.com`; this session has normal access and ran it for real, against the Entry-31 standard (*verify before trusting; raw Etherscan looked plausible but was 46.6x off for UNI and 10?? off for AAVE*).

What I did

1. **Read the required context first** — Decisions Log Entries 31 (Etherscan pilot rejection) and 32 (PQ waterfall), confirmed AAVE/LDO/GNS are still NaN in `pq_tokens.csv`, and read `dune_pilot_test.py`.

2. **Ran dune_pilot_test.py** (via python; the python3 alias here is a Windows Store stub). All three stopped at `needs_manual_column_mapping` — the auto-picker grabbed wrong tables: Aave **Polygon** `aave_evt_transfer`, Lido `steth_evt_submitted` (no USD), Gains per-trade table (no `amount_usd`).
3. **Manual mapping (dune_pilot_explore.py)** — catalog-searched for the correct **normalized (Spellbook)** abstractions with pre-priced USD. Found `lending.borrow` (cross-protocol spell, has `amount_usd`), `dune.gains.result_g_trade_stats_defi_llama` (Gains' own DeFiLlama-feed table), `prices.day` (price join), and the Lido withdrawal-queue table `lido_ethereum.withdrawalqueueerc721_evt_withdrawalclaimed`. Pulled schemas from the search payloads (no execute credits) before writing any SQL.
4. **Aggregations (dune_pilot_aggregate.py)** — trailing-30-day figures. (Note: `/sql/execute` rejected `performance="medium"` as “Invalid performance tier”; free tier only exposes `small`.)
5. **Verification (dune_pilot_verify.py)** — caught and fixed the one trap, and pulled external references.

Numbers (window 2026-05-27 -> 2026-06-25)

- **AAVE** — `lending.borrow`, Ethereum v3, `transaction_type='borrow'`: **\$4.286 B** origination (25,452 borrows). Repays/liquidations are negative `amount_usd`, excluded.
- **LDO** — submit (stake) + `withdrawalclaimed` (unstake), ETH x WETH price: **\$1.583 B** flow (stake \$938 M + unstake \$645 M; net +174 k ETH).
- **GNS** — Gains DeFiLlama-feed `daily_volume`, all chains: **\$1.178 B** (\$39.3 M/day, 84,929 trades).

The trap I caught (Entry-31 standard in action)

First LDO run used `prices.day` filtered by `symbol='WETH'` and implied a nonsensical **\$767/ETH**. Diagnostic showed `symbol='WETH'` on Ethereum matches **3 distinct contracts**, prices \$0.0000008 -> \$2,112 (avg \$795) — scam tokens sharing the ticker, with `AVG()` dragged down by them. Re-ran filtered to the **canonical WETH contract** `0xc02a7756cc2` -> **\$1,768/ETH** (30-day vwap), consistent with DeFiLlama spot **\$1,560**. The contaminated number *looked* fine; only the cross-check exposed it. This is the headline caveat for any hand-rolled price join at panel scale.

Cross-checks

- **AAVE**: origination \$4.286 B ? DeFiLlama Aave-v3 Ethereum borrowed-outstanding \$7.215 B = 0.59x/30d (~1.7-month avg loan life) — order-of-magnitude sane. PASS.
- **LDO**: vwap \$1,768 vs spot \$1,560 — consistent. PASS.
- **GNS**: DeFiLlama per-protocol derivatives endpoint **also 402** (re-confirmed the gap). Corroborated via internal consistency (`daily_volume` = component sum to 0.000%), historical \$/day range, per-trade sanity. An independent raw-table reconstruction (`result_gtrade_all_orders_daily_view`, `position_size_dai`) gave ~\$44 M (~27x low) — **logged as inconclusive (ambiguous column = margin/DAI-only, not notional), not a contradiction**, per the prompt's “don't guess at ambiguous columns” rule.

Cost

No Dune credit-balance endpoint/header exists (checked headers directly). Precise call count: **14 catalog searches** (not query-credit-metered) + **9 /sql/execute runs** (all `small`; ~130 datapoints total) + 1 rejected medium call (0 cost). Only 3 of the 9 executes were strictly necessary. ? on the order of **<100 of 2,500 free monthly credits (<4%)**. Cost is a non-issue at this scale.

Recommendation

(A) — **free tier is sufficient to prototype all three categories at panel scale**, with one gate. Spell tables are cross-protocol, so the whole token panel's history is a few `GROUP BY project, block_month` queries, not one-per-token — credit-cheap even for multi-year backfill. **The one risk** that could downgrade to (B): free tier only exposes the `small` engine (the LDO join already took 66 s); a multi-year full-panel scan

could hit per-query limits and need the paid **medium/large** engine. **Gate before requesting funds:** run a free-tier panel-scale dry-run over the real token list; only buy a plan if **small** times out — and then for the engine, not for credits. **Do not start Phase 2c, and do not buy a Dune subscription, before that dry-run and human review.**

Files

- New: 03_data/DUNE_PILOT_REPORT.md; 04_code/dune_pilot_explore.py, dune_pilot_aggregate.py, dune_pilot_verify.py; raw JSON in 03_data/raw/dune_pilot/.
- Updated: 04_code/DATA_DECISIONS_LOG.md (Entry 36), 06_documentation/time_log.md, this log.
- Unchanged: pq_tokens.csv and all Phase 2/2b panel outputs (diagnostic pilot only — no panel writes).

/newpage

session_016_2026-06-25_dune_dryrun.md

Session 016 — Dune Free-Tier Full-Panel Dry-Run (the funding-gate test)

Date: 2026-06-25 **Agent:** Claude Code / Opus 4.8 (local, normal network access) **Prompt:** session kickoff (Dune dry-run; continues DUNE_PILOT_REPORT.md ?5) **Scope:** Diagnostic only — coverage of the real 17-token NaN list + full-history **small**-engine feasibility. **Not** a Phase 2c build. **No panel outputs written.** **Deliverable:** 03_data/DUNE_DRYRUN_REPORT.md (+ this log, Decisions Log Entry 37, time_log).

Why this session existed

Session 015's pilot recommended **(A)** (free Dune tier solves the token-side PQ gap) but made it conditional on one untested risk and two unchecked caveats (DUNE_PILOT_REPORT.md ?5): 1. Free tier exposes only the **small** query engine; a single LDO join already took 66 s on 30 days. Could a **full-history, full-panel** scan time out and force a paid **medium/large** engine? 2. (a) Is every NaN token's protocol actually in Dune's spellbook? (b) Does a full multi-year grouped query complete on **small**? — both for the **real** token list, not the 3-token sample. Moazzam has university approval to buy a paid plan **if** this dry-run shows the free tier can't cope. This session produces that decision's actual evidence. (AI must not purchase anything.)

What I did

1. **Read context first** — Decisions Log Entries 31 (Etherscan rejection — verify before trusting) and 36 (Dune 3-token pilot), DUNE_PILOT_REPORT.md ?5, and the pilot scripts (dune_pilot_{test,explore,aggregate,verify}.py) to reuse the already-validated tables/columns.
2. **Re-derived inputs live** — NaN list from pq_tokens.csv (17 in-scope tokens confirmed all-NaN); panel range from universe_panel.csv (**2015-08 -> 2026-05, 130 months**); Dune key present.
3. **Step 1 coverage** (dune_dryrun_coverage.py, dune_dryrun_coverage2.py):
 - Catalog /datasets/search for the 14 unvalidated protocols (hit a **429 rate-limit** after ~14 calls -> added throttle+retry in coverage2). Found that text-search mostly returns **raw decoded ERC-20/vault contracts**, not pre-priced volume series.
 - Decisive test: **SELECT DISTINCT project** from the normalized spells **lending.borrow** (25 projects) and **perpetual.trades** (28 projects). This is the real coverage criterion (= what made AAVE/LDO/GNS usable).
4. **Step 2 full-history** (dune_dryrun_fullpanel.py): one grouped query per category, **full history**, monthly grouping, **small only**, with wall-clock/engine-ms/row-count and explicit timeout/cap/error watch. Reused the pilot's validated tables and the canonical-WETH price join.
5. **STRK sanity** — its net amount_usd came out slightly negative; verified via a **transaction_type** breakdown that it's a real protocol (\$224.3 M lifetime *borrow* origination), not an empty shell.

What I found

Engine (the gated risk) — disproven. All three full-history grouped queries completed on `small`: - Lending (`lending.borrow`, `aave+strike`, full history): **2.0 s wall / 0.4 s engine**, 138 rows. - Derivatives (Gains DeFiLlama-feed table, full history): **1.9 s / 0.2 s**, 54 rows, ? \$122.8 B notional. - Liquid staking (Lido two-event join x canonical-WETH price, full history): **48.8 s / 43.1 s**, 102 rows, ? stake \$61.1 B / unstake \$44.0 B. ? the worst case, still well inside limits. No timeouts, no row caps, no truncation, no errors. The 66 s pilot figure (30 d) was variance, not a scan that grows dangerously with history.

Coverage — most of the 17 are not on Dune (expected). Only **4 covered** in the normalized spell layer: AAVE + **STRK** (lending; STRK = `project='strike'`, a **net-new** find beyond the pilot), LDO (staking), GNS (derivatives). The other **13** (ANC, BZRX, OM, WXT; AVNT, DDX, HAKKA, HXRO, KP3R, LINA, MIR, MYX, NMR) appear only as raw token/vault contracts — no normalized notional/volume. MIR's only EVM hits are the unrelated `mirror.xyz` NFT product (MIR = Mirror **Protocol on Terra**).

Decision / recommendation

- **(A) for the 4 covered tokens** — free `small` engine handles full-panel full-history scans; **no upgrade needed**. The spellbook gap for the other 13 is a **coverage** gap, not an **engine** gap, so a paid plan would not help them — **(C)**: they stay flagged-NaN. **No (B) outcome.**
- **Net: do not buy a Dune plan.** Recommend (after human review) a Phase-2c build filling AAVE/STRK/LDO/GNS from Dune on the free tier, leaving the other 13 documented NaN. (*Recommendation only — Moazzam decides.*)

Cost

6 `small` executes (~600 datapoints) + catalog searches (not metered) ~ <1% of 2,500 free monthly credits.

Files

- `03_data/DUNE_DRYRUN_REPORT.md` (deliverable)
- `04_code/dune_dryrun_coverage.py`, `dune_dryrun_coverage2.py`, `dune_dryrun_fullpanel.py`
- `03_data/raw/dune_dryrun/*.json`
- Decisions Log **Entry 37**; `time_log.md` updated.

What NOT done (per scope)

No Phase-2c build; no write to `pq_tokens.csv` or any Phase 2/2b output; no Dune billing action. Stop after this report pending review.

/newpage

`session_017_2026-06-25_phase2c_diagnostic.md`

Session 017 — Phase 2c Diagnostic: TVL/Fees/Volume/APY Metadata Audit (111 NaN tokens)

Date: 2026-06-25 **Agent:** Claude Code / Opus 4.8 (local, normal network access) **Prompt:** `04_code/CLAUDE_CODE_PHASE2C_DIAGNOSTIC_PROMPT.md` (drafted in the Entry-38 Cowork session) **Scope:** **Diagnostic / feasibility map only.** For each currently-NaN, non-Gaming token: what free DeFiLlama data exists, and could any of it plausibly become a transacted-value PQ given the protocol's economic model? **No `pq_tokens.csv` write, no PQ value computed, nothing purchased.** **Deliverable:** `03_data/PHASE2C_DIAGNOSTIC_REPORT.md` (+ this log, Decisions Log Entry 39, `time_log`).

Why this session existed

The Entry-38 Cowork discussion challenged the Dune dry-run's 4-of-17 coverage as too thin and asked whether TVL (a stock) could be converted into a PQ flow via a calibrated "turnover" rate (the AK-model framing, $Y=A?K ? PQ=A?TVL$). This session is the live, local audit that decision required: walk the full 111-token NaN universe (not just the 17 Dune-scoped tokens), check what DeFiLlama actually has per token, and test whether TVLxturnover is statistically defensible against a real comparable-protocol cohort — before any build is authorized.

What I did

1. **Read context first** — DATA_DECISIONS_LOG Entries 30 (TVL is capital stock, not PQ), 31 (Ether-scan rejection — verify before trusting), 32 (fee->volume inversion rule), 36/37 (Dune pilot + dry-run; the 4 covered tokens AAVE/STRK/LDO/GNS), 38 (this session's framing); DUNE_DRYRUN_REPORT.md ?2 (the 25 lending + 28 perp DISTINCT project cohorts); pq_tokens.csv; dune_dryrun_fullpanel.py.
2. **Step 1 — re-derived the worklist live** (phase2c_defillama_metadata.py preamble): 127 tokens, 16 covered, **111 NaN**, ? 7 Gaming = **104 audited**. Category counts matched the kickoff exactly. NVT collision noted (slug nerve-staking, key by slug not ticker).
3. **Verified every endpoint shape live before relying on it** (not from docs): protocol/{slug} tvl[], summary/fees/{slug}?dataType=dailyFees|dailyRevenue, summary/{dexs,derivatives,options}/{slug}, yields.llama.fi/pools, bridges.llama.fi/*.
4. **Step 2 — per-token metadata for all 104** (phase2c_defillama_metadata.py): TVL/Fees/Revenue/direct-volume/APY presence + date ranges. Then **investigated the 7 Dexs-category tokens for slug mismatches** (the kickoff's "check direct volume FIRST" instruction), verifying protocol identity by cmcId.
5. **Step 3 — turnover-calibration cohort** (phase2c_turnover_cohort.py + phase2c_turnover_refine.py): pulled the full lending.borrow (25 proj, borrow-filtered) and perpetual.trades (28 proj) cohorts from Dune; mapped each to a DeFiLlama slug **individually verified by name/cmcId** (not string-matched); computed turnover = PQ/TVL per protocol-month; ran a refinement pass summing TVL across all version slugs (aave-v1+v2+v3+v4, ?) to separate true dispersion from version-scope artifact.
6. **Step 4 — per-token verdicts** (phase2c_verdicts.py) + wrote the report, this log, Entry 39, time_log.

What I found

Only 5 of 104 tokens have a defensible free-tier transacted-value path — and 4 were already known. - AAVE, STRK, LDO, GNS — the Entry-37 Dune-spell set (unchanged). - SUN — **the one genuinely-new find**. Its AMM is SunSwap; DeFiLlama's dexs vertical carries it under sunswap-v1/v2/v3 (the stored slug sun.io is not in the vertical -> why Phase 2 missed it). Direct volume 2020-08->2026-06, no proxy needed. Verified by identity (sun.io cmcId 10529; the AMM is SUN's).

Two symbol-collision traps correctly avoided (cmcId mismatch, exactly the NVT-collision hazard): - dexs 'nerve' = Nerve Finance (NRV, cmcId **8755**) != NVT/Nerve Network (cmcId **5906**). - dexs 'velodrome-*' = a different VELO (Optimism) != velo-finance (cmcId 7127, Stellar payments).

The TVL->PQ turnover idea is NOT defensible as a level (the session's central result): - **Lending** (816 protocol-months, 21 projects, scope-matched): pooled median turnover ~ 0.28/month, but per-project medians span **0.0008 -> 1.24 (~1,455x)**; core cluster still ~10x; the low tail (venus 0.0008) survived scope-matching, so it's real dispersion, not artifact. - **Perpetual** (436 protocol-months, 16 projects): per-project medians **0.0000 -> 108.8 — no central tendency** (partly real high perp turnover, partly artifact — gains undercounted in perpetual.trades, tigris matched a \$9k-TVL shell). Calibrating one perp rate is impossible.

TVLxAPY (path 3) collapses on availability: yields.llama.fi/pools is a *current* snapshot, so 25 of the 28 Farm/Yield tokens (dead/delisted) have no APY at all; only CVX/FARM/ZBU have a current rate,

and none has a free historical APY series -> weak (constant-APY assumption required).

Fee-inversion fired for zero tokens — 26 have a Fees series but none is a single fixed volume-linked rate (L2 gas, bridge per-transfer, lending reserve factors, variable perps), consistent with Entry 33.

Two NEW landmines (both change kickoff assumptions): 1. **bridges.llama.fi is fully 402-paywalled** (all endpoint shapes) — the kickoff assumed it was free. The 11 bridge tokens have a *valid* object (transfer volume) that's simply not free. 2. The nerve/NVT and velodrome/VELO collisions are live traps (handled above).

Verdict tally (104 tokens)

verdict	n
? Dune spell (AAVE, STRK, LDO, GNS)	4
? direct volume (SUN)	1
~ weak TVLxAPY (CVX, FARM, ZBU)	3
? turnover undefensible	14
? no APY (dead farm/yield)	25
? bridge volume paywalled	11
? symbol collision (NVT, VELO)	2
? no economic model	44

Recommendation handed back

Build (after review) only the **5 viable tokens** — AAVE/STRK/LDO/GNS (Dune free) + SUN (DeFiLlama dexs), zero subscription. **+1 net-new token (SUN) over Entry 37**. Report TVLxturnover and TVLxAPY in the paper as **explored-and-rejected** (the turnover-dispersion negative result is itself a ?6 methodological finding). Leave ~96 tokens documented-NaN. Two paid levers (DeFiLlama Pro for derivatives+bridges; a bridges-API tier) flagged for Moazzam's decision, **not acted on**.

Files

- Report: 03_data/PHASE2C_DIAGNOSTIC_REPORT.md
- Code: 04_code/phase2c_{defillama_metadata,turnover_cohort,turnover_refine,verdicts}.py
- Data (diagnostic): 03_data/phase2/phase2c_{worklist,metadata,verdicts,turnover,turnover_scopematched}.
- Raw JSON: 03_data/raw/phase2c/
- Decisions Log: Entry 39

No panel outputs written. Do not start the Phase 2c build before this report is reviewed.

/newpage

session_018_2026-06-26_repo_sync_and_lambda_tvl_scope.md

Session 018 — 2026-06-26 — Repo-Sync Incident + lambda/TVL Scope-Up Kickoff

Platform: Cowork **Participants:** Moazzam Khoja (human), Claude (Sonnet 4.6, Cowork mode)

Context

Continuation of an in-progress Cowork session. Earlier in the same session (not re-litigated here): confirmed the coin/token x-axis correction (coins use lambda/(1?lambda); tokens use raw lambda, not the ratio, under any label), confirmed NV/TVL as the primary valuation-ratio Y-axis (NV/Fee secondary), and confirmed

that category-wise quadrants/fixed effects and a continuous Fees/TVL covariate are complementary controls, not substitutes — one addresses between-category dispersion, the other within-category dispersion.

Part 1 — Data coverage audit (lambda, TVL, PQ-for-coins, NV)

Moazzam asked for a score of what real data currently exists for: (1) lambda for all tokens and coins, (2) TVL for all tokens, (3) PQ for all coins, (4) NV for all tokens and coins — and where to source the gaps. Audited directly against the live files:

- **lambda:** 52/1,081 in-universe assets have any lambda-channel data (`03_data/phase1/lambda_panel.csv`) — almost entirely token-side (Channel 1: 6 curated EVM vote-escrow tokens + ETH native staking; Channel 3: 55 assets via the curated 56/57-space Snapshot map). Real ceilings as of this session: 127/448 tokens have a DeFiLlama `dl_slug` match, 123/448 have an identified staking/lock contract address, 29/448 an auto-matched Snapshot space. For coins: only ETH (1/633) has any channel data — no PoS-coin staking source has been live-verified yet.
- **TVL:** zero persistent panel exists. `phase2c_defillama_metadata.py`'s `check_tvl()` already fetches each token's full `tvl[]` series from `api.llama.fi/protocol/{slug}` but only keeps presence/range/last-value metadata (104-token diagnostic worklist; 83 confirmed present) — the actual series is fetched and discarded.
- **PQ-for-coins:** 58/633 with real non-NaN `pq_usd` data (`03_data/phase2/pq_coins.csv`); 131 coins attempted via the Phase 2/2b source ladder (Entries 30–35).
- **NV:** 100% solved — `universe_panel.csv` has zero missing `market_cap` rows across all 156,838 asset-month observations.

Delivered as a coverage table + prose in chat.

Part 2 — Scope decision + drafting the next kickoff prompt

Moazzam's direction: build lambda and TVL datasets first; figure out coin-PQ sourcing afterward, in a separate, later session. While re-grounding the lambda/TVL ceilings to draft the kickoff prompt (re-reading `phase1_assemble_lambda.py`, `phase2c_defillama_metadata.py`, `asset_onchain_identity.csv`, `snapshot_space_map.csv`, and Decisions Log Entries 21–27, 30), ran a `git status` / byte-level check on the repo as a routine sanity pass.

Part 3 — Repo-sync incident (the open item from a prior session, now diagnosed and fixed)

`git status` showed **every** tracked file simultaneously staged-deleted and untracked, and a stale `.git/index.lock` (timestamped 2026-06-25 17:56 — i.e. around session 017) that could not be removed from the Cowork sandbox (Operation not permitted; the sandbox's own `ps aux` showed no live git process, but the lock sits on a path mounted in from Moazzam's actual machine, so a still-running process there — invisible to the sandbox — can't be ruled out).

Byte-level comparison of working-tree files against the already-pushed HEAD (`a7058f2`) resolved the scope of the problem:

- **Not real drift (false positives from a Windows CRLF checkout):** `phase1_assemble_lambda.py`, `lambda_panel.csv`, `pq_tokens.csv`, `PHASE2C_DIAGNOSTIC_REPORT.md`, `main.tex`, `DATA_SPECIFICATION.md` — different MD5 against the git blob, but byte-identical once line endings were normalized (`tr -d '\r'`), and identical in pandas-parsed row/coverage counts. No code or data was actually stale; the earlier coverage-audit numbers in Part 1 are accurate.
- **Real drift (genuine content missing from the local working copy):**
 - `04_code/DATA_DECISIONS_LOG.md` — local copy stopped at Entry 32; HEAD has Entries 33–39 (the Phase 2 NVT_GL build, coin PQ Step-B1/Phase 2b, and the full Dune pilot -> dry-run -> Phase 2c diagnostic arc).
 - `06_documentation/time_log.md` — local copy was missing the last ~6 rows (sessions 013–017).

Both files were restored verbatim via `git show HEAD:<path>` (a plain file overwrite — not a git operation, so the held index lock didn't block it) and verified byte-identical to `HEAD` afterward. **No data was actually lost** — both files were intact and fully committed on GitHub the whole time; only the local working-tree copies had silently fallen behind.

This is logged as Decisions Log **Entry 40** and resolves the long-pending “log the repo-sync incident” item.

Deliverable

Drafted `04_code/CLAUDE_CODE_PHASE1_SCALE_LAMBDA_AND_TVL_PANEL_PROMPT.md` for the next Claude Code session: (A) scale lambda Channel 1 (EVM vote-escrow/staking curation) and Channel 3 (voting space map) for tokens past their current ceilings, plus a live, Entry-21-style source verification pass for PoS-coin staking data (coins currently have only ETH); (B) convert the existing TVL presence-check into a real, persistent `tv1_panel.csv` for the 127 `dl_slug`-matched tokens. Coin PQ sourcing is explicitly out of scope — deferred to a separate future prompt. The new prompt opens with a mandatory git-hygiene check (confirm no live process before clearing `.git/index.lock`; re-verify `DATA_DECISIONS_LOG.md/time_log.md` against `HEAD` before appending) so the next session doesn't build on a stale local log.

Open items carried forward

- Coin PQ sourcing beyond 58/633 — explicitly deferred, not yet scoped.
- Confirm with Moazzam whether any process on his machine is still holding `.git/index.lock`; if not, it can be deleted directly and `git add -A` will resync the index cleanly (the working tree itself is correct, per the byte-level check above).
- Carried, untouched this session: Task 22 (bibliography sanity-check), `main.tex` coin/token x-axis edit (not yet authorized/made).

/newpage

`session_019_2026-06-26_lambda_scale_and_tv1_panel.md`

Session 019 — lambda Scale-Up (Channels 1 & 3, tokens + PoS coins) + Real TVL Panel

Date: 2026-06-26 **Agent:** Claude Code / Opus 4.8 (local, normal network access) **Prompt:** `04_code/CLAUDE_CODE_PHASE1_SCALE_LAMBDA_AND_TVL_PANEL_PROMPT.md` (drafted in the Entry-40 session) **Scope:** lambda scale-up + a real TVL panel for tokens. **Out of scope (unchanged):** PQ-for-coins, Channel 2 (holding duration), and the lambda z-scoring/standardizability/weighting logic. **Deliverable:** `03_data/PHASE1_LAMBDA_SCALE_AND_TVL_PANEL_REPORT.md` (+ this log, Decisions Log Entry 41, `time_log`).

Step 0 — repo hygiene (per the prompt's mandatory check)

- No git process running; no `.git/index.lock` present. `git status` clean, `main` up to date with origin (HEAD 84a21e7). No index rebuild needed (the Entry-40 stale-index incident did not recur locally).
- Re-derived the live next-numbers (the prompt warned twice about stale kickoff numbers): last Decisions Log entry = 40 -> next 41; last session file = 018 -> next 019; last time-log row = session 018. All matched the prompt's stated numbers, but were confirmed live, not trusted.

What I did

1. **Read context first** — `DATA_SPECIFICATION ?3/?2.5/?4`; `DATA_DECISIONS_LOG` Entries 21–30, 40; the five Phase-1 channel scripts; `phase2c_defillama_metadata.py::check_tv1()`. Confirmed live network reach to `api.llama.fi` (200), `hub.snapshot.org/graphql` (200), Etherscan V2 (OK) — the Cowork sandbox couldn't reach llama; this local session can.

2. **Re-derived all ceilings live:** 127/448 tokens dl_slug-matched, 123 has _address, 29 has _snapshot; curated voting map 56 spaces / 55 tokens; coin-side lambda = ETH only (1/633). Matched the prompt.
3. **Part B (built first — most concrete):** new phase2_build_tv1_panel.py — full monthly TVL series for all 127 dl_slug-matched tokens -> tv1_panel.csv. 97 non-empty, 4,895 asset-months, 0 fetch failures, 30 expected empties (aggregators/DAOs/chains with no protocol TVL). Stretch goal: verified loose symbol+name matches -> +AXL, +PERP (99 tokens / 4,999 asset-months); rejected CVP/POLS (cmcId-mismatch collision risk) and METIS/HONEY/PUMP/PYTH/WLFI (zero TVL).
4. **Part A.1 (token Channel 1):** live balanceOf sanity-check then getLogs reconstruction for the candidates. **VERIFIED + built:** PENDLE (vePENDLE 22.9%), LQTY (LQTYStaking 57.8%), 1INCH (St1inch 15.8%), RPL (RocketVault 47.3%, flagged shared-vault). **VERIFIED mechanism but series DEFERRED:** GMX (StakedGmxTracker ~65%, Arbitrum) — full-history getLogs over Arbitrum's millions-of-blocks/month is impractically slow on the free tier (>60 s/month); GMX row left commented with rationale, to be built later via account/tokenTx pagination. **REJECTED (documented):** MKR (DSChief holds 0.5% post-Sky), BAL (veBAL wrong token), COMP (delegation, no lock), RUNE (native L1, placeholder address), ANGLE (not in universe). New phase1_channel1_evm_locks_ext.py (4 assets, 214 asset-months); reconstructed latest-locked matched live balanceOf to rounding for all four.
5. **Part A.2 (token Channel 3):** governanceID cross-check yielded 0 new spaces (all 29 already in the curated map). Probed the Entry-25 gap list live on Snapshot (id_in + ranking search): **+ENA ethegovernance.eth** and **+PERP vote-perp.eth** — both official, active, token-weighted (erc20-balance-of on the canonical token); added to the curated map and re-ran channel3 (now 57 spaces / 53 vw_turnout assets). ONDO/WLD = spam-only; PENDLE top hit = StakeDAO's sdpendle (not official, and PENDLE is now C1-covered); MKR/LQTY/RUNE = no verifiable clean Governor -> no VoteCast reconstruction (no-guess rule).
6. **Part A.3 (PoS-coin Channel 1):** Entry-21-style live audit across the reference PoS chains. **BUILT:** ADA via Koios epoch_info.active_stake (70 months, ratio 49->74%) and XTZ via TzKT cycles.totalBakingPower (95 months, flagged Paris-2024 redefinition). **Live-verified gaps:** Cosmos (/pool current-only), SOL/HBAR (current-only), ICP (ic-api 404), DOT/KSM (keyed), AVAX/NEAR/ALGO/ TRX/EOS/SUI/APT (no free historical). New phase1_channel1_pos_coins.py (2 assets, 165 asset-months); no value guessed/interpolated.
7. **Reassembled lambda** (phase1_assemble_lambda.py, logic untouched — it auto-globbed the new channel1_*.csv + the updated channel3_voting.csv). Wrote the report, this log, Entry 41, time_log.

Outcome

- **lambda: 1,374 -> 1,688 observed asset-months; 52 -> 58 distinct assets** (coin 5->7 [+ADA,+XTZ]; token 43->47 [+PENDLE,+LQTY,+ENA,+PERP]; other 4). Channel 1: 7->13 assets (coins 1->3, EVM-escrow 6->10); Channel 3: 51->53. Coin-side staking is now standardizable (ETH+ADA+XTZ co-occur).
- **TVL panel: 03_data/phase2/tv1_panel.csv — 4,999 monthly asset-months across 99 tokens, 2019-12 -> 2026-05.** A valuation-multiple denominator (NV/TVL), NOT a PQ proxy (Entry 30 unchanged).
- Still NOT covered: coin PQ (deferred), Channel 2 (gap), GMX series (deferred), PoS coins beyond ADA/XTZ (no free historical), the 30 empty-TVL tokens (undefined by construction).

Notes / landmines for the next session

- phase1_assemble_lambda.py crashes on the final console print under a cp1252 Windows console (lambda-> chars) — **cosmetic only; the CSV is written before the print.** Run with PYTHONIOENCODING=utf-8 to see the coverage summary. (Not “fixed” — out of the don't-touch-assembly scope.)
- bitinfocharts / DeFiLlama-Pro / Subscan / Solana-Beach all remain paid/keyed — no purchase made.
- Inherited caveat surfaced (not introduced): the TVL panel uses the identity map's one-slug-per-cmcId choice, so multi-version protocols capture only the mapped version (AAVE->aave-v2). Flag for the NV/TVL build.

- **Stop after the report** (per the prompt): no coin-PQ work started, Channel 2 untouched.

/newpage

session_020_2026-06-26_bucket2_bucket3_recovery.md

Session 020 — lambda Recovery: Bucket 2 (non-EVM coin staking) + Bucket 3 (EVM token locks)

Date: 2026-06-26 **Agent:** Claude Code / Opus 4.8 (local, normal network access) **Prompt:** 04_code/CLAUDE_CODE_LAMBDA_BUCKET2_BUCKET3_RECOVERY_PROMPT.md **Scope:** build everything in Bucket 2 + Bucket 3 that verifies **live, response-body**, and upgrade Entry 42's docs-level findings to response-body-verified ones. **Out of scope (unchanged):** PQ-for-coins, Channel 2, lambda assembly logic, Bucket 1, Phase 3. **Deliverables:** 03_data/SESSION020_BUCKET2_BUCKET3_COVERAGE_ADDENDUM.md, Decisions Log Entries 43–49, this log, time_log, the two new build scripts + their CSVs.

Context read first (per the prompt)

DATA_DECISIONS_LOG Entries 26/41/42; PHASE1_LAMBDA_SCALE_AND_TVL_PANEL_REPORT ?2–4; DATA_SPECIFICATION ?0 + ?3.1; classification_table.csv + universe_panel.csv (to recompute the real uncovered-token list); the existing channel scripts (phase1_channel1_evm_locks_ext.py, phase1_channel1_pos_coins.py, phase1_assemble_lambda.py) and the Dune dry-run scripts.

Headline

lambda **1,688 -> 1,880 observed asset-months; 58 -> 62 distinct assets** (coin 7->9, token 47->49). Every build cross-checks to a live on-chain balanceOf/aggregate; every non-build is a response-body-verified rejection/gap, not a silent drop. **No paid tier used; no purchase made.**

Part A — Bucket 2 (non-EVM coin staking)

- **TRX — BUILT, keyless.** TronScan freezeresource returns full daily history with **no API key** (total_freeze_weight); 78 months. **Corrects Entry 42** (no signup needed).
- **SOL — BUILT, keyless.** validators.app /epochs?per=200&page=N exposes total_active_stake **without a token** for every epoch it recorded (begins ~2023-01; earlier epochs null = data vintage, not paywall); 40 months. Refines Entry 42 to “free **and keyless**, ~2023-01+ depth”.
- **DOT/KSM — NOT built (Entry-42 correction).** Subscan era_stat returns **HTTP 403** “API strictly requires an API key”; free key is self-serve but needs interactive email signup, not completable headless. So the docs-level “free-tier” could not be response-body confirmed. No Pro purchase.
- **HBAR / SUI — documented gaps (scoped).** Both have free current-state endpoints (HBAR network/stake 14.6B; SUI sum stakingPoolSuiBalance 7.23B) but **no historical series** without per-account / per-pool aggregation that is keyless-intractable. Not partial-shipped.
- **CELO — documented gap (reclassification confirmed, build failed cross-check).** CELO is on Etherscan V2 (chainid 42220, existing key) and LockedGold is live (balanceOf 85.65M, getTotalLockedGold 82.43M). But free getLogs reconstruction gives only 2.0M (GoldToken Transfer — native locking emits none) / 25.8M (LockedGold native events — L2-migration carried ~57M as state, no re-emitted events) vs 82.43M. Fails the Entry-26 cross-check -> **not shipped** (flag-don't-ship). Clean number needs archive/PRO eth_call.
- **ATOM/INJ/SEI/KAVA/AVAX/NEAR/EOS/ICP/APT — open gaps (access checked, no purchase).** Cosmos = contact-sales (Mintscan) / trial-only points (Bitquery); AVAX = ambiguous pricing page; NEAR = key-gated/ current-only; EOS/ICP/APT = no free historical source. ALGO unchanged structural gap.

Output: 03_data/phase1/channel1_pos_coins_bucket2.csv (TRX+SOL, 118 asset-months, 2 assets).

Part B — Bucket 3 (EVM token locks)

- **Candidate pool: 290** (asset_class='token', uncovered, gov/staking tag or DeFiLlama category, not the 5 final rejects). Entry-26 clean-single-escrow test is the real filter — most are DEX/lending/RWA/meme/chain tokens with no base-token custody.
- **Dune method correction:** the kickoff's balances_<chain>.daily_updates **does not exist** on the free tier; the correct free tables are tokens_<chain>.transfers (cumulate in/out of escrow) and tokens_<chain>.balances. BSC schema = tokens_bnb. Etherscan V2 free key **does not** cover Base/BSC (does cover Ethereum/Arbitrum/Celo), so AERO/CAKE were verified via keyless public RPC balanceOf and built through Dune.
- **BUILT (each cross-checks final cumulative vs live balanceOf): GMX** (Arbitrum StakedGmxTracker, recon 6,162,450 vs 6,160,000 = 0.04% — retires the Entry-41 deferral; built first as the method check), **AERO** (Base veAERO, 968.4M vs 968.4M = 0.00%, 50% of supply; ratio>1 vs CMC circulating flagged), **CAKE** (BNB veCAKE, 5.897M vs 5.897M = 0.00%; small ~1.5% share flagged, RPL standard).
- **AXS REJECT** (staking on Ronin, no free EVM index). **VELO DEFER** (v1 cmc 7127 vs v2-token escrow — cmcId/symbol collision the project forbids).

Output: 03_data/phase1/channel1_evm_locks_bucket3.csv (GMX+AERO+CAKE, 101 asset-months, 3 assets).

Assembly

Re-ran phase1_assemble_lambda.py (untouched; auto-globs channel1_*.csv). 1,880 asset-months / 62 assets; coin 9, token 49, other 4; 2-channel months 322->354 (GMX's voting+lock); Ch1 standardizable months 73->78. GMX upgraded 1->2-channel (it already had a Channel-3 Snapshot value).

New/changed files

- 04_code/phase1_channel1_pos_coins_bucket2.py (new) -> 03_data/phase1/channel1_pos_coins_bucket2.csv
- 04_code/phase1_channel1_evm_locks_bucket3.py (new) -> 03_data/phase1/channel1_evm_locks_bucket3.csv
- 04_code/_celo_event_check.py (kept: CELO reconstruction cross-check, reproducibility)
- 03_data/phase1/_bucket3_candidates.csv (the 290-candidate pool)
- 03_data/phase1/lambda_panel.csv (re-assembled)
- 03_data/SESSION020_BUCKET2_BUCKET3_COVERAGE_ADDENDUM.md; DATA_DECISIONS_LOG Entries 43-49

Stop point

Bucket 2 + Bucket 3 done. **Do not start Bucket 1 or Phase 3 without review.** Open items for Moazzam: free Subscan key (DOT/KSM), the VELO v1/v2 identity call, and the Tier-5 access gates (no purchase made).

/newpage

session_021_2026-06-29_token_bucket1_exhaustive_reaudit.md

Session 021 — Token Bucket-1 Exhaustive Re-Audit

Date: 2026-06-29 **Model:** Claude Code / Opus 4.8 **Working dir:** C:/AFA_2027_QTM_Crypto **Kickoff:** 04_code/CLAUDE_CODE_TOKEN_BUCKET1_EXHAUSTIVE_REAUDIT_PROMPT.md **Scope:** Convert all 398 token-side “unrecoverable” cmc_ids (+ reconfirm the 6 already-rejected + resolve VELO) into individually documented, source-checked verdicts. Token-side only. No paid tier, no purchase.

What was done

Stage 0 — worklist rebuilt live. In-universe `asset_class='token'` `cmc_ids` not in `lambda_panel.csv`, minus VELO = **exactly 398** (448 ? 49 in `lambda` ? 1 VELO; reconciled to the penny). Cached 290-file not trusted. Found AXS/RUNE/MKR are inside the 398; BAL/COMP are already in `lambda` via Channel-3 voting; ANGLE is out-of-universe.

Stage 1 — bulk DeFiLlama triage, all 398, by cmcId never symbol. Live `api.llama.fi/protocols` (7,742 protocols). 92 of 398 have a clean `cmcId`-matched DL protocol; 306 have no DL protocol and no contract on file (“no on-chain identity available to check”). Rejected the symbol-matched `defillama_categories` as the Entry-20 collision landmine (it falsely promoted DOT-cmc814, HONEY, DRIFT, VOLT, LAYER to “Liquid Staking”).

Stage 2 — per-token check of all 92 DL-matched (no sampling). - 2a: `/protocol/{slug} chainTvls` - > 36 expose a **staking** bucket carrying a raw **staked-token-quantity** series (stronger than the USD proxy the kickoff anticipated). Computed `staked-qty` ? circulating for all 36. - 2b: Dune `tokens_<chain>.balances/net-transfers top-holder` query for every one of the 36 — does ONE contract hold the base token in the DL-reported staked amount (the Entry-26 identity)? - 2c: Artemis re-tested live — `app.artemis.xyz` -> `classic.artemis.ai`, JS SPA, no free per-asset data (reconfirms Entry 2/14). Not signed up, not paid.

Result — 5 BUILDS, each reconstructed via the Bucket-3 Dune transfers method and cross-checked to live `balanceOf` at **0.00%**:

cmc	sym	escrow	ratio
7737	API3	Api3Pool (Ethereum)	~74%
3835	ORBS	StakingContract (Ethereum)	~42%
2930	IQ	HiIQ veIQ (Ethereum)	~9%
35509	VVV	Venice staking (Base)	~73%
7127	VELO	veVELO (Optimism)	~7.4%

VELO — Entry-48 deferral RESOLVED. DeFiLlama’s own Velodrome V2 *and* V3 entries both carry `address=optimism:0x9560e827?` — the exact `cmc-7127` token. So it is the current canonical token (not a defunct “v1”); CMC and DeFiLlama agree -> the documented mapping the kickoff required. `veVELO` holds 1.296B base VELO.

Rejections (393): 367 REJECT-mechanism (multi-contract restaking like EIGEN; multi-pool like ILV; native/ delegation/farming/Snapshot/meme/stablecoin/wrapped-LST), 29 REJECT-no-data (non-EVM chains outside the free EVM Dune method — HXRO/SUN/ORN/TLM/C98/BRISE; no single escrow reproduces the DL figure — BTCST/KAITO/ATH/?; DL bucket 0 — HEX/SFUND/ADF; WARP `cmcId-1166` collision artifact), 1 N/A (ANGLE out-of-universe). The 6 reconfirmed unchanged.

lambda impact: `phase1_channel1_evm_locks_bucket1.py` -> `channel1_evm_locks_bucket1.csv` (200 asset-months, 5 assets); re-ran `phase1_assemble_lambda.py` (logic untouched). **lambda 1,880 -> 2,080** **observed asset-months, 62 -> 67** **distinct assets (token 49 -> 54).**

Decisions / deliverables

- `03_data/phase1/token_bucket1_full_audit.csv` — 402 rows (398 + VELO + BAL/COMP/ANGLE), one specific reason each.
- `03_data/SESSION021_TOKEN_BUCKET1_EXHAUSTIVE_AUDIT.md` — coverage addendum.
- `04_code/DATA_DECISIONS_LOG.md` Entries 50–52.
- Dune free tier: ~36 **small** executes (~15% of monthly budget). No purchase, no paid tier. Every join on `cmcId`.

Flags / caveats

- API3 & ORBS show `staking_ratio>1` in some months (CMC circulating excludes pooled tokens — AERO/SOL artifact, Entry 49); kept un-capped & flagged, lambda uses z-scored rank not level.
- GATED = 0: no priced self-serve option would recover any of the 398.

Open follow-up

- The 456-coin Bucket-1 has NOT had this individual re-audit — candidate for the same treatment, not started.
- Do not start coin-side Bucket 1 or Phase 3 without review.

/newpage

session_022_2026-06-29_etherscan_nonEVM_lambda_channel_audit.md

Session 022 — Etherscan + non-EVM lambda-channel recoverability audit (contract-read + on-chain confirm)

Date: 2026-06-29 **Agent:** Claude Code / Opus 4.8 **Type:** feasibility / identification pass (NO lambda panel modified)

User's question (evolved over the session)

1. Can a (payable) Etherscan API recover lambda Channels 1/2/3 for the 402 token-Bucket-1 rejects? Read the smart contracts, find the staking/voting events, confirm on Etherscan.
2. (After pushback that the first answer read 0 contracts) — actually do it: read every contract, find the functions, then find the data on Etherscan.
3. Extend to the full token universe (~1,039 expected).
4. Extend to non-EVM chains — each chain has its own transaction repository; assess recoverability.
5. At 66% context, write status + transition + next-session prompt; if finished, write all docs + commit.

What was done

Built a resumable pipeline (`04_code/universe_lambda_pipeline.py`): resolve contract (CMC `detail.platforms[]` + identity map) -> **read verified contract** (Etherscan-V2 `getsourcecode`, proxy->impl followed, cached `03_data/raw/etherscan_src/`) -> classify lambda mechanism from ABI (Ch1 holder-lock events excl. admin/vesting; Ch3 `DelegateVotesChanged/VotingPowerChanged`) -> **compute keccak-256 topic0 and run getLogs** to confirm the event actually fires. ABI presence never accepted as evidence. Deep-read Solidity for HEX/NMR/AST/CORE/ILV/FLOKI to separate genuine from look-alike.

Key results

- **EVM universe (token+other = 1,306):** 901 reachable & contract-read. **Ch1 confirmed = 6** (HEX, NMR, stkAAVE, XAN amount-bearing; AKRO, VSL bare `Locked()`). **Ch3 active confirmed = 34** (UNI, ENS, SUSHI, COMP, GTC, KP3R, BTRST, EIGEN, ONDO, STRK, MNT, BLUR, ?). **Ch3 needs-paid = 15 + Ch1 needs-paid = 1. Ch3 dormant/negligible = 15. Ch2 = all 901.**
- **Measured paid-API gate:** free Etherscan `getsourcecode` works on all chains; `getLogs/tokentx` is **FREE only on Ethereum/Polygon/Arbitrum/Blast and PAID-only on BSC/Base/Avalanche**. This is the real trigger for buying Etherscan Pro (16 BSC/Base/Avax candidates + panel-scale Ch2 throughput).
- **Non-EVM (405 off-Etherscan):** 284 no on-chain identity (dead, unrecoverable); 92 on free-indexer chains (Solana 59, Tron 8, Cosmos/Osmosis/Kava 7, +tail) -> Ch2 recoverable per indexer, Ch1/Ch3 per-project low-yield (native staking/gov = gas coin, out of token-scope); 22 EVM-but-not-Etherscan

-> same method, different explorer. Live-verified free repos: Cosmos LCD, Tron TronGrid, Solana RPC, Cardano Koios.

Corrections this session forced (value of reading vs. asserting)

- Excluded 4 vesting/compliance look-alike “locks” (MVL/WLD/FST/FBTC).
- Excluded 15 dormant-but-declared governance tokens (CORE/SUPER/ILV/FLOKI/PENDLE/?) — infra in bytecode, never used; only `getLogs` reveals it.
- Discovered the chain-specific free/paid `getLogs` boundary (would have shipped false “DORMANT” verdicts on BSC/Base otherwise).

Honesty flags

- Many of the 34 Ch3-active tokens (UNI/ENS/SUSHI/COMP) likely already have a Snapshot Ch3 series (Entry 25) -> on-chain `DelegateVotesChanged` is a cross-check there; net-new only where no Snapshot turnout.
- AKRO/VSL emit amount-less `Locked()` -> need contract-balance reads, not the event alone.
- Non-EVM assessment is at chain-capability altitude (live-API-probed), not a per-program read of all 92.

Artifacts (all committed)

03_data/phase1/universe_lambda_channel_map.csv (1,306), etherscan_lambda_channel_map.csv (402), non_evm_lambda_recoverability.csv (405); 03_data/ETHERSCAN_LAMBDA_CHANNEL_EMPIRICAL.md, ETHERSCAN_LAMBDA_CHANNEL_FEASIBILITY.md, NON_EVM_LAMBDA_CHANNEL_ASSESSMENT.md; 04_code/universe_lambda_pi + per-event detail CSVs; raw caches under 03_data/raw/; Decisions Log Entry 53; status/transition/next-prompt in 06_documentation/SESSION022_STATUS_AND_NEXT_SESSION.md.

Status

Identification map complete (EVM + non-EVM). **Nothing built into lambda.** Next: dedupe the 34 Ch3 vs Snapshot, build the 6 Ch1 series, decide the Etherscan-Pro purchase for the 16 paid-chain candidates. No paid tier / no purchase this session.

/newpage

session_023_2026-06-29_hex_akro_reconciliation_survivorship.md

Session 023 — HEX/AKRO Reconciliation + Survivorship-Bias Documentation

Date: 2026-06-29 ? **Agent:** Claude Code / Opus 4.8 ? **Working dir:** C:/AFA_2027_QTM_Crypto Kickoff: 04_code/CLAUDE_CODE_HEX_AKRO_RECONCILIATION_AND_SURVIVORSHIP_PROMPT.md **Basis (stated, not re-litigated):** where session 022’s verified-contract-read + `getLogs` method conflicts with session 021’s Dune-top-holder single-escrow substitute, **022’s method governs.**

Required reading (done first)

- DATA_DECISIONS_LOG Entry 26 (curated single-escrow standard) and Entry 53 (session 022 feasibility map).
- SESSION021_TOKEN_BUCKET1_EXHAUSTIVE_AUDIT.md — HEX rejected “*staking internal to the HEX contract*” / “*DL staking bucket reads 0*”; AKRO rejected in the “*no single contract reproduces the DL staked figure*” cluster (KAITO/ATH/SUPER?).

- SESSION022_STATUS_AND_NEXT_SESSION.md ?2a — HEX “amount-bearing, build-ready”; AKRO “bare Locked(), need contract-balance reads.”
- Cached session-022 artifacts: `universe_lambda_channel_map.csv` (HEX `cmc 5015 StakeStart`, AKRO `cmc 4134 Locked()` `logs=1`), `_universe_lambda_findings.csv`, `03_data/raw/etherscan_src/` (verified source).

Part 1 — HEX (cmc 5015): REJECT -> BUILD

- Read verified HEX source: `stakeStart()` does `_lockedHeartsTotal += newStakedHearts` then `_burn(msg.sender, newStakedHearts) -> non-custodial burn`, no escrow holds the tokens (so 021’s single-escrow `balanceOf` probe and DeFiLlama correctly saw nothing). New construction path vs Entry 26.
- Custody resolved directly: staker `balanceOf` falls (burn), HEX contract `balanceOf` does not rise; staked quantity lives only in `lockedHeartsTotal`.
- Denominator checked, not assumed: contract `NatSpec — totalSupply() = circulating, excludes staked; allocatedSupply() = totalSupply + lockedHeartsTotal`. CMC circ mirrors `totalSupply -> same Entry-49 artifact as AERO/SOL/API3/ORBS`, not a new double-count. Live reads: locked 619.0B, liquid 57.0B, alloc 676.0B; within HEX’s observed window (2020-03..2024-05) `locked < circ` so ratio `< 1`.
- Reconstruction: only `StakeStart(+stakedHearts)` and `StakeEnd(?orig stakedHearts)` move `lockedHeartsTotal` (single `-= site`; `StakeGoodAccounting` doesn’t). Decoded on Dune `hex_ethereum.HEX_evt_StakeStart/_StakeEnd, stakedHearts=(data0>>40)&(2^72?1)` via exact `UINT256` arithmetic (verified == Python decode); `StakeEnd` amount via `stakeId` join.
- Cross-check: recon final **61,900,823,759,862,091,712 hearts == live `globalInfo()[0]` -> drift 0.000000%**.
- Built `04_code/phase1_channel1_hex_stake.py -> 03_data/phase1/channel1_hex_stake.csv` (50 asset-months, ratio 14.3%->43.5%). Re-assembled lambda: **2,080->2,130 asset-months, 67->68 assets**. Logged Entry 54.

Part 1 — AKRO (cmc 4134): REJECT reconfirmed

- Session 022’s “GENUINE Locked()” is a **false positive** of its name-matching Ch1 classifier. AKRO’s address is the **token contract itself** (`TokenProxy->AkropolisToken`); `Locked()` is from a `Lockable` base — `lock() onlyOwner { setLock(true); emit Locked(); }`, an **admin pause switch**, no amount, no escrow. Live `getLogs`: fired **exactly once** (block 8099298, `data=0x`).
- No mechanism to reconstruct -> **REJECT**; 021 stands, now with the contract-level reason. No 021-vs-022 contradiction remains. Logged Entry 55.

Part 2 — Survivorship-bias (Entry 56)

- Re-derived live: `non_evm_lambda_recoverability.csv` class NO-IDENTITY = **284**. Reconciled to the penny against `universe_lambda_channel_map.csv`: `1,306 = 901 reachable + 405 off-Etherscan`; the `405 == exactly the non-reachable rows (overlap 405/405, 0 outside, 0 dupes)`; `405 = 284 + 92 + 22 + 7`.
- Cohort: 83% `asset_class` “other”; 89% pre-2020; ~32 newer non-standard-chain tail. Documented as a **permanent, acknowledged survivorship-bias limitation**, with the bias *direction* (dead cohort is disproportionately low-conviction/no-governance -> truncates the low-lambda tail -> survivors conditionally higher-lambda). Wrote `03_data/SURVIVORSHIP_BIAS_NOTE.md` (paper-ready) + mirrored in `SESSION023_HEX_AKRO_RECONCILIATION.md ?3`.

Rules respected

No paid tier (free Etherscan key + free Dune small). Joins on `cmc_id`. No numerator guessed — HEX reconstructed and reconciled to live chain state at 0.0000%. Log append-only (Entries 1–53 untouched; continued 54/55/56). Did **not** touch the Ch3 dedup, the 16 BSC/Base/Avax paid-gated tokens, or the Ch2 prototype.

Artifacts

- New: 04_code/phase1_channel1_hex_stake.py, 03_data/phase1/channel1_hex_stake.csv, 03_data/raw/phase1_onchain/dune_locks/HEX_stake_recon.json, 03_data/SESSION023_HEX_AKRO_RECONCILIATION.md, 03_data/SURVIVORSHIP_BIAS_NOTE.md.
- Changed: 03_data/phase1/lambda_panel.csv (re-assembled), 04_code/DATA_DECISIONS_LOG.md (+54/55/56), 06_documentation/time_log.md.

/newpage

session_024_2026-06-30_lambda_free_build.md

Session 024 — lambda free-source BUILD (Phase 1)

Date: 2026-06-30 ? **Agent:** Claude Code / Opus 4.8 ? **Kickoff:** CLAUDE_CODE_SESSION024_LAMBDA_FREE_BUILD_PROMPT.md

Decisions Log: Entries 57–60 ? **Report:** 03_data/SESSION024_FREE_BUILD_REPORT.md

Mandate

Turn session-022's free-source identification map into actual lambda series, using ONLY free sources on free chains. Two-phase plan: this session = Phase 1 (free); Phase 2 (paid) explicitly out of scope. Baseline: 2,130 asset-months / 68 assets (session 023, b7ecb73).

Result

lambda 2,130 -> 2,699 observed asset-months; 68 -> 90 distinct assets (+569 / +22).

Tasks

- **A — Channel-3 on-chain delegation (NEW ch3_delegation channel): +560 asset-months / +21 assets.** Deduped 34 ERC20Votes-ACTIVE tokens vs Snapshot (10 overlap -> cross-check; 24 net-new). Built net-new from DelegateVotesChanged replay (delegated weight outstanding ? circulating). 21/24 shipped; TOMI/UXLINK/CYBER excluded (no in-window delegation, block-verified); ETHDYDX excluded (Aave auto-power, not opt-in delegation). Documented as a DISTINCT sub-channel (stock of governance activation vs Snapshot's turnout flow); primary-only into lambda -> no governance double-count.
- **B — Channel-1: XAN BUILT (+9 / +1).** Permanent non-custodial lock; cumulative Locked.value == live lockedSupply() at 0.000000%. VSL rejected (AKRO-pattern cyclic pause), NMR deferred (no Entry-26 anchor), stkAAVE excluded (double-counts AAVE's lock).
- **C — Channel-2: engine built + budget measured (no lambda change).** FIFO coin-age engine validated on MET; address-screen finding 90.8% -> 1.3%; RAD mid-size = 700+ calls, did not complete -> panel-scale blows free cap = Phase-2 trigger. Channel 2 stays lambda NaN, now measured.
- **D & E — time-boxed triage, 0 free clean builds.** EVM-non-Etherscan mostly wrapped/gas (out of token-scope); non-EVM indexers are current-state-only (archive wall) -> Phase 2.

Method compliance

Real events/balances only; Entry-26 cross-check cleared for XAN; cmc_id joins; logs-not-eth_call; assembler z-score/equal-weight logic untouched (channel inputs widened only); no paid tier; Decisions Log append-only (57–60). Context-discipline checkpoint honored — report + transition + Phase-2 kickoff written before commit.

Artifacts

phase1_channel3_onchain_delegation.py, phase1_channel1_freebuild.py, phase1_channel2_holding.py, phase1_channel2_budget_probe.py; channel3_onchain_delegation.csv, channel1_freebuild.csv,

channel2_holding.csv, _channel2_budget_probe.json; re-assembled lambda_panel.csv. Transition + Phase-2 worklist: 06_documentation/SESSION024_STATUS_AND_NEXT_SESSION.md.

/newpage

session_025_2026-07-01_etherscan_pro_channel2_build.md

Session 025 — Etherscan Pro BUILD: P2-2 + Channel-2 at panel scale

Date: 2026-06-30 -> 2026-07-01 ? **Agent:** Claude Code / Opus 4.8 ? **Kickoff:** CLAUDE_CODE_SESSION025_PRO_BUILD_PROMPT
Decisions Log: Entries 61–64 ? **Report:** 03_data/SESSION025_PRO_BUILD_REPORT.md

Mandate

The payoff of the Etherscan API Pro purchase (\$199/mo, 200k calls/day, all chains): (A) run the 16 BSC/Base/Optimism/Linea paid-gated Channel-1/3 candidates (P2-2), and (B) build Channel 2 (holding duration / coin-age) — the Entry-24 “single highest-value addition”, NaN since inception — at panel scale. Baseline: 2,699 asset-months / 90 assets (session 024).

Result

lambda 2,699 -> 6,097 observed asset-months; 90 -> 288 distinct assets (+3,398 / +198).
Channel 2 is no longer the lambda NaN column (Entry 24 closed for the <=3,000-holder cross-section).

What happened

- **Pro verified before any build:** getLogs returns real data on BSC/Base/Optimism/Linea/Avax (free tier refused these). Entry 61.
- **Task A — P2-2 (Entry 62):** mechanism-verified each token on-chain first. **5 BUILT** (AWE 15.25% delegation — material; CHEEL/FORM/LINEA/ZORA ~0% low-activation), ALT excluded (degenerate), **OP deferred** (verified firing, history too large for a correct full replay), 8 DORMANT (event never fired), TNC rejected (bare Locked(address), fails Entry-26). **ch3_delegation** 560->627 asset-months / 21->26 assets.
- **Task B — Channel 2 (Entries 63–64):**
 - **Denominator FIX:** prototype divided by CMC circulating -> shares >100% (RAD 148–398%); corrected to on-chain supply -> bounded [0,1]. Re-validated on RAD (screened 11–27%).
 - **All-month contract screen** (eth_getCode time-invariant -> one pass cleans the series).
 - **Metadata sizing (per Moazzam’s steer):** replaced fetch-until-cap hit-and-trial with a **tokenholdercount** pre-pass (1 call/token; all 793 sized). Build runs smallest-first, defers the tail by metadata, caches month-blocks per chain.
 - **Built 207 tokens / 3,413 asset-months** (median screened HODL 40.6%, 2/197 degenerate). The ~586 tokens > 3,000 holders are the resumable worklist.
- **Data-integrity incident:** an overnight network outage made the engine swallow DNS failures into empty getLogs results -> silently dropped block ranges = corrupted coin-age. Fixed with a robust getLogs (RAISES on network failure, aborts+retries the token), non-caching of failed block lookups, and a circuit-breaker. Deleted + cleanly re-fetched all 65 suspect checkpoints (0 network failures on the re-run). Lesson: a swallowed network error corrupts silently — worse than a crash.

Depth (the kickoff’s headline metric), honestly

2-channel asset-months 354 -> 435 (+81), but the 2-channel SHARE fell 13.1% -> 7.1% because breadth tripled with single-channel ch2. Channel 2 widened the panel (small/mid tokens) but added a 2nd channel to only 4 already-in-lambda assets — the large in-lambda governance/staking tokens are > 3,000 holders and were deferred. Lifting depth (toward 3-channel assets) is exactly what the deferred tail run delivers next.

Method compliance

Mechanism-verified-before-build (Entry 55); no partial/degenerate series shipped; Entry-26 bar applied to TNC (rejected); denominator fix makes the HODL share interpretable; `cmc_id` joins only; `logs-not-eth_call`; assembler z-score/equal-weight logic untouched (only a channel input added); Pro key only, no other paid tier. Decisions Log 61–64 (append-only).

Artifacts

`phase1_channel2_panel.py`, `phase1_channel2_sizeprobe.py`, `phase1_channel2_validate.py`, `phase1_p2p_probe.py`; `03_data/phase1/channel2_holding.csv` (207 tokens), `_channel2_sizes.csv`; `channel3_onchain_delegation.csv` (+P2-2); `lambda_panel.csv` (re-assembled).

Next session (resumable)

1. Channel-2 tail: ~586 tokens > 3,000 holders (esp. large in-lambda governance/staking -> creates 3-channel assets).
2. OP delegation (block-windowed).
3. Checkpoints store raw events -> `--aggregate --recompute` re-derives with no re-fetch.

/newpage

`session_026_2026-07-01_cowork_session025_review_and_plan.md`

Session 026 — Cowork: Session 025 Review + Session 026 Planning

Date: 2026-07-01

Type: Cowork (review + planning, no data build)

Participants: Moazzam Khoja, Claude (Sonnet 4.6 via Cowork)

Context

Session started from a summarized context (prior conversation exceeded context window). Summary covered sessions 001–025 and all standing project rules.

Issues resolved

1. Git state corruption

On arrival, the repo had an unusual git state: all tracked files staged for deletion (D prefix in `git status`) while the working-tree files on disk were an older version of the repo (`DATA_DECISIONS_LOG` at Entry 39, `lambda_panel` at 1,382 rows). The remote (GitHub) was intact at the correct state (HEAD = commit 68d1b79, Entry 60, 2,699 lambda asset-months from session 024).

The staged deletions had not been pushed, so GitHub was clean. Resolution: user ran `git restore --staged .` then `git restore .` from `C:/AFA_2027_QTM_Crypto`, which cleared the staged deletions and restored the working tree to HEAD state. Confirmed by `git status` producing no output.

Note: the sandbox (Cowork) mount remained stale throughout — the fix was verified via the user’s own terminal, which is the authoritative view of the repo state.

2. What session 025 actually accomplished

Read `03_data/SESSION025_PRO_BUILD_REPORT.md` in full. Key findings:

- **Etherscan Pro** verified working on BSC/Base/Optimism/Linea before any build.

- **P2-2 (16 paid-gated tokens):** 5 BUILT into ch3_delegation (AWE material at 15.25%, CHEEL/FORM/LINEA/ZORA at ~0%), ALT excluded (2 events net to 0), OP deferred (verified firing but too large to complete — >80 min/400MB killed), 8 DORMANT, TNC rejected (bare Locked(), AKRO/VSL pattern). Net: ch3_delegation 560->627 asset-months / 21->26 assets.
- **Channel-2 BUILT** at panel scale for 207 tokens / 3,413 asset-months. Key methodology corrections: denominator fix (on-chain supply, not CMC circulating), all-month contract screen. Data-integrity bug (overnight network outage -> silent transfer-history corruption) found and fixed; all 65 affected checkpoints deleted and re-fetched cleanly.
- **lambda 2,699 -> 6,097 asset-months / 90 -> 288 assets.** Breadth tripled; 2-channel share fell 13.1%->7.1% because ch2 additions are mostly new small/mid tokens, not the large governance tokens (those >3,000 holders, deferred by metadata). 3-channel assets: 0.

3. OP delegation explained

User asked what OP delegation means. Explained: OP (Optimism governance token, cmc 11840) uses ERC20Votes; every delegation change fires `DelegateVotesChanged`. Session 025 confirmed 46,974+ events but killed the full download at >80 min. Fix: block-windowed fetch (50k blocks per window). OP delegation enters lambda as ch3_delegation with role=cross_check (OP already has ch3_voting from Snapshot; governance waterfall rule from Entry 57 applies — only one governance channel enters lambda per asset).

Session 026 plan

User requested a session 026 prompt covering both the Channel-2 tail and OP delegation.

Task A — OP delegation (block-windowed): fetch `DelegateVotesChanged` in 50k-block windows from block 0; checkpoint; replay running-balance logic; role=cross_check, does NOT enter lambda as an additional channel.

Task B — Channel-2 tail (>3,000 holders, priority-ordered): - Tier 1 (3-channel targets, already have ch1 AND ch3): CVX, SUSHI, FRAX, GMX - Tier 2 (large governance/staking tokens): ORBS, RPL, LQTY, XAN, API3, CAKE, AAVE, CRV, ENS, LDO, ARB, UNI, COMP, GRT, BAL, YFI, PENDLE, EIGEN, BLAST, ENA, WLF, GNS, MNT, etc. - HEX (9M holders / ~724k getLogs calls): explicitly skipped, logged as permanent deferral. - Engine is already built and checkpointed; session 026 extends the run with higher HOLDER_MAX.

Prompt written and saved to 04_code/CLAUDE_CODE_SESSION026_CHANNEL2_TAIL_AND_OP_PROMPT.md.

Files produced this session

- 04_code/CLAUDE_CODE_SESSION026_CHANNEL2_TAIL_AND_OP_PROMPT.md (new)
- 06_documentation/ai_conversations/session_026_2026-07-01_cowork_session025_review_and_plan.md (this file)
- 06_documentation/time_log.md (to be appended)

Decision: session 025 outputs to be committed

Session 025 results were in the working tree but not committed (only the partial early commit 68d1b79 existed). This commit captures all session 025 outputs plus the session 026 prompt.

/newpage

session_026_channel2_tail_and_op.md

Session 026 — Channel-2 tail (3-channel creators) + OP delegation (block-windowed)

Date: 2026-07-01 ? **Working dir:** C:/AFA_2027_QTM_Crypto ? **Key:** Etherscan API Pro (200k/day)
Decisions Log: Entries 65–67 ? **Report:** 03_data/SESSION026_TAIL_BUILD_REPORT.md

Starting lambda (session 025 close, Entry 64): **6,097 asset-months / 288 assets**, 0 three-channel.

Objective

Two carried-forward Phase-2 items: - **Task A — OP delegation:** finish the one deferred P2-2 item (Entry 62). OP (cmc 11840, Optimism, ERC20Votes) verified firing but its full DelegateVotesChanged history was too large to complete in session 025 (>80 min, killed). Block-windowed incremental fetch is the fix. - **Task B — Channel-2 tail:** add ch2 (coin-age/HODL) to the >3,000-holder tokens deferred by the session-025 metadata filter — especially the tokens that already have ch1 and/or ch3, so ch2 turns them into 2-/3-channel assets (the depth the session-025 breadth expansion lacked).

What was done

Task A — OP delegation (phase1_op_delegation.py, NEW)

Root-caused the session-025 hang: the delegation builder binary-splits getLogs from block 0 to ~153M; the first calls span 100M+ blocks, which Optimism getLogs cannot serve in one shot -> 60-second timeouts cascading = the >80 min. Two fixes in a dedicated OP script: 1. **Block-windowing** — never issue getLogs over a range larger than BLOCK_WINDOW (500,000); within a window that hits the 1000-log page cap, binary-split (shallow, no timeout). 2. **Streaming replay** — OP's DelegateVotesChanged history is *enormous* (the “46,974” in Entry 62 was just the first 60 calls; the real count is **351,186 events / 23,138 delegates by block 15M alone — only 10% of the range**, projecting to ~1.5–2M events). Storing the full event list is the ~400MB/1GB memory wall session 025 also hit. Instead each event is folded into a per-delegate running balance AS IT ARRIVES and the total is snapshotted at each month-end block; the raw log is discarded. Memory is bounded by delegate count (~130–440MB observed), not event count. Windows are fetched in increasing block order and getLogs returns logs in (block, logIndex) order -> the stream is monotonic -> the incremental replay is identical to the batch replay. Checkpoint is incremental (every 30 windows) and resumable.

Timing decision (user-steered): at the calibrated rate (~1.2s/call on Optimism, with some connection resets retried), full OP completion projects to ~2–3.5 hours. OP is a **cross-check** (role=crosscheck; it does NOT enter lambda — OP stays on its Snapshot ch3_voting turnout per the governance waterfall, Entry 57). Because Task B (Channel-2 -> first 3-channel lambda assets) is the higher-value, lambda-moving work and the two cannot run concurrently (shared 10 calls/s limit), the user chose **Task B first, OP after**. OP was paused at block 15M (checkpoint resumable) and the wall-clock redirected to Task B. **OP FINAL STATUS: COMPLETED** — 11,722,683 events / 245,439 delegates; delegated/circ median 7.98% (range 3.54–13.05%), role=crosscheck (NOT in lambda; OP is in lambda via Snapshot ch3_voting). The serial resume STALLED at 68% on an Optimism ConnectionReset storm (fresh-connection-per-call churn on the very active recent-blocks region — 810k events in the first 2M blocks alone); fixed by adding the ch2 engine's pooled keep-alive Session + 8-way concurrency + raise-not-swallow retry, finishing the last 32% cleanly in ~59 min (CPU fell from ~4,000 s of reset churn to near-idle).

Task B — Channel-2 tail, 3-channel creators only (phase1_channel2_panel.py, extended)

Added a WORKLIST env: process an explicit priority-ordered cmc_id list (not smallest-first), bypassing the HOLDER_MAX metadata-defer for the listed tokens (they are deliberately requested).

Classification first (the key efficiency step, user-steered): verified per token which existing channels it has, so ch2 is fetched only where it creates the most value: - **3-CHANNEL creators (have ch1 AND ch3v):** RPL(12k), FRAX(18k), CVX(31k), YFI(52k), CRV(100k), 1INCH(112k), SUSHI(127k), AAVE(200k), GMX(299k). - **2-channel only (one channel):** DDX, ORBS, XAN, LQTY, CAKE, API3, PENDLE.

An initial run interleaved 2-channel tokens and hit **ORBS** — 9k holders but a *hidden giant* (holder_count under-counts transfer volume; ORBS is a heavily-traded 2019 token, >2M transfers, 442MB, still going after ~40 min) for ZERO 3-channel value. Stopped and **restricted the build to 3-channel creators only, smallest-first** (RPL done -> FRAX -> CVX -> YFI -> giants), with a per-token memory heartbeat to catch any further hidden giant within ~5 min. This is the “no wasted effort on 2-channel tokens” discipline.

Built this session (screened HODL-6m = supply held >6mo by non-contract addresses / on-chain supply; the session-025 denominator + contract-screen methodology, unchanged):

token	holders	transfers	getLogs	scr months	screened HODL-6m (range / median)	3-channel months
RPL	12,253	690,529	2,165	57	0.0–22.1% / 17.4%	28
DDX	2,969	42,367	143	32	0.0–62.9% / 49.2%	—
(2-ch)						
FRAX	17,726	1,310,744	3,967	54	~ / 10.9%	52
CVX	31,170	2,827,597	8,477	56	~ / 10.7% (last 21.3%)	56
YFI	52,242	2,152,448	14,916*	70	~ / 53.1% (last 68.8%)	29
CRV/1INCH/SUSHI/AAVE	200k					21/41/35/60
GMX	299,448	deferred to fresh quota day				32

*YFI gl is cumulative-at-completion in the stream run. YFI median HODL-6m 53% is high but economically legitimate (YFI’s small-supply, famously long-term holder base — not degenerate; it varies month-to-month). RPL/FRAX/CVX 10–17% match the RAD governance-token precedent.

STREAMING + CONCURRENT engine (phase1_channel2_stream.py, NEW — the giant-token unlock)

The session-025 panel engine loads a token’s ENTIRE Transfer history into RAM before replaying — fine to ~2M transfers (FRAX 1.3M ~ 460MB) but OOMs on the 100k+-holder giants (AAVE ~40M ~ 10GB, GMX ~60M ~ 15GB; the “did not complete” wall) and is serial (~1.5 calls/s, far under the Pro 10/s). New engine, **validated byte-identical** to the panel engine before use: - **Streaming FIFO replay:** events fetched in block-order batches, folded into per-address FIFO lots AS THEY ARRIVE, month-end HODL snapshots taken when a month-end block is crossed, raw events discarded. Memory bounded by live-address lot state (CVX 2.8M transfers held in only ~96k lots; the old engine would hold 2.8M events). Empty-deque pruning (PRUNE_EVERY) drops sold-out addresses (correctness-preserving) to bound churn/dust-heavy giants (YFI/AAVE/GMX). - **Concurrent windowed fetch:** ThreadPoolExecutor (8 workers) under a global ~9 calls/s token bucket, in block-ordered batches (each batch fully fetched + sorted before feeding the replay -> the stream stays monotonic -> identical to the batch replay). Thread-local pooled Session (HTTP keep-alive) — WITHOUT it, 8 workers x ~9 TLS handshakes/s burned ~6 CPU cores on crypto; with it the job is properly I/O-bound (near-idle CPU) at the full 9 calls/s. - **Validation gates (both passed, 0 diffs):** (1) `--validate 2943` — streaming replay vs panel. _replay on RPL’s stored events, IDENTICAL (incl. with pruning forced every 50k events); (2) `--checkfetch 7228` — full concurrent-fetch + stream pipeline re-fetching DDX vs its stored serial checkpoint, transfers 42,367 = 42,367, max|screened diff| = 0.

Intermediate lambda (RPL+DDX+FRAX+CVX+YFI folded; CRV/1INCH/SUSHI/AAVE still fetching)

FIRST 3-channel assets in the panel's history: n_channels 1->5558 / 2->374 / **3->165** (was 0). 3-channel assets: **RPL(28mo)**, **YFI(29)**, **FRAX(52)**, **CVX(56)** — all ch1_staking+ch2_holding+ch3_voting. 2+ channel share 7.1% -> **8.8%**. CRV/1INCH/SUSHI/AAVE add ~157 more 3-channel asset-months + 4 more 3-channel assets when they land.

RPL screened HODL-6m median 17.4% (raw incl-contracts 61.8%; 22 LP/staking/treasury contracts removed) — bounded and economically sensible, matching the RAD governance-token precedent.

lambda result (after --aggregate + phase1_assemble_lambda.py)

- **lambda 6,097 -> 6,021 observed asset-months / 288 -> 282 assets.** The count dips because the contamination fix removed 6 spam-only dead tokens (their sole channel was a now-nulled contaminated ch2). This session's value is DEPTH + INTEGRITY, not breadth.
- **n_channels: 1 -> 5,356 ? 2 -> 333 ? 3 -> 332** (was 5,662 / 435 / 0).
- **First 3-channel assets (9):** CVX(56), FRAX(52), 1INCH(41), AAVE(38), SUSHI(35), GMX(32), YFI(29), RPL(28), CRV(21) = **332 three-channel asset-months** (all ch1_staking+ch2_holding+ch3_voting). **2+ channel share 7.1% -> 11.0%.**
- Data-integrity: an address-poisoning spam / phantom-lot bug was caught (11 tokens) and fixed with a two-layer guard (per-event value cap VAL_CAP_MULT=100 + per-month exclusion CON-TAM_MULT= 100), re-validated byte-identical on RPL. AAVE's spam-contaminated 2024-08->2026-05 ch2 window is excluded (documented limitation; a per-token totalSupply cap would recover it — future work).

Budget / method compliance

cmc_id joins only; screened HODL denominator = on-chain supply (not CMC circulating); assembler z-score/equal-weight logic untouched (channel input only); OP delegation role=crosscheck (not in lambda); no additional paid subscriptions. getLogs budget well under 200k/day (wall-clock per token, not the daily quota, is the binding constraint — the session-025 finding, reconfirmed).

/newpage

session_027_tv1_expansion_and_ch2_tail.md

Session 027 — TVL expansion + Channel-2 tail completion (2026-07-02 -> 2026-07-04)

Model/interface: Claude Fable 5 via Claude Code (desktop). Kickoff prompt: 04_code/CLAUDE_CODE_SESSION027_TVL_EXPANSION (committed alongside). This structured log is the companion to the verbatim transcript (primary record, AFA rules).

Initial prompt (summary)

Two tasks in priority order against lambda = 6,021 asset-months / 282 assets (session 026 close): **Task A** — TVL expansion (Entry 68): slug discovery for CRV/YFI/FRAX/GMX (+ RPL direct), extend the TVL fetch to all lambda tokens with slugs, bulk cmcId discovery for the rest, chain-level TVL where applicable, rebuild tv1_panel.csv, plausibility-check the 5 priority tokens. **Task B** — Channel-2 tail (Entry 69): build ch2 for the 43 lambda assets with ch1 OR ch3 but no ch2 (holder_count <= 500k), smallest-first, session-026 streaming engine, 200k/day budget with 20k headroom, B2 integrity scan, re-aggregate + re-assemble, B4 sanity checks. Standing rules: cmc_id joins only, append-only decisions log (entries 68-70), no new paid subscriptions, commit+push at session end.

Timeline / what was done

2026-07-02 evening (local). Required reading (Entries 63–67, session-026 report, coverage map, both builders). Task A executed: DL /protocols cmcId lookup -> 0 hits for all four priority tokens AND all 207 no-slug non-coin lambda assets (DL cmcId covers only 1,709/7,770; earlier sessions had harvested every match). Fallback exact symbol+name scan -> the PARENT-protocol finding (CRV/YFI/FRAX/GMX all live under DL parents; parent slug = the token's whole-protocol claim; verified /protocol/{parent} serves aggregated series with correct launch dates). 23 slugs written to identity (incl. UNI 7083-not-4113 collision handling); XVS fixed to venus after venus-finance 400'd; ZORA rejected + reverted. Builder extended: OTHER_ADDS (RPL/SSV/BLUR/RAIN/MV — the 'other'-class filter gap that had silently excluded RPL despite its existing slug) and CHAIN_LEVEL (ARB/OP->'OP Mainnet'/MNT/APE/BLAST via historicalChainTvl; coins excluded — NVT framework; CYBER/PENGU/GBYTE rejected). Panel rebuilt 4,999->6,620 asset-months / 99->130 tokens, 0 failures; A5 gate passed on all five priority tokens; 331/332 three-channel asset-months have TVL (gap = SUSHI launch month). User set session to always-allow for autonomous overnight/day operation; first ch2 launch stopped at user request (~100 calls burned, no checkpoints), resumed next morning.

2026-07-03 (local). Entry 68 logged. Ch2 tail relaunched 14:57 UTC (fresh quota day, DAILY_CAP=140k in-process so a mid-flight giant cannot pass the 180k stop-rule). Run completed 40/43 tokens at 141,005 getLogs (cap-stop after APE, exactly as designed; zero aborts). B2 scan on the 40: clean (worst on-chain/circ ratio 31x HAKKA, legit Entry-49 band; 0 nulled). B4: big-token medians 13–48% bounded. XAN median-0 investigated -> age artifact (launch mid-2025; real 74–77% once lots can exceed 6m), coherent with the non-custodial XanV1 lock. Final 3 (ZRX/COMP/UNI) scheduled via a sleep-until-00:05-UTC background job.

2026-07-04 (UTC). ZRX/COMP/UNI built on the fresh quota day (41,871 calls; UNI 8.3M transfers, the largest of the tail). Aggregate -> 260 tokens / 7,823 rows. lambda re-assembled: **6,021 -> 7,051 asset-months / 282 assets; n_channels 1/2/3 = 5,331/1,388/332; 2+ share 11.0% -> 24.4%.** Entries 69–70 logged; SESSION027_TVL_AND_CH2_REPORT.md written; time log + memory updated; commit + push.

Key decisions (all logged in DATA_DECISIONS_LOG Entries 68–70)

1. Parent-protocol slugs as the canonical TVL denominator for multi-version protocols.
2. Chain-level TVL (chain: prefix) for canonical L2 governance tokens; coins never get TVL.
3. Treasury/Foundation/CEX/bridge slugs kept OUT of the denominator (OP/MNT foundations, ZORA bridge, GBYTE third-party DEX, PENGU).
4. Rejection list logged (NEST ambiguous parents, POWER/M0 unconfirmable, CVP re-rejected, WLFI no series) so future sessions don't re-try them.
5. Ch2 engine + guard thresholds untouched; the 43-token tail accepted into lambda; XAN flagged as an age artifact, not excluded; HEX never attempted (Entry 66 permanent deferral).
6. Budget discipline: in-process cap set BELOW the stop-rule so mid-flight overshoot is bounded; two quota days used (141,005 + 41,871 = 182,876 getLogs), both with headroom.

Artifacts

- 03_data/phase1/asset_onchain_identity.csv (+23 slugs, XVS fix, ZORA revert)
- 04_code/phase2_build_tvl_panel.py (OTHER_ADDS, CHAIN_LEVEL, fetch_chain)
- 03_data/phase2/tvl_panel.csv + tvl_panel_coverage.csv (6,620 asset-months / 130 tokens)
- 03_data/phase1/channel2_holding.csv (260 tokens / 7,823 rows)
- 03_data/phase1/lambda_panel.csv (7,051 asset-months / 282 assets)
- 04_code/DATA_DECISIONS_LOG.md Entries 68–70
- 03_data/SESSION027_TVL_AND_CH2_REPORT.md
- 43 new per-token checkpoints under 03_data/raw/phase1_onchain/holding/

/newpage

Session 028 — Coin Staking (ch1) Expansion + AAVE ch2 Fix

Date: 2026-07-04 ? **Model:** Claude Fable 5 (Claude Code desktop) ? **Kickoff:** 04_code/CLAUDE_CODE_SESSION028_COIN_STA
? **Log entries:** 71–75

What was asked

Two tasks: (A) expand coin Channel-1 staking for the 49 NVT-ready/lambda-absent coins — first the EVM chains on the Etherscan Pro key (BNB/GLMR/MOVR/S/BERA/XDC), then CELO balance-history, then AVAX/NEAR/Cosmos alternative-source probes; DOT/KSM deferred (Subscan signup blocked). (B) recover AAVE's 22 spam-nulled ch2 months.

How it ran (decisions and turning points)

1. **Chain coverage check first:** all 9 target chainids live on Etherscan V2 with the Pro key. pycryptodome for keccak topics; new `_s028_evm.py` helper module.
2. **BNB:** fetched StakeHub ABI -> corrected the kickoff's event-data order (`shares`, `bnbAmount`). Full event fetch (106k logs, 6 event types). Reconciliation table: `del?und+rew` = live ? `totalPooledBNB` over 53 StakeCredit contracts **to the integer** (25,717,017 BNB); adding `MigrateSuccess` -> +80% (migration double-emits Delegated). Series starts 2024-07 (BC-fusion completed 2024-07-14; earlier stake lived on the retired Beacon Chain — documented gap).
3. **Moonbeam/Moonriver:** precompile emits no logs, no aggregate getter; **discovered Etherscan proxy `eth_call` ignores historical tags** (`round()` constant at every tag) — kickoff's fallback impossible as written. Resolution: official RPCs are full archives AND answer Substrate state queries -> read `ParachainStaking.Total` (`twox128` storage key, `xxhash`) at each month-end block. Decode validated via the `eth_call` ? `getCandidateTotalCounted` superset relation (1.054 / 1.015).
4. **Sonic:** full SFC event fetch (246k logs). Naive replay drifted +0.86% and growing; **block-bisection to a single block (60,010,966) proved restakeRewards emits BOTH Delegated and Restake-dRewards** for the same amount. `del?und` alone = archive `totalStake()` at +/-0.0000% at six blocks across the whole history.
5. **XDC:** `XDCValidator` holds native XDC. Three-way tie-out: ? `getCandidateCap` (2.650B) <= event replay (2.668B; +18M resigned-pending) <= contract balance (2.701B; +32.625M CONSTANT genesis/eventless offset, stable at every probed block from 40M). Series = Pro `balancehistory` at month-ends (72 mo), flagged for the pending/genesis components.
6. **BERA:** deposits-only replay = 382.6M BERA > circulating — withdrawals are consensus-side, invisible in EL logs; no free CL archive API (`routescan/berascan/hub` probed). Documented gap.
7. **CELO:** Pro `balancehistory` works on 42220 but ONLY post-L2-migration (block >= ~31.06M). **forno.celo.org serves the FULL archive keyless** — historical `getTotalLockedGold()` (the exact number Entry 46 wanted) at every month-end back to 2020-07. 70 months, closes the Entry-46 gap for \$0.
8. **AVAX:** kickoff endpoints 404; the working shape is `metrics.avax.network/v2/networks/mainnet/metrics/{metric}` — **keyless**, daily to genesis. Semantics anchored to the chain: `P-Chain platform.getTotalStake ~ validatorWeight+delegatorWeight` (additive), not `validatorWeight` alone. 67 months. Resolves the Entry-42/47 AvaCloud ambiguity: free, no account at all.
9. **NEAR/Cosmos:** `NearBlocks` charts = 2,174 days of history but no staking field (`stats/validators` current-only); `StakingRewards GraphQL 401`; `Numia 401`. Gaps stand.
10. **AAVE ch2 — two-stage detective work:** kickoff's (b) denominator-swap reading rejected on mechanics (phantom lots also poison the NUMERATOR — they age at EOA-class addresses). First rebuild with the Entry-70 per-token `totalSupply` cap (16,000,000, verified live; the kickoff's token address ?DDaE8 was wrong — the identity map's ?DDaE9 is canonical) recovered the months but left a fake HODL jump at 2025-01 and supply still climbing to 62M. Forensics on the sharpest jump month (2024-07) found the REAL vector: **fake-value SELF-transfers** from a 0.0-balance address (incl. max-uint256

= Entry 66's 1.16e60, and a fake 10M that passes any cap) — the FIFO pop-then-append mints the shortfall as phantom supply. Fix: **self-transfer skip** (accounting identity) in all three engine variants + the cap as defense-in-depth; second full rebuild -> supply 16.0–18.0M all months, recovered HODL 20.0–31.3%, pre-spam months $\leq 1.9e-5$ from old. **Offline diagnostic over all 210 stored-event checkpoints: zero other tokens affected.** Old checkpoint kept as 7278_AAVE.entry66.bak.

Outcome

- New builder `phase1_channel1_pos_coins_evm.py` -> 358 asset-months / 7 coins (BNB 23, S 17, GLMR 52, MOVR 57, XDC 72, CELO 70, AVAX 67), all Entry-26-passing.
- `lambda`: 7,051 -> 7,409 asset-months / 282 -> 289 assets (coins 9 -> 16).
- Regression-ready coins (`lambda?NVT_GL`): 5 -> 12; 505 overlapping coin asset-months.
- AAVE ch2: all 22 nulled months recovered; 67/67 ch2 months; AAVE 3-channel through 2026-05; panel 3-channel asset-months 332 -> 354 (+22 = exactly the recovered window); 2+ channel share 23.2% (compositional dip from 24.4%: +358 single-channel coin months).
- Entries 71–75; report `03_data/SESSION028_COIN_STAKING_REPORT.md`; gap register with exact Moazzam actions (Subscan free key being the actionable one).

Session hygiene

Single Etherscan Pro quota day, ~45k calls total incl. two full AAVE re-fetches (16,555 + 16,557) — well under 200k. No signups, no purchases. Keyless sources used: Moonbeam/Moonriver official RPCs, `forno.celo.org`, `metrics.avax.network`, `api.avax.network` (P-Chain), `rpc.soniclabs.com`. Commit+push at session end.

/newpage

`session_028_2026-07-04_cowork_coverage_csv_and_session028_plan.md`

Session 028 — Cowork: Coverage CSV + Session 028 Planning

Date: 2026-07-04 **Type:** Cowork (review + planning + tooling, no data build) **Participants:** Moazzam Khoja, Claude (Sonnet 4.6 via Cowork)

Context

Session started from a summarized context (prior conversation exceeded context window). Summary covered sessions 001–027 and all standing project rules.

Work done this session

1. Universe coverage CSV generated

User requested a comprehensive table of all ~1,939 in-universe assets showing current data coverage and what remains needed per asset. Built `03_data/universe_coverage_status.csv` (1,939 rows) by merging:

- `classification_table.csv` (master asset list: `asset_class`, `sector`, `staking_start`, `classification_basis`)
- `lambda_panel.csv` (current `lambda` channel coverage: `has_ch1`, `has_ch2`, `has_ch3_v`, `has_ch3_d`, `lambda_months`, `lambda_n_channels`, `lambda_channels`)
- `nvt_gl_panel.csv` (PQ/NVT_GL coverage: `has_nvt_gl`, `nvt_months`, `pq_source`)
- `tv1_panel.csv` (TVL coverage: `has_tv1`, `tv1_months`)
- `universe_lambda_channel_map.csv` (EVM feasibility: `evm_chain`, `etherscan_reachable`, `ucm_ch1`, `ucm_ch2`, `ucm_ch3`)
- `_channel12_sizes.csv` (`holder_count`, `est_getlogs_calls`)

what_needed column logic: - Coins: `ch1_staking` (if PoS; inferred from `classification_basis` PoS tags) + `pq_nvtgl`; `ch2` and `ch3` excluded per framework decision (coins use NVT, not NV/TVL; `ch3` governance N/A)
- Tokens/other: `ch1_staking` (if `ucm_ch1` available) + `ch2_holding` (with holder count) + `ch3_governance` (if `ucm_ch3` available) + `tv1_defillama` - `coin_staking_type` column: `pos` / `pos_possible` / `pow_only` / `unknown` (derived from `classification_basis` tags)

coverage_status: `complete` / `partial` / `not_started` (stablecoins excluded from `in_universe`)

Key finding: universe is 1,939 assets (858 other, 633 coin, 448 token — no stablecoins in-universe per `classification_table.csv` filter).

Post-session-027 counts (from actual `lambda_panel.csv` + `tv1/nvt` files): - `lambda`: 282 assets (9 coin, 128 token, 145 other) - Token/other `lambda` ? TVL (regression-ready): 80 assets - Coin `lambda` ? NVT_GL (regression-ready): 5 assets - `n_channels`: 225 single / 48 two / 9 three

2. Session 027 review

Read `SESSION027_TVL_AND_CH2_REPORT.md` in full. Key outcomes: - TVL: 99 -> 130 tokens; all 9 three-channel assets now have TVL denominator - `ch2`: all 43 remaining `ch1`-or-`ch3`-no-`ch2` `lambda` tokens built (2+ channel share 11->24.4%) - `lambda`: 6,021 -> 7,051 asset-months (282 assets unchanged — depth session) - AAVE has 22 null `ch2` months (spam-excluded, 2024-08->2026-05) — identified as a fix target

3. Next-steps discussion

Key insight: coin side is the regression bottleneck. 49 coins have NVT_GL but no `lambda`. Priorities: 1. Coin staking (`ch1`) for PoS chains — primarily BNB/BSC (Etherscan Pro in hand), DOT/KSM (free Subscan key needed), CELO (Pro balance history endpoint probe), AVAX (AvaCloud gate check), NEAR (NearBlocks probe), Cosmos alternatives 2. AAVE `ch2` fix — per-token `totalSupply` denominator recovers 22 null months 3. Token breadth `ch2` (~500 not-yet-`lambda` tokens) — lower priority, single-channel adds

4. Session 028 prompt drafted

`04_code/CLAUDE_CODE_SESSION028_COIN_STAKING_AND_AAVE_FIX_PROMPT.md` created. Tasks: - A1: BNB via BSC StakeHub contract `getLogs` (Etherscan Pro, chainid 56) - A2: DOT/KSM via Subscan (check for key in `.api_keys.json`; flag if missing) - A3: CELO via Etherscan Pro `balancehistory` endpoint on chainid 42220 - A4: AVAX via AvaCloud Metrics API gate check - A5: NEAR via NearBlocks historical staking probe - A6: Cosmos (ATOM/INJ/SEI/KAVA) via StakingRewards.com GraphQL + Numia - B: AAVE `ch2` fix using ERC20 `totalSupply()` at month-end blocks as denominator

Files produced this session

- `03_data/universe_coverage_status.csv` (new — 1,939 asset coverage table)
- `04_code/CLAUDE_CODE_SESSION027_TVL_EXPANSION_AND_CH2_TAIL_PROMPT.md` (created earlier this session, committed with the session 028 prompt)
- `04_code/CLAUDE_CODE_SESSION028_COIN_STAKING_AND_AAVE_FIX_PROMPT.md` (new)
- `06_documentation/ai_conversations/session_028_2026-07-04_cowork_coverage_csv_and_session028_plan.md` (this file)
- `06_documentation/time_log.md` (to be appended)

Standing rules confirmed

All standing rules in force: - No paid tier without Moazzam executing the purchase himself - `cmc_id` joins only - Flag, don't guess - `DATA_DECISIONS_LOG` append-only - Etherscan Pro Standard active (\$199/mo, activated 2026-06-30)

/newpage

session_029_breadth_build_and_coin_probes.md

Session 029 — ch2 breadth build (47 TVL-ready tokens) + TVL slug verdicts + PoS coin probes

Date: 2026-07-04 **Agent:** Claude Code (Fable 5) **Kickoff:** 04_code/CLAUDE_CODE_SESSION029_BREADTH_BUILD_AND_COIN_P
Log entries: 76–79. **Report:** 03_data/SESSION029_BREADTH_AND_COIN_PROBE_REPORT.md

What was done (chronological)

1. **Task A0 (multi-chain adaptation):** read `phase1_channel2_stream.py` fully; found the engine already passes per-token `chainid` through every API call (`getLogs`, `decimals`, `month-blocks`, `contract screen`) — the only mainnet-era constraint was the `CHAIN_ID` chain-name lookup in `load_worklist`. Extended it with BSC (56) and Base (8453); no other engine change; guard thresholds untouched. Probed `getLogs` live on 56/137/42161/8453 — all answer. All 47 targets present in the channel map with addresses + `etherscan_reachable=yes` and observed universe months; zero pre-existing checkpoints.
2. **Task A build:** launched the full 47-token worklist (mainnet smallest-first -> BSC -> Polygon -> Base -> RAIN/Arbitrum last) at `WORKERS=8`, `RATE=8/s`, `DAILY_CAP=180000` (20k headroom rule). Ran through the session in the background; results in Entry 76.
3. **Task B (30 min):** probed all six orphaned `dl_slugs` — all six REJECT: DL returns 200 with protocol metadata but EMPTY `tvL/chainTvls/currentChainTvls` on every chain. COW double-checked against the full `/protocols` list + `parentProtocols` (only CoW-named TVL series is balancer-cow-amm under parent Balancer — not COW's claim). No panel rebuild.
4. **Task C1 (EVM coins):** Etherscan V2 chainlist fetched — NONE of Flare/Ronin/Kaia/Cronos/Chiliz/Lisk are covered (64-chain list). POL probed on Ethereum mainnet: `totalStaked()` is validator SELF-stake only (11.7M); the correct aggregate is `currentValidatorSetTotalStake()` (3.58B live). Etherscan proxy ignores historical tags (Entry-71), so free archive RPCs were probed: publicnode requires a personal token, llamarpc/meowrpc/flashbots refuse — **eth.drpc.org serves full mainnet archive eth_call keyless**. POL fetched 2020-06->2026-05 (`_s029_pol_fetch.py`). RON: official RPC pruned; **ronin.drpc.org is a keyless archive**; contract identities verified against axieinfinity/ronin-dpos-contracts; balance route REJECTED (+15.6% over candidate-stake sum) in favor of the getter route. KAIA: public-en reverts historically but **archive-en.node.kaia.io (official) answers klay_getStakingInfo at any block**. FLR: P-chain live getter found, then **PChainStakeMirror.totalSupply() on the official C-chain archive RPC** (registry-resolved) ties to the P-chain at ?0.18%. CRO: staking is Cosmos-side (crypto.org LCD live-only) -> ATOM-class gap. CHZ: 0x?1000 balance series archive-readable but semantics unanchorable keylessly -> gate-open w/ manual action. LSK: no staking contract locatable -> gap.
5. **Task C2 (native):** EGLD **BUILT** (tools.multiversx.com growth-api daily series to 2020-07, ?0.11% vs live economics). STRK **BUILT** (lava archive `starknet_call` on the real staking contract from docs.starknet.io — the kickoff's two addresses were the STRK token and the minting curve, both ABI-verified wrong). XRD **BUILT** (Babylon Gateway historical `at_ledger_state` — 245–287 validators/page, single page guarded). PEAQ **BUILT** (KILT-fork pallet storage `TotalCollatorStake`; Moonbeam's `.Total` is null). TON/FLOW/DFI/DASH/WAN/HYPE = documented gaps; CORE = key-gated (Moazzam action logged).
6. **Builders:** `pol_series()` added to `phase1_channel1_pos_coins_evm.py` (MATIC <=2024-08 / POL >=2024-09 listing handoff, no overlap month); new `phase1_channel1_pos_coins_native.py` emits RON/KLAY/FLR/EGLD/STRK/XRD/PEAQ (`channel1_pos_coins_native.csv`, picked up by the assembler glob).
7. **Task D:** new reusable `build_coverage_status.py` (replaces the pre-027 inline generation; `coin_staking_type` carried forward as static metadata). Final assemble + coverage rebuild + report + log entries + commit/push at session close.

Session outcome

lambda 7,409 -> 9,580 asset-months / 289 -> 337 assets (coins 16 -> 25); regression-ready 92 -> 138 assets (coins 12 -> 20 via lambda?NVT_GL, tokens/other 80 -> 118 via lambda?TVL). Task A: 40/47 ch2 targets built across two quota days (MDX 17.7M and MBOX 21.2M transfers — the two largest ch2 builds ever), 39 entered lambda, B2 clean, 7-token resume list pending. Task B: all six slugs rejected (no TVL on DL). Task C: 8 coin series / 9 lambda assets built, all state reads of chain-own aggregates with Entry-26 cross-checks; 10 gaps/gates documented. Coverage CSV regenerated via new reusable build_coverage_status.py: 149 complete / 236 partial / 1,554 not_started. Quota anomaly logged in Entry 76 (no daily-limit rejection at ~292k calls/day).

Landmines hit / recorded

- Polygon StakeManager `totalStaked()` != total staked (self-stake only; use `currentValidatorSetTotalStake()`).
- eth.drpc.org = keyless mainnet (and Ronin) archive; publicnode gates archive behind a personal token; blastapi retired its public Starknet endpoint.
- Kaia: only `archive-en.node.kaia.io` serves historical state; the regular public endpoint returns “execution reverted” for historical calls (NOT pruned-trie errors).
- peaq’s staking pallet is the KILT fork: storage item `TotalCollatorStake` (struct{collators, delegators}), not Moonbeam’s `Total`.
- Radix Babylon Gateway accepts `at_ledger_state={"timestamp":?}` on `/state/validators/list` — free historical validator stakes; Olympia era unreachable (old gateway retired).
- Starknet staking contract = 0x00ca1702e64c? (docs.starknet.io chain-info); 0x00ca1705e74? is the MINTING CURVE and 0x04718f5a? the STRK token.
- Ronin staking contract balance != staked (pending undelegations of revoked candidates, +16% at head): use `getManyStakingTotals` over `getValidatorCandidates`.
- DeFiLlama `/protocol/{slug}` can 200 with entirely empty `tv1+chainTvls` (protocol listed, TVL never tracked) — an empty-series REJECT is not an API failure.

/newpage

session_030_2026-07-05_task_a_resume_cut_short.md

Session 030 — Task-A Resume (cut short at user departure)

Date: 2026-07-05 **Model/interface:** Claude Fable 5 via Claude Code (desktop) **Kickoff prompt:** 04_code/CLAUDE_CODE_SESSION030_TASK_A_RESUME_PROMPT.md **Initial user message:** “read 04_code/CLAUDE_CODE_SESSION030_TASK_A_RESUME_PROMPT.md and proceed”

Conversation flow (structured companion; verbatim transcript is the primary record)

1. **User:** read the session 030 resume prompt and proceed. **AI:** located repo at `C:/AFA_2027_QTM_Crypto`, verified clean tree on post-029 state; confirmed none of the 7 WORKLIST tokens (SFUND/MYX/ADF/AVNT/KAITO/VV) had rows or checkpoints (clean fresh builds); verified channel map entries, Pro key, observed months (73 total); launched `phase1_channel12_stream.py` in background with `WORKLIST=8972,36410,24796,38299,35763,35509`, default guards untouched.
2. **User:** are the session objectives achieved? **AI:** no — 1/7 done. SFUND completed as a hidden giant (tf=3,415,005, gl=28,553 vs ~4k estimate); MYX mid-scan.
3. **User:** how are we doing now? **AI:** log stale 86 min at MYX batch 61/128; verified process alive (established Etherscan connections + CPU accrual) — silent binary-split storm in MYX’s Aug-2025 BSC launch region (blocks ~49M+), engine prints only every 5 batches. Not a hang.

4. **User:** current status? **AI:** still in the silent stretch (~2h20m); process verified alive again; running total estimated ~95–100k getLogs; flagged shrinking cap margin, token-boundary stop semantics.
5. **User:** leaving for airport at 5 PM CST, out for a week, resume Monday 7/13 — complete whatever we have by then. **AI:** set 3:45 PM cutoff timer + task-tracked wrap-up plan (stop engine at cutoff if mid-token; B2/B4; assemble; coverage; Entry 80; report; memory; commit+push).
6. **User:** how are we doing? **AI:** log advanced — MYX revealed as a MEGA-giant: batch 81/128, tf=15,438,541, gl=148,993 cumulative (~120k MYX alone), ~10h from finishing (~1 AM ETA). Asked the user: leave running unattended for the week vs. kill at cutoff. **User decision:** “just wrap up now. My departure time is now 4:30 PM... Create a transition report and next session prompt.”
7. **AI wrap-up (15:05–15:30):** killed PID 29972 (MYX aborted, no partial checkpoint by engine design); ran `panel.aggregate() -> channel2_holding.csv` 301 tokens / 9,817 rows (SFUND folded in); B2/B4 on SFUND (PASS/PASS: max contamination 1.94x; screened HODL-6m [12.3%, 24.9%] median 20.6%); `phase1_assemble_lambda.py -> lambda 9,603 asset-months / 338 assets`; `build_coverage_status.py -> 150 complete / 235 partial / 1,554 not_started`; regression-ready **139** (tokens/other lambda?TVL 119 assets / 4,217 months; coins 20 unchanged). Wrote Entry 80, `03_data/SESSION030_TASK_A_RESUME_REPORT.md` (transition report), `04_code/CLAUDE_CODE_SESSION031_TASK_A_RESUME_PROMPT.md` (small-chains-first day 1: ADF/VVV/KAITO/AVNT/RAIN; MYX alone on a dedicated day, budget ~250–300k calls). Commit+push at session end per standing authorization.

Key decisions

- Kill vs. leave-running: user chose kill at departure; ~120k MYX calls written off (quota daily-resetting and evidently non-binding per Entry 76 — cost is time only).
- Methodology lesson recorded (Entry 80): holder-count call estimates undershoot BSC tokens 7–30x; sequence small chains first, BSC last.

Quota

~149k getLogs (SFUND 28,553 + MYX ~120,440 aborted), single process, no API rejection.

/newpage

`session_031_2026-07-24_task_a_day1_complete.md`

Session 031 — Task-A Day-1 Resume (all 5 small tokens built)

Date: 2026-07-23 -> 2026-07-24 (build spanned two calendar days; overnight lost to OS sleep)

Model/interface: Claude Fable 5 via Claude Code (desktop) **Kickoff prompt:** `04_code/CLAUDE_CODE_SESSION031_TASK_A`

Initial user message: “Read `CLAUDE_CODE_SESSION031_TASK_A_RESUME_PROMPT.md` in `C:/AFA_2027_QTM_Crypto/04_code` and proceed from there.”

Conversation flow (structured companion; verbatim transcript is the primary record)

1. **User:** read the session 031 resume prompt and proceed. **AI:** verified state — last commit still session 030 (no intervening session despite the planned 7/13 date); none of the 6 tokens in the stream progress file; Pro key present. Launched day-1 build in background at 14:52 CDT 7/23 with `WORKLIST=24796,35509,35763,38299,38341` (small-chains-first per Entry 80); armed a log monitor on token boundaries/errors. Decision: do NOT start MYX after day-1 finishes — mid-afternoon start would repeat the session-030 abort risk; it keeps its dedicated day.
2. **Monitor events (7/23 afternoon):** ADF DONE in ~4 min (gl=1,499, under its 1,871 estimate). VVV started.

3. **User (next morning, 7/24):** “tell me the status of this session.” **AI:** log silent since 17:30 the previous evening (~16h) at VVV batch 51/59. Hang-check protocol: CPU sampling (bursty: 0.016s/25s then 19s/150s), 8 established connections, zero read/write I/O, then a **py-spy thread dump** — workers healthy (2 mid-HTTP on Etherscan, rest queued on the rate limiter, 8–10-level binary-split recursion). Declared alive-not-hung; let it run.
4. **User:** “does it make sense to remove sleep option while this task is running. It could be that the computer sleep broke the task.” **AI:** user’s theory confirmed — Kernel-Power event log showed Modern Standby cycles all night (sleep timeout was 20 minutes); the build had been frozen, not grinding. Code check: **robust_getlogs** has timeout=60 + 6 retries + drops broken pooled connections -> self-heals on wake, zero loss. **Set sleep+hibernate timeouts to NEVER (AC+DC) via powercfg** (user request). Warned lid-close still sleeps; user confirmed lid stays open.
5. **Status checks through 7/24 midday (user asked ~5x):** VVV DONE — **new all-time ch2 record: tf=27,792,859 / gl=82,066 vs ~1k estimate (82x)**, B4 median 5.7%. KAITO DONE — gl=25,870 (8.6x), 16 screened months, B4 flagged-high (median 64.9% / last 66.7%, airdrop-lockup, kept per protocol). AVNT DONE — gl=52,758 (13x), tf=17.6M, median 0.0% (10-month-old token, structural). RAIN started. AI flagged the emerging lesson: **Base undershoots like BSC** (VVV 82x / AVNT 13x / KAITO 8.6x); Polygon/Arbitrum estimates held.
6. **16:40 interruption:** Windows Update auto-restarted the machine (sleep=never does not prevent update restarts), killing RAIN at batch 536/587 (~90%, ~9,824 gl lost — no partial checkpoint by design). Monitor + background task died with it. **User:** “ok can you finish the task? I thought that the sleep was set to never but I saw my computer was restarted.” **AI:** diagnosed the reboot from the event log (16:40 boot), confirmed sleep settings survived, relaunched RAIN alone (WORKLIST=38341), recommended pausing Windows Update before the MYX day. RAIN rebuilt clean in ~35 min: gl=19,315 (1.4x est), B4 median 0.0% / last 17.2% (young token).
7. **AI close-out (7/24 evening, autonomous):** aggregate -> channel2_holding.csv **306 tokens / 9,857 rows**; B2 check all 5 PASS (max onchain/circ 8.2x vs 100x guard); B4 all in range (KAITO flagged); **phase1_assemble_lambda.py -> lambda 9,638 asset-months / 341 assets** (accounting verified against git HEAD: +3 net-new assets ADF/KAITO/AVNT; VVV’s 4 ch2 months coincide with its existing ch3 months = n_channels upgrade; RAIN +6 net); **build_coverage_status.py -> 153 complete / 232 partial / 1,554 not_started**; regression-ready **142** (coins 20 + tokens/other lambda?TVL 122 assets / 4,243 months). Wrote Entry 81, **03_data/SESSION031_TASK_A_RESUME_REPORT.md**, updated memory. Commit **8b2629e** pushed. Quota: ~191k getLogs total (~28.9k on 7/23, ~162.5k on 7/24 incl. the lost RAIN attempt); no API rejection.

Key decisions this session

- MYX deferred to its own quota day even though day-1 finished — 12–15h runway must start early in a day (session-030 lesson).
- Sleep set to never (user-initiated, AI-executed via powercfg); **pause Windows Update before the MYX run** (AI recommendation, left for the user to click).
- KAITO’s high-but-in-range B4 (64.9%) kept and flagged, not suppressed, per protocol.
- Base added to the BSC estimate-skepticism rule (Entry 81 methodology lesson).

Next session (032)

MYX (36410/BSC) alone: ~250–300k calls, 12–15h, start early, Windows Update paused, sleep=never confirmed. Reuse the Day-2 section of the session-031 prompt. Then Entry-79 open items (b)–(g).

/newpage

session_032_2026-07-24_myx_build_task_a_closed.md

Session 032 — MYX (36410/BSC) ch2 build; Task A fully closed

Date: 2026-07-24 (evening/overnight) **Model/interface:** Claude Fable 5 via Claude Code (desktop app, Windows) **Kickoff prompt file:** 04_code/CLAUDE_CODE_SESSION032_MYX_PROMPT.md **Mode:** fully autonomous overnight run (always-allow permissions; standing end-of-session commit+push authorization) **Commits:** e9c0378 (main results), 7bb3d88 + cbd9e7f (build-log fix), 8fae95a (retroactive session-031 logs)

Initial user prompt (verbatim)

read CLAUDE_CODE_SESSION032_MYX_PROMPT.md in C:/AFA_2027_QTM_Crypto/04_code
and then proceed for the session overnight

All subsequent work was autonomous per the kickoff file; no further human input during the session.

Structured event log

1. **Read kickoff prompt** (MYX 36410/BSC alone; budget ~250–300k getLogs / 12–15h; pause Windows Update first; B2/B4 -> assemble -> Entry 82 -> report -> commit+push).
2. **Pre-flight checks:**
 - Sleep/hibernate confirmed 0 (Never) on AC+DC (session-031 powercfg fix held).
 - **Windows Update pause NOT possible:** session shell is non-elevated; the pause values live under HKLM (.../WindowsUpdate/UX/Settings). Compensating verification: no pending reboot (RebootRequired + CBS flags both clear) and the July cumulative/.NET wave had already installed the same morning (08:00–08:12), so forced-restart risk was judged low. Proceeded with the risk documented.
 - Repo clean at 11e60ad; engine untouched; no stale MYX checkpoint; ch2 at 306 tokens.
3. **Build launched** ~17:58 CDT (WORKLIST=36410 PYTHONUTF8=1 python 04_code/phase1_channel2_stream.py, background task, log tee'd). A liveness watchdog (10-min samples of engine CPU accrual + established TCP connections; ~1h stale threshold per the kickoff's no-kill rule) ran alongside.
4. **Build completed 20:37 CDT — 2.7h, not 12–15h.** DONE line: tf=22,451,143, gl=67,233, contract screen 21/40, 10 screened months, screened HODL-6m median 3.0% / last 5.2%. The predicted silent stretch after batch 61 occurred but lasted ~1h (batch 61->66: 67s->3,659s). No OS interruption; watchdog never fired.
5. **Integrity checks:** B2 PASS — max onchain/circ 10.5x (launch month 2025-08), decaying to 5.8x by 2026-05, vs the 100x guard. B4 PASS — screened HODL-6m median 3.0%, last 5.2%, inside [0, 80%] (young-token structural low HODL, same pattern as AVNT/RAIN). Consistency check vs the session-030 abort: batch 81 tf=15,459,154 this run vs 15,438,541 at the 030 kill — the rebuild reproduces the aborted run's data.
6. **Assemble + coverage:** lambda 9,638 -> 9,648 asset-months / 341 -> 342 assets; ch2 306 -> 307 tokens / 9,867 rows; coverage 154 complete / 231 partial / 1,554 not_started. MYX lambda?TVL overlap = all 10 months (myx-finance) -> regression-ready 142 -> **143** (coins 20 + tokens/other 123; lambda?TVL months 4,243 -> 4,253).
7. **Records written:** Entry 82 appended to 04_code/DATA_DECISIONS_LOG.md; 03_data/SESSION032_MYX_REPORT.md written; committed and pushed.
8. **Repo hygiene find:** .gitignore line 7 (*.log) had silently excluded ALL build logs — session 031's report claimed its logs were committed but they never were. Force-added the session-032 log (03_data/phase1/_session032_stream.log) and retroactively the two session-031 logs. (Future sessions: git add -f for build logs, or amend .gitignore.)

Key findings / decisions

- **Call-count anomaly (favorable, documented in Entry 82):** session 030 logged ~120,440 gl for MYX by batch 81; this run reached batch 81 with 46,346 gl and identical tf. Engine unchanged, block density fixed -> 030's counter was most plausibly inflated by retry churn. **Treat session-030-derived call estimates as upper bounds.**
- MYX tf=22.45M is the **second-largest ch2 build ever** (beats MBOX 21.2M; VVV holds the record at 27.8M) and the largest BSC build.
- Task A (session 029) is **fully closed**: 47 targets = 41 built, 5 non-EVM skipped, 1 (WARP) deferred for identity-map review.
- Windows Update pausing requires an elevated session (or manual Settings action) — noted for future long builds; tonight it was not needed (no restart occurred).

Post-session state

lambda 9,648 / 342; regression-ready 143; ch2 307 tokens / 9,867 rows; coverage 154/231/1,554. Quota: 67,273 calls (67,233 getLogs + 40 getcode); no API rejection; DAILY_CAP never approached. **Next (session 033):** Entry-79 open items (b)-(g) — DOT/KSM Subscan key, CHZ anchor, CORE key, WARP identity review, non-TVL breadth ch2 (~500 tokens), MATIC NVT_GL probe.

Companion structured log; the verbatim transcript in the Claude Code session history is the primary record per the AFA record-keeping rules.

/newpage

session_033_2026-07-25_xtz_matic_nvt_probes_negative.md

Session 033 — XTZ/MATIC NVT_GL probes (both negative); JOE/SAFE closed

Date: 2026-07-25 **Model/interface:** Claude Fable 5 via Claude Code (desktop app, Windows) **Kickoff prompt file:** 04_code/CLAUDE_CODE_SESSION033_XTZ_MATIC_NVT_PROMPT.md **Mode:** autonomous (always-allow permissions; standing end-of-session commit+push authorization) **Commit:** 9d87213

Initial user prompt (verbatim)

read CLAUDE_CODE_SESSION033_XTZ_MATIC_NVT_PROMPT.md in C:/AFA_2027_QTM_Crypto/04_code and then proceed from there

All subsequent work was autonomous per the kickoff file; no further human input during the session.

Structured event log

1. **Read kickoff prompt** (Task A: XTZ PQ via TzKT totalTransferred; Task B: MATIC bitinfocharts probe; Task C: JOE/SAFE/DYDX classification review; Task D: rebuild; Entry 83; report; commit+push). Read phase2b_pq_coins.py, phase2_nvt_gl.py, and the existing XTZ/MATIC/DYDX NaN-marker rows in pq_coins.csv before writing anything.
2. **Task A premise disproven at first probe.** Live GET of api.tzkt.io/v1/statistics/daily returned supply/staking fields only — **there is no totalTransferred field**. Rather than build from a nonexistent series, pivoted to an exhaustive free-source sweep for Tezos historical transfer value:
 - TzKT full swagger (287 paths): no historical volume/sum endpoint anywhere; the Statistics schema confirmed supply/staking-only.
 - back.tzkt.io/v1/home (works only with browser UA+Referer; plain requests 403): current-day volume aggregates + 30-day price chart — no history.

- TzStats `api.tzstats.com`: DEAD (connection refused). TzPro: unreachable.
 - CoinMetrics community API: `xtz` exposes only `TxCnt/TxTfrCnt`; `TxTfrValUSD` -> 403 pro-gated. CM GitHub `csv/xtz.csv`: same community columns, no transfer value.
 - bitinfocharts `sentinusd-xtz`: EMPTY page (no series, not the BTC default).
 - Messari legacy keyless API: 404 (dead). CryptoCompare blockchain histo: 401 key-gated. Raw TzKT operation iteration forbidden (Entry 31/32). **Verdict: XTZ stays PQ=NaN**; marker refined to `NaN:xtz_no_free_native_series_s033` with the full probe list.
3. **Task B (MATIC)**: bitinfocharts `sentinusd-matic` returned 5,852 days starting 2010/07/17 with last value identical to the BTC reference — **BTC-default guard triggered** (Polygon launched 2020). CM community rejects value metrics for `matic`. **MATIC stays PQ=NaN**; marker refined to `NaN:polygon_native_volume_no_free_source`. Entry-79(g) “candidate 21st regression-ready coin” closed negative.
 4. **Task C**: verified against `classification_table.csv`, `lambda_panel.csv`, `tv1_panel.csv`: JOE (6 lambda mo) and SAFE (10 lambda mo) are governance tokens classified `coin` via staking tags, with no native chain and 0 TVL rows -> **permanently closed architectural gaps** (no NVT denominator exists in principle; no re-probe). DYDX legitimately a coin but 1 lambda month (2024-03) -> skip.
 5. **Task D**: re-ran `phase2_nvt_gl.py` + `build_coverage_status.py` after the two marker-note edits. Output identical to session 032: NVT_GL 2,526 asset-months / 67 assets; coins regression-ready (lambda?NVT_GL) 20 / 645 months; coverage 154/231/1,554; regression-ready total 143. No data changes.
 6. **Records**: DATA_DECISIONS_LOG Entry 83 appended (rewritten to reflect the actual negative outcome, not the prompt’s template); report `03_data/SESSION033_XTZ_MATIC_NVT_REPORT.md`; kickoff prompt committed.
 7. **Commit + push** 9d87213 to origin/main per standing authorization.

Deviations from the kickoff prompt

- Task A was **not executed as written** because the specified TzKT field does not exist. The prompt’s A2–A5 build steps were replaced by the source sweep above and a refined NaN marker. The prompt’s draft Entry 83 (which assumed a successful XTZ build) was rewritten accordingly.
- No API keys used; all probes free/keyless, read-only, and low-volume (~15 HTTP requests).

Outcome

No new PQ data. Two NaN markers made precise; three gaps (XTZ paid-only, MATIC paid-only, JOE/SAFE architectural) are now closed with live-verified documentation; DYDX deferred. Open items for session 034: Entry-79 (b)–(e) — DOT/KSM Subscan key, CHZ manual anchor, CORE key, WARP identity review, non-TVL breadth ch2.

/newpage

`session_034_2026-07-25_chz_blockchair_defi_batch1.md`

Session 034 — CHZ ch1 built; Blockchair XTZ/MATIC failed; EVM DeFi Breadth Batch 1

Date: 2026-07-25 **Model/interface:** Claude Fable 5 via Claude Code (desktop app, Windows) **Kickoff prompt file:** `04_code/CLAUDE_CODE_SESSION034_COMBINED_PROMPT.md` **Mode:** autonomous (always-allow permissions; standing end-of-session commit+push authorization) **Commit:** 861d1fd (+ record-keeping commit after)

Initial user prompt (verbatim)

read CLAUDE_CODE_SESSION034_COMBINED_PROMPT.md in C:/AFA_2027_QTM_Crypto/04_code
and proceed from there

Subsequent human input: five “status?” check-ins during the ~5h batch build; no decisions requested or given mid-session.

Structured event log

1. **Read kickoff prompt** (Task A CHZ ch1 via Chiliz RPC; Task B Blockchair XTZ/MATIC probe; Task C 102-token EVM DeFi ch2 batch + DeFiLlama TVL matching; Task D assemble/report/commit). Verified repo clean at 033 head (0036378).
2. **Windows Update paused until 2026-08-01** (HKLM PauseUpdatesExpiryTime — the shell had elevation this time, unlike session 032). Sleep already Never (031 fix).
3. **Task C launched first** (longest): 102-token WORKLIST on the unchanged session-026 streaming engine, background, log to 03_data/session034_batch1_build.log.
4. **Task A — CHZ ch1 BUILT** (04_code/session034_chz_ch1.py): chiliz.drpc.org responded first; binary-search month-end blocks (cached), native `eth_getBalance` of 0x...1000 at 35 month-ends 2023-07..2026-05. Cross-check: 2026-05 balance 2,391,774,380 vs anchor 2,416,757,292 -> **drift ?1.03%** (<5% gate). Staking ratio 2.35%->26.61% (~4x step-up 2024-06). `channel1_chz.csv` written. CHZ already had 21 non-NaN PQ months -> 21st regression-ready coin.
5. **Task B — Blockchair FAILED keyless** (04_code/session034_blockchair_probe.py): tezos/{calls,operations,transactions} and polygon/transactions with `?a=sum(...)` all HTTP 404; IP blacklisted (HTTP 430) after ~4 anonymous requests, incl. /stats. Flagged for Moazzam: paid key (~\$30/mo) might unblock but 404s suggest the aggregation tables may not exist for these chains at any tier — confirm with Blockchair support before paying. NOT subscribed. XTZ/MATIC stay PQ=NaN.
6. **Task C2 — TVL slug matching done during the build** (independent of it). Raw symbol match was ~40% wrong (Litentry->lighter-bridge, 2017-Jupiter->jupiter-lend, Wrapped Solana->solana-farm, youves/autofarm/jetswap/zipswap all namesake collisions caught by DL cmcId authority). Final: 24 token-class slugs into `asset_onchain_identity.csv`; OTHER_ADDS += MULTI/ORC/MUBI/FF; CHAIN_LEVEL += METIS, XAI (Entry-68 pattern). LST receipt tokens excluded on circularity (NV~TVL by construction). CEL/FTT have no DL entries (CeFi). `phase2_build_tvl_panel.py` rerun: **159 assets / 7,889 asset-months**.
7. **Task C build completed**: 101/102 built (MSOL already complete -> skipped), 154,049 getLogs (est ~147k), 47.56M transfers, 2,916 screened months. B2 clean; B4 flagged-high kept: META, TROY, SMT, YOU, BOXx2, WHITE, HOT, STRONG. Transient connection resets mid-build self-recovered (engine retries). Survivorship: CEL 54 scrMo, FTT 81 scrMo (HODL 34.7%->82.5%).
8. **Task D**: assemble -> **lambda 12,599 asset-months / 444 assets**; coverage -> 184/302/1,453; regression-ready **143->173** (coins 21, tokens/other 152; 29 new Batch-1 lambda?TVL tokens, 790 overlapped months). ch2 panel 408 tokens / 12,955 rows.
9. **Entry 84** appended; 03_data/SESSION034_COMBINED_REPORT.md + 03_data/session034_batch1_token_table.csv written; build log force-added (`.gitignore *.log`); committed 861d1fd and pushed.

Data-access notes

- Chiliz Chain 2.0: chiliz.drpc.org, keyless archive (`eth_getBalance` at historical blocks worked to genesis-era months).
- Blockchair: anonymous tier blacklists within ~4 requests (2026 behavior).
- Etherscan Pro V2: 154k getLogs in one run, no rejection (DAILY_CAP=185k in-process).
- DeFiLlama: api.llama.fi keyless as always; /protocols cmcId field used as match authority.

/newpage

session_035_2026-07-25_defi_batch2.md

Session 035 — EVM DeFi Breadth Batch 2 (13/13); SNX/LEND/ANKR TVL; KNCL window-clip

Date: 2026-07-25 (same day as session 034, same conversation) **Model/interface:** Claude Fable 5 via Claude Code (desktop app, Windows) **Kickoff:** user pasted the Batch-2 WORKLIST command from the 034 prompt directly **Mode:** autonomous after launch; standing end-of-session commit+push authorization **Commits:** a7aa885 (+ record-keeping commit after)

Initial user prompt (verbatim)

```
WORKLIST=3783,2120,5824,1982,1757,2586,2165,5266,2239,28568,9436,2308,2691 PYTHONUTF8=1  
python 04_code/phase1_channel2_stream.py
```

One “status?” check-in mid-build; no decisions requested.

Structured event log

1. **Launched Batch 2 in background** (13 tokens). Flagged the quota caveat (154k gl already spent same-day; 034 prompt suggested waiting for reset) — quota did NOT bind: full run completed same-day (credit-based, confirming the session-032 read).
2. **Pre-flight while building:** identity map showed ANKR slug present but other-class (0 panel rows — the RPL-style filter gap); AAVE holds aave-v2 so aave-v1 free for LEND; MKR has NO dl_slug; cmc 2691 “PNT” is Penta, NOT pNetwork (034 prompt note wrong).
3. **Build completed: 13/13**, 74,783 getLogs (0.63x of est 119k), 22.9M transfers, 592 screened months. B2 clean; B4 flag-high only PNT (97.9%, dead token, kept). FUN 105 scrMo (longest of the batch), ANKR 85, SNX 73, SLP 58, ELON 56, KNCL 47.
4. **Migration-window checks:** KNCL lambda 2017-09..2021-07 vs KNC-9444 lambda 2021-07+ -> single overlap month; MATIC/POL rule applied. MKR lambda = ZERO months (never built) -> no MKR/SKY double-count exposure; nothing assigned to MKR.
5. **TVL implementation:** SNX->synthetix parent (children-only in /protocols; parent fetchable, 2,545 daily points 2019-08+); LEND->aave-v1 (73 mo, 4 lambda-overlap 2020-05..2020-08 = the dead-token region); ANKR->OTHER_ADDS (cmcId exact, 67 mo); KNCL->kyberswap-classic clipped <=2021-06 via new CLIP mechanism added to phase2_build_tv1_panel.py (9 mo). SAI excluded on liability-token circularity (CDP stablecoin: NV~supply, NV/TVL = inverse collateral ratio); EETH excluded (LST receipt, Entry-84 rule); FUEL/FUN/ERC20/MLK/SLP/ELON/PNT have no protocol TVL.
6. **Rebuild + assemble:** tv1_panel 163 assets / 8,120 mo; lambda 13,191 / 457 assets; regression-ready 173->177 (coins 21, tokens/other 156; new: SNX 73, ANKR 66, KNCL 9, LEND 4 overlap months); ch2 421 tokens / 13,580 rows; coverage 188/311/1,440.
7. **Entry 85**, SESSION035_BATCH2_REPORT.md, token table, build log force-added; committed a7aa885, pushed.

Data-access notes

- Etherscan Pro V2: 229k getLogs total across the calendar day over two runs, no rejection — the account quota is credit-based, not a hard daily getLogs cap.
- DeFiLlama parent slugs (synthetix) fetchable at /protocol/{slug} even when absent from /protocols; maker parent is NOT (400) — Maker history lives under Sky children.

/newpage

session_036_2026-07-26_steth_meme_batch3a.md

Session 036 — EVM DeFi Breadth Batch 3a: stETH + MEME (lambda-only)

Date: 2026-07-26 **Model/interface:** Claude Fable 5 via Claude Code (desktop app, Windows) **Kickoff:** user pointed to CLAUDE_CODE_SESSION036_STETH_MEME_PROMPT.md (“proceed from there”) **Mode:** autonomous after launch; standing end-of-session commit+push authorization **Commits:** 287fe7e (+ record-keeping commit after)

Initial user prompt (verbatim)

read CLAUDE_CODE_SESSION036_STETH_MEME_PROMPT.md in C:/AFA_2027_QTM_Crypto/04_code and proceed from there

No mid-session check-ins or decisions requested.

Structured event log

1. **Windows Update paused** until 2026-08-02 via registry (elevated shell available this session, unlike session 032). Sleep already set to Never.
2. **Launched build in background** (WORKLIST=8085,28301). Completed in ~50 min — far under the 4–8h estimate: 22,444 getLogs actual vs ~110k estimated (~5x over-estimate; both tokens have long sparse pre-activity block ranges).
3. **stETH (8085):** 13,365 gl / 4,529,175 tf / 50 scrMo. B2 pass, B4 pass (HODL-6m median 13.4%, last 13.1%).
4. **MEME (28301):** 9,079 gl / 2,932,933 tf / 31 scrMo. B2 pass, B4 pass (median 3.7%, last 15.4%).
5. **TVL — stETH:** excluded per prompt (LST receipt circularity, Entry 84 rule); no stETH->lido mapping created; lambda retained for conviction-only panel.
6. **TVL — MEME (nuance vs prompt):** DeFiLlama DOES list a memecoin protocol with cmcId=28301 and the correct contract (0xb131...cd74, category Farm, Ethereum) — not a symbol clash (Conflux MemeDex is the unrelated symbol match) — but its TVL series has ZERO data points and empty currentChainTvls. No usable TVL -> lambda-only, same outcome the prompt expected via “no match”.
7. **Assemble + coverage:** lambda 13,191->13,272 asset-months / 457->459 assets; regression-ready 177 UNCHANGED (coins 21, tokens/other 156), as expected; ch2 423 tokens / 13,661 rows; coverage 188/313/1,438 (two new tokens moved not_started->partial).
8. **Entry 86, SESSION036_BATCH3A_REPORT.md;** committed 287fe7e, pushed.

Data-access notes

- getLogs estimates from holder-count heuristics continue to run high for older tokens (this session ~5x); treat as upper bounds (session-032 precedent).
- Etherscan quota nowhere near binding (22.4k gl).

Next

- Session 037 — SHIB (5994), ~128k gl est (likely high), lambda-only.
- Regression-ready movement requires user actions: Subscan key (DOT/KSM), CORE key, Cosmos key. WARP review and non-TVL breadth (~500 tokens) still open.

/newpage

session_037_2026-07-27_dot_ksm_core_ch1.md

Session 037 — DOT + KSM + CORE channel-1 (staking) builds

Date: 2026-07-27 **Model/interface:** Claude Fable 5 via Claude Code (desktop app, Windows) **Kickoff:** user pointed to CLAUDE_CODE_SESSION037_DOT_KSM_CORE_PROMPT.md (“proceed from there”) **Mode:** autonomous after launch; standing end-of-session commit+push authorization **Commits:** 7ca6b14 (+ record-keeping hash-fix commit after)

Initial user prompt (verbatim)

read CLAUDE_CODE_SESSION037_DOT_KSM_CORE_PROMPT.md in C:/AFA_2027_QTM_Crypto/04_code and proceed from there

No mid-session check-ins or decisions requested.

Structured event log

1. **Subscan plan failed on first probe:** era_stat is per-address (400 “address is a required field”); /api/scan/daily Bonded category returns all zeros with a ~2-month free-plan history window; subscan.io website mid-upgrade (charts unreachable). Entry-87 build plan unusable as written.
2. **Pivot designed and validated:** raw-storage-key reads of Staking.ErasTotalStake(ActiveEra) at month-end blocks (interpolation search on Timestamp.Now) from public archive RPCs, via a ~200-line websocket client (no metadata decode). Discovered both chains’ staking had migrated to Asset Hub (relay ActiveEra = null at head) -> AH fallback per month; parity AH endpoints verified archive to >=2022 (block-1 state absent -> binary-search state anchor added).
3. **DOT built:** 71 months 2020-08..2026-06; relay<=2025-10, AH from 2025-11 (boundary ?0.7%); latest 862.3M; drift vs fresh ?2.18% PASS; external anchor (Coinbase/StakingRewards ~881.9M, 52.0%) matches fresh to 0.05%.
4. **KSM built:** 76 months 2020-03..2026-06; dropped 2019-11..2020-02 — old runtimes only expose SlotStake = MIN validator backing, not a total (wrong metric); latest 8.38M; drift ?2.05% PASS; anchor (~8.5M, 46%) matches. 2020-07 ratio suppressed (CMC circulating 2.99M < staked — bad point).
5. **CORE probes:** prompt’s staking_summary path doesn’t exist; real API at staking-api.coredao.org is current-only; CoreScan proxy/balancehistory endpoints broken (empty 200s); rpc.coredao.org pruned; **Ankr + dRPC are archive.** Contract-balance approach measured and REJECTED (+6.4% vs official). Exact definition found and validated to the digit: stakedCoreAmount == ? active validators’ CoreAgent.candidateMap.amount.
6. **CORE built exact:** 42 months 2023-01..2026-06 via eth_call at month-end blocks; active set via getValidatorOps() (legacy: currentValidatorSet(i) walk); legacy stake via PledgeAgent.agentsMap word0 <=2024-10; drift ?2.55% PASS.
7. **Assemble + coverage:** lambda 13,272->13,449 asset-months / 459->462 assets; regression-ready 177->178 (coins 21->22: CORE); coverage 189/314/1,436. **Prompt premise wrong for DOT/KSM:** they have NO NVT_GL (pq_usd all null) -> partial (pq_nvtgl their only gap), not regression-ready. TRX pow_only mislabel identified (explains narrative/file coin-count +/-1).
8. **Entry 88, SESSION037_DOT_KSM_CORE_REPORT.md, committed + pushed.**

Data-access notes

- Subscan free API: no network-wide staking history at all (per-address era_stat; daily charts plan-gated to ~2 months). A paid plan was NOT needed — archive RPC state reads are free and authoritative.
- OnFinality public relay endpoints are full archive (block 1 state readable); parity Asset Hub endpoints archive from ~2022 snapshot; Dwellir/IBP/dotters DNS-blocked on this network.

- Ankr `rpc.ankr.com/core` and `core.drpc.org` are keyless archive for Core; everything CoreScan/official-RPC side is current-only, broken, or pruned.
- Windows Update pause was skipped (short session, no multi-hour unattended run); sleep already Never.

/newpage

session_038_2026-07-27_shib_ch2_batch3b.md

Session 038 — SHIB ch2 (EVM DeFi Breadth Batch 3b)

Date: 2026-07-27 **Model/interface:** Claude Fable 5 via Claude Code (desktop app, Windows) **Kickoff:** user pointed to `CLAUDE_CODE_SESSION038_SHIB_PROMPT.md` (“proceed”) **Mode:** autonomous after launch; standing end-of-session commit+push authorization **Commit:** 263655a

Initial user prompt (verbatim)

```
read CLAUDE_CODE_SESSION038_SHIB_PROMPT.md in C:/AFA_2027_QTM_Crypto/04_code
and proceed
```

No mid-session check-ins or decisions requested.

Structured event log

1. **Pre-flight:** sleep/hibernate/monitor set to Never via powercfg; Windows Update paused via registry until 2026-08-03 (elevated shell OK).
2. **Task A — SHIB (5994/Ethereum) ch2 built:** `WORKLIST=5994` on the unchanged stream engine (`VAL_CAP_MULT = CONTAM_MULT = 100`, self-transfer skip intact). 53,683 getLogs actual vs 128k estimate = **0.42x** (the sessions 034–036 holder-count-overestimate pattern holds; first ~11M blocks produced ~88 getLogs). 17,845,462 transfers; 61 screened months (2021-05..2026-05); contract screen 13/81 candidates; ~2.1h wall at 8 workers. **B2 pass** (zero months excluded by the 100x contamination guard). **B4 pass** (screened HODL-6m median 78.1% ≤ 80%; last 82.5%).
3. **Task B — TVL decision (verified, not just documented):** `tv1_panel` checked — the only shiba-related slug `shibaswap` is already assigned to BONE (cmc 11865, 59 months); SHIB has zero `tv1_panel` rows. SHIB excluded from NV/TVL regression — meme token, no direct protocol TVL; assigning `shibaswap` would double-count the DEX TVL against the wrong token. **SHIB -> lambda-only**. No slug written.
4. **Task C — assemble + coverage:** lambda 13,449->**13,510** asset-months / 462->**463** assets; `channel2_holding.csv` 423->424 tokens / 13,661->13,722 rows; coverage 189 complete / 315 partial / 1,435 not_started (SHIB not_started->partial); **regression-ready 178 unchanged** (coins 22, tokens/other 156 — recomputed from same-month lambda?NVT / lambda?TVL overlap to confirm).
5. **Entry 89, 03_data/SESSION038_SHIB_REPORT.md**, build log force-added (`.gitignore *.log` gotcha), committed + pushed (263655a).

Data-access notes

- Etherscan Pro key (`.api_keys.json` “etherscan”), 8 workers @ 8/s; 53,683 getLogs — well under the 185k daily cap and the 150k stop threshold.
- Engine gotcha (recurring): `phase1_assemble_lambda.py` final print crashes on the cp1252 console (lambda char) without `PYTHONUTF8=1`; CSV written correctly either way — rerun with the env var for the full summary.

Status after session

EVM DeFi breadth COMPLETE (batches 1, 2, 3a, 3b all done). Next (039): DOT/KSM PQ source probe; TRX coin_staking_type fix (pow_only->DPoS); WARP identity review.

/newpage

session_039_2026-07-28_dotksm_pq_trx_warp.md

Session 039 — DOT/KSM PQ probe; TRX staking-type fix; WARP identity review

Date: 2026-07-28 **Model/interface:** Claude Fable 5 via Claude Code (desktop app, Windows) **Kickoff:** user pointed to CLAUDE_CODE_SESSION039_DOTKSM_PQ_FIXES_PROMPT.md (“proceed”) **Mode:** autonomous after launch; standing end-of-session commit+push authorization **Commit:** 9172e42

Initial user prompt (verbatim)

```
read CLAUDE_CODE_SESSION039_DOTKSM_PQ_FIXES_PROMPT.md in C:/AFA_2027_QTM_Crypto/04_coc
and then proceed
```

No mid-session check-ins or decisions requested.

Structured event log

1. **Pre-flight:** repo clean (only the kickoff prompt untracked); WU pause from session 036/038 still in effect (until ~2026-08-02); no long build this session so no powercfg changes needed.
2. **Task A1 — Subscan /api/scan/daily: format=month** rejected (HTTP 400, only day/hour/6hour exist). **format=day** over any multi-year window -> HTTP 403 **history_window_exceeded** for ALL categories (transfer/extrinsic/ transaction/fee) on both polkadot and kusama endpoints. Window bisection: 30-day range returns data; 90-day fails -> free window ~ 2 months (same wall as session 037's “Bonded” probe). The in-window payload is also degenerate (2026-06-01: total=3 transfers / 5.98 DOT — not network-wide volume).
3. **Task A3 skipped by rule:** raw multi-year block iteration forbidden (Entry 31/32); not attempted.
4. **Task A4 — Blockchair keyless:** /polkadot/stats + /kusama/stats = 200 but **both indexes FROZEN** (best blocks 2025-05-26 / 2025-05-09); aggregation endpoint /{chain}/calls?a=date(time),sum(value)&q= = 404 on both (same no-aggregation-tables pattern as XTZ/MATIC, Entry 84). Stopped at 4 keyless calls (blacklist threshold). **Task A verdict: no free PQ source; DOT/KSM stay PARTIAL (gap = pq_nvtgl only); Blockchair ruled out permanently for DOT/KSM (stale index — paid key would not help); reopen only on Subscan Pro decision.**
5. **Task B — TRX (1958) label fix:** grep located coin_staking_type as a carry-forward column in universe_coverage_status.csv itself (build_coverage_status.py:33 reads the old file; static metadata from session 022). Edited the row pow_only -> pos, re-ran the builder. Verified: TRX lambda?NVT same-month overlap = 58 months -> stays complete via the proper pos path; regression-ready coins derive cleanly to **22 with TRX included** (33 complete coins ? 11 pow_only-NVT-only); Entry-88's +/-1 bookkeeping discrepancy resolved; headline 178 unchanged.
6. **Task C — WARP (1166) identity review:** checkpoint (raw/phase1_onchain/ holding/1166_WARP.json) shows 27,257 getLogs / **0 transfers** for 0x83e6...5Aa. Cached CMC detail (raw/cmc_detail/1166.json) is decisive: id 1166 = warpcoin.com, **category “coin”**, added 2016-02-03, **inactive** since 2018-05-08, own chain explorer — never an ERC-20. dl_slug polkastarter traced to DeFiLlama's registry carrying wrong cmcId 1166 (Polkastarter = POLS, 7208). **Closed as permanent identity mismatch:**

- (a) `BAD_DL_CMCID = {"1166"}` override added to `phase1_build_identity_map.py`; (b) identity CSV row cleared; (c) 54 bogus polkastarter TVL months purged from `tv1_panel` (8,120->8,066 mo, 163->162 assets); (d) coverage -> `not_started`. No lambda impact (0 lambda rows).
- 7. **Assemble + verify:** lambda **13,510 / 463 unchanged** (PYTHONUTF8=1); coverage 189/314/1,436; regression-ready **178 (coins 22, tokens/other 156)** re-derived from the coverage file.
- 8. **Record-keeping:** Entry 90 appended; `SESSION039_DOTKSM_PQ_FIXES_REPORT.md` written; this log + `time_log` entry; commit+push at session end.

Files touched

- `04_code/s039_probe_subscan.py`, `s039_probe_subscan2.py`, `s039_probe_blockchair.py` (new probe scripts)
- `04_code/phase1_build_identity_map.py` (`BAD_DL_CMCID` override)
- `03_data/phase1/asset_onchain_identity.csv` (1166 row cleared)
- `03_data/phase2/tv1_panel.csv` (54 WARP rows purged)
- `03_data/universe_coverage_status.csv` (TRX pos; rebuilt twice)
- `04_code/DATA_DECISIONS_LOG.md` (Entry 90)
- `03_data/SESSION039_DOTKSM_PQ_FIXES_REPORT.md` (new)

/newpage

`session_040_2026-07-28_cosmos_lcd_ch1.md`

Session 040 — 2026-07-28 — CRO/INJ/KAVA/SEI ch1 via Cosmos Archive LCD

Model: Claude Fable 5 (Claude Code desktop) **Prompt:** `04_code/CLAUDE_CODE_SESSION040_COSMOS_LCD_PROMPT.md`
Mode: fully autonomous; one unrelated user interjection (a WRDS/IBES connection error from a different project, user then said to ignore it). **Keys:** none — keyless LCD endpoints only.

Course of the session

1. Pre-flight: sleep=Never confirmed (AC/DC 0x0), Windows Update paused through 2026-08-03.
2. Wrote `04_code/session040_cosmos_lcd.py` (Tasks A–D in one script).
3. **Run 1:** prompt's CRO candidates DNS-dead; only KAVA passed via `cosmos.directory` proxy; crashed on wrong panel column name (`cmc_supply_circ` -> actual `circulating_supply`).
4. Refreshed candidates from `chains.cosmos.directory` registry (5–8 REST endpoints per chain).
5. **Run 2:** CRO found a real archive (`rest.mainnet.crypto.org`, 11 months built); INJ all pruned; KAVA died on HTTP 420 rate limits (not retried); SEI's `pocket.network` "pass" exposed as a **fake archive** (year-old bonded identical to live -> height header ignored). Run also crashed at final `to_csv` — pandas half-uninstalled by an interrupted upgrade elsewhere.
6. Fixes: fake-archive probe guard (old bonded == live bonded -> FAIL); binary-search guard for months predating the earliest stored block (KAVA's chain restarted at height 1 on 2022-05-25 — previously would have mis-attributed block-1 state to 2019–2022 months); 420 -> retry with 15–45 s backoff; no-retry on deterministic `invalid denom`; per-chain search pacing; `stdlib-csv` output. Reinstalled pandas (3.0.5) after finding the ~**andas** remnant.
7. **Run 3 (clean):** CRO 11 months, KAVA 17 months, 28 rows; CRO drift 1.81% PASS; KAVA drift 23.06% WARN.
8. **KAVA retry passes (2):** gentler pacing (1.5 s) recovered all 9 rate-limited months -> KAVA 26 months, CSV 37 rows. Drift diagnostic showed bonded 99.1M (Jun 30) -> 103.1M (Jul 14) -> 127.8M (live); genuine mid-July staking surge, WARN accepted.
9. Task E: CRO + KAVA `pos_possible` -> `pos`.
10. Task F: assemble + coverage rebuild ran clean under pandas 3.0.5.
11. Entry 91, this log, report, `time_log`; commit + push at session end.

Outcome

- lambda 13,510 -> **13,547** asset-months / 463 -> **465** assets
- Regression-ready 178 -> **180** (coins 22 -> 24: CRO, KAVA; tokens/other 156)
- Coverage 191 complete / 312 partial / 1,436 not_started
- INJ + SEI ch1: **closed as blocked** — no free archive LCD exists; SEI additionally hazarded by fake-archive gateways. Reopen only on paid indexer decision.

Gotchas recorded

- Cosmos public “archive” gateways can silently ignore `x-cosmos-block-height` — always verify old-state != live-state.
- **invalid denom**: from staking/pool at old heights = SDK codec boundary (deterministic; do not retry). CRO decodes from ~2025-06, KAVA from ~2024-Q2.
- `api.data.kava.io` rate-limits with HTTP **420** (not 429).
- `kava_2222-10` height reset (2022-05-25) means pre-2022-05 KAVA months are unreachable on the current chain regardless of archive depth.

Commit: 9587d0d (pushed to origin/main at session end)

/newpage

`session__041__2026-07-30__hxro__sxp__osmo.md`

Session 041 — 2026-07-30 — HXRO moot; SXP ch2; OSMO ch1; gap closures

Model: Claude Fable 5 (Claude Code desktop) **Prompt:** `04_code/CLAUDE_CODE_SESSION041_HXRO_SXP_OSMO_PROMPT.md`

Mode: fully autonomous, no check-ins. **Keys:** `.api_keys.json` “etherscan” (SXP probe + build) — last day of the Etherscan Pro subscription. OSMO keyless.

Course of the session

1. Pre-flight: sleep was set to 20 min -> set to Never (AC/DC 0); Windows Update pause confirmed through 2026-08-03.
2. **Task A (HXRO) launched first** (Etherscan-critical) — stream engine skipped it: “already complete”. Investigation found the prompt premise wrong twice:
 - Streamed checkpoints store `rows/mblocks` (24 months intact), not `monthly/last_block` — the “empty checkpoint” reading was a schema misread.
 - Deeper: HXRO panel months are `carried_forward` from 2022-10 (CMC top-1000 visibility lost) while TVL starts 2023-02, and `phase1_assemble_lambda.py` computes lambda on `status='observed'` rows only. Observed window ? TVL window = ? -> **no ch2 rebuild can ever make HXRO regression-ready under the current spec**. Task A closed moot, 0 quota spent. Reopen only via a Phase-3 lambda-on-observed rule change.
3. **Task B (SXP):** Ethereum probe PASS (ABI OK, supply 285.4M, getLogs clean). Root cause of the stale `non-EVM` flag: unprefixed Multi-Chain address in identity. Fixed identity (+`ethereum:` prefix) and `universe_lambda_channel_map.csv` (Ethereum/reachable/Transfer-log), ran ch2: 569,311 tf, 1,886 gl, 77 lambda months 2019-09->2026-02, B2 clean, B4 HODL med 27.0% pass. lambda?TVL 60 months -> complete. BSC fallback unused.
4. **Task C (OSMO):** prompt’s 3 LCDs pruned at 365d. Swept all 16 chain-registry REST endpoints + extras: `osmosis.api.pocket.network` is a FAKE ARCHIVE (height header ignored — SEI landmine family); only `osmosis-api.noders.services` is real. Wrote `session041_osmo_ch1.py` (session-040 pattern + binary-search earliest-block fallback). Built 2 months (2026-04, 2026-05), drift 1.77% PASS. Investigation of the shallow result: prompt’s 15,000 blocks/day is stale — Osmosis runs ~73,300 blocks/day (~1.2 s/block, epochs-module anchors), so the apparent “598-day retention” was ~122 days; every public

node's state+block floor sits at 2026-04-02/03 (chain-wide post-upgrade state-sync point). Nothing keyless goes deeper; 2021-06->2026-03 reopens only on a paid indexer (Numia/ Mintsan). Both built months overlap TVL -> complete. coin_staking_type NaN -> pos (inert for a token-class asset, recorded per prompt).

5. **Task D:** CASINO (Fantom not in Etherscan V2), RUNE (THORChain custom indexer), SUN (Tron engine gap) documented as permanent gaps in Entry 92.
6. **Task E:** assemble + coverage rebuild. lambda 13,547->13,626 / 465->467. Coverage 193/310/1,436. Regression-ready semantics recorded explicitly (complete AND lambda>0, excluding 11 pow_only NVT-only coins): **180 -> 182** (coins 24 unchanged; tokens/other 156->158 = SXP + OSMO).
7. Entry 92, SESSION041_HXRO_SXP_OSMO_REPORT.md, this log, time_log; commit f9cc64b pushed mid-session (data), record-keeping committed at session end.

Deviations from the prompt

- Task A not run at all (moot, see above) — the expected “HXRO partial -> complete” outcome is unreachable by construction.
- OSMO built 2 months, not the hoped ~53 — archive reality, method unchanged.
- Session prompt's OSMO blocks/day constant corrected in the record.

Guards / landmines confirmed

- Fake-archive guard (old-bonded == live-bonded) caught pocket.network again.
- .gitignore *.log — session logs force-added.
- PYTHONUTF8=1 used on all builder/assemble runs.

/newpage

session_042_2026-08-04_paper_restructure_tvlg1_phase3_design.md

Session 042 — 2026-08-04 — Cowork (Fable 5): Paper restructure, TVL*, Phase 3 design

What happened

1. **Full paper restructure** (continuing Moazzam's rewrite request from session 041.5): Theory split into coin framework (QTM -> SoV/MoE -> lambda/(1-lambda), H1a) and token framework (no MoE -> governance moat / costly signaling, H1b), common valuation conditioning (H2) and quadrant (H3). New Section 3 “Variable Construction” (lambda 3-channel composite, NVT_GL, NV/TVL). Data section rebuilt: sources -> sample funnel -> summary stats.
2. **Introduction restored:** Moazzam rejected the rewritten intro (“starts with a boring difference between coins and tokens”). Restored the original progression (motivation / DCF critique -> QTM -> SoV+MoE definitions -> lambda extension -> coin channels) from commit 5d4c575 with MINOR changes only: token paragraph now states the MoE denominator is absent so SoV/MoE is undefined and lambda conveys conviction directly; final paragraph gives two metrics per asset type. Abstract restored similarly.
3. **Growth-adjusted TVL** (Moazzam: “otherwise we will miss growth metric completely”): built phase2_tvlg1.py -> nv_tvlg1_panel.csv. TVL* = same DCF machinery/PARAMS as PQ*; TVL0 = trailing-12m mean level. 111 assets/3,125 mo; token regression sample now 101/2,771 (2021-01..2026-05); combined 125/3,489, \$451B = 19.0% of universe at 2026-05. Caveat: 44.6% of months hit the g cap (mostly -50% floor). Entry 93.
4. **Data 4.2 funnel:** new Table 1 walking 1,939 -> minus 858 other -> coin track 633 -> 24 (minus 323 PoW, 24 unverifiable, 256 no staking archive, 6 no PQ overlap) and token track 448 -> 101 (minus 299 no TVL, 3 no lambda, 45 short TVL history). POL reconciled (coverage file said ‘unknown’, actually PoS). Summary stats now Table 2.

5. **Bibliography:** fabricated entries dao2024governance + daoreview2025 removed (long-standing task #22); bianchi2022 undefined citation fixed.
6. **Phase 3 design** (Moazzam's 3-element plan): (1) H1/H2 FE panel regressions; (2) H3 Stars-minus-Avoid vs monthly LTW (2022 JF) factor analogs built from our own universe; (3) horse race vs raw NVT, Metcalfe (BitInfoCharts AA build), MVRV (checkpoint probe only — Etherscan lapsed), S2F, technicals. Spec: 04_code/PHASE3_ANALYSIS_SPECIFICATION.md; kickoff: 04_code/CLAUDE_CODE_PHASE3_KICKOFF_PROMPT.md. Entry 94.

Ops incident resolved

Git commits kept failing with stale .lock files. Root cause: Cowork sandbox may create/ write but NOT delete files in the mounted folder by default; git cannot remove its own lock files. Fixed by granting the folder delete permission (mcp allow_cowork_file_delete). One collateral commit (39f43b1) briefly recorded phase2_tvl_gl.py + nv_tvl_gl_panel.csv as deleted due to a stale index from the workaround period; amended to 9308322 with both files verified tracked at HEAD.

Open items

- Moazzam still unsatisfied with the introduction; deliberately deferred until empirical results exist (“no use changing it until we have empirical tests”).
- Phase 3 core session (Tasks A-D) ready to launch; Phase 3b horse race after.
- g-cap sensitivity for NV/TVL_GL queued in Phase 3 robustness.

Commits

687d724 (intro restore), 2b9ac26 (TVL*), 9308322 (funnel), + record-keeping commit.

Addendum — same Cowork conversation, continued through the full Phase 3/3b/3c cycle

After the Phase 3 design was committed, this conversation continued through: results review of sessions 043/044/045 (findings review docx + paper-style tables docx); exploratory token quintile sorts (Entry 97) and size-as-delta test (Entry 108, run inline); Phase 3b design + kickoff (Entries 98-99, M1-M4 mechanism framework into paper Section 2.3); DCF misattribution fix + P/F comparator (Entry 107); Phase 3c kickoff incl. technical battery completion (Entries 109-110); full results written into the paper after each results report (Entries 106, 115, 116); H3 re-run under alternative valuation axes (inline, phase3_explore_h3_altval.py — null everywhere); float-placement fix (placeins); Section 7 Summary of Findings with scorecard table. Paper stands at 45 pp with 13 tables, all hypotheses + extension + horse race + mechanisms written up, awaiting Moazzam's pruning review.

State at close (2026-08-04 evening)

- Paper: complete draft, all results in. NEXT: Moazzam reads and decides table pruning (candidates: pooled specs, delta-lambda nulls, Metcalfe + per-sector nulls to appendix); intro rework deferred until after pruning.
- Open tasks: bibliography final pass (old task 22, largely done via entry removals); push to GitHub (many local commits pending — commit_and_push.bat).
- Data/infra: Etherscan lapsed; delete permission enabled for the folder; git works normally from Cowork now.

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session_043_2026-08-04_phase3_core_tests.md

Session 043 — 2026-08-04 — Claude Code (Fable 5): Phase 3 core empirical tests

Kickoff: 04_code/CLAUDE_CODE_PHASE3_KICKOFF_PROMPT.md. Fully autonomous. Spec followed: 04_code/PHASE3_ANALYSIS_SPECIFICATION.md (pre-registered ladder, no tuning after results). Decisions: Entries 95–96. Full readings: 03_data/PHASE3_RESULTS_REPORT.md.

What happened

1. **Task A — regression panel.** phase3_panel.py -> 03_data/phase3/regression_panel.csv (3,489 rows). Funnel gate verified EXACTLY before proceeding: coins 24/718 (2020-12..2026-05, median MC \$823M, NVT_GL 0.017), tokens 101/2,771 (2021-01..2026-05, median MC \$98M, NV/TVL_GL 2.29 IQR 0.20–16.10). Key reconciliation: coin membership = coverage_status=='complete' & lambda_months>0 (a naive coin_staking_type=='pos' filter gives 11/396 — wrong). Coin conviction: 671 ch1 ln-odds months + 47 lambda_z fallback. Assert() guards the funnel in the builder.
2. **Task B — factors.** phase3_factors.py -> ltw_factors_monthly.csv (129 months). CMKT +6.0%/mo (t=2.9), corr 0.93 w/ BTC (sanity pass); CSMB +1.1% (t=0.6); CMOM ?4.6%/mo (t=?1.6) — monthly analog of LTW momentum is negative (short-horizon reversal), documented deviation.
3. **Task C — H1/H2.** phase3_regressions.py -> tables/h1h2_coefficients.csv, h1h2_fm_tokens.csv. **H1a REJECTED at t+1** (coin conv t <= 0.7 unconditional). **H2 SUPPORTED for coins:** convxval ?0.0169 (t=?3.54); split-diff ?0.042 (t=?3.99); conviction is +0.7%/SD in cheap coin-months, ?3.0%/SD in expensive. **H1b weak support:** tokens +0.6–0.8%/mo per SD, t=1.5–2.3 (best with sector FE; FM NW-3 t=1.77/2.19). **H2 REJECTED for tokens** (interaction +0.003, wrong sign). **Voting-weighted lambda NOT better than passive** (710-mo ch3 subsample, all |t|<0.6; low power — lambda_z itself is ~0 there). Caveat recorded: 24 coin clusters; quote the interacted diff test, not the s5_high t=?10.9.
4. **Task D — H3 portfolio.** phase3_portfolios.py -> portfolio_returns.csv, tables/h3_alphas.csv, h3_stats.csv. **H3 REJECTED:** no SMA variant has significant NW-3 alpha vs CMKT/CSMB/CMOM (best pooled VW +4.3%/mo, t=1.48; coin EW ?3.4%/mo, t=?1.5). Breadth guard (>=4/leg) kills 45/65 coin class-months -> 20 non-contiguous months. Token EW alpha post-2023 is negative. Only VW Sharpe (0.66–0.75) beats EW-universe (0.61); EW primary (0.33–0.36) does not. Costs immaterial (turnover 0.10–0.28/leg-month). Single-dimension comparators also dead.
5. **Task E — probes (Entries 95–96).** E1: BitInfoCharts activeaddresses ETH end-to-end OK (4,003 daily obs 2015-08..2026-08; HTML cached to raw/bitinfocharts); TRX/ADA/SOL = HTTP-200 STUB pages, zero data — only ETH of the 24 sample coins covered -> Metcalfe cannot enter the cross-sectional horse race. E2: realized-cap **NO for coin track** — the 24 coins have zero ch2 checkpoints (ch2 = EVM tokens); 211/426 token checkpoints (events-schema) retain full raw transfers and ARE replayable locally, but only 12/101 sample tokens qualify (80 streamed = aggregates only) -> MVRV dropped from horse race.

Environment

statsmodels 0.14.6 + linearmodels 7.0 installed (were absent). No paid services used; zero Etherscan calls; 6 BitInfoCharts fetches (free, keyless).

Open items -> Phase 3b / paper

- Write paper Section 5 from PHASE3_RESULTS_REPORT.md (H2-coins is the headline positive result; H1a/H2-tokens/H3 reported as nulls per honest-results clause).
- Horse race (Phase 3b): comparators reduce to raw NVT / raw NV/TVL, S2F, 52-wk high, MA cross, momentum family (Metcalfe ETH-only baseline; MVRV dropped).

- Robustness queue (spec sec. 7): `lambda_z` vs ln-odds, raw vs GL ratios, MRP 20/40%, B4-flag drop, tercile splits, ex-top-3 MC, g-cap sensitivity (Entry 93 caveat).
- Consider (paper discussion, not respecification): H3 breadth guard redesign for the thin coin cross-section would be a DEVIATION requiring pre-logging.

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`session_044_2026-08-04_phase3b_confirmatory_sorts.md`

Session 044 — 2026-08-04 — Claude Code (Fable 5): Phase 3b confirmatory sorts, coarse sectors, horse race, heterogeneity, mechanisms

Kickoff: `04_code/CLAUDE_CODE_PHASE3B_KICKOFF_PROMPT.md` (Task E / spec ?8.6 added mid-session by user note; picked up and run). Fully autonomous otherwise. Spec: `04_code/PHASE3_ANALYSIS_SPECIFICATION.md` ?8. Decisions: Entries 100–105 (Entries 98–99 were already written by the Cowork design session). Full readings: `03_data/PHASE3B_RESULTS_REPORT.md`.

What happened

1. **Task A — coarse sector remap.** `phase3b_sector_map.py` -> `03_data/phase3/sector_coarse_map.csv` + `tables/sector_coarse_sizes.csv`. Priority-rule keyword map (Entry 100): DEX 41, Other 21, Lending 16, Yield 15, Derivatives 6, Staking/LSD 2; median names/group-month DEX 25 / Other 9 / Lending 8. `phase3b_regressions_coarse.py` re-ran the token ladder both ways (`tables/h1h2_sector_fe_comparison.csv`): raw7 columns reproduce 043 exactly; **coarse FE attenuates the conviction slope** (`s6_1` t 2.30->1.95; `s6_2` 1.93->1.49) — part of the token premium is between-coarse-sector.
2. **Task B — confirmatory conviction sorts (headline).** `phase3b_sorts.py` -> `conv_sort_returns.csv`, `tables/convsort_{alphas,stats,spanning}.csv`. **q5_ew: alpha +1.71%/mo (t=2.18)**, net-50bps +1.53% (t=1.95), post-2023 +1.48% (t=1.79), Sharpe 0.86, ~10.6 names/leg. **Spanning (make-or-break): SURVIVES — alpha strengthens to +2.53% (t=3.18) with the reversal LS controlled and +2.34% (t=2.60) vs the full four-competitor battery** (conviction leg tilts to recent winners; winner-minus-loser itself has alpha ?4.68%, t=?3.02). Honest limits at equal prominence: decile DIES (t=0.02) — monotone-sharpening narrative does not extend; sector-neutral quintile DIES (t=0.39); per-sector terciles all null (DEX 0.89 / Other ?0.77 / Lending ?0.16); coin tercile analog flat (t=0.16).
3. **Task C — horse race.** `phase3b_signals.py` -> `horserace_signals.csv` (raw val, S2F dual build, 52wk-high, MA cross); `phase3b_horserace.py` -> `tables/horserace_{panel,spanning}.csv`. **Coin convxval survives the full joint comparator race (?0.0176, t=?2.83)**; token conv attenuates to t=1.10 jointly (only reversal survives, t=?2.46) — token conviction is a portfolio-extremes phenomenon, not a robust linear panel slope. Spanning both directions: q5 survives every single-competitor control; underpowered n=28/31 joint cells reported as such; no competitor has positive alpha on LTW+q5. `phase3b_metcalf.py` -> descriptive ETH+PoW panel (7 real series; BTC Metcalfe mean-reversion t=?3.69). **New BitInfoCharts landmine: unknown tickers redirect HTTP-200 to the default btc-ltc-eth chart** — page-title guard added (Entry 101); ZEC = stub.
4. **Task D — heterogeneity (run all, report all).** `phase3b_gov_classification.py` -> `token_gov_classification.csv` (101 tokens, ve 31/plain 70, fee 66/nofee 35, 32 low-confidence; dominant-regime rule; Entry 103; EPIC = Ethernity Chain). `phase3b_heterogeneity.py` -> `tables/heterogeneity.csv`, `het_portfolios.csv`. ?lambda: nothing (levels or sorts). **ve split REJECTED — wrong direction** (ve t=0.21 vs plain t=1.21; interaction negative). Fee split: no difference. Size flat; turnover gradient tilts WRONG way (hi t=1.41). Regimes: coin interaction is post-2023 (?0.0205, t=?4.32) and mildly bull-tilted. Measurement: coin interaction robust to MRP 20/40 (t~?3.5) and stronger ex-conv-fallback (t=?4.67), but raw NVT shrinks it (?0.0060, t=?1.90) and

ex-g-cap (40% of coin months) kills significance ($t=0.97$) — growth-levelization is load-bearing (Entry 104).

5. **Task E — mechanisms M1–M4.** In `phase3b_heterogeneity.py` -> `tables/mechanisms.csv`. **M1 NOT SUPPORTED (convxturnover positive, wrong sign); M2 rejected (no revival under sector-demeaned val, panel or portfolio); M3 rejected (token null not a measurement artifact); M4: coin result SURVIVES staking-yield controls (convxval 0.0188, $t=4.39$)** — not a seigniorage artifact. Paper 2.3 loses its M1 leg and must reframe (Entry 105). Staking-yield = `supply_g12 / logistic(conv)`, ch1 months.

Session facts

- No paid calls; 8 free keyless BitInfoCharts fetches (AA pages, cached under `03_data/raw/bitinfocharts/activeaddresses` bogus btg cache deleted).
- Phase 0–3 outputs untouched; all new outputs in `03_data/phase3/`, builders `04_code/phase3b_*.py`.
- Sanity checks passed: raw7 ladder == session 043; q5_ew == Entry 97 exploratory numbers exactly (+1.71%/mo, $t=2.18$).
- Known non-blocking noise: pandas4 concat-sort deprecation warnings in `phase3b_sorts.py`; linearmodels sqrt warnings on some absorbed sector dummies (reported coefficients all have finite SEs).

Next session candidates

- Paper integration: Section 5 sort battery + spanning table; Section 2.3 mechanism reframe (M1 reversed); coin-H2 caveat block (raw-NVT attenuation, g-cap subsample).
- Optional: logged-in verification pass on the 32 low-confidence governance classifications; per-sector tests once breadth grows.

/newpage

`session_045_2026-08-04_phase3c_fees_dcf_technicals.md`

Session 045 — 2026-08-04 — Claude Code (Fable 5): Phase 3c — fee/revenue DCF comparators (tokens only) + technical battery completion

Kickoff: `04_code/CLAUDE_CODE_PHASE3C_KICKOFF_PROMPT.md`. Fully autonomous. Spec: `04_code/PHASE3_ANALYSIS_SPECIFICATION.md` 78.7 (as amended: tokens only) + 78.8. Decisions: Entries 111–114 (109–110 were written by the Cowork design session). Full readings: `03_data/PHASE3C_RESULTS_REPORT.md`.

What happened

1. **Task A — fees/revenue panel.** `phase3c_fees_panel.py` -> `03_data/phase3/fees_revenue_panel.csv` (2,437 rows) + `tables/fees_coverage.csv`. DeFiLlama fees API verified live (free/keyless; parent slugs accepted by /summary even when the overview lists only version children — live probe per slug is the authority). Identity: `cmc_id` -> `dl_slug` from `tbl_panel.csv`; 4 chain-level tokens (ARB/METIS/APE/BLAST) on DL chain fee adapters, flagged (sequencer fees = DAO-accruing; Entry-68/84 precedent). **Coverage 55/101 tokens fees, 51/101 revenue**, median 44 mo, starts mostly 2021+. Six candidate slug resolutions probed and all REJECTED (metronome/rari/nerve/aurora/bounce.tech/thundercore — collision or zero usable months; Entry 111). `dydx-v3` adapter covers only the 2023-11..2024-10 wind-down (limitation logged).
2. **Task B — comparators.** `phase3c_comparators.py` -> `fee_comparators.csv`. `pf` = MC/trailing-12m fees (≥ 6 obs); `prev_gl` = MC/REV* and `pf_gl` = MC/F* via the EXACT `phase2_nvt_gl_PARAMS/pq_star` machinery, $g = 3y$ CAGR of the base (2y/1y fallback). Coverage on the token panel: `pf` 1,367/51, `pf_gl` 1,178/44, `prev_gl` 976/35 (+202 `pf_gl`-only rows = the fees-reported-revenue-not asymmetry, itself a finding). Median P/F 12.1; $\text{corr}(\ln P/F, \ln NV/TVL) = 0.48$ — distinct axis.

3. **Task C — the core question (C1) and the race.** `phase3c_tests.py` -> `tables/phase3c_{c1_feeval,race_panel}`. **C1 VERDICT: the token H2 null survives its sixth and final measurement candidate — $\text{conv} \times \ln P/F = ?0.0024$ ($t = ?0.45$); $\text{conv} \times \ln \text{prev_gl} = ?0.0035$ ($t = ?0.31$).** First-ever H2-signed point estimates and correctly-ordered splits, but a null; coverage-matched NV/TVL_GL baselines equally dead -> not a denominator effect. **New fact: $\ln P/F$ is the first valuation signal that works in the token panel** (singles $t = ?2.50/?2.70$ sub-2024; s4 level $t = ?3.21$; survives the completed joint battery $t = ?2.32$) — and the DCF transform destroys it (prev_gl $t = ?0.42$), while the quintile portfolio version earns nothing ($t = ?0.10$): panel-slope phenomenon, mirror image of conviction. Composition caveat: on the revenue-covered subsample the conviction slope flips negative ($?2.01$) before any fee variable enters. C3: neither fee comparator spans the conviction quintile; C4 sub-periods add nothing.
4. **Task D — technical battery completion.** `phase3c_technicals.py` -> `technical_signals.csv` (`ma_dist`, `vol12`, `ivol`, `amihud` [snapshot-volume caveat], `skew36`; 90–100% coverage). **KEY VERDICT: the token conviction quintile SURVIVES the completed battery — $+1.74\%/mo$ ($t = 2.45$) vs LTW + 12 competitor long-shorts; $+2.17\%$ ($t = 2.82$) with the fee comparators added.** The two underpowered 044 MA-cross spanning cells are superseded by continuous `ma_dist` and **RESOLVED: $+all\ 0.99t/n28 \rightarrow 2.73t/n43$; single cell $1.16t/n31 \rightarrow 2.46t/n50$.** Vol and skew (the last candidate spanners) do not span. No technical earns a significant token long-short. **Coin interaction survives the completed battery too: $\text{conv} \times \text{val} ?0.0235$ ($t = ?2.68$) full / $?0.0188$ ($t = ?3.17$) sub-2024;** `amihud` is the strongest new coin single ($t = ?2.65$) and doesn't dent it. Paper exclusions sentence (RSI/MACD/Bollinger redundant; daily-native infeasible) drafted in the report ?6.

Session facts

- 202 free keyless DeFiLlama fetches (cached under `03_data/raw/phase3c/fees/`); no paid calls, no Etherscan.
- Phase 0–3b outputs untouched; new outputs in `03_data/phase3/`, builders `04_code/phase3c_*.py`.
- Sanity checks passed: `q5_ew` reference alpha reproduces 044 exactly ($+1.71\%/mo$, $t = 2.18$, $n = 50$); funnel counts unchanged (101/2,771 tokens).
- Known non-blocking noise: `pandas4` concat-sort deprecation warnings in `phase3c_tests.py` spanning section.

Next session candidates

- Paper integration: Section 6 sixth-conditioner row set (C1); Section 5 horse race gains P/F (the practitioner-multiple-beats-its-own-DCF framing); spanning table upgraded to the completed battery with the 044 supersession footnote; coin H2 paragraph updated to the 12-comparator figure; limitations (fee coverage, revenue-subsample composition).
- Optional: VW variants of the fee-comparator sorts; P/F within-sector version (is the level effect between- or within-sector, mirroring the conviction decomposition).

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Part 2: Human Time Log

Human Time Log — AFA 2027 QTM Crypto Research

Per AFA rules, document all human activity: what you did, how long it took, and whether it was a prompt to AI or a direct contribution.

Date	Duration (min)	Activity Type	Description
2026-06-10	15	Prompt	Initial project scoping — shared AFA call and QTM framework write-up with AI
2026-06-10	10	Direct	Created GitHub repository; reviewed AI-generated README and folder structure
2026-06-10	60	Decision + Review	Theoretical framework discussion: convenience yield, coins vs. tokens, TVL disambiguation, rejected theta coefficient, finalized lambda framing and paper outline
2026-06-10	15	Review	Reviewed and approved full LaTeX introduction draft (Section 1)
2026-06-12	45	Review + Decision	Read introduction draft, identified 8 issues, reviewed revised draft, empirical design brainstorm (lambda measures, NVT/G, asset universe, portfolio construction)
2026-06-12	30	Decision	Finalized NVT/PMT leveled approach, asset category structure (coins/L1, L2, token groups), blockchain-agnostic design, HYPE coin vs. token classification
2026-06-12	30	Decision + Review	Hypothesis structure finalized: lambda as primary signal, NVT/f as conditioning variable, H1a/H1b/H2/H3 approved; reviewed Hypotheses section draft
2026-06-19	20	Decision + Review	Chose “applied theory” (light formalization) over full structural model for Section 2; decided to merge Theory and Hypotheses into one combined section; reviewed drafted propositions and theory section

Date	Duration (min)	Activity Type	Description
2026-06-19	25	Decision + Review	Identified flawed $F(s_t + /bar{c}_{t+1})$ aggregation step in Section 2.1; requested rigorous continuum-of-agents fix instead of a bare assumption; explored governance-token staking mechanics (vote-escrow vs. snapshot vs. fee-sharing); decided $/lambda$ should be a multi-channel measurement index (staking/holding/voting), construction deferred to data stage
2026-06-19	35	Decision + Review	Challenged the artificial coin/governance-token model split and the $/theta_{t+1}$ vs. random-walk simplification (accepted correction on the latter); approved unifying Sections 2.1/2.2 into one “Locking Decision” model; identified unjustified “L1-only” restriction in Hypothesis 1a and the implicit asset-index ambiguity in Proposition 1; reviewed and approved the revised, unified Section 2.1 and renumbered Propositions/Section 2 subsections

Date	Duration (min)	Activity Type	Description
2026-06-19	10	Decision	Identified that “NVT/fundamental” is a misleading name (implies division by an undefined “fundamental” term); evaluated candidate replacements and selected “Growth-Levelized NVT” over “Growth-Normalized NVT” on grounds of mechanical precision; approved renaming throughout the paper
2026-06-22	25	Decision + Review	Decided to write a data specification document (for a separate Claude Code session) before drafting Section 3 prose; answered clarifying questions on asset-universe scope (point-in-time rank screen, not fixed-dollar floor, to jointly avoid penny-token noise and survivorship bias), sample period (Aug 2015 onward, not Jan 2021), factor model (both CAPM and Liu-Tsyvinski-Wu, reported as mutual robustness), and lambda-index construction (equal-weighted average of standardized available channels); reviewed and approved 04_code/DATA_SPECIFICATION.md and 04_code/DATA_DECISIONS_LOG.md

Date	Duration (min)	Activity Type	Description
2026-06-22	~15 (est., confirm)	Prompt + Decision	Launched the Phase 0 pipeline session (Claude Code / Opus 4.8); answered two judgment-call questions when CoinGecko's 365-day historical limit was surfaced — chose the free CoinMarketCap historical endpoint as the ranking backbone, and chose to exclude stablecoins entirely from the universe; confirmed AFA record-keeping requirements and the GitHub repo, and authorized end-of-session commit+push
2026-06-22	30	Review + Decision	Reviewed Phase 0 outputs (coverage report, session status, decisions log) against Section 2/3 framing and prior asset-pricing precedent (Liu-Tsyvinski-Wu stablecoin exclusion); confirmed no main.tex changes needed yet; resolved death-return treatment (split presumed-failed vs. temporarily-out before deciding a formula), classification-review scope (clarified “thin” = months-observed persistence filter, not market cap; set ≥ 12 -month threshold), Artemis/paid CoinGecko access (deferred), and N=250 (confirmed)

Date	Duration (min)	Activity Type	Description
2026-06-22	15	Decision	Resolved meme-coin/NFT question (meme coins already correctly excluded from H1a/H1b/H3 via the ‘other’ classification; keep current handling, confirm NFT absence as a verification step) and the Quadrant Portfolio 2x2 construction method ($\lambda/(1-\lambda)$) and Growth-Levelized NVT median splits computed within coin and token subsamples separately, not pooled); approved a Phase 0 follow-up kickoff prompt (carry-forward split + classification confirmation pass) before Phase 1 begins
2026-06-22	15	Decision	Before committing, clarified that “class” should also mean a narrower economic-function/sector tag (L1/L2, DEX/Perpetuals, Lending/Staking) independent of the coin/token cut used in Entry 13; decided to capture this now as a sector field sourced from existing DeFiLlama categories + CMC layer-1/layer-2 tags (spec section 2.6, Decisions Log Entry 16), deferring which sector-level comparisons to test until later; revised the Phase 0 follow-up kickoff prompt to add this as a third deliverable

2026-06-22 | ~10 (est., confirm) | Prompt + Review | Launched the Phase 0B follow-up session (Claude Code / Opus 4.8) with the prepared follow-up kickoff prompt; reviewed the three completed deliverables (carry-forward presumed_failed/temporarily_out split with right-censoring counts; 16-name classification confirmation flips + meme/NFT confirmation; sector-field coverage); authorized updating project memory,

revising the Phase 1 kickoff prompt, and committing+pushing logs and prompt history per the project cadence |

2026-06-22 | 15 | Review + Decision | Independently re-read PHASE0B_FOLLOWUP_REPORT.md and Decisions Log Entries 17-20 (Cowork session, separate from the Claude Code session that produced them); verified the carry-forward subtype column, the 16 classification flips, and the sector field actually exist in universe_panel.csv/classification_table.csv; confirmed the 4 flagged open items (death-return formula, meme-token demotion, 16 gray-zone names, sector DeFiLlama symbol-collision) are non-blocking for Phase 1 (Phase 3-scoped, or explicitly meant to be informed by Phase 1 data); gave the go-ahead to start Phase 1 (lambda channels) via CLAUDE_CODE_PHASE1_KICKOFF_PROMPT.md |

Activity Types: - **Prompt** — writing/refining a prompt to the AI - **Direct** — human-authored code, writing, or edits - **Review** — reading and evaluating AI output - **Decision** — making a judgment call on research direction

2026-06-24 | ~10 (est., confirm) | Prompt + Decision | Launched session 012 (Claude Code / Opus 4.8) to run the Phase 2 PQ Etherscan feasibility pilot (CLAUDE_CODE_PHASE2_PQ_PILOT_PROMPT.md), since the Cowork session that wrote the prompt had no network route to Etherscan. Supplied that the free Etherscan key from the prior session lives in gitignored 04_code/.api_keys.json; authorized the AI to find it there and run the pilot end-to-end (pull all Transfer events for UNI/AAVE over a recent month, dollarize, cross-check vs DeFiLlama, extrapolate to panel scale) and to commit+push at session end. Pilot outcome: raw governance-token Transfer-log volume is the *wrong quantity* (UNI token transfers 46.6x smaller than Uniswap DEX swap volume, corr 0.30; AAVE raw sum corrupted by sentinel-value transfers) and full-panel multi-year raw extraction is ~weeks on the free key — so PQ source decided as DeFiLlama reported sector-volume with TVL/fees as side diagnostics (Decisions Log Entry 31; 03_data/PHASE2_PQ_PILOT_REPORT.md). Phase 2 build remains paused pending review of this recommendation. |

2026-06-23 | ~15 (est., confirm) | Prompt + Decision | Launched the Phase 1 (lambda channels) session (Claude Code / Opus 4.8) via CLAUDE_CODE_PHASE1_KICKOFF_PROMPT.md; mid-session redirected the data sourcing to canonical chain sources (“etherscan or other chain main data source”) instead of an aggregator, to ensure lambda numbers come from the real on-chain source; registered a free Etherscan V2 API key (“AFA Paper”) and supplied it for gitignored local storage; authorized end-of-session commit+push |

2026-06-23 | 20 | Review + Decision | Reviewed PHASE1_COVERAGE_REPORT.md and session_009_2026-06-23_phase1_lambda. clarified the free-vs-paid Etherscan question (free tier covers everything built; the real constraint is no free archive *state*, not cost). Decided the next session’s scope: one combined Claude Code session for Phase 1 finishing touches (finish ETH staking series, explore a free workaround for Channel 2 before considering any paid source, confirm N=250, revisit the 16 gray-zone names, fix the report’s internal asset-month count inconsistency) followed immediately by Phase 2 (NVT_GL); decided to keep ETH’s cumulative-deposit method as-is (no post-Shapella netting this pass). Drafted and reviewed CLAUDE_CODE_PHASE2_KICKOFF_PROMPT.md; logged Decisions Log Entry 28 |

2026-06-24 | ~10 (est., confirm) | Prompt + Decision | Launched session 010 (Claude Code / Opus 4.8) via CLAUDE_CODE_PHASE2_KICKOFF_PROMPT.md. Part A (Phase 1 finishing touches) ran to completion — reviewed at a high level. Mid-Part-B, when the AI surfaced the PQ-definition divergence (proposed TVL for tokens and on-chain transaction volume from Etherscan for coins; AI countered that PQ must be a throughput *flow* not a stock (TVL), and that coin transfer-volume hits the same free-archive wall as Phase 1), **decided to pause the build and take the two PQ decisions into Cowork** where the full project context lives, rather than resolve them inline. Directed the AI to generate a status report capturing the questions (06_documentation/PHASE2_PQ_DECISION_STATUS.md) and to commit for log/history integrity |

2026-06-24 | 40 (est., confirm) | Review + Decision | Read PHASE2_PQ_DECISION_STATUS.md (session 010’s fees-recommended PQ framing); requested research on how Artemis Analytics/Token Terminal/DeFiLlama define “economic activity” (initial read favored fees over TVL). Pushed back on that read: economic activity is the *value* a transaction generates, fees are the *cost* of facilitating it — same error as treating a country’s tax revenue as GDP. Re-derived PQ from the paper’s own $M^?V=P^?Q$ identity, walked through what TVL actually represents (capital stock that enables activity, not the activity itself), and landed on

sector-appropriate transacted value (DEX/swap volume, loan-origination flow, derivatives notional) as the corrected PQ definition — consistent with the original NVT ratio's own lineage. Asked whether Etherscan's existing Transfer-log mechanism could compute true transfer volume directly; confirmed the Cowork sandbox has no live network route to Etherscan, so decided to pilot-test feasibility (UNI, AAVE, 30-day window, cross-checked against DeFiLlama volume) in the next Claude Code session rather than resolve scale by extrapolation. Logged Decisions Log Entry 30; drafted `CLAUDE_CODE_PHASE2_PQ_PILOT_PROMPT.md`; logged this session as `session_011_2026-06-24_pq_theory.md` |

2026-06-24 | 35 (est., confirm) | Review + Decision | Read `PHASE2_PQ_PILOT_REPORT.md` (session 012's pilot finding: raw governance-token Transfer volume is the wrong quantity for tokens, not just costly). Pushed back on the project's own prior "archive access" claim for native-coin transfers ("doesn't Etherscan clearly show ETH transfer value for every transaction?"); researched the full-node-vs-archive-node distinction and confirmed the claim was a conflation — archive access is only needed for historical *state* (balance/coin-age, Entry 24), not for summing the **value** field already in ordinary block data, which Etherscan's free API already serves; real constraint is call-volume (block count), not access. Researched Artemis's "Total Economic Activity" metric in full (= Chain Fees & MEV + Settlement Volume + Application Fees) and found Artemis's own blog independently makes the same fees-are-not-GDP argument used in session 011. Asked for a consolidated recommendation and which source (raw Etherscan vs. DeFiLlama) is the "cleaner truth" for tokens; answered that DeFiLlama is cleaner on two independent grounds (right object — protocol pool events, not a side governance token's wallet transfers — and free of AAVE's sentinel-value corruption). Decided the token rule: DeFiLlama volume primary; if unavailable, back out volume from fee only when the fee is a confidently known single, stable rate (not multi-tier/variable/non-volume-linked), else flag PQ missing. Decided the coin rule (a per-asset ladder, not one global source): checked live whether DeFiLlama has a chain-level "activity" metric (`/overview/dexs/{chain}` — confirmed real but live-tested as degenerate for Bitcoin, \$419,825/24h vs. Bitcoin's actual base-layer settlement, driven by 3 niche bolted-on protocols) -> use it only where non-degenerate (DeFi-heavy chains) -> else Artemis Settlement Volume if live-verified free-tier-accessible -> else coin-specific native fallback (blockchain.com's free BTC "Estimated Transaction Value" series, or bounded native block iteration) -> chain fees only as an explicitly flagged last resort. Logged Decisions Log Entry 32, superseding Entry 31's coin-fallback line. |

2026-06-24 | ~10 (est., confirm) | Prompt + Decision | Launched session 013 (Claude Code / Opus 4.8) via `CLAUDE_CODE_PHASE2_PQ_BUILD_PROMPT.md` to build the Phase 2 NVT_GL on the resolved PQ definition (Entries 30–32). Confirmed the repo path (C:/AFA_2027_QTM_Crypto, C: root) when the session opened outside it, and that the tree held uncommitted Part-A work (Entries 30–32) to build on, not clobber. Mid-Step-B1, when the AI surfaced that **Artemis's REST API is paid-only** (free "Lite" tier = Terminal/Sheets only) and that this leaves **81 material coins** (XRP, DOGE, LTC, BCH, XMR, ZEC, DASH, ATOM, DOT, ?) with no free transacted-value PQ source — the build prompt's designated STOP-and-report point — **decided: "start on what you have and create a Phase 2b to find solutions for the 81 coins"** rather than auto-dropping them to a toll proxy. Authorized building the covered panel (16 tokens via DeFiLlama sector volume + 49 Rung-1 coins + BTC via blockchain.com), flagging the 81 coins PQ=NaN, writing the Phase 2b kickoff, and end-of-session commit+push. Outcome: NVT_GL for 1,821 asset-months / 59 assets (Decisions Log Entries 33–34; `03_data/PHASE2_COVERAGE_REPORT.md`). Phase 2 covered-panel build complete; Phase 2b (81 coins) and Phase 3 both pending review. |

2026-06-25 | ~10 (est., confirm) | Prompt + Decision | Launched session 014 (Claude Code / Opus 4.8) via `CLAUDE_CODE_PHASE2B_KICKOFF_PROMPT.md` to source native settlement-value PQ for the 81 deferred GAP-R2 coins. No Artemis key present (`.api_keys.json` has only Etherscan), so Rung 2 stayed closed and the session worked free sources only. Authorized building on free, live-verified sources and end-of-session commit+push. Outcome: **8 coins sourced** via bitinfocharts native "Sent in USD" (DOGE, LTC, BCH, DASH, ETC, BTG full; BSV/ZEC partial-stale) — flagged change-inflated (UTXO) / transparent-only (ZEC) per the kickoff's honesty rules; **73 remain PQ=NaN with live-verified reasons** (XRP — Ripple Data API dead, no free XRPL volume series; XMR — RingCT; 71 others — no free ready-made native series, Cosmos/Kaspa LCDs current-state only, Filfox is fees, Mintscan/Subscan need keys). NVT_GL extended **1,821 -> 2,526 asset-months, 59 -> 67 assets** (Decisions Log Entry 35; `03_data/PHASE2_COVERAGE_REPORT.md`). Phase 2b done for what free sources allow; Phase 3 pending review. |

2026-06-25 | ~10 (est., confirm) | Prompt + Decision | Launched session 015 (Claude Code / Opus 4.8)

via `CLAUDE_CODE_DUNE_PILOT_PROMPT.md` to run the bounded Dune Analytics feasibility pilot — does Dune’s decoded/Spellbook data supply the token-side PQ DeFiLlama lacks (Lending, Liquid Staking) or paywalls (Derivatives 402), *before* requesting university funds for a paid Dune plan. Free Dune key already in gitignored `.api_keys.json`; authorized running it for real and end-of-session commit+push. Outcome: **all 3 categories cleanly solved on the free tier** — AAVE 30d borrow origination \$4.29B (lending.borrow spell, cross-checked 0.59x outstanding), LDO 30d stake+unstake flow \$1.58B (Lido events x canonical-WETH price; caught and fixed a `symbol='WETH'` scam-token contamination that had implied \$767/ETH vs real \$1,768), GNS 30d perps notional \$1.18B (Gains’ own DeFiLlama-upstream table — fills the 402 gap for free). Cost trivial: 9 small SQL executes + 14 catalog searches, <100 of 2,500 free monthly credits. **Recommendation (A)** — free tier sufficient to prototype all three at panel scale (spell tables aggregate the whole panel per query), **gated on a free-tier panel-scale dry-run** to confirm the small engine survives multi-year scans before any subscription is bought (Decisions Log Entry 36; `03_data/DUNE_PILOT_REPORT.md`). No panel outputs written (diagnostic only). **Do not buy a Dune plan or start Phase 2c before the dry-run + review.** |

2026-06-25 | ~10 (est., confirm) | Prompt + Decision | Launched session 016 (Claude Code / Opus 4.8) — the Dune free-tier full-panel **dry-run** that session 015’s recommendation (A) was gated on. Authorized running it for real against the gitignored free Dune key and end-of-session commit+push; reiterated the AI must **not** purchase/subscribe to any plan (Moazzam has university approval to buy *if* the evidence requires it, but would do checkout himself). Outcome: **the gated engine risk is disproven -> unconditional (A)**. All three full-history, monthly-grouped queries completed on the free small engine (Lending 2.0 s, Derivatives 1.9 s, the join-heavy Liquid-Staking worst case 48.8 s — no timeouts/row-caps/errors; the pilot’s scary 66 s/30 d was variance). Coverage: only **4 of the 17** NaN tokens are in Dune’s normalized spellbook — AAVE, **STRK** (Strike Finance, a net-new find: \$224.3 M lifetime borrow origination), LDO, GNS; the other **13** appear only as raw token/vault contracts -> **(C)**, and crucially **a paid plan would not recover them** (the residual gap is spellbook coverage, which money can’t buy, not engine size, which it could). **Net: do not buy a Dune plan**; a future Phase-2c can fill those 4 from Dune on the free tier and leave the other 13 documented NaN (Decisions Log Entry 37; `03_data/DUNE_DRYRUN_REPORT.md`). No panel outputs written (diagnostic only). Phase 2c + any purchase remain paused pending human review. |

2026-06-25 | 45 (est., confirm) | Review + Decision | Pushed back on the Dune dry-run’s 4-of-17 coverage as too thin a cross-section to be useful, and asked whether TVL (a stock) can proxy PQ (a flow) via some “rate,” drawing a nominal-GDP/no-labor analogy; separately asked what “turnover” means and whether NVT’s T is TVL. Corrected: NVT’s T is on-chain transacted volume, never TVL; the existing TVL ratio is Market-Cap/TVL, unrelated. Mapped the “capital x rate, no labor” intuition onto the AK growth model ($Y=A^?K^?PQ=A^?TVL$, $A=\text{turnover}=\text{corporate-finance asset-turnover ratio}$), tying it to the paper’s own $M^?V=P^?Q$ framing. Surfaced DeFiLlama’s separate Fees/Revenue series as a more direct lead than turnover, via Entry 32’s already-decided fee->volume inversion rule (Volume ~ Fees ? a confidently known fixed fee rate), now generalized from 8 coins to the full token set. Live-verified the Cowork sandbox cannot reach `api.llama.fi` (403 blocked-by-allowlist) — must verify from local Claude Code. Re-derived the real 111-token NaN category breakdown (Yield 16, Derivatives 10, Farm 8, Gaming 7, Dexs 7, Services 7, Bridge 6, Lending 6, Launchpad 6, Chain 6, Yield Aggregator 4, +singles), materially larger/different than the 17-token Dune scope. Decided to scope the next session as a **diagnostic-only metadata audit** across all 111 NaN tokens (excluding the 7 Gaming tokens, which stay NaN by design): check what DeFiLlama data exists per token, judge the viable conversion path per protocol’s actual economic model, and calibrate/report turnover dispersion across the 25 lending + 28 perpetual Dune cohort (Entry 37) before recommending any TVLxturnover imputation. Logged Decisions Log Entry 38; drafted `04_code/CLAUDE_CODE_PHASE2C_DIAGNOSTIC_PROMPT.md`. No data pulled, no panel write this session. |

2026-06-25 | ~10 (est., confirm) | Prompt + Decision | Launched session 017 (Claude Code / Opus 4.8) via `CLAUDE_CODE_PHASE2C_DIAGNOSTIC_PROMPT.md` — the local, diagnostic-only metadata audit of all 104 non-Gaming NaN tokens that the Entry-38 TVL-as-flow discussion required. Authorized running free-tier DeFiLlama + the existing free Dune key for real; reiterated no `pq_tokens.csv` write, no PQ value, and **no purchase** (DeFiLlama Pro / bridges-API decisions are Moazzam’s). Outcome: **only 5 of 104 tokens have a defensible free transacted-value path, and 4 were already known** (AAVE/STRK/LDO/GNS, Entry 37); the one net-new find is **SUN** (its AMM is SunSwap -> DeFiLlama dexs `sunswap-v1/v2/v3`, missed in Phase 2 because the stored slug `sun.io` isn’t in the vertical). **TVLxturnover is statistically indefensible**

— the 25-lending/28-perp Dune cohort shows per-project turnover dispersion ~1,455x (lending) and unbounded (perps, no central tendency), even after version-scope-matching TVL -> report it as explored-and-rejected, don't build. TVLxAPY collapses (yields snapshot is current-only; 25/28 farm/yield tokens dead). Two new landmines: **bridges.llama.fi now fully 402-paywalled** (kickoff assumed free), and the nerve/NVT + velodrome/VELO symbol collisions (both ruled out on cmcId). Token-side PQ realistically grows 16->~21, not "most of 111" (Decisions Log Entry 39; 03_data/PHASE2C_DIAGNOSTIC_REPORT.md). No panel outputs written. Phase 2c build + any purchase remain paused pending review. |

2026-06-30 | 15 (est., confirm) | Purchase + Decision | Purchased Etherscan API Pro Standard Plan (\$199/month, 200k calls/day, all chains including BSC/Base/Avalanche, auto-renew monthly). Purchase triggered by session-024's measured finding that Channel-2 (coin-age/HODL) panel-scale build needs ~555k getLogs calls = 5.6x the free 100k/day cap. Standard Plan solves Channel-2 throughput AND unlocks the 16 P2-2 tokens (BSC/Base/Avax) from session-022 that were gated on paid getLogs. Drafted 04_code/CLAUDE_CODE_SESSION025_ETHERSCAN_PRO_BUILD_PROMPT.md for next Claude Code session: verify Pro key on BSC, run P2-2 (16 paid-gated tokens), build Channel-2 HODL-6m at panel scale (793 EVM free-chain tokens), wire into lambda assembler as ch2_holding, report new asset-month count. CEX address label feed (Nansen) deferred — contract screening via eth_getCode handles dominant contamination; CEX EOA limitation to be documented in paper. Manual step before session 025: retrieve Pro API key from etherscan.io/myapikey and update .api_keys.json["etherscan"] locally. |

2026-06-29 | 20 (est., confirm) | Review + Decision | Reviewed session 022 (Claude Code, run locally — full-universe lambda-channel feasibility pass: read verified contract source + confirmed mechanisms live via getLogs across 1,306 EVM+non-EVM assets; see 06_documentation/SESSION022_STATUS_AND_NEXT_SESSION.md). Verdict: a genuine methodological upgrade over session 021 — used Etherscan directly (contract-code reads + getLogs) rather than 021's Dune top-holder substitute, and measured the real free/paid boundary (getSourcecode free on every chain; getLogs/tokenTx free only on Ethereum/Polygon/Arbitrum/Blast, paid-only on BSC/Base/Avalanche). Flagged two contradictions before any build: **HEX** and **AKRO** are listed among session 022's "6 genuine Channel-1" tokens despite explicit session-021 rejections (HEX: "staking internal to the contract, no separate escrow"; AKRO: "no single contract reproduces the DL staked figure," grouped with KAITO/ATH/SUPER) — and AKRO's own parenthetical ("bare Locked(), need contract-balance reads") shows it hasn't actually cleared the Entry-26 bar yet despite being bundled into the "6." Also flagged the unreconciled 1,306-asset universe count and the pre-dedup "34 Channel-3" figure. Moazzam's call: trust session 022's contract-code-read + getLogs method over 021's Dune substitute as the tiebreaker, and confirmed the 1,306-asset universe is the full in-scope set (Claude re-derived it directly from the data folder after the 398 audit). Separately, directed that the ~284 dead 2014-2018 listings (no chain/contract identity on any explorer, EVM or non-EVM) be documented as a permanent survivorship-bias limitation, not a pending data gap. Drafted 04_code/CLAUDE_CODE_HEX_AKRO_RECONCILIATION_AND_SURVIVORSHIP_PROMPT.md for the next Claude Code session: reconcile HEX/AKRO using session 022's method (full reconstruction + balance cross-check, not just "an event fires"), log the resolution as a new DATA_DECISIONS_LOG entry naming which prior entry it supersedes, and write a citable survivorship-bias note for the paper's data section. Session 022's other open items (Channel-3 dedup, the 16 paid-gated tokens, Channel-2 prototype) explicitly out of scope for the next session. |

2026-06-26 | 35 (est., confirm) | Review + Prompt + Decision | Cowork session. Re-confirmed the coin/token lambda/(1-lambda) vs. raw-lambda x-axis correction, the NV/TVL-primary valuation-ratio choice, and that category fixed effects and a continuous Fees/TVL control are complements, not substitutes (carried from a prior session). Audited current real coverage for lambda, TVL, PQ-for-coins, and NV across the universe (lambda: 52/1,081 in-universe assets, almost all token-side; TVL: 0 in a persistent panel, only diagnostic presence metadata for 104 tokens; PQ-for-coins: 58/633 with real data; NV: 100% solved via market cap) and reported it in chat. Directed: build lambda and TVL datasets first; figure out coin PQ sourcing afterward, in a separate session. While grounding the next kickoff prompt, found **the local working copies of DATA_DECISIONS_LOG.md and time_log.md were stale** — missing Entries 33-39 and the last ~6 rows (sessions 013-017) that were already committed and pushed to HEAD — alongside a stale .git/index.lock (could not be removed from the Cowork sandbox; likely held by a process on Moazzam's machine, invisible to the sandbox) and an index that showed every tracked file as simultaneously staged-deleted and untracked. Verified byte-for-byte that this was **not data loss** (code/data CSVs matched HEAD once Windows CRLF line endings were normalized out) — isolated to the two

append-only logs, which were restored verbatim from `git show HEAD:<path>` (a plain file overwrite, not a git operation, so the held lock didn't block it). Logged the incident and the scope decision as Decisions Log Entry 40, drafted `CLAUDE_CODE_PHASE1_SCALE_LAMBDA_AND_TVL_PANEL_PROMPT.md` (lambda Channel 1/3 scale-up for tokens, a live PoS-coin source-verification pass for coins, and a real `tv1_panel.csv` build replacing the diagnostic-only TVL check; coin PQ explicitly out of scope), and logged this session as `session_018_2026-06-26_repo_sync_and_lambda_tv1_scope.md`. Flagged to Moazzam: confirm no git process is actually running on his machine, then manually clear `.git/index.lock` if needed before the next Claude Code session commits. |

2026-06-26 | ~10 (est., confirm) | Prompt + Decision | Launched session 019 (Claude Code / Opus 4.8) via `CLAUDE_CODE_PHASE1_SCALE_LAMBDA_AND_TVL_PANEL_PROMPT.md` to scale lambda coverage and build the first real TVL panel. Authorized running free, live-verified sources (DeFiLlama, Snapshot, Etherscan free key, chain-native explorers Koios/TzKT) and end-of-session commit+push; reiterated no paid tier / no purchase. Step-0 hygiene clean (no stale index.lock locally; next-numbers re-derived live = Entry 41 / session 019). Outcome: **lambda 1,374->1,688 observed asset-months, 52->58 assets** — coin staking went ETH-only -> ETH+ADA+XTZ (Koios `active_stake` / TzKT `totalBakingPower`, both free+keyless+historical; XTZ flagged for the 2024 Paris redefinition), four EVM vote-escrow/stake tokens added to token C1 (PENDLE/LQTY/1INCH/RPL, each Entry-26 verified; GMX verified-but-deferred for Arbitrum-getLogs perf; MKR/BAL/COMP/RUNE/ANGLE rejected with reasons), and two governance spaces the Entry-25 gap list actually had (ENA `ethenagovernance.eth`, PERP `vote-perp.eth`) added to C3. Separately built `03_data/phase2/tv1_panel.csv` — **a real persistent monthly TVL panel, 99 tokens / 4,999 asset-months, 2019-12->2026-05** (a valuation-multiple NV/TVL denominator, not a PQ proxy; Entry 30 unchanged). Decisions Log Entry 41; `03_data/PHASE1_LAMBDA_SCALE_AND_TVL_PANEL_REPORT.md`. Coin PQ + Channel 2 remain out of scope, pending review. |

2026-06-26 | 30 (est., confirm) | Prompt + Review | Cowork session (no Claude Code launch). Asked to verify, before any purchase, whether the “needs a paid chain indexer” gap flagged for ~170 non-EVM PoS coins (Entry 41's A.3 line) is actually true — i.e. whether the indexers even contain the needed historical series, paid or free, for the 17 representative chains (ATOM, INJ, SEI, KAVA, CELO, SOL, HBAR, ICP, DOT, KSM, AVAX, NEAR, ALGO, TRX, EOS, SUI, APT). Live web research only (WebSearch + targeted doc fetches), no keys obtained, no purchase made. Found Entry 41's native-RPC-only check had missed free third-party paths for 4 chains: **SOL** (validators.app, free signup, `/epochs` endpoint = historical `total_active_stake`) and **TRX** (TronScan `freezeresource`, free 60 req/hr tier) are now free-verified; **DOT/KSM** (Subscan era-statistics, untagged [PRO]) likely free but not pixel-schema-confirmed. Found **HBAR** and **SUI** have free official historical data (Hedera Mirror Node balance snapshots; Sui's on-chain per-epoch exchange-rate table) but assembling the network-wide series is unbuilt engineering work, not a payment gate. Found **ALGO** is a genuine structural gap — Algorand's own infra providers confirm no vendor, paid or free, has a precomputed participation series; it would require full transaction-history replay regardless of payment. Found **CELO** migrated to an Ethereum L2 in March 2025 and may now be EVM-getLogs-reconstructable like ETH/EVM tokens instead of needing a non-EVM indexer at all — flagged as an unverified reclassification candidate. **ATOM/INJ/SEI/KAVA** (Mintscan is contact-based/undisclosed pricing, Bitquery coverage unconfirmed per-appchain), **AVAX** (AvaCloud Metrics API exists, official, historical, pricing-page ambiguous), **NEAR** (Pikespeak/NearBlocks, pricing unconfirmed), and **EOS/ICP/APT** (no historical source found, APT's free indexer looks current-state-only) remain open, unresolved gaps. Logged Decisions Log Entry 42 with the full per-chain breakdown and a four-way classification (free-done / free-but-engineering / paid-unconfirmed / no-source-found); reported to Moazzam in chat. No purchase made or recommended. |

2026-06-26 | ~10 (est., confirm) | Prompt + Decision | Launched session 020 (Claude Code / Opus 4.8) via `CLAUDE_CODE_LAMBDA_BUCKET2_BUCKET3_RECOVERY_PROMPT.md` to BUILD everything in Bucket 2 (non-EVM coin staking) and Bucket 3 (EVM token locks) that verifies live (response body), upgrading Entry 42's docs-level findings. Authorized free, live-verified sources + the existing free Dune/Etherscan keys and end-of-session commit+push; reiterated no paid tier / no purchase / no Subscan-Pro. Outcome: **lambda 1,688->1,880 observed asset-months, 58->62 assets**. Coin staking 3->5: **TRX + SOL built KEYLESS** (TronScan `freezeresource` and validators.app `/epochs?per=200` both answer unauthenticated — corrects Entry 42's “free with signup”; SOL keyless depth ~2023-01+). Token locks 10->13: **GMX** (the Entry-41 Arbitrum deferral) + **AERO + CAKE** built via a validated Dune `tokens_<chain>.transfers` cumulative reconstruction, each cross-checked to live `balanceOf` at <0.1% (the

kickoff's named `balances_<chain>.daily_updates` table doesn't exist on free tier — corrected; Etherscan-V2 free key doesn't cover Base/BSC — used keyless RPC + Dune). Response-body-verified non-builds: **DOT/KSM** Subscan era_stat is HTTP-403 key-gated (free key needs interactive signup, not completable headless — corrects Entry 42); **CELO** EVM-reclassification confirmed (on Etherscan-V2 42220) but free get-Logs reconstruction undercounts the L2-migration-carried locked state 3x vs on-chain `getTotalLockedGold` -> documented gap, not shipped; **HBAR/SUI** free current-state only, historical aggregation keyless-intractable; **ATOM/INJ/SEI/KAVA/AVAX/NEAR/EOS/ICP/APT** access-gate-checked (contact-sales / trial-only / ambiguous / no-source), no purchase; **AXS** rejected (Ronin), **VELO** deferred (v1/v2 cmcId collision). Decisions Log Entries 43–49; 03_data/SESSION020_BUCKET2_BUCKET3_COVERAGE_ADDENDUM.md; session_020_2026-06-26_bucket2_bucket3_recovery.md. Bucket 2 + Bucket 3 done; Bucket 1 + Phase 3 paused pending review. |

2026-06-26 | 20 (est., confirm) | Review + Decision | Cowork session (no Claude Code launch). Reviewed session 020's three-bucket table (recomputed: tokens 49 recovered / 398 unrecoverable / 1 deferred) and challenged the "398 unrecoverable" figure directly — given how thin the paper's overall lambda sample already is (62 of 1,081 in-universe assets), recovering even a fraction of 398 written-off tokens would be material, so demanded certainty across Etherscan, Artemis, Dune, and DeFiLlama before accepting any of them as closed. On inspection, found the 398 is NOT one uniform check: only ~6 (MKR/BAL/COMP/RUNE/ANGLE/AXS) were individually verified with a specific documented mechanism reason; ~287 were rejected at the *category* level ("the overwhelming majority are DEX/lending/RWA/meme/chain tokens...") rather than contract-by-contract; ~111 were never individually opened at all — excluded purely by a tag/category pre-filter in `classification_table.csv`. Also flagged two untried avenues: DeFiLlama's category-level "Staking" TVL-to-token-quantity proxy, and Artemis's free Lite-tier dashboard/UI (distinct from its paid API) for manual spot-checks. Decided: commission a full individual re-audit of all 398 (+ light reconfirmation of the 6 already-rejected + VELO) before any of them is accepted as a final, paper-ready rejection — a triage funnel (cheap bulk DeFiLlama/identity pass for all 398, full 4-source deep check for everything that clears the cheap filter, fast-but-specific reason for everything that doesn't) so every token ends with its own documented verdict, none dropped silently. Drafted 04_code/CLAUDE_CODE_TOKEN_BUCKET1_EXHAUSTIVE_REAUDIT_PROMPT.md. Coin-side Bucket 1 (456 coins) flagged as a candidate for the same treatment later, not started. |

2026-06-29 | ~10 (est., confirm) | Prompt + Decision | Launched session 021 (Claude Code / Opus 4.8) via CLAUDE_CODE_TOKEN_BUCKET1_EXHAUSTIVE_REAUDIT_PROMPT.md — the exhaustive per-token re-audit of all 398 token-side "unrecoverable" cmc_ids the prior Cowork review challenged as resting on three uneven rigor levels. Authorized free, live-verified sources (DeFiLlama, Dune top-holder/transfers via the existing free key, Artemis re-test) and end-of-session commit+push; reiterated no paid tier / no purchase / cmcId-only joins. Outcome: **the 398 was confirmed exact (live re-derivation), and 5 recoverable tokens the session-020 category-level pass had written off were BUILT — API3 (~74%), ORBS (~42%), IQ (~9%), VVV (~73%) (all in the 398) plus VELO (~7.4%, the Entry-48 deferral RESOLVED** via DeFiLlama's own Velodrome V2/V3 entries both carrying the cmc-7127 token address 0x9560e827, agreeing with CMC — the documented mapping the kickoff required). Each is a clean single-contract base-token lock cross-checking to live `balanceOf` at **0.00%** (Entry-26 standard), reconstructed via the Bucket-3 Dune `tokens_<chain>.transfers` method. The funnel: Stage-1 cmcId-only DeFiLlama triage of all 398 (92 DL-matched, 306 no-on-chain-identity; symbol-matched categories rejected as the Entry-20 collision landmine), Stage-2a chainTvlS on all 92 (36 expose a staked-token-QUANTITY series — stronger than the USD proxy expected), Stage-2b Dune top-holder single-escrow test on every one of the 36 (only the 5 pass; EIGEN multi-contract restaking / ILV multi-pool / KAITO-ATH-etc treasury-dominated all fail the `balanceOf` identity — the CELO lesson), Stage-2c Artemis re-tested live (JS/login-gated, no free per-asset data, reconfirms Entry 2/14, not signed up). The other 393 each carry an individual reason (367 REJECT-mechanism, 29 REJECT-no-data incl. non-EVM HXRO/SUN/ORN/TLM and the WARP cmcId-1166 collision, 1 ANGLE out-of-universe); the 6 already-rejected reconfirmed unchanged; **GATED = 0** (nothing paywalled-but-buyable). **lambda 1,880 -> 2,080 observed asset-months, 62 -> 67 distinct assets (token 49 -> 54)**. Dune free tier ~36 small executes (~15% of the 2,500 monthly budget). Decisions Log Entries 50–52; 03_data/phase1/token_bucket1_full_audit.csv (402 rows); 03_data/SESSION021_TOKEN_BUCKET1_EXHAUSTIVE_AUDIT.md; session_021_2026-06-29_token_bucket1_exhaustive_re Token Bucket 1 done; **coin-side Bucket 1 (456 coins) flagged for the same treatment, not started; Phase 3 paused pending review.** |

2026-06-29 | 90 (est., confirm) | Build + Analysis + Decision | Session 022 (Claude Code / Opus 4.8). Asked whether a (payable) Etherscan API could recover lambda Channel 1/2/3 for the 402 token-Bucket-1 rejects, then for the whole token universe, then non-EVM. First produced a metadata-only feasibility note (ETHERSCAN_LAMBDA_CHANNEL_FEASIBILITY.md); on pushback that 0 contracts had actually been read, did the real work: built a resumable pipeline (04_code/universe_lambda_pipeline.py) that resolves each token's contract via CMC detail.platforms[] + identity map, **reads the verified contract** via Etherscan-V2 getsourcecode (proxy->implementation followed; raw cached under 03_data/raw/), classifies the lambda mechanism from the ABI, then **computes each candidate event's keccak-256 topic0 and runs getLogs** — accepting a channel only when logs actually fire, not on ABI presence. Ran it over the full **token+other universe (1,306)**: **901 EVM-reachable, contract-read; Ch1 genuine getLogs-CONFIRMED = 6** (HEX/NMR/stkAAVE/XAN amount-bearing, AKRO/VSL bare-Locked()), **Ch3 ACTIVE getLogs-CONFIRMED = 34** (UNI/ENS/SUSHI/COMP/GTC/KP3R/BTRST/EIGEN/ONDO/STRK/MNT/BLUR/... emitting DelegateVotesChanged), **Ch3 needs-paid = 15 + Ch1 needs-paid = 1** (TNC), **Ch3 dormant/negligible = 15, Ch2 = all 901**. Reading the source corrected three errors metadata reasoning would have shipped: vesting/compliance look-alike locks excluded (MVL/WLD/FST/FBTC), dormant-but-declared governance excluded (CORE/SUPER/ILV/FLOKI), and a **measured chain-gate** — free Etherscan getLogs works on Ethereum/Polygon/Arbitrum/Blast but is **PAID-only on BSC/Base/Avalanche** (“Free API access is not supported for this chain”), the concrete paid-API trigger (16 candidate tokens + panel-scale Ch2). Then assessed the **405 off-Etherscan tokens**: 284 have no on-chain identity (dead, unrecoverable); 92 on free-indexer non-EVM chains (Solana 59 dominant, Tron 8, Cosmos/Osmosis/Kava 7, +tail) -> Ch2 recoverable per chain indexer, Ch1/Ch3 per-project (chain-native staking/gov measures the gas coin, out of token-scope) low-yield; 22 EVM-but-not-Etherscan (KAIA/HyperEVM/...) recoverable by the same method on chain-specific explorers. Live-verified free non-EVM repositories: Cosmos LCD (bonded_tokens, gov/proposals), Tron TronGrid, Solana RPC, Cardano Koios. **NO lambda panel modified — identification map only**, feeds a future build. Caveat flagged: many of the 34 Ch3-active likely already have a Snapshot Ch3 series (Entry 25) -> on-chain DelegateVotesChanged is a cross-check, net-new only where no Snapshot turnout. Artifacts: 03_data/phase1/universe_lambda_channel_map.csv (1,306), etherscan_lambda_channel_map.csv (402), non_evm_lambda_recoverability.csv (405); 03_data/ETHERSCAN_LAMBDA_CHANNEL_EMPIRICAL.md + NON_EVM_LAMBDA_CHANNEL_ASSESSMENT.md; Decisions Log Entry 53; status/transition + next-session prompt in 06_documentation/SESSION022_STATUS_AND_NEXT_SESSION.md. No paid tier, no purchase. | 2026-06-29 | 25 (est., confirm) | Review + Prompt | Cowork session. Read the already-drafted session-023 kickoff (CLAUDE_CODE_HEX_AKRO_RECONCILIATION_AND_SURVIVORSHIP_PROMPT.md) to adopt its conventions, then drafted the **session-024 BUILD kickoff** (04_code/CLAUDE_CODE_SESSION024_LAMBDA_FREE_BUILD_PROMPT.md) that turns session 022's identification map into actual lambda series. Clarified scope with the user: the build is **two-phase — Phase 1 = every FREE source on every chain identified as free in session 022** (not just Etherscan: ETH/Polygon/Arbitrum/Blast getLogs + the 22 EVM-but-not-Etherscan chains via free Blockscout/native RPC + non-EVM free indexers Solana RPC/Tron TronGrid/Cosmos LCD/Cardano Koios), **Phase 2 = paid services** (Etherscan Pro for BSC/Base/Avax, paid non-EVM indexers, panel-scale Channel-2 if the free 100k/day cap binds). Prompt is dependency-gated on session 023 (HEX/AKRO BUILD/REJECT verdicts) and structured as ordered Tasks A–E: (A) dedup the 34 getLogs-confirmed active ERC20Votes tokens vs the existing Snapshot panel then build net-new on-chain DelegateVotesChanged turnout; (B) build the free-chain Channel-1 locks (NMR/stkAAVE/XAN/VSL + HEX/AKRO if 023 passed them) via event-replay cross-checked to balanceOf at ~0.00% (Entry 26), with the Entry-49 circulating-supply double-count check per token; (C) **prototype-then-scale Channel 2** (the Entry-24 “highest-value addition” — FIFO coin-age engine on one ETH token, MEASURE the call budget vs the free cap before scaling, new phase1_channel2_holding.py); (D) the 22 EVM-other tokens via free chain explorers; (E) time-boxed non-EVM free-indexer Channel-2, skipping the 284 dead no-identity tokens. Carried all standing method rules (cmc_id joins, logs-not-eth_call, no interpolation, append-only log, no paid tier in Phase 1, extend-don't-rewrite the phase1_channel* scripts + assembler glob). **Per user instruction, embedded a context-window discipline rule in the prompt: at ~2/3 (66%) context, stop new build work and write a session report + transition doc + next-session prompt + commit before the window closes.** No data built this session (prompt-drafting only); committed +

pushed. |
2026-06-29 | 55 (est., confirm) | Build + Analysis + Decision | Session 023 (Claude Code / Opus 4.8) via `CLAUDE_CODE_HEX_AKRO_RECONCILIATION_AND_SURVIVORSHIP_PROMPT.md`. Resolved the two 021-vs-022 contradictions by running session 022's verified-contract-read + getLogs method to completion. **HEX: REJECT->BUILD** — read the HEX source (staking is non-custodial `_burn`, tracked in `lockedHeartsTotal`; that is why 021's single-escrow `balanceOf` probe saw nothing), reconstructed the monthly staked series from `StakeStart(+)/StakeEnd(?)` decoded on Dune, `final == live globalInfo()[0]` at 0.0000% drift; shipped `channel1_hex_stake.csv` (50 asset-months). **lambda 2,080->2,130 asset-months / 67->68 assets**. Denominator resolved: CMC circ excludes staked HEX (contract `NatSpec`) = same Entry-49 artifact, but locked<circ in-window. **AKRO: REJECT reconfirmed** — 022's "GENUINE Locked()" is a false positive; the event is an owner-only admin pause switch on the token contract (fired once, no amount), not a stake; 021's verdict stands with the contract reason. **Survivorship:** re-derived live 284 NO-IDENTITY listings, reconciled to the penny (1,306 = 901 reachable + 405 off-Etherscan; 405 = 284+92+22+7), documented as an acknowledged survivorship-bias limitation. Logged Entries 54/55/56; wrote `SESSION023_HEX_AKRO_RECONCILIATION.md` + `SURVIVORSHIP_BIAS_NOTE.md`. No paid tier. Did not touch 022's other open items. Commit+push at session end |
2026-06-30 | 75 (est., confirm) | Build + Analysis + Decision | Session 024 (Claude Code / Opus 4.8) via `CLAUDE_CODE_SESSION024_LAMBDA_FREE_BUILD_PROMPT.md` — Phase 1 (FREE sources only) BUILD of lambda Channels 1/2/3 from session-022's identification map. Authorized free sources on free chains (ETH/Arbitrum/Blast getLogs via the existing Etherscan-V2 key) + end-of-session commit+push; Phase 2 (paid) explicitly out of scope. **lambda 2,130 -> 2,699 observed asset-months, 68 -> 90 distinct assets (+569 / +22)**. **Task A (Channel-3 on-chain delegation, +560 asset-months / +21 assets):** deduped the 34 getLogs-confirmed ERC20Votes-ACTIVE tokens vs the Snapshot panel (10 overlap -> cross-check; 24 net-new), built a monthly delegated-voting-weight series for the net-new from `DelegateVotesChanged` event replay; 21 of 24 shipped (TOMI/UMLINK delegation post-dates their panel window — block-verified; CYBER 0-delegated; **ETHDYDX excluded on mechanism** — Aave auto-power, ratio~1, not opt-in). **Construct decision documented:** delegated-weight = a STOCK of governance-activated supply, a DISTINCT sub-channel from Snapshot's turnout FLOW -> written to `channel3_onchain_delegation.csv`, enters lambda as new channel `ch3_delegation`, **primary-only (net-new) to avoid governance double-count** (verified: never co-occurs with `ch3_voting`; 2-channel count unchanged at 354). **Task B (Channel-1, +9 / +1): XAN BUILT** (XanV1 permanent non-custodial lock; cumulative `?Locked.value == live lockedSupply()` at 0.000000%, the Entry-26 bar; locked 7.5B > circ 2.5B -> Entry-49 artifact flagged, `locked_fraction_alloc` written); **VSL REJECT** (the AKRO pattern, as the kickoff warned — `Locked()` is a cyclic epoch transfer-pause flag, no amount/escrow); **NMR DEFER** (Numerai tournament `Staked` is per-(staker,tag) cumulative with destroy/release + burn-on-stake -> no aggregate/escrow anchor for the Entry-26 cross-check); **stkAAVE(36246) EXCLUDE** (Staked-AAVE wrapper double-counts AAVE's Safety-Module lock already in lambda via cmc 7278). **Task C (Channel-2 coin-age, prototype + budget, NO lambda change by design):** built a FIFO per-address coin-age engine (`phase1_channel2_holding.py`), validated end-to-end on MET; **address-class hygiene shown empirically — raw HODL-6m 90.8% collapses to 1.3% after removing 5 contract holders** (nearly all "long-held" supply is contract/LP/treasury, not EOA conviction); **call budget MEASURED** (`phase1_channel2_budget_probe.py`): MET small = 79 getLogs calls complete, **RAD mid-size = 204,428+ transfers / 700+ calls / DID NOT COMPLETE**, 793 free-chain tokens x tail cost -> **panel-scale Channel-2 blows the free ~100k/day cap = the documented Phase-2 trigger** (Task C2), Channel 2 stays the lambda NaN column but now backed by numbers. **Tasks D & E — time-boxed triage, 0 free clean builds:** 22 EVM-non-Etherscan are ~19 wrapped/native-gas (out of token-scope) + KSP/BORA (KAIA, flagged Phase 2); 92 non-EVM free indexers serve current-state-only (the Entry-21 archive wall on non-EVM) -> historical Channel-2 is paid (Phase 2); 284 NO-IDENTITY not chased (Entry 56). Assembler extended to read `ch3_delegation` (z-score/equal-weight logic untouched) and its cosmetic -> print crash fixed. Decisions Log Entries 57-60; `03_data/SESSION024_FREE_BUILD_REPORT.md`; Phase-2 worklist + kickoff `06_documentation/SESSION024_STATUS_AND_NEXT_SESSION.md`; `session_024_2026-06-30_lambda_free_build.md`. No paid tier, no purchase. |
2026-07-01 | 30 (est.) | Review + Decision + Prompt | Cowork session (session 026 planning). Resumed from summarized context. Diagnosed a git state corruption (all files staged for deletion + working tree at Entry-39

era; GitHub intact). User ran `git restore --staged .` then `git restore .` to fix local state. Read `SESSION025_PRO_BUILD_REPORT.md` in full: λ 2,699->6,097 / 90->288 assets; Channel-2 BUILT for 207 tokens; P2-2 5 BUILT / OP deferred / TNC rejected; data-integrity bug found and fixed. Clarified OP delegation concept. Drafted session 026 prompt (`CLAUDE_CODE_SESSION026_CHANNEL2_TAIL_AND_OP_PROMPT.md`): Task A = OP block-windowed delegation (role=cross_check), Task B = Channel-2 tail prioritizing the 4 tokens already with ch1+ch3 (CVX/SUSHI/FRAX/GMX -> first 3-channel assets) then large governance tokens; HEX explicitly deferred. Committed all session 025 outputs + session 026 prompt. Noted: user expressed frustration with manual session-prompt workflow; future Cowork sessions will run code directly where feasible. |

2026-07-01 | ~120 (est., confirm) | Prompt + Build + Analysis + Decision | Session 025 (Claude Code / Opus 4.8) via `CLAUDE_CODE_SESSION025_PRO_BUILD_PROMPT.md` — Phase-2 PAID build on the newly purchased **Etherscan API Pro** (Standard, \$199/mo, 200k calls/day, all chains; Moazzam purchased/activated 2026-06-30; supplied the key in gitignored `.api_keys.json`). Authorized end-of-session commit+push; no other paid subscription. **Verified Pro before building** (getLogs returns real data on BSC/Base/Optimism/Linea/Avax — free tier refused these). **Result: λ 2,699 -> 6,097 observed asset-months, 90 -> 288 distinct assets (+3,398 / +198).** **Task A (P2-2, the 16 paid-gated tokens):** mechanism-verified each on-chain first (the AKRO discipline) — 5 BUILT into `ch3_delegation` (AWE cmc 4006 with a material 15.25% delegated/circ ratio; CHEEL/FORM/LINEA/ZORA fire at ~0% = low governance activation), ALT excluded (2 events net to 0 delegated), **OP deferred** (verified firing but its OP-airdrop delegation history is too large to complete — a partial replay would be a wrong stock), 8 DORMANT (BAKE/BNX/EDG/ESPORTS/MCT/MDX/PONKE/TKO — ERC20Votes ABI present but DelegateVotesChanged never fired), **TNC rejected** (Locked(address) bare/no-amount = the AKRO/VSL false-positive, fails Entry-26); `ch3_delegation` grew 560->627 asset-months / 21->26 assets. **Task B (Channel-2 coin-age / HODL — the Entry-24 gap, NaN since inception, now BUILT):** two methodology corrections — (1) **denominator FIX** (the session-024 prototype divided by CMC circulating -> shares >100%, RAD 148-398%, because circulating excludes locked/treasury; corrected to on-chain supply -> bounded [0,1]; re-validated on RAD, screened HODL-6m 11-27%), (2) **all-month contract screen** (`eth_getCode` is time-invariant -> one pass cleans the whole series, fixing the last-month-only prototype). Per Moazzam's steer, replaced the wasteful fetch-until-cap approach with a **tokenholdercount metadata pre-pass** (1 call/token; all 793 sized: 113 <=1k holders ? 215 <=3k) so the build runs smallest-first, defers the high-volume tail BY METADATA (no wasted fetching), and caches month-blocks per chain. **Built 207 tokens / 3,413 asset-months** (median screened HODL 40.6%, only 2/197 degenerate); the ~586 tokens >3k holders are the resumable worklist. **Data-integrity bug found & fixed mid-session:** an overnight network outage made the engine swallow DNS failures into empty getLogs results -> silently dropped block ranges = corrupted coin-age; added a robust getLogs that RAISES on network failure (aborts+retries the token, never checkpoints a partial history) + backoff/circuit-breaker + non-caching of failed block lookups; DELETED and cleanly re-fetched all 65 possibly-affected checkpoints (0 network failures on the re-run). **Depth (headline metric):** 2-channel asset-months 354->435 (+81) but the 2-channel SHARE fell 13.1%->7.1% (breadth tripled with single-channel ch2); 3-channel still ~0 because the large in- λ governance tokens are >3k holders (deferred) — the tail run is what creates 3-channel assets. Assembler extended to read `ch2_holding` (z-score/equal-weight logic untouched). Decisions Log Entries 61-64; `03_data/SESSION025_PRO_BUILD_REPORT.md`; builders `phase1_channel2_panel.py` / `_sizeprobe.py` / `_validate.py` / `_p2p2_probe.py`; `session_025_2026-07-01_etherscan_pro_channel2_build.md`. Commit+push at session end. |

2026-07-01/02 | ~45 | Prompt + Decision + Steering | Launched session 026 (Channel-2 tail + OP delegation) via `CLAUDE_CODE_SESSION026_CHANNEL2_TAIL_AND_OP_PROMPT.md`. Steered key decisions across the session: (a) Task B (3-channel creators) before OP when OP proved ~2-3h and is only a cross-check; (b) classify-first / build-only-3-channel-creators efficiency discipline after ORBS wasted effort as a hidden giant; (c) approved the streaming + concurrent engine upgrade to build the AAVE/GMX giants that OOM the session-025 engine; (d) authorized fully-autonomous overnight completion incl. commit+push; (e) directed finishing OP this session (not spilling over), prompting the pooled-Session + concurrency fix for the Optimism ConnectionReset stall. Outcome: 9 first-ever 3-channel λ assets / 332 three-channel asset-months (from 0); OP delegation completed (11.7M events, median 7.98 | 2026-07-01/02 | ~45 | Prompt + Decision + Steering | Launched session 026 (Channel-2 tail + OP delegation) via `CLAUDE_CODE_SESSION026_CHANNEL2_TAIL_AND_OP_PROMPT.md`.

Steered key decisions: (a) Task B (3-channel creators) before OP once OP proved ~2-3h and is only a cross-check; (b) classify-first / build-only-3-channel-creators efficiency discipline after ORBS wasted effort as a hidden giant; (c) approved the streaming + concurrent engine upgrade to build the AAVE/GMX giants that OOM the session-025 engine; (d) authorized fully-autonomous overnight completion incl. commit+push; (e) directed finishing OP this session, prompting the pooled-Session + concurrency fix for the Optimism ConnectionReset stall. Outcome: 9 first-ever 3-channel assets / 332 three-channel asset-months (from 0); OP delegation completed (11.7M events, delegated/circ median 7.98 pct, cross-check); an address-poisoning contamination bug caught + fixed with a validated 2-layer guard. λ 6,097->6,021 asset-months / 288->282 assets (depth up, contaminated spam-only tokens removed). Decisions Log Entries 65-67; 03_data/SESSION026_TAIL_BUILD_REPORT.md. |

2026-07-04 | ~45 (est.) | Review + Analysis + Prompt | Cowork session (session 028 planning). Resumed from summarized context (session 027 complete). Read SESSION027_TVL_AND_CH2_REPORT.md: λ 6,021->7,051 / 282 assets; TVL 99->130 tokens (all 9 three-channel assets now have TVL); ch2 tail complete for all 43 remaining λ tokens (2+ channel share 11->24.4%); AAVE 22 null ch2 months identified. Generated universe_coverage_status.csv (1,939-asset coverage map: asset_class, λ channels, NVT_GL, TVL, holder_count, coverage_status, what_needed per asset). Key finding: coin regression sample is thin (5 coin λ ?NVT_GL); 49 PoS coins have NVT_GL but no λ — coin staking is the session 028 priority. Discussed next steps: (1) BNB via BSC StakeHub getLogs (Etherscan Pro in hand), (2) DOT/KSM via Subscan free key (Moazzam needs to sign up at pro.subscan.io), (3) CELO via Etherscan Pro balance history, (4) AVAX/NEAR gate probes, (5) Cosmos alternatives. AAVE ch2 fix (per-token totalSupply denominator) bundled as Task B. Drafted session 028 prompt; session log written. |

2026-07-02/04 | ~20 (est., confirm) | Prompt + Decision | Launched session 027 (Claude Fable 5 / Claude Code desktop) via CLAUDE_CODE_SESSION027_TVL_EXPANSION_AND_CH2_TAIL_PROMPT.md; authorized fully-autonomous multi-day operation (always-allow permissions) incl. end-of-session commit+push; one early stop/resume when permission mode could not be changed the first evening. Outcome: **Task A TVL expansion (Entry 68)** — DL cmcId lookup yielded ZERO hits (field covers 1,709/7,770; prior sessions had harvested all), so exact symbol+name verification found the four missing 3-channel tokens under DL PARENT protocols (CRV->curve-finance, YFI->yearn, FRAX->frax-finance, GMX->gmx; parent slug = the token whole-protocol claim) + 19 more verified slugs (UNI cmc-7083-not-4113 collision handled; XVS->venus after venus-finance 400d; ZORA rejected/reverted — bridge custody != protocol TVL); builder extended with OTHER_ADDS (RPL/SSV/BLUR/RAIN/MV — the other-class filter gap that had excluded RPL) + CHAIN_LEVEL via historicalChainTvl (ARB/OP=OP Mainnet/MNT/APE/BLAST, dl_slug=chain: prefix; coins excluded per NVT framework; CYBER/PENGU/GBYTE rejected); **tv1_panel 4,999->6,620 asset-months / 99->130 tokens, 0 failures; λ assets with TVL 49->80; 331/332 three-channel asset-months now have the NV/TVL denominator** (gap = SUSHI launch month); A5 gate passed on all five priority tokens. **Task B Channel-2 tail (Entry 69)** — ALL 43 ch1-or-ch3-no-ch2 λ tokens ($\leq 500k$ holders) built smallest-first with the unchanged session-026 streaming engine across two compliant quota days (141,005 getLogs day 1, clean cap-stop after APE at in-process DAILY_CAP=140k; 41,871 day 2 for ZRX/COMP/UNI after the 00:00 UTC reset; 182,876 total; 60.1M Transfer events; hidden giants UNI 8.3M/APE 5.9M/BNT 5.7M absorbed in bounded memory); B2 integrity scan CLEAN (0 months >100x, worst HAKKA 31x = legit Entry-49 band); B4 medians all bounded 13-51%; XAN median-0 explained as an age artifact (real 74-77% once lots can exceed 6m; non-custodial XanV1 lock ages in place). **λ 6,021->7,051 asset-months / 282 assets; n_channels 1/2/3 = 5,331/1,388/332; 2-channel months 333->1,388; 2+ share 11.0%->24.4%** (3-channel unchanged — none of the 43 had both ch1+ch3, by construction). Decisions Log Entries 68-70; 03_data/SESSION027_TVL_AND_CH2_REPORT.md; session_027_tv1_expansion_and_ch2_tail.md. Next: coin staking ch1 (non-EVM), ~500 breadth-only ch2 tokens, AAVE spam-window recovery, PQ expansion. Commit+push at session end. || 2026-07-04 | ~240 (est., confirm) | Prompt + Build + Analysis + Decision | Session 028 (Claude Fable 5 / Claude Code desktop) via CLAUDE_CODE_SESSION028_COIN_STAKING_AND_AAVE_FIX_PROMPT.md — coin Channel-1 expansion + AAVE ch2 recovery. **Task A: 7 coins BUILT (+358 asset-months), every series Entry-26-cross-checked at ~0% against the chain's own figure:** BNB (StakeHub replay del?und+rewards = live ? totalPooledBNB to the integer; post-BC-fusion window 2024-07+; corrected kickoff's event data order AND accounting — MigrateSuccess double-emits Delegated, +80% if added), S/Sonic (SFC replay;

block-bisection proved RestakedRewards double-emits Delegated -> del?und alone = archive totalStake at +/-0.0000% at 6 blocks), GLMR/MOVR (discovered Etherscan proxy eth_call IGNORES historical tags -> read the staking pallet's own aggregate ParachainStaking.Total via the official archive RPCs' Substrate state API, keyless; decode validated by the ? getCandidateTotalCounted superset relation), XDC (Pro balancehistory of the native-stake XDCValidator; 3-way tie-out caps 2.650B <= replay 2.668B <= balance 2.701B with a CONSTANT +32.625M genesis offset), CELO (**Entry-46 gap closed for \$0** — forno.celo.org serves FULL pre-L2-migration archive -> historical getTotalLockedGold() keyless back to 2020-07; Etherscan balancehistory on 42220 only covers post-migration), AVAX (**Entry-42/47 ambiguity resolved: metrics.avax.network is keyless-free**; validatorWeight+delegatorWeight additive semantics verified vs live P-Chain platform.getTotalStake). BERA = documented gap (consensus-side withdrawals invisible; cum deposits 382.6M > circulating). DOT/KSM deferred (no subscan key — the one actionable Moazzam gate); NEAR/Cosmos gates re-verified 401/key-gated. **Task B: AAVE's 22 spam-nulled ch2 months RECOVERED — root cause was NOT what the kickoff assumed: fake-value SELF-transfers (from==to) from a 0.0-balance address on the real AAVE contract (incl. max-uint256 = Entry 66's 1.16e60; sub-cap fakes like 10M pass ANY value cap); FIFO pop-then-append minted the shortfall as phantom supply AND phantom aged numerator lots. Fix = engine-wide self-transfer skip (accounting identity, all 3 engine variants) + PER_TOKEN_VAL_CAP={7278:16M} defense-in-depth (global thresholds untouched); full re-fetch -> supply 16.0–18.0M all 67 months (was 1.02e9), recovered HODL 20.0–31.3%, pre-spam months <=1.9e-5 from old; offline diagnostic re-played all 210 stored-event checkpoints -> ZERO other tokens affected. lambda 7,051 -> 7,409 asset-months / 282 -> 289 assets (coins 9->16); regression-ready coins (lambda?NVT_GL) 5 -> 12 / 505 coin asset-months; 3-channel asset-months 332 -> 354; AAVE 3-channel complete. ~45k Pro calls (single day, incl. two full AAVE re-fetches); no signups, no purchases. Entries 71–75; 03_data/SESSION028_COIN_STAKING_REPORT.md; builder phase1_channel1_pos_coins_evm.py + _s028_* stage/diagnostic scripts; session_028_2026-07-04_coin_staking_and_aave_fix.md. Commit+push at session end. |**

2026-07-04/05 | ~9 (est., confirm) | Prompt + Build | Launched session 029 (Claude Code / Fable 5) via CLAUDE_CODE_SESSION029_BREADTH_BUILD_AND_COIN_PROBES_PROMPT.md: ch2 breadth for the 47 TVL-covered non-lambda tokens, orphaned dl_slug verdicts, and unresearched PoS coin probes. Outcome: **lambda 7,409->9,580 asset-months / 289->337 assets; regression-ready 92->138** (coins 12->20 incl. new POL/MATIC/RON/KAIA/FLR/EGLD/STRK/XRD/PEAQ builds — all state reads of chain-own aggregates via keyless archives (drpc/lava/onfinality/archive-en.kaia/flare-api/multiversx/radix gateway); tokens+other 80->118 via 40 new ch2 builds incl. the two largest ever, MDX 17.7M + MBOX 21.2M transfers). All six orphaned TVL slugs rejected (DL tracks no TVL). 7-token ch2 resume list + CHZ/CORE gates documented. Entries 76-79; SESSION029_BREADTH_AND_COIN_PROBE_REPORT.md. |

2026-07-05 | ~15 (est., confirm) | Prompt + Decision | Launched session 030 (Claude Code / Fable 5) via CLAUDE_CODE_SESSION030_TASK_A_RESUME_PROMPT.md (7-token ch2 Task-A resume); periodic status checks; mid-session schedule decision (airport 4:30 PM, out 7/6-7/12): chose to kill the in-flight MYX mega-giant build (~120k calls written off, no partial checkpoint) rather than leave a week-long unattended process, and directed a transition report + session 031 pickup prompt. Outcome: SFUND BUILT (28,553 getLogs, 23 months, B2/B4 PASS, regression-ready); lambda 9,580->9,603 / 337->338 assets; regression-ready 138->139; 6 tokens roll to session 031 (2026-07-13, small-chains-first, MYX on its own quota day). Entry 80; SESSION030_TASK_A_RESUME_REPORT.md; session_030 log. Commit+push at session end. |

2026-07-23/24 | ~20 (est., confirm) | Prompt + Decision | Launched session 031 (Claude Code / Fable 5) via CLAUDE_CODE_SESSION031_TASK_A_RESUME_PROMPT.md (5-token day-1 ch2 resume); periodic status checks across two days; diagnosed the overnight stall as OS sleep (user's hypothesis — confirmed via Kernel-Power event log) and directed disabling sleep for the duration; after a Windows Update auto-restart killed RAIN at 90%, directed finishing the task (RAIN rebuilt). Outcome: ALL 5 tokens BUILT (ADF/VVV/KAITO/AVNT/RAIN; VVV = new all-time ch2 record tf=27.8M, 82x under-estimate; Base joins BSC as an undershoot chain); lambda 9,603->9,638 / 341 assets; regression-ready 139->142; ch2 306 tokens / 9,857 rows. MYX rolls to a dedicated day (session 032) with Windows Update paused. Entry 81; SESSION031_TASK_A_RESUME_REPORT.md; session_031 log. Commit+push at session end (8b2629e). |

2026-07-24 | ~5 (est., confirm) | Prompt | Launched session 032 (Claude Code / Fable 5) via

CLAUDE_CODE_SESSION032_MYX_PROMPT.md with a single overnight-authorization prompt; no further human input during the session. Outcome (autonomous): MYX (36410/BSC) ch2 BUILT in 2.7h / 67,233 getLogs (0.25x of the 250-300k budget - session 030's 120k-by-batch-81 counter was evidently retry-churn-inflated; 030-derived estimates are upper bounds); tf=22,451,143 = 2nd-largest ch2 build ever (VVV keeps the 27.8M record), largest BSC build; B2 PASS (max 10.5x), B4 PASS (median 3.0% / last 5.2%); 10/10 months, all TVL-overlapped -> regression-ready. Session-029 Task A FULLY CLOSED (47 targets: 41 built, 5 non-EVM skipped, 1 WARP deferred). lambda 9,638->9,648 / 342 assets; regression-ready 142->143; ch2 307 tokens / 9,867 rows. Windows Update pause was not possible (non-elevated shell; verified no pending reboot instead - no restart occurred); .gitignore *.log found to have silently dropped session-031 build logs (both sessions' logs force-added). Entry 82; SESSION032_MYX_REPORT.md; session_032 log. Commit+push at session end. |

2026-07-25 | ~10 (est., confirm) | Prompt | Launched session 034 (Claude Code / Fable 5) via CLAUDE_CODE_SESSION034_COMBINED_PROMPT.md; five brief status check-ins during the ~5h autonomous run; no mid-session decisions. Outcome: CHZ ch1 BUILT (35 mo, anchor drift -1.03%, 21st regression-ready coin); Blockchair XTZ/MATIC FAILED keyless (tezos/polygon aggregation 404 + IP blacklist after ~4 anonymous calls; paid-key decision flagged, not subscribed); EVM DeFi Breadth Batch 1: 101/102 ch2 tokens built (154k getLogs, 47.6M transfers, B2 clean, 9 B4 flagged-high kept; CEL 54 / FTT 81 scrMo survivorship builds); TVL matching with cmcId authority (raw symbol ~40% wrong) -> 24 token slugs + MULTI/ORC/MUBI/FF other-adds + METIS/XAI chain-level; tvl_panel 159 assets / 7,889 mo. lambda 9,648->12,599 / 342->444 assets; regression-ready 143->173 (coins 21, tokens/other 152). Entry 84; SESSION034_COMBINED_REPORT.md; session_034 log. Commit+push at session end (861d1fd). |

2026-07-25 | ~3 (est., confirm) | Prompt | Launched session 035 same-day by pasting the Batch-2 WORKLIST command (from the 034 prompt); one status check-in; no mid-session decisions. Outcome (autonomous): EVM DeFi Breadth Batch 2 13/13 built (74,783 gl = 0.63x est; 22.9M tf; 592 scrMo; B2 clean; PNT B4 flag-high 97.9% kept - cmc 2691 is dead Penta, not pNetwork as 034 prompt claimed); Etherscan quota did not bind same-day (229k gl total across two runs - credit-based confirmed); TVL: SNX->synthetix parent (73 overlap mo), LEND->aave-v1 (4 mo, dead-token region), ANKR->OTHER_ADDS (66 mo), KNCL->kyberswap-classic window-clipped <=2021-06 via new CLIP mechanism (MATIC/POL rule vs KNC-9444); MKR zero-lambda -> nothing assigned; SAI excluded (CDP-stablecoin liability circularity). lambda 12,599->13,191 / 444->457 assets; regression-ready 173->177 (coins 21, tokens/other 156); tvl_panel 163/8,120 mo; ch2 421/13,580 rows. Entry 85; SESSION035_BATCH2_REPORT.md; session_035 log. Commit+push (a7aa885). |

2026-07-26 | ~1 (est., confirm) | Prompt | Launched session 036 (Claude Code / Fable 5) via CLAUDE_CODE_SESSION036_STETH_MEME_PROMPT.md; fully autonomous, no check-ins. Outcome: stETH (8085) + MEME (28301) ch2 BUILT in ~50 min - 22,444 gl actual vs ~110k est (~5x over-estimate; sparse pre-activity blocks); stETH 4.53M tf / 50 scrMo (B2/B4 pass, HODL med 13.4%), MEME 2.93M tf / 31 scrMo (B2/B4 pass, med 3.7%). TVL: stETH excluded (LST receipt circularity, Entry 84; no lido mapping); MEME nuance - DL HAS a memecoin listing (cmcId 28301, correct contract) but its TVL series is EMPTY (0 points) -> lambda-only as expected. Windows Update paused via registry (elevated shell OK this time). lambda 13,191->13,272 / 457->459 assets; regression-ready 177 UNCHANGED (coins 21, tokens/other 156); ch2 423/13,661 rows; coverage 188/313/1,438. Entry 86; SESSION036_BATCH3A_REPORT.md; session_036 log. Commit+push at session end (287fe7e). |

2026-07-27 | ~1 (est., confirm) | Prompt | Launched session 037 (Claude Code / Fable 5) via CLAUDE_CODE_SESSION037_DOT_KSM_CORE_PROMPT.md; fully autonomous, no check-ins. Outcome: DOT + KSM + CORE ch1 ALL BUILT — but via a method pivot: Subscan era_stat is per-address (no network-wide history on free plan), so staked series were read from archive-RPC state (Staking.ErasTotalStake at month-end blocks; Asset Hub for post-AHM months). DOT 71 mo (drift -2.18% PASS, external anchor matches), KSM 76 mo (-2.05% PASS; 4 early SlotStake months dropped as wrong-metric), CORE 42 mo EXACT (definition validated to the digit vs staking-api; -2.55% PASS; balance-based approach rejected at +6.4%). lambda 13,272->13,449 / 459->462; regression-ready 177->178 (coins 22: CORE in; DOT/KSM stay partial — prompt premise wrong, they have NO NVT_GL/PQ). Entry 88; SESSION037 report; committed+pushed. |

2026-07-27 | ~1 (est., confirm) | Prompt | Launched session 038 (Claude Code / Fable 5) via CLAUDE_CODE_SESSION038_SHIB_PROMPT.md; fully autonomous, no check-ins. Outcome: SHIB

(5994/Ethereum) ch2 BUILT in ~2.1h - 53,683 getLogs actual vs 128k est (0.42x; holder-count overestimate pattern of 034-036 holds); 17.85M transfers, 61 screened months 2021-05..2026-05, B2 pass (zero contaminated months), B4 pass (HODL-6m median 78.1%). TVL: SHIB excluded from NV/TVL regression (meme token, no direct protocol TVL; shibaswap slug verified as belonging to BONE cmc 11865 with 59 months already in panel - assigning to SHIB would double-count) -> lambda-only. lambda 13,449->13,510 / 462->463 assets; ch2 424 tokens / 13,722 rows; coverage 189/315/1,435; regression-ready 178 UNCHANGED (coins 22, tokens/other 156). EVM DeFi breadth COMPLETE (all batches 1-3b). Entry 89; SESSION038_SHIB_REPORT.md; session_038 log. Commit+push at session end (263655a). |

2026-07-28 | ~1 (est., confirm) | Prompt | Launched session 039 (Claude Code / Fable 5) via CLAUDE_CODE_SESSION039_DOTKSM_PQ_FIXES_PROMPT.md; fully autonomous, no check-ins. Outcome: DOT/KSM PQ probe NEGATIVE - Subscan daily stats free window ~2 months (history_window_exceeded on all categories, format=month unsupported; in-window data degenerate) and Blockchair aggregation 404 with BOTH Substrate indexes frozen May-2025 (paid key would not help - DOT/KSM removed from any Blockchair consideration); DOT/KSM stay partial (pq_nvttl gap), reopen only on Subscan Pro. TRX (1958) coin_staking_type fixed pow_only->pos in universe_coverage_status.csv (the label is a carry-forward column in the CSV itself); regression-ready coins now derive cleanly to 22 with TRX included (Entry-88 +/-1 discrepancy resolved). WARP (1166) closed as PERMANENT identity mismatch: CMC 1166 = warppcoin.com 2016 standalone PoS coin (inactive 2018), never an ERC-20; BAD_DL_CMCID override added to identity builder, identity row cleared, 54 bogus polkastarter TVL months purged (tv1_panel 8,120->8,066 mo / 163->162 assets), coverage -> not_started. lambda 13,510 / 463 UNCHANGED; coverage 189/314/1,436; regression-ready 178 (coins 22, tokens/other 156). Entry 90; SESSION039_DOTKSM_PQ_FIXES_REPORT.md; session_039 log. Commit+push at session end. |

2026-07-28 | ~1 (est., confirm) | Prompt | Launched session 040 (Claude Code / Fable 5) via CLAUDE_CODE_SESSION040_COSMOS_LCD_PROMPT.md; fully autonomous (one unrelated WRDS interjection, told to ignore). Outcome: CRO + KAVA ch1 BUILT via keyless Cosmos archive LCD (pool @ month-end blocks, binary search) - CRO 11 mo 2025-07..2026-05 via rest.mainnet.crypto.org (prompt candidates DNS-dead; registry refresh), drift 1.81% PASS; KAVA 26 mo 2024-04..2026-05 via api.data.kava.io (HTTP-420 rate limits, 3 passes at 0.6-1.5s pacing), drift 23.06% WARN investigated-genuine (25M-KAVA staking surge mid-Jul 2026, verified at 2026-06-30 + 2026-07-14 probe blocks). INJ NEGATIVE (7/7 LCDs pruned) + SEI NEGATIVE (8/8 pruned; sei.api.pocket.network AND rest.cosmos.directory/sei are FAKE archives ignoring x-cosmos-block-height - caught by old-bonded==live-bonded guard, now permanent probe check). Codec boundaries: CRO state decodes ~2025-06+, KAVA ~2024-Q2+ (invalid denom = deterministic, no-retry); kava_2222-10 height reset 2022-05-25. Env incident: user-site pandas mid-upgrade-broken (~andas remnant) - reinstalled 3.0.5, assemble/coverage ran unmodified. lambda 13,510->13,547 / 463->465; regression-ready 178->180 (coins 22->24); coverage 191/312/1,436; CRO+KAVA pos_possible->pos. Entry 91; SESSION040_COSMOS_LCD_REPORT.md; session_040 log. Commit+push at session end. |

2026-07-30 | ~1 (est., confirm) | Prompt | Launched session 041 (Claude Code / Fable 5) via CLAUDE_CODE_SESSION041_HXRO_SXP_OSMO_PROMPT.md; fully autonomous, no check-ins. Outcome: HXRO Task A closed MOOT with 0 quota - checkpoint was never empty (rows/mblocks schema, 24 months intact) and, decisively, HXRO is carried_forward from 2022-10 while TVL starts 2023-02, and lambda is computed on observed rows only -> observed?TVL = empty set, no rebuild can help; permanent gap unless Phase 3 revisits the lambda-on-observed rule. SXP (4279) probe PASS on Ethereum (stale non-EVM flag = unprefixed Multi-Chain address) -> ch2 BUILT: 569,311 tf / 1,886 gl / 77 lambda months 2019-09..2026-02, B2 clean, B4 HODL med 27.0%, lambda?TVL 60 mo -> complete. OSMO (12220) ch1 BUILT 2 months (2026-04, 2026-05, drift 1.77% PASS) via osmosis-api.noders.services - the ONLY real archive of 16+ registry endpoints (osmosis.api.pocket.network = FAKE archive, SEI landmine family); prompt's 15k blocks/day stale (actual ~73.3k/day), all public nodes floor at 2026-04-02 chain-wide; deeper history needs paid indexer; both months overlap TVL -> complete; coin_staking_type -> pos. CASINO/RUNE/SUN documented permanent gaps. lambda 13,547->13,626 / 465->467; coverage 193/310/1,436; regression-ready 180->182 (coins 24, tokens/other 156->158). Etherscan Pro lapsed after this session; no remaining Etherscan-dependent work identified. Entry 92; SESSION041_HXRO_SXP_OSMO_REPORT.md; session_041 log. Commit+push f9cc64b + record-keeping commit at session end. |

2026-08-04 | ~3 (est., confirm) | Cowork | Session 042 (Cowork / Fable 5): full paper restructure + TVL* + Phase 3 design. (1) Theory bifurcated (2.1 coin SoV/MoE, 2.2 token governance moat, 2.3

valuation conditioning, 2.4 quadrant); new Section 3 Variable Construction; Data rewritten as sources -> funnel -> stats. (2) Introduction restored to original progression (motivation -> QTM -> SoV/MoE -> lambda) per review; coin/token split now enters AFTER lambda established; minor token-paragraph adjustments only. (3) NV/TVL_GL adopted: phase2_tvl_gl.py + nv_tvl_gl_panel.csv (111/3,125); token sample 101/2,771; combined 125/3,489/\$451B (Entry 93). (4) Data 4.2 funnel Table 1 (1,939 -> 858 other -> coin 633->24 / token 448->101), all counts verified vs coverage CSV + panel joins (POL reconciled). (5) Fabricated bib entries removed (dao2024governance, daoreview2025). (6) Phase 3 empirical design fixed (Entry 94): H1/H2 panel FE regressions, H3 vs self-built monthly LTW factors, horse race (raw NVT, Metcalfe, MVRV probe, S2F, technicals); PHASE3_ANALYSIS_SPECIFICATION.md + CLAUDE_CODE_PHASE3_KICKOFF_PROMPT.md written. (7) Ops: root-caused git lock failures = Cowork delete-permission default; permission granted, git now clean. Commits 687d724, 2b9ac26, 9308322 + this record-keeping commit. |

2026-08-04 | ~1 (est., confirm) | Prompt | Launched session 043 (Claude Code / Fable 5) via CLAUDE_CODE_PHASE3_KICKOFF_PROMPT.md; fully autonomous, no check-ins. Outcome: Phase 3 core tests COMPLETE. Task A regression_panel.csv funnel-gated EXACT (coins 24/718, tokens 101/2,771; coin membership = coverage complete & lambda>0, NOT staking_type=='pos' which gives 11/396). Task B monthly LTW analogs (CMKT +6.0%/mo t=2.9 corr 0.93 BTC; CSMB flat; CMOM NEGATIVE -4.6%/mo t=-1.6, monthly-reversal deviation documented). Task C: H1a REJECTED at t+1 (coin conv t<=0.7); H2 SUPPORTED coins (conv x val t=-3.54, split-diff t=-3.99: +0.7%/SD cheap vs -3.0%/SD expensive); H1b weak support tokens (+0.6-0.8%/mo per SD, t=1.5-2.3, best sector-FE/FM); H2 REJECTED tokens (wrong sign, t=0.8); voting-weighted lambda NOT better (all |t|<0.6). Task D: H3 REJECTED - no SMA alpha significant (best pooled VW +4.3%/mo t=1.48); breadth guard removes 45/65 coin months (20 left); only VW Sharpe beats EW-universe benchmark. Task E probes: E1 BitInfoCharts AA = ETH-only among 24 coins (TRX/ADA/SOL stub pages) -> Metcalfe out of cross-sectional race; E2 realized-cap NO for coins (zero ch2 checkpoints; 211/426 token checkpoints replayable but only 12/101 sample tokens) -> MVRV dropped. No paid calls; statsmodels/linearmodels installed. Entries 95-96; PHASE3_RESULTS_REPORT.md; session_043 log. Commit+push at session end. |

2026-08-04 | ~2 (est., confirm) | Cowork | Session 042 continued: reviewed session-043 Phase 3 results with Moazzam. Produced PHASE3_FINDINGS_REVIEW.docx (readable summary) + PHASE3_RESULTS_TABLES.docx (8 paper-style tables, stars). Key readings: H2-coins is the strong result (interaction t=-3.54, split-diff t=-3.99); H1b weak; H3 null power-limited; token signal is ONE-dimensional. User-directed exploratory sorts (Entry 97): token conviction quintile EW alpha +1.71%/mo t=2.18 Sharpe 1.01, post-2023 survives (+1.48%, t=1.79) - median split was diluting a linear signal. Sector labels too granular for within-sector signal (median 1 token/sector-month) -> coarse remap queued. Phase 3b design fixed (Entry 98, spec ?8): confirmatory quintile/decile sorts, coarse sectors, spanning vs reversal (make-or-break), horse race, heterogeneity batch. CLAUDE_CODE_PHASE3B_KICKOFF_PROMPT.md written. |

2026-08-04 | ~1 (est., confirm) | Prompt | Launched session 044 (Claude Code / Fable 5) via CLAUDE_CODE_PHASE3B_KICKOFF_PROMPT.md; one mid-session user note (Task E added to kickoff/spec 8.6 - picked up). Outcome: Phase 3b COMPLETE (Tasks A-E). A: coarse sector map (DEX 41/Other 21/Lending 16/Yield 15/Deriv 6/LSD 2); coarse FE ATTENUATES token conv slope (s6_1 t 2.30->1.95) - premium partly between-sector. B (headline): conviction quintile EW alpha +1.71%/mo t=2.18 CONFIRMED in pre-specified battery (net-50bps +1.53% t=1.95; post-2023 +1.48% t=1.79); SPANNING SURVIVES - alpha strengthens to +2.53% t=3.18 controlling reversal, +2.34% t=2.60 vs all four competitor LS; but decile DIES (t=0.02), sector-neutral quintile DIES (t=0.39), per-sector terciles all null - extremes-only, between-sector signal. C: coin convxval survives full joint horse race (-0.0176 t=-2.83); token conv attenuates to t=1.10 jointly (only reversal survives); Metcalfe descriptive 7 series (BTC t=-3.69); NEW LANDMINE: BitInfoCharts unknown tickers 200-redirect to default btc-ltc-eth chart (title guard added). D: ve/fee classification 101 tokens (31 ve / 66 fee, 32 low-conf); ve split REJECTED (wrong direction), fee split null, dlambda null, size flat, turnover tilts wrong way; coin interaction robust MRP 20/40 (t=-3.5) + stronger ex-fallback (t=-4.67) BUT raw-NVT attenuates (t=-1.9) and ex-g-cap (40% of coin months) kills significance (t=-0.97) - growth-levelization load-bearing. E: M1 NOT SUPPORTED (convxturnover wrong sign), M2/M3 rejected, M4 coin result survives staking-yield controls (t=-4.39) - paper 2.3 must reframe. Entries 100-105; PHASE3B_RESULTS_REPORT.md; session_044 log. Commit+push at session end. |

2026-08-04 | ~1 (est., confirm) | Prompt | Launched session 045 (Claude Code / Fable 5) via CLAUDE_CODE_PHASE3C_KICKOFF_PROMPT.md; fully autonomous, no check-ins. Outcome: Phase 3c COMPLETE (Tasks A-D). A: DeFiLlama fees/revenue panel for the 101-token sample (endpoints verified live; slug identity = tvl_panel map; ARB/METIS/APE/BLAST chain-level flagged; 6 candidate slug resolutions probed+rejected) - coverage 55/101 fees, 51/101 revenue, median 44 mo, mostly 2021+; fees-without-revenue asymmetry reported as finding. B: pf = MC/12m-fees, prev_gl = MC/REV* + pf_gl = MC/F* via EXACT phase2 PQ* machinery (pf 1,367 rows/51 tokens; prev_gl 976/35; median P/F 12.1). C1 VERDICT: token H2 null SURVIVES the sixth+final measurement candidate - conv x lnP/F -0.0024 (t=-0.45), conv x ln prev_gl -0.0035 (t=-0.31); first H2-signed estimates + right-ordered splits but null; coverage-matched TVL_GL baselines equally dead -> not a denominator effect. NEW FACT: ln P/F is the first working token valuation signal (singles t=-2.50, s4 level t=-3.21, survives completed joint t=-2.32) but DCF transform destroys it and quintile portfolio earns nothing - panel-level phenomenon; revenue-subsample conviction sign-flip caveat (t=-2.01) logged. D: technical battery (ma_dist/vol12/ivol/amihud[snapshot caveat]/skew36) - token conviction quintile SURVIVES the COMPLETED battery (+1.74%/mo t=2.45 vs 12 LS; +2.17% t=2.82 with fee LS); the two underpowered 044 MA-cross cells superseded by ma_dist and RESOLVED (0.99t/n28 -> 2.73t/n43); coin conv x val survives completed battery (-0.0235 t=-2.68 full, -3.17 sub-2024); amihud strongest new coin single (-2.65). 202 free DL fetches, no paid calls. Entries 111-114; PHASE3C_RESULTS_REPORT.md; session_045 log. Commit+push at session end. |

2026-08-04 | ~4 (est., confirm) | Cowork | Session 042 continuation (same conversation): full results cycle into paper. Reviewed 043/044/045 results; produced findings-review + paper-style-tables docx; inline analyses (token quintile Entry 97, size-as-delta Entry 108, H3 under P/F+REV* Entry 116); designed + kicked off 3b and 3c (Entries 98-99, 107, 109-110); M1-M4 into theory; wrote Results 5.1-5.5 + Robustness 6.1-6.3 + Summary-of-Findings scorecard (Sec 7) + conclusion + abstract/intro updates; three alt-valuation tables made visible (8-10); float placement fixed. Paper 45 pp / 13 tables, complete draft awaiting pruning review. Commits through d6137fd. |

2026-08-05 | ~4 (est., confirm) | Cowork | Sessions 042 continuation: full pruning + review + front-matter cycle. Review-round edits (Entry 121: delta levels, survivorship, FM explanation, guard relaxation, tercile removal, channel decomposition table incl. NEW finding HODL carries the token signal); figures created + WP ordering (Entry 122); DEX-quintile within-sector finding (Entry 117); VW spanning + P/F quadrant checks (Entry 118); long-only analysis — conviction is an exclusionary screen (Entry 125); proper summary stats table + narrative (Entry 120); sample-composition appendix; intro rewritten in 2 iterations (Entries 123-124: lambda bridge after QTM/tokenized framing, findings led by joint-race + spanning survival, contributions as prose); abstract finalized per Moazzam's edit; title -> "Skin in the Chain: Locked Supply and the Cross-Section of Cryptocurrency Returns" (Entry 126). Complete 58-pp draft awaiting fresh-eyes review. |

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Part 3: Kickoff Prompts (verbatim initial prompts to autonomous AI sessions)

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CLAUDE_CODE_DUNE_DRYRUN_PROMPT.md

Claude Code Pilot Prompt — Dune Free-Tier Panel-Scale Dry-Run

Paste the prompt below as the first message in a new Claude Code session opened with working directory C:/AFA_2027_QTM_Crypto (the repo root). This is the explicit gate Entry 36 / DUNE_PILOT_REPORT.md ?5 requires **before** any university funds are requested for a paid Dune plan. It is still a diagnostic dry-run, not a Phase 2c build.

You're working in the AFA 2027 QTM Crypto research repo. Before doing anything else, read in full:

1. 04_code/DATA_DECISIONS_LOG.md Entry 31 (Etherscan pilot rejection --- verify before trusting a number) and Entry 36 (Dune 3-token feasibility pilot --- AAVE/LDO/GNS all passed on the free tier, but only as single tokens over 30 days).
2. 03_data/DUNE_PILOT_REPORT.md, especially ?5: the one open risk is that the free tier only exposes the `small` query engine (a single LDO join already took 66s), and two coverage caveats were flagged but NOT yet checked: (a) whether every NaN token's protocol actually exists in Dune's spellbook (majors like Aave/Lido/Gains are confirmed; long-tail protocols may not be), and (b) whether a full multi-year, full-panel grouped query completes on `small` without hitting time/row limits. This session answers both, for the REAL token list, not the 3-token sample.
3. 03_data/phase2/pq_tokens.csv --- confirm the current full NaN lists (re-derive live, don't trust a stale copy):
 - Lending (6): AAVE, ANC, BZRX, OM, STRK, WXT
 - Liquid Staking (1): LDO
 - Derivatives (10): AVNT, DDX, GNS, HAKKA, HXRO, KP3R, LINA, MIR, MYX, NMR
4. 03_data/universe_panel.csv --- confirm the panel's full date range (was 2015-08 to 2026-05, 130 months, as of last check --- re-verify).
5. 04_code/dune_pilot_{test,explore,aggregate,verify}.py and 03_data/raw/dune_pilot/ --- the working code and raw query results from the 3-token pilot. Reuse the same tables/columns already validated there (`lending.borrow`, the Lido event tables + canonical-WETH-contract price join, `dune.gains.result\_g\_trade\_stats\_defi\_llama`) rather than re-discovering them.

The free-tier Dune key is in gitignored `04\_code/.api\_keys.json` under "dune" (2,500 free credits/month, resets monthly; ~100 were used by the 3-token pilot).

Funding context (read before deciding what to recommend)

Moazzam has university approval for a paid Dune plan **if** this dry-run shows the free tier can't handle full-panel scans. This removes the prior hard "do not upgrade" gate --- you may recommend a paid plan if the evidence supports it. It does NOT mean you (or any AI agent) should attempt to purchase or subscribe. If a paid plan is the right call, say so clearly in the report and stop --- Moazzam will do the actual checkout himself in Dune's billing UI. Do not enter payment details, do not create or modify any billing subscription, under any circumstance.

What to do

Step 1 --- Coverage check (catalog search only, cheap, not query-credit-metered).
For each of the 17 tokens above, confirm whether its protocol is actually findable in Dune's spellbook/decoded catalog (`/v1/datasets/search`). For the 3 already-validated protocols (Aave->AAVE, Lido->LDO, Gains->GNS) you already know the answer from the pilot --- don't re-spend calls confirming those. For the other 14 (ANC, BZRX, OM, STRK, WXT; AVNT, DDX, HAKKA, HXRO, KP3R, LINA, MIR, MYX, NMR), search and report found/not-found, with the matched `full\_name` if found. Do not guess --- if a search returns nothing relevant after 2-3 reasonable query phrasings, mark it `not\_found` and move on; do not force a match.

Step 2 --- Full-history, full-panel grouped query per category, `small` engine only.
For whichever tokens in each category ARE covered (per Step 1):

- **Lending:** one `lending.borrow` query, `GROUP BY project, block\_month`, filtered to

the covered lending protocols, full history (no 30-day window this time --- the spec's actual panel needs the whole series). Record wall-clock time and row count returned.

- **Liquid Staking (LDO only):** the same stake+unstake event tables from the pilot, full history, monthly grouping, with the price join on the **canonical WETH contract** (never ``symbol='WETH'`` --- Entry 36's trap). Record wall-clock time.
- **Derivatives:** GNS via ``dune.gains.result_g_trade_stats_defi_llama``, full history, monthly grouping. For any other derivatives tokens found covered in Step 1, the equivalent per-protocol table if one exists.

Note explicitly whether any query times out, hits a row cap, or returns a Dune error --- that is the actual data point this dry-run exists to produce. If a query is slow but completes, record the exact seconds.

Step 3 --- Report results, including coverage gaps. Most of the 17 tokens will likely turn out NOT to be in Dune's spellbook (long-tail protocols, per the report's own caveat) --- that is a useful, expected finding, not a failure of this dry-run. Report it plainly per token: covered+fast, covered+slow/timed-out, or not_found.

What NOT to do

- Do not build a panel-scale extraction pipeline (``phase2c_pq_tokens.py`` or similar).
- Do not write to ``pq_tokens.csv`` or any Phase 2/2b output --- diagnostic only.
- Do not purchase, subscribe to, or modify any Dune billing plan.
- Do not guess at column semantics or force a coverage match that isn't real --- same standard as Entry 31/36.

Deliverable

Write ``03_data/DUNE_DRYRUN_REPORT.md``: the Step 1 coverage table (17 tokens, found/not found, matched table if found), the Step 2 timing/row-count results per category, and a clear recommendation per category --- (A) free ``small`` engine handles the full panel, no upgrade needed, (B) free tier finds the tables but ``small`` can't finish full-history scans, recommend the paid plan **for the engine, not credits** (Moazzam must purchase it himself), or (C) most tokens in this category aren't on Dune at all regardless of engine tier --- note this means even a paid plan wouldn't help those specific tokens; they stay flagged-NaN. Log this session as ``06_documentation/ai_conversations/session_016_<date>dune_dryrun.md``, continue ``04_code/DATA_DECISIONS_LOG.md`` from **Entry 37**, and update ``06_documentation/time_log.md``. Commit and push to github.com/moazzamkhoja/AFA_2027_QTM_Crypto (main) at session end. Stop after the report --- do not start a Phase 2c build until this is reviewed.

/newpage

CLAUDE_CODE_DUNE_PILOT_PROMPT.md

Claude Code Pilot Prompt — Dune Analytics Feasibility Pilot (Lending/Staking/Perps)

Paste the prompt below as the first message in a new Claude Code session opened with working directory `C:/AFA_2027_QTM_Crypto` (the repo root). This is a bounded **diagnostic pilot**, not a Phase 2c build — it tests whether Dune's decoded tables can supply the token-side PQ categories DeFiLlama has no volume dimension for, before any university funds are requested for a paid Dune subscription.

You're working in the AFA 2027 QTM Crypto research repo. Phase 1 and Phase 2 (coins) are

complete; token-side PQ has a structural gap (see below). Before doing anything else, read in full:

1. 04_code/DATA_DECISIONS_LOG.md Entry 31 --- the Etherscan Transfer-log pilot. It looked plausible but was the WRONG QUANTITY: UNI raw transfer volume was 46.6x smaller than actual Uniswap DEX volume (corr 0.30), and AAVE's raw transfer sum was corrupted by sentinel-value transfers (off by ~10 orders of magnitude before cleaning). The standard this pilot must meet: don't trust a number just because a query returns one --- verify it against a real reference before reporting it as usable.
2. 04_code/DATA_DECISIONS_LOG.md Entry 32 --- the PQ source waterfall this gap sits inside (DeFiLlama volume primary; fee-backout only for confidently-known flat fee rates; else PQ flagged missing).
3. 03_data/phase2/pq_tokens.csv --- confirm AAVE (category "Lending"), LDO (category "Liquid Staking"), and GNS (category "Derivatives") are all still `pq\_usd` NaN, each because DeFiLlama has no transacted-value series for that category (Lending/Liquid Staking) or has paywalled it (Derivatives --- HTTP 402 on the perps dimension, affecting 10 tokens including GNS). These three are the test cases: one per category.
4. 04_code/dune_pilot_test.py --- already written, NOT yet run. It searches Dune's dataset catalog for the real decoded tables behind AAVE borrow events, Lido stake/unstake events, and GNS trade events, samples 5 raw rows to confirm actual column names (does NOT assume Spellbook naming guesses are correct), and attempts a trailing-30-day SUM only if an obvious USD-amount + timestamp column pair exists.

A free-tier Dune API key is already in gitignored `04\_code/.api\_keys.json` under "dune" (2,500 free credits). The Cowork session that wrote the script has no network route to api.dune.com (sandbox egress is allowlisted to dev-infra domains only) --- that's the only reason this wasn't already run. You have normal network access; run it for real.

What to do

1. Run `python3 04\_code/dune\_pilot\_test.py`. Read its console output and the raw JSON it writes to `03\_data/raw/dune\_pilot/`.
2. For any token where the script stopped at "needs_manual_column_mapping" (no obvious USD-amount/timestamp pair), look at the real sample columns it printed and write the correct aggregation SQL yourself --- e.g. raw token-unit amounts may need a join to a price table, or the right table may be a normalized abstraction (`lending.borrow`, `dex.trades`, or similar cross-protocol table) rather than the raw per-pool event table. Use Dune's `/v1/datasets/search` endpoint again with different query phrasing if the first candidate table was wrong.
3. For each of the three tokens, get a trailing-30-day (or most recent complete month, whichever is cleaner) USD volume figure:
 - AAVE: borrow/loan-origination volume (NOT TVL, NOT repayments-net --- origination flow)
 - LDO: stETH stake + unstake flow (submit + withdrawal events, summed as volume not net)
 - GNS: gTrade perps trading/notional volume
4. ****Cross-check each number against an independent reference**** before trusting it --- the whole point, per Entry 31's standard. Options: Dune's own public dashboards (`dune.com/gains/gtrade\_stats` for GNS; Lido's official stats page; Aave's own analytics page or DeFiLlama's AAVE TVL/fees as a sanity bound), or order-of-magnitude plausibility against the token's market cap / TVL. State the comparison explicitly --- ratio, and whether it's in a sane range (unlike the Etherscan pilot's 46.6x miss).
5. ****Note the credit cost.**** Record how many Dune credits/API calls this 3-token pilot used, out of the 2,500 free. This is the actual feasibility data point for the university-funding decision Moazzam is weighing --- be precise, not approximate.

What NOT to do

Do not build a panel-scale Dune extraction pipeline (`phase2c\_pq\_tokens.py` or similar) and do not attempt to backfill multi-year history. This is a 3-token, ~30-day feasibility check only. Do not guess at column semantics if a table's columns are ambiguous --- say so in the report rather than reporting an unverified number.

Deliverable

Write `03\_data/DUNE\_PILOT\_REPORT.md`: the three tables actually used (with full Dune `full\_name`), the verified 30-day volume figure for each (or "could not verify" with the reason), the cross-check comparison for each, total credits/calls consumed, and a clear recommendation --- (A) Dune's free tier is sufficient to prototype all three categories at panel scale, (B) free tier works for discovery/spot-checks but a paid plan is needed for full-panel history, or (C) Dune does not cleanly solve one or more of these categories (say which, and why). Log this session as `06\_documentation/ai\_conversations/session\_015\_2026-06-25\_dune\_pilot.md`, continue `04\_code/DATA\_DECISIONS\_LOG.md` from Entry 33, and update `06\_documentation/time\_log.md`. Commit and push to github.com/moazzamkhoja/AFA_2027_QTM_Crypto (main) at session end. Stop after the report --- do not start a Phase 2c build until this is reviewed.

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CLAUDE_CODE_HEX_AKRO_RECONCILIATION_AND_SURVIVORSHIP_PROMPT

Claude Code Kickoff Prompt — HEX/AKRO Reconciliation + Survivorship-Bias Documentation

Paste the prompt below as the first message in a new Claude Code session opened with working directory `C:/AFA_2027_QTM_Crypto`. This follows a Cowork review of session 022's full-universe feasibility pass (`06_documentation/SESSION022_STATUS_AND_NEXT_SESSION.md`), which read verified contract source code and confirmed mechanisms live via `getLogs` — a materially more rigorous, Etherscan-direct method than session 021's Dune top-holder substitute. Two of session 022's "6 genuine Channel-1" tokens, HEX and AKRO, directly contradict explicit session-021 rejections. The decision is to trust session 022's contract-code-read + `getLogs` method as the higher-fidelity standard and resolve the contradiction with it — not to referee 021 vs. 022 by assumption. A second, separate task documents the ~284 dead 2014-2018 listings (no chain/contract identity on any explorer) as an acknowledged survivorship-bias limitation rather than a pending data gap.

You're working in the AFA 2027 QTM Crypto research repo. Two things to resolve this session, in order. Read everything in "Required reading" before touching either.

Required reading

1. `04_code/DATA_DECISIONS_LOG.md` Entry 26 (the single-escrow custody standard) and Entry 53 (session 022's feasibility-finding entry) --- read the full text, not a summary.
2. `03_data/SESSION021_TOKEN_BUCKET1_EXHAUSTIVE_AUDIT.md` --- the exact rejection lines for HEX ("staking internal to the HEX contract", Stage 2b) and AKRO ("no single contract reproduces the DL staked figure", Stage 2b, grouped with KAITO/ATH/SUPER/etc.).
3. `06_documentation/SESSION022_STATUS_AND_NEXT_SESSION.md` section 2a --- HEX and AKRO are listed under "Channel 1 genuine lock/stake, `getLogs`-CONFIRMED: 6", but the parenthetical splits them: HEX is "amount-bearing, build-ready"; AKRO is "bare Locked(), need contract-balance reads" --- a weaker confirmation than the other 4 (NMR, `stkAAVE`, XAN, and the other amount-bearing one).
4. The raw per-event caches already produced by session 022 for these two `cmc_ids` --- pull their rows directly rather than re-querying Etherscan from scratch: `03_data/phase1/`

universe_lambda_channel_map.csv, 04_code/_universe_lambda_findings.csv (or the bucket-1 subset 04_code/_etherscan_lambda_findings.csv / 03_data/phase1/etherscan_lambda_channel_map.csv), and 03_data/raw/etherscan_src/ for the cached verified source.

Part 1 --- HEX and AKRO reconciliation (apply session 022's method, not session 021's)

For each of HEX and AKRO, run the FULL Entry-26 test using session 022's higher-fidelity method --- do not stop at "an event fires." The bar is the same one the 5 session-021 BUILDS cleared: a specific contract/event must let you reconstruct a monthly locked-quantity series that cross-checks to a live on-chain balance, not just confirm a mechanism exists.

****HEX:****

- Identify the exact staking event(s) in the cached verified source (likely `StakeStart`/`StakeEnd`-style events). Confirm via getLogs whether the event itself carries the staked amount directly (amount-bearing) --- if so, this is a genuinely different construction path than Entry-26's transfer-based reconstruction (no separate escrow contract needed), and that's fine, but it must be stated explicitly as a new method, not silently treated as equivalent.
- Resolve the custody question directly, don't infer it: does HEX's staking move the staked tokens out of the holder's wallet balance into the HEX contract's own balance (custodial, `balanceOf` of the contract rises), or does the holder's `balanceOf` stay unchanged while a separate internal struct/mapping just flags the stake (non-custodial bookkeeping)?
- If non-custodial: check HEX's actual CoinMarketCap `circulating\_supply` field/methodology directly (don't assume) --- does it already exclude staked HEX, or does it include it? This determines whether building this series double-counts against the existing AERO/SOL/API3/ORBS denominator artifact (Entry 49). State the answer plainly before deciding to build.
- Reconstruct the monthly staked-HEX series via event replay (sum of active stakes by month from `StakeStart`/`StakeEnd` amounts) and spot-check it against at least 2-3 months of the contract's own on-chain state (e.g. a public `globalInfo`/`stakeShares` read if one exists) the same way the 5 BUILDS were cross-checked to 0.00% drift in session 021.

****AKRO:****

- Locate the actual named staking/lock contract address from the cached verified source (the "bare Locked()" event implies a contract was found but not yet confirmed amount-bearing).
- Confirm via getLogs + a direct contract-balance read whether that ONE contract holds a meaningful, reproducible share of AKRO's base-token supply --- i.e. satisfy the SAME single-escrow test that rejected it in session 021, not a looser "a Locked event exists somewhere" bar.
- If the contract's balance does not reproduce a stable, sensible staked-supply figure (e.g. it's a treasury/multisig holding an unrelated allocation, the original session-021 suspicion for this whole cohort --- KAITO/ATH/SUPER etc.), say so explicitly and explain what the contract actually is.

****Resolution for each:****

- If it now passes the full standard: build it into lambda via the existing `phase1\_channel1\*.py` scripts (extend, don't rebuild), and log a NEW DATA_DECISIONS_LOG entry (continue from Entry 54 --- do not edit the session-021 entry) stating explicitly: "session 021's rejection of [HEX/AKRO] is superseded by Entry [N], using session 022's contract-code-read + getLogs method, which found [specific reason]." Name the superseded entry by number.
- If it still fails once fully reconstructed: log that explicitly too, with the specific reason the getLogs confirmation wasn't sufficient (e.g. "mechanism exists and is active, but is non-custodial / multi-contract / doesn't reproduce a usable figure") --- restate the session-021 rejection with the new evidence, don't leave it ambiguous between the two sessions' reports.
- Either way, write a short addendum note (append to a new `03\_data/SESSION023\_HEX\_AKRO\_RECONCILIATION.md`) so the contradiction between 021 and 022 has a documented resolution on record, not two conflicting reports sitting side by side.

Part 2 --- Survivorship-bias documentation (the dead 2014-2018 listings)

Session 022 found 284 of the 405 non-EVM/off-Etherscan tokens have NO chain/contract identity recoverable on any explorer checked (EVM or non-EVM) --- dead listings from 2014-2018. Re-derive this count live from `03\_data/phase1/non\_evm\_lambda\_recoverability.csv` (and cross-check against `03\_data/phase1/universe\_lambda\_channel\_map.csv`) rather than reusing the report's number without confirming it --- reconcile it to the penny against the documented universe the same way session 021 reconciled its 398.

These are not a pending data gap to keep chasing --- they are genuinely, permanently unrecoverable (no contract, no chain identity, on any free or paid source). Document them as an acknowledged survivorship-bias limitation:

1. A new DATA_DECISIONS_LOG entry (next number after the HEX/AKRO entries) stating the live-reconciled count, the criteria used to call a token "dead" (no chain/contract identity on any explorer, EVM or non-EVM, checked across sessions 021-022), and that no further recovery effort is planned for this group.
2. A short, paper-ready paragraph (3-5 sentences, citable) for the data/methodology section stating: these tokens were active CMC listings in 2014-2018 with no recoverable on-chain identity today (delisted, abandoned, or rug-pulled), their exclusion is a genuine survivorship-bias source rather than a methodological choice, and this should be named explicitly rather than absorbed silently into the broader "no-data" rejection category. Note for the paper's limitations discussion whether survivors are likely to differ systematically from this dead cohort in ways correlated with lambda or returns (e.g. dead tokens disproportionately being low-conviction/no-governance assets to begin with), since that's the actual bias-direction question a referee would ask.
3. Save the paragraph as a clearly-labeled section in `03\_data/SESSION023\_HEX\_AKRO\_RECONCILIATION.md` (or a separate `03\_data/SURVIVORSHIP\_BIAS\_NOTE.md` if cleaner) so it can be pulled directly into `05\_paper/` later without re-deriving it.

Rules (unchanged)

- Trust session 022's contract-code-read + getLogs method over session 021's Dune-substitute method wherever they conflict --- that's the explicit basis for this session, not a neutral re-litigation.
- Never sign up for or pay for any paid tier (Etherscan Pro included). Flag, don't act.
- Join everything on `cmc\_id`, never `symbol`.
- Never guess or interpolate a numerator --- reconstruct from real events/balances or flag the gap.
- DATA_DECISIONS_LOG.md is append-only --- never edit Entries 1-53. Continue at Entry 54.
- Do NOT touch the Channel-3 dedup, the 16 BSC/Base/Avax paid-gated tokens, or the Channel-2 prototype from session 022's other open items --- this session is scoped to HEX/AKRO + the survivorship-bias note only. Those remain separate, pending review.
- Update `06\_documentation/time\_log.md` and log this session as `06\_documentation/ai\_conversations/session\_023\_\*.md`. Commit and push to github.com/moazzamkhoja/AFA_2027_QTM_Crypto (main) at session end.

Stop when HEX and AKRO each have a final, explicitly-resolved verdict (BUILD or REJECT, with the 021-vs-022 contradiction named and settled) and the survivorship-bias note is written. Do not start any of session 022's other open items without review.

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CLAUDE_CODE_KICKOFF_PROMPT.md

Claude Code Kickoff Prompt — Phase 0 (Asset Universe)

Paste the prompt below as the first message in a new Claude Code session opened with working directory C:/AFA_2027_QTM_Crypto (the repo root). Run Phase 0 to completion, review the output, then come back for a kickoff prompt for Phase 1.

You're working in the AFA 2027 QTM Crypto research repo. Before doing anything else, read in full:

1. 04_code/DATA_SPECIFICATION.md --- the full data specification. Don't skim it.
2. 04_code/DATA_DECISIONS_LOG.md --- the append-only log you'll be writing to.
3. 05_paper/main.tex, Section 2 (the Locking Decision model) and Appendix A (Growth-Levelized NVT) --- for the theoretical context behind the variables.

Your task right now is PHASE 0 ONLY, per Section 7 of the spec: build the asset universe. Do not start Phase 1 (lambda channels) or any later phase in this session.

Deliverables for Phase 0:

1. A reproducible script (not a one-off manual pull, not hand-typed data) that builds a point-in-time monthly ranked universe of crypto assets from CoinGecko, DeFiLlama, and/or Artemis Analytics. Follow Section 2.1 exactly: rank by market cap each month using historical, point-in-time snapshots; an asset enters the panel the first month it's in the top 250 (this threshold is adjustable, report sensitivity if you change it); once in, it never leaves the panel even if it later falls out of the top 250, delists, or goes to zero. Before you build this, verify what these APIs actually support today (rate limits, whether a key is required, how far back historical snapshots go, whether delisted/dead coins are retrievable) --- don't assume from general knowledge.
2. A classification table (coin vs. governance token) per Section 2.3, with the supporting evidence you used for each asset, and explicit flags for ambiguous or transition cases (e.g., an asset whose locking mechanism changed over time). The ETH proof-of-work-to-proof-of-stake transition in the spec is one illustrative example to check, not the only one --- look for others as you go.
3. A coverage report: number of qualifying assets per month and per year across 2015-08 through the most recent complete month. Call out where the panel is thin. Separately note how much further back pre-2015 native coins (BTC chiefly) could be extended, in case a coins-only robustness subsample becomes useful later.
4. Update 04_code/DATA_DECISIONS_LOG.md as you go, not at the end, for every deviation from the spec, every proxy substitution, and every threshold or judgment call.

When Phase 0 is done, stop. Summarize what you built, what the coverage report shows, and any open questions or judgment calls that need a decision before Phase 1 starts. Do not proceed past Phase 0 without that review.

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CLAUDE_CODE_KICKOFF_PROMPT_PHASE0_FOLLOWUP.md

Claude Code Kickoff Prompt — Phase 0 Follow-Up (pre-Phase 1 cleanup)

Paste the prompt below as the first message in a new Claude Code session opened with working directory `C:/AFA_2027_QTM_Crypto` (the repo root). This is a short follow-up to Phase 0, not Phase 1 itself. Run it to completion, review the output, then come back for the Phase 1 (lambda channels) kickoff prompt.

You're working in the AFA 2027 QTM Crypto research repo. A prior session (logged as Session 006) completed Phase 0 (asset universe) and stopped for review. That review is done. Before doing anything else, read in full:

1. `04_code/DATA_SPECIFICATION.md` --- including new section 2.6 (sector/economic-function classification), added after this review.
2. `04_code/DATA_DECISIONS_LOG.md` --- read Entries 1-16. Entries 10-16 are the review decisions you're implementing in this session.
3. `03_data/PHASE0_COVERAGE_REPORT.md`
4. `06_documentation/PHASE0_SESSION_STATUS.md`

Your task right now is a FOCUSED FOLLOW-UP to Phase 0, not Phase 1. Do not start Phase 1 (lambda channels) in this session. There are three deliverables:

1. CARRY-FORWARD SPLIT (implements Decisions Log Entry 10).
For every asset-month currently marked `status='carried_forward'` in `03_data/universe_panel.csv`, classify it into one of two subtypes:
 - 'presumed_failed': the asset never reappears in the observed top-1000 snapshot for the remainder of the sample (2026-05-31).
 - 'temporarily_out': the asset drops out of the top-1000 snapshot but reappears in a later month.Add this subtype as a new column (don't overwrite the existing status column).
Report: total carried-forward asset-months, split by subtype; the count of distinct assets in each subtype; the distribution of gap lengths for `temporarily_out` (min/median/max months out); and the count of `presumed_failed` assets whose gap, if it had continued to the end of the sample, would have looked identical to a `temporarily_out` case still mid-gap as of 2026-05-31 (i.e. flag the right-censoring risk explicitly --- don't silently treat "never seen again so far" as equivalent to "permanently dead" for assets whose gap started recently).
Do NOT pick or implement a death-return formula. That decision is intentionally deferred --- just produce the split and the diagnostic counts so the next decision can be made with real numbers in front of it.
2. CLASSIFICATION CONFIRMATION PASS (implements Decisions Log Entry 11).
From `03_data/classification_table.csv`, select 'other'/ambiguous_flag=True assets with ≥ 12 observed asset-months in `03_data/universe_panel.csv` (count months where the asset is actually observed, i.e. exclude carried-forward months from this count --- carried-forward months aren't real observations of the asset's behavior).
For each selected asset, do a quick evidence-based review using whatever tags, DeFiLlama category data, and public knowledge of the project you can verify, and propose: keep as 'other', reclassify to 'coin', or reclassify to 'token' --- with a one-line rationale per asset. Do not invent classifications you can't support; if an asset is genuinely ambiguous after review, say so explicitly and leave it 'other' with a note, rather than forcing a label.

Within this pass, specifically check whether any of these reviewed names are meme coins (DOGE, SHIB, PEPE, WIF, BONK, FLOKI, and similar) or NFT-adjacent tokens. Per Decisions Log Entry 12, meme coins are correctly left as 'other' (no change needed) --- just confirm they're in fact landing there and aren't being mis-flagged as coin/token by the existing tag rules. Also explicitly check the universe for anything that looks like an actual NFT collection (not a fungible token) --- confirm none exist in the panel (the CMC fungible-listings endpoint shouldn't have any), and say so explicitly in your report rather than assuming it.

3. SECTOR/ECONOMIC-FUNCTION CLASSIFICATION (implements Decisions Log Entry 16, spec section 2.6). This is a SECOND, INDEPENDENT classification field --- do not merge it with or use it to revise the coin/token `asset\_class` field from deliverable 2 or from Section 2.3; it sits alongside it, not inside it.

For every asset in 03_data/classification_table.csv, add a `sector` column:

- Primary source: the existing `defillama\_categories` field (already populated for 574/2010 assets) --- these values (Dexs, Lending, Derivatives, Liquid Staking, CDP, Bridge, RWA, Gaming, NFT Marketplace, Synthetics, Options, Restaking, Staking Pool, etc.) are themselves usable sector tags; carry them into `sector` as-is (keep multi-value semicolon-separated strings rather than forcing one pick).
- Fallback source: CMC `tags`, specifically `layer-1` and `layer-2` (134 and 43 assets respectively) for base-layer coins that have no DeFiLlama protocol entry, plus any other clearly sector-like tags you find useful (oracle, privacy, depin, meme, etc.). If both sources give a signal for the same asset, keep both rather than discarding either.
- Leave `sector` blank/null where neither source gives a usable signal --- do not guess or infer a sector from the asset name alone.

Report coverage: total assets with a non-null `sector`, broken out by `asset\_class` (coin/token/other) --- i.e., what fraction of coins get a sector tag (expect this to lean on layer-1/layer-2 tags) versus what fraction of tokens do (expect this to lean on DeFiLlama categories), and the residual with no tag from either source.

Do NOT decide or propose which sector-level comparisons (e.g. DEX vs. Perpetuals, Lending vs. Staking, L1 vs. L2) will actually be used in the paper --- that decision is explicitly deferred (Decisions Log Entry 16). Your job here is only to get the field populated and report its coverage.

Update 04_code/DATA_DECISIONS_LOG.md as you go for any new judgment call (e.g. the exact rule you use to call something 'temporarily_out' vs 'presumed_failed' at the very end of the sample window, any specific reclassification you make, or any sector-tagging edge case). Produce a short written report (similar in format to PHASE0_COVERAGE_REPORT.md) covering all three deliverables --- save it as 03_data/PHASE0B_FOLLOWUP_REPORT.md.

When all three deliverables are done, stop. Summarize the carry-forward split numbers, list every reclassification you made (from -> to, with rationale), confirm meme-coin/NFT handling, report sector-tag coverage by asset_class, and flag anything that still needs a human decision. Do not proceed to Phase 1 without that review.

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CLAUDE_CODE_LAMBDA_BUCKET2_BUCKET3_RECOVERY_PROMPT.md

Claude Code Kickoff Prompt — lambda Recovery: Bucket 2 (non-EVM coin staking) + Bucket 3 (EVM token locks)

Paste the prompt below as the first message in a new Claude Code session opened with working directory `C:/AFA_2027_QTM_Crypto`. This builds on a Cowork session's live verification of Bucket 2 (Entry 42) and the three-bucket coverage taxonomy built earlier that session. Run to completion, review the output, then come back before touching Bucket 1 or Phase 3.

You're working in the AFA 2027 QTM Crypto research repo. Phase 0/0B/1/2/2b/2c are complete; session 019 scaled lambda to 58 assets / 1,688 asset-months. A Cowork session then decomposed the 1,023 uncovered coin+token assets (94.6% of the 1,081-asset universe) into three buckets:

- Bucket 1 (~783 assets) --- no plausible staking/locking/voting mechanism exists at all. Out of scope for this session. Do not touch.
- Bucket 2 (~170 coins) --- non-EVM PoS coins needing a chain-specific indexer. A live verification pass (`DATA_DECISIONS_LOG.md` Entry 42) found free paths for some of these that Part A.3's native-RPC-only check had missed, an engineering (not payment) gap for two more, a genuine structural dead-end for one, and left several unconfirmed.
- Bucket 3 (~73 tokens + GMX) --- EVM tokens with a plausible governance/staking tag where the bottleneck is finding/verifying a clean single-escrow lock contract, not payment.

This session's job is to BUILD everything in Bucket 2 and Bucket 3 that verifies live, and to upgrade Entry 42's docs-level Bucket-2 findings to response-body-verified ones. Before doing anything else, read in full:

1. `04_code/DATA_DECISIONS_LOG.md` Entries 26, 41, 42 --- the EVM clean-escrow standard, the Part A.1/A.3 verdict lists, and this session's starting point (Bucket 2 live verification).
2. `03_data/PHASE1_LAMBDA_SCALE_AND_TVL_PANEL_REPORT.md` ?2--4 --- Part A.1 (token Channel-1 verdicts: built vs. rejected vs. deferred) and Part A.3 (coin staking-source audit).
3. `04_code/DATA_SPECIFICATION.md` ?0 (flag, don't guess) and ?3.1 (Channel 1 definition).
4. `03_data/classification_table.csv` and `03_data/universe_panel.csv` --- the universe, tags, and which asset-months already have a Channel-1 or Channel-3 value (use this directly to compute the actual "not yet recovered by any channel" token list for Part B --- don't take 73 as exact).
5. `04_code/phase1_channel1_evm_locks.py`, `phase1_channel1_evm_locks_ext.py`, `phase1_channel1_pos_coins.py`, `phase1_channel1_eth_staking.py` --- extend these, don't rebuild.

Part A --- Bucket 2: non-EVM coin staking (~170 coins)

Goal: turn Entry 42's docs-level findings into built series or honest, response-body-verified rejections. Do not re-derive from scratch --- Entry 42 already did the desk research; this session does the live signup + call + build.

****Tier 1 --- build now (free, high confidence in Entry 42):****

- SOL: sign up for a free `validators.app` API token, call ``/api/v1/epochs/mainnet.json`` live, confirm the ``total_active_stake`` field and per-epoch depth, build the monthly bonded-stake series.
- TRX: sign up for TronScan's free API tier (60 req/hr, auto-approved), call ``freezeresource`` with date ranges, build the monthly frozen-TRX series.

****Tier 2 --- verify live FIRST, build only if confirmed (medium confidence in Entry 42):****

- DOT, KSM: sign up for a free Subscan API key, call the validator era-statistics endpoint live.

Confirm the response body actually contains per-era bonded totals --- Entry 42's "free" read was inferred from the absence of a `[PRO]` tag in the docs, never pixel-confirmed. If the live call is gated or empty, log that as a correction to Entry 42 and stop. Do not pay for Subscan Pro.

****Tier 3 --- engineering, not payment (build the aggregation):****

- HBAR: iterate Hedera Mirror Node `/api/v1/accounts` (and the balances endpoint's `timestamp` fallback to 15-minute snapshot files) to sum `staked\_node\_id`/`staked\_account\_id`-linked balances per historical snapshot, producing a network-wide staked-total series. Free, public, no key. Scope the compute cost (accounts x snapshots) before committing to a full build; if genuinely intractable, log it as a documented gap rather than ship a partial series.
- SUI: query the `staking\_pool` module's per-epoch exchange-rate-history table via free public RPC/GraphQL for every active validator pool since each pool's activation epoch; sum across pools per epoch for the network-wide total. Free, public, no key.

****Tier 4 --- reclassification check, not a coin-side build:****

- CEL0: confirm whether the legacy LockedGold/Election locking contract is still live after the March 2025 L2 migration, and whether Celoscan exposes its Transfer/event logs the way Etherscan does for EVM tokens. If yes, build CEL0 with the SAME `getLogs` method as Entry 26 instead of treating it as a non-EVM coin. If no, leave it as an open Bucket-2 gap.

****Tier 5 --- check access live, build/pay nothing without Moazzam (ATOM, INJ, SEI, KAVA, AVAX, NEAR):****

- Visit the actual signup/pricing flow (not just docs or search results) for Mintsan/Cosmostation (ATOM/INJ/SEI/KAVA), Bitquery, AvaCloud Metrics API (AVAX), and Pikespeak/NearBlocks (NEAR). Attempt a free-tier signup only where one is genuinely self-serve and free. Report exactly what each gate looks like (self-serve free / self-serve paid with disclosed price / contact-sales only). Do NOT purchase or commit to any paid tier --- that decision is Moazzam's alone.

****Tier 6 --- keep searching (EOS, ICP, APT):**** no historical source found yet, free or paid. If nothing turns up this session, leave as a documented gap.

****Excluded (ALGO):**** confirmed structural gap, Entry 42 --- no vendor, paid or free, has this. Don't spend time here.

Part B --- Bucket 3: EVM tokens with a plausible governance/staking tag (~73 + GMX)

Already built (Entry 26 + Part A.1): CRV, CVX, FXS, SUSHI, AAVE, YFI, PENDLE, LQTY, 1INCH, RPL.
Already rejected, final (Part A.1, re-read before excluding): MKR, BAL, COMP, RUNE, ANGLE.

1. From classification_table.csv, take every `asset\_class='token'` row not already recovered by any channel in universe_panel.csv and not one of the 5 final rejects above, that carries a staking/governance/vote/lock-relevant tag or `defillama\_categories` value. Log the resulting count --- don't force it to match the ~73 estimate.
2. For each candidate, apply the Entry-26 standard: a single, verified contract that holds a meaningful share of token supply via direct custody --- not a wrapped/composite asset (like veBAL's 80/20 BPT), not in-wallet delegation (like COMP), not a collateral/C-ratio system (like SNX). Use the protocol's own docs, its Defillama page, and the verified contract on the relevant chain's explorer. Classify each: BUILD / REJECT (specific reason) / DEFER (specific blocker).
3. For every BUILD candidate, reconstruct the historical locked-supply series from Dune's free curated `balances\_<chain>.daily\_updates` table (confirmed free-tier accessible across 20 EVM chains including Arbitrum) --- query the escrow contract's balance over time directly from this table, NOT via Etherscan `getLogs` (which hit the free-tier result-window cap on Arbitrum for GMX). Use the existing Dune key in `04\_code/.api\_keys.json` under `"dune"`; stay inside the free-tier query-credit budget (2,500/month, prior usage <4%) and flag any query that would

meaningfully eat into it instead of just running it.

4. GMX specifically: mechanism already verified (StakedGmxTracker, Arbitrum, ~65% of supply, Part A.1) --- skip straight to step 3 for this one. Build it first, as a confidence check on the Dune-Balances method before working the rest of the queue.

Rules (unchanged from the spec and prior entries)

- Never sign up for or pay for any paid tier of anything. Flag a priced, self-serve option and report back to Moazzam --- he executes any purchase himself.
- Verify every source live (actual signup + actual call + actual response body) before treating it as built. Docs language is not sufficient --- that's exactly the gap between Entry 42's inference and a real verification.
- Join everything on `cmc\_id`, never `symbol`.
- Never guess or interpolate a numerator. Flag as a documented gap if it can't be verified live.
- DATA_DECISIONS_LOG.md is append-only --- never edit or delete Entries 1--42. Continue at Entry 43, one entry per chain/token group (not one giant entry).

Deliverables

1. Built series for whichever Bucket-2 chains verify live, in `phase1\_channel1\_pos\_coins.py` (or a new `phase1\_channel1\_pos\_coins\_bucket2.py`).
2. Built series for whichever Bucket-3 tokens verify BUILD, plus GMX, in `phase1\_channel1\_evm\_locks\_ext.py` (or a new file), using the Dune-Balances method.
3. Re-run `phase1\_assemble\_lambda.py` to fold the new Channel-1 series into lambda; report new asset/asset-month counts against the session-019 baseline (58 assets / 1,688 asset-months).
4. A short coverage addendum (markdown) listing, per chain/token: built / rejected-with-reason / deferred-with-reason / still-open-gap --- the response-body-verified successor to Entry 42.
5. DATA_DECISIONS_LOG.md continued from Entry 43. Log this session as `06\_documentation/ai\_conversations/session\_020\_\*.md`; update `06\_documentation/time\_log.md`.
6. Commit and push to github.com/moazzamkhoja/AFA_2027_QTM_Crypto (main) at session end.

Stop when Bucket 2 and Bucket 3 are done. Do not start Bucket 1 work or Phase 3 without review.

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CLAUDE_CODE_PHASE1_KICKOFF_PROMPT.md

Claude Code Kickoff Prompt — Phase 1 (lambda Channels)

Paste the prompt below as the first message in a new Claude Code session opened with working directory C:/AFA_2027_QTM_Crypto (the repo root). Run Phase 1 to completion, review the output, then come back for a Phase 2 kickoff prompt.

You're working in the AFA 2027 QTM Crypto research repo. Phase 0 (asset universe) AND the Phase 0B follow-up (carry-forward split, classification confirmation, sector field) are both complete. Before doing anything else, read in full:

1. 04_code/DATA_SPECIFICATION.md --- Sections 1, 3 (the lambda index), and 7 (phasing).
2. 04_code/DATA_DECISIONS_LOG.md --- the append-only log (20 entries so far); you'll add to it starting at Entry 21.
3. 06_documentation/PHASE0_TO_PHASE1_TRANSITION.md --- the handoff brief: repo state, stable keys, what Phase 1 must produce, and verified source pointers. READ ITS ?0 ADDENDUM FIRST (Phase 0B changes).
4. 03_data/PHASE0B_FOLLOWUP_REPORT.md --- the carry-forward split, the 16 reclassifications, the meme/NFT confirmation, and the sector-field coverage.

5. 06_documentation/PHASE0_SESSION_STATUS.md --- what Phase 0 built and the open questions.
6. 05_paper/main.tex Section 2 (the Locking Decision model) for the theory behind lambda.

Your task is PHASE 1 ONLY, per Section 7 of the spec: construct the lambda locking/conviction index per asset-month. Do not start Phase 2 (NVT_GL) or later in this session.

Build on the frozen Phase 0 universe --- do not rebuild it. Join everything on `cmc\_id` (never `symbol`). Compute lambda only for `status='observed'` asset-months in 03_data/universe_panel.csv; leave `carried\_forward` tails alone (the new `carry\_forward\_subtype` column splits those tails into presumed_failed/temporarily_out --- that is for the Phase 3 death-return policy, not for lambda). Use the post-confirmation `asset\_class` in 03_data/classification_table.csv (coin 633 / token 448 / other 858); the pre-confirmation label is preserved in `asset\_class\_original` if you need to audit a flip.

Deliverables for Phase 1:

1. A reproducible script (cache raw API pulls, like Phase 0 did) producing a per-asset-month table of the three lambda channels and the combined lambda_t:
 - Channel 1 --- staking/locking ratio (staked-or-locked / circulating-or-total).
 - Channel 2 --- holding duration (on-chain HODL/coin-age proxy; document the exact computation and limits per chain).
 - Channel 3 --- voting engagement (voters / eligible supply, monthly).lambda_t = equal-weighted average of the standardized (z-scored within each monthly cross-section) values of whichever channels are observable for that asset-month. Do NOT impute missing channels --- average only over channels that exist, and record how many and which contributed.
2. Respect time-varying mechanisms: read staking_start / transition_note from 03_data/classification_table.csv (e.g. ETH has no staking channel before 2020-12-01). Watch for and log other PoW->PoS or mechanism changes you encounter, with dates.
3. Before building each channel, VERIFY the current free access of every source you plan to use (per-chain explorers, beaconcha.in-style staking dashboards, DeFiLlama locked supply, Snapshot/Tally/Boardroom for voting) --- don't assume from general knowledge. Log dead/paywalled sources and the proxy you used instead.
4. A Phase 1 coverage report: per-asset-month channel-availability breakdown (how many of the three channels are populated, by year and by asset class), where each channel is thin or absent, and which hypotheses that gates. Update DATA_DECISIONS_LOG.md as you go (next entry is 21), not at the end. Log this session as 06_documentation/ai_conversations/session_009_*.md (007 = Phase 0 review, 008 = Phase 0B follow-up) and update 06_documentation/time_log.md.

When Phase 1 is done, stop. Summarize what you built, what the channel-availability report shows, and any open questions before Phase 2. Commit and push to github.com/moazzamkhoja/AFA_2027_QTM_Crypto (main) at session end. Do not proceed past Phase 1 without review.

/newpage

CLAUDE_CODE_PHASE1_SCALE_LAMBDA_AND_TVL_PANEL_PROMPT.md

Claude Code Kickoff Prompt — Scale lambda Coverage + Build a Real TVL Panel

Paste everything below the line into a new Claude Code session with working directory `C:/AFA_2027_QTM_Crypto`.

You're working in the AFA 2027 QTM Crypto research repo (github.com/moazzamkhoja/AFA_2027_QTM_Crypto).

Step 0 — Repo hygiene check (do this before anything else)

A prior session left a stale `.git/index.lock` and the local working copies of `04_code/DATA_DECISIONS_LOG.md` / `06_documentation/time_log.md` had silently fallen behind HEAD (Entries 33–39 and the last several time-log rows were committed and pushed, but missing from the local files — diagnosed and fixed in Cowork as Decisions Log Entry 40; see that entry for the full account). Before touching any data:

1. Confirm no git process is actually running on this machine (check Task Manager / any open terminal running git), then run `git status`. If you see `.git/index.lock` and no process is running, delete it.
2. Run `git status` again. If every tracked file shows as simultaneously staged-deleted and untracked, the index itself is stale — run `git add -A` (this rebuilds the index from the working tree; it does **not** delete anything, since the files are already present on disk) and confirm `git status` afterward shows either a clean tree or a small, sensible diff. If `git add -A` instead produces a huge, every-file-changed diff, stop and report it — don't commit blindly.
3. Read `04_code/DATA_DECISIONS_LOG.md` in full and confirm live what the actual last entry number is (it should be 40, per Entry 40 itself, but verify — don't trust this prompt's number). Do the same for the last session number in `06_documentation/ai_conversations/` (should be 018) and the last row of `06_documentation/time_log.md`. This project has twice now had a kickoff prompt state a stale next-number; always re-derive it live.

Required reading (in full)

1. `04_code/DATA_SPECIFICATION.md` — Section 3 (lambda: all three channels, including 3.2's acknowledgment that holding-duration is the hardest to source), Section 2.5 (per-asset source map: chain explorer per native chain, Snapshot/Tally/Boardroom/DeepDAO for voting), Section 4.1–4.2 (TVL's role as a *denominator*, not a PQ proxy).
2. `04_code/DATA_DECISIONS_LOG.md` — Entries 21 (live free-access audit methodology — the template to repeat for PoS coins below), 22 (DeFiLlama as a registry/address-book only, never the lambda measurement itself), 23 (ETH native staking via beacon deposit-contract logs), 24 (Channel 2 holding-duration — still NOT BUILT, still out of scope this session), 25 (Channel 3: Snapshot GraphQL + the curated space map + the token-weight guard, and the explicit “not on Snapshot” gap list), 26 (Channel 1: the curated EVM vote-escrow set and *why* veBAL and SNX were deliberately excluded, not silently proxied), 27 (the z-score/equal-weight/ ≥ 2 -asset-standardizability assembly rule — unchanged this session), 30 (TVL is a stock, not a PQ flow — still true; this session uses TVL only as a valuation-multiple denominator, never as a PQ substitute), 40 (this session's own framing and the repo-hygiene incident from Step 0).
3. `03_data/phase1/asset_onchain_identity.csv` and the script that built it, `04_code/phase1_build_identity_map.py` — the authoritative `cmc_id -> dl_slug / token_address / snapshot_space` registry. Re-derive the current counts live (don't trust numbers below) — as of 2026-06-26: 127/448 tokens have a `dl_slug` match, 123/448 have `has_address=True`, 29/448 have `has_snapshot=True` (auto-matched only).
4. `03_data/phase1/snapshot_space_map.csv` (the broader, curated 56/57-space map — 56 spaces covering 55 of the 448 tokens in `channel3_voting.csv`, wider than the 29 auto-matched above) and `04_code/phase1_channel3_voting.py`.

5. `04_code/phase1_channel1_evm_locks.py` and `04_code/phase1_channel1_eth_staking.py` — the existing, working event-log-reconstruction method (Etherscan `getLogs`) for Channel 1. The current curated EVM vote-escrow set is 6 tokens: `veCRV`, `vlCVX`, `veFXS`, `xsUSHI`, `stkAAVE`, `veYFI`.
6. `04_code/phase1_assemble_lambda.py` — the channel-combination logic. **Do not change the z-scoring / standardizability / equal-weight logic.** This session only widens the channel *input* files (`channel1_*.csv`, `channel3_voting.csv`, and a new `channel1_pos_coins.csv` — see Part A.3); the assembly script should pick up wider inputs automatically since it globs `channel1_*.csv`.
7. `04_code/phase2c_defillama_metadata.py` — specifically `check_tv1()` (currently fetches `api.llama.fi/protocol/{slug}`'s full `tv1[]` series and discards everything except presence/range/last-value). This session turns that into a real panel write — see Part B.

Funding / cost context (same standing rule as every prior session)

DeFiLlama's TVL data and Snapshot's GraphQL API are free and keyless. Etherscan V2 requires the existing free API key already stored gitignored at `04_code/.api_keys.json` ("AFA Paper" key, ~60 EVM chains via `chainid`) — free tier only, no archive-state access (Entry 21: historical `eth_call/balanceOf` silently returns latest-state, not historical state; only `getLogs` gives genuine point-in-time data). **Do not sign up for, key into, or use any paid tier of any service this session — DeFiLlama Pro, Etherscan Pro/archive, beaconcha.in, Boardroom, Tally, or anything else — under any circumstance.** If a real source turns out to be paywalled, log it as a gap and stop there; do not work around it by paying. Moazzam handles any purchase decision himself, after reviewing evidence, same as the standing rule for Dune and Artemis.

Scope boundary — read this twice

This session is **lambda scale-up + a real TVL panel for tokens. Nothing else.**

- **Do not build PQ for coins.** Coin PQ sourcing (currently 58/633 with real data) is explicitly deferred to a separate, later, explicitly-authorized session. If you notice a promising PQ-for-coins lead while working on lambda, write it down in the deliverable report as a flagged observation — do not chase it or build it this session.
- **Do not touch Channel 2 (holding duration).** Still a documented gap (Entry 24). Out of scope.
- **Do not re-derive or change the lambda assembly logic** in `phase1_assemble_lambda.py` — only add channel inputs.
- Join everything on `cmc_id`, never `symbol` (Entry 25's `veBAL/SNX` exclusions and Entry 39's `NVT/nerve`, `VELO/velodrome` symbol collisions are exactly the failure mode this guards against).

Part A — Scale lambda coverage

A.1 — Token Channel 1 (staking/locking): widen the curated EVM vote-escrow/staking set

The current set (6 tokens: `CRV`, `CVX`, `FXS`, `SUSHI`, `AAVE`, `YFI`) was built by individually verifying, per protocol, that the escrow-balance-equals-locked-supply identity holds cleanly (Entry 26) — explicitly excluding protocols like `veBAL` (locks an 80/20 BPT, not `BAL`) and `SNX` (a C-ratio system, not a lock) **by design**. Using the *same individual-verification standard* (not broad automated matching), check the other DeFiLlama-matched tokens (`dl_slug` non-null, `has_address=True`, ideally cross-referenced against DeFiLlama's `*-staking/*-vesting` chainTvl buckets in the `/protocol/{slug}` payload, which Entry 21 already confirmed exist) for additional EVM tokens where a vote-escrow or staking-lock contract cleanly equals locked supply of *that* token. Plausible candidates to check (verify each individually — do not assume any of these work without checking): `GMX` (escrowed `GMX`), `RPL` (Rocket Pool), `COMP`, `MKR`, `BAL` itself (not `veBAL`), `PENDLE` (`vependle`), `LQTY`, `RUNE`, `ANGLE`, `1INCH`. For each one added, use the same `getLogs`-based event-log reconstruction already proven for the existing 6 — don't invent a new method. For each one checked and rejected, write down why (same honesty standard as the `veBAL/SNX` exclusions).

A.2 — Token Channel 3 (voting): widen past the current 55-asset build

`snapshot_space_map.csv` has 56/57 curated spaces; `channel3_voting.csv` covers 55 of the 448 tokens. Two concrete widening paths:

- Cross-check `asset_onchain_identity.csv`'s `governanceID` field (DeFiLlama's own auto-matched Snapshot space, currently only surfaced for 29 tokens) against the full 127-token `dl_slug`-matched set for any space not yet in the curated map, and add it if a direct lookup against Snapshot's GraphQL API (hub.snapshot.org/graphql) confirms it's a real, active space.
- Entry 25 documented a specific “not on Snapshot, on-chain governance only” gap list: MKR, LQTY, PENDLE, RUNE, PERP, WLD, ONDO, ENA. For any of these that are EVM-based with a Governor-style on-chain voting contract, attempt on-chain `VoteCast` event-log reconstruction (the same `getLogs` method already used for Channel 1, applied to a different event signature) rather than Snapshot. Verify each protocol's actual governance contract address and ABI before assuming Governor Bravo compatibility — don't guess a log signature.

A.3 — Coin Channel 1 (staking/locking): live source-verification pass for PoS coins

Today only ETH (1 of 633 coins) has any lambda-channel data. Before building anything, do an Entry-21-style **live** free-access audit (not assumed from general knowledge) for the highest-market-cap PoS/staking coins in the 633-coin roster. Derive your own candidate list from `classification_table.csv` (coins with a known PoS/DPoS/NPoS consensus mechanism) rather than taking any example list at face value — but for reference, chains worth checking first: SOL, ADA, DOT, ATOM, AVAX, NEAR, ALGO, TRX, XTZ, EOS, ICP, KSM, CELO, KAVA, HBAR, INJ, SEI, SUI, APT. For each chain, check whether its own free, keyless block explorer or API publishes a staking-ratio / bonded-supply time series (e.g. a chain-native explorer's stats endpoint — verify the actual current endpoint and access tier live, the same discipline Entry 21 applied to beaconcha.in/Boardroom and found them now paywalled). Build a `channel1_pos_coins.csv` (same shape as the existing `channel1_*.csv` files: `cmc_id`, `month_end`, `staking_ratio`) for whatever is genuinely free and historical; for everything else, log it as a documented gap with the specific reason (paywalled, current-state-only, no public API at all) — same standard as Entry 21/24. **Do not guess a value or interpolate a current ratio backward in time.**

Part B — Build a real, persistent TVL panel (tokens)

`check_tvl()` in `phase2c_defillama_metadata.py` already calls `api.llama.fi/protocol/{slug}` and receives the full `tvl` array (`[{date: unix_ts, totalLiquidityUSD: float}, ...]`) in the response — it currently discards everything except presence flag, date range, and the last value. Extend this (or write a new script alongside it — your call, but reuse the working fetch/cache/retry logic, don't rebuild it) to persist the **full series**, at monthly grain (last observation per calendar month, matching `universe_panel.csv`'s convention), for **every token with a confirmed `dl_slug` match** in `asset_onchain_identity.csv` — not just the 104/111-token NaN-PQ worklist subset used by the Phase 2c diagnostic. That's 127 tokens as of this session; re-derive live.

Output: `03_data/phase2/tvl_panel.csv` with columns `cmc_id`, `symbol`, `dl_slug`, `month_end`, `ym`, `tvl_usd`. Cache raw per-token JSON under `03_data/raw/phase2/tvl/` (same caching discipline as every prior phase). Use `time.sleep(0.2)` between calls per the existing script's pattern.

Optional, time-permitting, secondary to the core 127-token build: the 127-token ceiling comes from `phase1_build_identity_map.py`'s `cmcId`-keyed join against DeFiLlama's `/protocols` list. Check whether any of the 321 currently-unmatched tokens can be picked up via a looser match (e.g. name/symbol similarity against DeFiLlama's protocol list) — but **verify each candidate match individually before adding it** (same standard as every identity-mapping decision in this log; a wrong slug match would silently corrupt the panel). If this turns out to be slow or low-yield, stop and note it in the report rather than grinding on it — it's a stretch goal, not the deliverable.

Re-affirm, in the report, that this panel is a valuation-multiple denominator (NV/TVL), not a PQ substitute — Entry 30's stock-vs-flow distinction is unchanged by this build.

What NOT to do

- No PQ-for-coins work. No Channel 2 (holding duration) work.
- No paid tier of anything (DeFiLlama Pro, Etherscan archive, Boardroom, Tally, beaconcha.in).
- No symbol-only joins — always `cmc_id`.
- No silently proxying an excluded escrow/staking relationship (the veBAL/SNX standard applies to every new candidate in A.1 and A.3 equally).
- No guessing a staking ratio, a Snapshot space, or a slug match — verify live or flag the gap.
- No editing `phase1_assemble_lambda.py`'s z-scoring/standardizability/weighting logic.
- No financial/purchase action of any kind.

Deliverable

Write `03_data/PHASE1_LAMBDA_SCALE_AND_TVL_PANEL_REPORT.md` containing: - Before/after lambda coverage: by channel, by asset class (coin/token), asset-month counts and distinct-asset counts, same format as the existing `PHASE1_COVERAGE_REPORT.md`. - Every new Channel-1 candidate checked in A.1, with a verified/rejected verdict and why. - The Channel-3 widening result from A.2 (new spaces added, on-chain `VoteCast` attempts and outcomes for the Entry-25 gap list). - The live PoS-coin source audit from A.3: which chains had a free historical staking series, which were paywalled/current-state-only/absent, and the resulting coin-side Channel 1 coverage. - TVL panel coverage: token count, date range, any tokens where the fetch failed or returned an empty series, and the result of the optional slug-widening attempt if you did it. - A clear closing note on what's still NOT covered and why, so the next session (coin PQ) starts from an accurate picture.

Then: - Log this session as `06_documentation/ai_conversations/session_019_<date>_<topic>.md` (verify 019 is actually next per Step 0.3). - Continue `04_code/DATA_DECISIONS_LOG.md` from the live-verified next entry number (per Step 0.3 — should be 41, confirm). - Add a row to `06_documentation/time_log.md`. - Commit and push to `main` on github.com/moazzamkhoja/AFA_2027_QTM_Crypto — but only after re-running `git status` per Step 0 and confirming the diff looks sane (the real files you intentionally changed, nothing more).

Stop after the report. Do not start coin PQ sourcing, and do not touch Channel 2, before this is reviewed.

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CLAUDE_CODE_PHASE2C_DIAGNOSTIC_PROMPT.md

Claude Code Kickoff Prompt — Phase 2c Diagnostic: TVL/Fees/Volume Metadata Audit (111 NaN Tokens)

Paste everything below the line into a new Claude Code session with working directory `C:/AFA_2027_QTM_Crypto`.

You're working in the AFA 2027 QTM Crypto research repo (github.com/moazzamkhoja/AFA_2027_QTM_Crypto).

Required reading (do this first, in full)

1. `04_code/DATA_DECISIONS_LOG.md` — Entry 30 (TVL is capital stock, not PQ — the original objection to treating TVL as a flow), Entry 31 (Etherscan token-transfer rejection — measuring the wrong quantity; verify every number before trusting it), Entry 32 (the token-side PQ source waterfall, including the **already-decided fee-inversion rule**: “back out volume from fee only when the fee is a confidently known single, stable rate (not multi-tier/variable/non-volume-linked), else flag PQ missing”), Entry 36 and Entry 37 (Dune pilot + full-panel dry-run: 4 of the original 17 token-side NaN tokens resolved via Dune — AAVE, STRK, LDO, GNS; the other 13 are absent from Dune's spellbook and money can't fix that), and Entry 38 (this session's framing — read this one closely, it sets the rules below).

2. 03_data/DUNE_DRYRUN_REPORT.md ?2 for the full DISTINCT project lists — **25 lending projects** and **28 perpetual.trades projects** — this is your turnover-calibration cohort in Step 3 below.
3. 03_data/phase2/pq_tokens.csv — **re-derive live, do not trust a stale count.** Columns: cmc_id, symbol, dl_slug, dl_category, month_end, pq_usd, pq_source, note. A token counts as “covered” if it has at least one non-NaN pq_usd month. As of the last live check: 127 total tokens, 16 covered, 111 NaN, breaking into categories: Yield 16, Derivatives 10, Farm 8, Gaming 7, Dexs 7, Services 7, Bridge 6, Lending 6, Launchpad 6, Chain 6, Yield Aggregator 4, Canonical Bridge 3, Cross Chain Bridge 2, Developer Tools 3, Token Locker 2, plus 17 singleton categories (1 token each). **Re-verify these counts yourself before proceeding** — this file may have changed since last checked.

Funding / cost context

DeFiLlama’s relevant data (TVL, Fees, Revenue, dex/derivatives volume, yields/APY) is on the **free, keyless api.llama.fi / yields.llama.fi / bridges.llama.fi** tier per their public docs (docs.llama.fi/api) — no subscription should be needed for anything in this session. DeFiLlama also sells a **Pro API at ~\$300/mo** (pro-api.llama.fi) for higher rate limits / extra endpoints. **Do not sign up for, key into, or use the Pro API under any circumstance this session — flag it as a gap and stop, don’t work around it by paying.** Moazzam handles any purchase decision himself, after reviewing evidence, same as the standing rule for Dune.

The conceptual framework — read before doing anything, it defines what this session is actually checking

This is a metadata/feasibility audit, not a PQ build. For each of the 111 NaN tokens, the question is “*what data exists, and given this protocol’s actual economic model, could any of it plausibly become a PQ (transacted-value) measure?*” — not “compute the number.” **No pq_tokens.csv writes this session.**

Background the spec didn’t previously need: PQ is a *flow* (transacted value per period); TVL is a *stock* (locked capital). Entry 30 already established you cannot treat them as the same thing. The question this session explores is whether a **calibrated conversion** from the stock to a flow proxy is defensible — the same structure as the no-labor AK growth model, $Y = A \cdot K$, where A (“turnover”) is flow ? stock, identical to the corporate-finance asset-turnover ratio. If $PQ \sim A_{\text{protocol}} \times TVL_{\text{protocol}}$ and A can be estimated with reasonable confidence, that is a usable proxy; if A is wildly dispersed across comparable protocols, it isn’t, and that finding should be reported plainly rather than forced.

Rank conversion paths in this order of preference (most direct and least assumption-laden first):

(1) Direct volume — no conversion needed at all. pq_tokens.csv was originally populated by checking DeFiLlama’s summary volume for each protocol’s primary category, but DeFiLlama has *separate* verticals for different activity types that may not have been checked: **dexs** (DEX swap volume), **derivatives** (perps/options notional), **bridges** (cross-chain transfer volume). A token whose category is Dexs, Derivatives, Bridge, Canonical Bridge, or Cross Chain Bridge may already have a directly-reported volume series sitting in one of these verticals — check this FIRST, before reaching for any proxy. If found, the token may not need a proxy at all (flag it as **direct_volume_available**, separate from the proxy-feasibility verdicts below).

(2) Fee inversion — Entry 32’s existing rule, now extended from coins to the rest of the token set. If DeFiLlama’s **summary/fees/{slug}** has a Fees series but no direct volume exists, and the protocol’s fee structure is a single, fixed, **confidently citable** rate (e.g., a stated 0.3% swap fee documented in the protocol’s own docs) — not variable, tiered, or usage-dependent — then $Volume \sim Fees \cdot fee_rate$. Do not guess a fee rate from a generic “typical AMM fee” assumption; either find the protocol’s actual documented rate or flag **fee_rate_not_confidently_known** and stop there for that token.

(3) TVL x APY — for Farm / Yield / Yield Aggregator tokens specifically (28 tokens, the single largest bucket). For protocols where the entire “product” is a published yield on locked capital, realized APY is a directly observed, protocol-specific rate (check **yields.llama.fi/pools**, matched by protocol/slug) — a far better-grounded A than a cross-sectional turnover average, since it’s specific to *that* protocol rather than borrowed from others. $PQ_{\text{farm}} \sim TVL \times APY$.

(4) TVL x calibrated turnover — for Lending, Derivatives, Liquid Staking, and any other category where the protocol’s activity genuinely flows through locked capital (NOT for Gaming, Launchpad, Chain, Developer Tools, Services, Token Locker — these have no comparable capital->flow channel; don’t force this model where the economic model doesn’t support it, leave NaN instead). This path is only as credible as the calibration in Step 3 below — do not propose a turnover rate for any token without first checking whether the comparable-cohort dispersion (Step 3) is tight enough to trust.

Gaming (7 tokens) is explicitly excluded from this entire audit. Skip them. They stay NaN by design — this was a deliberate decision (Entry 38), not an oversight to fix.

What to do

Step 1 — Re-derive the worklist. From `pq_tokens.csv`, pull `symbol + dl_slug + dl_category` for every currently-NaN token. Drop the 7 Gaming tokens (out of scope per above) — you should have 104 tokens left to actually check. Group them by category for the steps below. **Watch for the NVT symbol collision:** one of the NaN Dexs-category tokens has ticker NVT (Nerve Network) — this is unrelated to the NVT *metric* used throughout this paper. Always key by `dl_slug`, never by bare ticker symbol, when there’s any ambiguity.

Step 2 — Per-token DeFiLlama metadata check (all 104). Before calling anything, confirm the *current* exact endpoint shapes against `docs.llama.fi/api` — don’t assume the paths below haven’t shifted; verify, the same discipline as every prior session in this log. Starting points to verify: - **TVL:** `https://api.llama.fi/protocol/{slug}` — `tv1` time series present? Date range? - **Fees / Revenue:** `https://api.llama.fi/summary/fees/{slug}?dataType=dailyFees` (and `dailyRevenue`) — present? Date range? - **Direct volume**, where category-appropriate: the `dexs` vertical for Dexs-category tokens, the `derivatives` vertical for Derivatives-category tokens, `bridges.llama.fi` for the Bridge/Canonical Bridge/Cross Chain Bridge tokens (11 total). - **APY/yields**, for Farm/Yield/Yield Aggregator tokens (28 total): `yields.llama.fi/pools`, matched by `protocol/slug` (confirm the current matching key — it may be by pool ID, not slug directly). Use only the free `api.llama.fi` / `yields.llama.fi` / `bridges.llama.fi` surface. Record, per token: TVL Y/N + date range, Fees Y/N + date range, Revenue Y/N + date range, direct Volume Y/N + date range (and which vertical), APY Y/N + date range.

Step 3 — Build the turnover-calibration cohort and test its dispersion. Reuse the validated query pattern from `dune_dryrun_fullpanel.py`, but pull the **full DISTINCT project** lists this time — all 25 lending projects and all 28 perpetual.trades projects (not just `aave/strike/gains_network`), full history, GROUP BY project, month. For each project, find its matching DeFiLlama slug — **Dune’s project string and DeFiLlama’s slug will not always match exactly; verify each mapping individually, do not assume a string match is correct** — and pull that protocol’s TVL history. Compute `turnover = PQ / TVL` per protocol-month. Report the distribution **separately for lending vs. perpetual**: median, interquartile range, min/max, and how many projects you could and couldn’t confidently slug-match. This dispersion is the actual evidence for whether category-level turnover imputation (path 4 above) is defensible. **If it spans more than roughly an order of magnitude, say so plainly** — a wide, unusable dispersion is a legitimate and useful finding, not a failure to fix.

Step 4 — Per-token feasibility verdict (the actual deliverable). For each of the 104 tokens, report: what data exists (from Step 2), which conversion path (if any) looks viable given that protocol’s specific economic model, and a one-line rationale referencing the actual data found — not a blanket category assumption. **Do not compute or insert an actual PQ value for any token this session.** This is a feasibility map; an actual Phase 2c build is a separate, later, explicitly-authorized session.

What NOT to do

- Do not write to `03_data/phase2/pq_tokens.csv` (or any panel CSV) — no PQ values, no `pq_source` changes, nothing. This is diagnostic-only, same standard as the Dune pilot and dry-run sessions.
- Do not sign up for, key into, or use DeFiLlama’s Pro API — free tier only.

- Do not guess: not a fee rate, not a Dune-project-to-DeFiLlama-slug mapping, not a turnover figure. Flag low-confidence items and move on rather than forcing a number.
- Do not build the actual TVLxturnover (or TVLxAPY) imputation pipeline yet — that’s a follow-on session, gated on this report’s findings and human review, exactly like Entry 36 gated Entry 37.
- Do not touch the 7 Gaming-category tokens.
- Do not purchase anything or take any financial action.

Deliverable

Write `03_data/PHASE2C_DIAGNOSTIC_REPORT.md` containing: the per-token feasibility table (104 rows, grouped by category, per Step 4), the turnover-dispersion analysis from Step 3 (lending and perpetual reported separately), and a clear closing recommendation — which categories/tokens look promising for a real Phase 2c build and via which path, which look hopeless given the economic model, and whether the TVLxturnover idea is statistically defensible at all given the observed dispersion (or only defensible for some categories and not others).

Then: - Log this session as `06_documentation/ai_conversations/session_017_<date>_phase2c_diagnostic.md` (follow the existing session-log format/structure in that directory). - Continue `04_code/DATA_DECISIONS_LOG.md` from **Entry 39** (Entry 38 is the last existing entry). - Add a row to `06_documentation/time_log.md`. - Commit and push to main on github.com/moazzamkhoja/AFA_2027_QTM_Crypto.

Stop after the report. Do not start an actual Phase 2c panel build, and do not write any PQ values, before this diagnostic is reviewed.

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CLAUDE_CODE_PHASE2_KICKOFF_PROMPT.md

Claude Code Kickoff Prompt — Phase 1 Finishing Touches + Phase 2 (NVT_GL)

Paste the prompt below as the first message in a new Claude Code session opened with working directory `C:/AFA_2027_QTM_Crypto` (the repo root). This session has two parts: close out Phase 1’s open items, then move straight into Phase 2 in the same session.

You're working in the AFA 2027 QTM Crypto research repo. Phase 0, Phase 0B, and Phase 1 (lambda channels) are complete. Before doing anything else, read in full:

1. `03_data/PHASE1_COVERAGE_REPORT.md` --- what Phase 1 built and its open items.
2. `04_code/DATA_DECISIONS_LOG.md` Entries 21-28 --- the lambda-channel judgment calls and why, including Entry 28 (the scoping decisions behind this exact prompt).
3. `04_code/DATA_SPECIFICATION.md` Section 4 (NVT_GL) and Section 7 (phasing) --- for Part B.

This session has two parts. Do Part A first, then continue straight into Part B --- these are finishing touches plus the next phase, not a from-scratch rebuild of either.

PART A --- Phase 1 finishing touches

1. ****Finish the ETH staking series.**** ``phase1_channel1_eth_staking.py`` is resumable via monthly checkpoints under ``03_data/raw/phase1_onchain/`` --- re-run it to completion through 2026-05. Keep the cumulative-deposit method as-is (do NOT add post-Shapella withdrawal netting --- that needs a consensus-layer/beacon-chain data source, which is new scope, not a finishing touch). Keep documenting it as a monotone upper-envelope conviction proxy, not a net-stake figure, exactly as Entry 23 already does.

2. ****Channel 2 (coin-age)** --- explore a free workaround before concluding it's infeasible.** Entry 24 found no free aggregator/API serving HODL-wave data across the panel. Before leaving it there, specifically check (verify live, don't assume): whether any Bitcoin-chain explorer (mempool.space, blockstream.info, blockchair) exposes a free UTXO-age or "coins last moved" metric usable as a coin-age proxy for BTC --- BTC is the single highest-value pre-2020 coin to unlock. If something usable exists for BTC, check whether the same or a comparable explorer covers other major pre-2020 coins (LTC, XRP, DOGE, etc.) before building anything. If nothing free and credible turns up, leave Channel 2 as a documented gap (per the spec's "flag, don't guess" principle) and say so plainly in the updated coverage report --- do not procure a paid source in this session; that decision is deferred.
3. ****Confirm N=250.**** Per the coverage report's own ?7 point 5: lambda density is governance-token-driven, and tightening the universe wouldn't fix the coin-side gap (it's a source problem, not a universe-size problem). No action needed --- just state this confirmation explicitly in the updated coverage report so it's on record.
4. ****Revisit the 16 gray-zone classifications**** (OP, MNT, MANTA, IMX, RPL, ANKR, SSV, STRD, EWT, GBYTE, FCT, BLUR, LOOKS, ME, PNK, PTS --- `classification\_table.csv`, Entry 18) now that real Channel-1 lock data exists. Check whether any of these now show a clean security-staking lock (vs. governance-only/no lock) and reclassify with justification per the existing `classification\_basis` convention if so. If the evidence is unchanged, leave them as `other` and say why in one line each.
5. ****Fix the internal inconsistency in `PHASE1\_COVERAGE\_REPORT.md`**** --- ?5's headline ("1,326 asset-months, 52 distinct assets") doesn't match ?8 and Decisions Log Entry 27 (1,308 asset-months, 51 assets). Reconcile every section of the report to whatever the final, post-ETH-resume numbers actually are.
6. ****Optional, lower priority:**** if time allows, enrich the identity map from CMC `detail.platforms[]` (Entry 22's noted next step) to widen Channel 1's EVM token coverage beyond the current curated 6. Not required before moving to Part B.

Log all of this in `DATA\_DECISIONS\_LOG.md` (next entry is 29 --- Entry 28 is already used by the Cowork-side scoping decision behind this prompt) and re-run `phase1\_assemble\_lambda.py` if any of the above changes the underlying channel data.

PART B --- Phase 2: Growth-Levelized NVT (NVT_GL)

Per spec Section 4: $NVT_GL = MC / PQ^*$, where PQ^* levelizes projected transaction throughput into a present-value annuity (PMT-style) using a CAGR growth rate g , a CAPM-style discount rate r_e , a terminal growth rate g_{inf} , and a horizon n .

Build, per asset-month, on the frozen Phase 0 universe (join on `cmc\_id`, `status='observed'` rows only):

1. ****MC**** --- market cap, same source as Phase 0 (already in `universe\_panel.csv`).
2. ****PQ**** --- nominal economic throughput.
 - Coins: on-chain transaction volume. VERIFY live free access of a transaction-volume source per chain before building (continue the project's preference for canonical chain data over aggregators where feasible --- e.g. transaction volume directly from an explorer's logs --- but DeFiLlama fees/revenue data is an acceptable source for protocol-level throughput per the spec; document whichever you use per asset). Flag

- and prefer an "adjusted" volume series netting out exchange-internal/self-churn transfers if available; otherwise use raw volume and flag it (a known NVT weak point).
- Governance tokens: protocol throughput (DEX volume, fees, or active users --- DeFiLlama fees/revenue is fine here). Use one consistent definition where possible; document where it had to vary across tokens.
- 3. ****g**** --- trailing 3-year CAGR of PQ; flag and document a shorter window for assets with less history.
- 4. ****r_e**** --- CAPM-style beta against a crypto market index (cap-weighted index or BTC as a simple alternative) plus a risk-free rate. Keep this discount-rate use of "CAPM" distinct in naming/code from the portfolio-level CAPM control in Phase 3 (spec ?5) --- don't conflate the two.
- 5. ****g_inf**** --- a single macro-anchored constant in the 2-4% range, not estimated per asset; flag explicitly as a robustness parameter for later sensitivity testing.
- 6. ****n**** --- projection horizon, default 10 years; also a robustness parameter, not estimated.

Output: NVT_GL per asset-month, plus the underlying g, r_e, and PQ series (so the g_inf/n/k assumptions can be varied later without rebuilding). Verify every new source's live free access before building on it, exactly as Phase 1 did, and log dead/paywalled sources plus the substitute you used.

Produce a Phase 2 coverage report (coverage by year/asset class, where PQ had to fall back to raw vs. adjusted volume, any source landmines hit) before stopping. Log this session as `06\_documentation/ai\_conversations/session\_010\_\*.md`, update `DATA\_DECISIONS\_LOG.md` (continuing from wherever Part A left off) and `06\_documentation/time\_log.md`. Commit and push to github.com/moazzamkhoja/AFA_2027_QTM_Crypto (main) at session end. Do not start Phase 3 without review.

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CLAUDE_CODE_PHASE2_PQ_BUILD_PROMPT.md

Claude Code Kickoff Prompt — Phase 2 Build: PQ (tokens + coins), resuming Part B

Paste the prompt below as the first message in a new Claude Code session opened with working directory C:/AFA_2027_QTM_Crypto (the repo root). This resumes Part B of CLAUDE_CODE_PHASE2_KICKOFF_PROMPT.md (session 010), which paused at the PQ-definition question. That question is now fully resolved (Entries 30-32) — this session builds on it.

You're working in the AFA 2027 QTM Crypto research repo. Phase 0/0B/1 are complete. Phase 2 (NVT_GL) Part A (token PQ theory/pilot) is complete; this session resumes Part B (the actual build) of CLAUDE_CODE_PHASE2_KICKOFF_PROMPT.md, now that PQ is fully resolved. Before doing anything else, read in full:

1. 04_code/DATA_DECISIONS_LOG.md Entries 30, 31, 32 --- the full PQ resolution: fees rejected on theoretical grounds (toll, not transacted value); tokens resolved to DeFiLlama sector-appropriate reported volume (governance-token Transfer-log volume piloted and rejected --- wrong object, e.g. UNI 46.6x off DeFiLlama's Uniswap volume, AAVE corrupted by sentinel-value transfers); coins resolved to a per-asset source ladder (Entry 32), not one global source.
2. 03_data/PHASE2_PQ_PILOT_REPORT.md --- the token pilot's full empirical findings (session 012).

3. 06_documentation/PHASE2_PQ_DECISION_STATUS.md --- original framing, now superseded by the entries above; read for context only, do not follow its ?3-4 recommendations.
4. CLAUDE_CODE_PHASE2_KICKOFF_PROMPT.md Part B --- the still-valid spec for the rest of NVT_GL (MC, g, r_e, g_inf, n) that this session resumes. Only step 2 (PQ) in that Part B is replaced --- by the logic below --- everything else there still applies as written.
5. 04_code/DATA_SPECIFICATION.md Section 4 (NVT_GL) and `03\_data/classification\_table.csv` (the `sector` and asset-class fields you'll route on).

PART A --- Tokens: build directly (fully resolved, no further verification needed)

PQ = DeFiLlama sector-routed reported volume, per Entry 31/32:

- DEX/swap volume for AMMs, perps notional for derivatives, borrow/origination flow for lending --- routed by the `sector` field (Entry 16). Verify the live DeFiLlama endpoint per protocol before pulling (confirm slugs, don't assume).
- ****Fee-to-volume backout (Entry 32), only when DeFiLlama has no volume series for a protocol-month:**** check whether that protocol's fee is a confidently known, SINGLE, STABLE rate over the window (e.g. a documented flat swap fee) --- if so, `notional = fee / rate`, use as PQ, and log the rate and its source inline per protocol. Do NOT do this when the rate is multi-tier (e.g. Uniswap V3's several fee-tier pools --- the blended rate is itself unknown without volume, circular), governance-adjustable/variable across the window, or not a simple function of notional (e.g. a lending reserve factor, which is a % of interest, not of loan volume). If neither a DeFiLlama series nor a confident rate exists for a protocol-month, ****flag PQ as missing (NaN)**** --- do not substitute the raw fee directly as PQ.
- TVL and fees stay as side diagnostic columns (+ Volume/TVL turnover), per Entry 30/31. Raw Etherscan Transfer logs are an occasional spot-check only, never primary.

PART B --- Coins: verify first, then build, same session

****Step B1 --- bounded live verification (do this before writing any coin extraction code):****

1. ****Artemis access.**** Check artemisanalytics.com/products/api and the free "Lite" tier live --- actually attempt the documented free-tier flow, don't infer from marketing copy. Determine: (a) is Settlement Volume exposed as a standalone series (not just bundled inside Total Economic Activity), (b) historical depth, (c) how many of the panel's coins/chains it actually covers. Phase 0 (Entry 2) found Artemis's API dead; a relaunch was found by web search but never live-verified --- this is that verification.
2. ****DeFiLlama chain-dexs coverage per coin.**** For every coin in the panel, check `/overview/dexs/{chain}` live. Classify each as ****non-degenerate**** (material DeFi/DEX activity relative to the chain's real economic footprint --- safe as Rung 1) vs. ****degenerate**** (negligible, bolted-on activity unrelated to the chain's real throughput --- already confirmed for Bitcoin: \$419,825/24h, \$18.9M/30d, driven by 3 niche protocols, nowhere near BTC's actual settlement). Propose and state an explicit materiality threshold for this split (e.g. relative to market cap or a known tx-count baseline) rather than eyeballing case by case.
3. ****BTC-specific fallback.**** Verify blockchain.com's Charts API "Estimated Transaction Value (USD)" series live (endpoint, history depth, update cadence, confirm it excludes change outputs as documented).
4. ****Produce a per-coin rung-assignment table**** (Rung 1/2/3/4, per Entry 32's ladder) before any Part-B extraction code. ****If Artemis turns out paid-only, or coverage is ambiguous for any coin that's material to the panel (large market cap or long history), STOP and report rather than guessing**** --- this is the one part of this session that should pause for human review if it isn't clean. If the picture is clean (Artemis usable as scoped, or the DeFiLlama/BTC-fallback combination covers the panel adequately), proceed straight into B2 without waiting for a review round-trip --- the ladder logic itself is already decided.

****Step B2 --- build, applying Entry 32's ladder per coin:****

- ****Rung 1**** --- DeFiLlama chain DEX volume, where non-degenerate.
- ****Rung 2**** --- Artemis Settlement Volume, where Rung 1 is degenerate/unavailable and Artemis is confirmed accessible at adequate coverage/depth from B1. Use Settlement Volume ONLY --- do not adopt the full Total Economic Activity composite (it bundles in Chain Fees & MEV and Application Fees, which are toll measures --- the same flaw already rejected for token fees).
- ****Rung 3**** --- coin-specific native fallback: blockchain.com's series for BTC; for other chains with no equivalent ready-made series, bounded native `value` block iteration over a ****recent window only**** (the archive-access objection was wrong per Entry 32, but full multi-year history per chain hits the same call-volume wall the token pilot found --- just block-count-driven instead of log-count-driven; do not attempt it).
- ****Rung 4**** --- DeFiLlama chain fees, last resort only, for any coin where 1-3 are all unavailable. Explicitly flag these rows in the output (a column noting "fallback: toll-based proxy, not transacted value") --- do not let this blend in silently with Rungs 1-3.
- Log which rung every coin actually used, with the reason, in the coverage report.

What NOT to do

Don't attempt full multi-year native block iteration across the whole coin panel. Don't adopt Artemis's full Total Economic Activity composite. Don't silently fall to Rung 4 without flagging it in the data. Don't resolve an ambiguous Artemis/coverage finding by guessing --- stop and report per B1 instead.

Deliverable

Build the PQ construction for both tokens (Part A) and coins (Part B), then continue CLAUDE_CODE_PHASE2_KICKOFF_PROMPT.md Part B's remaining steps (MC join, g, r_e, g_inf, n) to produce NVT_GL per asset-month. Produce a Phase 2 coverage report: per-asset PQ source/rung, where the token fee-backout rule fired and what rate was used, coin rung assignments and why, and any coverage gaps (flagged NaN, not guessed). Log this session as `06\_documentation/ai\_conversations/session\_013\_\*.md`, continue `DATA\_DECISIONS\_LOG.md` from Entry 33, and update `06\_documentation/time\_log.md`. Commit and push to github.com/moazzamkhoja/AFA_2027_QTM_Crypto (main) at session end. Do not start Phase 3 without review.

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CLAUDE_CODE_PHASE2_PQ_PILOT_PROMPT.md

Claude Code Pilot Prompt — PQ Definition: Etherscan Transfer-Volume Feasibility Pilot

Paste the prompt below as the first message in a new Claude Code session opened with working directory C:/AFA_2027_QTM_Crypto (the repo root). This is a bounded **diagnostic pilot**, not a full Phase 2 build — it answers one empirical question before Decision 1 and Decision 2 in 06_documentation/PHASE2_PQ_DECISION_STATUS.md can be finalized.

You're working in the AFA 2027 QTM Crypto research repo. Phase 0/0B/1 are complete; Phase 2 (NVT_GL) is paused on a PQ-definition question. Before doing anything else, read in full:

1. 06_documentation/PHASE2_PQ_DECISION_STATUS.md --- the original Decision 1/2 framing

- (recommended fees for both coins and tokens; now superseded, see #2).
2. 04_code/DATA_DECISIONS_LOG.md Entry 30 --- the Cowork-side theoretical resolution: fees is REJECTED as the PQ proxy. PQ = P?Q = nominal value of goods/services transacted (a flow) --- fees are the cost/toll of facilitating that transaction, not the transaction's value itself, which is structurally the same error as treating a government's tax revenue as a proxy for GDP. TVL is also rejected as PQ (it's the capital stock that enables activity --- an AMM's pooled inventory, a lending pool's loanable funds, a staking protocol's AUM --- not the activity itself). The corrected target is **transacted value**: swap/DEX volume for AMMs, loan-origination/borrow flow for lending, notional volume for derivatives --- sector-appropriate via the existing `sector` field in `03\_data/classification\_table.csv` (Entry 16). TVL and fees are demoted to side-column diagnostics (capital-stock control; cost-of-intermediation), not PQ itself.
 3. 04_code/phase1_channel1_evm_locks.py --- the existing, working Etherscan `getLogs` Transfer-log pattern (recursive block-range bisection on the API's 1,000-log-per-call cap). This pilot reuses that exact mechanism with one change: NO counterparty-address filter (you want ALL transfers of the token, not just transfers to/from one escrow).

Why this pilot exists

The open question is **scale**, not mechanism. Channel 1's escrow approach is fast because filtering for one specific counterparty address is sparse --- most blocks have zero matching logs. Pulling every Transfer event for an actively-traded token removes that filter, so the same 1,000-log cap will be hit far more often, forcing much finer block-range bisection. Entry 24 found full-supply Transfer enumeration "orders of magnitude beyond" the free rate-limited key for a **different** purpose (HODL/coin-age, full multi-year history) --- but that finding has never been re-tested for **this** purpose (a recent-window swap-volume proxy) or quantified with real numbers. The Cowork session that produced this prompt has no network route to api.etherscan.io (confirmed directly, sandbox egress is allowlisted), so it could not get those numbers itself. Get real numbers instead of extrapolating.

What to do

Pick **two** representative tokens already in the universe with active trading and DeFiLlama coverage: **UNI** (Uniswap governance token) and **AAVE** (cmc_id 7278, already in the Channel-1 escrow table). Verify both are in `universe\_panel.csv` with `status='observed'` before proceeding.

For each token, over a recent, bounded **30-day window** (most recent complete month):

1. **Pull** all Transfer events, no counterparty filter. Same `getLogs` mechanism as `phase1\_channel1\_evm\_locks.py` (topic0 = Transfer signature, address = token contract, fromBlock/toBlock = the 30-day block range), but drop the topic1/topic2 address filter. Reuse the existing recursive bisection on the 1,000-log cap.
2. **Record, precisely:** total API calls made, total wall-clock time, total raw Transfer event count, max bisection depth reached. If the call count or time balloons to a point where extrapolating to full multi-year history x the relevant token universe is clearly infeasible, you can stop early for that token and report the partial numbers --- don't burn the whole rate-limit budget proving a point that's already clear, but say explicitly what "infeasible" means numerically (e.g. "X calls/token/month implies Y calls for the full panel, which would take Z hours even at the rate limit").
3. **Dollarize.** Sum raw transfer amounts (scaled by token decimals) by day, multiply by that day's USD price (already in `universe\_panel.csv` from the Phase 0 CMC backbone) to get a daily dollar-transfer-volume series for the 30-day window.
4. **Cross-check against DeFiLlama.** Pull DeFiLlama's reported DEX/swap volume for the same

protocol over the same 30-day window (verify the correct live endpoint, don't assume it). Compare the two series: ratio, correlation, and whether raw-Transfer-log volume is a multiple of DeFiLlama's reported volume (expected, since routing hops, wrapping, and exchange-internal transfers inflate raw Transfer counts --- quantify the multiplier, don't just note that it exists).

5. ****Extrapolate, with real numbers.**** From the measured 30-day call count, estimate calls needed for (a) that token's full history (since launch/DeFiLlama-coverage start) and (b) scaling to however many tokens in the universe would need this approach (DEX/governance tokens with no better source). State plainly whether full-panel raw-Transfer-volume PQ is realistically buildable on the free rate-limited key, or should be ruled out in favor of DeFiLlama's own reported volume/fees series (with raw-Transfer-log spot-checks as a periodic sanity check rather than the primary source).

What NOT to do

This is a diagnostic pilot, not Phase 2. Do ****not**** build `phase2\_pq.py`, `phase2\_nvt\_gl.py`, or attempt full-history/full-panel extraction. Do not commit any large raw-log dumps --- cache only what's needed to support the numbers in your report, same caching discipline as Phase 1.

Deliverable

Write `03\_data/PHASE2\_PQ\_PILOT\_REPORT.md`: the measured call counts/timing for both tokens, the dollarized 30-day volume series, the DeFiLlama cross-check and noise-multiplier estimate, the full-panel extrapolation, and a clear recommendation --- (A) raw Etherscan Transfer-log volume is buildable at panel scale, (B) it is not, use DeFiLlama's reported volume/fees as the working PQ source with TVL as a side column, or (C) something in between (e.g. buildable for a small curated flagship subset only). Log this session as `06\_documentation/ai\_conversations/session\_012\_\*.md` (011 is the Cowork session that produced this prompt), continue `DATA\_DECISIONS\_LOG.md` from Entry 31, and update `06\_documentation/time\_log.md`. Commit and push to github.com/moazzamkhoja/AFA_2027_QTM_Crypto (main) at session end. Stop after the report --- do not resume the Phase 2 build until this is reviewed.

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CLAUDE_CODE_PHASE3B_KICKOFF_PROMPT.md

Claude Code Kickoff Prompt — Phase 3b: Confirmatory sorts, coarse sectors, horse race, heterogeneity

Paste the prompt below as the first message in a new Claude Code session, working directory C:/AFA_2027_QTM_Crypto. Phase 3 core (session 043) is complete; results reviewed with Moazzam (Entries 95–97). This session runs spec ?8: the confirmatory conviction-only sorts, the coarse sector remap, the horse race, and the pre-specified heterogeneity batch.

You're working in the AFA 2027 QTM Crypto research repo. Phase 3 core is complete (session 043; regression_panel.csv, ltw_factors_monthly.csv, portfolio_returns.csv and tables all built). Before doing anything else, read in full:

1. 04_code/PHASE3_ANALYSIS_SPECIFICATION.md --- including ?8 (Phase 3b), which is this session's spec. Where ?8 and earlier sections conflict, ?8 wins.
2. 04_code/DATA_DECISIONS_LOG.md Entries 93--97 --- especially Entry 96 (MVRV dropped, Metcalfe restricted) and Entry 97 (the EXPLORATORY token quintile finding this

session must treat confirmatorily: sharper sorts monotonically strengthen the token conviction signal; quintile EW alpha +1.71%/mo t=2.18, post-2023 +1.48%/mo t=1.79).

3. 03_data/PHASE3_RESULTS_REPORT.md --- session 043 results and caveats.

Standing rules (unchanged)

- NO paid services; NO new Etherscan getLogs (Pro lapsed). Free keyless web fetches (BitInfoCharts pattern) are allowed.
- Joins on cmc_id only. Phase 0--3 outputs are read-only; new outputs to 03_data/phase3/, builders 04_code/phase3b_*.py.
- Log every decision (next entry: 98). Honest-results clause applies: run everything in the spec, report everything, tune nothing.

Task A --- Coarse sector remap (spec ?8.1) [prerequisite]

Deterministic keyword mapping of raw DeFiLlama category strings to {DEX, Lending, Yield, Derivatives, Staking/LSD, Other}; output 03_data/phase3/sector_coarse_map.csv; report group sizes per month. Then re-run the token regression ladder (043 specs s6_1--s6_4) with coarse-sector FE and report side-by-side with the raw-sector FE versions.

Task B --- Confirmatory conviction-only token sorts (spec ?8.2) [the headline task]

Quintile EW primary; decile, tercile, VW secondary; min 3/leg; NW-3 alphas vs LTW factors; sub-periods; 25/50bps haircuts; turnover. Sector-neutralized quintile (demean conv within coarse sector-month where group n>=3, else within class-month). Per-sector tercile long-shorts for the 2--3 largest coarse groups (single sort --- this is Moazzam's by-category power test; no valuation dimension). Coin tercile analog, descriptive. CRITICAL: spanning regressions of the quintile SMA on LTW factors + identically-built long-shorts on r_1m (reversal), mom_3m, 52-wk-high, size --- the token reversal effect is strong (FM t ~ ?3.4) and the conviction quintile must be shown distinct from it, or the result dies here and that is the finding.

Task C --- Horse race (spec ?8.3 / ?5)

Panel race per track: each comparator signal singly then jointly (raw NVT, raw NV/TVL, S2F, 52-wk-high, MA cross, momentum family --- all derivable from existing panels). Spanning tests both directions including the quintile portfolio. Metcalfe: ETH + PoW baselines descriptive appendix only. MVRV: skip (Entry 96).

Task D --- Heterogeneity batch (spec ?8.4) [run ALL, report ALL]

(1) ?lambda 1m/3m levels+changes regressions and ?lambda quintile sort; (2) vote-escrow vs plain governance classification of the 101 tokens (from protocol documentation; log each with source) and split regressions; (3) fee-share vs no-fee split; (4) size and turnover tercile splits; (5) bull/bear (CMKT sign) and pre/post-2023 regime splits; (6) measurement robustness: raw NV/TVL, g-cap-excluded, B4-excluded, conv_source-excluded, MRP 20/40 re-derivations.

Task E --- Mechanism discrimination M1--M4 (spec ?8.6) [feeds paper Section 2.3 + Robustness]

The paper now states four mechanisms for the coin/token H2 asymmetry with discriminating predictions. Run the ?8.6 table exactly: M1 conv x turnover interaction for tokens (turnover = volume_24h/MC); M2 token interaction with valuation demeaned within coarse sector-month; M3 raw NV/TVL + g-cap-excluded interaction (reuse Task D(6)); M4 coin spec s4 plus staking-yield level and conv x staking-yield interaction (yield approximated from issuance; log the construction as an Entry). Report the M1--M4 verdict table explicitly in the results report --- this section is what the paper's mechanism narrative depends on, and a negative verdict on M4 (coin result is a b_t artifact) must be reported with the same prominence as a positive one.

Deliverables

03_data/PHASE3B_RESULTS_REPORT.md (every table + one-paragraph reading; negative results at equal prominence; the paper will disclose Entry 97's exploratory origin --- do not launder it into a pre-registered result). Decisions log entries; session log (session_044...); time_log row; commit and push at session end.

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CLAUDE_CODE_PHASE3C_KICKOFF_PROMPT.md

Claude Code Kickoff Prompt — Phase 3c: Fee/Revenue DCF Comparators (TOKENS ONLY)

Paste the prompt below as the first message in a new Claude Code session, working directory C:/AFA_2027_QTM_Crypto. Phase 3 core (session 043) and 3b (session 044) are complete. This session builds the DCF-style comparators the paper's introduction names as its foil — price-to-fees and a revenue-DCF ratio — and runs them through the horse race and the H2 interaction. TOKENS ONLY (Moazzam's directive: fee-DCF is not conceptually valid for coins, where fees are the validator toll rejected as a fundamental in Entry 30).

You're working in the AFA 2027 QTM Crypto research repo. Phases 0--3b complete. Before doing anything else, read in full:

1. 04_code/PHASE3_ANALYSIS_SPECIFICATION.md ?8.7 --- this session's spec (as amended: tokens only, coin chain-fee build dropped).
2. 04_code/DATA_DECISIONS_LOG.md Entries 106--109 --- current paper state, the DCF misattribution fix, size-as-delta results (Entry 108: the token H2 null has now survived five conditioning variables; THIS session provides the sixth and last measurement candidate).
3. 03_data/PHASE3B_RESULTS_REPORT.md ?3 --- the horse race machinery these comparators slot into.

Standing rules (unchanged)

- NO paid services; NO new Etherscan getLogs. DeFiLlama endpoints are free/keyless --- verify each live before building; log dead/changed endpoints.
- Joins on cmc_id only. Fees/revenue matched via the existing dl_slug identity map (03_data/ identity files used by phase2_build_tv1_panel.py) --- NEVER by symbol.
- New outputs to 03_data/phase3/; builders 04_code/phase3c_*.py; log every decision (next entry: 110). Honest-results clause applies.

Task A --- Fee and revenue panel build

For each of the 101 regression-sample tokens with a dl_slug: pull daily protocol fees AND daily revenue from the DeFiLlama fees API (check live: api.llama.fi /summary/fees/{slug}?dataType=dailyFees and dataType=dailyRevenue; also the /overview/fees listing to resolve slugs). Sum to monthly. Distinguish and keep BOTH series: fees = what users pay the protocol; revenue = the share accruing to the protocol/treasury/holders. Output 03_data/phase3/fees_revenue_panel.csv (cmc_id, month_end, fees_usd, revenue_usd, dl_fees_slug, source_notes). Report coverage explicitly (tokens covered, months per token, start dates); fee histories mostly begin 2021+ and many protocols report fees but not revenue --- that asymmetry is a finding, report it.

Task B --- The two comparator measures (tokens only)

1. P/F (practitioner multiple): $pf = MC / \text{trailing-12m fees } (>=6 \text{ monthly obs})$.
2. Revenue DCF ratio: $prev_gl = MC / REV^*$, where REV^* applies the EXACT PQ^*/TVL^* machinery (PARAMS of phase2_nvt_gl.py: rf 4%, MRP 30%, g_inf 3%, n 10, g cap [-50%,+200%], beta36, re_floor) to a base of trailing-12m revenue with $g = \text{trailing 3y CAGR (2y/1y fallback)}$. Where revenue is unreported but fees are, compute a fees-based variant $pf_gl = MC / F^*$ and flag; report both but keep $prev_gl$ primary (DCF discounts what holders could claim, not what users pay).

Both ln-transformed, winsorized 1/99, standardized within token-month.

Task C --- Tests (the order matters; C1 is Moazzam's core question)

- C1. ****H2 with a fee-anchored delta proxy****: token spec s4 (conv + controls + val + convxval, month FE, two-way clustered) with val = ln P/F and separately val = ln $prev_gl$. This is the sixth and cleanest measurement test of the token H2 null: fees/revenue are earned in external assets and are NOT mechanically linked to the token's own price the way TVL is. Also run the split-sample version and the coverage-matched baseline (NV/TVL_GL interaction re-estimated on the SAME subsample) so any revival is attributable to the denominator, not the sample.
- C2. Horse race singles + joint: add ln P/F and ln $prev_gl$ columns to the token panel race; report whether conviction's slope changes on the fee-covered subsample.
- C3. Portfolios: quintile long-shorts on P/F and $prev_gl$ (cheap minus expensive, EW, min 3/leg) vs LTW factors; spanning both directions against the conviction quintile (does the DCF comparator span conviction, or vice versa).
- C4. Sub-periods (post-2023) for anything significant.

Task D --- Technical battery completion (both tracks; all from universe_panel, no new data)

Add five signals to the horse-race panel and portfolio machinery (spec ?8.8):

1. $ma_dist = price/MA10 ? 1$ (continuous; REPLACES the binary MA-cross in spanning --- the binary version's long-shorts had only 28--31 overlapping months in session 044; re-run the two affected spanning cells with ma_dist quintile long-shorts and note the supersession).
2. $vol12 = \text{trailing 12m monthly-return SD } (>=8 \text{ obs})$.
3. $ivol = \text{residual SD from trailing 36m regression of asset return on CMKT } (>=12 \text{ obs})$.
4. $amihud = \text{trailing 12m mean of } |r_month| / (\text{month-end volume}_{24h})$, ln-transformed. FLAG in output and report: $volume_{24h}$ is a month-end snapshot, not a monthly aggregate --- noisy proxy, log as a caveat (Entry).
5. $skew36 = \text{trailing 36m monthly return skewness } (>=18 \text{ obs})$.

All standardized within class-month. Run: (a) horse-race singles + joint columns per track; (b) quintile long-shorts per signal; (c) ADD $vol12$, $ivol$, $amihud$, $skew36$, ma_dist long-shorts to the conviction-quintile spanning battery (the key question: does the token conviction quintile survive the completed battery); (d) one paper sentence justifying exclusions: monthly RSI/MACD/Bollinger are transformations of the momentum/MA-distance/volatility set; daily-native signals (MAX, true RSI) infeasible without daily price histories (free-tier depth limits) --- note as limitation.

Deliverables

03_data/PHASE3C_RESULTS_REPORT.md --- lead with the C1 verdict (H2-in-tokens under fee-anchored valuation: revived or still dead), then the completed-battery spanning verdict for the conviction quintile (Task D), coverage table, all tables with one-paragraph readings, negative results at equal prominence. Decisions log entries; session log (session_045...); time_log row; commit and push at session end.

/newpage

CLAUDE_CODE_PHASE3_KICKOFF_PROMPT.md

Claude Code Kickoff Prompt — Phase 3: Empirical Tests (H1, H2, H3 core)

Paste the prompt below as the first message in a new Claude Code session, working directory `C:/AFA_2027_QTM_Crypto`. Phase 3 turns the completed data panels into the paper's empirical results. This session covers the **core tests** (regression panel, LTW factors, H1/H2 regressions, H3 portfolio). The horse race (Test 3) is Phase 3b — this session only runs its two data probes at the end if quota/time allows.

You're working in the AFA 2027 QTM Crypto research repo. Phases 0--2 are complete: lambda panel (467 assets), NVT_GL panel (coins), NV/TVL_GL panel (tokens) are built and committed. This session runs the core empirical tests. Before doing anything else, read in full:

1. `04_code/PHASE3_ANALYSIS_SPECIFICATION.md` --- the complete spec for this session. Follow it; where it is silent, decide conservatively and log the decision.
2. `04_code/DATA_DECISIONS_LOG.md` Entries 92--94 --- current state, Etherscan lapse, NV/TVL_GL adoption, Phase 3 design decisions.
3. `05_paper/main.tex` Sections 2--4 --- the hypotheses (H1a, H1b, H2, H3) exactly as stated, and the sample definitions the results must match (24 coins/718 mo; 101 tokens/2,771 mo).

Standing rules

- NO paid services. Etherscan Pro lapsed 2026-07-30 --- no new `getLogs` calls at all. Stored data under `03_data/raw/` may be read freely.
- All joins on `cmc_id`, never symbol.
- Do not modify any Phase 0--2 output file. New outputs go to `03_data/phase3/` and builders to `04_code/phase3_*.py`.
- Returns come from `universe_panel` observed rows only (never `carry_forward` rows).
- Log every methodological decision in `DATA_DECISIONS_LOG.md` (next entry: 95).

Task A --- Regression panel (spec ?1--2)

Build `03_data/phase3/regression_panel.csv`: one row per asset-month in the coin (24/718) and token (101/2,771) samples with forward returns r_{t+1} (and cumulative $t+3$, $t+6$), conviction (coins: $\ln$ -odds of `raw_ch1_staking` with `lambda_z` fallback + `conv_source` flag; tokens: `lambda_z`), valuation ($\ln$ NVT_GL / $\ln$ NV_TVL_GL, winsorized 1/99), controls ($\ln$ MC, r_{1m} , mom_{3m} , mom_{12_2} , $beta_{36}$), sector, class-month standardized versions. Verify row counts reproduce the paper's Table 1 funnel numbers exactly before proceeding; if they do not, STOP and reconcile first --- do not run regressions on a panel that disagrees with the paper.

Task B --- Monthly LTW factors (spec ?4.1)

Build `03_data/phase3/ltw_factors_monthly.csv` (CMKT, CSMB, CMOM as monthly analogs of Liu--Tsyvinski--Wu 2022 JF, constructed from our 1,939-asset universe; MC $\geq$ \$1M filter, winsorized constituent returns). Report factor summary stats and correlation with BTC.

Task C --- H1/H2 regressions (spec ?3)

Run the specification ladder per track (pooled OLS, month FE, two-way clustered SEs; tokens add sector FE and the voting-weighted-lambda column; FM secondary for tokens only). Output coefficient tables to `03_data/phase3/tables/` as CSV. Key cells: sign/significance of conviction (H1a, H1b) and of the conviction x valuation interaction (H2, expected negative), plus the below/above-median-valuation split.

Task D --- H3 portfolio (spec ?4.2--4.3)

Median-split quadrants within class-month (≥ 4 assets per leg guard), Stars-minus-Avoid EW

and VW, monthly rebalanced. Time-series alpha vs the Task-B factors (Newey-West 3 lags), Sharpe vs EW benchmark, sub-periods pre/post 2023-01, turnover and 25/50 bps cost haircuts, and the conviction-only / valuation-only single-dimension comparators.

```
## Task E (only if time/quota remain) --- Phase 3b probes (spec ?5.3)
E1: BitInfoCharts active-addresses scrape feasibility for the 24 sample coins (one test
coin end-to-end, then stop and report scale estimate). E2: ch2 checkpoint realized-cap
probe --- determine from 03_data/raw/phase1_onchain/ checkpoint schema whether last-move
attribution is recoverable without new API calls; answer YES/NO with evidence, build
nothing yet. Log both as Entry 95 (or 95--96).
```

```
## Deliverables and record-keeping (house style)
- 03_data/PHASE3_RESULTS_REPORT.md --- every table with a one-paragraph reading; report
  negative and anomalous results with the same prominence as positive ones. Do NOT tune
  specifications to produce significance; the spec's ladder is the pre-registration.
- DATA_DECISIONS_LOG.md entries; session log in 06_documentation/ai_conversations/
  (session_043_...); time_log.md row.
- Commit and push at session end (delete permission for the folder is enabled; git works
  normally; commit_and_push.bat exists if needed).
```

```
## Honest-results clause
```

The paper's hypotheses are falsifiable and the results section will report whatever comes out. If conviction does not predict returns, or the interaction has the wrong sign, that is a finding --- document it cleanly, do not iterate on variable definitions until something works. Any deviation from the spec must be logged with its rationale BEFORE results are produced, not after.

/newpage

CLAUDE_CODE_SESSION024_LAMBDA_FREE_BUILD_PROMPT.md

Claude Code Kickoff Prompt — Session 024: BUILD lambda Channels 1/2/3 from FREE sources (all chains)

Paste the prompt below as the first message of a new Claude Code session, working directory C:/AFA_2027_QTM_Crypto. This is the **build** continuation of session 022's full-universe identification map (06_documentation/SESSION022_STATUS_AND_NEXT_SESSION.md, Decisions Log Entry 53), which read verified contract code and confirmed mechanisms live via `getLogs`. Session 022 produced an identification map but built nothing. This session turns the **free-source-confirmed** rows into actual lambda series.

This is PHASE 1 of a two-phase build: PHASE 1 = every FREE source on every chain identified as free (not just Etherscan). PHASE 2 (a later session) = paid services. Do not sign up for or use any paid tier this session — Phase 2 is explicitly out of scope here.

Dependency — DONE: session 023 (CLAUDE_CODE_HEX_AKRO_RECONCILIATION_AND_SURVIVORSHIP_PROMPT.md) has already run (commit b7ecb73, Entries 54/55/56). Its verdicts are **final: HEX = BUILD (already shipped — this is NOT a rebuild target)** — 03_data/phase1/channel1_hex_stake.csv via 04_code/phase1_channel1_hex_stake.py, already in the assembler glob; **AKRO = REJECT** (its Locked() is an owner-only admin pause switch, not a stake — do NOT attempt it). Read 03_data/SESSION023_HEX_AKRO_RECONCILIATION.md + Entries 54/55/56 before building Channel 1 so you don't re-derive or duplicate HEX. **Current lambda baseline going into this session: 2,130 observed asset-months / 68 distinct assets** — measure your deltas from there.

You're working in the AFA 2027 QTM Crypto research repo. This session BUILDS lambda series from free on

sources across ALL chains identified as free in session 022 --- Etherscan-free EVM chains, EVM-but-not-Etherscan chains via their own free explorers, and non-EVM free indexers. Read everything in "Required reading" before building anything.

Required reading (full text, not summaries)

1. 04_code/DATA_DECISIONS_LOG.md --- Entry 21 (logs not eth_call; the no-archive rule), Entry 23 (ETH beacon-deposit getLogs event-replay --- the Channel-1 reconstruction template), Entry 24 (Channel 2 unbuilt = "highest-value addition"), Entry 25 (Channel 3 = Snapshot + token-weight guard), Entry 26 (single-escrow custody standard + cross-check-to-balanceOf at ~0.00% drift), Entry 49 (the circulating_supply staked-token double-count artifact), Entry 53 (session 022 feasibility), and **Entries 54 (HEX BUILD --- a THIRD Channel-1 template: non-custodial burn-and-track, reconstructed StakeStart(+)/StakeEnd(?) decoded-event amounts, no escrow; see `phase1\_channel1\_hex\_stake.py`), 55 (AKRO REJECT --- the "an event named Locked() is not necessarily a lock" lesson), 56 (the 284 dead-1 survivorship-bias note)** from session 023. **Next free log entry number = 57.**
2. 03_data/ETHERSCAN_LAMBDA_CHANNEL_EMPIRICAL.md and 03_data/NON_EVM_LAMBDA_CHANNEL_ASSESSMENT.md --- the per-channel findings and the measured free/paid chain gate.
3. The three maps, joined on cmc_id, are your build worklist:
 - 03_data/phase1/universe_lambda_channel_map.csv (1,306 rows; ch1/ch2/ch3 verdict + event signature)
 - 03_data/phase1/non_evm_lambda_recoverability.csv (405 rows; chain, repository, per-channel status)
 - 04_code/_universe_lambda_findings.csv (per-event topic0, log counts, first block)
4. The existing build scripts you will EXTEND (do not rebuild): phase1_channel1_eth_staking.py, phase1_channel1_evm_locks*.py, phase1_channel3_voting.py, phase1_assemble_lambda.py. New Channel-1 batches land as a new 03_data/phase1/channel1_*.csv (the assembler globs channel1_*.csv); Channel 3 appends to channel3_voting.csv. There is NO channel2_*.csv yet --- you are creating that pipeline.
5. 03_data/phase1/snapshot_space_map.csv and channel3_voting.csv --- the EXISTING Channel-3 panel, need for the dedup in Task A.

FREE-source inventory (from session 022 --- what "free" means per chain)

- Etherscan V2 free key (04_code/.api_keys.json): getsourcecode on ALL chains; getLogs/tokentx FREE only on Ethereum(1)/Polygon(137)/Arbitrum(42161)/Blast(81457). BSC/Base/Avalanche getLogs are PAID -> PHASE1
- EVM-but-not-Etherscan (22 tokens: KAIA/Klaytn, HyperEVM, Manta, X Layer, EOS-EVM, IOTA-EVM, Velas, Viction, Flare, PulseChain, smartBCH, Chiliz, ...): free via each chain's own Blockscout/native JSON-RPC eth_getLogs --- same contract-read+event-replay method, different endpoint.
- Non-EVM free indexers (live-verified session 022): Solana JSON-RPC (api.mainnet-beta.solana.com) + Solscan; Tron TronGrid/TronScan; Cosmos LCD (lcd.osmosis.zone etc. /cosmos/staking, /cosmos/gov); Cardano Koios. Keyless where session 022 confirmed it.

Build tasks, in priority order (highest value / lowest effort first)

Task A --- Channel 3: dedup the 34 active ERC20Votes tokens vs the Snapshot panel, then build net-new. The 34 getLogs-CONFIRMED active-voting tokens (universe_lambda_channel_map.csv, ch3=='ERC20Votes-ACTIVE COMP, MET, DDX, FORTH, CVP, RAD, DMG, FLUID, ETHDYDX, BIT, EUL, T, ONDO, BONE, CYBER, PEAK, W, EIGEN, W BLAST, UNI, ENS, SUSHI, GTC, KP3R, BTRST, RGT, HFT, STRK, MNT, BLUR, RAIN, TOMI, UXLINK).

1. Join on cmc_id against snapshot_space_map.csv + channel3_voting.csv. Split into (i) already-have-a-Snapshot-Channel-3-series (cross-check only) and (ii) NET-NEW (no Snapshot turnout in the panel).
2. For the NET-NEW set, build a monthly Channel-3 series from on-chain `DelegateVotesChanged` event reported delegated voting weight outstanding at each month-end / circulating supply (the same token-weighted turnout definition as Entry 25, but sourced on-chain). All are on free chains (ETH + Blast).
 - Token-weight validity guard (Entry 25): null spaces where weight isn't token-denominated; flag.
 - Decide & document: is on-chain delegated-weight the same construct as Snapshot voter-turnout, or a distinct sub-channel? State it; do not silently merge two different measures into channel3_voting.
3. Append to channel3_voting.csv (or a clearly-named channel3_onchain_*.csv if the construct differs), re-run phase1_assemble_lambda.py, report the lambda asset-month / asset-count delta.

Task B --- Channel 1: build the free-chain confirmed locks

Confirmed genuine, amount-bearing (build via getLogs event-replay -> monthly locked / total-or-circulating supply, cross-checked to a live contract balanceOf/global read at ~0.00% drift, Entry 26): **NMR, stkAA, XAN.** (**HEX is already BUILT in session 023 --- do NOT rebuild it; AKRO is REJECT --- do NOT attempt)

- **Before reconstructing ANY token, read its verified source and confirm the event is a genuine holder lock/stake --- not an admin/pause/vesting flag.** Session 023's AKRO finding is the cautionary case: `Locked()` was an owner-only `Lockable.lock()` pause switch that fired once with no amount, and session 022's name-matching classifier wrongly flagged it "GENUINE." **Apply this directly to VSL below** --- `Locked()` may be the same false positive; verify the mechanism before building.
- VSL: bare `Locked()` (no amount in event) -> ONLY if the source confirms it's a real lock, reconstruct from the lock contract's balanceOf at each month-end block, not event amounts; flag as the weaker construction. If `Locked()` is an admin/pause flag (the AKRO pattern), REJECT it with the contract reason.
- Apply the Entry-49 denominator check per token: does CMC circulating_supply already exclude the staked amount (ratio can exceed 1)? State per token; keep un-capped + flagged, use z-scored rank not level. (HEX is the worked example --- its contract NatSpec exposed `allocatedSupply = totalSupply + locked`, clean `locked/(locked+circ)` fraction was written alongside `locked/circ`; do the same where a token's contract exposes a staked-inclusive supply.)
- For high-event-count tokens, Dune's **free** decoded-event tables (`<project>\_<chain>.<CONTRACT>\_evt\_` small engine) are an acceptable getLogs substitute and were used for HEX (Entry 54) and the Bucket-builds (Entries 48/51) --- same "logs not eth_call" rule, far fewer calls than windowed Etherscan getLogs.
- Land as a new 03_data/phase1/channel1_freebuild.csv (picked up by the assembler glob). Extend phase1_channel1_evm_locks*.py, don't rewrite. Re-run the assembler; report the delta.

Task C --- Channel 2: holding-duration, the Entry-24 gap --- PROTOTYPE then scale on free chains
This is the single highest-value addition (Entry 24) and the most-recoverable channel (793 free EVM + ~100 non-EVM). Build it carefully:

1. Prototype on ONE Ethereum token's FULL Transfer-log history (free tier): build a FIFO per-address coin-age engine (acquisition-lot tracking), then compute the Channel-2 metric at each month-end --- share of supply unmoved > a threshold window (state the window; HODL-wave style). Validate on a token with known behavior.
2. MEASURE the call budget: full Transfer history windowed by block vs the free 100k-calls/day cap. Report the per-token call count and extrapolate to the free-chain EVM population BEFORE scaling --- if it blows the free cap, that is itself the Phase-2 (paid) trigger; document it, don't silently stall.
3. Address-class hygiene: exclude exchange/contract/LP wallets from "holder conviction" where identifiers (Etherscan labels are incomplete --- document the filter as a judgment call per Entry-24/spec ?0).
4. Scale across free-chain EVM tokens as budget allows; new pipeline phase1_channel2_holding.py -> 03_data/phase1/channel2_holding.csv; wire it into phase1_assemble_lambda.py (Channel 2 currently always NaN --- adding it changes the channel-availability counts; report them).

Task D --- EVM-but-not-Etherscan (22 tokens): same method, free chain explorers

Run the session-022 contract-read+getLogs method against each chain's free Blockscout/native RPC eth_getLogs for the 22 tokens (non_evm_lambda_recoverability.csv class='EVM-non-Etherscan'). Recover any genuine Ch1 lock / active Ch3 voting the same way; fold into the channel1_*/channel3_* outputs.

Task E --- Non-EVM free indexers: scoped, low-expected-yield, time-boxed

For the 92 non-EVM-indexed tokens (Solana 59 dominant), Channel 2 is the realistic target via free indexers (Solana getProgramAccounts token-account ages; Tron/Cosmos/Cardano transfer history). Channel 1/3 require per-project program reads (Anchor/Realms/CosmWasm) and are low-yield --- time-box this; build only what free indexer cleanly supports, flag the rest for Phase 2. Do NOT chase the 284 no-identity tokens (dead session 023 documents them as survivorship bias).

Method rules (unchanged from the project standard)

- Reconstruct numerators from real events/balances; NEVER guess or interpolate. Flag gaps (spec ?0).
- Channel 1 must clear the Entry-26 bar: a specific contract/event reconstructs a monthly locked-quantity.

series cross-checked to a live on-chain balance (~0.00% drift) --- not just "a mechanism exists."

- Join everything on `cmc_id`, never symbol (the Entry-20/39 collision landmine).
- Logs not historical `eth_call` (Entry 21). Respect each asset's `staking_start`; pre-start months = NaN n
- NO paid tier, no purchase, no signup --- Etherscan Pro and paid indexers are PHASE 2. Flag, don't act
- `DATA_DECISIONS_LOG.md` is append-only --- continue at the next free entry number; never edit prior ent
- Standardize/assemble exactly as `phase1_assemble_lambda.py` already does (z-score per channel per month equal-weight available channels, record channel count) --- extend, don't change its logic.
- KNOWN GOTCHA (session 023): ``phase1_assemble_lambda.py`` writes ``lambda_panel.csv`` correctly, then can throw a cosmetic ``UnicodeEncodeError`` on a ``->`` in its final ``print`` under the Windows cp1252 console CSV is already written when it throws. Run it with ``PYTHONUTF8=1`` (or read the file to confirm the cor rather than assuming the assembly failed. A one-char fix to that print is fine if you touch it.

CONTEXT-WINDOW DISCIPLINE (required)

Track your context usage as you work. ****When you reach ~2/3 (about 66%) of the context window**, STOP** starting new build work and instead, before you run out:

1. Write a session report of what was built vs. left (counts, lambda delta, which tasks A--E are done/p
 2. Write a transition document capturing exact state (which CSVs written, which scripts extended, what v mid-flight, any cross-check still pending).
 3. Write the NEXT-session prompt to continue THIS build from where you stopped (same two-phase framing) as a new numbered session.
 4. Commit and push everything before the window closes --- a clean checkpoint beats a lost in-flight bu
- If you finish all free-source tasks before 2/3, proceed to write the PHASE 2 (paid) kickoff prompt as th next session (see below), then log + commit.

PHASE 2 preview (for the transition / next prompt, NOT this session)

Phase 2 builds from paid services: Etherscan Pro for getLogs on BSC/Base/Avalanche (the 16 paid-gated candidates: ALT, AWE, BAKE, BNX, CHEEL, EDG, ESPORTS, FORM, LINEA, MCT, MDX, OP, PONKE, TKO, ZORA, TNC) panel-scale Channel-2 throughput if the free 100k/day cap binds (Task C2), and paid non-EVM indexers for the Channel-1/3 gaps Task E flagged. Phase 2 requires explicit human purchase approval --- do not pre-e

Close-out (every session)

Update `06_documentation/time_log.md` and log this session as
`06_documentation/ai_conversations/session_024_*.md`. Commit and push to
github.com/moazzamkhoja/AFA_2027_QTM_Crypto (main) at session end (or at the 2/3 checkpoint).

Stop when either (a) all free-source tasks A--E are built/flagged and the Phase-2 prompt is written, or (b) you hit the 2/3 context checkpoint and have written the report + transition + next-session prompt and committed. Either way, end with the lambda panel re-assembled and the asset-month/asset-count delta rep

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CLAUDE_CODE_SESSION025_ETHERSCAN_PRO_BUILD_PROMPT.md

Claude Code Kickoff Prompt — Session 025: Etherscan Pro Channel-2 + P2-2 Build

Paste the prompt below as the first message in a new Claude Code session opened with working directory `C:/AFA_2027_QTM_Crypto`. Moazzam has purchased Etherscan API Pro (Standard Plan, \$199/month, 200k calls/day, activated 2026-06-30). The Pro key is already in `.api_keys.json` under "etherscan". This session is the direct payoff of that purchase: build Channel 2 (holding duration) at panel scale, and run the 16 BSC/Base/Avax paid-gated tokens (P2-2) through the session-022 getLogs method.

You're working in the AFA 2027 QTM Crypto research repo. Moazzam has purchased Etherscan

API Pro (Standard Plan: 200k calls/day, 10 calls/sec, all chains including BSC/Base/Avax). The Pro key is in ``.api_keys.json`` under "etherscan". Before anything else:

1. Verify the key works and is Pro-grade: call ``eth_blockNumber`` on Ethereum AND a `getLogs` probe on BSC (chain 56) --- if BSC returns data rather than "Free API access is not supported for this chain", the Pro upgrade is confirmed. Log the result explicitly; do not assume the key is upgraded without verifying.

Read before starting work:

- `04_code/DATA_DECISIONS_LOG.md` Entries 24, 57--60 (Channel-2 design decisions, session-024 budget measurement, assembled lambda state)
- `06_documentation/SESSION024_STATUS_AND_NEXT_SESSION.md` (Phase-2 worklist, gotchas)
- `03_data/SESSION024_FREE_BUILD_REPORT.md` (what was built, what was measured)
- `04_code/phase1_channel2_holding.py` and `phase1_channel2_budget_probe.py` (the engine and budget measurement already built in session 024 --- extend, don't rewrite)
- `03_data/phase1/universe_lambda_channel_map.csv` (1,306-row identification map from session 022 --- use to drive the P2-2 token list)
- `04_code/.api_keys.json` (confirm "etherscan" key before any API call)

Task A --- P2-2: the 16 BSC/Base/Avax paid-gated Channel-1/3 tokens

These tokens were identified in session 022 as having ERC20Votes or staking contracts on BSC/Base/Avalanche, but `getLogs` on those chains required a paid plan. Now that the Pro key covers those chains, run them through the same session-022 pipeline:

Tokens: ALT, AWE, BAKE, BNX, CHEEL, EDG, ESPORTS, FORM, LINEA, MCT, MDX, OP, PONKE, TKO, ZORA (Channel-3 candidates), TNC (Channel-1 candidate).

For each:

- a. Pull its row from `universe_lambda_channel_map.csv` to get chain + contract address.
- b. Read the verified contract source (`getsourcecode` --- already cached in `03_data/raw/etherscan_src/` if session 022 got it; re-fetch only if missing).
- c. Confirm the mechanism via `getLogs` (same method as session 022/024). For Channel-3: confirm `DelegateVotesChanged` actually fires and is non-trivial. For TNC Channel-1: apply the full Entry-26 test (single escrow, amount-bearing, cross-check to live balance).
- d. Build any that pass (same scripts as session 024: `phase1_channel3_onchain_delegation.py` for delegation, `phase1_channel1_freebuild.py` for any Channel-1 pass). Skip/log those that are DORMANT or fail the mechanism test --- do not force builds.
- e. Log verdicts and any builds in `DATA_DECISIONS_LOG` (continue from Entry 61).

Task B --- Channel 2 panel-scale build (the main event)

The engine (`phase1_channel2_holding.py`) was proven on MET in session 024. The budget probe showed RAD (mid-size) needed 700+ calls and didn't complete on the free 100k/day cap. With 200k/day Pro, a mid-size token completes in one day; the full 793-token panel runs in ~3 days. Do this as a resumable, checkpointed build --- do NOT try to do it in one session.

****Step B1 --- scope the build:****

From `universe_lambda_channel_map.csv`, extract the 793 free-chain EVM tokens (those with chain in Ethereum/Polygon/Arbitrum/Blast AND `reachable=True`). Cross-check against the current `lambda_panel.csv` to identify which of those 793 already have ANY lambda channel (exclude them from Channel-2 build only if they already contribute on Ch1 or Ch3 --- Channel 2 can ADD to an existing asset's lambda). The correct logic: build Channel 2 for ALL 793 where Transfer log history exists, regardless of existing channels --- it widens the channel count.

****Step B2 --- run the build, respecting the budget:****

- Use `phase1_channel2_holding.py` with the Pro key. The session-024 gotchas apply: checkpoint per token; delete poisoned checkpoints (`getblocknobytime->None` on last month); retry transient `getLogs` errors.
- Stay within 200k calls/day. Track consumed calls and checkpoint after each token so the build is resumable across sessions. Do NOT burn through the monthly quota in one run.
- Contract-holder screening (`eth_getCode`): already implemented in session 024 --- use it. CEX EOA wallets are NOT separately screened (no paid label feed); document this as a named limitation ("CEX custodial wallets not screened; contract addresses excluded via `eth_getCode`") in the output and flag column. Do not attempt CEX screening with any free heuristic that could introduce worse bias than the known limitation.
- HODL metric: share of supply held by EOA addresses unmoved for ≥ 6 months at each month-end, as a fraction of circulating supply. Compute at monthly granularity (month-end block, same as the rest of `lambda`).

****Step B3 --- address-class hygiene check (spot validation):****

Before wiring into `lambda`, run the MET validation again (it had a v2 migration muddying the 12m window per session-024 note --- pick a cleaner single-deployment token to validate against; CRV or AAVE are good candidates since we already have their Channel-1/3 series). Confirm the contract-screen step still collapses the contamination to a sensible range. If the validated HODL series looks economically reasonable (not 95%+ or 0.1% for a normal governance token), proceed to wire into `lambda`. If it looks degenerate, STOP and report --- do not silently fold bad data into the panel.

****Step B4 --- wire into `lambda` assembler:****

Extend `phase1_assemble_lambda.py` to accept channel ``ch2_holding`` as an input (same z-score/equal-weight logic, do not change assembler structure --- add an input, don't rewrite). Re-run the assembler and report the new asset-month count, `n_channels` distribution, and the share of asset-months that now have 2 or 3 channels (the cross-section depth improvement is the headline metric for the paper's data section).

Rules (unchanged)

- `cmc_id` joins only, never symbol.
- Entry-26 bar for any Channel-1 build: single contract, amount-bearing, cross-checked to live balanceOf at 0.0x% drift.
- `DATA_DECISIONS_LOG.md` append-only. Continue from Entry 61. One entry per major decision cluster (not one per token). Log the Etherscan Pro purchase itself as an entry (date, plan, what it unlocks, cost) --- it is a data-sourcing decision.
- No additional paid subscriptions. Do not sign up for Nansen, Glassnode, CoinMetrics, Alchemy, or any other service.
- Track `getLogs` call budget actively. If the build is consuming calls faster than expected, checkpoint and STOP rather than blowing the monthly quota. Report consumed vs remaining.
- Run `PYTHONUTF8=1` (the `->` print crash fix from session 024).
- Update `06_documentation/time_log.md`. Log session as `06_documentation/ai_conversations/session_025_*.md`. Commit and push at end of session.

Deliverables

1. P2-2 verdict table: all 16 tokens with chain, contract, verdict (BUILD/REJECT/DORMANT), reason. Any builds folded into `lambda`.
2. `03_data/phase1/channel2_holding.csv`: per-asset-month HODL-6m series for all tokens where the build completed this session (may be partial --- document what completed).
3. Updated `lambda_panel.csv` with `ch2_holding` wired in for completed tokens. New asset-month count and `n_channels` distribution reported.

4. 03_data/SESSION025_PRO_BUILD_REPORT.md: parallel structure to SESSION024 report --- what was built, what was deferred (tokens not yet completed in the channel-2 batch), budget accounting (calls consumed vs remaining monthly quota), any anomalies.
5. DATA_DECISIONS_LOG.md Entries 61+, time_log.md updated, session log written.
6. Commit and push.

STOP at end of session or when the daily call budget approaches its limit (whichever comes first). If Channel 2 is not complete in one session, checkpoint and report how many tokens were completed --- the next session continues from the checkpoint. Do not start any other Phase-2 items (NMR, KAIA, non-EVM) without review.

/newpage

CLAUDE_CODE_SESSION026_CHANNEL2_TAIL_AND_OP_PROMPT.md

Claude Code Kickoff Prompt — Session 026: Channel-2 Tail + OP Delegation

Paste the prompt below as the first message in a new Claude Code session opened with working directory C:/AFA_2027_QTM_Crypto.

Context: Session 025 built Channel-2 for 197 tokens ($\leq 3,000$ holders) and P2-2 delegation for 5 tokens. lambda is now 6,097 asset-months / 288 assets. The single biggest remaining lambda gain is adding ch2 to the 72 tokens that already have ch1 or ch3 — especially the 4 with BOTH ch1 and ch3 (CVX, SUSHI, FRAX, GMX), which become **3-channel assets** the moment ch2 lands. All of these were deferred by the session-025 HOLDER_MAX=3,000 metadata filter. OP delegation (46,974+ events, killed at >80 min) is the one unfinished P2-2 item.

You're working in the AFA 2027 QTM Crypto research repo. lambda is currently 6,097 asset-months / 288 assets (session 025, Entry 64). The Etherscan Pro key (200k calls/day) is in ``.api_keys.json`` under "etherscan". Two tasks this session, in order.

Required reading before starting

- 04_code/DATA_DECISIONS_LOG.md Entries 59, 62--64 (Channel-2 design, P2-2 results, session 025 close-out)
- 03_data/SESSION025_PRO_BUILD_REPORT.md (full session 025 account --- read entirely)
- 04_code/phase1_channel2_panel.py (the panel builder --- extend, don't rewrite)
- 04_code/phase1_channel3_onchain_delegation.py (the delegation builder)
- 03_data/phase1/_channel2_sizes.csv (793 tokens sized; the deferred tail starts at holder_count > 3,000)
- 03_data/phase1/lambda_panel.csv (current lambda panel --- use to identify priority tokens)

Priority token list (derive this before starting work)

From ``.lambda_panel.csv``, identify tokens with ch1 OR ch3 but NO ch2. Cross-reference against ``.channel2_sizes.csv``. This is the priority ordering for Task B:

****Tier 1 --- 3-channel targets (already have BOTH ch1 AND ch3):****

CVX (31k holders), SUSHI (127k), FRAX (18k), GMX (299k)

Adding ch2 to these creates the first 3-channel observations in the panel.

****Tier 2 --- high-value single upgrades (large governance/staking tokens):****

DDX (3k), ORBS (9k), RPL (12k), LQTY (14k), XAN (13k), API3 (24k), CAKE (21k),
AAVE (200k), CRV (100k), ENS (68k), LDO (65k), ARB (64k), UNI (385k),
COMP (220k), GRT (173k), BAL (49k), YFI (52k), PENDLE (73k), EIGEN (223k),
BLAST (273k), ENA (98k), WLF1 (100k), GNS (21k), MNT (30k), CVX (31k), BAL (49k)

****Explicit skip --- too large for any practical build:****

HEX (cmc_id 5015): 9,054,606 holders / ~724k getLogs calls. Do not attempt. Log as permanently deferred in DATA_DECISIONS_LOG (the HEX Channel-2 gap is an acknowledged limitation --- its staking is already captured in ch1, so the absence of ch2 is a minor gap).

Task A --- OP delegation (block-windowed incremental fetch)

OP (cmc_id 11840, Optimism chain 10, ERC20Votes) was verified firing in session 025 (46,974+ `DelegateVotesChanged` events confirmed) but the full history download was killed at >80 min due to volume. The fix is a block-windowed fetch.

- a. Add a `BLOCK\_WINDOW` parameter to `phase1\_channel3\_onchain\_delegation.py` (or a separate OP-specific incremental script). Default 50,000 blocks per window; tune down if needed.
- b. Fetch DelegateVotesChanged in windows from block 0 to current, stitching results. Store the raw event list in a checkpoint so the full replay doesn't re-fetch on rerun.
- c. From the complete event list, replay the running-balance logic (same as all other delegation builds): track each delegate's latest `newBalance` via DelegateVotesChanged, sum at each month-end block = delegated weight outstanding, ? circulating supply.
- d. The `DELEG\_CAP` guard added in session 025 prevents a future repeat of this. For OP's backfill specifically, the guard must be lifted or set very high (OP is a verified build --- just slow).
- e. Fold the resulting series into `channel3\_onchain\_delegation.csv` (same schema) and re-run `phase1\_assemble\_lambda.py`. OP already has ch3_voting (Snapshot); this is the ch3_delegation sub-channel, which is governance-channel-waterfall: OP stays on Snapshot turnout as its primary (role=cross_check for the delegation series). Do NOT add OP's delegation as a second governance input to lambda --- the waterfall rule from Entry 57 applies.
- f. Log in DATA_DECISIONS_LOG (Entry 65): OP delegation completed, method, event count, ratio.

Task B --- Channel-2 tail (>3,000 holders, priority-ordered)

The engine is already built (`phase1\_channel2\_panel.py`) and the per-token checkpoints in `03\_data/raw/phase1\_onchain/holding/` let completed tokens re-aggregate instantly without re-fetching. DO NOT re-run tokens already completed in session 025 --- load their checkpoints and skip.

****Step B1 --- run the tail in priority order:****

Set `HOLDER\_MAX` high enough to include the Tier 1 and Tier 2 tokens (start with `HOLDER\_MAX=250000` which covers everything except UNI, COMP, EIGEN, BLAST, GMX). Run smallest-first within the worklist. Monitor calls/wall-clock and checkpoint.

The key targets to reach this session (in order of value):

1. CVX, FRAX, SUSHI, GMX -> 3-channel assets (highest priority)
2. DDX, ORBS, RPL, LQTY, XAN, API3, CAKE -> moderate size, ch1 governance tokens
3. CRV, AAVE, LDO, ENS, BAL, YFI, GNS, MNT, CVX, PENDLE -> large, high-visibility tokens
4. ARB, EIGEN, BLAST, ENA, WLF1, UNI, COMP, GRT -> very large, if budget allows

If the daily budget approaches DAILY_CAP before completing all tiers, checkpoint and stop. Report how far down the priority list you reached.

****Step B2 --- data integrity check (the overnight-outage lesson):****

Before aggregating, verify no new checkpoints have the network-failure corruption signature (a checkpoint that shows 0 transfers for a mid-history block range on a token known to have volume). The `\_robust\_getlogs` fix from session 025 should prevent this, but verify explicitly by checking that completed tokens' checkpoint transfer counts are plausible (not suspiciously low for large tokens). Delete and re-fetch any suspect checkpoint.

****Step B3 --- aggregate and assemble:****

Re-run `phase1\_channel2\_panel.py --aggregate` to rebuild `channel2\_holding.csv` from all checkpoints (session 025 + session 026). Then re-run `phase1\_assemble\_lambda.py` to fold the updated ch2 into `lambda\_panel.csv`.

Report the new asset-month count, `n_channels` distribution, and specifically:

- How many 3-channel asset-months now exist?
- Which assets achieved 3-channel coverage?
- What is the new 2+ channel share?

****Step B4 --- per-asset screened HODL-6m sanity check on large tokens:****

For any Tier 1 or large Tier 2 token newly completed, spot-check its screened HODL-6m series looks economically reasonable (not 95%+ or 0.1% every month). If a series looks degenerate, DO NOT fold it into lambda --- log as anomaly and report.

Rules (unchanged)

- `cmc_id` joins only, never symbol.
- Entry-26 bar for any Channel-1 build (none here, but carry the rule).
- `DATA_DECISIONS_LOG.md` append-only. Continue from Entry 65. One entry per major decision cluster. Log: OP delegation (65), Channel-2 tail results including 3-channel count (66), session close-out with new lambda count (67).
- No additional paid subscriptions.
- Track `getLogs` call budget. STOP and checkpoint before hitting `DAILY_CAP` (leave 20k headroom). Per-token checkpoints make every stop resumable.
- Run `PYTHONUTF8=1`.
- Update `06_documentation/time_log.md`. Log session as `06_documentation/ai_conversations/session_026_*.md`. Commit and push at session end.

Deliverables

1. OP delegation series in `channel3\_onchain\_delegation.csv`; role=cross_check (NOT added to lambda as a channel per the governance waterfall rule).
2. Updated `channel2\_holding.csv` with all newly completed tail tokens.
3. Updated `lambda\_panel.csv` with new ch2 additions + OP delegation cross-check noted.
4. New lambda asset-month count, `n_channels` distribution, ****first 3-channel asset list****.
5. `03\_data/SESSION026\_TAIL\_BUILD\_REPORT.md` (parallel structure to SESSION025 report).
6. `DATA_DECISIONS_LOG` Entries 65--67. `time_log.md` updated. Session log written.
7. Commit and push.

STOP at end of session or when `DAILY_CAP` approaches. If Channel-2 tail is not fully complete, checkpoint and report how many priority-tier tokens were completed. Do not start other Phase-2 items (NMR, KAIA, non-EVM, Phase 2c NVT_GL) without review.

/newpage

Claude Code Kickoff Prompt — Session 027: TVL Expansion + Channel-2 Tail (continued)

Paste the prompt below as the first message in a new Claude Code session opened with working directory C:/AFA_2027_QTM_Crypto.

Context: Session 026 built the first 3-channel assets (332 asset-months across 9 tokens) and completed OP delegation as a crosscheck. lambda is now 6,021 asset-months / 282 assets. A fresh universe coverage CSV (03_data/universe_coverage_status.csv) maps all 1,939 in-universe assets with current data status and what's needed per asset. Two tasks this session, in priority order: (A) TVL expansion, and (B) ch2 holding tail continued.

You're working in the AFA 2027 QTM Crypto research repo. lambda is 6,021 asset-months / 282 assets (session 026, Entry 67). Etherscan Pro key (200k calls/day, 10/s) is in `.api\_keys.json` under "etherscan". DeFiLlama API is free/keyless. Two tasks this session, in priority order.

Required reading before starting

- 04_code/DATA_DECISIONS_LOG.md --- read Entries 63--67 (ch2 methodology, TVL scope, session 026 close). Continue from Entry 68.
- 03_data/SESSION026_TAIL_BUILD_REPORT.md --- full session 026 account (read entirely)
- 03_data/universe_coverage_status.csv --- NEW: coverage map for all 1,939 universe assets. Columns: cmc_id, symbol, name, asset_class, lambda_months, lambda_n_channels, lambda_channels, has_ch1, has_ch2, has_ch3_v, has_ch3_d, has_nvt_gl, has_tvl, holder_count, coverage_status, what_needed. Use this to drive priority ordering in both tasks.
- 03_data/phase1/asset_onchain_identity.csv --- dl_slug column is the DeFiLlama protocol slug
- 04_code/phase2_build_tvl_panel.py --- existing TVL panel builder (idempotent, cached per slug)
- 04_code/phase1_channel2_stream.py --- streaming ch2 engine (session 026, validated)
- 03_data/phase1/lambda_panel.csv --- current lambda panel

TASK A --- TVL Expansion (Entry 68)

Motivation

TVL is the denominator in the tokens' valuation multiple (NV/TVL). Five 3-channel assets (CRV/YFI/FRAX/GMX/RPL) have full lambda but no TVL --- they cannot enter the regression without it. 76 more 1--2-channel tokens also have no TVL. TVL fetches are free (api.llama.fi, keyless), cached per slug, and take <1s each --- this is the highest-ROI task in the session.

Step A1 --- Slug discovery for priority tokens

Four of the five 3-channel tokens have NO dl_slug in asset_onchain_identity.csv:

- CRV (cmc_id 6538, Curve DAO Token)
- YFI (cmc_id 5864, Yearn Finance)
- FRAX (cmc_id 6953, Frax / prev. FXS)
- GMX (cmc_id 11857, GMX)

RPL (cmc_id 2943) already has dl_slug = "rocket-pool" -> skip discovery, fetch directly.

For each of the four missing tokens:

- a. Call `https://api.llama.fi/protocols` (returns the full DeFiLlama protocol list as JSON). Match by CMC ID first (field `cmcId`), then by name/symbol if CMC ID is absent or stale.

- The protocol list is ~2000 entries; cache it locally so you don't re-fetch per token.
- Confirm the match is the right protocol (name + TVL series looks right --- not a chain-level entry or an unrelated protocol with the same ticker).
 - Candidate slugs to expect: "curve-dex" or similar for CRV; "yearn-finance" for YFI; "frax" for FRAX; "gmx" for GMX. Verify against the list, do not assume.
 - Write the confirmed slug back into `asset_onchain_identity.csv` (`cmc_id` join only, never symbol; update `dl_slug` column, also update `dl_matched=True` if that column exists).
 - Log the slug decisions in `DATA_DECISIONS_LOG` (Entry 68), noting any slug that was ambiguous or could not be confirmed.

Step A2 --- TVL fetch for all lambda tokens with missing TVL

After confirming slugs for the 5 priority tokens, extend the fetch to ALL lambda assets missing TVL that have a confirmed `dl_slug`:

From `asset_onchain_identity.csv`, collect all rows where:

- `cmc_id` is in `lambda_panel.csv` (has_lambda)
- `dl_slug` is not null
- `cmc_id` NOT already in `tv1_panel.csv` (no need to re-fetch existing TVL rows)

The existing script ``phase2_build_tv1_panel.py`` already does this derivation. Read it fully before running --- the join logic uses `IDENT` (`asset_onchain_identity.csv`) to derive the slug list. Run it; it will skip already-cached protocol JSONs and only fetch new ones.

Expected coverage gain: ~23 tokens immediately (have existing slugs), + 4 more if slugs confirmed above. Report how many new token-TVL months were added.

Step A3 --- Slug discovery for the broader lambda-but-no-TVL list

After the priority tokens, attempt slug discovery for the remaining lambda assets without TVL (about 200 tokens have no `dl_slug` and no TVL). Method:

- For each token, try ``https://api.llama.fi/protocols`` matched by CMC ID (``cmcId`` field). This is a BULK operation --- fetch the protocol list ONCE, build a dict keyed by `cmcId`, then look up each token's `cmc_id` directly.
- For hits: confirm the name/symbol match is correct (same spot-check as A1), write slug to `asset_onchain_identity.csv`, queue for TVL fetch.
- For misses (`cmcId` not in DeFiLlama list): check if the token is a CHAIN-level protocol on DeFiLlama (i.e. the TVL is at chain level, not protocol level). If so, try ``https://api.llama.fi/v2/chains`` for the chain TVL series. This applies to:
 - Coin lambda assets (ETH, SOL, TRX, ADA --- their "TVL" is the chain's DeFi TVL, not a protocol). NOTE: for COINS we use NVT (not NV/TVL) per the framework, so chain TVL is NOT needed in the denominator --- skip coins for TVL.
 - Governance tokens whose protocol TVL is tracked at chain level (e.g. ARB, OP).
- Log total: how many new slugs confirmed, how many still unmatched, how many were coins (excluded from TVL requirement).

A4 --- Rebuild `tv1_panel.csv`

After all slug updates, re-run ``phase2_build_tv1_panel.py`` to build the updated TVL panel. Report: old rows / new rows, distinct tokens before/after.

A5 --- Cross-check priority token TVL plausibility

For the 5 three-channel tokens (CRV, YFI, FRAX, GMX, RPL), spot-check the TVL series:

- Series should span at least 24 months
- No month should show TVL ~ 0 for a major protocol except during genuine early-launch
- TVL should be in USD (not raw token units --- the DeFiLlama API returns USD)

Report any anomalies; do NOT fold into panel if series looks wrong.

TASK B --- Channel-2 Tail continued (Entry 69)

Motivation

43 assets in lambda have ch1 OR ch3 but NO ch2. Adding ch2 upgrades them from 1-channel to 2-channel (or 2-channel to 3-channel). All are EVM/Ethereum tokens with holder counts <=500k and buildable via the session-026 streaming engine.

Priority list (sorted smallest-first, build in this order)

Derived from universe_coverage_status.csv: cmc_id IN lambda_panel AND has_ch2=False AND (has_ch1=True OR has_ch3_v=True) AND holder_count <= 500000.

Key tokens in approximate holder-count order:

- NFTX (8191, ~3.7k), HAKKA (6622, ~4k), IQ (2930, ~4.1k), BZRX (5810, ~5k)
- COW (19269, ~7k), MC (13523, ~7k), SSV (12999, ~7.4k), OHM (9067, ~8.4k)
- KP3R (7535, ~8.5k), ORBS (3835, ~9k), RGT (7486, ~9.4k), SYN (12147, ~10.4k)
- PNT (5794, ~10.8k), STRK-token (8911, ~11.9k), XAN (38481, ~13k)
- LQTY (7429, ~14.4k), ALCX (8613, ~14.5k), PERP (6950, ~14.6k)
- HFT (22461, ~16.2k), TRU (7725, ~16.2k), FARM (6859, ~16.2k)
- CAKE (7186, ~20.6k), GNS (13663, ~20.6k), LINA (7102, ~22.8k), API3 (7737, ~23.9k)
- MNT (27075, ~29.8k), BADGER (7859, ~30.3k), BNT (1727, ~38.5k), STG (18934, ~39.2k)
- BAL (5728, ~48.8k), REN (2539, ~59.6k)
- ARB (11841, ~63.9k), LDO (8000, ~65.5k), ENS (13855, ~67.6k)
- PENDLE (9481, ~72.8k), GTC (10052, ~89.6k), ENA (30171, ~97.8k)
- LRC (1934, ~167k), GRT (6719, ~173k), APE (18876, ~188k), ZRX (1896, ~190k)
- COMP (5692, ~220k), UNI (7083, ~385k)

Do NOT re-run tokens already completed in session 026. Load their checkpoints and skip. Use `phase1\_channel2\_stream.py` (the streaming engine from session 026, NOT the old panel engine). Do NOT attempt HEX (cmc_id 5015, 9M holders --- permanently deferred, Entry 66).

B1 --- Run the tail smallest-first

Set HOLDER_MAX high enough to cover the full list (HOLDER_MAX=500000 or unlimited since all targets are pre-screened here). Process in holder-count order. Monitor memory heartbeat for any hidden giant (a token with low holder_count but unexpectedly large Transfer history --- like ORBS in session 026). If a token's est_getlogs_calls from `\_channel2\_sizes.csv` exceeds 50k, log it and evaluate budget before proceeding.

Daily budget: DAILY_CAP = 200,000 getLogs calls. Stop and checkpoint before hitting the cap (leave 20k headroom). All stops are clean because the streaming engine checkpoints per-token.

B2 --- Address-poisoning two-layer guard (already in the engine)

The session-026 engine already has:

1. Per-event VAL_CAP_MULT=100 (skip events exceeding 100x max circulating supply)
2. Per-month CONTAM_MULT=100 (emit NULL for months where on-chain supply > 100x circulating)

Do NOT change these thresholds. Do run the B2 full-panel integrity scan on completion (reconstructed on-chain supply vs circulating) to catch any new contaminated tokens.

B3 --- Aggregate and assemble

After all newly completed tokens:

- a. Re-run `phase1\_channel2\_panel.py --aggregate` to rebuild channel2_holding.csv
- b. Re-run `phase1\_assemble\_lambda.py` to update lambda_panel.csv
- c. Report: new asset-month count, n_channels distribution, new 2-channel and 3-channel assets

B4 --- Screened HODL-6m sanity check

For any token with >50k holders newly completed, spot-check: median screened HODL-6m should be >0.1% and <95%. Flag any degenerate series.

Rules (unchanged from session 026)

- cmc_id joins ONLY, never symbol.
- DATA_DECISIONS_LOG.md append-only. Continue from Entry 68. One entry per task cluster:
 - Entry 68 = TVL expansion (slug discovery + fetch results + panel rebuild stats)
 - Entry 69 = ch2 tail results (tokens completed, 3-channel count, integrity scan)
 - Entry 70 = session close-out (new lambda count, new TVL count, next priorities)
- No additional paid subscriptions. DeFiLlama is free/keyless; Etherscan Pro for ch2.
- Track getLogs call budget for Task B. STOP before DAILY_CAP - 20k.
- Per-token checkpoints make every stop resumable with no data loss.
- PYTHONUTF8=1.
- Update 06_documentation/time_log.md.
- Write session log to 06_documentation/ai_conversations/session_027_*.md.
- Write 03_data/SESSION027_TVL_AND_CH2_REPORT.md (parallel structure to session 026 report).
- Commit and push at session end.

Deliverables

1. Updated `03\_data/phase1/asset\_onchain\_identity.csv` --- new dl_slug entries for priority tokens
2. Updated `03\_data/phase2/tvl\_panel.csv` --- expanded TVL coverage
3. Updated `03\_data/phase1/channel2\_holding.csv` and `03\_data/phase1/lambda\_panel.csv`
4. DATA_DECISIONS_LOG entries 68--70
5. `03\_data/SESSION027\_TVL\_AND\_CH2\_REPORT.md`
6. time_log.md updated; session_027 log written
7. Commit and push

Not in scope this session

- Coin staking (ch1 for AVAX/BNB/NEAR/INJ/SUI/APT etc.) --- multiple non-EVM chains, separate research effort; defer to session 028.
- PQ/NVT_GL expansion --- defer.
- Any Phase-3 or regression work --- defer.
- HEX ch2 --- permanently deferred (Entry 66).

STOP at end of session or when DAILY_CAP approaches. If ch2 tail is not fully complete, checkpoint and report how far down the priority list you reached.

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CLAUDE_CODE_SESSION028_COIN_STAKING_AND_AAVE_FIX_PROMPT.md

Claude Code Kickoff Prompt — Session 028: Coin Staking (ch1) Expansion + AAVE ch2 Fix

Paste the prompt below as the first message in a new Claude Code session opened with working directory C:/AFA_2027_QTM_Crypto.

Context: Sessions 025–027 built Channel-2 for 260 tokens, created the first 3-channel assets, and expanded TVL to 130 tokens. lambda is 7,051 asset-months / 282 assets. The coin side of the regression is thin: only 5 coin assets have BOTH lambda AND NVT_GL (ETH, TRX, ADA, SOL + 1 more). 49 coins have NVT_GL but NO lambda. This session expands coin ch1: first via EVM chains already covered by the Etherscan Pro key (no new signups), then via non-EVM probes. Secondary task: AAVE ch2 spam-window fix.

NOTE: Moazzam's Etherscan Pro Standard Plan (activated 2026-06-30) covers the following chains that are relevant to coin staking this session — all accessible with the existing "etherscan" key in .api_keys.json: chainid 56 = BNB Smart Chain (BNB staking) chainid 1284 = Moonbeam (GLMR staking via Parachain-Staking precompile) chainid 1285 = Moonriver (MOVR staking via ParachainStaking precompile) chainid 146 = Sonic Mainnet (S/Sonic staking via SFC contract) chainid 80094 = Berachain (BERA Proof-of-Liquidity staking contract) chainid 50 = XDC Network (XDC validator staking) chainid 5000 = Mantle Mainnet (MNT staking — lower priority, verify mechanism first) chainid 42220 = Celo Mainnet (CELO — getLogs failed Entry-26 cross-check; try balance history) chainid 43114 = Avalanche C-Chain (AVAX — note: AVAX staking is P-Chain, NOT C-Chain; this chainid does NOT help with AVAX staking) DOT/KSM Subscan signup failed (code verification error at pro.subscan.io) — skip this session.

You're working in the AFA 2027 QTM Crypto research repo. lambda is 7,051 asset-months / 282 assets (session 027, Entry 70). Two tasks this session, in priority order.

Required reading before starting

- 04_code/DATA_DECISIONS_LOG.md Entries 41--47, 70 --- the full coin-staking audit:
 - Entry 41 = first live coin-staking pass (ADA/XTZ built; gaps documented)
 - Entry 42 = Bucket-2 four-way gap reclassification (free/engineering/paid/none)
 - Entries 43--47 = per-chain build/reject/gap outcomes (TRX/SOL built; DOT/KSM key-gated; HBAR/SUI engineering; CELO EVM-reclassified but cross-check failed; ATOM/INJ/SEI/KAVA/AVAX/NEAR/EOS/ICP/APT gates documented)
 - Entry 70 = session 027 close-out, coin staking listed as top next priority
- 03_data/universe_coverage_status.csv --- the NEW post-027 coverage map; filter to asset_class='coin' AND has_nvt_gl=True AND has_ch1=False to get the 49-coin target list.
These are coins with the denominator ready but no lambda --- each one built enters regression.
- 04_code/phase1_channel1_pos_coins.py --- ADA, XTZ builder (extend, don't rewrite)
- 04_code/phase1_channel1_pos_coins_bucket2.py --- TRX, SOL builder (extend, don't rewrite)
- 04_code/.api_keys.json --- check for keys: "etherscan" (Pro, 200k/day, all chains incl. BSC), "subscan" (may or may not be present --- see Task A2)
- 03_data/SESSION027_TVL_AND_CH2_REPORT.md --- full session 027 account

Continue DATA_DECISIONS_LOG from Entry 71. One entry per coin/chain group.

Work through the chains below in priority order. For EACH chain: (1) verify the source live before building --- an actual API call, not docs inference; (2) cross-check the reconstructed series against a live on-chain reference (the Entry-26 bar applied to coins --- e.g. compare the reconstructed month-end total to the chain's own reported staked supply for the same date); (3) build only if cross-check passes; (4) log the result in DATA_DECISIONS_LOG regardless of whether you build or document a gap.

- Use ``eth_getLogs`` (via Etherscan Pro V2 API, ``?chainid=<id>``) on the staking contract
- Replay events to reconstruct net staked total at each month-end block
- Cross-check reconstructed total vs live on-chain ``eth_call`` to the staking contract's getter function (e.g. ``totalStaked()``, ``totalPooledBNB()``, etc.) at ~0% drift
- Denominator = circulating supply from `universe_panel.csv` (`cmc_id` join), NOT `max_supply`
- Extend ``phase1_channel1_pos_coins_bucket2.py`` (or a new ``_evm.py``) --- do not rewrite

[illegible]

- ### A2 --- GLMR / MOVR via Moonbeam/Moonriver (Etherscan Pro, cmc id 6836 / 9285)

[illegible]

- a. Probe getLogs on the ParachainStaking precompile for staking-relevant events. Likely candidates: ``Delegation(...)``, ``DelegationIncreased(...)``, ``DelegationDecreased(...)``, ``DelegatorLeft(...)``. Try a short recent block range first to confirm events fire and amounts are in GLMR/MOVR (18 decimals).
- b. If events confirmed and net staked balance is reconstructable: build the monthly series from chain genesis (Moonbeam ~2021-12, Moonriver ~2021-08) to 2026-05.
- c. Cross-check: compare reconstructed total staked against ``totalStake()`` or equivalent getter on the precompile, or Moonscan's reported staked supply.
- d. If getLogs on the precompile fails or events are insufficient to reconstruct net staked (e.g. events track per-delegator changes but no aggregate is reconstructable): try querying ``totalStake()`` directly via ``eth_call`` at each month-end block on Etherscan Pro. This is ~70 calls for Moonbeam's history --- cheap and clean.
- e. Log result (Entry 71).

A3 --- S (Sonic) via Sonic Mainnet (Etherscan Pro, cmc_id 32684)

Sonic Mainnet (chainid 146) uses the SFC (Special Fee Contract / Staking Facility Contract) for validator staking --- the same contract family as Fantom Opera (Sonic's predecessor). SFC contract address on Sonic: likely ``0xFC00FACE000000000000000000000000000000000000000000000000000000000000`` (verify on sonic.ftmscan.com / sonicscan.org before using).

- a. Probe getLogs on the SFC contract for staking events. Key candidates: ``Delegated(address indexed delegator, uint256 indexed toStakerID, uint256 amount)`` and ``Undelegated(...)``. A recent short block range first.
- b. If events confirmed: replay Delegated/Undelegated to reconstruct net staked total per month-end. SFC also has ``totalActiveStake()`` getter for cross-check.
- c. Cross-check vs ``totalActiveStake()`` via ``eth_call`` at a recent month-end block.
- d. Log result (Entry 71).

A4 --- BERA via Berachain (Etherscan Pro, cmc_id 24647)

Berachain (chainid 80094) uses Proof of Liquidity. BERA staking involves the BeaconDeposit contract or a dedicated validator staking contract.

- a. Identify the correct staking contract on Berachain (check beratrail.io or berascan.com for the main staking/deposit contract address).
- b. Probe getLogs for deposit/stake events (likely ``Deposit(...)`` on a deposit contract, or similar). If Berachain launched recently (~2025), the history may be short --- that's fine, build what's available.
- c. Cross-check vs reported total staked BERA.
- d. If the staking mechanism is purely Cosmos-side (Berachain is a Cosmos-EVM hybrid) and getLogs cannot reconstruct it: document as gap and move on.
- e. Log result (Entry 71).

A5 --- XDC via XDC Network (Etherscan Pro, cmc_id 2634)

XDC Network (chainid 50) uses a Proof-of-Stake-XinFin consensus. Masternodes stake XDC. Check xdcscan.com for the staking/masternode contract.

- a. Identify the staking contract (XDC uses a ``XDCValidator`` contract for masternode staking; likely at a known address --- check xdcscan.com or XDC's official docs).
- b. Probe getLogs for ``Staked(...)`` / ``Unstaked(...)`` or equivalent events.
- c. Cross-check vs reported total staked XDC.
- d. Log result (Entry 72).

A6 --- DOT / KSM (DEFERRED --- Subscan signup failed)

DOT (cmc_id 6636) and KSM (cmc_id 5034): Subscan free-tier signup failed at verification step (code delivery issue at pro.subscan.io). Log as "deferred --- Subscan free key needed; signup blocked by verification code failure; retry later." Do NOT attempt to build without a key. Do NOT purchase Subscan Pro.

Note: if a "subscan" key IS found in `api\_keys.json` at runtime, build DOT/KSM using the method in Entry 44 (`era\_stat` endpoint, paginate, cross-check). Otherwise log and skip.

A7 --- CELO via Etherscan Pro balance history (cmc_id 5567)

Entry 46: getLogs reconstruction underperformed (3x undercount) because native CELO locking does not emit ERC20 Transfer logs. Etherscan Pro has a `balancehistory` endpoint (`/api?module=account&action=balancehistory&address=<contract>&blockno=<block>&chainid=42220`) that returns the native coin balance of an address at a specific block --- i.e. the LockedGold contract's native CELO balance at each month-end block, without needing to reconstruct from events.

- a. Probe Celoscan (chainid 42220) balance history: for the LockedGold contract `0x6cC083Aed9e3ebe302A6336dBC7c921C9f03349E`, call `balancehistory` at the most recent month-end block. Confirm the endpoint is accessible with the Etherscan Pro key on chainid 42220 and returns a non-zero CELO balance in wei.
- b. If the endpoint works: fetch month-end CELO balance for the LockedGold contract at each month-end block from 2020-06 (CELO mainnet) to 2026-05. Cross-check: latest month-end balance vs `getTotalLockedGold()` live call (Entry 46 found live balance of 82.43M CELO). Must reproduce at ~0% drift.
- c. If endpoint works and cross-check passes: build the series, add to pos_coins_bucket2.py.
- d. If endpoint is blocked on chainid 42220 or returns all zeros: document as still-open gap (the ONLY fix is a paid balance-history endpoint or archive eth_call node), move on.

A8 --- AVAX via AvaCloud Metrics API (cmc_id 5805)

Entry 42/47: AvaCloud has a historical "Staking Information" feature but the free vs paid gate was genuinely unclear from public pricing pages.

- a. Make a direct GET to the AvaCloud Metrics API staking endpoint: `https://glacier-api.avax.network/v1/chains/mainnet/metrics/staking` (try both with and without an `x-glacier-api-key` header). Also try: `https://metrics.avax.network/v2/network/staking/historic` if that exists. Record: (1) HTTP status code, (2) whether it requires a key, (3) whether a free key can be obtained self-serve without purchasing.
- b. If a free endpoint exists: fetch the historical total staked AVAX series, build, cross-check against the Avalanche Explorer's reported staked supply.
- c. If key-gated with a self-serve free signup: do NOT sign up (non-interactive session). Log the exact gate and URL so Moazzam can complete the signup in 5 minutes.
- d. If paid-only or contact-sales: log as documented gap, no purchase.

A9 --- NEAR via NearBlocks (cmc_id 6535)

Entry 47: NearBlocks confirmed current `/v1/stats` only; historical staking endpoint and pricing unconfirmed.

- a. Probe ``https://api.nearblocks.io/v1/stats`` --- confirm it returns current staking data (total staked NEAR, validators, etc.).
- b. Check if NearBlocks has a ``/v1/staking/history`` or similar historical endpoint. Try: ``https://api.nearblocks.io/v1/validators/stats?from=2021-01-01&to=2021-12-31`` or similar date-windowed queries. Also check ``https://api3.nearblocks.io/v1/stats`` (v3).
- c. If a free historical endpoint exists: build the monthly series from NEAR genesis (~2020-10) to 2026-05. Cross-check against NEAR's reported total staked.
- d. If not available free: log the specific gate and move on.

A10 --- Cosmos chains: ATOM, INJ, SEI, KAVA (alternative free sources)

Entry 47: Mintscan is contact-sales; Bitquery has a trial-only free tier; public LCD is current-state-only. Try two alternative sources NOT checked before:

- a. ****StakingRewards.com API**** (``https://api.stakingrewards.com/public/query``): StakingRewards has an undocumented or documented GraphQL API. Try:
 - Query for ATOM staking history: ``{ assets(where: { slug: "cosmos" }) { metrics { rewardRate stakedTokens } } }`` or similar. Determine if the response includes historical time-series data (not just current snapshot).
 - If historical data is available for ATOM, try INJ/SEI/KAVA with their respective slugs.
 - If only current state: document and move on.
- b. ****Numia (open Cosmos data)**** (``https://data.numia.xyz``): Numia provides indexed Cosmos data as BigQuery public datasets. Check if they have a REST endpoint for historical bonded tokens per chain, or if data is BigQuery-only (requires a Google account, outside scope of this non-interactive session).
- c. If neither source provides free historical Cosmos staking: leave ATOM/INJ/SEI/KAVA as documented gaps (Entry 47 stands). Do NOT contact Mintscan.

A11 --- Assemble and report

After all attempts:

- a. Re-run ``phase1_channel1_pos_coins.py`` and ``phase1_channel1_pos_coins_bucket2.py`` to regenerate the Channel-1 pos-coin CSVs.
- b. Re-run ``phase1_assemble_lambda.py`` to fold new coin ch1 series into `lambda_panel.csv`.
- c. Report:
 - Which coins were newly built (cmc_id, source, cross-check drift, months built)
 - Which gates remain open (chain, gate type, what Moazzam needs to do)
 - New lambda asset-month count, new coin lambda count, new coin lambda?NVT_GL count (regression-read)

TASK B --- AAVE ch2 spam-excluded window recovery (Entry ~72)

Context (Entry 66): AAVE's 2024-08->2026-05 ch2 months were nulled by the contamination fix because AAVE's address-poisoning spam pushed its reconstructed on-chain supply to 1.16e60 tokens in that window. The per-event VAL_CAP_MULT=100 fix removed phantom lots and restored real data for months where per-event capping was enough --- but some months still exceeded the CONTAM_MULT=100 threshold and were nulled. The Entry-66 note flagged a per-token totalSupply approach as a more precise fix.

AAVE currently has 67 ch2 months (max n_channels=3), with 22 nulls in the recent window. Recovering them gives AAVE 3-channel coverage through 2026-05 (the full panel window).

- a. Check: for AAVE (cmc_id 7278), inspect the ch2 checkpoint (in

- `03\_data/raw/phase1\_onchain/holding/`). Identify which months have null ch2.
- The fix: use AAVE's actual ERC20 `totalSupply()` at each month-end block (via `eth\_call`) as the denominator instead of the reconstructed lot-sum (which is polluted by phantom lots from spam). The CONTAM_MULT check compares reconstructed supply to CMC circulating; with the real totalSupply as denominator, the HODL share is anchored to actual on-chain supply even when phantom lots exist in the numerator.
NOTE: this is a PER-TOKEN fix for AAVE only --- do NOT change the global thresholds (VAL_CAP_MULT and CONTAM_MULT) or apply this approach universally.
 - For each null month: call `eth\_call` on AAVE's token contract (`0x7Fc66500c84A76Ad7e9c93437bFc5Ac33E2DDaE8`, mainnet) at the month-end block to get `totalSupply()`. Use this as the denominator. Recompute screened HODL-6m = (sum of lots >6m old held by non-contract addresses) / totalSupply.
 - Cross-check: for a month where ch2 was not null, verify that using totalSupply vs the reconstructed lot-sum gives similar HODL shares (they should be close for clean months, since totalSupply ~ on-chain lot-sum when no phantom lots exist).
 - If the recovered series looks economically reasonable (0.5%--80%, not 100% every month), write the updated rows back. Use `--recompute` if supported; otherwise patch the ch2 checkpoint and re-aggregate.
 - Re-run `phase1\_assemble\_lambda.py` after AAVE fix. Report: months recovered, HODL-6m range for the recovered months.

Rules (unchanged)

- cmc_id joins ONLY, never symbol.
- Entry-26 cross-check bar: reconstructed series must reproduce the on-chain state at ~0% drift before being folded into lambda. A series that fails cross-check is NOT shipped --- it becomes a documented gap. "Flag, don't guess" (spec ?0).
- DATA_DECISIONS_LOG.md append-only. Continue from Entry 71.
Entry 71 = EVM coin builds: BNB + GLMR/MOVR + Sonic/S + BERA + XDC (built or gap per chain)
Entry 72 = DOT/KSM deferral note + CELO balance-history probe
Entry 73 = AVAX/NEAR/Cosmos probes
Entry 74 = AAVE ch2 fix
Entry 75 = session close-out (new lambda count, new coin regression-ready count)
- No additional paid subscriptions. Etherscan Pro (existing) only.
- Do NOT sign up for anything (non-interactive session). If a free self-serve key is needed, document the exact URL and what Moazzam needs to do.
- Do NOT pay for Subscan Pro, AvaCloud paid tier, Pikespeak, Mintscan.
- PYTHONUTF8=1.
- Update 06_documentation/time_log.md.
- Write session log to 06_documentation/ai_conversations/session_028_*.md.
- Write 03_data/SESSION028_COIN_STAKING_REPORT.md.
- Commit and push at session end.

Deliverables

- Extended `phase1\_channel1\_pos\_coins.py` or `\_bucket2.py` with any newly built coin series
- Updated `lambda\_panel.csv` with new coin ch1 + AAVE ch2 fix
- DATA_DECISIONS_LOG entries 71--75
- `03\_data/SESSION028\_COIN\_STAKING\_REPORT.md` --- per-chain: built / gate-open / gap
- Gap register update: exact gate and Moazzam action for each open chain
(e.g. "Subscan free key: sign up at pro.subscan.io; drop key in .api_keys.json under 'subscan'; re-run the builder")

6. time_log.md updated; session_028 log written
7. Commit and push

Not in scope this session

- ch2 for the ~500 breadth-only (non-lambda) tokens --- defer.
- PQ/NVT_GL expansion --- defer.
- Any Phase-3 / regression work --- defer.
- EOS, ICP, APT --- Entry 47 documents no source exists; do not re-check unless a new source is found in the Cosmos/alternative probe above.
- HBAR, SUI --- engineering-intractable without a keyed indexer (Entry 45); defer.
- ALGO --- structural gap confirmed (Entry 42); do not retry.

STOP at end of session or when all viable coin chains have been attempted. Report the gap register with enough detail that Moazzam can action the open items (Subscan signup, etc.).

/newpage

CLAUDE_CODE_SESSION029_BREADTH_BUILD_AND_COIN_PROBES_PROMPT.

Claude Code Kickoff Prompt — Session 029: Token ch2 Breadth Build + PoS Coin Probes

Paste the prompt below as the first message in a new Claude Code session opened with working directory C:/AFA_2027_QTM_Crypto.

Context: Session 028 added 7 PoS coins (+358 coin asset-months) and recovered AAVE's 22 spam-nulled ch2 months. lambda is now 7,409 asset-months / 289 assets; 90 assets are regression-ready (12 coin lambda?NVT_GL + 78 token/other lambda?TVL). Three high-value tasks this session, in priority order: (A) build ch2 for the 47 non-lambda EVM tokens that already have confirmed TVL — these enter the NV/TVL regression immediately on lambda assignment; (B) verify and fetch TVL for the 9 lambda tokens that have a DeFiLlama slug but no TVL yet; (C) probe every unresearched PoS coin for free historical staking data and build what passes the Entry-26 cross-check. Deferred to a later session by design: DOT/KSM (Subscan signup pending), NEAR, ATOM/INJ/SEI/KAVA (Cosmos — Moazzam sourcing keys separately).

You're working in the AFA 2027 QTM Crypto research repo. lambda is 7,409 asset-months / 289 assets (session 028, Entry 75). Regression-ready: 12 coin (lambda?NVT_GL) + 78 token/other (lambda?TVL) = 90. Etherscan Pro key (200k calls/day, 10/s, all EVM chains) is in `api\_keys.json` under "etherscan". DeFiLlama free/keyless. Three tasks this session in priority order.

Required reading before starting

- 04_code/DATA_DECISIONS_LOG.md --- read Entries 68--75 (session 027-028 full account). Continue from Entry 76.
- 03_data/SESSION028_COIN_STAKING_REPORT.md --- full session 028 account
- 03_data/phase1/asset_onchain_identity.csv --- token contract addresses (erc20_address column or equivalent), chain info, dl_slug for TVL
- 04_code/phase1_channel2_stream.py --- the validated streaming engine (sessions 025-027). READ IT FULLY before Task A. Do not change guard thresholds (VAL_CAP_MULT/CONTAM_MULT=100). The AAVE self-transfer skip (`from == to`) added in session 028 is already in the engine.
- 03_data/phase1/lambda_panel.csv --- current lambda panel (post-028)
- 03_data/phase2/tvl_panel.csv --- current TVL panel
- 04_code/phase1_channel1_pos_coins_evm.py --- session 028 EVM coin builder (reference for

```

multi-chain Etherscan Pro API pattern: `?chainid=<id>`)

## TASK A --- ch2 for non-lambda EVM tokens that already have TVL (Entry 76)

### Why these are the priority

47 tokens are in the universe, have confirmed TVL in `tvl_panel.csv`, but have ZERO lambda
channels. Building ch2 (HODL-wave via Transfer event replay) assigns them lambda with one channel.
Since they already have NV (from universe_panel.csv) and TVL, they are regression-ready the
moment lambda is assigned. Up to +47 assets enter the NV/TVL regression with no further work.

### A0 --- Multi-chain adaptation

The streaming engine was built for Ethereum mainnet. Several tokens in this list are on other
chains. Before building:
a. Check if `phase1_channel2_stream.py` accepts a `chainid` parameter. If not, add a
  `--chainid` CLI argument that is passed through to the Etherscan Pro API call
  (`?chainid=<id>`). Default = 1 (Ethereum mainnet). Use the same API key for all chains.
b. Verify Etherscan Pro getLogs coverage for the non-mainnet chains used below:
  - BSC (chainid 56): already confirmed (session 028 BNB build)
  - Polygon (chainid 137): probe one recent block to confirm
  - Arbitrum (chainid 42161): probe one recent block to confirm
  - Base (chainid 8453): probe one recent block to confirm
  Log which chains are confirmed in the DATA_DECISIONS_LOG entry.
c. For multi-chain tokens: get the correct token contract address for THAT chain from
  asset_onchain_identity.csv. If missing, look up via Etherscan's token-info endpoint
  (`?module=token&action=tokeninfo&contractaddress=...&chainid=<id>`).

### A1 --- Target list (ordered by est_getlogs_calls within each chain group, smallest first)

Build in the order below. Load checkpoints and skip if a token was already completed.
Do NOT build HEX (permanently deferred, Entry 66).

**Ethereum mainnet (chainid 1)** --- 36 tokens:

| cmc_id | symbol | est_getlogs_calls | notes |
|-----|-----|-----|-----|
| 7242 | CORE (cVault) | 277 | DL slug: cvault-finance |
| 14783 | MAGIC | 465 | DL slug: treasure |
| 9119 | TLM | 466 | DL slug: alien-worlds |
| 11079 | BRISE | 300 | DL slug: bitgert |
| 14803 | AURORA | 611 | DL slug: aurora-plus |
| 6929 | HEGIC | 602 | DL slug: hegic |
| 8602 | AUCTION | 744 | DL slug: bounce-finance |
| 10903 | C98 | 426 | DL slug: coin98 |
| 7654 | RFOX | 773 | DL slug: rfox |
| 3748 | HXRO | 534 | DL slug: hxro-network |
| 1552 | MLN | 759 | DL slug: enzyme-finance |
| 8615 | EPIC | 753 | DL slug: ethernity-chain |
| 5601 | STAKE | 708 | DL slug: xdai-stake-bridge |
| 2933 | BMX | 696 | DL slug: bitmart |
| 2982 | MVL | 939 | DL slug: mvl-staking |
| 4134 | AKRO | 1526 | DL slug: akropolis |
| 1104 | REP | 1628 | DL slug: augur |

```

7617	SFI	978	DL slug: saffron-finance
5566	KEEP	949	DL slug: keep-network
5957	YFII	1159	DL slug: yfii
5829	SWAP	1390	DL slug: team-finance
17799	AXL	982	Axelar; check addr in aoi.csv
9543	BICO	1403	DL slug: hyphen
3814	CELR	1832	DL slug: cbridge
1166	WARP	2543	DL slug: polkastarter
7857	MIR	2317	DL slug: mirror
1732	NMR	3126	DL slug: erasure
8290	SUPER	3407	DL slug: superfarm
30083	ATH	4320	DL slug: aethir
7232	ALPHA	924	DL slug: stella
34812	BIO	894	DL slug: bio-protocol
5631	ORN	784	DL slug: dedust
8719	ILV	4276	DL slug: illuvium
5845	BTCST	---	BSC item listed below
6945	AMP	8106	DL slug: flexa; large
10804	FLOKI	8199	DL slug: flokifi-locker; large

****BSC (chainid 56)**** --- 7 tokens (probe getLogs on BSC first):

cmc_id	symbol	notes
8335	MDX	DL slug: mdex
8938	EPS	DL slug: ellipsis-finance
7064	BAKE	DL slug: bakeryswap
8891	BTCST	DL slug: btcst
9175	MBOX	DL slug: mobox
8972	SFUND	DL slug: seedify
36410	MYX	DL slug: myx-finance

****Polygon (chainid 137)**** --- 1 token (probe first):

cmc_id	symbol	est_getlogs_calls	notes
24796	ADF	1871	DL slug: artdefinance

****Arbitrum (chainid 42161)**** --- 1 token (probe first):

cmc_id	symbol	est_getlogs_calls	notes
38341	RAIN	13637	DL slug: rain; LARGEST --- build last

****Base (chainid 8453)**** --- 3 tokens (probe coverage first; build only if getLogs confirmed):

cmc_id	symbol	notes
38299	AVNT	DL slug: avantis
35763	KAITO	DL slug: kaito
35509	VVV	DL slug: venice

****Skip (non-EVM, cannot build with current APIs):****

SXP (4279), RUNE (4157), OSMO (12220), SUN (10529), CASINO (1573)

A2 --- Build protocol

For each token:

- Get the ERC-20 contract address from `asset_onchain_identity.csv`. If missing, use Etherscan's ``tokeninfo`` endpoint for the relevant chain.
- Run the streaming engine with ``--chainid <id>`` (or equivalent). Same guard thresholds: `VAL_CAP_MULT=100`, `CONTAM_MULT=100`. Per-token checkpoint saves automatically.
- If a token is a known scam/dead/zero-transfer contract (0 Transfer events in last 100k blocks), log it and move on --- do not spend `getLogs` budget.
- After completing each chain group, verify B2 integrity scan: reconstructed on-chain supply vs CMC circulating. Log any contaminated months (there should be few --- these tokens were pre-screened).

A3 --- Budget management

DAILY_CAP = 200,000 `getLogs` calls. Reserve 20k headroom. Checkpoint per-token.
Build smallest-first so large tokens (AMP 8k, FLOKI 8k, RAIN 14k) land at the end of each day.
If daily cap approaches, stop cleanly at the current token boundary --- the checkpoint makes resumption zero-loss. Session 028 used only ~45k calls --- this session has a full 200k budget.

A4 --- Aggregate and assemble (after Task A completes)

- Run ``phase1_channel2_panel.py --aggregate`` to rebuild `channel2_holding.csv`
- Run ``phase1_assemble_lambda.py`` to update `lambda_panel.csv`
- Report: new lambda asset-months, how many of the 47 targets newly entered lambda, how many were skipped (dead/non-EVM/LUT), new NV/TVL regression-ready count

TASK B --- TVL for lambda tokens with orphaned `dl_slugs` (Entry 77)

Nine lambda tokens have a `dl_slug` in ``asset_onchain_identity.csv`` but are absent from ``tvl_panel.csv``. Three were explicitly rejected in session 027 (GBYTE/oswap-amm = third-party DEX, not protocol TVL; PENGU/pudgy-penguins = not Abstract's fee token; CYBER/cyberconnect = chain TVL ~\$0). Do NOT re-attempt those three.

The remaining 6 to check (probe DeFiLlama, accept if TVL is non-trivial and continuous):

cmc_id	symbol	dl_slug	notes
19269	COW	cowswap	CoW Protocol --- batch-auction DEX; should have genuine TVL
1896	ZRX	Ox-aggregator	Ox limit-order system; may have low TVL
9421	FORTH	forth-dao	Ampleforth governance; probably minimal TVL
13855	ENS	ens	ENS is a naming service, not a DeFi protocol --- likely \$0 TVL
10052	GTC	gitcoin	Gitcoin grants --- likely \$0 TVL
23054	CHEEL	cheelee	Verify if DeFiLlama reports non-trivial TVL

For each:

- GET ``https://api.llama.fi/protocol/{slug}`` --- inspect the TVL series.
- Accept if: TVL series exists, at least 6 months non-zero, values in USD (not fractions of a cent). Add to `tvl_panel.csv` and `asset_onchain_identity.csv` (`dl_matched=True`).
- Reject if: TVL is \$0 or near-zero throughout, or the protocol is clearly a different entity than the token. Log the rejection reason.

d. After any adds: re-run ``phase2_build_tvl_panel.py`` and re-run ``phase1_assemble_lambda.py`` so newly TVL-covered lambda tokens reflect in the regression-ready count.

TASK C --- Probe and build unresearched PoS coins (Entry 78)

Goal: find free historical staking data for PoS coins not previously researched, build any that pass the Entry-26 cross-check (~0% drift vs the chain's own on-chain figure). Deferred scope (do NOT probe this session): DOT/KSM, NEAR, ATOM/INJ/SEI/KAVA.

The Entry-26 bar applies to coins exactly as to tokens: the reconstructed series must reproduce the chain's own on-chain staked total at ~0% drift at ≥ 3 probe points before being accepted. If a source is found but drift is $>5\%$ and unexplained, document as a gap. "Flag, don't guess."

Build in sub-task order below. EVM-accessible chains first (highest probability of success with the Etherscan Pro key already in hand), then native-chain API probes.

C1 --- EVM-accessible PoS coins (probe Etherscan Pro + chain RPC)

****POL (cmc_id 28321, Polygon 2.0 token)****

Polygon PoS validator staking is managed by the ``StakeManager`` contract on Ethereum MAINNET at ``0x5e3Ef299fDDf15eAa0432E6e66473ace8c13D908`` (chainid 1). Before 2024, MATIC was the staked token; after the Polygon 2.0 migration (Sept 2023) POL replaced MATIC, but the staking contract on Ethereum still accepts both via migration contracts.

- a. Probe StakeManager getLogs on Ethereum mainnet for staking events. Key events:
``Staked(address indexed signer, uint256 indexed validatorId, uint256 nonce, uint256 indexed activationEpoch, uint256 amount, uint256 total, address indexed contractAddress)`` and ``UnstakeInit(...)`` / ``Unstaked(...)``. The ``total`` field in Staked may give running aggregate directly (spot-check).
- b. If reconstruction is feasible: build the monthly series from 2020-06 (Polygon PoS launch) to 2026-05. Cross-check: reconstructed month-end total vs live ``currentValidatorSetSize()`` or ``totalStaked()`` on StakeManager.
- c. Denominator: POL/MATIC circulating supply from `universe_panel.csv` (cmc_id 28321 or MATIC cmc_id 3890 --- check which is in the universe; join on cmc_id only).
- d. If getLogs approach fails (too complex event accounting, MigrateSuccess-like inflation): try Etherscan's ``balancehistory`` on StakeManager for native MATIC balance --- this would give total staked MATIC at each block.

****FLR (cmc_id 7950, Flare Network, chainid 14)****

Flare has FTS02 staking (direct validator staking) and delegation contracts. Staking launched ~2023. Check whether chainid 14 is covered by Etherscan Pro (probe a recent getLogs).

If covered:

- a. Identify the staking contract on Flare (check `flare-explorer.flare.network` or `flare.network/developers` for ValidatorRewardManager or StakingManager address).
- b. Probe getLogs for Stake/Unstake events. Build if reconstructable.
- c. Cross-check vs live total staked reported by the chain explorer.

****RON (cmc_id 14101, Ronin Network, chainid 2020)****

Ronin is an Ethereum sidechain. Check if chainid 2020 is covered by Etherscan Pro.
If covered:

- a. Identify Ronin's staking contract (`RoninValidatorSet` or `StakingContract` --- check app.roninchain.com or docs.roninchain.com).
- b. Probe getLogs for Staked/Unstaked events. Build if cross-check passes.

****KLAY/KAIA (cmc_id 4256, Kaia blockchain, chainid 8217)****

Klaytn rebranded to Kaia. Kaia uses a governance council model. Check if chainid 8217 is in Etherscan Pro. If so, probe the staking/governance contract for getLogs.

Note: Kaia's GC staking may be managed through off-chain governance --- if no staking events exist on-chain, document and move on.

****CRO (cmc_id 3635, Cronos EVM, chainid 25)****

Cronos EVM is an EVM chain but CRO staking is primarily on the Crypto.org chain (Cosmos SDK), not on Cronos EVM. Probe chainid 25 for a staking contract; if none found, note this and log as gap (Cosmos-side, same gate as ATOM).

****CHZ (cmc_id 4066, Chiliz chain, chainid 88888)****

Chiliz 2.0 has an EVM-compatible chain. Check if chainid 88888 is in Etherscan Pro. If so, probe for validator staking events. If not covered, check if Chiliz has a free native API (chiliscan.com API) with historical staking totals.

****LSK (cmc_id 1214, Lisk)****

Lisk migrated to an Ethereum L2 (Lisk chain, chainid 1135, built on OP Stack) in 2024. LSK staking may now exist as an EVM contract on chainid 1135. Check Etherscan Pro coverage for chainid 1135; also check lisk.com/documentation for a staking contract address. If staking exists and the chain is covered: probe and build.

C2 --- Native-chain API probes (non-EVM)

For each coin below: make at least one live API call to confirm the endpoint structure, then determine if historical monthly data is freely available. Build if yes; document gate if no.

****TON (cmc_id 11419, The Open Network)****

- Primary: `https://tonapi.io/v2/blockchain/masterchain/stats`` or `https://toncenter.com/api/v3/statistics`` --- check for a `staking` or `total\_stake` field.
- Historical: try `https://tonapi.io/v2/rates/history?tokens=TON&start=<unix>&end=<unix>`` to see what history is served; separately check if ton.org/v3 has a staking timeseries endpoint.
- If historical staking tonnage available: build monthly series from TON mainnet (~2021-12). Cross-check vs tonviewer.com reported staked TON.

****EGLD (cmc_id 6892, MultiversX / formerly Elrond)****

- Primary: `https://api.multiversx.com/economics`` --- confirmed to return a `staked` field (check live). This is the current snapshot.
- Historical: try `https://api.multiversx.com/economics?timestamp=<unix>`` or check if the MultiversX data API serves historical economics snapshots. Also try: `https://tools.multiversx.com/analytics-api/query?series=totalValueStaked&resolution=month`` (the Grafana-style analytics endpoint that powers their explorer charts).
- If historical series available: build from EGLD mainnet launch (~2020-09).

****FLOW (cmc_id 4558, Flow blockchain)****

- Primary: ``https://rest-mainnet.onflow.org/v1/network/parameters`` and ``https://flowscan.io/api/staking`` (or similar).
- Historical: try ``https://flowdiver.io/api/v1/staking/history?interval=monthly`` or the Flow Access API's ``GetAccountAtLatestBlock`` for the staking contract.
- The Flow staking contract is at ``0x8624b52f9ddcd04a``. Try querying ``totalCommitted()`` via the Flow Access API Cadence script at historical block heights.
Block height ? timestamp mapping via ``https://rest-mainnet.onflow.org/v1/blocks/{height}``.
- If historical series feasible: build from FLOW launch (~2020-10).

****STRK (cmc_id 22691, Starknet)****

- Starknet staking launched ~2024-11 via the staking contract at ``0x04718f5a0fc34cc1af16a1cdee98ffb20c31f5cd61d6ab07201858f4287c938d`` on Starknet mainnet.
- Try ``https://alpha-mainnet.starknet.io/feeder_gateway/get_state_update?blockNumber=<n>`` or ``https://starknet.io/api/staking/total`` (if it exists).
- Starknet IS an Ethereum L2 but uses its own transaction format (not EVM). Etherscan Pro does NOT cover Starknet. Try the official Starknet API or Starkscan API (``https://api.starkscan.co/api/v0/statistics/staking``) for historical total staked.
- Note: if staking series only covers 2024-11 onward (~8 months), still build --- a short series is better than no series.

****DFI (cmc_id 5804, DeFiChain)****

- DeFiChain uses its own blockchain (not EVM). Staking via masternodes.
- Try ``https://ocean.defichain.com/v0/mainnet/stats`` --- check if it includes a staking/masternode total. Also try ``https://defiscan.live/api/stats``.
- Historical: check ``https://ocean.defichain.com/v0/mainnet/stats?timestamp=<unix>`` or look for a DeFiChain analytics API with time-series staking data.

****DASH (cmc_id 131, Dash)****

- Dash masternodes require a 1,000 DASH collateral lock. This is economically equivalent to staking for the purposes of Channel 1 (capital locked in consensus participation).
- Try ``https://insight.dash.org/insight-api/masternode/count`` for current count.
- Historical: check ``https://stats.masternode.me/api/history`` or the official Dash data API for monthly masternode count + collateral (1000 DASH x count = total locked).
- Cross-check: reconstructed total locked DASH vs live masternode count x 1000.
- If monthly masternode count history is available: build the series. Flag that the mechanism is masternode-collateral (not PoS delegation) in the DATA_DECISIONS_LOG.

****Additional low-probability probes (attempt if budget allows, document gate if not):****

XRD (cmc_id 11948, Radix): try ``https://babylon-mainnet-gateway.radixdlt.com/statistics/validators/uptime`` or ``https://radix-babylon-gateway.api.radixdlt.com`` for historical staking statistics.

PEAQ (cmc_id 14588): check `peaq.network/docs/API` for historical staking series.

CORE (cmc_id 23254, Core blockchain): check ``https://openapi.coredao.org`` for historical total staked CORE.

HYPE (cmc_id 32196, Hyperliquid): Hyperliquid is a new chain (~2024). Check if ``https://api.hyperliquid.xyz/info`` returns staking data, and if history is available.

WAN (cmc_id 2606, Wanchain): try ``https://www.explorewanchain.org/api/staking/history`` or similar; Wanchain has PoS since ~2020.

For any coin where data is found but the series is <6 months: build anyway (short series still enters the cross-section; annotate in the flag column).

C3 --- EVM coin build protocol (same as session 028)

If an EVM staking source is found:

- a. Verify live: make the actual API call, check response shape and magnitude.
- b. Build from chain genesis (or earliest available month) to 2026-05.
- c. Entry-26 cross-check: reconstructed month-end total vs chain's own live figure at ~0% drift (<=5% accepted if the source of drift is fully explained and documented).
- d. Denominator: coin circulating supply from universe_panel.csv (cmc_id join only).
- e. Add to `phase1\_channel1\_pos\_coins\_evm.py` or create `phase1\_channel1\_pos\_coins\_native.py` for non-EVM native-API coins.
- f. The assembler's `channel1\_\*.csv` glob picks up any new `channel1\_\*.csv` file automatically.

TASK D --- Assemble, update coverage CSV, report (Entry 79)

After all tasks complete:

- a. Final assemble: `phase1\_assemble\_lambda.py` -> updated lambda_panel.csv
- b. Rebuild TVL panel if any new slugs confirmed: `phase2\_build\_tvl\_panel.py`
- c. **Regenerate `03\_data/universe\_coverage\_status.csv`**: the existing file is stale (generated pre-session-027, reflects none of sessions 027/028 progress). Run the same generation script (or recreate it inline) using the updated lambda_panel, nvt_gl_panel, and tvl_panel. Columns must include: cmc_id, symbol, name, asset_class, sector, coin_staking_type, lambda_months, lambda_n_channels, lambda_channels, has_ch1, has_ch2, has_ch3_v, has_ch3_d, has_nvt_gl, nvt_months, pq_source, has_tvl, tvl_months, holder_count, est_getlogs_calls, evm_chain, etherscan_reachable, coverage_status, what_needed.
- d. Report in `03\_data/SESSION029\_BREADTH\_AND\_COIN\_PROBE\_REPORT.md`:
 - Task A: which tokens built, which skipped (reason), new lambda count, new regression-ready count
 - Task B: TVL slug verdicts (accepted/rejected + reason for each)
 - Task C: per-coin probe outcome (built / gate-open / gap-confirmed), with the exact gate and Moazzam action for any open items
 - Final regression-ready tally: coin (lambda?NVT_GL) + token/other (lambda?TVL) = total

Rules (unchanged)

- cmc_id joins ONLY, never symbol.
- Entry-26 cross-check bar for all new coin series: ~0% drift vs the chain's own figure. A series that fails cross-check is NOT shipped --- it becomes a documented gap.
- DATA_DECISIONS_LOG.md append-only. Continue from Entry 76:
 - Entry 76 = Task A results (ch2 breadth build --- tokens built, skipped, budget used)
 - Entry 77 = Task B results (TVL slug verification verdicts)
 - Entry 78 = Task C results (coin probe outcomes --- built / gate / gap per coin)
 - Entry 79 = session close-out (new lambda count, new regression-ready count, gap register)
- No additional paid subscriptions. Etherscan Pro (existing) + keyless sources only.
- Do NOT sign up for anything. If a source requires a free self-serve signup, log the exact URL and what Moazzam needs to do --- do NOT complete the signup in a headless session.
- Deferred this session (Moazzam sourcing separately): DOT/KSM (Subscan), NEAR (Pikespeak),

```

    ATOM/INJ/SEI/KAVA (Cosmos Mintscan/Numia), BERA (no beacon-kit archive API).
- PYTHONUTF8=1.
- Update 06_documentation/time_log.md.
- Write session log to 06_documentation/ai_conversations/session_029_*.md.
- Write 03_data/SESSION029_BREADTH_AND_COIN_PROBE_REPORT.md.
- Commit and push at session end.

## Deliverables

1. Updated `03_data/phase1/channel2_holding.csv` --- new ch2 rows for Task A tokens
2. Updated `03_data/phase1/lambda_panel.csv` --- all new lambda assignments
3. Updated `03_data/phase2/tvl_panel.csv` --- any new TVL series from Task B
4. New/extended `04_code/phase1_channel1_pos_coins_evm.py` or `_native.py` --- any new coin
   ch1 series from Task C
5. Updated `03_data/universe_coverage_status.csv` --- fresh post-029 coverage map
6. DATA_DECISIONS_LOG entries 76--79
7. `03_data/SESSION029_BREADTH_AND_COIN_PROBE_REPORT.md`
8. Gap register: for each open item, the exact gate and Moazzam action (e.g., "sign up at
   X, get key Y, drop in .api_keys.json under 'Z', re-run script W")
9. time_log.md updated; session_029 log written
10. Commit and push

## Not in scope this session

- DOT/KSM --- deferred (Subscan signup pending; Moazzam will retry and run a dedicated session)
- NEAR --- deferred (Pikespeak gate; Moazzam sourcing)
- ATOM/INJ/SEI/KAVA --- deferred (Cosmos keys; Moazzam sourcing)
- BERA --- permanent gap unless Berachain ships a public CL archive API
- HBAR/SUI/ALGO/EOS/ICP/APT --- unchanged from Entry 45/47; do not re-check
- Phase-3 / regression work --- defer
- PQ/NVT_GL expansion --- defer
- ch3 for new tokens --- defer (ch2 single-channel is the priority; ch3 can be added later)

STOP at end of session or when daily getLogs budget approaches. All stops are clean via
per-token checkpoints. Resume with the next uncompleted token in the priority list.

/newpage

```

CLAUDE_CODE_SESSION030_TASK_A_RESUME_PROMPT.md

Claude Code Session 030 — Task A Resume: 7 Remaining ch2 Tokens

Date: 2026-07-05

Estimated getLogs calls: ~30–45k (well within one quota day)

Etherscan Pro key: in 04_code/.api_keys.json under "etherscan"

Context

Session 029 built ch2 (HODL-wave) for 40 of 47 non-lambda EVM tokens that already have TVL. Seven tokens hit the daily quota boundary before completion. Checkpoints are intact — this session is a zero-loss resume. No methodology changes, no new chains, no new engines.

Starting state (post-029 assembler): - lambda: 9,580 asset-months / 337 assets - regression-ready: 138 (coins 20, tokens 118) - 03_data/phase1/channel2_holding.csv: 300 tokens

After this session completes: ~307 tokens in channel2_holding.csv; 7 new lambda assets; regression-ready should reach ~145 (all 7 have TVL).

Files to read before starting

- 04_code/phase1_channel2_stream.py — the multi-chain streaming engine (validated in sessions 025–029; multi-chain --chainid support added in session 029). **Do not change guard thresholds (VAL_CAP_MULT = CONTAM_MULT = 100). Do not remove the from == to self-transfer skip (added session 028, engine-wide).**
- 03_data/phase1/channel2_holding.csv — current ch2 panel (check which of the 7 have any check-point; skip any already complete).
- 03_data/phase2/tvl_panel.csv — TVL panel (all 7 have TVL; no changes needed here).
- 03_data/phase1/lambda_panel.csv — current lambda panel (post-029).
- 04_code/phase1_assemble_lambda.py — assembler (run after all 7 are built).

Task — Build the 7 remaining tokens

Build in the order below (smallest est_getlogs_calls first within each chain; RAIN last). Load checkpoints and skip any token already completed (check channel2_holding.csv).

BSC (chainid 56)

cmc_id	symbol	address	DL slug	est_getLogs
8972	SFUND	0x477bc8d23c634c154061869478bce96be6045d12	seedify	~4k
36410	MYX	0xd82544bf0dfe8385ef8fa34d67e6e4940cc63e16	myx-finance	~4k

Polygon (chainid 137)

cmc_id	symbol	address	DL slug	est_getLogs
24796	ADF	0x6BD10299f4f1d31b3489Dc369eA958712d27c81b	artdefinance	1,871

Base (chainid 8453)

cmc_id	symbol	address	DL slug	est_getLogs
38299	AVNT	0x696f9436b67233384889472cd7cd58a6fb5df4f1	avantis	~4k
35763	KAITO	0x98d0baa52b2d063e780de12f615f963fe8537553	kaito	~3k
35509	VVV	0xacfe6019ed1a7dc6f7b508c02d1b04ec88cc21bf	venice	~1k

Arbitrum (chainid 42161) — build last

cmc_id	symbol	address	DL slug	est_getLogs
38341	RAIN	0x25118290e6A5f4139381D072181157035864099d	rain	13,637

All four chains (BSC, Polygon, Base, Arbitrum) were verified live in session 029 and return proper log lists via Etherscan Pro V2 (?chainid=<id>). No re-verification needed.

Build protocol

1. For each token, invoke `phase1_channel2_stream.py` with the correct `--chainid` and `--address`. The script loads its checkpoint automatically.
 2. Apply the standard B2/B4 integrity checks on each completed series:
 - **B2:** no month above the 100x contamination threshold.
 - **B4:** screened HODL-6m in [0, 80%]; flag but do not suppress values outside that range (document reason).
 3. After all 7 are built, run the assembler (`phase1_assemble_lambda.py`) to regenerate `lambda_panel.csv`.
 4. Run `04_code/build_coverage_status.py` to regenerate `universe_coverage_status.csv`.
-

DATA_DECISIONS_LOG entry (Entry 80)

After the build and assemble, append to `04_code/DATA_DECISIONS_LOG.md`:

```
### Entry 80 --- Session 030: Task-A resume --- 7 remaining ch2 tokens BUILT; lambda close-out
```

Resume of session 029's Task A. Tokens built: SFUND (8972/BSC), MYX (36410/BSC), ADF (24796/Polygon), AVNT (38299/Base), KAITO (35763/Base), VVV (35509/Base), RAIN (38341/Arbitrum). [Fill in actual getLogs calls, any anomalies, screened HODL-6m medians, and whether each passed B2/B4.] All 7 have TVL in `tvl_panel.csv` -> all are regression-ready immediately. Post-assemble lambda: [fill]. Regression-ready: [fill].

Session report

Write `03_data/SESSION030_TASK_A_RESUME_REPORT.md` covering: - Per-token outcome table (symbol | `cmc_id` | chain | getLogs calls | months built | screened HODL-6m | B2/B4 pass | notes) - Any anomalies (hidden giants, empty series, contamination) - Post-assemble lambda totals (asset-months / assets / regression-ready) - Quota used

Commit

At session end, commit and push all new/modified files: - `03_data/phase1/channel2_holding.csv` - `03_data/phase1/lambda_panel.csv` - `03_data/universe_coverage_status.csv` - `04_code/DATA_DECISIONS_LOG.md` - `03_data/SESSION030_TASK_A_RESUME_REPORT.md` - Any checkpoint files under `03_data/raw/phase1_onchain/`

Commit message: `session 030: ch2 Task-A resume --- 7 tokens built (SFUND/MYX/ADF/AVNT/KAITO/VVV/RAIN)`
`/newpage`

CLAUDE_CODE_SESSION031_TASK_A_RESUME_PROMPT.md

Claude Code Session 031 — Task A Resume: 6 Remaining ch2 Tokens (small-first; MYX gets its own day)

Planned date: Monday 2026-07-13 (user back from a week away) **Etherscan Pro key:** in `04_code/.api_keys.json` under "etherscan" **Predecessor:** session 030 (Entry 80, `03_data/SESSION030_TASK_A_RESUME_R`

— SFUND built; MYX aborted mid-build with NO checkpoint; 5 tokens untouched.

Context

Session 030 built SFUND (28,553 getLogs — 7x its estimate) and had to kill MYX at user departure after ~120k calls (batch 81/128, tf=15.4M, ~10h from finishing). The streamed engine keeps no partial checkpoint, so MYX restarts from scratch.

Starting state (post-030, commit pushed 2026-07-05): - lambda: 9,603 asset-months / 338 assets; regression-ready 139 (coins 20, tokens/other 119) - 03_data/phase1/channel2_holding.csv: 301 tokens / 9,817 rows - Engine unchanged: 04_code/phase1_channel2_stream.py (do NOT touch VAL_CAP_MULT = CONTAM_MULT = 100; keep the from == to self-transfer skip)

LESSON FROM 029/030 (Entry 80): holder-count estimates undershoot BSC 7–30x. Build small chains FIRST, BSC LAST, and give MYX a dedicated quota day.

Day 1 — the 5 small tokens (~25–30k est, treat as lower bound)

WORKLIST=24796,35509,35763,38299,38341 PYTHONUTF8=1 python 04_code/phase1_channel2_stream.py

order	cmc_id	symbol	chain (chainid)	address	DL slug	est_getLogs
1	24796	ADF	Polygon (137)	0x6BD10299f4f1d31b3489Dc36DeA958712d27c81b	ADF	35763
2	35509	VVV	Base (8453)	0xacfe6019ed1a7dc6f7b508a02d1b04ec88cc21bf	VVV	38299
3	35763	KAITO	Base (8453)	0x98d0baa52b2d063e780de3bf615f963fe8537553	KAITO	38341
4	38299	AVNT	Base (8453)	0x696f9436b07288384889471cd7cd58a6fb5df4f1	AVNT	38341
5	38341	RAIN	Arbitrum (42161)	0x25118290e645f4139381D07C31157035864099d	RAIN	38341

The engine auto-skips any token whose checkpoint already exists. All 5 have TVL in 03_data/phase2/tvl_panel.csv -> regression-ready on completion.

Day 2 (or whenever day-1 quota allows) — MYX alone

WORKLIST=36410 PYTHONUTF8=1 python 04_code/phase1_channel2_stream.py

- MYX (36410) BSC 0xd82544bf0dfe8385ef8fa34d67e6e4940cc63e16, DL slug myx-finance.
- **Budget ~250–300k calls and ~12–15h wall-clock.** Session 030 observed tf=15.4M by batch 81/128; expect >20M total (MBOX-record class).
- Expect a MULTI-HOUR SILENT stretch after batch ~61 (blocks >=49M = the Aug-2025 launch region; the engine prints every 5 batches and binary-splits densely there). Verify liveness via CPU accrual / established connections, not log freshness.
- Start it EARLY in the day. Do not kill it unless truly hung (no CPU accrual AND no established connections over several minutes).

Build protocol (unchanged)

1. B2: no month above the 100x contamination guard. B4: screened HODL-6m in [0, 80%]; flag but don't suppress (document).
2. After all builds: `python 04_code/phase1_assemble_lambda.py`, then `python 04_code/build_coverage_status.py`.
3. Append Entry 81 to 04_code/DATA_DECISIONS_LOG.md (per-token gl/months/medians/B2-B4, post-assemble lambda and regression-ready).

4. Write 03_data/SESSION031_TASK_A_RESUME_REPORT.md (same table format as the 030 report).
5. Commit + push at session end: session 031: ch2 Task-A close-out --- <N> tokens built (<symbols>)

After this completes: ch2 panel ~307 tokens; regression-ready ~145; session 029's Entry-79 open items (b)-(g) — DOT/KSM key, CHZ anchor, CORE key, WARP review, non-TVL breadth, MATIC NVT probe — are then the queue.

/newpage

CLAUDE_CODE_SESSION032_MYX_PROMPT.md

Claude Code Session 032 — MYX ch2 build

Date: 2026-07-24 (overnight run) **Etherscan Pro key:** 04_code/.api_keys.json under "etherscan"

Context

MYX (36410) is the last remaining token from the session 029 Task-A list. It was aborted in session 030 at batch 81/128 (tf=15.4M, ~10h from finishing) due to user departure. It was NOT started in session 031 (reserved for a dedicated day). No checkpoint exists — clean start.

Starting state (post-031): - lambda: 9,638 asset-months / 341 assets; regression-ready 142 (coins 20, tokens/other 122) (*Note: coverage_status.csv may show 153 complete — coin count includes PoW coins tagged complete; functionally 20 PoS coins with lambda?NVT_GL*) - 03_data/phase1/channel2_holding.csv: 306 tokens / 9,857 rows - Engine: 04_code/phase1_channel2_stream.py — DO NOT change guard thresholds (VAL_CAP_MULT = CONTAM_MULT = 100) or the from == to self-transfer skip.

BEFORE STARTING: Pause Windows Update (Settings -> Windows Update -> Pause for 7 days). Sleep/hibernate is already set to Never from the session 031 powercfg fix.

Task — Build MYX

WORKLIST=36410 PYTHONUTF8=1 python 04_code/phase1_channel2_stream.py

cmc_id	symbol	chain	chainid	address	DL slug
36410	MYX	BSC	56	0xd82544bf0dfe8385ef8fa34d67e6e4940cc63e16	myx-finance

Budget: ~250–300k getLogs calls / ~12–15h wall-clock.

Expected behavior during the run

- Session 030 observed tf=15.4M by batch 81/128; expect >20M total (MBOX-record class).
- **Expect a MULTI-HOUR SILENT stretch after batch ~61** (BSC blocks >=49M = the Aug-2025 MYX launch region; the engine binary-splits densely there and prints only every 5 batches). This is NOT a hang. Verify liveness via CPU accrual or active network connections. Do not kill it unless CPU is idle AND no established connections for several minutes.
- The API has shown no hard daily-cap rejection at these volumes (Entry 76/80: evidently credit-based). Run until completion.

After MYX completes

1. B2 / B4 integrity checks

- B2: confirm no month exceeds the 100x contamination threshold.
- B4: confirm screened HODL-6m median is in [0, 80%]. Flag but do not suppress if outside.

2. Assemble

```
python 04_code/phase1_assemble_lambda.py
python 04_code/build_coverage_status.py
```

3. DATA_DECISIONS_LOG — Entry 82

Append to 04_code/DATA_DECISIONS_LOG.md:

```
### Entry 82 --- Session 032: MYX (36410/BSC) ch2 BUILT; Task-A fully closed

MYX ch2 (HODL-wave): [getLogs actual] getLogs / [months built]/[observed window],
[screened months]. screened HODL-6m median [X]% / last [Y]%. B2: [max multiplier].
B4: [PASS/FLAG]. tf=[N]. Expect BSC mega-giant pattern (>20M transfers).
MYX has TVL (myx-finance, 27 months) -> regression-ready on completion.
Post-assemble: lambda [X] asset-months / [N] assets. Regression-ready: [N].
Session 029 Task A fully closed (all 47 targets resolved: 41 built, 5 non-EVM skipped,
1 WARP deferred for identity review).
```

4. Session report

Write 03_data/SESSION032_MYX_REPORT.md: - Token outcome table (same format as session 031 report) - Anomalies (esp. if tf > 21.2M MBOX record) - Post-assemble lambda totals - Quota used

5. Commit + push

```
git add -A
git commit -m "session 032: MYX ch2 built --- Task-A fully closed"
git push
/newpage
```

CLAUDE_CODE_SESSION033_XTZ_MATIC_NVT_PROMPT.md

Claude Code Session 033 — XTZ and MATIC NVT_GL (PQ probe and build)

Date: 2026-07-25 **No API key required** — all sources in this session are free and keyless.

Context

Five coins have lambda but no NVT_GL denominator (pq_nvtgl gap):

cmc_id	symbol	lambda months	why gapped
2011	XTZ	78	Phase 2b logged “no free native series” — but TzKT.io was never tried

cmc_id	symbol	lambda months	why gapped
3890	MATIC	50	Same note; MATIC ends 2024-08 at POL handoff
21585	SAFE	10	Governance token, no native chain — see Task C
11396	JOE	6	DEX governance token, no native chain — see Task C
28324	DYDX	1	dYdX L1 — 1 month too short for regression; skip

The Phase 2b default note (“Cosmos LCD / api.kaspa.org expose only current state?”) was copy-pasted across all non-bitinfocharts coins. XTZ is Tezos, not Cosmos. TzKT.io is the official Tezos indexer with a free, keyless, aggregate daily-statistics API going back to mainnet genesis (2018-06). This session corrects that oversight.

Entry 31/32 rule: “Raw multi-year native block iteration forbidden.” TzKT’s /v1/statistics/daily endpoint returns pre-aggregated daily totals — this is the same pattern as blockchain.com for BTC and bitinfocharts for UTXO chains. It is NOT raw block iteration. It is permitted.

Task A — XTZ PQ via TzKT.io (priority)

A1. Read the existing build infrastructure first

Read these files before writing any code: - 04_code/phase2b_pq_coins.py — how pq_coins.csv is structured and updated - 04_code/phase2_nvt_gl.py — how pq_coins.csv feeds into NVT_GL (columns needed: cmc_id, symbol, month_end, pq_usd, pq_source, rung, note) - 03_data/phase2/pq_coins.csv — the existing file (XTZ row is a single NaN-marker row) - 03_data/universe_panel.csv — for XTZ month-end prices (cmc_id=2011, status=‘observed’)

A2. Fetch TzKT daily statistics

Endpoint: <https://api.tzkt.io/v1/statistics/daily> Parameters: - limit=10000 (gets full history in one call — Tezos has ~2,100 days since mainnet) - sort.asc=date

Key field: **totalTransferred** — total tez transferred per day, in **mutez** (1 XTZ = 1,000,000 mutez).

Convert: `pq_xtz_day = totalTransferred / 1_000_000`

If the endpoint does not return all history in one page, paginate using **offset** parameter. Cache the raw JSON to 03_data/raw/tzkt/xtz_statistics_daily.json (gitignored path).

A3. Cross-check the series

Before building, verify the series is not degenerate: 1. **Start date:** first date should be ~2018-06-30 (Tezos mainnet genesis). 2. **Recent magnitude:** sum the last 30 available days of **totalTransferred** / 1e6 = recent monthly XTZ volume. Convert at recent XTZ price. Compare to the DeFiLlama chain DEX figure for Tezos (~\$16k/30d from **rung_table.csv**). The TzKT native volume should be **orders of magnitude larger** than DEX-only (this is the whole point — the **mat_ratio=0.000045** in the rung table means DEX captured only 0.0045% of cap, but native settlement value is a real number). If the TzKT figure is implausibly small (in the thousands USD/month), flag and investigate before building. 3. **No BTC-default contamination** (not applicable here — TzKT is Tezos-specific, not a multi-chain chart site that silently substitutes BTC).

A4. Aggregate to monthly and compute PQ in USD

```
import calendar, json
import pandas as pd

# Load raw
with open("03_data/raw/tzkt/xtz_statistics_daily.json") as f:
    data = json.load(f)

# Daily XTZ volume
rows = []
for r in data:
    date_str = r["date"][:10] # 'YYYY-MM-DD'
    xtz = r["totalTransferred"] / 1_000_000.0
    rows.append({"date": date_str, "pq_xtz": xtz})
daily = pd.DataFrame(rows)
daily["date"] = pd.to_datetime(daily["date"])

# Month-end bucket
daily["month_end"] = daily["date"].apply(
    lambda d: pd.Timestamp(d.year, d.month, calendar.monthrange(d.year, d.month)[1])
)
monthly_xtz = daily.groupby("month_end")["pq_xtz"].sum().reset_index()
monthly_xtz["month_end"] = monthly_xtz["month_end"].dt.strftime("%Y-%m-%d")

# Join XTZ price from universe_panel
panel = pd.read_csv("03_data/universe_panel.csv")
panel["month_end"] = pd.to_datetime(panel["month_end"]).dt.strftime("%Y-%m-%d")
xtz_price = panel[(panel["cmc_id"] == 2011) & (panel["status"] == "observed")][["month_end", "price"]]

merged = monthly_xtz.merge(xtz_price, on="month_end", how="inner")
merged["pq_usd"] = merged["pq_xtz"] * merged["price"]
merged = merged[merged["pq_usd"] > 0]

# Columns for pq_coins.csv
merged["cmc_id"] = 2011
merged["symbol"] = "XTZ"
merged["pq_source"] = "tzkt_total_transferred"
merged["rung"] = "R3-tzkt"
merged["note"] = (
    "Tezos native totalTransferred (aggregate daily XTZ transferred on-chain, "
    "in mutez / 1e6 = XTZ). Source: api.tzkt.io/v1/statistics/daily (official "
    "TzKT indexer, free, keyless, historical). Summed daily->monthly, multiplied "
    "by month-end CMC price. This is native settlement value, not DEX-only volume."
)
```

A5. Update pq_coins.csv

Remove the existing XTZ NaN-marker row and append the new monthly rows:

```
pq = pd.read_csv("03_data/phase2/pq_coins.csv")
pq = pq[pq["cmc_id"] != 2011] # drop old XTZ row(s)
new_xtz = merged[["cmc_id", "symbol", "month_end", "pq_usd", "pq_source", "rung", "note"]]
pq = pd.concat([pq, new_xtz], ignore_index=True)
pq = pq.sort_values(["cmc_id", "month_end"], na_position="last")
```

```

pq.to_csv("03_data/phase2/pq_coins.csv", index=False)
print(f"XTZ rows added: {len(new_xtz)} months, range {new_xtz['month_end'].min()} -> {new_xtz['month_end'].max()}")
print(f"PQ range: ${new_xtz['pq_usd'].min()/1e6:.1f}M -- ${new_xtz['pq_usd'].max()/1e6:.1f}M/month")

```

Task B — MATIC PQ probe

MATIC (cmc_id=3890) lambda runs 2020-06 -> 2024-08 (50 months; ends at POL handoff).

B1. Try bitinfocharts

The Phase 2b script only tested 13 known tickers. Try matic live:

```

import urllib.request, re

url = "https://bitinfocharts.com/comparison/sentinusd-matic.html"
html = urllib.request.urlopen(
    urllib.request.Request(url, headers={"User-Agent": "Mozilla/5.0"}), timeout=30
).read().decode("utf-8", "ignore")

rows = re.findall(r'new Date/("(\/d{4})\/(\/d{2})\/(\/d{2})"/), (null|[0-9.eE+-]+)', html)
valid = [(f"{y}/{m}/{d}", float(v)) for y,m,d,v in rows if v != "null"]

# BTC-default guard: fetch BTC reference and assert MATIC differs
btc_url = "https://bitinfocharts.com/comparison/sentinusd-btc.html"
btc_html = urllib.request.urlopen(
    urllib.request.Request(btc_url, headers={"User-Agent": "Mozilla/5.0"}), timeout=30
).read().decode("utf-8", "ignore")
btc_rows = re.findall(r'new Date/("(\/d{4})\/(\/d{2})\/(\/d{2})"/), (null|[0-9.eE+-]+)', btc_html)
btc_last = float([v for _,_,v in btc_rows if v != "null"][-1])

if valid:
    matic_last = valid[-1][1]
    if abs(matic_last - btc_last) < 1e-6:
        print("MATIC series == BTC default -> bitinfocharts does not cover MATIC")
    else:
        print(f"MATIC series found: {len(valid)} days, recent ${matic_last/1e6:.1f}M/day")
else:
    print("MATIC: empty or unparseable -> bitinfocharts does not cover MATIC")

```

B2. If bitinfocharts fails: mark still-gapped

If MATIC is not on bitinfocharts, update its NaN-marker row with a more precise note:

```

pq_source: "NaN: polygon_native_volume_no_free_source"
note: "MATIC/Polygon: bitinfocharts does not expose 'sentinusd-matic' (tested live
      2026-07-25 --- returns BTC default or empty). Polygon native MATIC transfer value
      requires PolygonScan full-history export (no free API) or Artemis (paid).
      DeFiLlama Polygon chain DEX volume degenerate (not settlement value). lambda window
      ends 2024-08 at POL handoff --- short window reduces regression value.
      MATIC stays PQ=NaN."

```

Do NOT try PolygonScan raw block iteration — forbidden by Entry 31/32.

Task C — JOE and SAFE classification review (no build needed)

JOE (cmc_id=11396)

Classification_table says: `asset_class=coin, classification_basis="consensus tags: ['staking']"`. JOE is the Trader Joe DEX governance token on Avalanche/Arbitrum — not a native gas coin for any chain. Its lambda (6 months) comes from a vote-escrow staking mechanism (veJOE).

Finding: There is no valid PQ (chain settlement value) for JOE — it is not a chain's native coin. It also has no TVL in the DeFiLlama panel. This token is stuck regardless of classification. Document in the DATA_DECISIONS_LOG that JOE's `pq_nvtgl` gap is permanent (governance token misclassified as coin; no applicable NVT denominator; no TVL path either). Do not attempt a PQ probe.

SAFE (cmc_id=21585)

Same situation: Safe governance token, not a chain native. `classification_basis="consensus tags: ['staking']"`. No valid PQ path and no TVL. Document same conclusion.

DYDX (cmc_id=28324)

dYdX is the native token of the dYdX Chain (a Cosmos L1) — legitimately a coin. However it has only 1 month of lambda (far too short for regression). Note it and skip for now; revisit if lambda extends via a Cosmos key.

Task D — Rebuild NVT_GL panel

After updating `pq_coins.csv` (Task A + any B update):

```
python 04_code/phase2_nvt_gl.py
python 04_code/build_coverage_status.py
```

Verify XTZ appears in `nvt_gl_panel.csv` with non-null `nvt_gl` rows for the months where both lambda and NVT_GL overlap (should be the full 78-month lambda window since TzKT goes back to 2018-06).

Print the overlap count:

```
nvt = pd.read_csv("03_data/phase2/nvt_gl_panel.csv")
lam = pd.read_csv("03_data/phase1/lambda_panel.csv")
xtz_nvt = nvt[(nvt["cmc_id"]==2011) & nvt["nvt_gl"].notna()]
xtz_lam = lam[lam["cmc_id"]==2011]
print(f"XTZ NVT_GL months: {len(xtz_nvt)}")
print(f"XTZ lambda months: {len(xtz_lam)}")
overlap = set(xtz_nvt["month_end"]) & set(xtz_lam["month_end"])
print(f"XTZ regression-ready overlap: {len(overlap)} months")
```

DATA_DECISIONS_LOG — Entry 83

Append to `04_code/DATA_DECISIONS_LOG.md`:

```
### Entry 83 --- Session 033: XTZ NVT_GL BUILT via TzKT.io; MATIC still gapped; JOE/SAFE closed
```

```
XTZ (2011): Phase 2b's GAP-R2 note was incorrect --- Cosmos LCD is not the XTZ indexer.
TzKT.io (official Tezos indexer) serves free, keyless aggregate daily statistics via
/v1/statistics/daily including `totalTransferred` (native settlement value in mutez).
[N] months of PQ built ([date_range]); PQ range $[X]M--$[Y]M/month; regression-ready
overlap with lambda: [N] months. pq_source = "tzkt_total_transferred", rung = "R3-tzkt".
```

MATIC (3890): bitinfocharts does not cover 'matic' ticker (BTC-default guard triggered / empty). Polygon native transfer value has no free keyless source. lambda window ends 2024-08 at POL handoff. MATIC stays PQ=NaN; note updated with precise reason.

JOE (11396) and SAFE (21585): Permanently closed. Classified as coin via staking tag but are DEX/service governance tokens with no native chain. No valid PQ denominator exists (not a chain gas coin); also no TVL path. Gap is architectural, not a data-sourcing problem. DYDX (28324): 1-month lambda too short for regression; skip until lambda extends via Cosmos key.

Post-rebuild: coins regression-ready (lambda?NVT_GL): [20->N].

Session report

Write 03_data/SESSION033_XTZ_MATIC_NVT_REPORT.md: - XTZ: months fetched, date range, PQ USD range, cross-check passed/failed, regression overlap - MATIC: bitinfocharts probe result, final verdict - JOE/SAFE/DYDX: classification notes - Post-rebuild NVT_GL counts and regression-ready total

Commit

```
git add -A
git commit -m "session 033: XTZ NVT_GL built via TzKT; MATIC/JOE/SAFE gaps closed"
git push
/newpage
```

CLAUDE_CODE_SESSION034_COMBINED_PROMPT.md

Claude Code Session 034 — CHZ ch1 + Blockchair XTZ/MATIC + EVM DeFi Breadth Batch 1

Date: 2026-07-25 **Etherscan Pro key:** 04_code/.api_keys.json under "etherscan" (needed for Task D only) **No key needed:** Tasks A, B, C use public RPC / Blockchair free tier

Starting state (post-033): - lambda: 9,648 asset-months / 342 assets; regression-ready 143 (coins 20, tokens/other 123) - channel2_holding.csv: 307 tokens / 9,867 rows - No changes to guard thresholds; from == to self-transfer skip stays

BEFORE STARTING: Sleep/hibernate already set to Never (session 031 fix). Pause Windows Update (Settings -> Windows Update -> Pause for 7 days) before Task D.

Task A — CHZ ch1 staking (Chiliz Chain RPC, ~1–2h)

A1. Context and anchor

CHZ (cmc_id=4066) has 0 lambda months — ch1_staking[pos_possible] gate was the “semantics anchor” check (Entry 79 / SESSION029 probe). That gate is now OPEN:

Anchor confirmed 2026-07-25: staking.chiliz.com shows **2,416,757,292 CHZ** staked, which matches the balance of the Chiliz Chain staking contract 0x00 (2.38B last read, 1.5% drift = expected growth). This is a native-balance staking contract (same pattern as XDC's 0x...0088), NOT an event-replay target. Build via balance reads at month-end blocks.

Chain: Chiliz Chain 2.0 (chainid=88888). NOT on Etherscan Pro V2 (confirmed in Entry 78 — probed live). Use the public Chiliz RPC directly.

A2. RPC connection

Try in order:

```
CHILIZ_RPCS = [
    "https://chiliz.drpc.org",          # drpc pattern (same as ronin.drpc.org in prod)
    "https://rpc.ankr.com/chiliz",      # Ankr fallback
    "https://rpc.chiliz.io",           # official fallback
]
```

Test with a simple `eth_blockNumber` call. Use the first that responds.

A3. Month-end block lookup (binary search)

Etherscan Pro's `getblocknobytime` does NOT cover chainid=88888. Use binary search via `eth_getBlockByNumber` to find the last block of each month-end (UTC 23:59:59):

```
import calendar, time, json, requests

def chiliz_rpc(rpc_url, method, params):
    r = requests.post(rpc_url, json={"jsonrpc": "2.0", "id": 1, "method": method, "params": params},
                      timeout=30)
    return r.json()["result"]

def block_at_ts(rpc_url, target_ts, lo=0, hi=None):
    """Binary search for block closest to target_ts (from below)."""
    if hi is None:
        hi = int(chiliz_rpc(rpc_url, "eth_blockNumber", []), 16)
    while lo < hi - 1:
        mid = (lo + hi) // 2
        b = chiliz_rpc(rpc_url, "eth_getBlockByNumber", [hex(mid), False])
        if b is None:
            hi = mid; continue
        if int(b["timestamp"], 16) <= target_ts:
            lo = mid
        else:
            hi = mid
    return lo

def month_end_blocks(yms, rpc_url, cache_file):
    """yms = ["2023-07", "2023-08", ...]; returns {ym: block_number}."""
    import os, json
    cache = json.loads(open(cache_file).read()) if os.path.exists(cache_file) else {}
    for ym in yms:
        if ym in cache and cache[ym]:
            continue
        y, m = int(ym[:4]), int(ym[5:7])
        last_day = calendar.monthrange(y, m)[1]
        ts = int(calendar.timegm(
            time.strptime(f"{ym}-{last_day:02d} 23:59:59", "%Y-%m-%d %H:%M:%S")))
        b = block_at_ts(rpc_url, ts)
        # verify the block we found is actually <= target_ts
        blk = chiliz_rpc(rpc_url, "eth_getBlockByNumber", [hex(b), False])
```


API. Free tier: ~30 req/day. The one-time pull needs ~96 monthly points for XTZ and ~48 for MATIC — doable over a few minutes with 1s sleep between calls.

No Blockchain key required for free tier. If a key exists in `.api_keys.json` under "blockchair", use `?key=<key>` parameter; otherwise proceed keyless.

B1. XTZ — Tezos native transfer volume

Blockchair's Tezos model uses `calls` (internal operations) and `operations`. The field for value transferred is `amount` (in mutez = 1e-6 XTZ).

Probe one month first to confirm the endpoint shape:

```
import requests, time

def blockchair_get(url, params=None):
    r = requests.get(url, params=params, timeout=30,
                     headers={"User-Agent": "academic-research/1.0"})
    return r.json()

# Probe: Tezos - try multiple entity types to find native transfer value
for entity in ["calls", "operations", "transactions"]:
    url = f"https://api.blockchair.com/tezos/{entity}"
    r = blockchair_get(url, {"a": "sum(amount)", "q": "time(2024-01)",
                             "limit": 1})
    print(f"tezos/{entity}: status={r.get('context', {}).get('code', '?')} "
          f"data={r.get('data')}")
    time.sleep(1)
```

Pick the entity type that returns a non-zero `sum(amount)`. Then build monthly series:

```
rows = []
for ym in month_list("2018-06", "2026-05"): # XTZ mainnet genesis 2018-06
    y, m = ym[:4], ym[5:]
    r = blockchair_get(f"https://api.blockchair.com/tezos/{ENTITY}",
                       {"a": f"sum(amount)", "q": f"time({ym})"})
    val_mutez = r["data"][0].get("sum(amount)", 0)
    rows.append({"month_end": month_end_str(ym), "pq_xtz": val_mutez / 1e6})
    time.sleep(1.2) # free tier rate limit
```

BTC-default guard: after fetching, compare a 2024-01 XTZ value to the Bitcoin equivalent call. If XTZ `sum(amount)` matches BTC to 6 decimal places: abort.

B2. MATIC — Polygon native transfer volume

Polygon is EVM. Native MATIC transferred is in the `value` field (Wei = 1e18).

```
# Probe one month
r = blockchair_get("https://api.blockchair.com/polygon/transactions",
                   {"a": "sum(value)", "q": "time(2024-01)", "limit": 1})
print(f"polygon/transactions sum(value): {r.get('data')}")
```

If that works, build series for `month_list("2020-06", "2024-08")` (MATIC lambda window).

B3. If both work: update `pq_coins.csv` and rebuild `NVT_GL`

Remove existing NaN-marker rows for XTZ (`cmc_id=2011`) and/or MATIC (`cmc_id=3890`), append new monthly rows, sort, save. Then:

```
python 04_code/phase2_nvt_gl.py
python 04_code/build_coverage_status.py
```

Print regression-ready overlap counts for XTZ and MATIC.

B4. If Blockchair rate-limits or requires paid key

Document in the session report exactly what was tried and what error was returned. If the only blocker is a paid key (~\$30/month), flag it clearly in the report so Moazzam can decide: “Blockchair key needed for XTZ and/or MATIC — \$30/month plan covers both in a one-time pull.”

Do NOT subscribe without explicit approval.

Task C — EVM DeFi Breadth Batch 1 (102 tokens, ~147k getLogs, ~3–8h)

These are DeFi-sector EVM tokens (DEX, Lending, Liquid Staking, Yield, Bridge, Derivatives, etc.) in the not_started universe, sorted by estimated getLogs ascending (smallest first). All are buildable via the existing Etherscan Pro V2 multi-chain engine. No engine changes.

C1. Run the build

WORKLIST=11461,33981,1888,2593,38417,5830,23177,3367,3418,17050,1882,2576,7672,35818,1768,39125,2503,29

Notable tokens (survivorship bias targets): - CEL (2700, ~2.4k gl) — Celsius, DEAD, Yield Aggregator — include historical TVL check - MULTI (17050, ~337 gl) — Multichain, DEAD, Bridge — include historical TVL check - FTT (4195, ~2.5k gl) — FTX Token, DEAD — Exchange Token/Derivatives - LEND (2239 — in Batch 2) — old Aave v1, MIGRATED — Lending, historical TVL on DeFiLlama - SAI (2308 — in Batch 2) — old MakerDAO single-collateral, DEAD — CDP

Engine rules (unchanged): - VAL_CAP_MULT = CONTAM_MULT = 100 — do NOT change - from == to self-transfer skip — do NOT remove - B2: no month exceeds 100x contamination guard - B4: screened HODL-6m in [0, 80%]; flag but do not suppress if outside

Expected behavior: - Most tokens have est_gl < 2k -> very fast (seconds each) - Tokens with est_gl 3k–5k -> a few minutes each - Total wall-clock ~3–8h depending on rate-limit behavior - No long silences expected (these are small Ethereum mainnet tokens)

C2. DeFiLlama TVL slug lookup (after build completes)

For every token successfully built in Batch 1, check DeFiLlama for a protocol slug. This step determines which ch2 builds become regression-ready (lambda?TVL).

```
import requests, json, time
from pathlib import Path

# Fetch full protocol list
protos = requests.get("https://api.llama.fi/protocols", timeout=60).json()

# For each built token, try symbol match then address match
built_ids = [...] # list of cmc_ids that were just built
id_map = {...} # cmc_id -> {symbol, address, chainid}

matches = []
for cmc_id in built_ids:
    sym = id_map[cmc_id]["symbol"].lower()
    addr = id_map[cmc_id].get("address", "").lower()
```

```

candidates = [p for p in protos
               if p.get("symbol","").lower() == sym
               or addr in str(p.get("address","")).lower()]
if candidates:
    # pick the one with most TVL history
    best = max(candidates, key=lambda p: p.get("tvl", 0))
    matches.append({"cmc_id": cmc_id, "dl_slug": best["slug"],
                   "dl_name": best["name"], "tvl": best.get("tvl",0)})
    print(f" MATCH: cmc_id={cmc_id} {sym} -> {best['slug']} (TVL ${best.get('tvl',0)/1e6:.1f}M)")
else:
    print(f" NO MATCH: cmc_id={cmc_id} {sym}")

print(f"/n{n(len(matches))/len(built_ids)} tokens matched on DeFiLlama")

```

For each match, fetch TVL history and append to 03_data/phase2/tvl_panel.csv using the same schema as existing TVL rows. DeFiLlama historical TVL endpoint: <https://api.llama.fi/protocol/{slug}> -> tvl array with {date, totalLiquidityUSD}.

Dead protocol note: DeFiLlama archives historical TVL even for dead/bankrupt protocols (CEL, MULTI, LEND, SAI). Fetch their history exactly like live protocols. This is the survivorship bias defense — the low-TVL, low-lambda region of the regression.

Task D — Assemble and rebuild

```
python 04_code/phase1_assemble_lambda.py
python 04_code/build_coverage_status.py
```

Print: - New lambda asset-months and asset count - New regression-ready count (tokens with both ch2 and TVL) - How many of the 102 Batch 1 tokens found TVL slugs

DATA_DECISIONS_LOG — Entry 84

Append to 04_code/DATA_DECISIONS_LOG.md:

```

### Entry 84 --- Session 034: CHZ ch1 BUILT; Blockchair XTZ/MATIC [probe result];
EVM DeFi Breadth Batch 1 (102 tokens)

```

```

CHZ (4066): ch1 staking built via Chiliz Chain 2.0 public RPC (eth_getBalance on
0x...1000 staking contract at month-end blocks; anchor drift [X]% vs 2,416,757,292
CHZ confirmed 2026-07-25). Window: [start]..2026-05 ([N] months). Staking ratio range
[X%]..[Y%]. pq_source suffix: chiliz-chain-pubRPC.

```

```

Blockchair XTZ: [BUILT via blockchair/tezos/{entity} sum(amount) mutez / 1e6 |
FAILED: <reason>]. [If built]: [N] months, PQ range $[X]M..[Y]M/month. Coins
regression-ready [20->21].

```

```

Blockchair MATIC: [BUILT | FAILED: <reason>]. [If built]: [N] months.

```

```

EVM DeFi Breadth Batch 1: [N] of 102 tokens built (B2/B4 pass/flag per token below
or in session report). [M] tokens matched DeFiLlama TVL slugs. [K] new
regression-ready tokens (lambda?TVL). Dead protocol builds: CEL, MULTI, [others].

```

```

Post-assemble: lambda [X] asset-months / [N] assets. Regression-ready [143->N].

```

Remaining EVM DeFi breadth batches:

Batch 2 (session 035): 13 tokens, ~119k getLogs --- WORKLIST provided in 034 prompt

Batch 3 (session 036): stETH + MEME, ~110k getLogs

Batch 4 (session 037): SHIB alone, ~128k getLogs

Session report

Write 03_data/SESSION034_COMBINED_REPORT.md: - CHZ: months built, staking ratio range, cross-check result - Blockchair XTZ: probe result (pass/fail, months if pass, PQ range) - Blockchair MATIC: probe result - EVM Batch 1: token-level table (symbol, cmc_id, getLogs actual, months built, B2/B4 status, TVL slug found Y/N) - Post-assemble lambda and regression-ready totals

Commit

```
git add -A
git commit -m "session 034: CHZ ch1 built; Blockchair XTZ/MATIC probe; EVM DeFi batch 1 (102 tokens)"
git push
```

Reference: remaining EVM DeFi batches (for follow-on sessions)

Session 035 — Batch 2 (13 tokens, ~119k getLogs, run immediately after quota resets):

WORKLIST=3783,2120,5824,1982,1757,2586,2165,5266,2239,28568,9436,2308,2691 PYTHONUTF8=1 python 04_code/

Notable: SNX (Synthetix, 7k gl), LEND (old Aave v1, 10.7k gl), SAI (old MakerDAO, 14.1k gl), EETH (EtherFi, 12.8k gl), PNT (pNetwork bridge, 14.3k gl)

Session 036 — Batch 3a (2 tokens, ~110k getLogs):

WORKLIST=8085,28301 PYTHONUTF8=1 python 04_code/phase1_channel2_stream.py

stETH (49k gl) — Lido's core staked ETH token, massive TVL; MEME (61k gl)

Session 037 — Batch 3b (1 token, ~128k getLogs):

WORKLIST=5994 PYTHONUTF8=1 python 04_code/phase1_channel2_stream.py

SHIB (Shiba Inu, 128k gl) — largest meme/DeFi-hybrid build; no protocol TVL expected

/newpage

CLAUDE_CODE_SESSION036_STETH_MEME_PROMPT.md

Claude Code Session 036 — EVM DeFi Breadth Batch 3a: stETH + MEME

Date: 2026-07-25 **Etherscan Pro key:** 04_code/.api_keys.json under "etherscan"

Starting state (post-035): - lambda: 13,191 asset-months / 457 assets; regression-ready 177 (coins 21, tokens/other 156) - channel2_holding.csv: 421 tokens / 13,580 rows - Engine unchanged: 04_code/phase1_channel2_stream.py

BEFORE STARTING: Pause Windows Update. Sleep/hibernate already set to Never.

Task A — Build

WORKLIST=8085,28301 PYTHONUTF8=1 python 04_code/phase1_channel2_stream.py

cmc_id	symbol	chain (chainid)	est_getLogs	sector
8085	stETH	Ethereum (1)	~49k	Liquid Staking;Restaking
28301	MEME	Ethereum (1)	~61k	Dexs;Farm;NFT;Meme

Total estimate: ~110k getLogs, well under the 185k daily cap. Expected runtime: 4-8h.

Engine rules (unchanged): - VAL_CAP_MULT = CONTAM_MULT = 100 — do NOT change - from == to self-transfer skip — do NOT remove - B2: no month exceeds 100x contamination guard - B4: flag but do not suppress if screened HODL-6m median > 80%

Task B — TVL slug check

stETH (8085)

stETH IS the Lido liquid staking receipt token. Its market cap ~ the ETH staked in Lido by construction — **NV/TVL is circular** (same rule as wstETH, weETH, cbETH, rETH excluded in session 034). Do NOT assign a TVL slug.

Document in the session report: “stETH excluded from NV/TVL regression (LST receipt circularity — NV~TVL by construction, Entry 84 rule). lambda months retained for the conviction-only analysis.”

The correct Lido TVL slug (lido) is already referenced by other tokens if applicable. Do not create a stETH -> lido mapping.

MEME (28301)

Check DeFiLlama: <https://api.llama.fi/protocols> — search for MEME by symbol and by any known contract address. Expect no match (no dedicated protocol TVL for a meme/farm token). If a match is found, verify it is not a coincidental symbol clash before accepting.

Task C — Assemble and rebuild

```
python 04_code/phase1_assemble_lambda.py
python 04_code/build_coverage_status.py
```

Print new lambda totals and regression-ready count. Regression-ready is NOT expected to change (both tokens are lambda-only adds due to TVL circularity/no-match).

DATA_DECISIONS_LOG — Entry 86

Append to 04_code/DATA_DECISIONS_LOG.md:

```
### Entry 86 --- Session 036: stETH + MEME ch2 built (lambda-only; no TVL regression entry)
```

```
stETH (8085/Ethereum): [N] getLogs / [tf] transfers / [scr] screened months.
B2 pass. B4 [pass/flag]. TVL excluded: LST receipt circularity (NV~Lido TVL by
construction, Entry 84 rule). lambda months retained for conviction-only panel.
```

MEME (28301/Ethereum): [N] getLogs / [tf] transfers / [scr] screened months.
B2 pass. B4 [pass/flag]. TVL: no DeFiLlama protocol match confirmed.

Post-assemble: lambda [X] asset-months / [N] assets. Regression-ready [177->N] (no change expected).

Remaining EVM breadth: Session 037 --- SHIB (5994), ~128k getLogs, lambda-only.

Session report

Write 03_data/SESSION036_BATCH3A_REPORT.md: - Per-token: getLogs actual, transfers, screened months, B2/B4, TVL decision - Post-assemble lambda and regression-ready totals - Note: regression-ready unchanged; next count movement requires user actions (Subscan key -> DOT/KSM, CORE key, Cosmos key)

Commit

```
git add -A
git commit -m "session 036: stETH + MEME ch2 built (lambda-only; no TVL entry)"
git push
/newpage
```

CLAUDE_CODE_SESSION037_DOT_KSM_CORE_PROMPT.md

Claude Code Session 037 — DOT/KSM ch1 (Subscan era_stat) + CORE ch1 probe/build

Date: 2026-07-27 **Keys in use:** 04_code/.api_keys.json -> "subscan" (DOT/KSM), "coredao" (CORE)

Starting state (post-036): - lambda: 13,272 asset-months / 459 assets; regression-ready 177 (coins 21, tokens/other 156) - DOT (cmc 6636) + KSM (cmc 5034): have NVT_GL; need ch1 -> each becomes regression-ready - CORE (cmc 23254): 40 NVT_GL months; needs ch1 -> regression-ready - Expected post-session: coins 21 -> 24 regression-ready (DOT + KSM + CORE if ch1 confirmed)

BEFORE STARTING: Pause Windows Update. Sleep/hibernate set to Never.

Task A — DOT ch1 (Subscan era_stat)

A0: Probe response structure (1 API call)

```
import json, requests
key = json.loads(open("04_code/.api_keys.json").read())["subscan"]
r = requests.post(
    "https://polkadot.api.subscan.io/api/scan/staking/era_stat",
    headers={"Content-Type": "application/json", "X-API-Key": key},
    json={"row": 2, "page": 0}
)
print(json.dumps(r.json(), indent=2))
```

Print the full response. Identify: - The field name for total bonded (likely `bonded_total`, `total_bonded`, or similar) — it will be a string in Planck units - Whether `start_block_num` / `end_block_num` are present (for

timestamp calculation) - Whether any direct timestamp field is present - Confirm `data.count` gives total era count

Then fetch genesis timestamp:

```
r0 = requests.post(
    "https://polkadot.api.subscan.io/api/scan/block",
    headers={"Content-Type": "application/json", "X-API-Key": key},
    json={"block_num": 0}
)
print("DOT genesis block:", json.dumps(r0.json()["data"], indent=2))
```

Record `block_timestamp` from the response as `DOT_GENESIS_TS`.

A1: Paginate all DOT eras

```
import datetime as dt

DOT_GENESIS_TS = <from_probe>    # Unix timestamp of block 0
DOT_BLOCK_SECS = 6                # Polkadot target block time

all_eras = []
page = 0
while True:
    r = requests.post(
        "https://polkadot.api.subscan.io/api/scan/staking/era_stat",
        headers={"Content-Type": "application/json", "X-API-Key": key},
        json={"row": 100, "page": page},
        timeout=30
    )
    data = r.json()["data"]
    batch = data.get("list", [])
    if not batch:
        break
    all_eras.extend(batch)
    print(f" page {page}: {len(batch)} eras, running total {len(all_eras)}/{data['count']}")
    if len(all_eras) >= data["count"]:
        break
    page += 1
    time.sleep(0.2)    # polite pacing (free plan: 30 req/s)
print(f"Total DOT eras fetched: {len(all_eras)}")
```

Expected: ~2000–2200 eras (22 pages). Runtime < 30 seconds.

A2: Map eras to months, convert Planck->DOT

```
from collections import defaultdict

bonded_field = "<field_name_from_probe>"    # e.g. "bonded_total"
block_field  = "end_block_num"              # or "start_block_num" if end not present

by_ym = defaultdict(lambda: (0, -1))      # ym -> (bonded_planck, era_num)

for e in all_eras:
    era_num    = e["era"]
    end_block  = e[block_field]
    bonded_str = e[bonded_field]
```

```

    if not bonded_str or bonded_str in ("0", None):
        continue
    bonded_planck = int(bonded_str)
    ts = DOT_GENESIS_TS + end_block * DOT_BLOCK_SECS
    ym = dt.datetime.utcnow().timestamp(ts).strftime("%Y-%m")
    # Keep last era in each month (highest era_num = most recent snapshot)
    cur_bonded, cur_era = by_ym[ym]
    if era_num > cur_era:
        by_ym[ym] = (bonded_planck, era_num)

# Convert to DOT (1 DOT = 10^10 Planck)
dot_series = {ym: bonded / 1e10 for ym, (bonded, _) in by_ym.items()}

# Drop pre-staking months (NPoS active from ~2020-08)
dot_series = {ym: v for ym, v in dot_series.items() if ym >= "2020-08"}
print(f"DOT months: {min(dot_series)}..{max(dot_series)}")
print(f"Latest staked: {dot_series[max(dot_series)]:.0f} DOT")

```

A3: Cross-check (DOT)

```

# Fetch the latest era fresh (bypass any cache)
r_fresh = requests.post(
    "https://polkadot.api.subscan.io/api/scan/staking/era_stat",
    headers={"Content-Type": "application/json", "X-API-Key": key},
    json={"row": 1, "page": 0} # most recent era
)
latest_era = r_fresh.json()["data"]["list"][0]
latest_fresh = int(latest_era[bonded_field]) / 1e10

latest_ym = max(dot_series)
latest_ours = dot_series[latest_ym]
drift = (latest_ours - latest_fresh) / latest_fresh
print(f"DOT cross-check: ours={latest_ours:.0f} fresh={latest_fresh:.0f} drift={drift:.2%}")
# ACCEPT if abs(drift) < 5% (Entry-26 standard)
# REJECT if drift > 5% --- investigate pagination order / off-by-one

```

Task B — KSM ch1 (same algorithm, different constants)

B0: Probe KSM genesis timestamp

```

r0_ksm = requests.post(
    "https://kusama.api.subscan.io/api/scan/block",
    headers={"Content-Type": "application/json", "X-API-Key": key},
    json={"block_num": 0}
)
print("KSM genesis:", json.dumps(r0_ksm.json()["data"], indent=2))
KSM_GENESIS_TS = <from_response>

```

Also verify era count:

```

r_ksm_probe = requests.post(
    "https://kusama.api.subscan.io/api/scan/staking/era_stat",
    headers={"Content-Type": "application/json", "X-API-Key": key},
    json={"row": 1, "page": 0}
)

```

```
)
total_ksm_eras = r_ksm_probe.json()["data"]["count"]
print(f"KSM total eras: {total_ksm_eras}")    # expect ~8,000--11,000
```

B1: Paginate all KSM eras

Same loop as DOT but `kusama.api.subscan.io`. Expected ~100 pages. KSM has more eras (~6-hour era vs DOT's 24-hour era), so allow ~2 minutes.

B2: Map to months

Same algorithm. Constants: - **KSM**: 1 KSM = 10^{12} Planck (NOT 10^{10} — KSM never redenominated) - KSM_BLOCK_SECS = 6 - Drop months before 2019-09 (KSM staking pre-NPoS era unreliable)

```
ksm_series = {ym: bonded / 1e12 for ...}
```

B3: Cross-check (KSM)

Same as DOT cross-check but `kusama.api.subscan.io`.

Task C — CORE ch1 probe + conditional build

CORE = Core DAO native coin, `cmc_id=23254`. Chain launched 2023-01-14. Key stored as "coredao" in `.api_keys.json`. Entry 78 probe target: `/api/stats/staking_summary` on `openapi.coredao.org`.

C0: Probe the `staking_summary` endpoint

```
core_key = json.loads(open("04_code/.api_keys.json").read())["coredao"]
```

```
# Try both common auth patterns
import requests
```

```
# Pattern A: apikey as query param (Etherscan-compatible)
r = requests.get(
    "https://openapi.coredao.org/api/stats/staking_summary",
    params={"apikey": core_key},
    timeout=30
)
print("Pattern A:", r.status_code, r.text[:500])
```

```
# Pattern B: X-API-Key header
r2 = requests.get(
    "https://openapi.coredao.org/api/stats/staking_summary",
    headers={"X-API-Key": core_key},
    timeout=30
)
print("Pattern B:", r2.status_code, r2.text[:500])
```

Read the response carefully:

- If the response contains a time-series array (daily/weekly totals going back to 2023): use it directly — extract `staked_CORE` per period, bucket to months, convert units to CORE, build `channel1_core.csv`. Go to step C1-timeseries.
- If the response is current-only (single snapshot, no history): pivot to block-level reads. Go to step C1-blocklevel.

- **If 401/403 even with key:** try alternate base URL `https://scan.coredao.org/api/stats/staking_summary` with same auth patterns. If all fail, log CORE as “API probe failed — investigate key format” and skip CORE build this session.

C1-timeseries (if historical series available)

Parse the returned series. Map each data point to a calendar month (last point in month = snapshot). Identify the unit of the staked amount (likely in CORE or in wei). Convert to CORE if needed (1 CORE = 10^{18} wei if EVM, or check decimals field in response).

Cross-check: latest month in series vs live `module=stats&action=coinsupply` or similar endpoint.

Output: `03_data/phase1/channel1_core.csv`

Source string: `"openapi.coredao.org:staking_summary"`

C1-blocklevel (if current-only)

Identify the CORE staking contract — try these in order:

Step 1 — check if Core (chainid=1116) is on Etherscan Pro V2:

```
ev2 = requests.get("https://api.etherscan.io/v2/chainlist",
    params={"apikey": <etherscan_key>}, timeout=20).json()
core_in_v2 = any(str(c.get("chainid")) == "1116" for c in ev2.get("result", []))
print("Core (1116) in Etherscan V2:", core_in_v2)
```

Step 2 — find the staking/pledge contract on CoreScan: Core DAO’s Satoshi Plus uses a `PledgeAgent` contract that holds delegated CORE. Try the CoreScan verified contracts API:

```
r_src = requests.get(
    "https://openapi.coredao.org/api",
    params={"module": "contract", "action": "getsourcecode",
        "address": "0x0000000000000000000000000000000000000000000000000000000000000000",
        "apikey": core_key},
    timeout=20
)
print(r_src.status_code, r_src.text[:300])
```

(System contract `0x...1001` is a common address for staking controllers on EVM chains. Try also `0x...1000`, `0x...1002`.)

If a verified staking/pledge contract is found, get its address. Then:

Step 3 — block-level balance at month-end blocks:

For each month-end from 2023-01 to 2026-05: 1. Get an approximate month-end block number: - CORE genesis: 2023-01-14. Block time: ~3 seconds. - Block at month-end M: $(M_{ts} - CORE_GENESIS_TS) / 3$ 2. Use CoreScan `module=account&action=balance&address=<staking_contract>&tag=<hex_block>&apikey=<key>` if available, OR `eth_getBalance` via JSON-RPC. 3. Convert from wei (divide by $1e18$) to get CORE staked.

Cross-check: latest block balance vs live `eth_getBalance` call.

Output: `03_data/phase1/channel1_core.csv`

Source string: `"openapi.coredao.org:PledgeAgent.balance(month-end block)"`

C2: If CORE cannot be built this session

Log: “CORE (23254) ch1 probe: [describe what endpoint returned]. Block-level approach: [describe outcome]. Deferred to Session 038 for manual staking contract identification.”

Do NOT ship a partial or guessed series.

Task D — Assemble and rebuild coverage

```
python 04_code/phase1_assemble_lambda.py
python 04_code/build_coverage_status.py
```

Print new lambda totals and regression-ready counts. - If DOT + KSM built: coins regression-ready 21 -> 23 - If CORE also built: coins regression-ready -> 24 - Tokens/other should be unchanged.

DATA_DECISIONS_LOG — Entry 87

Append to 04_code/DATA_DECISIONS_LOG.md:

```
### Entry 87 --- Session 037: DOT + KSM ch1 BUILT via Subscan era_stat; CORE ch1 probe
```

```
**DOT (6636):** [N] eras paginated across [P] pages.
Bonded_field = [field_name]. Genesis_ts = [ts].
[N_months] months built ([start]..[end]). Latest staked: [X] DOT.
Cross-check: ours=[X] / fresh=[Y] / drift=[D%] --- [PASS/FAIL].
Source: subscan-polkadot:era_stat.bonded_total / 1e10.
DOT enters regression-ready (has NVT_GL): coins [21->22] or [21->23] (with KSM).
```

```
**KSM (5034):** [N] eras paginated across [P] pages.
KSM_genesis_ts = [ts]. 1 KSM = 1e12 Planck.
[N_months] months built ([start]..[end]). Latest staked: [X] KSM.
Cross-check: ours=[X] / fresh=[Y] / drift=[D%] --- [PASS/FAIL].
Source: subscan-kusama:era_stat.bonded_total / 1e12.
KSM enters regression-ready: coins [22->23].
```

```
**CORE (23254):** Probe result: [describe staking_summary response].
[BUILT / DEFERRED / FAILED] --- [brief description].
If built: [N_months] months ([start]..[end]). Latest staked: [X] CORE.
Cross-check: [result]. Source: [source string].
If deferred: [reason + next action].
```

```
Post-assemble: lambda [13,272->X] asset-months / [459->N] assets.
Regression-ready coins: [21->N] (DOT, KSM[, CORE] now complete).
```

Session report

Write 03_data/SESSION037_DOT_KSM_CORE_REPORT.md: - Per-asset: eras fetched, months built, bonded field, Planck conversion, drift - CORE: full probe trace and outcome (built or deferred) - Post-assemble lambda and regression-ready totals - Next steps

Commit

```
git add -A
git commit -m "session 037: DOT + KSM ch1 built (Subscan era_stat); CORE ch1 [built/deferred]"
git push
```

Technical notes

DOT Planck: 1 DOT = 10^{10} Planck. Polkadot redenominated in August 2020 (100 old DOT -> 1 new DOT). The Planck unit itself did not change. The `bonded_total` from Subscan is expressed in Planck throughout — dividing by $1e10$ always gives post-redenomination DOT.

KSM Planck: 1 KSM = 10^{12} Planck. No redenomination ever occurred on Kusama.

Subscan pagination order: eras are returned oldest-first (era 0 at page 0). Paginate forward to collect all history. The final page will have fewer than 100 entries.

Rate limiting: Subscan free plan supports 30 requests/second. With `time.sleep(0.2)` between pages, actual rate is ~5 req/s — well within limits. Total wall-clock for DOT (~22 pages) + KSM (~100 pages) should be under 5 minutes.

Month-end bucketing convention: all other ch1 files use the last calendar day of the month as `month_end` (e.g., 2024-01-31). Use `calendar.monthrange(yr, mo)[1]` to get the last day and format as YYYY-MM-DD. The `staked_native` value for that row is the last-era snapshot within that calendar month.

Cross-check standard (Entry 26): drift < 5% = PASS; 5–10% = PASS with flag; > 10% = REJECT (do not ship, investigate).

schema (matches all other `channel1_*.csv` files):

`cmc_id, symbol, month_end, staked_native, circulating_supply, staking_ratio, source, flag`

Join `circulating_supply` from `03_data/universe_panel.csv` on (`cmc_id, month_end`) using YYYY-MM matching. `staking_ratio = staked_native / circulating_supply` (null if `circulating_supply` missing or zero).

CORE chain genesis: Core DAO mainnet launched 2023-01-14. The earliest valid ch1 month is 2023-01. NVT_GL has 40 months for CORE 23254; typical window overlap will be ~29 months (2023-01->2026-05).

Worklist for Sessions 038+: - Session 038: SHIB (5994) ch2, ~128k getLogs est (likely over-estimate), lambda-only - WARP (1166): identity-map review (built-but-empty, Entry 79) - Cosmos key -> CRO, INJ, SEI, KAVA ch1 (+4 coins) - Blockchair support email -> XTZ/MATIC NVT_GL (before paying ~\$30/mo) - Task #22: bibliography sanity-check

/newpage

CLAUDE_CODE_SESSION038_SHIB_PROMPT.md

Claude Code Session 038 — SHIB ch2 (EVM DeFi Breadth Batch 3b)

Date: 2026-07-27 **Etherscan Pro key:** `04_code/.api_keys.json` under "etherscan"

Starting state (post-037): - lambda: 13,449 asset-months / 462 assets; regression-ready 178 (coins 22, tokens/other 156) - `channel2_holding.csv`: 423 tokens / 13,661 rows - Engine unchanged: `04_code/phase1_channel2_stream.py`

BEFORE STARTING: Pause Windows Update. Sleep/hibernate set to Never.

Task A — Build

WORKLIST=5994 PYTHONUTF8=1 python 04_code/phase1_channel2_stream.py

cmc_id	symbol	chain (chainid)	contract	est_getLogs	sector
5994	SHIB	Ethereum (1)	0x95aD61b0a150d79819dCF64E1E6Cc0f8B464cE	128k	DeFi-Basic

Estimate note: sessions 034–036 ran 3–5x below holder-count estimates due to long sparse pre-activity blocks. Expect actual getLogs ~25–35k; budget 2–4h not 10h+. If actual getLogs exceeds 150k, stop and report — do not continue past the daily cap.

Engine rules (unchanged): - VAL_CAP_MULT = CONTAM_MULT = 100 — do NOT change - from == to self-transfer skip — do NOT remove - B2: no month exceeds 100x contamination guard - B4: flag but do not suppress if screened HODL-6m median > 80%

Task B — TVL slug check

SHIB is a meme token with **no direct protocol TVL**. Do NOT assign a TVL slug.

The only DeFiLlama slug containing “shib” is **shibaswap** — the ShibaSwap DEX. That protocol’s governance token is **BONE** (cmc_id=11865), not SHIB. Assigning **shibaswap** to SHIB would double-count the DEX TVL against the wrong token.

Document in the session report: “SHIB excluded from NV/TVL regression — meme token, no direct protocol TVL. **shibaswap** slug belongs to BONE (cmc_id=11865), not SHIB.”

SHIB is lambda-only.

Task C — Assemble and rebuild

```
python 04_code/phase1_assemble_lambda.py
python 04_code/build_coverage_status.py
```

Print new lambda totals and regression-ready count. Regression-ready is NOT expected to change (SHIB is lambda-only).

DATA_DECISIONS_LOG — Entry 89

Append to 04_code/DATA_DECISIONS_LOG.md:

```
### Entry 89 --- Session 038: SHIB ch2 built (lambda-only; no TVL regression entry)
```

```
SHIB (5994/Ethereum): [N] getLogs / [tf] transfers / [scr] screened months.
B2 pass. B4 [pass/flag]. Actual getLogs vs 128k estimate: [ratio]x.
```

```
TVL: no protocol TVL for SHIB. `shibaswap` slug in tvl_panel belongs to BONE
(cmc_id=11865, ShibaSwap governance token) --- assigning to SHIB would be wrong.
SHIB -> lambda-only.
```

```
Post-assemble: lambda [X] asset-months / [N] assets. Regression-ready [178->N]
(no change expected).
```

```
EVM DeFi breadth complete (all batches 1--3b done). Next: Session 039 --- DOT/KSM PQ
source probe; TRX coin_staking_type fix; WARP identity review.
```

Session report

Write 03_data/SESSION038_SHIB_REPORT.md: - getLogs actual vs 128k estimate (note the overestimate ratio)
- transfers, screened months, B2/B4 - TVL decision with BONE/shibaswap disambiguation - Post-assemble
lambda and regression-ready totals

Commit

```
git add -A
git commit -m "session 038: SHIB ch2 built (lambda-only; no TVL entry)"
git push
/newpage
```

CLAUDE_CODE_SESSION039_DOTKSM_PQ_FIXES_PROMPT.md

Claude Code Session 039 — DOT/KSM PQ probe + TRX fix + WARP review

Date: 2026-07-28 **Keys in use:** 04_code/.api_keys.json -> "subscan", "etherscan"

Starting state (post-038): - lambda: 13,510 asset-months / 463 assets; regression-ready 178 (coins 22, tokens/other 156) - DOT (6636) + KSM (5034): ch1 BUILT (session 037), but pq_usd ALL NULL in nvt_gl_panel.csv — not yet regression-ready. Finding a PQ source for either would add 1–2 coins immediately. - TRX (1958): mislabeled `pow_only` in coverage logic; DPoS, has 78 ch1 months. - WARP (1166): ch2 built-but-empty (Entry 79 identity mismatch).

Task A — DOT/KSM PQ source probe

DOT and KSM need annualized on-chain transfer volume in USD (`pq_usd`) to compute `NVT_GL`. The goal here is to find a free source with multi-year history. Run probes in order and stop at the first that returns usable data.

A1: Subscan daily stats — “transfer” category (same key as ch1)

The session-037 pivot notes the `/api/scan/daily` endpoint returned `history_window_exceeded` for the “Bonded” category (staking). The “transfer” category may have a different (longer) history window. Probe it:

```
import json, requests

key = json.loads(open("04_code/.api_keys.json").read())["subscan"]

# DOT probe --- last 90 days of transfer data to check depth
r = requests.post(
    "https://polkadot.api.subscan.io/api/scan/daily",
    headers={"Content-Type": "application/json", "X-API-Key": key},
    json={"start": "2020-08-01", "end": "2026-06-30",
         "format": "month", "category": "transfer"}
)
print("DOT transfer status:", r.status_code)
d = r.json()
print(json.dumps(d, indent=2)[:1000])
```

Also try categories "extrinsic", "transaction", "fee".

If historical monthly transfer data returns (multi-year): proceed to A2. **If history_window_exceeded:** try A3.

A2: Build DOT + KSM PQ series from Subscan daily stats

If Subscan returns usable monthly transfer amounts:

- Field to use: whichever of `transfer_amount`, `amount`, `volume_usd`, or similar contains native-token transfer volume (inspect the response keys).
- Convert to USD using `universe_panel.csv` price column for that month-end.
- Sum or take the month-end value as appropriate.
- Run for both `polkadot.api.subscan.io` and `kusama.api.subscan.io`.
- Output: append rows to `03_data/phase2/pq_coins.csv` and rebuild `03_data/phase2/nvt_gl_panel.csv` via `python 04_code/phase2_nvt_gl.py`.
- Cross-check: the resulting NVT_GL ratio for DOT/KSM should be broadly consistent with other L1 coins (typical range 10–200).

A3: Polkadot archive RPC — enumerate transfer extrinsics (BLOCKED by Entry 31/32)

Raw multi-year block iteration is **FORBIDDEN** (Entry 31/32 rule). Do NOT attempt to scan blocks one-by-one for transfer extrinsics. If Subscan daily stats fail, move to A4.

A4: Blockchair Polkadot/Kusama aggregation probe

Blockchair supports DOT and KSM natively (not EVM). Try the chain-level stats and daily aggregation endpoints:

```
# Current stats (free, keyless)
r_stats = requests.get("https://api.blockchair.com/polkadot/stats", timeout=20)
print("Blockchair DOT stats:", r_stats.status_code, r_stats.text[:500])

# Daily aggregated transfer volume (the same endpoint that failed for XTZ/MATIC
# in session 034 --- but Polkadot may be implemented differently)
r_agg = requests.get(
    "https://api.blockchair.com/polkadot/calls",
    params={"a": "date(time),sum(value)", "q": "type(transfer)"},
    timeout=30
)
print("Blockchair DOT aggregation:", r_agg.status_code, r_agg.text[:500])
```

If aggregation returns multi-year daily/monthly data: parse it, convert to USD, append to `pq_coins.csv`, rebuild NVT_GL panel.

If 404 or blocked: note it, log CORE result in Entry 90 as “DOT/KSM PQ no free source found — candidates exhausted, Blockchair paid tier or Subscan Pro required”, and continue to Task B.

Task B — TRX coin_staking_type fix

TRX (`cmc_id=1958`) is classified `pow_only` in `universe_coverage_status.csv` but TRON uses DPoS (Delegated Proof of Stake) and has 78 ch1 staking months already built. This mislabel makes the coin count in coverage reports inaccurate.

Find where `coin_staking_type` is set. It is likely either: 1. A column in `03_data/universe_panel.csv` (or a static metadata file) 2. Computed inline in `04_code/build_coverage_status.py`

```
import subprocess
result = subprocess.run(
    ["grep", "-rn", "pow_only/|coin_staking_type/|1958", "04_code/", "--include=*.py"],
    capture_output=True, text=True
)
print(result.stdout)
```

Once found, change TRX's entry from `pow_only` to `pos` (DPoS is a PoS variant). Re-run `python 04_code/build_coverage_status.py` and confirm: - TRX moves from the PoW bucket to PoS - Regression-ready coin count is consistent with 22 (TRX already has both `ch1` and `NVT_GL`, so fixing the label may shift it from “partial” to “complete”)

Important: do NOT change any logic that affects whether TRX's `ch1` data is used in the `lambda` panel — that data is already correctly included. This is a metadata label fix only.

Task C — WARP identity review

WARP (`cmc_id=1166`) has a `ch2` checkpoint but all months are “pre-history” (zero transfers in every observed month) because the stored contract address (`0x83e6f1E41cdd28eAcEB20Cb649155049Fac3D5Aa` or similar) postdates the CMC listing's active window (2016..2018-02).

Check:

```
import json
from pathlib import Path
```

```
# Find the WARP checkpoint
```

```
ck_files = list(Path("03_data/raw/phase1_ch2").glob("1166_*.json"))
print("WARP checkpoints:", ck_files)
```

```
if ck_files:
```

```
    ck = json.loads(ck_files[0].read_text())
```

```
    print("address:", ck.get("address"))
```

```
    print("chainid:", ck.get("chainid"))
```

```
    print("n_transfers:", ck.get("n_transfers"))
```

```
    print("months with data:", [m for m, v in ck.get("monthly", {}).items() if v.get("screened_hodl_6m")])
```

Then look up `cmc_id=1166` on CoinMarketCap (via web or the universe panel) to confirm what “WARP” actually is: - If the listing refers to a pre-2018 dead project that never deployed an ERC-20 token, close as **permanent identity mismatch** — no usable data possible. - If the listing refers to a live project with a different contract address, update the identity mapping and rebuild.

In either case: document the decision in Entry 90 and update coverage status to `not_started` (if dead/mismatch) or trigger a rebuild (if remapped).

DATA_DECISIONS_LOG — Entry 90

Append to `04_code/DATA_DECISIONS_LOG.md`:

```
### Entry 90 --- Session 039: DOT/KSM PQ probe; TRX label fix; WARP review
```

```
**DOT/KSM PQ probe:**
```

```
Subscan /api/scan/daily category=transfer: [result --- returned/history_exceeded/error].
```

```
Blockchair polkadot/calls aggregation: [result --- returned/404/blocked].
```

```
Decision: [BUILT: append to pq_coins + rebuild NVT_GL -> coins +2 regression-ready]
```

```
[or: no free PQ source found; Subscan Pro or Blockchair paid required]
```

```

**TRX (1958) coin_staking_type fix:**
Label was pow_only; corrected to pos (DPoS, has ch1 78 months).
Source of label: [file/line]. build_coverage_status.py re-run.
TRX coverage status: [partial->complete if NVT_GL present, else unchanged].

**WARP (1166) identity review:**
Contract [address]. Observed window [dates]. Transfer count [N].
Decision: [permanent identity mismatch, closed as not_started]
          [or: remapped to [new address], rebuild triggered].

Post-assemble: lambda [X] / [N] assets. Regression-ready [178->N].

```

Session report

Write 03_data/SESSION039_DOTKSM_PQ_FIXES_REPORT.md: - DOT/KSM probe: exact endpoints tried, response content (first 200 chars), verdict - TRX fix: where the label lived, what changed, new coverage counts - WARP: checkpoint address, transfer count, identity verdict - Post-assemble totals

Commit

```

git add -A
git commit -m "session 039: DOT/KSM PQ probe; TRX staking-type fix; WARP identity review"
git push

/newpage

```

CLAUDE_CODE_SESSION040_COSMOS_LCD_PROMPT.md

Claude Code Session 040 — CRO / INJ / KAVA / SEI ch1 via Cosmos Archive LCD

Date: 2026-07-28 **Keys in use:** none — approach is free and keyless **API keys file:** 04_code/.api_keys.json (not needed for this session)

Starting state (post-039): - lambda: 13,510 asset-months / 463 assets; regression-ready 178 (coins 22, tokens/other 156) - All four target chains have NVT_GL (pq_usd non-null) but zero ch1 lambda months - Building ch1 for all four immediately makes them regression-ready -> expected 178 -> 182 - Output file naming convention: 03_data/phase1/channel1_cosmos_lcd.csv (same schema as channel1_dot_ksm.csv: cmc_id, symbol, month_end, staked_native, circulating_supply, staking_ratio, source, flag)

Chain registry

cmc_id	symbol	Cosmos chain	denom	decimals	genesis (approx)	blocks/day
3635	CRO	crypto-org-chain-mainnet-1	basecro	8	2021-03-25	~14,400 (~6s/block)
7226	INJ	injective-1	inj	18	2021-11-08	~28,800 (~3s/block)
4846	KAVA	kava_2222-10	ukava	6	2019-11-15	~14,400 (~6s/block)
23149	SEI	pacific-1	usei	6	2023-08-15	~216,000 (~0.4s/block)

CRO decimal warning: CRO uses 8 decimals (basecro), NOT 6. Do NOT divide by 10^6 . **INJ decimal warning:** INJ uses 18 decimals, same as Ethereum ERC-20 native wei.

Task A — Probe: find an archive-capable LCD for each chain

For each chain, test the LCD candidates in the order listed below. Stop at the first one that passes BOTH the liveness check and the archive depth test.

Candidate LCD endpoints (try in this order)

```
CHAIN_CONFIG = {
  "CRO": {
    "cmc_id": 3635,
    "symbol": "CRO",
    "denom": "basecro",
    "decimals": 8,
    "lcd_candidates": [
      "https://rest.crypto.org",
      "https://api-cryptoorgchain-ia.cosmosia.notional.ventures",
      "https://rest.cosmos.directory/cryptoorgchain",
    ],
    "rpc": "https://rpc.crypto.org",
    "genesis_date": "2021-03-25",
    "blocks_per_day": 14400,
  },
  "INJ": {
    "cmc_id": 7226,
    "symbol": "INJ",
    "denom": "inj",
    "decimals": 18,
    "lcd_candidates": [
      "https://sentry.lcd.injective.network",
      "https://injective-api.polkachu.com",
      "https://rest.cosmos.directory/injective",
    ],
    "rpc": "https://sentry.tm.injective.network",
    "genesis_date": "2021-11-08",
    "blocks_per_day": 28800,
  },
  "KAVA": {
    "cmc_id": 4846,
    "symbol": "KAVA",
    "denom": "ukava",
    "decimals": 6,
    "lcd_candidates": [
      "https://api.kava.io",
      "https://kava-api.polkachu.com",
      "https://rest.cosmos.directory/kava",
    ],
    "rpc": "https://rpc.kava.io",
    "genesis_date": "2019-11-15",
    "blocks_per_day": 14400,
  },
}
```

```

"SEI": {
    "cmc_id": 23149,
    "symbol": "SEI",
    "denom": "usei",
    "decimals": 6,
    "lcd_candidates": [
        "https://rest.sei-apis.com",
        "https://sei-api.polkachu.com",
        "https://rest.cosmos.directory/sei",
    ],
    "rpc": "https://rpc.sei-apis.com",
    "genesis_date": "2023-08-15",
    "blocks_per_day": 216000,
},
}

```

Probe script

```

import requests, json, time
from datetime import datetime, timezone

def probe_chain(sym, cfg, timeout=20):
    """Try each LCD candidate. Return (lcd_url, latest_height) for first archive-capable node."""
    for lcd in cfg["lcd_candidates"]:
        # Step 1: liveness check --- current pool
        try:
            r = requests.get(f"{lcd}/cosmos/staking/v1beta1/pool", timeout=timeout)
            if not r.ok:
                print(f" {lcd} pool: {r.status_code} --- skip")
                continue
            live_bonded = r.json()["pool"]["bonded_tokens"]
            print(f" {lcd} LIVE --- bonded={live_bonded[:20]}...")
        except Exception as e:
            print(f" {lcd} UNREACHABLE: {e}")
            continue

        # Step 2: get latest block height
        try:
            rb = requests.get(
                f"{lcd}/cosmos/base/tendermint/v1beta1/blocks/latest", timeout=timeout
            )
            latest_height = int(rb.json()["block"]["header"]["height"])
            print(f" {lcd} latest height={latest_height}")
        except Exception as e:
            print(f" {lcd} cannot read latest block: {e}")
            continue

        # Step 3: archive depth test --- probe block ~365 days ago
        old_height = max(1, latest_height - cfg["blocks_per_day"] * 365)
        try:
            ra = requests.get(
                f"{lcd}/cosmos/staking/v1beta1/pool",
                headers={"x-cosmos-block-height": str(old_height)},
                timeout=timeout,
            )

```

```

    )
    if ra.ok:
        bonded_old = ra.json()["pool"]["bonded_tokens"]
        print(f" {lcd} ARCHIVE OK @ height {old_height}: bonded={bonded_old[:20]}...")
        return lcd, latest_height
    else:
        print(f" {lcd} archive probe FAILED @ {old_height}: {ra.status_code} {ra.text[:80]}")
except Exception as e:
    print(f" {lcd} archive probe error: {e}")
time.sleep(0.5)

return None, None # no archive node found

for sym, cfg in CHAIN_CONFIG.items():
    print(f"/n=== {sym} ===")
    lcd, height = probe_chain(sym, cfg)
    if lcd:
        print(f" --> USE: {lcd} (latest height {height})")
    else:
        print(f" --> NO ARCHIVE NODE FOUND for {sym}")

```

If no archive node found for a chain: document in Entry 91 and skip that chain. The remaining chains with archive nodes proceed to Task B.

Task B — Binary search for month-end blocks

For each chain with a confirmed archive LCD, find the block height at the last moment of each month-end date within the observed window (from `universe_panel.csv`).

```

import pandas as pd
from datetime import datetime, timezone, timedelta
import calendar

def parse_block_time(lcd, height, timeout=15):
    """Return UTC datetime for block at given height."""
    r = requests.get(
        f"{lcd}/cosmos/base/tendermint/v1beta1/blocks/{height}",
        timeout=timeout
    )
    t_str = r.json()["block"]["header"]["time"]
    # RFC3339 with nanoseconds: "2021-03-31T23:45:12.345678901Z"
    # Truncate nanoseconds to microseconds for fromisoformat compatibility
    t_str = t_str[:26].rstrip('Z') + '+00:00'
    return datetime.fromisoformat(t_str)

def find_month_end_block(lcd, year, month, lo, hi, timeout=15):
    """
    Binary search for the last block at or before the last second of month-end.
    lo, hi: known block heights bracketing the target.
    Returns (block_height, block_time_utc).
    """
    # Target: last second of the last day of the month
    last_day = calendar.monthrange(year, month)[1]
    target = datetime(year, month, last_day, 23, 59, 59, tzinfo=timezone.utc)

```

```

while hi - lo > 1:
    mid = (lo + hi) // 2
    t = parse_block_time(lcd, mid, timeout)
    if t <= target:
        lo = mid
    else:
        hi = mid
    time.sleep(0.05) # be gentle

# Verify lo is at or before target, hi is after
t_lo = parse_block_time(lcd, lo, timeout)
return lo, t_lo

# Build month list for each chain
def get_observed_months(cmc_id):
    """Month-ends where universe_panel has an observed row for this asset."""
    panel = pd.read_csv("03_data/universe_panel.csv")
    rows = panel[(panel["cmc_id"] == cmc_id) & (panel["status"] == "observed")]
    months = sorted(rows["month_end"].str[:7].unique())
    return months # list of "YYYY-MM" strings

```

Efficiency: binary search takes ~17 iterations per month-end ($2^{17} = 131,072$ blocks). At 0.05s per call, ~1 second per month. For 60 months -> ~60s per chain. Acceptable.

Use the `latest_height` from Task A as the upper bound. Use 1 as the lower bound (or the known genesis block if faster). If the genesis block postdates the first observed month, skip months before genesis.

Task C — Build the ch1 series

For each chain x each observed month:

```

def build_chain_series(sym, cfg, lcd, latest_height):
    """Build staking series for one chain. Return list of row dicts."""
    cmc_id = cfg["cmc_id"]
    decimals = cfg["decimals"]
    denom = cfg["denom"]

    # Load circulating supply from universe panel
    panel = pd.read_csv("03_data/universe_panel.csv")
    panel_cid = panel[
        (panel["cmc_id"] == cmc_id) & panel["cmc_supply_circ"].notna()
    ][["month_end", "cmc_supply_circ"]].set_index("month_end")

    months = get_observed_months(cmc_id)
    from datetime import date
    import calendar

    rows = []
    lo = 1
    for ym in months:
        year, mon = int(ym[:4]), int(ym[5:7])
        # Skip months before chain genesis
        genesis = datetime.fromisoformat(cfg["genesis_date"]).replace(tzinfo=timezone.utc)

```

```

last_day = calendar.monthrange(year, mon)[1]
month_end_dt = datetime(year, mon, last_day, 23, 59, 59, tzinfo=timezone.utc)
if month_end_dt < genesis:
    continue

# Find month-end block via binary search
block_h, block_t = find_month_end_block(lcd, year, mon, lo, latest_height)
lo = block_h # monotonic: next month's lo >= this month's block

# Query pool at that block
r = requests.get(
    f"{lcd}/cosmos/staking/v1beta1/pool",
    headers={"x-cosmos-block-height": str(block_h)},
    timeout=20,
)
if not r.ok:
    print(f" {sym} {ym}: pool query failed at height {block_h}: {r.status_code}")
    continue

bonded_raw = int(r.json()["pool"]["bonded_tokens"])
staked_native = bonded_raw / (10 ** decimals)

# Circulating supply from panel
month_end_str = f"{year}-{mon:02d}-{last_day:02d}"
if month_end_str not in panel_cid.index:
    print(f" {sym} {ym}: no circulating supply in panel --- skip")
    continue
circ = float(panel_cid.loc[month_end_str, "cmc_supply_circ"])
staking_ratio = staked_native / circ if circ > 0 else None

rows.append({
    "cmc_id": cmc_id,
    "symbol": sym,
    "month_end": month_end_str,
    "staked_native": round(staked_native, 4),
    "circulating_supply": round(circ, 4),
    "staking_ratio": round(staking_ratio, 6) if staking_ratio else None,
    "source": f"cosmos_lcd_pool@{block_h}",
    "flag": f"{denom} / 10^{decimals}; block_time={block_t.isoformat()[:19]}Z",
})
print(f" {sym} {ym}: height={block_h}, staked={staked_native:,.0f}, ratio={staking_ratio:.4f}")

return rows

```

Collect all rows from all chains that passed the archive probe. Write to output:

```

import csv
from pathlib import Path

all_rows = []
for sym, cfg in CHAIN_CONFIG.items():
    # only process chains where probe succeeded
    if sym not in archive_confirmed:
        continue
    lcd, latest_height = archive_confirmed[sym]

```

```

chain_rows = build_chain_series(sym, cfg, lcd, latest_height)
all_rows.extend(chain_rows)
print(f"/n{sym}: {len(chain_rows)} months built")

if all_rows:
    out_path = Path("03_data/phase1/channel1_cosmos_lcd.csv")
    df_out = pd.DataFrame(all_rows)
    df_out.to_csv(out_path, index=False)
    print(f"/nWrote {len(df_out)} rows to {out_path}")
    print(df_out.groupby("symbol")[["month_end", "staking_ratio"]].agg(
        {"month_end": ["count", "min", "max"], "staking_ratio": "mean"}
    ))

```

Task D — Cross-check (Entry-26 standard)

For each chain built, compare the latest month's `staked_native` against the live current pool endpoint (no height header — returns the present-day value):

```

for sym, cfg in CHAIN_CONFIG.items():
    if sym not in archive_confirmed:
        continue
    lcd = archive_confirmed[sym][0]
    decimals = cfg["decimals"]

    # Live pool (current)
    r_live = requests.get(f"{lcd}/cosmos/staking/v1beta1/pool", timeout=15)
    live_bonded = int(r_live.json()["pool"]["bonded_tokens"]) / (10 ** decimals)

    # Our latest built month
    df_sym = df_out[df_out["symbol"] == sym]
    last_row = df_sym.sort_values("month_end").iloc[-1]
    built_val = last_row["staked_native"]

    # Drift: built is month-end of last panel month (may be 1--3 months ago)
    # so some drift is expected from staking in/out since then.
    # A <5% drift PASS confirms we're reading the right chain + denom.
    # For chains where staking has shifted significantly, note it but don't hard-fail.
    drift_pct = abs(built_val - live_bonded) / live_bonded * 100
    status = "PASS" if drift_pct < 5 else "WARN (staking shifted since last panel month)"
    print(f"{sym}: built_latest={built_val:,.0f} live={live_bonded:,.0f} drift={drift_pct:.2f}% {sta

```

If drift > 20% for any chain, investigate — likely a wrong denom or decimal error. A drift of 5–15% is acceptable given that live values are 1–3 months after our last panel month.

Task E — Coverage label fix

CRO (3635) and KAVA (4846) are currently labeled `pos_possible` in `03_data/universe_coverage_status.csv`. Now that we have confirmed archive staking data, update them to `pos`.

First check current labels for all four:

```

cov = pd.read_csv("03_data/universe_coverage_status.csv")
print(cov[cov["cmc_id"].isin([3635, 7226, 4846, 23149])][

```

```
["cmc_id", "symbol", "coin_staking_type", "coverage_status"]
])
```

For any that show `pos_possible` and for which we successfully built `ch1`, update:

```
# Edit universe_coverage_status.csv directly:
# for each chain we built, set coin_staking_type = "pos" (it IS a PoS chain)
for cid in [cid for cid in [3635, 7226, 4846, 23149] if cid in built_cids]:
    cov.loc[cov["cmc_id"] == cid, "coin_staking_type"] = "pos"
cov.to_csv("03_data/universe_coverage_status.csv", index=False)
```

Do NOT change the `coverage_status` column manually — it will be recomputed in Task F.

Task F — Assemble + rebuild coverage

```
python 04_code/phase1_assemble_lambda.py
python 04_code/build_coverage_status.py
```

Expected outcomes: - `lambda` panel: +N asset-months (where N = total months built across all 4 chains) -
Each built chain: `coverage_status partial` -> `complete` (has `ch1` + `NVT_GL pq_usd`) - Regression-ready:
178 -> **182** (if all 4 chains built; fewer if some archive probes fail)

Print the new regression-ready breakdown:

```
cov = pd.read_csv("03_data/universe_coverage_status.csv")
print("Regression-ready:", (cov["coverage_status"] == "complete").sum())
print(cov[cov["cmc_id"].isin([3635, 7226, 4846, 23149])][
    ["cmc_id", "symbol", "coverage_status", "lambda_months"]
])
```

DATA_DECISIONS_LOG — Entry 91

Append to `04_code/DATA_DECISIONS_LOG.md`:

```
### Entry 91 --- Session 040: CRO/INJ/KAVA/SEI ch1 via Cosmos Archive LCD
```

```
**Approach:** free, keyless Cosmos SDK LCD `cosmos/staking/v1beta1/pool` with  

`x-cosmos-block-height` header at month-end blocks found via binary search on  

`cosmos/base/tendermint/v1beta1/blocks/{height}`.
```

```
**Archive probe results:**
```

```
- CRO: [archive node URL] --- PASS / FAIL
- INJ: [archive node URL] --- PASS / FAIL
- KAVA: [archive node URL] --- PASS / FAIL
- SEI: [archive node URL] --- PASS / FAIL
```

```
**Decimal notes:**
```

```
- CRO: basecro, 108 (NOT 106 like standard Cosmos chains)
- INJ: inj, 1018 (EVM-compatible native denom)
- KAVA: ukava, 106
- SEI: usei, 106
```

```
**Cross-check results (latest month vs live pool):**
```

```
- CRO drift: [X]% --- [PASS/WARN]
- INJ drift: [X]% --- [PASS/WARN]
```

```

- KAVA drift: [X]% --- [PASS/WARN]
- SEI drift: [X]% --- [PASS/WARN]

**Coverage label fixes:** coin_staking_type pos_possible -> pos for
[list chains updated].

**Post-assemble:** lambda [X] asset-months / [N] assets.
Regression-ready [178 -> N] (coins [22 -> N], tokens/other 156 unchanged).

Output: 03_data/phase1/channel1_cosmos_lcd.csv

```

Session report

Write 03_data/SESSION040_COSMOS_LCD_REPORT.md: - Archive probe table: each chain, endpoint tried, liveness result, archive result - Months built per chain + date range - Cross-check table: built vs live, drift % - Coverage label changes - Post-assemble lambda and regression-ready totals - Any chains that failed archive probe (with reason)

Commit

```

git add -A
git commit -m "session 040: CRO/INJ/KAVA/SEI ch1 via Cosmos archive LCD; regression-ready 178->N"
git push

```

Failure-mode guidance

Failure	Diagnosis	Action
All LCD candidates return 404/503	Chain may have migrated endpoints	Check official chain docs; try https://chains.cosmos.directory registry for updated URLs
Archive probe returns “height X was not found”	Node is pruned (non-archive)	Move to next candidate
Archive probe returns “request failed” with state mismatch	Migration / chain upgrade at that height	Try a slightly different height (+/-5,000 blocks)
<code>bonded_tokens</code> field missing in pool response	Different Cosmos SDK version	Check for <code>not_bonded_tokens</code> + <code>bonded_tokens</code> in response structure; some chains nest differently
Decimal mismatch (staking ratio > 1.0)	Wrong decimals used	Re-check chain denom registry; CRO=8, INJ=18, others=6
SEI binary search timeout (216k blocks/day)	Large height range	Narrow lo/hi bracket by estimating from known timestamps before searching

/newpage

CLAUDE_CODE_SESSION041_HXRO_SXP_OSMO_PROMPT.md

Claude Code Session 041 — HXRO ch2 extend + SXP probe + OSMO ch1

Date: 2026-07-30 **Keys in use:** 04_code/.api_keys.json -> "etherscan" (Tasks A + B) **ETHERSCAN SUBSCRIPTION NOTE:** Today is the last day of the paid Etherscan Pro subscription. Run Tasks A and B before the subscription lapses. Task C (OSMO) is free/keyless and can run in any order.

Starting state (post-040): - lambda: 13,547 asset-months / 465 assets; regression-ready **180** (coins 24, tokens/other 156) - 6 tokens have TVL but are not regression-ready: HXRO, SXP, CASINO, OSMO, RUNE, SUN - Of these, only HXRO has a confirmed Etherscan-reachable EVM address (chainid 1)

Task A — HXRO (3748) ch2 re-run / extend ? HIGHEST PRIORITY

Why re-run: HXRO has 24 ch2 months in channel2_holding.csv (2020-09-30 -> 2022-09-30) but its TVL in tvl_panel.csv starts 2023-02-28. Zero overlap -> stuck at partial. The checkpoint 03_data/raw/phase1_onchain/holding/3748_HXRO.json has monthly: [] (empty) and last_block: null, so a fresh run will rebuild from genesis and naturally extend through 2026-05-31 — creating 40 months of overlap with TVL and making HXRO regression-ready.

Etherscan address: 0x4bd70556ae3f8a6ec6c4080a0c327b24325438f3 (chainid 1, Ethereum) **Estimated getLogs:** ~534 (tiny — expect < 30 minutes wall time)

WORKLIST=3748 PYTHONUTF8=1 python 04_code/phase1_channel2_stream.py

Expected output: HXRO monthly series 2020-09 -> 2026-05 (old + new months). After build, check checkpoint monthly keys include 2023-02 onward.

```
import json
ck = json.loads(open("03_data/raw/phase1_onchain/holding/3748_HXRO.json").read())
months = sorted(ck.get("monthly", {}).keys())
print("HXRO months built:", len(months), months[:3], "...", months[-3:])
# Expect >= 2023-02-28 to appear
overlap_months = [m for m in months if m >= "2023-02-28"]
print("Overlap with TVL:", len(overlap_months))
```

Guards (unchanged): - VAL_CAP_MULT = CONTAM_MULT = 100 — do NOT change - from == to self-transfer skip — do NOT remove - B2: no month exceeds 100x contamination guard - B4: flag but do not suppress if screened HODL-6m median > 80%

Do NOT run assemble yet — wait until all tasks are done.

Task B — SXP (4279) manual probe

universe_coverage_status.csv shows SXP as evm_chain: non-EVM, etherscan_reachable: no but asset_onchain_identity.csv lists address 0x8ce9137d39326ad0cd6491fb5cc0cba0e089b6a9 with no chain prefix, which is the known Ethereum ERC-20 deployment of SXP (Swipe). The non-EVM flag likely came from the coverage builder not recognising an unprefixed multi-chain address.

Probe: confirm if this address is a live EVM ERC-20 on Ethereum (chainid 1)

```
import json, requests
```

```
key = json.loads(open("04_code/.api_keys.json").read())["etherscan"]
```

```

addr = "0x8ce9137d39326ad0cd6491fb5cc0cba0e089b6a9"

# 1. Contract ABI check (confirms it's a contract on Ethereum)
r_abi = requests.get(
    "https://api.etherscan.io/v2/api",
    params={"chainid": 1, "module": "contract", "action": "getabi",
           "address": addr, "apikey": key},
    timeout=15
)
print("ABI status:", r_abi.json().get("status"), r_abi.json().get("message"))

# 2. Token supply check (confirms ERC-20)
r_supply = requests.get(
    "https://api.etherscan.io/v2/api",
    params={"chainid": 1, "module": "stats", "action": "tokensupply",
           "contractaddress": addr, "apikey": key},
    timeout=15
)
print("Supply:", r_supply.json().get("result", "n/a")[:20])

# 3. Quick getLogs probe (most recent 2000 blocks)
import web3 # or use eth_getLogs via etherscan if web3 unavailable
r_logs = requests.get(
    "https://api.etherscan.io/v2/api",
    params={"chainid": 1, "module": "logs", "action": "getLogs",
           "address": addr, "fromBlock": "latest", "toBlock": "latest",
           "topic0": "0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef",
           "apikey": key},
    timeout=15
)
print("getLogs probe:", r_logs.json().get("status"), len(r_logs.json().get("result", [])), "events")

```

Decision logic: - If ABI + supply both return OK AND getLogs probe returns without error: **Run SXP ch2**
— update `asset_onchain_identity.csv` to add prefix `ethereum:0x8ce9137d39326ad0cd6491fb5cc0cba0e089b6a9`
and set `evm_chainid=1`, then `WORKLIST=4279 PYTHONUTF8=1 python 04_code/phase1_channel2_stream.py`
- If address not found or ABI status “NOTOK”: **Document as permanent gap** — SXP’s ERC-20 may have been migrated/deprecated. Move to BSC probe below.

BSC fallback probe (if Ethereum fails):

```

# SXP BSC address (known Binance-chain deployment)
bsc_addr = "0x47BEAd2563dCBf3bF2c9407fEa4dC236fAbA485A"
r_bsc = requests.get(
    "https://api.etherscan.io/v2/api",
    params={"chainid": 56, "module": "contract", "action": "getabi",
           "address": bsc_addr, "apikey": key},
    timeout=15
)
print("BSC ABI:", r_bsc.json().get("status"), r_bsc.json().get("message"))

```

If BSC address is live, run ch2 on chainid 56 instead.

If both fail: close SXP as a permanent gap in Entry 92.

Task C — OSMO (12220) ch1 via Cosmos Archive LCD

OSMO is the native staking + governance token of Osmosis (Cosmos SDK chain). It has 60 TVL months (2021-06-28 -> 2026-05-31) and 0 lambda months. Building ch1 = bonded OSMO via `cosmos/staking/v1beta1/pool` at month-end blocks, exactly the same approach as CRO and KAVA in session 040.

Field	Value
cmc_id	12220
symbol	OSMO
Cosmos chain	osmosis-1
denom	uosmo
decimals	6 (1 OSMO = 10 ⁶ uosmo)
Genesis	~2021-06-18
blocks/day	~15,000 (~5.75 s/block)

LCD candidates (probe in order): 1. <https://osmosis-api.polkachu.com> — Polkachu (known archive provider) 2. <https://lcd.osmosis.zone> — official Osmosis LCD 3. <https://rest.cosmos.directory/osmosis> — cosmos.directory aggregator

CometBFT RPC (for block timestamp lookups): <https://rpc.osmosis.zone>

C1 — Archive probe (same as session 040 Task A)

```
import requests, time

lcd_candidates = [
    "https://osmosis-api.polkachu.com",
    "https://lcd.osmosis.zone",
    "https://rest.cosmos.directory/osmosis",
]

BLOCKS_PER_DAY = 15000

for lcd in lcd_candidates:
    # Liveness
    try:
        r = requests.get(f"{lcd}/cosmos/staking/v1beta1/pool", timeout=20)
        if not r.ok:
            print(f"{lcd}: pool {r.status_code} --- skip"); continue
        live_bonded = r.json()["pool"]["bonded_tokens"]
        print(f"{lcd} LIVE --- bonded={live_bonded[:20]}...")
    except Exception as e:
        print(f"{lcd} UNREACHABLE: {e}"); continue

    # Latest block
    try:
        rb = requests.get(f"{lcd}/cosmos/base/tendermint/v1beta1/blocks/latest", timeout=20)
        latest = int(rb.json()["block"]["header"]["height"])
        print(f"  latest height={latest}")
    except Exception as e:
        print(f"  cannot read latest: {e}"); continue

    # Archive depth test (~365 days ago)
```

```

old_height = max(1, latest - BLOCKS_PER_DAY * 365)
ra = requests.get(
    f"{lcd}/cosmos/staking/v1beta1/pool",
    headers={"x-cosmos-block-height": str(old_height)},
    timeout=20
)
if ra.ok:
    bonded_old = ra.json()["pool"]["bonded_tokens"]
    # Fake-archive guard: if bonded_old == live_bonded digit-for-digit, header is ignored
    if bonded_old == live_bonded:
        print(f" FAKE ARCHIVE (header ignored) --- skip")
    else:
        print(f" ARCHIVE CONFIRMED @ {old_height}: bonded={bonded_old[:20]}...")
        print(f" --> USE {lcd}")
        break
else:
    print(f" archive FAIL @ {old_height}: {ra.status_code} {ra.text[:60]}")
time.sleep(0.5)

```

C2 — Build series (only if archive confirmed)

Use the same binary-search + pool-query pattern from session 040. Month range: observed months in `universe_panel.csv` for `cmc_id=12220` that are at or after 2021-06-18 (Osmosis genesis).

Output: `03_data/phase1/channel1_cosmos_osmo.csv` (separate file so it doesn't overwrite session 040's `channel1_cosmos_lcd.csv`).

Columns: `cmc_id`, `symbol`, `month_end`, `staked_native`, `circulating_supply`, `staking_ratio`, `source`, `flag`

```

# circulating_supply: from universe_panel.csv cmc_supply_circ column
# staked_native: bonded_tokens / 1e6
# source: f"cosmos_lcd_pool@{block_height}"
# flag: f"uosmo / 10^6; block_time={block_t_utc}"

```

C3 — Cross-check

Compare latest built month `staked_native` vs live pool value. Drift < 5% -> PASS. 5-20% acceptable with note (staking changes since last panel month). > 20% -> investigate decimals.

C4 — Coverage label

Update `coin_staking_type` for OSMO (12220) from NaN -> pos (Osmosis uses standard Cosmos SDK staking / delegation).

Task D — Close remaining TVL gaps (no data builds needed)

For each of the remaining 3 tokens with TVL but no viable build path, document their permanent-gap status in Entry 92:

cmc_id	symbol	chain	Why gap
1573	CASINO	Fantom (250)	etherscan_reachable: no — Fantom not in Etherscan Pro V2 coverage; no free archive alternative identified
4157	RUNE	THORChain	Non-EVM, non-Cosmos; native THORChain chain requires custom indexer
10529	SUN	Tron	Non-EVM; TronScan API available but ch2 engine not adapted for Tron

These stay partial. No action required beyond documenting in Entry 92.

Task E — Assemble + rebuild coverage

Run after ALL builds above are complete:

```
python 04_code/phase1_assemble_lambda.py
python 04_code/build_coverage_status.py
```

Expected outcomes (minimum scenario = HXRO only): - HXRO: partial -> **complete** (ch2 months now overlap TVL; regression-ready +1 -> 181)

Expected outcomes (maximum scenario = HXRO + SXP + OSMO all succeed): - Regression-ready: 180 -> up to **183**

Print final counts:

```
import pandas as pd
cov = pd.read_csv("03_data/universe_coverage_status.csv")
print("Regression-ready:", (cov["coverage_status"] == "complete").sum())
targets = [3748, 4279, 12220, 1573, 4157, 10529]
print(cov[cov["cmc_id"].isin(targets)][
    ["cmc_id", "symbol", "coverage_status", "lambda_months", "tvl_months"]
].to_string())
```

DATA_DECISIONS_LOG — Entry 92

Append to 04_code/DATA_DECISIONS_LOG.md:

```
### Entry 92 --- Session 041: HXRO ch2 extend; SXP probe; OSMO ch1; gap closures
```

```
**HXRO (3748) ch2 extend:**
```

```
Checkpoint was empty (monthly: []); ch2 re-run rebuilt from genesis through 2026-05.
Old coverage: 2020-09 -> 2022-09 (no TVL overlap). New coverage: [first] -> 2026-05.
Overlap with TVL (2023-02 -> 2026-05): [N] months. B2 [pass]. B4 [pass/flag].
Actual getLogs: [N] (est 534).
```

```
**SXP (4279) probe:**
```

```
Ethereum address 0x8ce9137d39326ad0cd6491fb5cc0cba0e089b6a9:
[ABI probe result + getLogs probe result]
```

Decision: [built on chainid 1 / built on BSC chainid 56 / permanent gap --- reason]

****OSMO (12220) ch1:****

Archive LCD: [node URL] --- PASS / FAIL

[N] months built, range [start] -> 2026-05. Drift: [X]% [PASS/WARN].

Denom: uosmo / 10⁶. coin_staking_type updated: NaN -> pos.

[or: NO ARCHIVE NODE --- permanent gap]

****Permanent gap closures:****

CASINO (1573): Fantom chainid 250 --- not in Etherscan V2 coverage, no free alternative.

RUNE (4157): THORChain --- non-EVM, non-Cosmos; custom indexer required.

SUN (10529): Tron --- ch2 engine not adapted for Tron; TronScan API deferred.

****Post-assemble:**** lambda [X] / [N] assets. Regression-ready [180 -> N].

Note: Etherscan Pro subscription lapsed after this session.

Remaining Etherscan-dependent work: none identified.

Session report

Write 03_data/SESSION041_HXRO_SXP_OSMO_REPORT.md: - HXRO: getLogs count, month range extended, TVL overlap months, B2/B4 - SXP: probe result (ABI status, getLogs status, decision) - OSMO: archive probe table, months built, cross-check drift - Gap closures table - Post-assemble totals

Commit

```
git add -A
```

```
git commit -m "session 041: HXRO ch2 extended; SXP probe; OSMO ch1; gap closures; regression-ready 180-
```

```
git push
```

```
/newpage
```

CLAUDE_CODE_TOKEN_BUCKET1_EXHAUSTIVE_REAUDIT_PROMPT.md

Claude Code Kickoff Prompt — Exhaustive Re-Audit of the 398 “Unrecoverable” Tokens

Paste the prompt below as the first message in a new Claude Code session opened with working directory C:/AFA_2027_QTM_Crypto. This follows a Cowork session’s challenge to session 020’s token-side Bucket-1 figure (398 tokens with no lambda Channel-1 series): that figure rests on three very different levels of rigor, not one uniform check, and the paper’s sample size is thin enough that no token should be written off without an individually documented, source-checked reason. Run to completion, review the output, then come back before touching coin-side Bucket 1 or Phase 3.

You're working in the AFA 2027 QTM Crypto research repo. Session 020 left the token universe (448 tokens) at: 49 recovered (lambda Channel 1 built), 1 deferred (VELO, identity ambiguity), and 398 marked unrecoverable. A Cowork review of that 398 found it is NOT one uniform level of checking:

- ~6 tokens (MKR, BAL, COMP, RUNE, ANGLE, AXS) --- individually verified, each with a specific documented protocol-design reason (no clean single-escrow lock exists, full stop). HIGH

- confidence these are genuinely unrecoverable by any vendor, free or paid.
- ~287 tokens --- were in the 290-candidate pool (``03_data/phase1/_bucket3_candidates.csv``) and were rejected, but the rejection was written at the CATEGORY level ("the overwhelming majority are DEX/lending/RWA/meme/chain tokens whose governance is delegation/farming/Snapshot") --- not confirmed contract-by-contract for every single one. MEDIUM confidence at best.
- ~111 tokens --- never opened individually at all. They were excluded purely because ``classification_table.csv`` tags / ``defillama_categories`` didn't suggest a staking mechanism. That's a pre-filter heuristic, not a check. LOW confidence.

This session's job is to convert ALL 398 (+ a final confirmation pass on the 6 already-rejected + resolve VELO's identity question) into individually documented verdicts, checked against every named source (Etherscan, Dune, DeFiLlama, Artemis), before any of them are accepted as a real, final rejection. Every one of the 398 must end this session with its own row and its own specific reason --- no category-level write-offs, no silent drops. Before doing anything else, read in full:

1. `04_code/DATA_DECISIONS_LOG.md` Entries 26, 41, 48, 49 --- the Entry-26 clean-escrow standard, the Part A.1 rejection list, and session 020's Bucket-3 build + the category-level rejection language that triggered this re-audit.
2. `03_data/SESSION020_BUCKET2_BUCKET3_COVERAGE_ADDENDUM.md` Part B --- the exact candidate-pool construction and the "~287 others" line being re-audited here.
3. `04_code/DATA_SPECIFICATION.md` ?0 (flag, don't guess) and ?3.1 (Channel 1 definition).
4. `03_data/phase1/_bucket3_candidates.csv` (290 rows) and `03_data/classification_table.csv` --- the 290-candidate pool plus the ~111 tokens that pool excluded. Recompute the full 398 list directly from `universe_panel.csv` (every ``asset_class='token'`` `cmc_id` with zero Channel-1 months) --- don't trust 398 as exact; confirm it live against the current panel.
5. `04_code/phase1_channel1_evm_locks_bucket3.py`, `phase1_channel1_evm_locks.py`, `phase1_channel1_evm_locks_ext.py` --- the Dune-transfers build method; extend, don't rebuild.
6. `04_code/.api_keys.json` --- confirm what's already there (etherscan, dune) before any new signup.

Method --- a triage funnel, not equal-depth effort on all 398

398 individual deep-dives across 4 sources each is not a uniform-depth task; do it as a funnel so effort concentrates where it can change the outcome, while every token still gets logged.

****Stage 0 --- rebuild the worklist live.**** Recompute the exact 398 (or whatever the live number is) from `universe_panel.csv` + `classification_table.csv`. Do not reuse the cached 290-candidate file without re-deriving it --- confirm it's still accurate against the current panel.

****Stage 1 --- cheap bulk triage for ALL 398 (must cover every row).****

- Pull DeFiLlama's bulk ``https://api.llama.fi/protocols`` (one call, already cached by ``phase1_build_identity_map.py`` --- refresh it). For every token's mapped protocol (by ``cmcId``, NEVER symbol), check the ``category`` field. Flag HIGH if category is "Staking" or "Liquid Staking" or similar; flag MEDIUM if category is ambiguous but the protocol has ANY tag/category suggesting governance/locking; flag LOW if the protocol is clearly DEX/lending/meme/bridge/L1-L2/RWA with no staking signal anywhere in DeFiLlama's metadata.
- For the ~111 tokens with no DeFiLlama protocol mapping at all (no ``cmcId`` match in ``/protocols``), do a second cheap pass: check the token's own ``classification_table.csv`` row for any docs URL, website, or platform/contract address already on file; if a contract address exists, that's enough to route into Stage 2. If NEITHER a DeFiLlama mapping NOR a contract address exists, log it explicitly as "no on-chain identity available to check at all" --- that is itself a specific, honest reason, not a silent drop.
- Log every one of the 398 with its Stage-1 label. This stage must finish for all 398 before Stage 2 starts.

****Stage 2 --- real per-token check for everything labeled HIGH or MEDIUM in Stage 1 (no sampling --- every HIGH/MEDIUM token gets this, not a subset).**** For each:

- a. ****DeFiLlama**** --- open the protocol's `/protocol/{slug}` endpoint; check `chainTvl`s for a named staking/locking pool breakdown. If one exists, note the USD value and flag it as a VALUE-based proxy for locked supply (distinct from a literal token-quantity reconstruction) --- do not treat a USD figure as equivalent to the existing Channel-1 token-quantity standard without saying so.
- b. ****Etherscan (or the relevant EVM chain explorer under the existing free key)**** --- find the token's contract; check its verified-contract / top-holders surface for any contract plausibly named escrow/vault/stake/gauge/lock holding a meaningful share of supply. If a candidate contract is found, apply the EXACT Entry-26 standard: single contract, direct custody of the BASE token, not a wrapped/composite asset, not in-wallet delegation, not a collateral system.
- c. ****Dune**** --- only for tokens that pass 2b with a real escrow candidate: confirm `tokens\_{chain}.transfers` or `balances` covers that chain/contract (per the session-020 correction --- `tokens\_bnb` not `tokens\_bsc`, etc.) before committing to a build. Track query credit usage against the free 2,500/month budget; flag, don't silently burn through it.
- d. ****Artemis**** --- test live, unauthenticated, whether artemis.xyz exposes a public per-asset terminal/dashboard page (not the API) showing a staking-ratio or locked-supply chart. Try it on 2-3 known-staking tokens (e.g. CRV, AAVE) and 2-3 known non-staking tokens first to learn the access pattern before deciding whether it's worth checking per-token at scale. If it requires login or a paid plan for the data you need, stop --- do not sign up or pay.

Classify each Stage-2 token: BUILD (ship it, same as Entry-26 builds) / REJECT-mechanism (specific contract/protocol reason, money can't fix it) / REJECT-no-data (mechanism plausible but no source, free or paid, covers it) / DEFER (specific blocker, e.g. an identity collision like VELO) / GATED (a real paid option exists --- name it, price it if disclosed, do NOT subscribe; report to Moazzam for his own purchase decision).

****Stage 3 --- fast confirmation pass for everything labeled LOW in Stage 1.**** Don't deep-dive these, but don't silently drop them either: for each, write one specific line (e.g. "wrapped/composite asset, no underlying lock," "meme token, no protocol, no contract evidence of any lock," "CEX/custodial token, governance is off-chain"). If anything in this pass looks ambiguous on a second look, promote it back into Stage 2 rather than force a quick verdict.

****Stage 4 --- final pass on the 6 already-rejected (MKR, BAL, COMP, RUNE, ANGLE, AXS) and VELO.**** Light-touch only: confirm via DeFiLlama + a quick Etherscan/Ronin check that nothing has changed since Entry 26/41/48 (no new escrow contract shipped, no protocol redesign). For VELO, attempt to resolve the v1/v2 identity question directly --- check CMC's own platform/contract field for cmc_id 7127 and DeFiLlama's protocol entry for Velodrome v2 to see if there's a documented, non-guessed mapping between the in-universe v1 contract and the v2 lock; if no clean mapping exists, leave deferred with that stated.

Rules (unchanged from the spec and prior entries)

- Never sign up for or pay for any paid tier of anything (Etherscan Pro/archive, Artemis Enterprise, Dune paid, Subscan Pro, etc.). Flag a priced, self-serve option and report back to Moazzam --- he executes any purchase himself, even with confirmed funding available.
- Join everything on `cmc\_id`, never `symbol`.
- Never guess or interpolate a numerator. A USD-value proxy (Stage 2a) must be labeled as a proxy, never silently treated as the token-quantity standard.
- Every one of the 398 (+6 reconfirm +1 VELO) gets its own row with a specific reason. No category-level verdict, no row left blank.
- DATA_DECISIONS_LOG.md is append-only --- never edit or delete Entries 1--49. Continue at Entry 50, one entry per stage/cluster of findings (not one giant entry covering all 398).
- Watch Dune's free query-credit usage given the larger scale of this session; flag if approaching a meaningful fraction of the monthly 2,500 budget instead of assuming there's always headroom.

Deliverables

1. A single CSV, `03\_data/phase1/token\_bucket1\_full\_audit.csv`, one row per token in the 398 (+6 reconfirmed +1 VELO) with columns: `cmc_id`, `symbol`, `name`, `defillama_protocol_slug`, `defillama_category`, `stage1_label`, `stage2_etherscan_check`, `stage2_dune_check`, `stage2_defillama_tvl_check`, `stage2_artemis_check`, `final_verdict` (BUILD / REJECT-mechanism / REJECT-no-data / DEFER / GATED), `reason` (specific, non-generic text).
2. Built series for every new BUILD verdict, folded into lambda via the existing `phase1\_channel1\_evm\_locks\*.py` scripts (extend, don't rebuild) and re-run `phase1\_assemble\_lambda.py`; report new asset/asset-month counts against the session-020 baseline (62 assets / 1,880 asset-months).
3. A coverage addendum, `03\_data/SESSION021\_TOKEN\_BUCKET1\_EXHAUSTIVE\_AUDIT.md`, summarizing: how many of the 398 ended in each final-verdict bucket, how many were GATED (with named source + price if disclosed), and an explicit statement of what was and was not checked for each confidence tier --- written so it can support the paper's methodology section as evidence that every reasonable free avenue was exhausted before any token was excluded.
4. DATA_DECISIONS_LOG.md continued from Entry 50. Log this session as `06\_documentation/ai\_conversations/session\_021\_\*.md`; update `06\_documentation/time\_log.md`.
5. Commit and push to github.com/moazzamkhoja/AFA_2027_QTM_Crypto (main) at session end.

Note: this session is token-side only (the 398 + 6 + VELO). The 456-coin Bucket-1 has NOT had this level of individual re-audit and may warrant the same treatment in a future session --- flag that as an open follow-up in the addendum, but do not start it here without review.

Stop when all 398 (+6 +1) have a logged, specific, individual verdict. Do not start coin-side Bucket 1 or Phase 3 without review.

/newpage

Part 4: Data Decisions Log

Data Decisions Log

Append an entry every time the empirical pipeline deviates from 04_code/DATA_SPECIFICATION.md — a proxy substitution, a classification judgment call, a threshold that needed adjusting, a source that didn't pan out, or any data-availability gap that forced a choice. This is the audit trail behind Section 3's eventual prose and is part of the AFA-required documentation of the AI-assisted research process.

Do not edit or delete past entries — append only, even if a later entry supersedes one.

Entry Template

Date: Spec section affected: **Asset(s)/period affected:** What the spec wanted: What was actually available: **Decision made:** **Rationale:** **Downstream impact** (what should be re-checked if this decision changes):

Entries

Entry 1 — Primary market-cap source: CoinMarketCap, not CoinGecko

Date: 2026-06-22 **Spec section affected:** 2.1, 2.4 (historical rankings / candidate sources) **Asset(s)/period affected:** all assets, entire 2015-08->2026-05 history **What the spec wanted:** point-in-time monthly top-N rankings (incl. delisted coins) built primarily from CoinGecko historical data. **What was actually available:**

CoinGecko’s *public* API caps historical queries at the past 365 days (verified live: error 10012, “Public API users are limited to querying historical data within the past 365 days”). Full history requires a paid plan; no API key was available in the environment. CoinMarketCap’s free `data-api/v3/cryptocurrency/listings/historical` endpoint, by contrast, serves daily point-in-time ranked listings back to ~2013-04-28, requires no key, includes assets as ranked on that date (delisted/ dead coins included), and returns market cap, price, volume24h, circulating/total/max supply, and tags. Limit ≥ 1000 confirmed. **Decision made:** use the CMC free historical listings endpoint as the ranking backbone for Phase 0. (Human approved this substitution explicitly when the constraint was surfaced.) **Rationale:** it is the only freely available source that satisfies the Section 2.1 anti-survivorship requirement (point-in-time rankings including delisted assets) over the full sample without a paid key. **Downstream impact:** if a CoinGecko Pro key is later obtained, cross-check CMC market caps against CoinGecko for discrepancies (Section 2.4 asks for a two-source cross-check; only CMC was usable here). All universe membership decisions rest on CMC mcap values.

Entry 2 — Artemis Analytics: no accessible free API (deferred)

Date: 2026-06-22 **Spec section affected:** 2.4 (sources), 4 (PQ for tokens, Phase 2) **Asset(s)/period affected:** n/a (source availability) **What the spec wanted:** Artemis used as a cross-chain on-chain fundamentals source. **What was actually available:** `api.artemisxyz.com` returned HTTP 410 Gone; `api.artemis.xyz / app.artemisanalytics.xyz` failed DNS resolution. No documented free public API endpoint responded. **Decision made:** defer Artemis to a later phase; do not block Phase 0 on it. **Rationale:** Phase 0 needs only rankings + classification, which CMC + DeFiLlama cover. **Downstream impact:** if Artemis fundamentals are wanted for PQ in Phase 2, this needs authenticated/paid access to be revisited.

Entry 3 — Monthly snapshot date = last calendar day (daily data available)

Date: 2026-06-22 **Spec section affected:** 2.2 (monthly frequency) **Asset(s)/period affected:** all months **What the spec wanted:** month-end monthly frequency. **What was actually available:** CMC’s historical endpoint serves *daily* snapshots (verified: distinct payloads for consecutive dates), not just weekly. **Decision made:** pull the snapshot dated the last calendar day of each month; if that exact date is empty, walk back up to 6 days and stamp the served date in `_meta`. **Rationale:** exact month-ends are available, removing the need to approximate with the nearest weekly snapshot. **Downstream impact:** none expected; served-vs-requested date is auditable per file.

Entry 4 — CMC tags are current metadata, not point-in-time

Date: 2026-06-22 **Spec section affected:** 2.3 (classification) **Asset(s)/period affected:** all assets (classification evidence) **What the spec wanted:** time-aware coin/token classification with evidence. **What was actually available:** CMC’s tags reflect the asset’s *current* metadata (e.g. ETH is tagged `pos` today; BTC carries present-day VC-portfolio tags). They are not snapshot-specific, so they describe the asset’s end-state, not its state in month t . **Decision made:** use the union of tags across snapshots as classification *evidence* (asset-class is treated as static), but handle known time-varying cases explicitly with dated overrides, and flag the limitation in the coverage report. **Rationale:** tags are still the best available structured signal of asset function; the end-state label is correct for most assets and the few transition cases are handled. **Downstream impact:** Phase 1 staking-channel availability must respect transition dates (e.g. ETH staking only from 2020-12), not the static coin label.

Entry 5 — ETH PoW->PoS handled as a dated transition

Date: 2026-06-22 **Spec section affected:** 2.3 (ambiguous/transition cases) **Asset(s)/period affected:** ETH (`cmc_id 1027`) **What the spec wanted:** explicit handling of ETH’s staking onset, dates verified. **What was actually available:** established public record: Beacon Chain genesis 2020-12-01 (staking first possible via the deposit contract opened Nov 2020); Merge to full PoS 2022-09-15. **Decision made:** classify ETH as coin but attach `staking_start=2020-12-01` and a `transition_note` recording no staking-based lambda channel before 2020-12 and full PoS at the 2022-09-15 Merge. **Rationale:** the staking lambda channel is undefined for ETH in most of the sample; a static label would mislead Phase 1. **Downstream**

impact: apply the same dated-onset logic to any other PoW->PoS transition discovered in Phase 1; ETH is the worked example, not necessarily the only one.

Entry 6 — Stablecoins excluded from the panel (ranking-inclusive)

Date: 2026-06-22 **Spec section affected:** 2.1, 2.3 **Asset(s)/period affected:** all stablecoins reaching top-250 (71 names) **What the spec wanted:** a coin-vs-token universe; stablecoins are not addressed directly by the functional cut. **What was actually available:** stablecoins occupy real top-250 rank slots but have ~0 returns and degenerate lambda, fitting neither H1a nor H1b. **Decision made:** (human decision) exclude stablecoins entirely from the universe. They are detected by the *exact* stablecoin CMC tag (substring matching wrongly catches stablecoin-protocol governance tokens and stablecoin-algorithmically-stabilized = LUNC, the Terra coin). Ranking is computed *inclusive* of stablecoins so the top-250 cutoff reflects true market structure; stablecoins are removed only from the output panel. **Rationale:** keeps the cutoff economically meaningful while keeping degenerate assets out of the tests. The functional cut is preserved: UST (stablecoin) excluded, LUNA/LUNC (coin) retained including its collapse. **Downstream impact:** the observed top-250 cross-section runs a few-to-low-tens below 250 in later years (stablecoins consume slots); reported in the coverage report.

Entry 7 — Top-N threshold = 250; carry-forward for unobserved in-panel months

Date: 2026-06-22 **Spec section affected:** 2.1 **Asset(s)/period affected:** all **What the spec wanted:** top-N entry (default 250), permanent retention, failures as return realizations not missing data. **What was actually available:** snapshots pulled to the top 1000, so an asset that drifts below rank 250 but stays in the top 1000 is still observed; only below rank 1000 is price visibility lost. **Decision made:** N=250 (sensitivity at 200/250/300 reported). Once an asset enters it stays in the panel; months where it is in-panel but absent from the top-1000 snapshot are marked `status='carried_forward'` with the last observed price retained. **Rationale:** satisfies the anti-survivorship rule while flagging (not hiding) the visibility gap. **Downstream impact:** Phase 3 must decide the death-return treatment for carried-forward tails (final delisting return vs. constant carry). Flagged in coverage report ?9.

Entry 8 — Classification: tag-rules + DeFiLlama + curated native-coin override

Date: 2026-06-22 **Spec section affected:** 2.3 **Asset(s)/period affected:** all (esp. major L1 coins with thin/contaminated tags) **What the spec wanted:** functional coin/token classification with evidence and flags. **What was actually available:** CMC tags are inconsistent — some L1 coins (ATOM, AVAX, DOT, FTM) carry defi ecosystem tags but no clean consensus tag; some governance tokens (UNI) now carry `layer-1` (Unichain); LST governance tokens (LDO) carry LST ecosystem tags. DeFiLlama category matching by symbol false-promotes some coins (DOT->Liquid Staking). **Decision made:** rules engine with (a) exact consensus-tag set -> coin; (b) `layer-1/ smart-contracts` -> coin only when no `governance/dao` tag; (c) explicit `governance/ dao` -> token (wins over ecosystem tags); (d) DeFiLlama promotion to token only on genuine DeFi categories (Dexs/Lending/CDP/Yield/Derivatives?); (e) a curated `NATIVE_COIN_OVERRIDE` (22 ids verified against the data as each symbol's highest-peak-mcap instance) forcing major native coins (XRP, XLM, etc.) to coin. Every non-clean case carries `ambiguous_flag=True`. **Rationale:** no single tag rule is reliable given CMC's metadata; combining rules + DeFiLlama + a small documented manual override correctly classifies all spot-checked majors while flagging the long tail for review. **Downstream impact:** the `other/ambiguous` bucket (large by raw name count, small by observed asset-months) needs a manual confirmation pass for any asset that actually enters the tests. LINK is left `other` (oracle token — neither a clean coin nor a governance token).

Entry 9 — Ranking recomputed by market cap among the returned set

Date: 2026-06-22 **Spec section affected:** 2.1 **Asset(s)/period affected:** all **What the spec wanted:** “rank all tracked assets by market cap” each month. **What was actually available:** CMC returns a stored `cmcRank` plus market caps. **Decision made:** recompute rank by descending market cap among the returned set rather than trusting `cmcRank`; drop rows with `null/<=0` market cap. **Rationale:** matches the

spec wording exactly and is robust to any stored-rank quirks. **Downstream impact:** ranks are internally consistent with the market-cap values used for entry decisions.

Entry 10 — Carry-forward death-return treatment: split before deciding

Date: 2026-06-22 **Spec section affected:** 2.1 (retention rule); Phase 3 (returns) **Asset(s)/period affected:** all `status='carried_forward'` asset-months **What the spec wanted:** Phase 0 flagged this as open (PHASE0_COVERAGE_REPORT.md ?9.1) rather than resolving it — Phase 3 needs a final death-return policy. **What was actually available:** no breakdown yet of how many carried-forward asset-months are presumed-failed (asset never reappears in the top-1000 through the end of the sample) versus temporarily-out (asset reappears later — a visibility gap, not a failure). **Decision made:** before any return-treatment formula is chosen, produce the presumed-failed vs. temporarily-out split with counts (and the longest temporarily-out gap observed). Decide the formula only after seeing those counts. This is a precursor task, run before Phase 1 channel work, not deferred all the way to Phase 3. **Rationale:** choosing a death-return convention blind, before knowing whether carried-forward months are mostly real failures or mostly short visibility gaps, risks picking a treatment that’s wrong for the dominant case. **Downstream impact:** Phase 1 (lambda channels) and later phases should not compute return-dependent statistics over carried-forward months until this split is reviewed.

Entry 11 — Manual classification review scoped by persistence, not market cap

Date: 2026-06-22 **Spec section affected:** 2.3 (classification) **Asset(s)/period affected:** `other/ambiguous_flag=True` assets only **What the spec wanted:** a manual confirmation pass for classifications that will actually enter the tests. **What was actually available:** 874 names are `other`; reviewing all of them by hand is not tractable, and most are one-or-two-month pump-and-dump blips that already passed the top-250 market-cap gate but contribute negligible weight to any regression. **Decision made:** scope the manual review to `other/ambiguous` names with ≥ 12 **observed asset-months** (a persistence filter, separate from and applied after the existing market-cap-based top-250 entry rule). Names below this persistence threshold are left unreviewed. **Rationale:** concentrates review effort on the names that could actually move a result; the unreviewed short-lived tail is, by construction, too thin to matter statistically. **Downstream impact:** any `other/ambiguous` name below the 12-month threshold that later turns out to matter (e.g., because of a methodology change) should be reviewed at that point, not assumed correctly classified.

Entry 12 — Meme coins: no change, current ‘other’ handling confirmed

Date: 2026-06-22 **Spec section affected:** 2.3 (classification), 2.1 (ranking) **Asset(s)/period affected:** meme coins (DOGE, SHIB, PEPE, etc.) and similar non-economic-output assets **What the spec wanted:** a coin/token functional cut; meme coins/NFTs were raised as a potential third category not addressed directly by that cut. **What was actually available:** meme coins already land in `other` (no staking mechanism, no governance mechanism) and are already excluded from H1a/H1b/H3 by that classification; they still count toward the top-250 ranking, the same treatment already applied to stablecoins. NFTs are presumed absent from the CMC fungible-token listings source entirely (not the same product as CMC’s NFT tracking) — to be explicitly confirmed in Phase 1, not assumed. **Decision made:** keep current handling — no Phase 0 changes. Confirm NFT absence as a Phase 1 verification step. **Rationale:** meme coins are already functionally excluded from the hypothesis tests by the existing classification; stripping them from the panel entirely (vs. leaving them visible as `other`) would have no effect on results and would reduce audit transparency. **Downstream impact:** none expected. If NFT-like entries are found in Phase 1, log and exclude them explicitly rather than assuming this entry covers it.

Entry 13 — Quadrant median splits computed within asset class, not pooled

Date: 2026-06-22 **Spec section affected:** 4 (Growth-Levelized NVT), main.tex Section “The Quadrant Portfolio” (H3); affects Phase 4 design, not Phase 1 build work directly **Asset(s)/period affected:** all coin and token observations entering H3 **What the spec wanted:** H3 reports coins and governance tokens separately; the spec did not specify whether the $\lambda/(1-\lambda)$ and Growth-Levelized NVT median splits

defining the Star/Avoid quadrants should be computed pooled or within each class. **What was actually available:** coins (security-staking norms) and governance tokens (DeFi vote-escrow/governance-staking norms) plausibly have structurally different $\lambda/(1-\lambda)$ distributions for reasons unrelated to conviction, which could let one class mechanically dominate “high λ ” under a pooled median. **Decision made:** compute the high/low median splits for both $\lambda/(1-\lambda)$ and Growth-Levelized NVT separately within coins and within tokens each month, not pooled. **Rationale:** keeps the Star/Avoid sort meaningful as a within-class signal rather than an artifact of cross-class composition differences. **Downstream impact:** this is a Phase 4 (portfolio assembly) specification, recorded now so it isn’t decided ad hoc later; revisit if Phase 1’s actual λ distributions show the two classes don’t differ enough to matter.

Entry 14 — Artemis Analytics / paid CoinGecko access deferred

Date: 2026-06-22 **Spec section affected:** 2.4, 4 (PQ for tokens, Phase 2) **Asset(s)/period affected:** n/a (source access) **What the spec wanted:** a decision on whether to pursue paid access now. **What was actually available:** no free API for either source (see Entry 2); not needed until Phase 2 (PQ construction) or for the Section 2.4 two-source market-cap cross-check. **Decision made:** defer; revisit only when Phase 2 actually needs it. **Rationale:** no Phase 0/1 work depends on it; premature procurement. **Downstream impact:** Phase 2 kickoff should re-raise this explicitly rather than assuming it’s been resolved.

Entry 15 — Universe size $N=250$ confirmed, no change

Date: 2026-06-22 **Spec section affected:** 2.1 **Asset(s)/period affected:** all **What the spec wanted:** confirmation of the default top- N threshold. **What was actually available:** rank-sensitivity table (200/250/300) showing the live cross-section is not drastically different across these thresholds. **Decision made:** keep $N=250$. **Rationale:** no evidence yet that a different threshold is needed; can revisit once Phase 1 λ -channel coverage is known. **Downstream impact:** none; flagged for revisit only if Phase 1 coverage suggests otherwise.

Entry 16 — Sector/economic-function classification added as a second, independent dimension

Date: 2026-06-22 **Spec section affected:** new 2.6 (added); clarifies that Entry 13’s “class” (coin vs. token) is a separate dimension from this one **Asset(s)/period affected:** all assets **What the spec wanted:** prior to this entry, the spec only defined one classification axis — the binary, functional coin/token cut (2.3), which Entry 13 then used as “the” class for the H3 quadrant median splits. **What was actually available:** the user clarified that “class” should also mean something narrower than coin/token — e.g., L1 vs. L2, DEX vs. Perpetuals/Derivatives, Lending vs. CDP vs. Liquid Staking — a sector/economic-function tag, not a refinement or replacement of the coin/token cut. A quick check of the existing Phase 0 output shows the raw ingredients for this already exist as classification evidence: 574/2010 assets in `classification_table.csv` carry a `defillama_categories` value (154 DEXs, 78 Yield, 59 Derivatives, 53 Lending, 34 Canonical Bridge, 20 Liquid Staking, 17 CDP, etc.), and 134/2010 carry a `layer-1` CMC tag with 43 carrying `layer-2` — i.e., DeFiLlama categories give DEX/Lending/Derivatives/Staking-type resolution for protocol tokens, and CMC tags give L1/L2 resolution mostly for base-layer coins that DeFiLlama doesn’t track. Neither source alone covers the whole universe; coverage is partial either way. **Decision made:** add a second, independent classification field (`sector/category`) populated from DeFiLlama categories + CMC tags, captured now as a coverage/data task (new spec section 2.6). Explicitly defer deciding which sector-level comparisons (DEX vs. Perp, Lending vs. Staking, L1 vs. L2, or others) will actually be tested in the paper — that choice is a later judgment call once Phase 1-4 data exists, not a Phase 0/1 data-capture decision. Entry 13’s coin/token median splits are unaffected by this entry; “class” in Entry 13 refers only to the coin/token cut. **Rationale:** the marginal cost of capturing this field now (it’s mostly already present in classification evidence pulled in Phase 0) is far lower than the cost of re-deriving it later if a sector-level comparison turns out to matter; deferring the analytical decision (which comparisons to run) avoids designing an analysis around a taxonomy that hasn’t been validated against real coverage yet. **Downstream impact:** the Phase 0 follow-up session should add a sector/category field and report coverage by `asset_class` (coin/token/other) per new spec section 2.6, before Phase 1 (λ channels) begins. Section 3 prose, once written, should describe both classification dimensions as independent.

Entry 17 — Carry-forward split rule: subtype by last-observation, right-censoring flagged

Date: 2026-06-22 **Spec section affected:** 2.1 (retention); implements Entry 10 **Asset(s)/period affected:** all `status='carried_forward'` asset-months (89,535) **What the spec wanted:** Entry 10 asked for a presumed_failed vs temporarily_out split with counts, before any death-return formula is chosen. **What was actually available:** the panel records each asset-month's status; whether a carried_forward month is a closed visibility gap vs a terminal failure is recoverable from the asset's own observed-month timeline. **Decision made:** for asset j with last observed month L_j , a carried_forward month at t is temporarily_out iff $t < L_j$ (a later observation exists, so the gap closes) and presumed_failed iff $t \geq L_j$ (trailing tail with no later observation). Stored in a new column `carry_forward_subtype` (the `status` column is untouched). The right-censoring risk is flagged explicitly: a presumed_failed asset whose *terminal* gap (months from L_j+1 to the 2026-05-31 sample end) is no longer than the longest observed temporarily_out gap could, in principle, still be a case that would have reappeared had the sample run longer — so “never seen again so far” is NOT silently equated with “permanently dead.” Reported at several thresholds (terminal gap $\leq$ max/median temporarily_out gap; $\leq 3/6/12$ months; and gaps that only started in the last 6/12/24 months). Script: `04_code/carry_forward_split.py`. **Rationale:** the rule is exactly equivalent to a gap analysis but vectorisable and unambiguous at the sample boundary, which is precisely where the death-vs-gap call is hardest. Producing the right-censoring counts (rather than a single death/alive label) keeps Entry 10's deferral honest. **Downstream impact:** the death-return formula (Phase 3) should be chosen with these counts in hand. Key numbers: 81,167 presumed_failed vs 8,368 temporarily_out asset-months; 2,107 closed gaps (median 1 mo, max 47 mo); 1,399 presumed_failed assets of which 98 have a trailing gap ≤ 6 months (started after 2025-11-30) and 166 ≤ 12 months — these are the right-censoring-exposed names a naive “dead = never seen again” rule would misclassify.

Entry 18 — Classification confirmation pass: 16 conservative reclassifications

Date: 2026-06-22 **Spec section affected:** 2.3; implements Entry 11 **Asset(s)/period affected:** other/ambiguous assets with ≥ 12 observed asset-months (630 candidates) **What the spec wanted:** a manual confirmation pass proposing keep-other / coin / token with a one-line rationale, not inventing classifications that can't be supported. **What was actually available:** the 630 candidates are overwhelmingly genuine utility/sector tokens (AI, gaming, payments, DePIN, storage, identity, exchange, meme). A small set were mis-left as other by the first-pass tag rules: (a) native PoS/DPoS chains whose CMC tags carried 'platform'/ecosystem labels or a non-standard consensus tag (e.g. Symbol's 'posplus'), or whose DeFiLlama category mislabelled the base chain (Kusama->“Liquid Staking”, Kujira->“Liquidations”, Polygon/Dymension->“Chain”); and (b) one DeFi bridge governance token (STG, veSTG fee-share) in a bridge category the first pass didn't promote. **Decision made:** flip 16 names with individually verifiable mechanisms — 15 other->coin (KSM, POL, DYM, KUJI, XYM, IOST, STEEM, ARDR, QKC, VLX, WICC, NEBL, UOS, CENNZ, WTC) and 1 other->token (STG) — applied to `classification_table.csv` (original label preserved in new column `asset_class_original`; reason in `confirmation_basis`; `ambiguous_flag` cleared on flips). A further 16 genuinely ambiguous names are left other WITH a note (`confirmation_basis` “gray-zone”): L2 gas/governance tokens with no security-staking and no vote-escrow lock (OP, MNT, MANTA, IMX), LST-protocol tokens kept out per Entry 8 (RPL, ANKR, SSV, STRD), weak/edge native chains (EWT PoA, GBYTE DAG no-reward, FCT), NFT-market governance tokens (BLUR, LOOKS, ME), a juror-staking work token (PNK), and a symbol-collision case (PTS). The remaining 598 stay other with an auto-generated sector rationale. Full per-asset proposals: `03_data/classification_confirmation_review.csv`. Script: `04_code/classify_confirmation_pass.py`. **Rationale:** the spec explicitly forbids inventing unsupportable labels and asks ambiguous names to be left other with a note; flipping only mechanism-verifiable native chains + one clear veToken, and documenting the gray zone, maximises correctness without contaminating H1a/H1b with forced labels. **Downstream impact:** in-universe class counts move coin 618->633, token 447->448, other 874->858. Re-running `classify_assets.py` regenerates the table from scratch and must be followed by `classify_confirmation_pass.py` to re-apply this layer. The gray-zone 16 (esp. OP/MNT/IMX/MANTA L2 tokens and the LST tokens) should be revisited in Phase 1 once staking/lock data is actually pulled.

Entry 19 — Meme/NFT handling confirmed; meme over-promotions flagged (not changed)

Date: 2026-06-22 **Spec section affected:** 2.3; confirms Entry 12 **Asset(s)/period affected:** meme-tagged assets (84 in-universe) and NFT-tagged assets **What the spec wanted:** confirm meme coins land in **other** and aren't mis-flagged as coin/token, and confirm no actual NFT collection exists in the panel. **What was actually available:** of 84 meme-tagged in-universe names, 58 are **other**, 21 **token**, 5 **coin**. The 5 coins (DOGE, MONA, MEME[1191], TRUMP[1185], M=MemeCore) are genuinely mineable/PoS coins — **coin** is functionally CORRECT (they earn mining/staking seigniorage), not a mis-flag. Of the 21 tokens, some are real DeFi protocols with meme-style names (SUSHI) where **token** is correct, but others (SHIB, FLOKI, BabyDoge, ELON, MEW, SNEK, ?) are memes promoted to **token** via an *attached* DEX/farm DeFiLlama category — arguably over-promoted under a strict Entry-12 reading. For NFTs: 112 names carry 'collectibles-nfts'/NFT tags, but all are FUNGIBLE tokens of NFT-ecosystem projects (MANA, SAND, APE, BLUR, IMX, ?); none is a non-fungible collection. The CMC fungible listings source does not list NFT collections (that is CMC's separate NFT product). **Decision made:** per Entry 12, make NO meme reclassifications in this pass (memes among the 630 **other**/ambiguous candidates correctly stay **other**; the mineable meme-coins correctly stay **coin**). The meme tokens over-promoted via attached DEX/farm categories are FLAGGED for human review but left as-is (demoting existing coin/token is out of this deliverable's scope, which only reviews **other** candidates). Confirm explicitly: no NFT collection (non-fungible) is present in the panel. **Rationale:** the mineable meme-coins legitimately differ from pure ERC-20 memes; a blanket "meme -> other" would wrongly strip DOGE/MONA of a real seigniorage mechanism. The DEX-attached meme-token promotions are a genuine edge worth a human decision but not a clear error to auto-correct here. **Downstream impact:** if a strict "all memes -> other regardless of attached protocol" rule is later preferred, the flagged token-promoted memes (SHIB, FLOKI, BabyDoge, ELON, MEW, SNEK, PONKE, VOLT, ?) are the names to revisit. NFT absence is now confirmed, closing the Entry 12 Phase-1 verification item.

Entry 20 — Sector field: DeFiLlama categories + curated CMC sector-tag whitelist

Date: 2026-06-22 **Spec section affected:** new 2.6; implements Entry 16 **Asset(s)/period affected:** all assets in `classification_table.csv` **What the spec wanted:** a second, independent sector field from DeFiLlama categories (primary) + CMC tags (fallback), both kept where both fire, blank where neither, with coverage reported by `asset_class`. **What was actually available:** `defillama_categories` already populated for 566/1939 in-universe assets; CMC tags carry sector-like signals (layer-1 134, layer-2 43, privacy 91, depin 62, meme 83, ?) mixed with non-sector noise (ecosystem/portfolio/listing/ governance tags). **Decision made:** `sector` = DeFiLlama categories carried in as-is (multi-value, semicolon-separated) UNION a curated CMC-tag->label whitelist (`SECTOR_TAG_MAP` in `04_code/build_sector_classification.py`), deduped. The whitelist deliberately EXCLUDES governance-axis tags ('governance','dao','defi') and all ecosystem/portfolio/listing tags — those are not sectors. Blank where neither source fires (no name-based guessing). Coverage: 1113/1939 (57.4%) get a sector; by class coin 54.5% (leans on CMC L1/L2/ smart-contract tags — 184 CMC-only), token 89.7% (leans on DeFiLlama — 146 DL-only + 175 both), other 42.7% (282 CMC-only); residual 826 (42.6%) with no signal — exactly the lean the spec anticipated. **Rationale:** matches 2.6's intent (capture the field now, defer which comparisons get tested); keeping both sources and excluding the governance axis preserves orthogonality to the coin/token cut. **Downstream impact / CAVEAT:** the DeFiLlama join is by ticker SYMBOL (inherited from `classify_assets.py`), not a unique protocol id, so short/common tickers over-attribute categories — e.g. BTC inherits "SoFi;Reserve Currency;Canonical Bridge" from unrelated protocols sharing the symbol. This adds noise to a few coins' DeFiLlama-sourced sector tags (the CMC-tag-sourced parts, e.g. BTC's Layer-1/Privacy, are clean). If sector tags are used analytically in a later phase, de-noise the symbol-matched DeFiLlama categories for base coins (or re-join DeFiLlama by protocol id/chain) before relying on them.

Entry 21 — Phase 1 source verification: live free-access audit of all lambda-channel sources

Date: 2026-06-23 **Spec section affected:** 3 (lambda channels), 2.5 (per-asset data sources); spec Section 3 step 3 explicitly requires verifying current free access of every source *before* building on it. **Asset(s)/period affected:** n/a (source-availability audit gating the whole of Phase 1) **What the spec wanted:** per-chain explorers (Etherscan/beaconcha.in, Solscan, ?), staking dashboards, DeFiLlama locked supply, and

Snapshot/Tally/Boardroom for voting — each verified live, not assumed. **What was actually available (probed live 2026-06-23):** - **CoinMarketCap historical listings** (Phase 0 backbone) — still serves month-end circulating/total supply per asset; this is the denominator source for the lambda ratios. - **DeFiLlama /protocols + /protocol/{slug}** — free, keyless, WORKS. Per-protocol payload carries token-denominated ***-staking** chainTvl buckets (e.g. **Ethereum-staking** -> {"CRV": 855M}), a numeric **cmcId**, and a **governanceID** (Snapshot space). USED ONLY AS A REGISTRY/ADDRESS-BOOK (see Entry 22), not as the lambda measurement. - **Snapshot GraphQL** (hub.snapshot.org/graphql) — free, keyless, WORKS. Returns proposals with **created/start/end/state/votes/scores_total** per space. This IS the canonical source for off-chain DAO voting (gasless signed messages stored by Snapshot), not an aggregator — adopted for Channel 3. - **Etherscan V2** (api.etherscan.io/v2/api) — V1 is fully DEPRECATED/dead (“switch to V2”). V2 REQUIRES AN API KEY. A free key (one key spans ~60 EVM chains via **chainid**) was obtained from the user (“AFA Paper” key) and stored gitignored at **04_code/.api_keys.json**. Verified working: **account/balance**, **block/getblocknobytime**, **proxy/eth_call**, **logs/getLogs** all return 200/OK. - **beaconcha.in** (ETH staking dashboard) — now returns 401 “valid API key required”; free no-key access GONE. **Boardroom** (api.boardroom.info) — 401 Unauthorized; needs a key. Both logged as paywalled; not used. - **Free public RPCs** (cloudflare / publicnode / ankr / llamapc / 1rpc) — serve only *latest* state; EVERY ONE blocks archive/historical queries (“Archive requests require a paid plan” / “must authenticate”). Confirms no keyless historical state path. - **CRITICAL — Etherscan V2 free eth_call is silently latest-only:** historical **eth_call** at a past **tag** (block) returns the *current* value with no error (beacon deposit **get_deposit_count()** and a UNI **balanceOf** were byte-identical at the merge block, an early-2021 block, and latest). So historical *state reads* are unavailable on the free tier; only **getLogs** (immutable event history) gives genuine point-in-time data. **Decision made:** (1) Channel 1 historical staked/locked supply must be reconstructed from **event logs (getLogs)**, never from historical **eth_call/balanceOf** state reads. (2) Channel 3 voting from Snapshot GraphQL (canonical). (3) beaconcha.in/Boardroom/Tally treated as paywalled — ETH staking comes from the deposit contract’s on-chain Deposit event logs instead of beaconcha.in. (4) DeFiLlama is a metadata registry only. **Rationale:** satisfies the spec’s “verify before building” rule and the user’s directive to source the lambda numbers from the canonical chain/Snapshot data rather than an aggregator, while honestly recording which originally-named sources are now paywalled. **Downstream impact:** the no-archive constraint caps Channel 1 to assets reachable by **event-log reconstruction on an EVM chain** (ETH native staking via the deposit contract; EVM vote-escrow tokens via Transfer logs into a known escrow contract). Non-EVM native PoS coins (Solana, Cosmos, Cardano, Tron, XRPL, ?) are NOT reachable with this key and become a documented coverage gap (Entry 24). If a paid archive-RPC/Etherscan-Pro tier is later obtained, historical **eth_call/balancehistory** would allow direct point-in-time **balanceOf** reads and broaden Channel 1 substantially.

Entry 22 — Asset->on-chain-identity map: DeFiLlama used as a registry, not a data source

Date: 2026-06-23 **Spec section affected:** 2.5 (per-asset source identification); precursor to all lambda channels **Asset(s)/period affected:** all in-universe assets (1,939) **What the spec wanted:** identify, per asset, the canonical chain explorer / token contract / governance venue from which to pull the lambda series. **What was actually available:** to reconstruct on-chain numbers we first need each **cmc_id**’s token contract address + chain (for Channel 1 log reconstruction) and its Snapshot space (Channel 3). Two keyless registries provide this metadata: (a) DeFiLlama **/protocols** carries a numeric **cmcId**, a token **address+chain**, and a **governanceID** (snapshot space); (b) CMC’s own **data-api/v3/cryptocurrency/detail?id=** returns **platforms[]** with **contractAddress+chain** per asset. **Decision made:** build **03_data/phase1/asset_onchain_identity.csv** joining the universe to DeFiLlama **/protocols** on **cmcId** (script **phase1_build_identity_map.py**). DeFiLlama is used STRICTLY as an address-book/registry — the lambda NUMBERS come from the chain (Etherscan logs) and Snapshot, never from DeFiLlama TVL. Raw registry cached at **03_data/raw/defillama/protocols.json**. **Rationale:** satisfies the user’s directive to source lambda from canonical chain data while still using a keyless directory to discover *which* contracts/spaces to read. **Downstream impact / coverage:** DeFiLlama’s **cmcId** is sparse — only 241/1,939 in-universe assets matched, of which 206 carry a token address and just 35 a Snapshot space (token-class 123/448 addressed, 29/448 with a space; coins lean to native chains DeFiLlama doesn’t list as protocols). The thin auto-mapping is WHY the Channel-3 space map is hand-extended (Entry 25)

and Channel-1 locks are a curated set (Entry 26). A future improvement is to enrich the map from CMC `detail.platforms[]` (cleaner per-asset contract coverage) to widen Channel 1 to more EVM tokens without per-protocol curation.

Entry 23 — Channel 1 (ETH native staking): beacon deposit-contract event-log reconstruction

Date: 2026-06-23 **Spec section affected:** 3.1 (staking/locking ratio); 2.3/2.5 (ETH transition, explorers) **Asset(s)/period affected:** ETH (cmc_id 1027), 2020-12 onward **What the spec wanted:** ETH staked supply per month from a staking dashboard (beaconcha.in-style), respecting `staking_start=2020-12-01`. **What was actually available:** beaconcha.in now requires a paid key (Entry 21); free-tier historical `eth_call` is latest-only. The canonical, free, historical-correct source is the Beacon Chain deposit contract's on-chain `DepositEvent` logs. **Decision made:** reconstruct month-end cumulative staked ETH from the deposit contract (0x0000?705Fa) `DepositEvent` logs via Etherscan V2 `getLogs`, parsing the 8-byte little-endian gwei `amount` from each event and cumulative-summing to each month-end block (`block/getblocknobytime`). Pre-`staking_start` months are emitted as NaN (PoW, no channel), NOT 0. Script `phase1_channel1_eth_staking.py` (resumable monthly checkpoints under `03_data/raw/phase1_onchain/`). Validated against known levels: ~2.17M ETH staked at 2020-12-31 (67,906 deposits), ~5.2M at 2021-05-31 — both match public record. **Rationale:** uses the chain itself; respects the dated transition; immutable logs sidestep the no-archive constraint. **Downstream impact / CAVEAT:** the deposit contract only RECEIVES ether — post-Shapella (2023-04) validator exits/withdrawals are consensus-layer and do NOT debit it, so cumulative-deposited OVERSTATES net active stake after 2023-04 (it is an upper envelope, a monotone on-chain conviction proxy). For an exact net-staked series, a consensus-layer (beacon) data source or a paid execution archive would be needed. Flagged in the coverage report and in the output `note` column.

Entry 24 — Channel 2 (holding duration / coin-age): NOT BUILT this phase (documented gap)

Date: 2026-06-23 **Spec section affected:** 3.2 (holding-duration channel) **Asset(s)/period affected:** all assets, all months **What the spec wanted:** an on-chain HODL-wave / coin-age proxy (share of supply unmoved over a window, or average coin-age of moved supply) per asset-month. **What was actually available:** computing coin-age requires the FULL transfer/UTXO history of each chain (every address's last-active balance, or UTXO ages for BTC) reconstructed to each month-end. For account chains that is the entire Transfer-log set of every token (orders of magnitude beyond the targeted escrow-only logs used in Channel 1); for BTC it is the full UTXO set age distribution. No free API serves ready HODL-wave series across the panel's chains (Glassnode/CoinMetrics/Artemis are paid; the keyless explorers cap `getLogs/archive` as in Entry 21). **Decision made:** do NOT build Channel 2 this phase. Keep the `ch2_holding` column in the lambda schema (always NaN) so the structure is explicit, and flag it as the single largest lambda-channel gap. lambda is therefore assembled from Channels 1 and 3 only in Phase 1. **Rationale:** the spec (Operating Principle, ?0) forbids silently substituting a weak proxy; a credible coin-age series needs either a paid fundamentals API or a per-chain full-history indexer, both out of scope for this pass. Better to flag the gap than ship an unsupportable number — especially as ?3.2 notes this would be the *only* channel for pre-2020 coins, so its absence is exactly what gates early-sample coin lambda. **Downstream impact:** pre-2020 coins get NO lambda at all in Phase 1 (no staking pre-PoS, no voting, no coin-age). If a paid source (Glassnode/CoinMetrics) or a BTC/ETH full-node indexer is later obtained, Channel 2 is the highest-value addition for early-sample coin coverage.

Entry 25 — Channel 3 (voting): Snapshot GraphQL + curated space map + token-weight guard

Date: 2026-06-23 **Spec section affected:** 3.3 (voting engagement), 2.5 (governance venues) **Asset(s)/period affected:** governance tokens with a Snapshot space (55 mapped) **What the spec wanted:** monthly participation = voters / eligible supply, from Snapshot/Tally/Boardroom for off-chain DAOs. **What was actually available:** Snapshot GraphQL is free/keyless and IS the canonical store of off-chain votes (Entry 21); Tally/Boardroom now need keys. DeFiLlama `governanceID` mapped only 35 in-universe

assets to spaces. **Decision made:** (1) space map = DeFiLlama governanceID spaces (35) ? a curated, name-verified set of 27 major DAOs keyed by EXPLICIT cmc_id (avoiding the symbol-collision trap — e.g. Uniswap is 7083, not the symbol-matched 4113), written to 03_data/phase1/snapshot_space_map.csv (56 spaces). (2) Per space, page ALL closed proposals (created-asc cursor) and aggregate by the month a proposal’s voting ENDS. (3) Channel-3 value = token-weighted turnout = $\text{mean}(\text{scores_total})/\text{circulating supply}$ (eligible base from the Phase 0 panel). (4) **Token-weight validity guard:** spaces whose median voting-power-per-voter ($\text{scores_total}/\text{votes}$) < 10 are 1-person-1-vote/ticket spaces where **scores_total** is NOT token-denominated; their **vw_turnout** is nulled (flagged **token_weighted=False**). This caught snxgov.eth, enzymefinance.eth, ilvgov.eth. Script **phase1_channel3_voting.py**; raw proposals cached per space. Result: 55 distinct assets, 51 with a valid token-weighted turnout, 1,598 asset-months, 2020-07->2026-06. **Rationale:** Snapshot is the real source, not an aggregator; explicit-cmc_id curation keeps the join correct; the token-weight guard prevents a strategy artifact from contaminating the z-scored channel. **Downstream impact:** on-chain-only DAOs (Compound Governor beyond comp-vote, MakerDAO, and tokens that vote purely on-chain: MKR, LQTY, PENDLE, RUNE, PERP, WLD, ONDO, ENA, ?) are NOT on Snapshot and are a documented gap — adding them needs on-chain Governor event reconstruction (VoteCast logs), a later extension. Voting is absent pre-2020 and for pure coins, as the spec anticipated.

Entry 26 — Channel 1 (EVM vote-escrow/staking locks): curated escrow set, log-reconstructed

Date: 2026-06-23 **Spec section affected:** 3.1 (locking ratio for vote-escrow tokens) **Asset(s)/period affected:** CRV, CVX, FXS, SUSHI, AAVE, YFI (6 EVM governance tokens) **What the spec wanted:** locked supply for vote-escrow tokens (veCRV/veBAL-style) ? supply. **What was actually available:** no free per-protocol “locked supply” time series (DeFiLlama gives a *-staking USD bucket but that is the aggregator, not the chain; and historical balanceOf is latest-only). Canonical method: locked supply at month-end = cumulative (Transfer INTO the escrow) ? (Transfer OUT of the escrow), from the base token’s on-chain Transfer logs (verified the Etherscan multi-topic filter **topic2+topic0_2_opr=and** isolates escrow-directed transfers). **Decision made:** reconstruct locked supply for a CURATED, high-confidence set of 6 escrows where the contract holds the BASE token directly (so $\text{balanceOf}(\text{escrow})=\text{locked}$): veCRV, vICVX, veFXS, xSUSHI, stkAAVE, veYFI (script **phase1_channel1_evm_locks.py**, addresses + mechanism in the script header). EXCLUDED by design and documented (not silently proxied): **veBAL** (locks an 80/20 BPT, not BAL) and **SNX** (collateral C-ratio system, not a simple lock). xSUSHI and stkAAVE are reward-staking rather than pure vote-escrow but are a genuine locked/committed-supply signal and are kept in Channel 1 flagged via **mechanism**. **Rationale:** restricts the on-chain reconstruction to cases where the escrow-balance = locked-supply identity holds cleanly, maximizing correctness over coverage; gives the vote-escrow tokens BOTH a Channel-1 (locked) and a Channel-3 (voting) value, the only way any asset gets a multi-channel lambda in Phase 1. **Downstream impact:** Channel 1 token coverage is intentionally small (6) this pass; widening it requires per-protocol escrow curation (each lock contract verified individually) or a paid archive tier for direct historical balanceOf. The veBAL/SNX exclusions should be revisited if those assets matter to a result.

Entry 27 — lambda assembly: monthly z-score, equal-weight, >=2-asset standardizability rule

Date: 2026-06-23 **Spec section affected:** 3 (lambda construction), 3.4 (output) **Asset(s)/period affected:** all observed asset-months with >=1 standardizable channel **What the spec wanted:** lambda_t = equal-weighted average of the standardized (z-scored within each monthly cross-section) values of whichever channels are observable; no imputation; record how many/which channels contributed. **What was actually available:** the three channel series built this phase — **ch1_staking** (ETH + 6 EVM locks), **ch2_holding** (none — Entry 24), **ch3_voting** (51 tokens). **Decision made:** **phase1_assemble_lambda.py** (1) z-scores each channel within each (month, channel) cross-section, (2) averages the available z-scores per asset-month with equal weight, (3) records **n_channels + channels**. **Standardizability rule:** a channel enters a given month’s lambda only if >=2 **observed assets** have a finite value that month AND the cross-sectional std > 0 — a single-asset channel cannot be z-scored (z would be 0/NaN) and is dropped for that month, with the fact counted in **_lambda_channel_diagnostics.csv**. Raw per-channel values are carried alongside **lambda_z** for audit. lambda is computed on **status='observed'** rows only. **Rationale:** the spec’s “standardize within the monthly cross-section” is only defined for a cross-section of >=2; making the rule explicit prevents a

degenerate single-asset channel (e.g. ETH alone on Channel 1 before the ve-tokens enter) from contributing a spurious $z=0$. Equal-weighting and no-imputation follow the spec verbatim. **Downstream impact:** result = 1,308 asset-months, 51 assets, 2020-08->2026-05; 253 asset-months are 2-channel (the 6 vote-escrow tokens), the rest 1-channel. λ is currently a **standardized score (λ_z)**, not a **[0,1] locking fraction** — the SoV/MoE map $\lambda/(1-\lambda)$ in the theory needs a level, not a z-score, so Phase 4 must decide how to convert (e.g. use the raw staking/locking ratio as the level where Channel 1 exists, and treat the z-scored multi-channel index as the cross-sectional conviction *ranking* the hypotheses actually test). Flagged for the Phase 2/4 kickoff.

Entry 28 — Phase 1 close-out scope and Phase 2 kickoff direction (Cowork review, pre-build)

Date: 2026-06-23 **Spec section affected:** 3 (λ channels, finishing touches), 4 (NVT_GL), 7 (phasing) **Asset(s)/period affected:** n/a (session-scoping decision, not a data change itself) **What the spec wanted:** a written coverage report at the end of each phase before scope is adjusted (?) — Phase 1’s own report (?) left several open items needing a human call before Phase 2 should start. **What was actually available:** three open items from PHASE1_COVERAGE_REPORT.md ?7 needed a decision: (1) whether to finish Phase 1 loose ends before or in parallel with Phase 2 — NVT_GL doesn’t depend on λ density, so it isn’t a hard blocker either way; (2) whether to procure a paid source (Glassnode/CoinMetrics/Artemis) for Channel 2 (coin-age) now; (3) how to handle ETH’s post-Shapella cumulative-vs-net staking question when resuming the staking series. **Decision made:** (1) one combined Claude Code session does Phase 1 finishing touches (finish ETH series, confirm $N=250$, revisit the 16 gray-zone names with real lock data, reconcile the report’s internal asset-month/asset-count inconsistency) followed immediately by Phase 2 (NVT_GL) in the same session — see 04_code/CLAUDE_CODE_PHASE2_KICKOFF_PROMPT.md. (2) Channel 2: explore a free workaround first (check live whether any BTC-chain explorer exposes a usable UTXO-age/“coins last moved” metric) before considering any paid source; if nothing free and credible turns up, the gap stays documented and the paid-source question stays open for a later call. (3) ETH staking: keep the cumulative-deposit method as-is when resuming the series — do NOT add post-Shapella withdrawal netting this pass, since that needs a new consensus-layer data source (new scope, not a finishing touch); keep it documented as a monotone upper-envelope proxy per Entry 23. **Rationale:** NVT_GL’s inputs (MC, PQ, g , r_e) are independent of λ , so gating Phase 2 on λ density would cost time without a real dependency. The Channel 2 paid-source question is a real cost decision and shouldn’t be made by inertia — checking a free option first costs nothing and may partially close the gap. The ETH netting question would expand this session’s scope into a new data source for a one-line accuracy improvement on an already-flagged, already-documented caveat — not worth doing before the series is even fully resumed. **Downstream impact:** if the free BTC coin-age check in Part A succeeds, it should be generalized to other major pre-2020 coins before the paid-source question is revisited. If ETH’s post-Shapella overstatement turns out to matter materially to a result later, the netting question returns as a Phase 3/4 item once a consensus-layer source is identified.

Entry 29 — Phase 1 close-out: ETH series finished, numbers reconciled, Channel 2 BTC re-checked, $N=250$ confirmed, gray-zone revisited

Date: 2026-06-24 **Spec section affected:** 3 (λ channels — finishing touches), 2.1 (universe size), 2.3 (classification), 7 (per-phase coverage report) **Asset(s)/period affected:** ETH staking series; the λ index headline counts; the 16 gray-zone names; BTC (coin-age source audit) **What the spec wanted:** Phase 1’s coverage report (?) left five open items requiring a human call before Phase 2; Entry 28 scoped them into this combined session. This entry records how each was resolved. **What was actually available / what was done:**

1. **ETH staking series — FINISHED.** Re-ran `phase1_channel1_eth_staking.py` to completion via its monthly checkpoints (`03_data/raw/phase1_onchain/eth_staking_monthly.json`). Full series is now **66 month-ends, 2020-12 -> 2026-05**, cumulative deposited ETH rising from 2.17M (ratio 0.019) to 86.16M (ratio 0.714). The cumulative-deposit method was kept **as-is** — post-Shapella withdrawal netting was deliberately NOT added (it needs a consensus-layer/beacon source = new scope, not a finishing touch; Entry 28). Still documented as a **monotone upper-envelope conviction proxy, not a net-stake figure** (the >0.45 ratios in 2025–26 are the documented overstatement; `note` column

carries it).

2. **Channel 2 (coin-age) — free BTC workaround checked live (2026-06-24), still a gap.** Before leaving Entry 24's conclusion in place, probed free Bitcoin explorers specifically:
 - **mempool.space / blockstream.info (Esplora):** no aggregate coin-age / HODL / CDD metric — only per-tx/address/UTXO data (reconstructing coin-age from them = the full UTXO-set indexing Entry 24 ruled out).
 - **blockchair /stats:** free/keyless and carries `cdd_24h + hodling_addresses`, but only as a **current 24h snapshot**; historical CDD chart data is not on a free/keyless endpoint (404 / 401 + bot-protection).
 - **bitcoin-data.com / bgeometrics (/v1/cdd, /v1/ancient-supply, dormancy, HODL waves):** a genuine free, keyless coin-age API — the closest usable thing found — BUT
 - (i) free tier serves only a **trailing ~4 years** (2022-06->present; 1,458 daily records), so it cannot reach the **pre-2020 depth that is the whole point**; (ii) hard rate limit **10 requests/hour** (`RATE_LIMIT_HOUR_EXCEEDED`); (iii) full history is paywalled — i.e. the same paid-aggregator category Entry 24 flagged, with only a shallow free slice. Nothing free *and* credible *and* panel-usable turned up, so the breadth check on LTC/XRP/DOGE was not pursued (depth + rate caps disqualify the approach first; XRP/DOGE aren't UTXO-CDD chains anyway). **No paid source procured — decision deferred (Entry 28).**
3. **N=250 — confirmed, no change.** Per the report's own ?7 point 5: lambda density is governance-token-driven, and tightening (or widening) the universe would not close the coin-side gap, which is a **source problem (no free coin-age / non-EVM staking data), not a universe-size problem.** The N=250 rank screen stays as set in Phase 0 (Entries 7, 15). Stated explicitly on record per the directive; no code change.
4. **16 gray-zone names — revisited with real lock data, all stay other.** The only Channel-1 lock series built (Entry 26) are the 6 curated escrows (CRV/CVX/FXS/SUSHI/AAVE/ YFI) — **none of the 16 gray-zone names is among them**, so no security-staking lock series exists for any of them. Checked each against the actual channel data:
 - **OP, MNT, RPL, SSV** — have a Snapshot **voting** space (so they DO appear in lambda as **other**, voting-only: SSV 31, RPL 28, MNT 10, OP 7 asset-months) but **no security lock**. OP/MNT are L2 gas+governance (security leans on Ethereum); RPL is a node-operator *collateral* bond (like SNX's excluded C-ratio system, Entry 26), SSV an operator bond for distributed-validator infra — neither is chain-security staking. Governance != a coin security lock and != a clean vote-escrow token -> stay **other**.
 - **MANTA, IMX** — L2; staking exists but security leans on Ethereum; no lock series, no voting -> **other**.
 - **ANKR, STRD** — liquid-staking protocol/appchain tokens (Entry 8 keeps the LST sector out of the coin/token cut); STRD is non-EVM (Cosmos), unreachable with the EVM key (Entry 21); no lock series, no voting -> **other**.
 - **EWI** (PoA, weak seigniorage), **GBYTE** (DAG, historically no staking reward), **FCT** (entry-credit, no security-staking reward) -> **other**.
 - **BLUR, LOOKS, ME** (NFT-marketplace governance/incentive tokens) -> **other**.
 - **PNK** (Kleros juror work/fee-share stake — not network security; no lock series) -> **other**, still "pending review" as in Entry 18. **PTS** (obscure symbol-collision, likely the mineable ProtoShares but low-confidence) -> **other**. **Net: zero reclassifications; classification_table.csv** is unchanged, so lambda assembly was not re-run on account of this item. Evidence is unchanged from Entry 18, exactly as the report ?7.4 anticipated.
5. **Report internal inconsistency — reconciled.** PHASE1_COVERAGE_REPORT.md ?5 had previously shown **1,326 asset-months / 52 assets** while ?8 and Entry 27 showed **1,308 / 51**. After the ETH resume the *correct, final* figures are **1,374 asset-months / 52 distinct assets** (2020-08 -> 2026-05); the +66 over the 1,308 baseline is exactly ETH's now-complete 66-month series (up from 18 partial months). Every section of the report (headline, ?4.1a, ?5 with its `n_channels` table and by-year table,

?6, ?7, ?8) was updated to these numbers. Final `n_channels` split: 1-channel 1,121 (voting-only 924, staking-only 197 incl. ETH's 66) + 2-channel 253 (the 6 vote-escrow tokens) = 1,374.

6. **(Optional) CMC `detail.platforms[]` identity-map enrichment — deferred.** Explicitly lower-priority and “not required before Part B” per the kickoff prompt; left for a later session to widen Channel 1’s EVM token coverage beyond the curated 6. Not done this session.

Decision made: resolve items 1/3/4/5 as above (finish ETH as upper-envelope; confirm $N=250$; keep all 16 gray-zone names **other**; reconcile the report to 1,374/52); keep item 2 (Channel 2) a documented gap with the free-source audit on record and the paid-source question deferred; defer item 6. **Rationale:** these are finishing touches, not new scope — each either completes an already-validated build (ETH), records a confirmation the report itself asked for ($N=250$), re-examines labels against now-available evidence without inventing unsupportable ones (gray-zone), fixes a bookkeeping inconsistency (report), or honestly reports a live source audit that came up empty (Channel 2) per the spec’s “flag, don’t guess” principle. **Downstream impact (what to re-check if this changes):** if a paid coin-age source or a consensus-layer ETH source is later procured, items 1 (net-stake ETH) and 2 (Channel 2) both reopen and `lambda` must be re-assembled. If the CMC `platforms[]` enrichment (item 6) is later run, re-run `phase1_build_identity_map.py` -> any new curated escrows -> `phase1_assemble_lambda.py`, and the 1,374/52 headline will move. The gray-zone names should be revisited again only if a security-staking lock series is actually built for one of them (e.g. RPL collateral, SSV operator bond) — at which point the coin/token call changes, not before.

Entry 30 — PQ definition (NVT_GL): fees rejected on theoretical grounds, not just feasibility; corrected to sector-appropriate transacted value

Date: 2026-06-24 **Spec section affected:** 4, 4.1 (NVT_GL — PQ definition) **Asset(s)/period affected:** n/a (methodological decision; applies to every coin and token going into Phase 2’s PQ series) **What the spec wanted:** ?4.1 lists “on-chain transaction (transfer) volume” for coins and “protocol throughput (DEX volume, total fees, or active-user counts)” for tokens — fees named as one acceptable option among several, not a preferred one. **What was actually available:** `PHASE2_PQ_DECISION_STATUS.md` (session 010) recommended **fees** (DeFiLlama protocol/chain fees) as the working PQ proxy for both coins and tokens, on feasibility grounds (free, keyless, deep history) — the human had proposed **TVL** as an alternative for tokens. An initial Cowork-side literature check (Artemis Analytics, Token Terminal, DeFiLlama definitions of “economic activity”) read as support for fees over TVL, since all three keep TVL as a metric separate from their “economic activity”/fees framing. **Decision made:** Reject fees as PQ — not on feasibility grounds, on theoretical ones. PQ in this paper’s own $M?V = P?Q$ identity is **nominal GDP**: the dollar value of goods/services exchanged, a flow. Fees are the **cost** of facilitating that exchange (a discretionary, governance-set rate — fee-switch votes, fee-tier choices), not the value of what was exchanged — structurally identical to treating a government’s tax revenue as a proxy for GDP. (Re-read literally, even DeFiLlama’s own description — fees “show how much the protocol is *facilitating* in economic activity” — supports this correction: a toll booth facilitates billions in freight without its toll revenue measuring that freight’s value. The initial literature check above read “facilitating” as “measuring”; that was an error, not a difference of opinion.) TVL is also rejected as PQ, on the reasoning already on record (stock, not flow) but reframed more precisely: TVL is the **capital stock** that enables activity — an AMM’s pooled inventory, a lending pool’s loanable funds, a staking protocol’s AUM — i.e. the K in a production function, not the output Y . A high-TVL pool with zero trading produces zero realized economic activity that period.

The corrected PQ definition is **transacted value**: the dollar value of what actually moved through the contract — swap/DEX volume for AMMs, loan-origination/borrow flow for lending, notional volume for derivatives — **sector-appropriate**, using the project’s own **sector** field (Entry 16) to route to the right flow per protocol type, rather than forcing one universal proxy (this also resolves the “DEX volume doesn’t generalize past DEX-type tokens” objection). This is, not coincidentally, what the *original* NVT ratio (Willy Woo’s Bitcoin metric, this paper’s direct namesake) used on the coin side: on-chain transaction *value*, never fees. TVL and fees are both **retained as secondary diagnostic columns** (capital-stock control; cost-of-intermediation/take-rate), plus a new **Volume/TVL turnover diagnostic** — a protocol-level restatement of $M?V=PQ$ with TVL standing in for M .

This correction is not yet fully implementable: whether true on-chain transfer/swap volume can be built at panel scale on the free Etherscan key (vs. falling back to DeFiLlama’s reported volume series) is an open empirical question, addressed by the pilot in Entry 31 / 04_code/CLAUDE_CODE_PHASE2_PQ_PILOT_PROMPT.md. **Rationale:** PQ is an accounting identity (P times Q), not a modeling choice — substituting a cost/toll variable for it embeds the protocol’s own discretionary pricing policy into the paper’s dependent variable, contaminating it for reasons unrelated to actual usage. The tax-revenue/GDP disanalogy and the capital-stock/production-function framing of TVL both come directly from re-deriving PQ from the paper’s own theoretical namesake (the quantity theory of money) rather than from how aggregators happen to label their dashboards. Full discussion in 06_documentation/ai_conversations/session_011_2026-06-24_pq_theory.md. **Downstream impact:** Supersedes the “(A) fees [RECOMMENDED]” lines for both Decision 1 and Decision 2 in PHASE2_PQ_DECISION_STATUS.md ?3–4. phase2_pq.py should NOT be built on fees. Whether it is built on true Etherscan-derived transfer/swap volume or on DeFiLlama’s reported volume series (with TVL/fees as side columns either way) depends on the pilot’s findings (Entry 31, once logged). If the pilot finds raw-Transfer-log volume infeasible at panel scale, DeFiLlama’s reported DEX/perps volume becomes the working source and the noise-multiplier estimate from the pilot should be carried into the paper’s methodology section as a documented limitation (spec ?6, the classic NVT wash-trading caveat).

Entry 31 — PQ source decided: DeFiLlama reported volume (raw Etherscan Transfer-log PQ piloted and rejected — wrong quantity, not just cost)

Date: 2026-06-24 **Spec section affected:** 4, 4.1, 6 (NVT_GL — PQ source & methodology limitation) **Asset(s)/period affected:** n/a (methodological; sets the PQ source for every Phase 2 asset). **Pilot evidence:** UNI (cmc_id 7083) and AAVE (cmc_id 7278), May 2026 (31-day window). **What the spec wanted:** ?4.1 lists “on-chain transaction (transfer) volume” as the ideal coin/token throughput measure; Entry 30 corrected PQ to sector-appropriate *transacted value* and left open whether that is buildable from raw Etherscan logs vs DeFiLlama’s reported volume (this Entry). **What was actually available / what the pilot found** (full report 03_data/PHASE2_PQ_PILOT_REPORT.md, code 04_code/phase2_pq_pilot.py + phase2_pq_pilot_diag.py, session 012): - **Cost is cheap for a recent window, contra a naive read of Entry 24.** Reusing the Channel-1 getLogs bisection with the counterparty filter dropped (ALL transfers): UNI = 133,350 transfers in **381 calls** (345.6 s); AAVE = 116,910 in **309 calls** (305.1 s) — ~345 calls/token-month, ~11 calls/day, ~0.9 s/call, far under the free 5 req/s & 100k/day caps. Entry 24’s “orders of magnitude” wall is the full *multi-year* regime, not a recent window. - **Extrapolation:** 1 token full history ~23.5k calls (~6 h); 1 recent month x 127 DeFi-slug tokens ~44k calls (~11 h) — both feasible. **Full history x 127 tokens ~1.75M calls (~17.5 days @ 100k/day); x 241 sluggish assets ~3.3M (~33 days)** — infeasible as a routine/repeatable build. - **Decisive: the governance token’s own Transfer events are the WRONG quantity.** UNI token-transfer volume = **\$0.79B** vs DeFiLlama Uniswap DEX swap volume = **\$36.75B** (swap **46.6x** larger, daily corr only **0.30**). AAVE raw sum was **physically impossible** (\$8.2x10?? vs 15.4M-token supply): **6** sentinel-value transfers, one of **10?? tokens = 6.5x10??x supply**; cleaned = **\$2.75B**, still unrelated to Aave lending throughput. Correct on-chain swap volume would require enumerating each protocol’s pool Swap events (= re-implementing DeFiLlama’s adapters) — out of scope on a free key. **Decision made: PQ source = DeFiLlama’s reported, sector-appropriate protocol volume** (DEX/swap for AMMs, perps notional for derivatives, borrow/origination for lending), routed by the **sector** field (Entry 16). **TVL and fees stay as side diagnostic columns** + the Volume/TVL turnover diagnostic, exactly as Entry 30 specified. Raw Etherscan Transfer logs are demoted to an **occasional spot-check**, never the primary source. This resolves the Entry 30 open question (and confirms the “if infeasible at panel scale -> DeFiLlama” branch of Entry 30’s downstream-impact note), and finalizes both Decision 1 and Decision 2 of PHASE2_PQ_DECISION_STATUS.md on the DeFiLlama-volume basis. **Rationale:** Two independent reasons, validity first. (1) **Validity:** a governance token’s Transfer events are not the protocol’s transacted value — empirically 47x off and barely correlated for UNI, and corrupted by non-economic sentinel transfers for AAVE (spec ?6 wash/internal-churn caveat in extreme form). (2) **Cost:** full-panel multi-year raw extraction is weeks of continuous runtime on the free key. DeFiLlama already computes sector-correct volume from the right per-protocol pool events. Option (A) [raw at panel scale] rejected on both grounds; pure Option (C) [flagship raw as a *source*] rejected because it still measures the wrong object — flagship raw kept only as a spot-check. **Downstream impact:** phase2_pq.py to be built on DeFiLlama reported volume (sector-routed),

with TVL/fees side columns — **after human review of the pilot**, not before. Carry into the paper’s methodology/limitations (spec ?6): reported aggregator volume is itself subject to the NVT wash-trading caveat; note that an independent raw-log reconstruction was piloted and found to measure a different quantity, so the aggregator series is adopted deliberately. Coin-side PQ (ETH/BTC *native* transfers) was NOT in this pilot and still faces the archive-state wall (Entries 21/24); DeFiLlama chain-level data remains the coin fallback. Full discussion: 06_documentation/ai_conversations/session_012_2026-06-24_pq_pilot.md.

Entry 32 — PQ source waterfall finalized: token fee->volume backout rule + coin source ladder (Cowork decision, supersedes Entry 31’s coin-fallback line and ?4 of PHASE2_PQ_DECISION_STATUS.md)

Date: 2026-06-24 **Spec section affected:** 4, 4.1, 6 (NVT_GL — PQ source, both Decision 1 refinement and Decision 2 resolution) **Asset(s)/period affected:** n/a (methodological; sets the PQ source-selection rule for every Phase 2 asset, tokens and coins). **Context.** Entry 31 settled tokens on DeFiLlama reported volume but left two things unresolved: (a) what to do when a protocol has *no* DeFiLlama volume series, and (b) Decision 2 (coins), which Entry 31’s closing line punted to “DeFiLlama chain-level data” — almost certainly meaning chain fees, which carries the identical toll-vs-value flaw Entry 30 already rejected for token fees. The human pushed back on the “archive access” framing for native-coin transfers (correct: that wall applies to historical *state* queries — coin-age/HODL-wave, Entry 24 — not to summing the *value* field already in ordinary block/transaction data, which any full node retains forever and Etherscan’s free API already serves), and asked specifically whether DeFiLlama has a coin-side “activity” metric before reaching for Artemis or raw Etherscan. Both points are resolved below with live, keyless verification (Cowork web-fetch, not the Cowork sandbox’s bash egress, which remains blocked). **What was checked and found: - DeFiLlama chain-level volume exists but is chain-structure-dependent.** `/overview/dexs/{chain}` (verified live) returns a real, aggregator-cleaned chain-level DEX-volume series. For Ethereum this is substantial (consistent with its DeFi-heavy structure). For **Bitcoin**, fetched live just now: total24h = **\$419,825**, total30d = **\$18.9M**, totalAllTime (since DeFiLlama started tracking) = **\$2.17B** — driven entirely by three niche bolted-on protocols (Bisq P2P exchange, Garden cross-chain bridge, LN Exchange Spot), not Bitcoin’s base-layer settlement, which is orders of magnitude larger. **Conclusion: DeFiLlama chain-DEX-volume is valid only for chains where DeFi/DEX activity is a material share of real economic activity (Ethereum, Solana, Avalanche-style smart-contract platforms). For payment/P2P-dominant coins (BTC, LTC, DOGE, etc.) it is degenerate and would silently understate true activity by orders of magnitude if used uniformly across the panel.** This is the same lesson as Entry 31 (aggregators clean data, but you must check the aggregator is measuring the *right object* for that specific asset) applied to a new case. - **Artemis Settlement Volume** (Entry 30/prior Cowork session): theoretically the right object for payment-dominant coins (P2P + DEX + NFT, explicitly includes native/token/stablecoin transfers, explicitly *not* a toll measure) but **access is unverified** — Phase 0 Entry 2 found Artemis’s API dead (`api.artemisxyz.com` -> HTTP 410); a fresh check found a relaunched API product and free “Lite” tier, but whether Settlement Volume is exposed standalone on the free tier, at what historical depth, and for how much of the ~250-asset coin panel, has **not** been live-tested from this sandbox. - **blockchain.com Charts API:** confirmed (web search) to expose a free, keyless, long-history daily “Estimated Transaction Value (USD)” series for BTC specifically, which already excludes change outputs (pre-cleaned of the classic UTXO change-inflation problem). Real candidate, BTC-only — does not generalize to the rest of the coin panel without a per-chain equivalent. **Decision made — two rules:** 1. **Token PQ fallback (refines Entry 31).** Primary source stays DeFiLlama sector-routed reported volume. Where a protocol has **no** DeFiLlama volume series, do **not** fall back to its fee as PQ directly (that reintroduces the Entry 30 toll-vs-value error). Instead, **only if** the protocol’s fee is a confidently known, *stable, single-rate* function of notional volume over the window in question (e.g. a documented flat swap fee, so **notional = fee / rate**), back out volume algebraically from the fee and use that as PQ. **Do not** apply this when the rate is multi-tier (e.g. Uniswap V3’s several fee-tier pools, where the blended rate is itself unknown without volume — circular), governance-adjustable/variable across the window, or not a simple function of notional (e.g. lending reserve factors, which are a % of *interest*, not of loan volume). If neither DeFiLlama volume nor a confident fee-rate backout exists for a protocol-month, **flag PQ as missing (NaN)** rather than substitute fees directly — per spec ?0 (“flag, don’t guess”). 2. **Coin PQ source ladder (resolves Decision 2), evaluated per coin/chain,**

not globally: - **Rung 1** — DeFiLlama chain-level DEX volume (/overview/dexs/{chain}), for chains where this is non-degenerate (material DeFi activity — confirm per-chain before using, not just for majors). - **Rung 2** — **Artemis Settlement Volume**, for chains where Rung 1 is degenerate/unavailable (payment- dominant coins), *if and only if* live-verified as free-tier-accessible with adequate panel coverage and historical depth (not yet confirmed — next session’s job). - **Rung 3** — **coin-specific native fallback:** blockchain.com’s Estimated Transaction Value series for BTC; for other chains without an equivalent ready-made series, raw native-value block iteration (now known to be a call-volume problem, not an access wall — feasible for a recent window, infeasible for full multi-year history per-chain, same shape as Entry 31’s token finding). - **Rung 4 (last resort, explicitly flagged, not a default)** — DeFiLlama chain fees, used only if no asset on rungs 1–3 is available for that coin, and documented in the paper’s limitations as a theoretically weaker, toll-based substitute, exactly parallel to how token fees were rejected. Aggregators (DeFiLlama, then Artemis) are preferred over raw reconstruction wherever they validly cover the asset, because they already absorb data-cleaning that raw logs require ad hoc (Entry 31’s AAVE sentinel-value problem) — but which rung applies must be checked per asset, not assumed uniformly, per the Bitcoin DEX-volume finding above. **Rationale:** Same first-principles standard as Entry 30 (transacted value, not toll) applied consistently to both the token fallback case and the coin case; the Bitcoin live check is the empirical guardrail that stops “DeFiLlama has a volume number for this chain” from being treated as automatically valid — exactly the AAVE-sentinel-value lesson, generalized. **Downstream impact:** Next Claude Code session should (a) live-verify Artemis Settlement Volume’s free-tier access, coverage, and historical depth; (b) live-check /overview/dexs/{chain} per coin in the panel to sort each into Rung 1 vs. needs-Rung-2/3; (c) build phase2_pq.py (tokens, with the fee-backout rule coded explicitly and rate-confidence judgment calls documented per protocol) and phase2_nvt_gl.py (coins, per the ladder above), logging every per-asset rung decision and every fee-backout rate used as it goes. Supersedes the “DeFiLlama chain-level data remains the coin fallback” line in Entry 31 and finalizes PHASE2_PQ_DECISION_STATUS.md ?4. Full discussion: this Cowork session (no transcript file written; captured here and in chat).

Entry 33 — Phase 2 NVT_GL built: token PQ (16 assets) + covered-coin PQ (50 assets) + full NVT_GL machinery; perps/derivatives volume found paywalled

Date: 2026-06-24 **Spec section affected:** 4, 4.1, 4.2 (NVT_GL build); 6 (landmines) **Asset(s)/period affected:** all assets entering Phase 2; PQ series + $g/r_e/PQ//NVT_GL$ per asset-month **What the spec wanted:** $NVT_GL = MC/PQ/$ per asset-month, with PQ = transacted value (Entries 30–32), g = trailing-3y CAGR of PQ , r_e = CAPM-style discount rate, g_inf/n robustness constants; emit all intermediates so assumptions vary without rebuild (spec ?4.2). **What was actually built (session 013):** - **Token PQ (Part A, phase2_pq_tokens.py)** — DeFiLlama sector-routed reported volume. Routed each of the 127 slugged tokens by its DeFiLlama category to the matching free volume dimension: /summary/dexs (11 tokens), /summary/aggregators (4, flagged as routed/double-counting), /summary/options (1). **16 tokens get a monthly volume PQ.** The other 111 are flagged NaN with explicit reasons — 93 have no transacted-value object (Yield/Farm/Lending/Gaming/Services/Bridge/Chain tokens have no swap/notional flow), 8 are slug-absent/ambiguous (SunSwap version-split, VELO/SXP symbol collisions — left NaN not guessed), and **10 are perps whose DeFiLlama volume dimension is now PAID-GATED (HTTP 402 at both /overview/derivatives and /summary/derivatives/{slug})** — a new landmine; open-interest is free but is a stock not a flow, so not valid PQ. **Fee->volume backout (Entry 32) fired for 0 protocols** — no token protocol-month met the strict documented-single-stable-rate test (all candidates are multi-tier DEX fees, variable perps fees, or lending reserve factors, explicitly excluded), so none were filled with raw fee. - **Coin PQ (Part B covered rungs, phase2_pq_coins.py)** — per the B1 rung table (Entry 34). **49 coins via Rung 1** (DeFiLlama /overview/dexs/{chain} daily->monthly, materiality $\geq 1\%$ monthly DEX/mcap) + **BTC via Rung 3** (blockchain.com Estimated Transaction Value, change-excluded, 2010->present). **Rung 4 (chain fees) auto-applied to ZERO coins** — it is a flagged toll proxy requiring explicit approval. The 81 GAP-R2 coins carry $PQ=NaN$, deferred to Phase 2b (Entry 34). - **NVT_GL (phase2_nvt_gl.py)** — $PQ0$ = trailing-12m sum of monthly PQ (annual throughput); g = trailing 3y CAGR of $PQ0$ ($\geq 1y$ fallback flagged in g_window_years ; capped [75%,+200%]); β = trailing-36m vs **BTC** (the spec’s simple market proxy — the cap-weighted index is numerically unusable, penny-token returns $\rightarrow inf$); $r_e = rf + \beta \cdot MRP$ with $rf=4\%$, $MRP=30\%$ as documented robustness parameters (NOT the realized ~114%/yr BTC pre-

mium, which is not a forward required return), floored at 0.05; $g_{inf}=3\%$, $n=10$. $PQ/* = \text{spec } ?4.1 \text{ levelized annuity}$; $NVT_GL = MC/PQ/$. **Result: 1,821 asset-months, 59 assets (46 coins, 13 tokens), 2016-08->2026-05. No pathologies (0 inf, 0 non-positive PQ/).** - **Diagnostics (phase2_pq_diagnostics.py)** — TVL (capital-stock control) + Volume/TVL turnover per Entry 30/31: 2,534 asset-months, median turnover 1.15x. **Decision made:** build NVT_GL on exactly the assets with a defensible transacted-value PQ ; flag every gap (NaN with reason) rather than substitute a toll/proxy; keep $MRP/rf/g_{inf}/n$ as emitted robustness parameters. **Rationale:** faithful to Entries 30–32 (transacted value, not toll) and spec ?0 (flag, don't guess) and ?4.2 (emit intermediates). Using BTC as the market index is the spec's named simple alternative and is forced by the cap-weighted index's numerical fragility. **Downstream impact (re-check if this changes): g-cap binds on 43.4% of NVT_GL rows**, and $PQ/*$ scales with $(1+g)^n$ so NVT_GL spans many orders of magnitude driven by g — **NVT_GL is reliable as a cross-sectional RANK/conditioning variable (how H2/H3 use it), not a cardinal level**; g_{cap} and n are the first sensitivity knobs (spec ?5). If perps volume access is later obtained, the 10 derivatives tokens reopen. If MRP/rf are re-chosen, re-derive r_e from the emitted beta — no rebuild. Outputs: 03_data/phase2/{pq_tokens,pq_coins,nvt_gl_panel,pq_diagnostics}.csv. Full numbers: 03_data/PHASE2_COVERAGE_REPORT.md. Session: 06_documentation/ai_conversations/session_013_2026-06-24_phase2_build.md.

Entry 34 — Coin PQ Step-B1 verification: Artemis paid-only; 81 material coins deferred to Phase 2b (human decision)

Date: 2026-06-24 **Spec section affected:** 4.1 (coin PQ source, Decision 2 / Entry 32 ladder); 7 (phasing — new Phase 2b) **Asset(s)/period affected:** the coin side of the panel; 81 material coins (peak mcap $\geq$ \$1B) flagged GAP-R2 **What the spec wanted:** Entry 32 set a coin source ladder (R1 DeFiLlama chain DEX -> R2 Artemis Settlement Volume -> R3 native -> R4 chain fees) and required a live B1 verification of Artemis access + per-coin DeFiLlama coverage before building, with an explicit instruction to STOP and report rather than guess if Artemis is paid-only or coverage is ambiguous for any material coin. **What was actually found (B1, live, session 013 — full report 03_data/PHASE2_COIN_PQ_VERIFICATION_B1.md):** - **Artemis REST API is PAID-ONLY.** Old hosts dead (`api.artemisxyz.com`->410, others DNS-fail, confirming Entry 2). Current product: `settlement_volume` *exists* as a standalone metric (right object), but the free “Lite” tier exposes only Terminal + a Google-Sheets plugin (100k ART calls/mo, not a scriptable REST path); Pro (\$300/mo) does not list REST API access; no free self-serve REST tier. **Rung 2 is closed for a reproducible pipeline.** - **DeFiLlama chain DEX coverage** (/overview/dexs/{chain}, 134 chains live): with an explicit materiality threshold (**30-day chain DEX volume ? market cap $\geq$ 0.01**; BTC 9x10?? fails, ETH 0.143/SOL 1.04 pass), **only 49 coins (40 material) are Rung-1 valid.** - **blockchain.com** Estimated Transaction Value (USD) for **BTC confirmed** (Rung 3, change-excluded, 2010->present) — BTC-specific, does not generalize. - **Net: 81 material coins** (XRP, DOGE, LTC, BCH, XMR, ZEC, DASH, ATOM, DOT, MATIC, ?) are left with **no free transacted-value PQ source** — their only ladder option was the now-paywalled Artemis. The DeFiLlama/BTC combination does **not** cover the coin panel adequately. **Decision made:** This is the B1 STOP-and-report condition. Reported to the human; the human's call (logged): **proceed now with the covered panel (16 tokens + 49 R1 coins + BTC) and stand up a Phase 2b to source the 81 coins later** — rather than auto-dropping them to Rung 4 (toll proxy) or silently NaN-ing them. Phase 2b kickoff written: 06_documentation/CLAUDE_CODE_PHASE2B_KICKOFF_PROMPT.md (XRPL APIs for XRP; bitinfocharts/ blockchair “sent-in-USD” for the UTXO payment coins; Artemis-paid option if access procured; **XMR noted a permanent gap — RingCT hides amounts**). Until Phase 2b, those 81 coins carry $PQ=NaN$ in `pq_coins.csv` / `nvt_gl_panel.csv`, flagged GAP:artemis_paid_only. **Rationale:** Entry 32's ladder is decided, but the B1 instruction explicitly designates the Artemis-paid / material-coverage-gap case as a human-review pause point — guessing a toll proxy for XRP/DOGE/LTC/etc. is exactly what the rule forbids. Proceeding with the covered panel keeps the session productive while the gap is honestly documented and scheduled, not papered over (spec ?0). **Downstream impact:** Phase 2b must fill these before any coins-only or full-panel H2/H3 result leans on coin NVT_GL coverage; the rung table 03_data/phase2_coin_rung_table.csv (rung==‘GAP-R2’) is the worklist. If Artemis API access is procured, Rung 2 reopens for most of them at once (Settlement Volume only — never the Total Economic Activity composite, which bundles toll measures).

Entry 35 — Phase 2b: 8 GAP-R2 coins sourced via bitinfocharts native settlement value; XRP/XMR permanent gaps; 71 others NaN (all live-verified, no Artemis key)

Date: 2026-06-25 **Spec section affected:** 4.1 (coin PQ source, Entry 32 ladder Rung 3-native); 6 (landmines); 7 (phasing — Phase 2b) **Asset(s)/period affected:** the 81 GAP-R2 coins (peak mcap $\geq$ \$1B) deferred from Phase 2 (Entry 34) **What the spec wanted:** Phase 2b (06_documentation/CLAUDE_CODE_PHASE2B_KICKOFF_PROMPT.md) — source *native settlement value* (on-chain payment/transfer value in USD, the coin-side analogue of Bitcoin's NVT denominator; NOT fees/DEX-volume/TVL) for the 81 coins, verifying free access live, flagging not guessing, and never doing raw multi-year block iteration. Artemis only if a key was procured. **What was actually available (live, keyless, 2026-06-25 — 04_code/.api_keys.json has only etherscan, so no Artemis; Rung 2 stayed closed):** - bitinfocharts “Sent in USD” (/comparison/sentinusd- $\{$ ticker $\}$.html) — free, keyless, daily, long history; summed daily->monthly (matching the BTC handling in phase2_pq_coins.py). **Critical landmine: unrecognised slugs silently serve BITCOIN's series** (verified: bch/bsv/btg/nano/peercoin/komodo/? via the $\{$ coin-name $\}$ -sentinusd alias all returned an identical BTC series), so the build is **ticker-keyed** and **BTC-default-guarded** (asserts each covered series' latest value $\neq$ BTC's). bitinfocharts exposes only 13 tickers [btc eth xrp zec doge ltc xmr bch dash etc bsv vtc btg]; the GAP-R2 overlap is **DOGE, LTC, BCH, DASH, ETC, BTG** (current through 2026-06) plus **BSV** (stale, ends 2021-08) and **ZEC** (stale, ends 2022-05). XRP's bitinfocharts page exists but is **empty** (not a UTXO chain; no “sent in USD” computed). - **XRP (cmc_id 52, highest-value GAP coin):** no free keyless historical XRPL payment-volume series. Checked live — data.ripple.com (Ripple Data API v2, which served payment_volume) -> **403/dead**; api.xrpscan.com -> account endpoints work but **no** historical-volume endpoint (docs+live); xrplmeta (s1.xrplmeta.org) -> token-metadata/cli node, not volume; api.xrpldata.com -> **XRPL NFT** API; bithomp -> 403 (key); data.xrplf.org -> nginx default / 404. Raw full-history ledger iteration (~21.6k ledgers/day) is the forbidden call-volume wall. - **XMR (cmc_id 328):** RingCT cryptographically hides amounts -> native transacted value unobservable on any source. Permanent gap, per the kickoff's explicit STOP. - **Next tier (ATOM/Cosmos, KAS/Kaspa, DOT-KSM/Polkadot, FIL, THETA, XTZ, VET, IOTA, NEO, ?):** probed live — Cosmos public LCD (cosmos-rest.publicnode.com) and api.kaspa.org return only **current state** (supply/network); Filfox returns **base-fee** (a toll, not value); Mintscan (apis.mintscan.io) and Subscan require **API keys**. No free, keyless, ready-made historical USD transacted-value series. - **blockchair** has **no** free historical charts API (/charts/... -> 404); only current /stats. **Decision made:** Fill PQ for the **8 bitinfocharts-covered coins** (pq_source=bitinfocharts_sentinusd, rung=R3-bitinfo), with explicit per-coin honesty flags in the note column: UTXO “Sent in USD” is total *output* value and therefore **change-INFLATED** (opposite of BTC's change-excluded series); ETC is account-model (no change); ZEC is **transparent-pool only** (shielded amounts hidden) and stale; BSV stale. Leave the other **73 coins PQ=NaN** with refined, source-specific reasons (XRP no_free_xrpl_volume_series, XMR xmr_ringct_unobservable, the rest no_free_native_series_p2b). No toll/fee proxy (Rung 4) applied to any coin; nothing guessed. New script 04_code/phase2b_pq_coins.py (idempotent post-process on pq_coins.csv; raw HTML cached gitignored under 03_data/raw/bitinfocharts/); must run AFTER phase2_pq_coins.py and BEFORE phase2_nvt_gl.py. **Result (re-ran phase2_nvt_gl.py + phase2_pq_diagnostics.py):** PQ asset-months **2,557 -> 3,358** (73 assets); NVT_GL **1,821 -> 2,526** asset-months, **59 -> 67** assets (**54 coins, 13 tokens**), 2016-08 -> 2026-05; no pathologies. g-cap still binds ~43% — the “rank, not cardinal level” caveat (?2a) is reinforced by the change-inflated UTXO PQ (e.g. LTC median NVT_GL ~ 5x10??). **Rationale:** native on-chain transfer/output value is exactly the coin-side object the original NVT used (Entry 30) and is what bitinfocharts “Sent in USD” measures; aggregators are preferred over raw reconstruction where they validly cover the asset (Entry 32). Every covered series is flagged for its known bias rather than silently blended; every uncovered coin is flagged with a live-verified reason rather than dropped to a toll proxy — per spec ?0 (“flag, don't guess”) and the kickoff's explicit rules. **Downstream impact (re-check if this changes):** if Artemis Settlement Volume or a Subscan/Mintscan/Glassnode key is procured, Rung 2 reopens for most of the 71 no_free_native_series_p2b coins at once (Settlement Volume only, never the Total-Economic-Activity composite) and the panel widens further. The 8 change-inflated UTXO series should be sensitivity-checked (or, for BTC-comparability, a change-excluded equivalent sought) if a coins-only NVT_GL *level* — not rank — ever drives a result. Re-running

phase2_pq_coins.py regenerates the old GAP markers, so phase2b_pq_coins.py must be re-run after it. Outputs unchanged in schema: 03_data/phase2/{pq_coins,nvt_gl_panel,pq_diagnostics}.csv. Session: 06_documentation/ai_conversations/ session_014_2026-06-25_phase2b_coins.md. **Do not start Phase 3 without review.**

Entry 36 — Dune feasibility pilot: free tier cleanly solves all 3 token-side PQ gap categories (Lending/Liquid-Staking/Perps); recommend (A) gated on a free-tier panel-scale dry-run

Date: 2026-06-25 **Spec section affected:** 4, 4.1 (token PQ source — the Entry-32 “flag PQ missing” branch for categories with no DeFiLlama series); 6 (landmines) **Asset(s)/period affected:** the token-side PQ gap — Lending, Liquid Staking, and the perps/derivatives 402-paywall (10 tokens incl. GNS). Pilot evidence: AAVE/LDO/GNS, trailing 30d (2026-05-27 -> 2026-06-25). **What the spec wanted:** test, *before* requesting university funds for a paid Dune subscription, whether Dune’s decoded/Spellbook tables can supply the protocol-level transacted value DeFiLlama has no (or paywalled) volume dimension for — one token per NaN category. Diagnostic pilot only, **not** a Phase 2c build. **What was actually available / what the pilot found** (full report 03_data/DUNE_PILOT_REPORT.md, code 04_code/dune_pilot_{test,explore,aggregate,verify}.py, raw JSON 03_data/raw/dune_pilot/, session 015): - The pre-written dune_pilot_test.py auto-picker stopped all 3 at **needs_manual_column_mapping** — it grabbed the **wrong** tables (Aave Polygon transfers, Lido submit w/o USD, a Gains per-trade table w/o amount_usd). The value was in the **manual mapping** to normalized abstractions, exactly as the prompt anticipated. - **AAVE (Lending) -> lending.borrow** (cross-protocol spell, pre-priced amount_usd): 30d Ethereum-v3 borrow origination = **\$4.286 B** (25,452 borrows; repays/liquidations are negative amount_usd, excluded). Cross-check: ? DeFiLlama Aave-v3 Ethereum borrowed-outstanding \$7.215 B = **0.59x/30d** (~1.7-mo avg loan life) — order-of-magnitude sane (cf. Entry 31’s 46.6x / 10?? misses). **PASS.** - **LDO (Liquid Staking) -> lido_ethereum.steth_evt_submitted + ?withdrawalqueueerc721_evt_withdrawalclaimed x prices.day:** 30d stake+unstake flow = **\$1.583 B** (stake \$938 M + unstake \$645 M). **The one trap, caught:** prices.day filtered by symbol='WETH' matched **3 contracts** (\$0.0000008–\$2,112, avg \$795) -> nonsensical **\$767/ETH**; switching to the **canonical WETH contract 0xc02a7756cc2** gave **\$1,768/ETH** vwap, consistent with DeFiLlama spot **\$1,560**. **PASS** — and the Entry-31 lesson recurred in a new form (clean-looking number, wrong; only the cross-check exposed it). Headline caveat for any hand-rolled price join at panel scale: filter by contract address, never ticker. - **GNS (Derivatives) -> dune.gains.result_g_trade_stats_defi_llama** (Gains’ own table that *feeds* DeFiLlama, = the dune.com/gains/gtrade_stats dashboard data): 30d notional = **\$1.178 B** (\$39.3 M/day, 84,929 trades, all chains). DeFiLlama per-protocol derivatives endpoint **re-confirmed 402** — the exact paywall that created this gap; Dune retrieves the same series **for free**. Cross-check is internal+historical (component sum = **daily_volume** to 0.000%; \$/day in gTrade’s known \$20–60 M band; per-trade \$13,871 sane). A raw-table reconstruction (**result_gtrade_all_orders_daily_view.position_size_dai**) gave ~\$44 M (~27x low) — **logged inconclusive (ambiguous column = margin/DAI-only, not notional)**, per “don’t guess at ambiguous columns,” **not** treated as a contradiction. **PASS (plausibility).** - **Cost:** no Dune credit-balance endpoint/header exists. Precise counts: **14 catalog /datasets/search (not query-metered) + 9 /sql/execute (all small; free tier rejects medium; ~130 datapoints total); only 3 executes strictly necessary.** ? on the order of <100 of 2,500 free monthly credits (<4%); **pool resets monthly.** Cost is a non-issue at this scale. **Decision made: Dune cleanly solves all three token-side PQ gap categories on the free tier -> recommendation (A)** (free tier sufficient to prototype all three at panel scale), with **one gate before any funding request**. Spell tables are cross-protocol, so the whole token panel’s history is a handful of **GROUP BY project, block_month** queries (not one-per-token), keeping even multi-year backfill credit-cheap. **The single risk that could downgrade to (B):** free tier only exposes the **small query engine** (the LDO join already took 66 s); a multi-year full-panel scan with joins could hit per-query time/row limits and require the paid **medium/large engine** — for the *engine*, not for credits. Two coverage caveats: confirm each panel protocol is in Dune’s spellbook (majors are; long-tail may not), and apply canonical-contract discipline to every price join. **Rationale:** same Entry-30/31 standard — measure the right object (protocol transacted value, not a side token’s transfers or a fee/toll) and verify every number against an independent reference before trusting it. The normalized spell tables already absorb the data-cleaning that raw Etherscan logs required ad hoc (Entry 31’s AAVE sentinel problem), and DeFiLlama’s

own upstream Gains table fills the 402 gap directly. **Downstream impact (do this before requesting funds / before Phase 2c):** run a **free-tier panel-scale dry-run** — one grouped `lending.borrow` / `perpetual.trades` (or `dex.trades`) / Lido-events query over full history across the real token list — and observe whether the **small** engine completes. Completes -> free tier fully sufficient, **no subscription needed (A)**; times out on multi-year scans -> modest paid plan justified **only** for the engine (B). This is the actual feasibility data point for the funding decision Moazzam is weighing. **Do not buy a Dune subscription, and do not start a Phase 2c build, before this dry-run and human review.** No panel outputs were written this session (`pq_tokens.csv` and all Phase 2/2b CSVs unchanged — diagnostic pilot only). Session: `06_documentation/ai_conversations/session_015_2026-06-25_dune_pilot.md`.

Entry 37 — Dune free-tier full-panel dry-run: small engine survives full-history scans (engine risk DISPROVEN -> (A)); 13 of 17 NaN tokens are not in Dune’s spellbook (coverage gap, not engine gap -> (C), unfixable by a paid plan)

Date: 2026-06-25 **Spec section affected:** 4, 4.1 (token PQ source — the Dune branch for DeFiLlama-absent/paywalled categories); 6 (landmines) **Asset(s)/period affected:** the 17 token-side NaN tokens in the 3 pilot categories — Lending {AAVE, ANC, BZRX, OM, STRK, WXT}, Liquid Staking {LDO}, Derivatives {AVNT, DDX, GNS, HAKKA, HXRO, KP3R, LINA, MIR, MYX, NMR}. Full history, panel 2015-08->2026-05 (130 months). **What the spec/Entry-36 wanted:** Entry 36 made recommendation (A) conditional on a **free-tier panel-scale dry-run** answering two open caveats for the *real* token list (not the 3-token sample): (a) is each NaN token’s protocol actually in Dune’s normalized spellbook (majors confirmed; long-tail unknown), and (b) does a **full-history, full-panel** grouped query complete on the free **small** engine (the LDO join took 66 s on just 30 days — the one risk that could downgrade to (B), a paid engine). Diagnostic only — **not** a Phase 2c build. **What was actually found** (full report `03_data/DUNE_DRYRUN_REPORT.md`, code `04_code/dune_dryrun_{coverage,coverage2,fullpanel}.py`, raw `03_data/raw/dune_dryrun/`, session 016): - **Engine risk (b) — DISPROVEN; (A) holds.** All three full-history, monthly-grouped queries completed on **small** with no timeout / row cap / truncation / error: **Lending** (`lending.borrow` GROUP BY project, month, `aave+strike`) **2.0 s** wall / 0.4 s engine, 138 rows; **Derivatives** (`dune.gains.result_g_trade_stats_defi_llama` GROUP BY month) **1.9 s** / 0.2 s, 54 rows, ? notional \$122.8 B; **Liquid Staking** (Lido submit + withdrawalclaimed x canonical-WETH `prices.day`, GROUP BY month) — the join-heavy worst case — **48.8 s** wall / 43.1 s engine, 102 rows, ? stake \$61.1 B / unstake \$44.0 B. The 66 s pilot figure (30 d) was variance, not a scan that grows dangerously with history (full history was *faster*). **small is sufficient for the full panel; no paid engine needed.** - **Coverage (a) — most of the 17 are not on Dune; the gap is in the spellbook, not the engine.** Decisive test = does the protocol appear as a project in the normalized cross-protocol spells (`lending.borrow`, 25 projects; `perpetual.trades`, 28 projects), the layer that made AAVE/LDO/GNS usable — *not* mere catalog text-search, which returns only raw decoded ERC-20/vault contracts. **Only 4 of 17 covered:** AAVE (`project='aave'`), STRK (`project='strike'`, Strike Finance — a NET-NEW find beyond the pilot: \$224.3 M lifetime borrow origination once filtered to `transaction_type='borrow'`; its unfiltered net was ?\$0.02 B because repays/liquidations carry negative `amount_usd`), LDO (Lido events + canonical-WETH join), GNS (Gains DeFiLlama-feed table + `gains_network` in `perpetual.trades`). **The other 13** (ANC, BZRX, OM, WXT; AVNT, DDX, HAKKA, HXRO, KP3R, LINA, MIR, MYX, NMR) appear only as raw token/vault contracts with no pre-priced notional/volume (e.g. ANC = `anchor_ethereum.anchorvault_*` ETH bridge only; MIR’s only EVM hits are the unrelated `mirror.xyz` NFT product — MIR = Mirror Protocol on Terra; MYX has decoded router/positionmanager calls but is absent from `perpetual.trades`). - **Cost:** 6 **small** executes (~600 datapoints) + catalog searches (not query-metered) ~ <1% of the 2,500 free monthly credits. Hit a transient **429 rate-limit** on `/datasets/search` after ~14 quick calls -> added throttle+retry (landmine: the catalog search endpoint rate-limits at a few calls/sec). **Decision made: No paid Dune plan is needed.** Recommendation per category: **(A)** Lending, Liquid Staking, Derivatives — for the **4 covered** NaN tokens (AAVE, STRK, LDO, GNS), the free **small** engine handles the full panel (worst case 48.8 s). **(C)** for the **13 uncovered** tokens — they are absent from Dune’s normalized spells entirely, so a **paid plan would not recover them** (paid tiers buy a bigger engine, not more spellbook coverage); they stay **flagged-NaN** with their existing documented reasons. **No (B) outcome** — the engine never struggled. This converts Entry 36’s *conditional* (A) into an *unconditional* (A) for

the covered set and resolves Moazzam’s funding question: don’t buy. **Rationale:** same Entry-30/31/36 standard — measure the right object (normalized protocol transacted value, not a side token’s transfers, not a raw-event hand-reconstruction = the rejected AAVE-sentinel path) and verify before trusting (the STRK negative-net check; the canonical-WETH contract join, never `symbol='WETH'`). The key reframing: the residual limitation is a **coverage** (spellbook) gap, which money cannot close, not an **engine** gap, which money could — so the funding lever is pointed at the wrong constraint. **Downstream impact:** After human review, a Phase-2c build may fill **AAVE, STRK, LDO, GNS** from Dune on the **free tier** (filter lending to `transaction_type='borrow'`; drive the Lido query off the submit table or filter to `>=` first Lido month; canonical-contract discipline on every price join), and leave the other 13 documented NaN — net +4 token PQ series (incl. the one bonus, STRK) at zero subscription cost. **Do not purchase any Dune plan; do not start the Phase 2c build before this report is reviewed.** No panel outputs were written this session (`pq_tokens.csv` and all Phase 2/2b CSVs unchanged — diagnostic only). Session: `06_documentation/ai_conversations/session_016_2026-06-25_dune_dryrun.md`.

Entry 38 — TVL->PQ stock-to-flow conversion framework (AK-model turnover); scoping the full 111-token NaN universe (not just the 17 Dune-scoped tokens) as a diagnostic-only Phase 2c metadata audit

Date: 2026-06-25 **Spec section affected:** 4, 4.1 (token PQ source) — extends, does not reverse, Entry 30 (TVL != PQ) and Entry 32 (the fee->volume inversion rule), generalizing both from the 17 Dune-pilot tokens to the full 111-token NaN universe. **Asset(s)/period affected:** all 111 NaN tokens in `pq_tokens.csv` (the 17 Dune-scoped tokens resolved/documented per Entry 37, plus the other 94 never previously examined). Live-reverified category breakdown: Yield 16, Derivatives 10, Farm 8, Gaming 7, Dexs 7, Services 7, Bridge 6, Lending 6, Launchpad 6, Chain 6, Yield Aggregator 4, Canonical Bridge 3, Cross Chain Bridge 2, Developer Tools 3, Token Locker 2, plus 17 singleton categories (1 token each). **What was discussed (Cowork session, not Claude Code):** Moazzam challenged the Dune dry-run’s 4-of-17 result as too thin a cross-section for the paper to be useful, and asked whether TVL (a stock) can be converted into a PQ proxy (a flow) via some “rate,” drawing an explicit analogy to nominal GDP = capital x an efficiency rate in a no-labor economy, and separately asked what “turnover” means and whether NVT’s “T” is itself TVL. **What was found:** - Corrected a specific factual claim: NVT’s “T” (Woo 2017) is on-chain transacted USD volume — a flow — never TVL; TVL is a separate, later concept (locked capital, a stock). The existing TVL-based ratio in practitioner use is Market-Cap/TVL (a price-to-book analog), structurally unrelated to NVT. - The user’s “capital -> economic activity at some rate, no labor” intuition maps directly onto the **AK endogenous-growth model** ($Y = A?K$): $PQ = A_protocol \times TVL$, where A (“turnover”) is the same object as the corporate-finance asset/capital-turnover ratio (flow ? stock). This ties directly to the paper’s own $M?V = P?Q$ framing ($M=TVL$ is structurally analogous). - DeFiLlama tracks **Fees and Revenue as separate, directly observed flow series** (distinct from TVL and from the Dune-verified Volume/spell layer) across 7,000+ protocols. Where a fixed, known fee rate exists, $Volume \sim Fees \times fee_rate$ — this is **not a new idea**, it is Entry 32’s already-decided rule (“back out volume from fee only when the fee is a confidently known single, stable rate, else flag missing”), now being extended from the 8 bitinfocharts coins to the broader 94-token set. - Attempted live DeFiLlama verification (myx-finance, avantis, linear-finance, ooki, mantra-dao, nereus-finance, loopring, velo-finance, hakka-finance) from the Cowork sandbox; blocked — `api.llama.fi` returns 403 **blocked-by-allowlist** from this sandbox’s network proxy. Live verification must happen in a local Claude Code session (normal network access), same as the Dune work. - The natural calibration cohort for a TVLxturnover rate is the **25 lending + 28 perpetual.trades** Dune **DISTINCT project** lists already pulled in Entry 37 — $n=25/28$ with real PQ, pairable against DeFiLlama TVL for the same protocols — not just the 4 directly-covered target tokens ($n=4$ is too thin to calibrate anything from). **Decision made:** Scope the next session as a **diagnostic-only metadata audit** (“Phase 2c diagnostic”), not a build. Per token (not blanket per-category), check what DeFiLlama data exists (TVL, Fees, Revenue, direct Volume via the dexs/derivatives/bridges verticals, APY via yields) and judge which conversion path is plausible given that protocol’s actual economic model: direct volume (no conversion needed) > fee inversion (Entry 32’s rule) > TVLxAPY (Farm/Yield/Yield-Aggregator, 28 tokens, APY is a directly observed protocol-specific rate) > TVLxcalibrated-turnover (Lending/Derivatives/Liquid-Staking, gated on the dispersion check below) > none. **Gaming (7 tokens) is explicitly out of scope and stays**

NaN — in-game activity isn't capital-driven and forcing this model there would be its weakest application. Before trusting any TVLxturnover number, compute the actual turnover distribution across the 25+28 comparable cohort and report its dispersion plainly — only recommend it if defensibly tight. **Rationale:** Same Entry-30/31/32/36/37 standard — measure the right object, verify before trusting, don't force a match where the economic model doesn't support one. This is an extension of the existing framework (calibrated stock->flow conversion via turnover, fee-inversion generalized), not new methodology invented from nothing. **Downstream impact:** No data pulled, no panel write, no PQ values computed — feasibility/coverage map only. Drafted 04_code/CLAUDE_CODE_PHASE2C_DIAGNOSTIC_PROMPT.md for the next Claude Code session. **Do not start an actual Phase 2c panel build before this diagnostic report is reviewed.**

Entry 39 — Phase 2c diagnostic executed: only 5 of 104 NaN tokens have a defensible free PQ path (4 known + SUN net-new); TVLxturnover statistically indefensible; bridges.llama.fi now 402-paywalled

Date: 2026-06-25 **Spec section affected:** 4, 4.1 (token PQ source — the Entry-38 metadata audit); 6 (landmines) **Asset(s)/period affected:** all 104 non-Gaming NaN tokens in `pq_tokens.csv` (the 111 NaN minus the 7 Gaming, which stay NaN by Entry-38 design). Diagnostic only. **What the spec/Entry-38 wanted:** a per-token feasibility map answering, for each NaN token, *what free DeFiLlama data exists and could any of it plausibly become a transacted-value PQ given the protocol's economic model?* — ranking paths direct-volume > fee-inversion > TVLxAPY > TVLxcalibrated-turnover > none — plus a turnover-dispersion test on the 25-lending + 28-perp Dune cohort to decide whether TVLxturnover is defensible. **No `pq_tokens.csv` write, no PQ value, no purchase.** **What was actually found (live, local Claude Code, full report 03_data/PHASE2C_DIAGNOSTIC_REPORT.md, code 04_code/phase2c_{defillama_metadata,turnover_cohort,turnover_refine,verdicts}.py, session 017):** - **Worklist re-derived live:** 127 tokens / 16 covered / 111 NaN / 104 audited — matches the kickoff exactly. All endpoint shapes were live-verified before use. - **Only 5 tokens have a defensible free transacted-value path; 4 were already known.** The Entry-37 Dune-spell set (AAVE, STRK, LDO, GNS) plus **one net-new find: SUN.** SUN.io's AMM is SunSwap, whose DEX volume sits in DeFiLlama's dexs vertical under `sunswap-v1/v2/v3` (the stored slug `sun.io` is absent from the vertical, which is why Phase 2 missed it) — **direct volume, no proxy, 2020-08->2026-06, identity-verified by `cmcId`.** - **TVLxcalibrated turnover (path 4) is NOT defensible as a level.** Pulled the full Dune cohorts (`lending.borrow` borrow-filtered, 25 proj / 816 protocol-months; `perpetual.trades`, 28 proj / 436 protocol-months), matched each to a DeFiLlama slug **verified individually by `cmcId/name`** (`lending` 21/25, `perps` 16/28; unmatched = dead/Terra/non-EVM), summed TVL across *all* version slugs to remove scope artifact, and computed turnover = PQ/TVL. **Lending per-project median turnover spans 0.0008->1.24 (~1,455x; ~10x even in the core cluster; the venus low tail survives scope-matching = real); perps span 0.0000->108.8 with no central tendency.** A borrowed category turnover rate would fabricate a number whose error dwarfs the signal (spec ?0). At most lending's pooled median (~0.28) is a coarse order-of-magnitude rank, never a level, never for perps. - **TVLxAPY (path 3) collapses on availability:** `yields.llama.fi/pools` is a *current* snapshot, so 25 of the 28 Farm/Yield tokens (dead/delisted) have no APY; only CVX/FARM/ZBU have a current rate and none has a free historical APY series -> weak (constant-APY assumption required), not built. - **Fee-inversion (path 2) fired for zero tokens** — 26 have a Fees series but none is a single fixed volume-linked rate (L2 gas, bridge per-transfer, lending reserve factors, variable perps) — consistent with Entry 33. - **Two NEW landmines:** (1) **bridges.llama.fi is fully 402-paywalled** across all endpoint shapes (`/bridges`, `/bridge/{id}`, `/bridgevolume`, `/bridgedaystats`) — the kickoff assumed it was free; the 11 bridge tokens thus have a *valid* object (transfer volume) that is merely not free, distinct from “no object.” (2) The `nerve/NVT` (`cmcId` 8755 != 5906) and `velodrome/VELO` (different VELO) **symbol collisions are live traps**, both ruled out on `cmcId` mismatch — exactly the NVT-collision hazard the kickoff flagged. - **Verdict tally (104):** ?`dune_spell` 4, ?`direct_volume` 1 (SUN), ~weak `ttl_apy` 3 (CVX/FARM/ZBU), ?`turnover_undefensible` 14, ?`no_apy` 25, ?`bridge_vol_paywalled` 11, ?`symbol_collision` 2, ?`no_economic_model` 44. **Decision made (diagnostic recommendation, for human review — nothing built or written this session):** A real Phase-2c build is worth running **only for the 5 viable tokens** — AAVE/STRK/LDO/GNS via Dune free tier + SUN via DeFiLlama dexs (sum `sunswap-v1/v2/v3`), **+1 net-new (SUN) over Entry 37, zero subscription. Report**

TVLxturnover and TVLxAPY in the paper as explored-and-rejected, not built — the turnover-dispersion negative result is itself a ?6 methodological finding. Leave ~96 tokens documented-NaN. The token-side PQ cross-section realistically grows from 16 covered to ~21, not “most of 111.” **Rationale:** Same Entry-30/31/32/36/37 standard — measure the right object, verify identity before trusting (the SUN/NVT/VELO cmcId checks; the version-scope-matched turnover), and report a wide/unusable dispersion plainly rather than forcing a number (spec ?0). The AK-model TVL-as-flow intuition (Entry 38) is theoretically clean but founders on empirical turnover dispersion. **Downstream impact (re-check if this changes):** A future explicitly-authorized Phase-2c build may fill the 5 viable tokens (Dune free for 4; DeFiLlama dexs for SUN) and leave the rest documented-NaN. Two paid levers that *would* extend coverage — **DeFiLlama Pro** (reopens the derivatives vertical for the 9 remaining derivatives tokens + the bridges vertical for the 11 bridge tokens) and a **bridges-API tier** — are flagged for Moazzam’s decision, **not acted on** (same standing rule as Dune/Artemis). If bridges access is obtained, the 11 `bridge_vol_paywalled` tokens reopen at once. No panel outputs written — `pq_tokens.csv` and all Phase 2/2b CSVs unchanged. Session: `06_documentation/ai_conversations/session_017_2026-06-25_phase2c_diagnostic.md`. **Do not start the Phase 2c build before this report is reviewed.**

Entry 40 — Sequencing decision: scale lambda + build a real TVL panel before sourcing coin PQ; local repo-sync incident found and fixed

Date: 2026-06-26 **Spec section affected:** 3 (lambda channels), 4 (TVL as a valuation-multiple denominator, distinct from PQ); process/documentation integrity (this log + `time_log.md` themselves). **Asset(s)/period affected:** n/a (scoping decision + a repo-hygiene fix, not a data build). **What happened:** Moazzam asked for a coverage audit of lambda/TVL/PQ/NV across the full universe (delivered in chat, Cowork, 2026-06-26), then said: build lambda and TVL first, decide how to source coin PQ afterward. Before drafting the next Claude Code kickoff prompt, re-verified the real ceilings this would have to work within: **127/448 tokens** have a confirmed DeFiLlama `dl_slug` match (`03_data/phase1/asset_onchain_identity.csv`) — the TVL build ceiling; **123/448 tokens** have an identified staking/lock contract address and **29/448** an auto-matched Snapshot space (vs. 55 assets actually built in `channel3_voting.csv` via the broader 56/57-space curated map, Entry 25) — the Channel-1/3 token ceilings; for coins, only **ETH** has any lambda-channel data today (1/633) — staking/voting data for other PoS coins is a live-verification gap, not yet audited per-chain the way EVM sources were in Entry 21. **Repo-sync incident found while grounding this (logged here per the open item from the prior Cowork session):** `git status` showed every tracked file as simultaneously staged-deleted and untracked, and a stale `.git/index.lock` (dated 2026-06-25 17:56, i.e. mid/post session 017) could not be removed from the Cowork sandbox (permission denied on the mounted path — consistent with a process on Moazzam’s actual machine still holding it, invisible to the sandbox). Byte-level comparison of working-tree files against the already-pushed HEAD (`a7058f2`) showed this was **not data loss**: code and data CSVs (`phase1_assemble_lambda.py`, `lambda_panel.csv`, `pq_tokens.csv`, `PHASE2C_DIAGNOSTIC_REPORT.md`) were byte-identical to HEAD once CRLF line endings were normalized out (a Windows-checkout artifact, not content drift). Two files were genuinely stale, missing real content present in HEAD: `04_code/DATA_DECISIONS_LOG.md` (working copy stopped at Entry 32; HEAD has Entries 33–39) and `06_documentation/time_log.md` (working copy was missing the last ~6 rows, sessions 013–017). Both were restored verbatim from `git show HEAD:<path>` (plain file overwrite, not a git operation, so the held index lock was irrelevant) and verified byte-identical to HEAD afterward. **Decision made:** (1) Scope the next Claude Code session to lambda-coverage scale-up (tokens: widen Channel 1’s curated EVM escrow set and Channel 3’s voting-space map past their current ceilings; coins: a live source-verification pass for PoS coins beyond ETH, Entry-21-style) and a **real, persistent TVL panel for the 127 dl_slug-matched tokens** (converting `phase2c_defillama_metadata.py`’s existing `check_tvl()` presence-check, which already fetches the full series and discards it, into an actual panel write) — drafted as `04_code/CLAUDE_CODE_PHASE1_SCALE_LAMBDA_AND_TVL_PANEL_PROMPT.md`. Coin PQ sourcing is explicitly deferred to a separate, later prompt, per Moazzam’s own sequencing. (2) The new prompt opens with a mandatory git-hygiene check (confirm no live git process before touching `.git/index.lock`; re-verify `DATA_DECISIONS_LOG.md/time_log.md` against HEAD before appending) so a future session doesn’t silently build on, or re-corrupt, a stale local copy of either log. **Rationale:** Matches the project’s own “verify before

building” discipline (spec ?0, Entry 21) applied to the repo’s own bookkeeping, not just external data sources — an append-only audit log that silently lost entries would defeat its own purpose. TVL is being built as a **valuation-multiple denominator (NV/TVL)**, not a PQ proxy — Entry 30’s stock-vs-flow rejection of TVL-as-PQ stands unchanged. **Downstream impact (what should be re-checked if this decision changes):** If the index.lock turns out to reflect a genuinely still-running process on Moazzam’s machine, the next session’s `git add -A/commit` will fail loudly (safe) rather than corrupt anything further. Coin PQ sourcing remains fully open and unscoped — to be addressed in a dedicated future prompt once lambda/TVL are in hand.

Entry 41 — Session 019 executed: lambda scaled to 1,688 asset-months / 58 assets (coin staking ETH->ETH+ADA+XTZ; +4 token lambda assets); real TVL panel built (99 tokens / 4,999 asset-months)

Date: 2026-06-26 **Spec section affected:** 3 (lambda Channels 1 & 3), 4.1–4.2 (TVL as a valuation-multiple denominator, not PQ). **Asset(s)/period affected:** lambda panel (`lambda_panel.csv`) and a new TVL panel (`tv1_panel.csv`), 2018–2026 monthly, observed asset-months. **What happened:** Ran the Entry-40 kickoff (`CLAUDE_CODE_PHASE1_SCALE_LAMBDA_AND_TVL_PANEL_PROMPT.md`). Step 0 hygiene: no git process, no `index.lock`, clean tree — the Entry-40 stale-index incident did **not** recur locally; live-re-derived next-numbers (Entry 41, session 019, both confirmed not trusted). **lambda assembly logic was not touched** — only channel input files were widened; `phase1_assemble_lambda.py` auto-globbered them. **(A.1 — token Channel 1, individual Entry-26 verification, live balanceOf + getLogs reconstruction).** Each candidate accepted only if a single contract holds the BASE token directly (`balanceOf == locked supply`). **VERIFIED + built** (`phase1_channel1_evm_locks_ext.py`, 4 assets / 214 asset-months; reconstructed latest-locked matched live `balanceOf` to rounding): PENDLE (vePENDLE, 22.9%), LQTY (LQTYStaking, 57.8%), 1INCH (St1inch, 15.8%), RPL (RocketVault, 47.3% — FLAGGED shared-vault, kept under the same standard as xSUSHI/stkAAVE). **VERIFIED mechanism but series DEFERRED:** GMX (StakedGmxTracker ~65%, Arbitrum) — full-history `getLogs` over Arbitrum’s millions-of-blocks/month is impractically slow on the free tier (>60 s/month); row left commented with rationale, to be built later via `account/tokentx` pagination. **REJECTED (documented, not silently proxied — the veBAL/SNX standard):** MKR (DSChief holds 0.5% post-Sky migration), BAL (veBAL locks an 80/20 BPT, not BAL), COMP (in-wallet delegation, no lock), RUNE (native THORChain L1, placeholder address), ANGLE (not in universe). **(A.2 — token Channel 3).** `governanceID` cross-check: 0 new (all 29 auto-matched spaces already in the curated map). Probed the Entry-25 “not on Snapshot” gap list live (`id_in + ranking(search:)`): **two were actually on Snapshot and were missed — ENA -> ethenagovernance.eth and PERP -> vote-perp.eth**, both official, active, token-weighted (erc20-balance-of on the canonical token); added to the curated map (now 57 spaces / 53 vw_turnout assets). ONDO/WLD = impostor/spam spaces only; PENDLE’s top hit `sdependle.eth` is StakeDAO’s third-party locker, not Pendle’s own governance (and PENDLE is now C1-covered); MKR/LQTY/RUNE expose no verifiable clean Governor -> no `VoteCast` reconstruction (no-guess rule). **(A.3 — coin Channel 1, Entry-21-style LIVE audit).** Only two PoS chains publish a free, keyless, historical staked-supply series: ADA via Koios `epoch_info.active_stake` (built, 70 months, 49->74%, Shelley-gated to 2020-08) and XTZ via TzKT `cycles.totalBakingPower` (built, 95 months, FLAGGED for the 2024 Paris baking-power redefinition, analogous to ETH’s Shapella caveat). New `phase1_channel1_pos_coins.py` (2 assets / 165 asset-months; no value guessed or interpolated). **Live-verified gaps:** Cosmos `/staking/pool` current-only (ATOM/INJ/SEI/KAVA/CELO — confirms Entry 24), SOL/HBAR current-only, ICP ic-api 404, DOT/KSM keyed (Subscan), AVAX/NEAR/ALGO/TRX/EOS/SUI/APT no free historical. **(B — real TVL panel).** New `phase2_build_tv1_panel.py` converts Phase-2c’s discard-everything `check_tv1()` into a full monthly-grain panel for all 127 dl_slug-matched tokens -> `03_data/phase2/tv1_panel.csv`: 97 non-empty, 4,895 asset-months, 0 fetch failures, 30 expected empties (aggregators/DAOs/chains with no protocol TVL). Stretch goal (low yield, as anticipated): of 321 unmatched tokens, only 9 exact symbol+name matches; verified-and-added AXL (axelar, \$135M, `cmcId` null — clean join miss) and PERP (perpetual-protocol, \$0.4M, DeFiLlama `cmcId` stale=1301 but slug unambiguously PERP) -> **99 tokens / 4,999 asset-months, 2019-12->2026-05**; rejected CVP/POLS (`cmcId`-mismatch collision risk, the Entry-39 landmine) and METIS/HONEY/PUMP/PYTH/WLFI (zero TVL). **Decision made:** Accept the four mainnet C1 tokens,

the two C3 spaces, and the two PoS coins into lambda; accept the TVL panel as the NV/TVL denominator. Defer GMX's series and all other PoS coins as documented free-tier gaps. **lambda before->after: 1,374->1,688 observed asset-months; 52->58 distinct assets (coin 5->7, token 43->47, other 4).** Full account in 03_data/PHASE1_LAMBDA_SCALE_AND_TVL_PANEL_REPORT.md. **Rationale:** Every addition cleared the project's existing standards — Entry-26 single-contract base-token-lock verification for A.1 (with explicit rejections, not silent proxies), live Snapshot strategy verification + the token-weight guard for A.2, Entry-21 live free-access auditing for A.3, and cmcId-only joins throughout (the AXL/PERP loose matches were each verified individually before acceptance, guarding the VELO/velodrome collision mode). TVL is a stock used only as a valuation-multiple denominator (Entry 30 unchanged). **Downstream impact (what to re-check if this changes):** GMX's C1 series is the one accepted-but-unbuilt item — build via `tokenTx` pagination next. Coin staking beyond ADA/XTZ and coin PQ both remain open and need either a keyed indexer (a Moazzam purchase decision) or native block iteration. The TVL panel inherits the identity map's one-slug-per-cmcId choice (AAVE->aave-v2 only) — flag for whoever builds the NV/TVL ratio. Channel 2 (holding duration) still 0, still a gap (Entry 24). Coin PQ untouched — next session, separately authorized.

Entry 42 — “Bucket 2” verification (17 non-EVM PoS coins): SOL/DOT/KSM/TRX upgraded to free-verified, no purchase needed; HBAR/SUI reframed as free-data-but-needs-engineering; CELO flagged as a possible EVM-reclassification; ATOM/INJ/SEI/KAVA/AVAX/NEAR/EOS/ICP/APT remain open gaps; ALGO confirmed a structural (not a money) gap. Refines/corrects Entry 41's A.3 quick-pass line for SOL, TRX, HBAR, SUI.

Date: 2026-06-26 **Spec section affected:** 3.1 (staking/locking ratio) — extends Entry 21/41's “verify live before building” discipline from chain-native RPC checks to third-party/official indexer products, for the non-EVM coins Entry 41's A.3 pass flagged as “no free historical” or “current-only” after checking only each chain's own RPC/LCD. **Asset(s)/period affected:** the 17 named coins behind Entry 41's A.3 gap list — ATOM, INJ, SEI, KAVA, CELO, SOL, HBAR, ICP, DOT, KSM, AVAX, NEAR, ALGO, TRX, EOS, SUI, APT — representative of the ~170-coin non-EVM PoS portion of the universe still missing a lambda Channel-1 series. Diagnostic only; nothing built, no key obtained, no purchase made. **Context:** Moazzam asked, after the lambda/TVL coverage audit, to verify this specific gap before any purchase decision — i.e. check live whether the paid chain-indexers Entry 41 gestured at actually contain the needed historical bonded-stake series at all, and whether a free path was missed. **What was found, live, this session (WebSearch + targeted doc fetches; no API keys used or obtained):** - **SOL — free, verified, upgrade.** `validators.app` (Solana, operated by Block Logic LLC) requires only a free signup + API token (Token header); no paid tier found anywhere in its docs. Its `/api/v1/epochs/:network.json` endpoint returns one record per epoch with `total_active_stake` and `total_rewards`, depth 169+ epochs in the documented example — a genuine historical total-active-stake series. Corrects Entry 41's “SOL current-only,” which checked only Solana's own RPC, not this third-party indexer. - **DOT / KSM — free, verified (medium confidence), upgrade.** Subscan's documented endpoint list tags certain endpoints with a literal [PRO] prefix (e.g. “[PRO] List multichain account assets”); the entire Staking section, including “List validator era statistics” (per-era stake), carries no such tag — i.e. it reads as free-tier. Could not confirm the exact JSON schema directly (docs are JS-rendered; raw fetch only returns sidebar nav), so this is an inference from the tagging convention, not a pixel-verified response body. Recommend a 5-minute live key-signup-and-call check before relying on it in a build. - **TRX — free, verified, upgrade.** TronScan's public `freezeresource` endpoint (`start_day/end_day` params) explicitly returns historical frozen-TRX/freeze-ratio for a date range. TronScan's API-key tiers (mandatory since 2024) are Free (60 req/hr, all read endpoints, auto-approved signup, no payment) / Developer (600 req/hr) / Partner (unlimited) — the Free tier covers this endpoint. Corrects Entry 41's “TRX no free historical,” which had not found this endpoint. - **HBAR — free data exists, but assembling the network series is an engineering task, not a purchase.** Hedera's own official, free, public Mirror Node REST API has `/api/v1/network/stake` (current aggregate) and the `balances` endpoint accepts a `timestamp` param that falls back to 15-minute balance-file snapshots — i.e. point-in-time historical account state is natively free. There is no single endpoint that already sums “total staked across all accounts” per past snapshot; building that series means iterating accounts' `staked_node_id/staked_account_id` at each snapshot. Corrects Entry 41's “HBAR current-only” only insofar as the underlying data is historical; the *aggregation* is unbuilt. - **SUI — free data exists on-chain, same engineering caveat as HBAR.** Sui's

`staking_pool` module maintains an on-chain exchange-rate-history table keyed by epoch (since each pool's activation epoch), queryable via the standard free public RPC/GraphQL — no third-party vendor needed. Total-network stake-per-epoch requires summing across all validator pools, an engineering task, not a payment gate. - **ALGO — structural gap, confirmed not a “buy a vendor” problem.** A Nodely (Algorand infra provider) blog post states plainly that no Algorand ledger dataset, public or commercial, contains precomputed/sampled participation data; a historical online-stake series has to be computed by replaying the chain's full transaction history from scratch. This is qualitatively different from the other gaps here: paying for *any* indexer would not solve it — the indexer itself would have to build the same from-scratch replay. - **CELO — possible reclassification candidate, not yet built.** Celo migrated its mainnet to an Ethereum L2 (OP Stack rollout) on 2025-03-26. Confirmed staking/validator-election mechanics (CELO locking + voting for validators) continue post-migration, just decoupled from consensus (validators now run community RPC nodes; sequencer fees are separate from staker rewards). Because Celo is now EVM-compatible with its own Etherscan-family explorer (Celoscan), its locked-CELO contract may be reconstructable via the same `getLogs` method already used for ETH/EVM tokens (Entry 21/26) instead of needing a non-EVM chain indexer at all — i.e. CELO may not belong in this bucket going forward. Not verified: whether the legacy LockedGold/Election contract is still the live locking mechanism post-migration, or whether Celoscan exposes its logs the way Etherscan does. Flagged for a follow-up identity-map check (Entry 22-style), not acted on. - **ATOM, INJ, SEI, KAVA (Cosmos-SDK appchains) — no change, still an open gap.** Mintscan/Cosmostation's official API is contact-based (`api@cosmostation.io`), pricing undisclosed — an Artemis-Enterprise-style sales process, not a self-serve purchase decision. Bitquery has a disclosed free-dev-tier + usage-based paid tier and confirmed general “Cosmos” coverage, but per-appchain coverage of Injective/Sei/Kava specifically, and whether its Cosmos staking data is a ready pool-level historical bonded-total vs. raw per-tx delegation events needing aggregation, is unconfirmed. Per-chain dashboards (`injscan.com`, `Seistream`, `kava.mintscan.io`) show only current-state snapshots. - **AVAX — promising official first-party candidate, pricing tier unconfirmed.** Ava Labs' own AvaCloud Metrics API has a dedicated “Staking Information” feature (validator/delegator counts, staking weights) explicitly described as historical and as powering the official Avalanche Explorer's own graphs — the strongest *type* of source in this whole list (official, first-party, purpose-built). AvaCloud's disclosed pricing structure found in search (Starter/Pro/Enterprise, a “\$999/mo Builder” plan) reads as being for its L1-deployment/infrastructure product line, not necessarily gating pure read access to the Metrics API — genuinely unclear without checking the API-key signup flow directly. - **NEAR — two unconfirmed candidates.** Pikespeak (registration/API-key gated, explicitly advertised as supporting historical validator/delegator data, pricing undisclosed) and NearBlocks (official-style explorer API; current `/v1/stats` confirmed; historical-staking-specific endpoint and pricing not confirmed). - **EOS, ICP — no source found.** No historical staked/vote-weight (EOS) or neuron-stake (ICP) API located, free or paid, in this pass. - **APT — free official API exists but looks current-state-only.** Aptos Labs' public Indexer GraphQL API (`api.mainnet.aptoslabs.com/v1/graphql`) is free and historical for transactions/assets generally, but the one staking-specific table found, `current_delegated_staking_pool_balances`, is named (and appears to behave) as current-state-only, the same limitation Entry 21 found for free-tier `eth_call`. A historical-staking-balances table was not located. **Decision made:** No purchase made or recommended yet. Reclassify the working picture of this gap from a flat “170 coins need a paid indexer, unverified” into four groups: (1) **free and verified, no purchase needed** — SOL, TRX, plus DOT/KSM at medium confidence (schema not pixel-confirmed); (2) **free data exists but needs engineering, not money** — HBAR, SUI, and ALGO (ALGO is the hardest of the three: full transaction replay, not just multi-pool summation); (3) **a paid/contact-based path may exist but is unconfirmed** — ATOM/INJ/SEI/KAVA (Mintscan contact-sales or unconfirmed Bitquery coverage), AVAX (AvaCloud, pricing-page ambiguity), NEAR (Pikespeak/NearBlocks); (4) **no source found yet, free or paid** — EOS, ICP, APT (APT has a free official API but it looks current-state-only for staking specifically). CELO is provisionally pulled out of this bucket pending confirmation that its post-L2-migration locking contract is EVM-`getLogs`-reconstructable the way ETH/EVM tokens already are. **Rationale:** Matches the project's standing rule (spec ?0, Entry 21) to verify access live before treating a gap as “needs a purchase” — several of these chains turned out to have a free path Entry 41's shallower native-RPC-only check missed, while ALGO turned out to be a case where money would not even help. Keeping the four-way split (free-done / free-but-engineering / paid-unconfirmed / no-source) instead of a single verdict avoids either overstating progress or sending Moazzam toward a purchase that wouldn't solve the actual problem (ALGO) or duplicating a path that's actually

free (SOL/TRX/DOT/KSM). **Downstream impact (what to re-check if this decision changes):** Before any Phase-1-style build: (a) live key-signup-and-call test for Subscan’s era-statistics endpoint and validators.app’s epochs endpoint, to move DOT/KSM/SOL from “verified via docs” to “verified via response body,” matching the A.1/A.2/A.3 standard in Entry 41; (b) the HBAR/SUI aggregation jobs are real (if modest) engineering work, not zero-cost — scope before committing; (c) CELO’s EVM-reconstructability needs a direct identity-map check (does Celoscan expose the legacy LockedGold/Election contract’s logs) before it can be moved out of this bucket for real; (d) AvaCloud/Pikespeak/NearBlocks/Mintscan/Bitquery pricing-page ambiguities all need a direct signup-flow check (not just search results) before Moazzam is asked to make any purchase call; (e) EOS/ICP/APT-historical remain genuinely unresolved — no recommendation to make here yet. No purchase has been made; nothing in this entry authorizes one.

Entry 43 — Bucket 2 Tier 1: TRX + SOL coin staking BUILT, keyless (no signup); corrects Entry 42’s “free with signup”

Date: 2026-06-26 **Spec section affected:** 3.1 (staking/locking ratio, coins). **Asset(s)/period affected:** TRX (cmc 1958) 2019-12->2026-06; SOL (cmc 5426) 2023-02->2026-06. **What the spec wanted:** staked-or-locked / circulating supply at month-end, from each chain’s canonical free source. **What was actually available (live, response-body verified this session):** - **TRX — keyless.** TronScan `apilist.tronscanapi.com/api/freezeresource?start_day=YYYY-MM-DD&end_day=YYYY-MM-DD` returns full **daily** history with **NO API key** (field `total_freeze_weight` = total frozen TRX, already in TRX not sun). History starts ~2020-05 (2019 returns `total:0`). This **corrects Entry 42**, which said TRX needed a free TronScan signup — the read endpoint answers unauthenticated. Built 78 months, ratio 26%->56% (latest 49%). - **SOL — keyless, ~2023-01+ depth.** `validators.app/api/v1/epochs/mainnet.json?per=200&page=N` returns `total_active_stake` (lamports / 1e9 = SOL) **without an API token** for every epoch validators.app recorded stake for — which begins ~epoch 414 (~2023-01); earlier epochs return `total_active_stake=null` on the free/keyless tier. The null pattern is **time-based, not all-or-nothing**, i.e. a data-vintage limit (validators.app did not collect the figure pre-2023), not a key paywall — so a token would not retroactively add it. This **corrects/refines Entry 42’s** “free, verified” to “free **and keyless**, but only ~2023-01+ depth.” Built 40 months, ratio ~74%. **Decision made:** Accept both into lambda Channel 1. New script `phase1_channel1_pos_coins_bucket2.py` -> `03_data/phase1/channel1_pos_coins_bucket2.csv` (118 asset-months, 2 assets; picked up by the `channel1_*.csv` glob). No value guessed/interpolated; denominator = panel circulating supply (cmc_id+month), same convention as ADA/XTZ (Entry 41). **Rationale:** Both cleared the project’s “verify access live, response body not docs” rule (spec ?0, Entry 21/42) — and turned out **more open** than Entry 42 inferred (keyless, no signup), so no purchase or even signup is needed. **Downstream impact (re-check if this changes):** SOL pre-2023 months stay NaN (no free source has them); a few early SOL months have `staking_ratio>1` (active stake includes CMC-non-circulating tokens) — kept un-capped and flagged, since lambda z-scores on relative rank, not level. If validators.app ever gates `total_active_stake` behind a token, this build breaks and needs the key.

Entry 44 — Bucket 2 Tier 2: DOT/KSM Subscan era-stat is key-gated (HTTP 403 unauth); corrects Entry 42’s docs-level “free-tier”

Date: 2026-06-26 **Spec section affected:** 3.1 (staking ratio, coins); access-verification discipline (spec ?0, Entry 21/42). **Asset(s)/period affected:** DOT (cmc 6636), KSM (cmc 5034) — not built this session. **What the spec wanted (kickoff Tier 2):** sign up for a free Subscan key, call the validator era-statistics endpoint live, confirm per-era bonded totals in the response body; build only if confirmed; do not pay for Pro. **What was actually found (live):** `polkadot.api.subscan.io/api/scan/staking/era_stat` (POST) returns **HTTP 403** `{"code":403,"message":"Subscan API strictly requires an API key. Unauthenticated access is disabled."}`. Subscan’s support/pricing confirms a **Free Plan** exists and is self-serve (register at `pro.subscan.io` to generate a key), and **all** API requests require a key. So Entry 42’s inference — that `era_stat` reads as free-tier because it carries no [PRO] tag — could **not** be response-body confirmed: the endpoint is unreachable without *any* key, and the free key requires an interactive, email-verified web signup that could not be completed in this non-interactive session. **Decision made:** Do **not** mark DOT/KSM built. Record this as a correction to Entry 42 (from “free, medium-confidence via docs” to

“key-gated; a free self-serve key likely covers era_stat but is unconfirmed and was not obtainable headless”). The build path is otherwise ready: a free Subscan key dropped into 04_code/.api_keys.json would let phase1_channel1_pos_coins_bucket2.py be extended to call era_stat and reconstruct per-era bonded totals. **No Subscan Pro purchase, per the rule. Rationale:** The project’s standing rule is response-body verification, not docs inference (the very gap this session was created to close). A 403 is a response body that says “key required” — honest to report as a gate, dishonest to treat as built. **Downstream impact:** Flagged for Moazzam to obtain the **free** (not Pro) Subscan key via the self-serve signup; once present, DOT/KSM become a ~5-minute extension. No purchase is authorized by this entry.

Entry 45 — Bucket 2 Tier 3: HBAR + SUI free data exists but the network-series aggregation is keyless-intractable (documented gaps, scoped)

Date: 2026-06-26 **Spec section affected:** 3.1 (staking ratio, coins). **Asset(s)/period affected:** HBAR (cmc 4642), SUI (cmc 20947) — not built. **What the spec wanted (kickoff Tier 3):** build the aggregation if tractable; if genuinely intractable, log a documented gap rather than ship a partial series. **What was found (live):** - **HBAR.** Hedera Mirror Node /api/v1/network/stake is free/keyless but returns only the **current** aggregate (stake_total = 1.461e18 tinybars ~ 14.6B HBAR) and takes **no** timestamp param. A historical network-staked series would require iterating /api/v1/accounts (and the balances endpoint’s 15-min snapshot files) to sum staked_node_id/staked_account_id-linked balances at each past snapshot — millions of accounts x hundreds of snapshots, intractable keyless. Confirms Entry 42’s engineering framing. - **SUI.** suix_getLatestSuiSystemState is free/keyless and gives the **current** total stake (sum of stakingPoolSuiBalance across 129 active validators = 7.23B SUI), but RPC exposes **only the current** system state. A historical per-epoch total needs reading every validator pool’s on-chain exchange_rates table object across all epochs since each pool’s activation — engineering work, keyless-intractable. **Decision made:** Leave both as **documented open gaps** (scoped, not partial-shipped). No interpolation, no current-value-carried-backward. **Rationale:** Shipping a one-point current snapshot as if it were a time series, or back-filling it, would violate the no-guess rule and the “don’t ship a partial series” instruction. **Downstream impact:** Both are recoverable with real (if modest) engineering — an account-snapshot batch job (HBAR) or a multi-pool exchange-rate crawl (SUI) — or a keyed indexer; neither is a free, ready-made series today.

Entry 46 — Bucket 2 Tier 4: CELO EVM-reclassification CONFIRMED, but free getLogs reconstruction fails the Entry-26 cross-check (3x undercount) -> documented gap, not shipped

Date: 2026-06-26 **Spec section affected:** 3.1 (locking ratio); Entry-26 single-clean-escrow standard + its balanceOf==locked cross-check. **Asset(s)/period affected:** CELO (cmc 5567) — checked, not built. **What the spec wanted (kickoff Tier 4):** confirm the legacy LockedGold/Election contract is still live post-2025-03 L2 migration and whether Celoscan exposes its logs; if yes, build with the SAME getLogs method as Entry 26; if no, leave as an open Bucket-2 gap. **What was found (live):** Reclassification **confirmed** — Celo is on Etherscan V2 (chainid 42220, covered by the existing free multichain key; confirmed in /v2/chainlist), and the legacy LockedGold 0x6cC083Aed9e3ebe302A6336dBC7c921C9f03349E is still the live custody: live balanceOf 85.65M CELO, on-chain getTotalLockedGold() 82.43M CELO (the ~3.2M gap is pending-withdrawal CELO still in the contract). **BUT** the free getLogs reconstruction does **not** reproduce that on-chain total, failing the Entry-26 cross-check by ~3x: - GoldToken (0x471E?438) Transfer in/out of LockedGold -> only **2.0M** CELO. Celo’s **native** CELO locking (lock() sends native value) does not emit a standard ERC-20 Transfer on GoldToken, so Transfer logs miss almost all of it. - LockedGold’s own native events cumsum(GoldLocked)+cumsum(GoldRelocked)?cumsum(GoldUnlocked) -> only **25.8M** CELO (lock 676M / unlock 662M churn nets tiny) vs the 82.43M target. The ~57M shortfall is locked CELO carried over as **state** in the 2025-03 L2 migration with no re-emitted lock event on the indexed chain. The only clean number is historical getTotalLockedGold()/balanceOf, i.e. archive eth_call / Etherscan balancehistory — a PRO endpoint, not free. **Decision made:** Do **not** ship CELO (it fails its own cross-check). Record the reclassification as confirmed (CELO is an EVM chain on the existing key, not a non-EVM-indexer problem) but the **free-tier log reconstruction as inadequate**. Documented open gap. Check code preserved in phase1_channel1_pos_coins_bucket2.py (celo_series) and 04_code/_celo_event_check.py. **Rationale:** Spec ?0 / the flag-don’t-ship rule: a 3x undercount that misses the migration-carried state is not a defensible locking ratio; better an honest gap than a

wrong number entering lambda. **Downstream impact:** CELO becomes buildable only with historical `getTotalLockedGold()` (archive/PRO `eth_call`) or a Celo indexer that exposes it. The Tier-4 instruction's premise ("same `getLogs` method as Entry 26") does not hold for a native-asset L2-migrated lock.

Entry 47 — Bucket 2 Tier 5/6: live access-gate check for ATOM/INJ/SEI/KAVA/AVAX/NEAR/EOS/ICP/AP — no self-serve free path, no purchase

Date: 2026-06-26 **Spec section affected:** 3.1 (coins); access-verification discipline. **Asset(s)/period affected:** ATOM (3794), INJ (7226), SEI (23149), KAVA (4846), AVAX (5805), NEAR (6535), EOS (1765), ICP (8916), APT (21794) — none built. **What the spec wanted (kickoff Tier 5/6):** visit the actual signup/pricing flow, attempt a free self-serve signup only where genuinely free, report the gate type per chain, **do not purchase**. **What was found (live pricing/signup pages, this session):** - **ATOM/INJ/SEI/KAVA (Cosmos):** Mintscan/Cosmostation API is **contact-sales**, pricing undisclosed (not self-serve). Bitquery's Developer tier is a **trial only** — 10K points for the first month, then upgrade/contact — not a sustained free tier; per-appchain historical bonded-total coverage and points cost remain unconfirmed. No self-serve free path. - **AVAX:** AvaCloud Metrics API has a historical "Staking Information" feature, but the free-read vs paid gate is **not determinable** from the public pricing pages (ambiguous, as Entry 42 found) — needs a direct API-key signup-flow check. - **NEAR:** Pikespeak (key-gated, pricing undisclosed) and NearBlocks (current `/v1/stats` only; historical-staking endpoint + pricing unconfirmed). No confirmed free historical path. - **EOS/ICP/APT:** no free, keyless, historical staked/neuron-stake/delegation series found (APT's free Indexer GraphQL `current_delegated_staking_pool_balances` is current-state-only). **Decision made:** No signup completed (none was genuinely self-serve **and** free **and** confirmed to carry the historical series), **no purchase made or recommended**. All nine stay open gaps with their gate type recorded. **Rationale:** The rule is verify-live-then-report, purchase is Moazzam's alone. None of these cleared the "self-serve + free + has the series" bar; the honest output is the gate map, not a build. **Downstream impact:** Any of these advancing requires a Moazzam-side signup/purchase decision against the recorded gate (contact-sales for Cosmos, ambiguous pricing page for AVAX, key-gated for NEAR). Nothing here authorizes a purchase.

Entry 48 — Bucket 3 BUILT: GMX + AERO + CAKE via a validated Dune curated-transfers reconstruction; corrects the kickoff's Dune table premise; AXS rejected, VELO deferred

Date: 2026-06-26 **Spec section affected:** 3.1 (token locking ratio); Entry-26 single-clean-escrow standard. **Asset(s)/period affected:** GMX (11857, Arbitrum), AERO (29270, Base), CAKE (7186, BNB) — built; AXS (6783) rejected; VELO (7127) deferred. **Candidate derivation:** from `classification_table.csv`, `asset_class='token'` rows not already in any channel (61 live-recomputed covered `cmc_ids`), not one of the 5 final rejects (MKR/BAL/COMP/RUNE/ANGLE), with a governance/staking tag or `deffillama_categories` value `->` **290 candidates** (`03_data/phase1/_bucket3_candidates.csv`) — logged, not forced to the ~73 estimate. The Entry-26 clean-single-escrow test is the real filter: the vast majority are DEX/lending/RWA/meme/chain tokens whose governance is delegation, MasterChef farming, or off-chain Snapshot — not base-token custody. **Dune method correction (response-body verified):** the kickoff's named table `balances_<chain>.daily_updates` **DOES NOT EXIST** on the Dune free tier (query FAILED: "does not exist or it is private"). The correct free curated tables are `tokens_<chain>.transfers` (cumulate inflow?outflow of the escrow per month) and `tokens_<chain>.balances` (historical snapshots). BSC's schema is `tokens_bnb`, not `tokens_bsc`. Also: the free **Etherscan V2** key does **not** cover Base/BSC ("Free API access is not supported for this chain. Please upgrade your api plan") — it does cover Ethereum, Arbitrum, Celo — so AERO/CAKE escrows were verified via **keyless public RPC** `balanceOf` and built entirely through Dune. **What was built (each cross-checked: reconstructed final cumulative vs live on-chain balanceOf — the Entry-26 balanceOf==locked identity):** - **GMX** — `StakedGmxTracker 0x908C79dD4` (Arbitrum), the Entry-41 deferral, built **first** as the method confidence-check: `recon 6,162,450 vs balanceOf 6,160,000 = 0.04%`. 44 lambda months, ratio 57%->84%. This retires the Entry-41 "series DEFERRED for Arbitrum `getLogs` perf" — Dune's pre-indexed transfers table makes the full-history scan trivial. - **AERO** — `veAERO VotingEscrow 0xeBf47e6B4` (Base): `recon 968,405,575 vs balanceOf 968,403,885 = 0.00%`. 50.3% of total AERO supply locked. 26 lambda months. **FLAG:** `staking_ratio>1` vs CMC circulating (CMC excludes `veAERO`-locked AERO); ~50% vs total

supply — kept un-capped & flagged. - **CAKE** — veCAKE 0x569271bAB (BNB): recon 5,896,692 vs balanceOf 5,896,692 = **0.00%**. **FLAG:** veCAKE adoption fell post-2024 -> small ~1.5% lock share; clean single-contract lock kept flagged (the RPL/xSUSHI standard). 31 lambda months. Built with a `block_time > 2023-10-01` floor so Dune prunes pre-escrow CAKE history and the query finishes inside the free-tier 2-min limit (the escrow held ~0 before deployment, so no locked supply is dropped). **Rejected/deferred (documented, not silently dropped):** **AXS REJECT** — AXS staking lives on the **Ronin** appchain, not indexed by any free EVM tool (no Etherscan-V2 free coverage, no Dune curated schema); legacy Ethereum staking contract abandoned. **VELO DEFER** — v1->v2 migration split: the in-universe cmc 7127 maps to the **v1** token 0x9560e827?, while the live veVELO locks the **v2** token 0x3c8B6502? (a different contract); joining a v1 cmc_id to a v2-token lock is the exact cmcId/symbol collision the project forbids (Entry 39 landmine), so it is deferred pending an identity-map resolution. **Decision made:** Accept GMX/AERO/CAKE into lambda Channel 1. New script `phase1_channel1_evm_locks_bucket3.py` -> `03_data/phase1/channel1_evm_locks_bucket3.csv` (101 asset-months, 3 assets; picked up by the `channel1_*.csv` glob). Free Dune key only; query budget cost is negligible (a handful of `small` executes). **Rationale:** Each cleared the Entry-26 standard (single contract holding the base token directly, cross-checked to live balanceOf at <0.1%) on cmc_id joins. The Dune transfers-cumsum is the historically-correct, free, fast successor to block-range getLogs for chains the Etherscan free key can't reach or where getLogs is too slow. **Downstream impact:** VELO is the one accepted-pending item (needs the v1/v2 identity call). The 287 non-clean-escrow candidates are not Channel-1 locks by construction (many already have a Channel-3 voting value). The `tokens_<chain>.transfers/block_time-floor` method generalizes to any future escrow on a Dune-covered EVM chain.

Entry 49 — Session 020 lambda assembly: 1,688 -> 1,880 observed asset-months, 58 -> 62 distinct assets

Date: 2026-06-26 **Spec section affected:** 3 (lambda index assembly — counts only; assembly logic untouched). **Asset(s)/period affected:** `03_data/phase1/lambda_panel.csv`, observed asset-months. **What happened:** Re-ran `phase1_assemble_lambda.py` (unchanged) after dropping the four new `channel1_*.csv` series (TRX/SOL from Bucket 2; GMX/AERO/CAKE from Bucket 3) into its auto-glob. **lambda before->after: 1,688 -> 1,880 observed asset-months; 58 -> 62 distinct assets.** Coin 7->9 (+TRX, +SOL), token 47->49 (+AERO, +CAKE; **GMX** was already in lambda via its Channel-3 Snapshot space, so its new Channel-1 lock **upgraded it to 2-channel** rather than adding an asset — which is why Channel-1 gained 3 token entrants but the distinct-asset total rose by 2 on the token side). 2-channel asset-months 322->354 (+32). Ch1 standardizable months 73->78. **Decision made:** Accept the assembled panel. Stop at Bucket 2 + Bucket 3 as instructed; do not start Bucket 1 or Phase 3 without review. **Rationale:** Only channel input files were widened; the z-scoring/standardizability/equal-weight/no-imputation logic was not touched (it auto-globs `channel1_*.csv` + `channel3_voting.csv`). **Downstream impact (re-check if this changes):** the AERO/SOL `staking_ratio>1` flag (CMC-circulating denominator artifact) and the CAKE small-share / GMX-Arbitrum-via-Dune notes ride along in the per-series `flag` columns; lambda uses z-scored relative rank, not the level, so a >1 ratio ranks correctly but should not be read as a literal locked fraction. Full account: `03_data/SESSION020_BUCKET2_BUCKET3_COVERAGE_ADDENDUM.md`.

Entry 50 — Session 021: Token Bucket-1 exhaustive re-audit, the funnel & method (398 confirmed exact, cmcId-only DL triage, DL token-quantity discovery)

Date: 2026-06-29 **Spec section affected:** 3.1 (Channel-1 token locking ratio); spec Section 0 (flag, don't guess) + Section 7 (exhaust free avenues before excluding). **Asset(s)/period affected:** the 398 token-side “unrecoverable” cmc_ids from session 020, re-audited individually. **Context:** A Cowork review challenged session 020's “398 unrecoverable” as resting on three different rigor levels (6 individually verified, ~287 rejected at the CATEGORY level via `_bucket3_candidates.csv`, ~111 never opened) rather than one uniform per-token check. Kickoff: `CLAUDE_CODE_TOKEN_BUCKET1_EXHAUSTIVE_REAUDIT_PROMPT.md`. **What was actually available / done:** - **Stage 0 (worklist rebuilt live, not from cache):** in-universe `asset_class='token'` cmc_ids NOT in `lambda_panel.csv`, minus VELO = **exactly 398** (448 - 49 in lambda - 1 VELO; reconciles to the penny). The cached 290-candidate file was not trusted. Of the 6 already-rejected, only **AXS/RUNE/MKR** are inside the 398; **BAL/COMP** are already in lambda via

Channel-3 voting (outside the 398); **ANGLE** is out-of-universe entirely. - **Stage 1 (bulk DeFiLlama triage, ALL 398, by cmcId NEVER symbol):** live `api.llama.fi/protocols` (7,742 protocols; 1,706 carry a cmcId). **92** of 398 have a clean cmcId-matched DL protocol; **306** have no cmcId DL protocol AND no contract address on file, logged explicitly as “no on-chain identity available to check” (the honest specific reason, per kickoff). The symbol-matched `defillama_categories` in `classification_table.csv` were deliberately NOT used for promotion – they are the Entry-20 collision landmine (DOT-cmc814, HONEY, DRIFT, VOLT, LAYER all carried spurious “Liquid Staking” from unrelated same-symbol protocols; a naive union produced 10 false HIGH + 15 false MEDIUM that strict cmcId matching collapsed to 1+1). - **Stage 2a (DeFiLlama /protocol/{slug} chainTvls, all 92):** **36** expose a `staking chainTvls` bucket – and crucially that bucket carries a raw `staked-TOKEN-QUANTITY` series (`chainTvls['staking']['tokens']`), not just the USD value the kickoff anticipated. Computed DL-staked-qty / panel circulating for all 36 (`_stage2_ratios.csv`). **Decision made:** Treat the DL token-quantity series as a *locator* (to find the escrow), not as the shipped numerator. Route every one of the 36 staking-bucket tokens into a Stage-2b on-chain single-escrow test (no sampling). Log every one of the 398 with a token-specific reason – no category verdicts. Staging artifacts: `_token_bucket1_worklist.csv`, `_stage1_triage.csv`, `_stage2a_dl_tv1.csv`, `_stage2_ratios.csv`. **Rationale:** Matches the kickoff’s funnel and the project’s cmcId-only rule; concentrates effort on the 36 where a base-token lock is even plausible while still giving the other 362 an individual line. **Downstream impact:** Stage-2b verdicts in Entry 51 (builds) and Entry 52 (rejections). The cmcId-only-vs-symbol distinction is the reason the HIGH/MEDIUM set is small but clean.

Entry 51 — Session 021: five Bucket-1 BUILDS (API3, ORBS, IQ, VVV) + VELO deferral RESOLVED; lambda 1,880->2,080 / 62->67

Date: 2026-06-29 **Spec section affected:** 3.1 (Channel-1 locking ratio); Entry-26 single-clean-escrow standard; Entry-48 Dune curated-transfers method. **Asset(s)/period affected:** API3 (7737), ORBS (3835), IQ (2930), VVV (35509), VELO (7127) – all built into lambda Channel 1. **What was available (Stage 2b, live, response-body verified):** For each of the 36 staking-bucket tokens, a Dune `tokens_<chain>.balances/net-transfers` top-holder query tested the Entry-26 identity: does ONE contract hold the base token in the DL-reported staked amount? **Five pass cleanly** (top holder == DL staked qty -> single-contract base-token custody), then the full series was reconstructed by cumulating `tokens_<chain>.transfers` in/out of that escrow and cross-checked to live `balanceOf` (the Bucket-3 method): - **API3** – `Api3Pool 0x6dd655...c76d76` (Ethereum), recon 64,354,124 == `balanceOf`, 0.00%. ~74%. Reward-staking (xSUSHI/stkAAVE-style). - **ORBS** – `StakingContract 0x01d59a...656c3` (Ethereum), recon 1,841,060,162 == `balanceOf`, 0.00%. ~42%. (Top holder was a treasury; the 3rd holder matched DL exactly – the escrow.) - **IQ** – `HiIQ veIQ 0x1bf545...e16ba` (Ethereum), recon 2,416,000,459 == `balanceOf`, 0.00%. ~9%. Curve-style vote-escrow of IQ. - **VVV** – `Venice staking 0x321b7f...f340ff` (Base), recon 33,279,865 == `balanceOf`, 0.00%. ~73%. (Base balances table times out on free tier -> escrow located via net-transfers.) - **VELO** – `veVELO VotingEscrow 0xfaf8fd...06787d` (Optimism), recon 1,295,615,052 == `balanceOf`, 0.00%. ~7.4%. **This resolves the Entry-48 deferral.** Entry 48 deferred VELO believing cmc 7127’s token `0x9560e827...` was a defunct “v1” token distinct from the v2 lock. DeFiLlama’s OWN Velodrome **V2** and **V3** entries BOTH carry `address=optimism:0x9560e827...` – i.e. it is the CURRENT canonical token; CMC (7127->`0x9560e827`) and DeFiLlama (V2/V3->`0x9560e827`) agree. That is the documented, non-guessed mapping the kickoff required; no collision remains. **Decision made:** Accept all five into lambda. New script `phase1_channel1_evm_locks_bucket1.py` -> `03_data/phase1/channel1_evm_locks_bucket1.csv` (200 asset-months, 5 assets; picked up by the `channel1_*.csv` glob). Re-ran `phase1_assemble_lambda.py` (logic untouched). **lambda before->after: 1,880 -> 2,080 observed asset-months; 62 -> 67 distinct assets (token 49->54).** 2-channel asset-months unchanged at 354 (none of the five had a prior Channel-3 value). **Rationale:** Each cleared the Entry-26 standard (single contract holding the base token directly, cross-checked to live `balanceOf` at 0.00%) on cmcId joins – the exact xSUSHI/stkAAVE/GMX bar. The DL token-quantity series only located the escrow; the shipped numerator is the on-chain reconstruction. **Downstream impact (re-check if this changes):** **FLAG** – API3 and ORBS show `staking_ratio>1` in some months (CMC `circulating_supply` excludes the pooled/staked tokens, the AERO/SOL artifact, Entry 49) – kept un-capped and flagged; lambda uses z-scored rank not level. VELO has 37 months of lock data but only 11 observed lambda months (panel visibility). The VELO resolution overrides Entry 48’s

DEFER; if a future identity-map pass disputes the 0x9560e827 canonical-token finding, re-open it.

Entry 52 — Session 021: the 393 rejections (clustered), Artemis re-test, 6-reconfirm, Dune budget, follow-ups

Date: 2026-06-29 **Spec section affected:** 3.1; Section 0 (flag-don't-ship); access-verification discipline. **Asset(s)/period affected:** the 393 of 398 that did NOT build, + the 6 reconfirm + ANGLE. **What was found / decided (every token has its own row + reason in 03_data/phase1/token_bucket1_full_audit.csv, 402 rows):** - **REJECT-mechanism (367):** specific protocol-design reasons no single-contract base-token lock exists. The Stage-2b multi-contract/native cases (the CELO lesson – an honest gap beats a wrong number): **EIGEN** (EigenLayer restaking spread across strategy contracts; top holder 184M vs DL 296M), **ILV** (multi-pool core staking), **KEEP** (staking migrated to the T token), **HEX** (staking internal to the HEX contract, no separate escrow; DL bucket reads 0). Plus the 362 non-staking-bucket / no-identity tokens, each given a sector-specific line (DEX/lending/derivatives/gaming/L1-L2/DePIN/meme/wrapped-LST/governance-only/no-identity) rather than a category verdict. - **REJECT-no-data (29):** plausible mechanism, no free source. No single contract reproduces the DL staked figure (**BTCST, ZBU, PEAK, KAITO, MBOX, TIME, ATH, AKRO, SUPER, EPS, AUCTION, MVL, MAGIC, BAKE, RFOX, MYX, SFI**; treasury/LP holders dominate balances); DL staking bucket reads 0 (**SFUND, ADF, FLEX, CASINO**); non-EVM staking outside the free EVM Dune curated-transfers method (**HXRO** Solana, **SUN** Tron, **ORN** TON, **TLM** WAX, **C98** TomoChain, **BRISE** Bitgert); cmcId-collision artifact (**WARP** – DL maps slug polkastarter to cmcId 1166 -> impossible 1764% ratio, the Entry-39 landmine). - **Artemis (Stage 2c), re-tested live:** app.artemis.xyz 308-redirects to classic.artemis.ai, which serves a JS-rendered SPA with no server-side per-asset staking data retrievable without login. No free per-asset staking-ratio surface – reconfirms Entry 2/14. **Not signed up, not paid.** - **The 6 reconfirmed unchanged:** MKR (DSChief in-wallet voting), BAL (veBAL locks an 80/20 BPT not BAL), COMP (in-wallet delegation), RUNE (native THORChain L1 bonding), AXS (Ronin appchain, no free EVM index); none exposes a cmcId-matched DL staking bucket live. ANGLE confirmed out-of-universe. **Decision made:** Accept all 393 rejections as individually documented, plus the reconfirms. **GATED = 0:** no priced self-serve option surfaced that would let a purchase recover any of the 398 – the rejections are mechanism- or non-EVM-data-bound, not paywalled. **Rationale:** Every avenue named in the kickoff (DeFiLlama, Etherscan-equivalent via Dune top-holder, Dune, Artemis) was checked per token before acceptance; the honest output is a specific reason per row, not a category write-off. **Downstream impact / resource accounting:** Dune free tier ~36 small executes this session (~350-400 of the 2,500 monthly credits, ~15%) – within headroom, flagged here per the kickoff rather than assumed. No Etherscan PRO, no paid tier, no purchase. **Open follow-up:** the 456-coin Bucket-1 has NOT had this individual re-audit and may warrant the same treatment in a future session – do not start without review. Full account: 03_data/SESSION021_TOKEN_BUCKET1_EXHAUSTIVE_AUDIT.md.

Entry 53 — Session 022: Etherscan/non-EVM contract-read feasibility map for lambda Channels 1/2/3 (universe-wide, identification only)

Date: 2026-06-29 **Spec section affected:** 3 (lambda channels), 2.5 (per-asset sources); feasibility/identification, NOT a build. **Asset(s)/period affected:** the full token+other universe (1,306) on the EVM side + the 405 off-Etherscan tokens (non-EVM). NO lambda panel modified; Entries 21-26 still govern. **What was done (real reads, not metadata reasoning):** For every token+other asset: (1) resolved an on-chain contract via CMC data-api detail.platforms[] (cached 03_data/raw/cmc_detail/) + the identity map; (2) read the verified contract via Etherscan-V2 getsourcecode (cached 03_data/raw/etherscan_src/, proxy->implementation followed); (3) classified the lambda mechanism from the ABI (Ch1 holder-lock events {StakeStart,Staked,Locked,BalanceLocked} excluding admin/vesting; Ch3 {DelegateVotesChanged,VotingPowerChanged}); (4) computed each candidate event's keccak-256 topic0 and ran logs/getLogs over full history – ABI presence was NOT accepted, only emitted logs. Pipeline: 04_code/universe_lambda_pipeline.py (resumable). **What was found:** - 901/1,306 EVM-reachable (793 on free-getLogs chains, 108 paid-only). Ch1 genuine getLogs-CONFIRMED = 6 (HEX,NMR,stkAAVE,XAN amount-bearing; AKRO,VSL bare Locked() need balance reads). Ch3 ACTIVE getLogs-CONFIRMED

= **34** (UNI,ENS,SUSHI,COMP,GTC,KP3R,BTRST,EIGEN,ONDO,STRK,MNT,BLUR,... all emit DelegateVotesChanged). **Ch3 ABI-present-but-needs-paid = 15** + Ch1 paid = 1 (TNC). **Ch3 infra-but-DORMANT/negligible = 15** (CORE,SUPER,ILV,FLOKI,PENDLE,...). **Ch2 = all 901**. - **MEASURED Etherscan free-tier chain gate:** getsourcencode free on all chains; **getLogs/tokenTx FREE only on Ethereum/Polygon/Arbitrum/Blast, PAID-only on BSC/Base/Avalanche** (“Free API access is not supported for this chain”). This is the concrete paid-API trigger: 16 BSC/Base/Avax candidates + panel-scale Ch2 throughput. - **Non-EVM (405):** 284 have NO chain/contract identity (dead, unrecoverable); 92 on free-indexer non-EVM chains (Solana 59 dominant, Tron 8, Cosmos/Osmosis/Kava 7, +tail) -> Ch2 recoverable per chain indexer, Ch1/Ch3 need per-project Anchor/CosmWasm/Realms reads (chain-native staking/gov = gas-coin, out of token-scope) with low expected yield; 22 EVM-but-not-Etherscan (KAIA/HyperEVM/...) recoverable by the SAME method on chain-specific explorers. Live-verified free APIs: Cosmos LCD (bonded_tokens,gov/proposals), Tron TronGrid, Solana RPC, Cardano Koios. **Decision made:** This is an identification map, not a source decision – no panel changed. Recorded as feasibility. The corrected finding vs the earlier same-day meta-analysis (ETHERSCAN_LAMBDA_CHANNEL_FEASIBILITY.md, written before any read): the bucket is NOT uniformly mechanism-dead – 24 bucket-1 tokens (and 40 universe-wide) carry getLogs-retrievable Ch1/Ch3 data the Snapshot+DeFiLlama audit never looked for. Overlap caveat: many of the 34 Ch3-active (UNI/ENS/SUSHI/COMP) likely already have a Snapshot Ch3 series (Entry 25) -> on-chain DelegateVotesChanged is a cross-check there, net-new only where no Snapshot turnout exists. **Rationale:** Satisfies the spec “verify before building” + the user directive to source lambda from canonical chain data; every verdict is backed by an actual contract read and an on-chain log query, with free/paid status measured live per chain. **Downstream impact (re-check if this changes):** Artifacts: 03_data/phase1/universe_lambda_channel_map.csv (1,306), etherscan_lambda_channel_map.csv (402), non_evm_lambda_recoverability.csv (405), 03_data/ETHERSCAN_LAMBDA_CHANNEL_EMPIRICAL.md, NON_EVM_LAMBDA_CHANNEL_ASSESSMENT.md. Next steps + next-session prompt in 06_documentation/SESSION022_STATUS_AND_NEXT_SESSION.md. NOTHING built – before building any series, dedupe Ch3 vs Snapshot panel and decide the Etherscan-Pro purchase for the 16 paid-chain candidates.

Entry 54 — Session 023: HEX Channel-1 BUILT (supersedes the session-021 HEX rejection) via session-022’s contract-read + getLogs event-replay method

Date: 2026-06-29 **Spec section affected:** 3.1 (locking ratio for staking tokens), 2.5 (per-asset sources). **Asset(s)/period affected:** HEX (cmc_id 5015), 50 observed asset-months 2020-03..2024-05. 03_data/phase1/channel1_hex_stake.csv (new), 03_data/phase1/lambda_panel.csv (re-assembled). **What the spec wanted:** locked/staked supply ? supply for a staking token, reconstructed from canonical chain data, cross-checked to on-chain state (the Entry-26 standard). **What was actually available — and how this supersedes session 021:** Session 021 (SESSION021_TOKEN_BUCKET1_EXHAUSTIVE_AUDIT.md, logged under Entry 50/52) **rejected HEX** at “Stage 2b” with the reason “*staking internal to the HEX contract*” + “*DL staking bucket reads 0*” — i.e. its Dune-top-holder single-escrow probe found no escrow contract whose balanceOf reproduced a DeFiLlama staked figure (and DeFiLlama reported 0). **That rejection is superseded by this entry.** Using session 022’s higher-fidelity method — read the verified HEX source (cached 03_data/raw/etherscan_src/1_0x2b591e997.json) and confirm via getLogs — the actual mechanism is now resolved directly: HEX staking is **non-custodial**. stakeStart() calls _burn(msg.sender, newStakedHearts), so staked HEX is **burned out of the ERC20 totalSupply** (no escrow contract holds it — which is exactly *why* session 021’s single-escrow balanceOf probe and DeFiLlama both saw nothing), and the staked quantity lives only in the contract’s internal lockedHeartsTotal global. This is a **genuinely different construction path than Entry 26’s transfer-into-escrow reconstruction**, logged here explicitly as a new method, not silently treated as equivalent. **Decision made (BUILD):** Reconstruct the monthly staked series exactly from the contract’s own accounting: only StakeStart (+stakedHearts) and StakeEnd (-original stakedHearts) move lockedHeartsTotal (verified: a single _lockedHeartsTotal -= site; StakeGoodAccounting does not touch it). So lockedHeartsTotal(t) = ? StakeStart.stakedHearts<=t ? ? StakeEnd.(orig)stakedHearts<=t, decoded on Dune (hex_ethereum.HEX_evt_StakeStart/_StakeEnd; stakedHearts=(data0>>40)&(2^72?1)

via exact UINT256 integer arithmetic, verified == Python decode; StakeEnd amount recovered by joining stakeId back to the StakeStart decode). **Cross-check: reconstructed final = 61,900,823,759,862,091,712 hearts == live globalInfo()[0] read (2026-06-29) at drift 0.000000%** — the same 0.00% bar the five session-021 BUILDS (API3/ORBS/IQ/VVV/VELO) cleared. Script: 04_code/phase1_channel1_hex_stake.py (its own builder, parallel to eth_staking/pos_coins, because there is no escrow to feed the evm_locks_* transfer scripts). Output picked up by the channel1_*.csv glob. **Denominator (resolved explicitly, not assumed):** The HEX contract’s own NatSpec states “*ERC20 totalSupply() is the circulating supply and does not include any staked Hearts. allocatedSupply() includes both*” (allocatedSupply = totalSupply + lockedHeartsTotal). CMC’s circulating_supply mirrors the on-chain ERC20 totalSupply and therefore **excludes staked HEX** — the SAME denominator artifact as the Entry-49 AERO/SOL/API3/ORBS series, NOT a new double-count. We ship `staking_ratio = locked/circulating` (panel basis) for cross-sectional comparability with the other Channel-1 token series (lambda z-scores within month — rank, not level — Entry 27/49); within HEX’s observed window (2020-03..2024-05) `locked < circulating` every month so the ratio is well-behaved (14.3%->43.5%, latest 33.6%, never >1). The theoretically-clean fraction `locked/(locked+circ)` (12.5%->30.3%) is written alongside as `locked_fraction_alloc` for audit. **Rationale:** Per the session directive, where session 022’s contract-code-read + getLogs method conflicts with session 021’s Dune-substitute method, the former governs. Here it does not merely “find an event” — it reconstructs a full monthly locked-quantity series that reconciles to live on-chain state at 0.0000%, clearing the same bar as the shipped BUILDS. **Downstream impact (re-check if this changes):** lambda: **2,080 -> 2,130 observed asset-months, 67 -> 68 distinct assets** (+HEX, single-channel Ch1). If CMC ever revises HEX circulating to an allocated-supply basis, switch the denominator to `locked_fraction_alloc` (already in the CSV). Full account: 03_data/SESSION023_HEX_AKRO_RECONCILIATION.md.

Entry 55 — Session 023: AKRO Channel-1 REJECT reconfirmed (session-022’s Locked() flag is a false positive; session-021 rejection stands, now with the contract reason)

Date: 2026-06-29 **Spec section affected:** 3.1 (locking ratio), 2.5 (per-asset sources). **Asset(s)/period affected:** AKRO (cmc_id 4134). No panel changed; AKRO remains absent from lambda. **What the spec wanted:** a single staking/lock contract holding a reproducible share of AKRO supply (the Entry-26 single-escrow test). **What was actually available — 021 vs 022 reconciled:** Session 021 rejected AKRO in the “no single contract reproduces the DL staked figure” cluster (with KAITO/ATH/SUPER/etc. — the “treasury dominates balances” suspicion). Session 022’s universe map flagged AKRO as “**Ch1 GENUINE, Locked(), needs contract-balance reads**” — an apparent contradiction. Running session 022’s own higher-fidelity method to completion (read the verified implementation, not just the ABI event name) **resolves it in favour of REJECT, and explains 022’s flag as a false positive:** AKRO’s address 0x8ab7404063ec? is the **AKRO token contract itself** (TokenProxy -> impl AkropolisToken), and its Locked() event comes from an OpenZeppelin-style Lockable base contract — function `lock() public onlyOwner { setLock(true); emit Locked(); }`. It is an **owner-only admin pause switch** (disables restricted methods), carries **no amount** (event Locked() has no parameters), and escrows **no tokens**. Live `getLogs` over full history confirms it fired **exactly once** (block 8099298, data=0x) — the one-time owner `lock()` call, not a staking series. Session 022’s Ch1 classifier matched on the event *name* Locked(); the full contract read shows the event is not a holder lock/stake at all. **Decision made (REJECT, reconfirmed):** AKRO has no staking/escrow mechanism to reconstruct. Session 021’s rejection **stands**, now upgraded from “no single contract matches the DL figure” to the precise contract-level reason: *the only Locked() event is an admin pause flag on the token contract, not a stake*. There is no 021-vs-022 contradiction once the contract is read — both reject; 022’s map row was an over-eager name-match the full Entry-26 test overturns. **Rationale:** The session’s standard is to trust 022’s method where it conflicts with 021 — applied here, 022’s method (read to completion) agrees with 021’s verdict and supplies the better reason. A mechanism merely *named* “Locked” is not a lock; ABI-event presence was explicitly never the bar (Entry 53). **Downstream impact (re-check if this changes):** None to lambda. If AKRO’s broader ecosystem (Akropolis had separate Sparta/Delphi staking pools) is ever brought in scope, those are *different contracts* (not cmc 4134’s token address) and would need their own identification + single-escrow cross-check. Full account: 03_data/SESSION023_HEX_AKRO_RECONCILIATION.md.

Entry 56 — Session 023: the 284 no-on-chain-identity listings documented as an acknowledged survivorship-bias limitation (not a pending data gap)

Date: 2026-06-29 **Spec section affected:** 2 (universe/coverage), 7 (limitations); paper data/methodology section. **Asset(s)/period affected:** 284 token+other assets with no recoverable chain/contract identity. No panel changed. **What the spec wanted:** a clear, reconciled account of which universe assets carry no lambda data and why, distinguishing permanent unrecoverability from pending effort. **What was actually available (re-derived live, reconciled to the penny):** From 03_data/phase1/non_evm_lambda_recoverability.csv (405 rows): **class NO-IDENTITY = 284** (chain and tx_repository both null; ch1=ch2=ch3=no by construction). Universe reconciliation against 03_data/phase1/universe_lambda_channel_map.csv: **1,306** token+other assets = **901** Etherscan-reachable + **405** off-Etherscan; the 405 off-Etherscan = exactly the 405 etherscan_reachable!=yes rows (overlap 405/405, 0 outside the map, 0 duplicate cmc_ids); the 405 = **284 NO-IDENTITY + 92 non-EVM-indexed + 22 EVM-non-Etherscan + 7 obscure**. Criteria for “dead”: no contract address resolvable via CMC detail.platforms[] + the identity map, and no chain/explorer (EVM or non-EVM) on which any lambda channel can be queried, checked across sessions 021–022. Cohort character: **83% are asset_class “other”** (non-DeFi/non-governance), **89% (252/284) are pre-2020 listings** (191 with cmc_id<2000, the earliest 2013-2017 era), with a ~32-asset tail of newer non-standard-chain assets (BRC-20/ordinals, BCH-ABC fork, etc.) whose identity the pipeline can’t resolve. **Decision made:** Classify the 284 as a **permanent, acknowledged survivorship-bias limitation, not a pending data gap** — no further recovery effort is planned (no contract, no chain identity, on any free or paid source). Paper-ready write-up saved to 03_data/SURVIVORSHIP_BIAS_NOTE.md and mirrored in 03_data/SESSION023_HEX_AKRO_RECONCILIATION.md ?3. **Rationale:** These were live CMC listings in (predominantly) 2014-2018 that are now delisted/abandoned/rug-pulled with no on-chain identity to query. Folding them silently into the broad “no-data” rejection bucket would understate a genuine survivorship bias a referee will ask about. Naming it explicitly — and noting its likely *direction* (the dead cohort is disproportionately low-conviction, no-governance, no-staking “other” assets that would sit at the bottom of the lambda distribution even had they survived, so their exclusion truncates the low-lambda tail and makes survivors conditionally higher-lambda) — is the honest treatment. **Downstream impact (re-check if this changes):** None to the built panel. Feeds the paper’s limitations/robustness discussion. If a paid multi-chain identity source is ever adopted, re-run the identity resolution before treating any of the 284 as recoverable.

Entry 57 — Session 024 (Task A): Channel-3 on-chain governance DELEGATION sub-channel built from DelegateVotesChanged replay (21 net-new assets)

Date: 2026-06-30 **Spec section affected:** 3.3 (voting/governance engagement), 2.5 (governance venues). **Asset(s)/period affected:** 21 net-new ERC20Votes-ACTIVE tokens (no Snapshot turnout series). New file 03_data/phase1/channel3_onchain_delegation.csv; phase1_assemble_lambda.py extended to read it as channel ch3_delegation. **What the spec wanted:** governance participation per asset-month for the 34 getLogs-CONFIRMED ERC20Votes-ACTIVE tokens (session-022 map), deduped against the existing Snapshot Channel-3 panel (Entry 25). **What was actually available / done:** Joined the 34 on cmc_id vs snapshot_space_map.csv+channel3_voting.csv: **10 already have a Snapshot turnout series** (GTC/ENS/MNT/COMP/SUSHI/UNI/RGT/KP3R/STRK/HFT) -> cross-check only; **24 are NET-NEW**. Built a monthly series for the net-new from on-chain DelegateVotesChanged event replay (phase1_channel3_onchain_delegation.py): replay all logs in (block,logIndex) order tracking each delegate’s latest newBalance; at each month-end block, delegated voting weight outstanding = sum of latest newBalance over all delegates; / circulating supply. Decimals read once via immutable decimals(); free getLogs chains only (ETH/Arbitrum/Blast). Topics: uint256 variant 0xdec2bacd... (22 tokens), uint96 variant 0x664ef4a2... (DDX). **Result: 21 of 24 net-new produce a ratio (560 asset-months); 3 do not** – TOMI/UXLINK (delegation events all post-date their observed panel window -> 0 in-window, verified by block comparison) and CYBER (2 events, 0 delegated). One of the 24, **ETHDYDX, was excluded on mechanism** (DelegatedPowerChanged/Aave-power model where every holder auto-carries voting power -> ratio~1, NOT opt-in delegation; flagged, not shipped – the AKRO/Entry-55 “verify the mechanism” discipline). **Decision made (CONSTRUCT – stated explicitly, not silently merged):** On-chain delegated voting weight is a **DISTINCT** sub-channel from Snapshot voter-turnout, NOT the same measure sourced

differently. Snapshot `vw_turnout` is a per-proposal participation FLOW; delegated-weight-outstanding is a STOCK of governance-ACTIVATED supply (tokens only count as votes once delegated, incl. self-delegation). So it is written to its own file and enters lambda as a separate channel `ch3_delegation`, z-scored in its own monthly cross-section. **Governance-channel waterfall (no double-count):** only `role=='primary'` rows (the net-new tokens) enter lambda; the 10 overlap tokens are computed as `role=='crosscheck'` (validation only, excluded from lambda – a token already represented in governance-lambda via Snapshot turnout is not also counted via delegation). Verified in the assembled panel: `ch3_delegation` never co-occurs with `ch3_voting` (2-channel asset-months unchanged at 354). **Rationale:** Sources lambda from canonical chain data (Entry 21 “logs not `eth_call`”); the construct distinction + primary-only rule prevent conflating two different governance measures or double-weighting governance. **Downstream impact (re-check if this changes): FLAG** – DDX shows `delegation_ratio>1` (204%: CMC circulating excludes delegated locked/treasury supply, the Entry-49 pattern) – kept un-capped & flagged, lambda z-scores rank not level. The empirical per-token cross-check of the 10 overlap tokens was DEFERRED (their full `DelegateVotesChanged` histories – GTC/ENS/UNI/COMP/SUSHI/STRK – are `getLogs`-heavy and did not complete in the session’s budget; the construct-level distinction stands and their lambda is unchanged, still Snapshot turnout). A `CROSSCHECK=light/all` env gate is wired in the builder for a future pass. lambda contribution folded into Entry 60.

Entry 58 — Session 024 (Task B): Channel-1 free-build — XAN BUILT; VSL/NMR/stkAAVE rejected/deferred with contract reasons

Date: 2026-06-30 **Spec section affected:** 3.1 (locking ratio for staking/locking tokens), 2.5 (per-asset sources). **Asset(s)/period affected:** XAN (cmc 38481) BUILT -> `03_data/phase1/channel1_freebuild.csv` (`phase1_channel1_freebuild.py`, picked up by the `channel1_*.csv` glob). VSL/NMR/stkAAVE not built (documented). **What the spec wanted:** locked/staked supply / supply for the session-022 map’s “Ch1 GENUINE” set (HEX/AKRO done in 023; NMR/stkAAVE/XAN/VSL remaining), each cross-checked to live on-chain state at ~0.00% (Entry-26 bar). **What was actually available — verified-source reads (cached `03_data/raw/etherscan_src/`, `proxy->impl` followed):** - XAN — BUILD. `XanV1` (Anoma, 2025; `proxy 0xcdbea37->impl 0x03997b56`). `lock(value)->_lock` does `lockedSupply += value`; `emit Locked(account,value)`; NatSpec “Permanently locks ... until upgraded”; `lockedSupply` is only ever incremented (no unlock/decrement). So `locked` series = cumulative sum of `Locked.value` (the HEX-style internal-accounting path, Entry 54), and **cross-check: reconstructed final 7,500,000,010 == live `lockedSupply()` (0xca5c7b91) at drift 0.000000%** – the Entry-26 bar. **DENOMINATOR (Entry-49):** locked (7.5B) EXCEEDS CMC circulating (2.5B) -> CMC circulating excludes the locked foundation/ecosystem allocation (the HEX/AERO/API3 artifact), so `staking_ratio>1` (300%); kept un-capped & flagged, lambda z-scores rank; `locked_fraction_alloc = locked/(locked+circ) = 75%` written alongside for audit (HEX precedent). 9 observed asset-months. - VSL — REJECT (the AKRO pattern, exactly as the kickoff anticipated). Direct contract `Token/Lockable`; `Locked()` is a BARE parameterless event fired by the `checkLock` modifier as a CYCLIC transfer-pause (each 30-day epoch: 25 days unlocked / 5 days locked; `Locked()/Unlocked()` on the flip). No amount, no escrow – a contract-wide transfer freeze, not a holder conviction lock. (39 logs = the periodic flips.) Same lesson as AKRO (Entry 55). - NMR — DEFER to Phase 2. `NumeraireBackend` (`proxy->UpgradeDelegate`) tournament staking; `Staked(staker,tag,totalAmountStaked,...)` amount is a per-(`staker,tag`) CUMULATIVE running total, decremented elsewhere by `StakeDestroyed/StakeReleased`, and the modern Erasure flow BURNS NMR on stake. No aggregate staked global and no escrow balance -> NO Entry-26 cross-check anchor. Flagged for a proper Phase-2 Erasure treatment rather than shipped unanchored. - **stkAAVE (cmc 36246) — EXCLUDE.** cmc 36246 is the Staked-AAVE wrapper token; its supply already represents AAVE locked in the Safety Module, ALREADY captured in lambda via AAVE (cmc 7278, Entry 26). Building it separately double-counts the same escrowed AAVE, and as its own asset locked/circulating~1 (degenerate). **Decision made:** Accept XAN into lambda Channel 1; reject VSL, defer NMR, exclude stkAAVE — each with the contract-level reason. **Rationale:** Where session-022’s name-level “Ch1 GENUINE” flag conflicts with the full verified-source read, the read governs (the session standard) – here it overturns VSL (admin/cyclic pause), confirms XAN (real monotone lock with a live global cross-check), and finds NMR/stkAAVE structurally unfit for a clean single-anchor reconstruction. **Downstream impact:** lambda contribution folded into

Entry 60. If CMC ever revises XAN circulating to an allocated-supply basis, switch its denominator to `locked_fraction_alloc` (already in the CSV).

Entry 59 — Session 024 (Task C): Channel-2 (coin-age/HODL) PROTOTYPE built + free-tier call-budget MEASURED -> confirmed Phase-2 trigger

Date: 2026-06-30 **Spec section affected:** 3.2 (holding-duration channel) — the Entry-24 gap. **Asset(s)/period affected:** Prototype only (MET, cmc 2873) -> `03_data/phase1/channel2_holding.csv`; budget probe -> `03_data/phase1/_channel2_budget_probe.json`. **lambda panel UNCHANGED by Channel 2 (still NaN – the gap persists, now measured).** **What the spec wanted (Entry 24):** an on-chain HODL-wave / coin-age series (share of supply unmoved > a window) per asset-month — flagged as “the single highest-value addition” but requiring each token’s FULL Transfer history. **What was actually built / measured:** - **Engine (`phase1_channel2_holding.py`):** a FIFO per-address coin-age engine — replay all Transfer logs (each log carries its own `timeStamp`, no extra calls) maintaining per-address acquisition lots; mint=push, burn=FIFO-pop, transfer=FIFO-pop sender + push one time-stamped lot to receiver; at each month-end, HODL share = supply in lots older than a window / circulating (6m and 12m reported). Validated end-to-end on MET (small enough to fetch completely). - **ADDRESS-CLASS HYGIENE finding (Entry-24 caveat, empirically demonstrated):** on MET’s last month, raw HODL-6m = **90.8%** but after screening out the CONTRACT addresses among the top >6m holders (`eth_getCode` non-empty; 5 of top 40) it COLLAPSES to **1.3%** – i.e. nearly all apparent “long-held” supply is contract-held (LP/treasury/staking), not EOA conviction. This is decisive: a raw coin-age channel without contract/CEX screening is dominated by non-holder addresses, and free Etherscan labels don’t cover CEX EOAs -> a residual bias that itself is a Phase-2 paid-label task. - **CALL BUDGET (`phase1_channel2_budget_probe.py`), the Phase-2 decision input:** MET (small) = 24,636 transfers, **79 getLogs calls**, full history, ~80s. RAD (mid-size) = **204,428+ transfers, 700+ getLogs calls, DID NOT COMPLETE** (capped, ~12 min). Extrapolation: free-chain EVM population = 793 tokens (Entry 53); even at RAD’s mid-size cost 793x700 ~ 555k calls ~ 5.6 free-days, and the high-volume tail (UNI/USDC-class, 100-1000x RAD’s transfer count) dominates -> **panel-scale Channel-2 blows the free ~100k getLogs/day cap.** **Decision made:** Build & validate the engine but DO NOT scale Channel 2 this session and DO NOT wire the single-token prototype into lambda (one asset can’t be z-scored anyway). **Per Task C2, the free-cap breach IS the documented Phase-2 (paid) trigger** – Channel 2 remains the lambda schema’s NaN column (Entry 24 unchanged in lambda terms) but is now backed by a working engine + a measured budget, not an open question. **Rationale:** Spec section 0 forbids silently stalling or shipping a weak proxy; the honest output is “method proven, free budget insufficient at panel scale, here are the numbers.” **Downstream impact:** MET’s mid-history token migration (v2 contract) makes its 12m window read 0 (window-length artifact) – a clean single-deployment token should be chosen for the eventual paid build. Phase 2 needs either a paid archive/indexer or a paid getLogs throughput tier for the high-volume tail, plus a paid address-label feed for CEX-EOA screening.

Entry 60 — Session 024 close-out: lambda 2,130 -> 2,699 asset-months / 68 -> 90 assets; Tasks D/E time-boxed triage (no free clean builds)

Date: 2026-06-30 **Spec section affected:** 3 (lambda assembly — counts only; z-score/standardizability/equal-weight logic untouched), plus Tasks D/E feasibility. **Asset(s)/period affected:** `03_data/phase1/lambda_panel.csv` re-assembled. **lambda DELTA (this session): 2,130 -> 2,699 observed asset-months (+569); 68 -> 90 distinct assets (+22).** Sources: `ch3_delegation` 560 asset-months / 21 net-new governance assets (Entry 57) + XAN Channel-1 9 asset-months / 1 asset (Entry 58). New channel `ch3_delegation` standardizable in 69 months; no change to 2-channel count (354) – confirming no governance double-count. Panel now spans 2019-12->2026-05 (earlier reach via DMG-era delegation). By class: token 73, coin 9, other 8. The assembler’s cosmetic -> `UnicodeEncodeError` print was fixed (one char). **Tasks D & E — TIME-BOXED TRIAGE (the kickoff’s instructed treatment for low-yield channels; no free clean builds shipped):** - **D (22 EVM-non-Etherscan):** ~19 are WRAPPED/native-gas tokens (WCHZ/WEOS/WVLX/wIOTA/OKB/WFLR/WHYPE/WPLS/KHYPE/BBTC/MANTA/MTL/REEF/MEGA/SAROS/A/ whose conviction belongs to the underlying gas coin, NOT the token -> out of token-scope (the Entry-53 logic). Only ~2-3 are genuine DEX/governance tokens (KSP/KlaySwap, BORA — both on KAIA) that might carry a token-level lock/vote; building them needs per-chain KAIA Klaytnscope/Blockscout getLogs verification

-> flagged as Phase-2 candidates. Net free clean builds this session: 0. - **E (92 non-EVM-indexed; Solana 53, Neo 9, Tron 6, Osmosis 4, ...)**: free indexers (Solana RPC `getProgramAccounts`, TronGrid, Cosmos LCD, Koios) serve CURRENT state, not historical month-end snapshots — the SAME archive wall as Entry 21, now on non-EVM. So a historical Channel-2 coin-age series (the realistic target) is effectively paid on these chains; Channel-1/3 need per-project Anchor/Realms/CosmWasm reads (gas-coin staking/gov, out of token-scope) with low yield. Flagged for Phase 2. The 284 NO-IDENTITY listings were NOT chased (Entry 56, survivorship bias). Net free clean builds this session: 0. **Decision made:** Accept the re-assembled panel. Tasks A/B/C delivered the session's lambda gain and the Channel-2 method+budget; D/E are honestly triaged as no-free-clean-build, contributing specific Phase-2 worklists rather than category write-offs. **Rationale:** Matches the kickoff's two-phase framing (every FREE source this phase; paid = Phase 2) and the project standard (a documented per-bucket reason beats a chased low-yield build). **Downstream impact:** Full account: 03_data/SESSION024_FREE_BUILD_REPORT.md. Phase-2 worklist + kickoff: 06_documentation/SESSION024_STATUS_AND_NEXT_SESSION.md.

Entry 61 — Session 025: Etherscan API Pro purchased (the Phase-2 data-sourcing decision) — unlocks BSC/Base/Optimism/Linea getLogs + panel-scale Channel-2

Date: 2026-06-30 **Spec section affected:** 2.5 (per-asset sources), 3 (lambda channels) — a paid data-sourcing decision, the documented Phase-2 trigger from Entries 24/53/59. **Asset(s)/period affected:** Sourcing capability only; enables the P2-2 (Entry 62) and Channel-2 (Entry 63) builds. **What the spec wanted:** the two measured paid triggers — (a) `getLogs` on BSC/Base/Avalanche for the 16 paid-gated Channel-1/3 candidates (Entry 53), and (b) panel-scale `getLogs` throughput for Channel-2 coin-age, where the free ~100k/day cap binds (Entry 59). **What was actually done:** Moazzam purchased **Etherscan API Pro — Standard Plan, \$199/month, 200,000 calls/day, 10 calls/sec, all chains** (activated 2026-06-30). Key stored in `gitignored 04_code/.api_keys.json` under "etherscan". **VERIFIED Pro-grade before any build** (`phase1_p2p2_probe.py` + direct probes): `eth_blockNumber` returns on Ethereum (chainid 1); `getLogs` returns real log data on BSC (chainid 56), Base (8453), Optimism (10), Linea (59144), Avalanche (43114) — the free tier returned "Free API access is not supported for this chain" on these. The upgrade is confirmed, not assumed. **Decision made:** Accept the purchase as the Phase-2 paid-source for lambda. NO other paid subscription (no Nansen/Glassnode/CoinMetrics/Alchemy) — the kickoff's standing constraint. Budget is actively tracked per run; builds checkpoint and stop before the 200k/day quota. **Rationale:** Both paid triggers were measured, not assumed (Entries 53, 59); a single \$199/mo Etherscan tier covers both, vs a per-metric data vendor. `cmc_id` joins and the Entry-26 cross-check bar continue to govern any build done on the Pro key. **Downstream impact (re-check if this changes):** The 200k/day cap and ~10/s rate are the active budget constraints for every Pro-key build; the high-volume **wall-clock** per token (not just daily quota) turns out to bind hardest (Entries 62 OP, 63 RAD/AAVE/CRV) — see those entries.

Entry 62 — Session 025 (Task A / P2-2): the 16 paid-gated Channel-1/3 candidates run on the Pro key — 5 BUILT, 1 deferred, 9 rejected/dormant

Date: 2026-06-30 **Spec section affected:** 3.3 (governance/delegation), 3.1 (locking), 2.5 (sources). **Asset(s)/period affected:** the 16 session-022 paid-gated tokens (15 Channel-3 ERC20Votes candidates on BSC/Base/Optimism/Linea + TNC Channel-1 on BSC). Builds folded into 03_data/phase1/channel3_onchain_delegation via `phase1_channel3_onchain_delegation.py` (P2-2 block appended to TOKENS). **What the spec wanted:** for each, confirm the mechanism actually fires on-chain (the AKRO/Entry-55 discipline) before building, then build the passers to the same standard as session 024. **What was actually done / found (mechanism VERIFIED first, phase1_p2p2_probe.py):** - **5 BUILT (net-new ch3_delegation primary, DelegateVotesChanged replay): AWE** (`cmc` 4006, Base; 208,687 events; final delegated/circ = 15.25% — a substantial, real governance-activation signal), and **CHEEL** (23054, BSC), **FORM** (23635, BSC), **LINEA** (27657, Linea), **ZORA** (35931, Base) — all fire with non-zero delegated weight but at ~0% ratio (governance technically active, almost no supply delegated; included as genuine low-activation rows, consistent with the existing BLAST/BLUR/EIGEN ~0% precedent, z-scored on rank). - **1 EXCLUDED on degeneracy: ALT** (10897, BSC) — 2 DelegateVotesChanged events netting to 0 delegated weight -> no ratio (the CYBER/session-024 pattern). Not shipped. - **1 DEFERRED on throughput: OP** (11840, Optimism) — mechanism VERIFIED FIRING (46,974+ events in the first 60 capped calls) but the

full DelegateVotesChanged history is pathologically large (the native OP-airdrop self-delegation count); a full-history getLogs replay did not complete in >80 min / 400 MB+ and was killed. A PARTIAL replay would be a wrong delegated-weight stock, so OP is NOT built (same “no partial series” discipline as Channel-2). Commented out in the builder with a note; re-run alone with a block-windowed incremental fetch. A per-token DELEG_CAP guard (default 6000 calls) was added so a future giant defers gracefully instead of hanging. - **8 DORMANT Channel-3 (NOT built):** BAKE (7064), BNX (9891), EDG (5274), ESPORTS (37414), MCT (16946), MDX (8335), PONKE (29150), TKO (9020) — all carry the ERC20Votes ABI but **DelegateVotesChanged has never fired** (0 logs on both the uint256 and uint96 topics over full history). No governance-activation signal exists to measure; not forced. - **1 REJECT Channel-1 (TNC, cmc 5524, BSC):** its cached verified source declares **event Locked(address indexed account)** — **bare, no amount**, exactly the AKRO/VSL false-positive the kickoff anticipated. With no amount word there is no escrow quantity and nothing to cross-check to a balance (fails the Entry-26 bar). The amount-bearing Locked(address,uint256) topic returns 0 logs. Rejected, same lesson as AKRO (Entry 55) / VSL (Entry 58). **Decision made:** Build the 5 firing net-new delegation tokens; exclude ALT (degenerate), defer OP (throughput), reject TNC (no amount), skip the 8 dormant. P2-2 net lambda contribution: the ch3_delegation primary set grows **560 asset-months / 21 assets -> 627 / 26 (+67 / +5)**. **Rationale:** Mechanism-verify-before-build (Entry 55) plus “don’t ship a partial or degenerate series” — every verdict is backed by an actual on-chain log query on the Pro key, not ABI presence. **Downstream impact (re-check if this changes):** OP is the one carried-forward P2-2 item (verified, deferred on volume — not a rejection). The 8 dormant tokens are settled (no signal), not pending. CHEEL/FORM/LINEA/ZORA enter lambda at ~0% (low-rank governance activation); AWE is the only economically material P2-2 add. Builder env gate DELEG_CAP documents the giant-delegation deferral path.

Entry 63 — Session 025 (Task B): Channel-2 (coin-age / HODL) BUILT at panel scale — engine denominator FIX + metadata-driven build (197 assets)

Date: 2026-07-01 **Spec section affected:** 3.2 (holding-duration channel) — the Entry-24 “single highest-value addition”, NaN since inception. **Asset(s)/period affected:** 197 free-chain EVM tokens -> 03_data/phase1/channel2_holding.csv (3,413 asset-months with a screened value); builders phase1_channel2_panel.py (+ phase1_channel2_sizeprobe.py, phase1_channel2_validate.py), reusing the session-024 FIFO engine. **What the spec wanted (Entry 24):** an on-chain HODL-wave / coin-age series (share of supply unmoved > a window) per asset-month — needs each token’s FULL Transfer history; the free getLogs cap blocked it (Entry 59), now unblocked by the Pro key (Entry 61). **What was actually built / the two methodology corrections:** - **DENOMINATOR FIX (decisive):** the session-024 prototype divided the coin-age numerator by **CMC circulating supply**, producing shares >100% (RAD raw 148–398%) because CMC circulating EXCLUDES locked/treasury/vested tokens (the Entry-49 pattern) while the numerator counts all on-chain tokens. Corrected to divide by **on-chain supply** (sum of all live FIFO lots at the month-end block) — the supply whose age the channel measures — so the share is a proper fraction in [0,1]. Re-validated on RAD (clean single-deployment governance token; AAVE/CRV were swapped out — their 1M+ transfer histories are the same throughput wall as OP and did not complete): post-fix RAD screened HODL-6m 11–27% (latest 22%), bounded and economically sensible -> gate PASSED. - **ALL-MONTH contract screen:** eth_getCode is time-invariant (an address is a contract in every month), so one classification pass over the union of each month’s top >6m holders cleans the WHOLE series (fixing the prototype’s last-month-only screen). Raw share (incl. contracts) vs screened (EOA-only) both written; the screened share is the lambda input. **CEX custodial EOAs are NOT screened** (no free label feed) -> named residual limitation, cex_screened=False flag. - **METADATA-driven sizing (replaced hit-and-trial):** the first cut fetched each token until a per-token call cap, then deferred — wasting ~cap calls per giant just to discover it was too big. Replaced with a pre-pass (phase1_channel2_sizeprobe.py): Etherscan Pro tokenholdercount = 1 call/token, a monotone proxy for Transfer volume (MET 748 holders/79 calls; RAD 7,785/1,357; XAN 12,963/giant). All 793 sized (_channel2_sizes.csv): 113 <=1k holders, 165 <=2k, 215 <=3k, 295 <=5k, 404 <=10k. The builder now processes SMALLEST-FIRST, DEFERS BY METADATA any token > HOLDER_MAX (skipped without fetching), and caches month-end blocks once per chain. Deterministic and budget-predictable. - **Built:** with HOLDER_MAX=3000, **207 tokens completed** (197 with a screened HODL series; the

rest 0-transfer / pre-history-only), 5,064 asset-month rows. Data quality: **median screened HODL-6m = 40.6%**, 80.6% of token-months < 95%, only **2 of 197 assets degenerate** (~100% every month = dead/illiquid). The illiquidity caveat (a non-trading token shows HODL~100% = inactivity, not conviction) is real but token-specific, not systemic; handled by the assembler's cross-sectional z-score (std>0, n>=2 guard drops degenerate all~100% months). The ~586 tokens > 3,000 holders (incl. major DeFi/governance) were DEFERRED BY METADATA — the documented resumable worklist (a dedicated higher-cap run gives them ch2). **DATA-INTEGRITY bug found & fixed (the overnight-outage incident):** during an unattended overnight run the machine's network dropped intermittently; the engine's `api()` swallowed the DNS failure into an EMPTY result, and `fetch_capped` treated the empty `getLogs` range as "0 transfers here" -> **silently dropping a block range = a corrupted coin-age series**. Fixed: a panel-local `_robust_getlogs` that RAISES `NetworkError` on connection failure (vs returning empty), so a dropped connection ABORTS+retries the token instead of checkpointing a partial history; `month_block` no longer caches a failed (None) lookup (which would poison every later token for that month); `main()` has a network-failure backoff + circuit-breaker. All 65 possibly-affected overnight checkpoints were DELETED and re-fetched with the robust code; the clean re-run logged **0 network failures** for all 207 tokens. RAD (validated during a stable session) was kept. **Decision made:** Wire the screened HODL-6m series into lambda as channel `ch2_holding` (assembler z-score/equal-weight logic untouched — only a channel input added). Ship the 197-asset panel with the illiquidity caveat documented; do NOT silently apply a liquidity filter (flag it for downstream analysis instead). Defer the >3,000-holder tail as the resumable next-session worklist. **Rationale:** Numerators from real Transfer events (no guessing); the denominator fix makes the share interpretable; metadata sizing spends the Pro budget deterministically; the network-failure fix guarantees no token is ever checkpointed with a partial history. The honest output is "channel built and validated for the small/mid cross-section; the high-volume tail is measured-and-deferred, not open." **Downstream impact (re-check if this changes):** `ch2_holding` uses on-chain supply as denominator (NOT CMC circulating — do not "reconcile" the two). The illiquidity caveat means a downstream liquidity screen is advisable before treating small-token ch2 as conviction. The >3,000-holder tokens (incl. the large in-lambda governance/staking assets) lack ch2 -> that's why 3-channel asset-months are ~0; a dedicated tail run would lift depth. lambda counts folded into Entry 64.

Entry 64 — Session 025 close-out: lambda 2,699 -> 6,097 asset-months / 90 -> 288 assets (Channel-2 panel + P2-2 delegation)

Date: 2026-07-01 **Spec section affected:** 3 (lambda assembly — counts only; z-score/standardizability/equal-weight logic untouched). **Asset(s)/period affected:** `03_data/phase1/lambda_panel.csv` re-assembled with `ch2_holding` added and the P2-2 `ch3_delegation` additions. **lambda DELTA (this session): 2,699 -> 6,097 observed asset-months (+3,398); 90 -> 288 distinct assets (+198).** Sources: `ch2_holding` 3,413 asset-months / 197 assets (Entry 63) + P2-2 `ch3_delegation` +67 asset-months / +5 assets (Entry 62, AWE/CHEEL/FORM/LINEA/ZORA). Month range now 2016-12 -> 2026-05 (older tokens' Transfer histories reach further back). Channel coverage (asset-months): `ch2_holding` 3,331 single + 937 `ch1_staking` + 834 `ch3_voting` + 560 `ch3_delegation` + 354 (`ch1+ch3_voting`) + 67 (`ch2+ch3_delegation`) + 14 (`ch2+ch3_voting`). `ch2_holding` is standardizable in 114 monthly cross-sections (the most of any channel). By class: token 128, other 151, coin 9. **Depth (the kickoff's headline metric):** 2-channel asset-months **354 -> 435 (+81)**; but the 2-channel SHARE FELL 13.1% -> 7.1% because breadth tripled with single-channel ch2 assets. This is an honest tradeoff: Channel 2 massively WIDENED the panel (mostly new small/mid tokens) but added a 2nd channel to only 4 already-in-lambda assets — because the large in-lambda governance/staking tokens are >3,000 holders and were metadata-deferred. 3-channel asset-months remain ~0 (no token yet has `ch1+ch2+ch3`). Lifting depth is exactly what the deferred >3,000-holder tail run would do. **Decision made:** Accept the re-assembled panel. Channel 2 is no longer the lambda NaN column (Entry 24 CLOSED for the <=3,000-holder cross-section, measured-and-deferred for the tail). **Rationale:** Matches the kickoff (build ch2 for ALL free-chain tokens where Transfer history exists; report breadth + depth). Every number from real events; assembler logic unchanged. **Downstream impact:** Full account `03_data/SESSION025_PRO_BUILD_REPORT.md`. Budget: `sizeprobe` 793 calls + build ~15-20k `getLogs/getCode` + P2-2 delegation — all well under the 200k/day Pro quota; the binding constraint was per-token WALL-CLOCK (full Transfer history), not the daily quota. Resumable worklist: the ~586 tokens

> 3,000 holders (`_channel2_sizes.csv` + per-token checkpoints store raw events for `--recompute`). No additional paid subscriptions.

Entry 65 — Session 026 (Task A): OP on-chain delegation COMPLETED — block-windowed streaming + concurrent pooled-Session fetch resolves the session-025 deferral

Date: 2026-07-02 **Spec section affected:** 3.3 (governance) — the one carried-forward P2-2 item (Entry 62). **Asset(s)/period affected:** OP (cmc 11840, Optimism chain 10, ERC20Votes 0x4200..0042), role=**crosscheck** -> `03_data/phase1/channel3_onchain_delegation.csv` (47 months); builder `phase1_op_delegation.py` (NEW). NOT in lambda. **What the spec wanted:** complete OP's DelegateVotesChanged replay (verified firing in session 025 but killed at >80 min). **Root causes + fixes (two, layered):** (1) the session-025 hang was over-large getLogs ranges (block 0->153M) timing out on Optimism -> **block-windowing** (never span a timeout-inducing range) + **streaming replay** (fold each event into a per-delegate running balance, snapshot at month-ends, discard the raw log -> memory bounded by delegate count, not the 11.7M-event volume). (2) the serial resume from 68% then STALLED on an Optimism **ConnectionReset** 10054 storm — opening a fresh TCP/TLS connection per call, while hammering the very active recent-blocks region (810k events in the first 2M blocks), made the server drop connections (CPU burned ~4,000 s on reconnect churn). Fixed by reusing the ch2 engine's **pooled keep-alive Session + 8-way concurrency + raise-not-swallow retry** (a genuinely failed call RAISES so no block window is silently skipped): CPU fell to near-idle and the last 32% finished cleanly in ~59 min at ~7.6 calls/s. **Result:** **11,722,683 DelegateVotesChanged events / 245,439 delegates** (the “46,974” in Entry 62 was only the first 60 capped calls; the true history is ~250x larger). **Delegated/circulating: median 7.98%, range 3.54–13.05%** (final 76.2M OP delegated = 3.54%) — a substantial, REAL governance-activation signal, far above the ~0% low-activation P2-2 tokens. **Decision made:** role=**crosscheck** per the governance waterfall (Entry 57) — OP already has a Snapshot `ch3_voting` turnout series, so on-chain delegation is recorded for comparison but the assembler folds ONLY role==‘primary’ delegation into lambda. Verified: lambda unchanged at 6,021/282, OP `n_channels`=1 (its lambda membership is via `ch3_voting` alone). **Rationale:** mechanism verified (Entry 62); the windowed+streaming+concurrent fetch computes the exact stock series without a partial (wrong) replay or a silent-gap corruption. **Downstream impact:** the pooled-Session + concurrent + raise-not-swallow template is now the canonical fetch for any pathologically-large or flaky-RPC on-chain replay (`ch2` or `ch3`).

Entry 66 — Session 026 (Task B): Channel-2 tail BUILT via a validated STREAMING + CONCURRENT engine -> first 3-channel lambda assets; address-poisoning contamination caught & fixed

Date: 2026-07-02 **Spec section affected:** 3.2 (holding-duration channel); 3 (lambda depth). **Asset(s)/period affected:** the deferred >3,000-holder tokens that already have `ch1/ch3` -> `03_data/phase1/channel2_holding.csv`; new engine `phase1_channel2_stream.py`, panel builder `phase1_channel2_panel.py` extended (WORKLIST env, per-token value cap, contamination exclusion). **What the spec wanted (session-025 worklist):** add `ch2` to the large in-lambda governance/staking tokens (the `ch1+ch3` set), creating the panel's first 3-channel assets — the depth the session-025 breadth expansion lacked. **Efficiency discipline (user-steered):** classified each candidate FIRST by which channels it already has, and built `ch2` ONLY for the 3-channel creators (tokens with `ch1` AND `ch3`). An initial interleaved run hit ORBS (9k holders but a hidden giant, >2M transfers — `holder_count` under-counts transfer volume) for zero 3-channel value -> restricted to 3-channel creators, smallest-first, with a memory heartbeat to catch hidden giants. **The engine (the giant-token unlock):** the session-025 panel engine loads a token's ENTIRE Transfer history into RAM before replaying — fine to ~2M transfers but OOMs on the 100k+-holder giants (AAVE/CRV/SUSHI/GMX at 4–60M transfers) and is serial. `phase1_channel2_stream.py` fixes both, **validated BYTE-IDENTICAL to the panel engine before use:** (1) STREAMING FIFO replay (fold events into per-address lots as they arrive, snapshot at month-ends, discard raw events; empty-deque pruning bounds churn/dust giants — CVX's 2.8M transfers held in ~96k lots, not 2.8M events); (2) CONCURRENT windowed fetch (8 workers, global ~8 calls/s token bucket, block-ordered batches -> monotonic stream -> identical to batch replay; pooled thread-local Session eliminates the per-call TLS-handshake CPU that otherwise burned ~6 cores). Validation gates (0 diffs): `--validate` 2943 (streaming replay == `panel._replay`

on RPL, incl. with pruning + value cap) and `--checkfetch 7228` (full concurrent pipeline re-fetch of DDX == its serial checkpoint, transfers identical). **Built (10 tokens):** RPL, DDX (2-channel), FRAX, CVX, YFI, CRV, 1INCH, SUSHI, AAVE, GMX. Screened HODL-6m medians 10.7–54.5% — all economically bounded (governance/liquid tokens 10–17% match the RAD precedent; YFI/SUSHI ~53% higher but legitimate for their long-term holder bases; none degenerate). **DATA-INTEGRITY (address-poisoning phantom lots) — a two-layer fix (B2 finding).** A full-panel scan (reconstructed on-chain supply vs circulating) caught 11 tokens contaminated by fake huge-value Transfer logs that, replayed through FIFO, become phantom lots dominating the coin-age denominator (AAVE read up to 1.16e60 tokens vs real 16M; 10 small/dead tokens at 50–42,000x circ). A universal threshold cannot separate spam from legit high-supply meme tokens, and a per-event cap alone misses accumulated sub-cap spam, so the fix is two layers: (1) PER-EVENT value cap (skip Transfers > VAL_CAP_MULT=100 x max circulating — a real transfer cannot exceed supply; 100x spares heavily-locked tokens whose on-chain supply exceeds CMC circulating, the Entry-49 pattern); (2) PER-MONTH exclusion (null months where on-chain supply > CONTAM_MULT=100 x circulating — 100x not 50x so legit early heavy-lock months like CRV's launch month, on-chain 1.3B/circ 26M/total 3B, are RETAINED). Both re-validated byte-identical on RPL. 10 event-storing tokens fixed by `--recompute` (no re-fetch); AAVE re-fetched clean. Result: 0 residual contaminated rows; 6 spam-only dead tokens correctly dropped from lambda. **lambda DELTA:** 6,097 -> 6,021 asset-months / 288 -> 282 assets (the COUNT dips because the contamination fix nulled 6 spam-only dead tokens whose sole channel it was — this session's value is DEPTH + INTEGRITY, not breadth). $n_channels\ 1/2/3 = 5,356 / 333 / \mathbf{332}$ (was 5,662 / 435 / 0). **First 3-channel assets (9):** CVX(56), FRAX(52), 1INCH(41), AAVE(38, recent spam-months excluded), SUSHI(35), GMX(32), YFI(29), RPL(28), CRV(21) = **332 three-channel asset-months**, all `ch1_staking+ch2_holding+ch3_voting`. **2+ channel share 7.1% -> 11.0%.** **Decision made:** wire the streamed screened HODL-6m into lambda (assembler untouched — channel input only); the streaming engine is the canonical path for the high-volume tail; the value cap + contamination exclusion make the coin-age channel spam-robust. **Rationale:** every number from real Transfer events; the streaming engine is proven byte-identical to the validated engine, so the giants are built to the same standard as the small/mid cross-section; depth is created exactly where the session-025 report predicted (the `ch1+ch3` tokens). **Downstream impact (re-check if this changes):** `phase1_channel2_stream.py` makes the entire remaining >3,000-holder tail tractable. Streamed checkpoints store rows but not raw events (`streamed:True`) -> `--recompute` unavailable for them (rows are final). AAVE's spam-contaminated 2024-08->2026-05 `ch2` window is EXCLUDED (the one coverage gap); a per-token totalSupply value cap (vs the circ-multiple heuristic) would recover it — noted refinement.

Entry 67 — Session 026 close-out: lambda 6,097 -> 6,021 asset-months / 288 -> 282 assets; first 3-channel depth (0 -> 332)

Date: 2026-07-02 **Spec section affected:** 3 (lambda assembly — counts/depth only; z-score/equal-weight logic untouched). **lambda DELTA (this session):** 6,097 -> 6,021 observed asset-months (-76); 288 -> 282 assets (-6) — the COUNT dips because the address-poisoning contamination fix (Entry 66) nulled the `ch2` of 6 spam-only dead tokens whose sole channel it was; the contribution is DEPTH + INTEGRITY, not breadth. $n_channels\ 1/2/3 = 5,356 / 333 / \mathbf{332}$ (was 5,662 / 435 / 0). **3-channel asset-months 0 -> 332 across 9 assets** (CVX/FRAX/1INCH/AAVE/SUSHI/GMX/YFI/RPL/CRV, all `ch1_staking+ch2_holding+ch3_voting`) — the panel's FIRST 3-channel observations, exactly the tokens the session-025 report predicted the >3,000-holder tail would lift. **2+ channel share 7.1% -> 11.0%.** **Budget:** Channel-2 tail + AAVE re-fetches ~200k getLogs via the streaming+concurrent engine (~8 calls/s under the 10/s Pro ceiling, DAILY_CAP-gated, crossing the 00:00 UTC quota reset); OP delegation ~26k getLogs this session (full history 11.7M events). The binding constraint remained per-token wall-clock, now cut ~6x by concurrency; the streaming engine removed the memory wall (giants held in bounded lot state, not full event lists). No additional paid subscriptions. **Decision made:** accept the re-assembled panel. Channel-2 depth built for all 9 three-channel creators + DDX; OP delegation completed as a cross-check; the value-cap + contamination exclusion make coin-age spam-robust. **Rationale:** every number from real on-chain events; assembler z-score/equal-weight logic untouched (only channel inputs added); the governance waterfall (OP crosscheck) prevents double-counting. **Downstream impact:** Full account 03_data/SESSION026_TAIL_BUILD_REPORT.md; session log 06_documentation/ai_conversations/session_026_channel2_tail_and_op.md. Resumable

worklist for the next Cowork data-collection plan: (a) the ~506 >3,000-holder 2-channel-only / breadth tokens; (b) the 62 tail tokens that already have ch1/ch3 (would add depth); (c) a per-token totalSupply value cap to RECOVER AAVE's spam-excluded 2024-08->2026-05 ch2 window. All buildable via `phase1_channel2_stream.py`.

Entry 68 — Session 027 (Task A): TVL denominator expansion — 23 new verified DeFiLlama slugs (incl. all four missing 3-channel tokens) + reviewed other-class adds + chain-level TVL for L2 governance tokens

Date: 2026-07-03 **Spec section affected:** Phase 2 valuation-multiple denominator (NV/TVL; Entry 30/40 semantics UNCHANGED — TVL is a STOCK denominator, not a PQ substitute, not a lambda channel). **Asset(s)/period affected:** `03_data/phase2/tvl_panel.csv` rebuilt 4,999 -> 6,620 asset-months / 99 -> 130 distinct tokens (2018-09 -> 2026-05); `asset_onchain_identity.csv` +23 dl_slug entries; builder `phase2_build_tvl_panel.py` extended (OTHER_ADDS, CHAIN_LEVEL). **What was needed:** the five 3-channel assets (CRV/YFI/FRAX/GMX/RPL) had full lambda but no TVL -> could not enter the NV/TVL regression; 76 more 1-2-channel lambda tokens likewise. Four of the five had NO dl_slug. **Slug discovery method + the PARENT-protocol finding:** DeFiLlama /protocols cmcId lookup found ZERO of the four (and zero of the 207 no-slug non-coin lambda assets) — every cmcId-matchable asset was already harvested in earlier sessions; DL's cmcId field covers only 1,709/7,770 protocols. Fallback = exact symbol+name match (the session-019 LOOSE_ADDS precedent), individually verified. The decisive pattern: all four priority tokens live under DL PARENT protocols with per-version children (curve-dex/llamalend/crvusd under parent curve-finance, etc.). Since the token governs the WHOLE family, the parent slug is the correct denominator; verified `/protocol/{parent}` serves the aggregated `tvl[]` series with correct launch dates (curve-finance 2020-02, yearn 2020-02, frax-finance 2020-12, gmx 2021-09). **CRV -> curve-finance, YFI -> yearn, FRAX -> frax-finance, GMX -> gmx. 19 more name-verified slugs (same bar):** AERO->aerodrome, ALCX->alchemix, BAL->balancer, BLUR->blur, BNT->bancor, BONE->shibaswap, CAKE->pancakeswap, COMP->compound-finance, ENA->ethena, EUL->euler, FLUID->fluid, LQTY->liquity, ONDO->ondo-finance, STG->stargate-finance, T->threshold-network, UNI(7083 ONLY — cmc 4113 /“UNI COIN/” is a symbol collision, NOT Uniswap)->uniswap, VVS->vvs-finance, XVS->venus (parent slug; /“venus-finance/” 400s), and RPL already had rocket-pool. **Builder extensions (two, both cmc_id-keyed):** (1) OTHER_ADDS — the token-class filter excluded `asset_class==‘other’` lambda assets; 5 reviewed adds: RPL/rocket-pool, SSV/ssv-network, BLUR/blur, RAIN/rain, MV/gensokishi. (2) CHAIN_LEVEL — canonical L2 governance tokens whose DL protocol entry is a Foundation/treasury or an empty parent get CHAIN DeFi TVL via `/v2/historicalChainTvl/{chain}`, recorded as `dl_slug=‘chain:{name}’` so downstream can distinguish: ARB->Arbitrum, OP->‘OP Mainnet’ (DL renamed; ‘Optimism’ is a dead \$0 alias), MNT->Mantle, APE->ApeChain, BLAST->Blast. COINS deliberately excluded from TVL (they use NVT per the framework). **Rejected (logged so they are not re-tried):** ZORA (parent ‘zora’ 400s; only child with TVL is a canonical bridge — bridged custody is not protocol TVL for the token claim; identity slug REVERTED); GBYTE (oswap-amm is a third-party DEX on Obyte, not GBYTE's protocol); PENGU (not Abstract's fee token — no defensible TVL); CYBER (Cyber chain TVL ~\$0); NEST (two conflicting DL parents — ambiguous); POWER cmc 39042 (DL ‘m0’ symbol matches but identity unconfirmable); CVP (session-019 rejection re-affirmed: DL cmcId 7368 mismatch); WLFI (name-exact but no TVL series); MNT/OP Foundation slugs NOT fetched as protocol TVL (treasury holdings != protocol TVL). Genuinely-no-TVL protocols left NaN: ENS, GTC, COW, FORTH, ZRX, CHEEL, APE-protocol (apecoin), CYBER-protocol (cyberconnect). **Result / plausibility gate (A5):** all five priority tokens PASS (56-76 months, no post-launch near-zeros, USD magnitudes sane — CRV peak \$23.1B Jan-2022, YFI peak \$5.9B, RPL last \$1.1B). Lambda assets with TVL 49 -> 80 (of 282); lambda asset-months with same-month TVL 2,103 of 6,021; **3-channel asset-months with TVL: 331 of 332** (the one gap = SUSHI 2020-08, its launch month, before DL's sushiswap series begins — genuine early-launch, per the A5 rule). 0 fetch failures in the final rebuild. **Decision made:** ship the expanded panel; chain-level rows carry the ‘chain:’ prefix so any downstream analysis can include/exclude them explicitly; parent-slug aggregation is the canonical treatment for multi-version protocols. **Rationale:** every slug verified name+series before writing (cmc_id joins only); the parent-vs-child choice matches the token's whole-protocol governance claim; treasury/bridge/foundation TVL explicitly kept OUT of the denominator. **Downstream impact:** NV/TVL is now computable for all 9

three-channel assets (SUSHI missing only its launch month). The ~140 remaining lambda-no-TVL assets are mostly Snapshot-governance/dead/small tokens with no DL protocol entry — a name-match pass found no further defensible hits; do NOT loosen the match bar to chase them.

Entry 69 — Session 027 (Task B): Channel-2 tail COMPLETE for all 43 ch1-or-ch3 lambda tokens $\leq 500k$ holders — every one now multi-channel; 2+ channel share 11.0% -> 24.4%

Date: 2026-07-04 **Spec section affected:** 3.2 (holding-duration channel); 3 (lambda depth). **Asset(s)/period affected:** the 43 lambda assets with ch1 OR ch3_voting but NO ch2 and holder_count $\leq 500k$ (from universe_coverage_status.csv), built smallest-first into 03_data/phase1/channel2_holding.csv via the validated session-026 streaming engine (phase1_channel2_stream.py, unchanged — no code edits, thresholds untouched). **What was built:** ALL 43 COMPLETED, zero network aborts, zero errors: NFTX HAKKA IQ BZRX COW MC SSV OHM KP3R ORBS RGT SYN PNT STRK XAN LQTY ALCX PERP HFT TRU FARM CAKE GNS(Polygon) LINA API3 MNT BADGER BNT STG BAL REN ARB LDO ENS PENDLE GTC ENA LRC GRT APE ZRX COMP UNI. 60,100,476 Transfer events replayed; ch2 panel 217 -> 260 tokens / 5,663 -> 7,823 rows. Hidden giants absorbed in bounded memory (holder_count under-predicts transfer volume, the ORBS lesson): UNI 8.3M transfers, APE 5.9M, BNT 5.7M (2017-era history), GNS 3.5M, ENA/PENDLE 3.3M each. **Budget (two quota days, both compliant):** day 1 (2026-07-03 UTC) 141,005 getLogs — the engine cap-stopped after APE at DAILY_CAP=140k (in-process cap set BELOW the 180k stop-rule so a mid-flight giant cannot overshoot past it; APE finished at 141,005, day total ~141.1k, 59k headroom); day 2 (2026-07-04 UTC) 41,871 for ZRX/COMP/UNI after the 00:00 UTC reset. Total 182,876. The per-token checkpoint made the cap-stop a clean resume, exactly as designed. **B2 integrity scan (full 43):** reconstructed on-chain supply vs circulating — **0 months above the 100x contamination threshold**; worst ratio 31x (HAKKA), inside the legitimate Entry-49 heavy-lock band; 0 months nulled. The two-layer guard (VAL_CAP_MULT / CONTAM_MULT = 100, Entry 66) was NOT modified. **B4 sanity (>50k-holder tokens):** screened HODL-6m medians all economically bounded — REN 42.5%, LRC 44.0%, APE 47.9%, ZRX 51.3%, UNI 50.9%, GRT 37.2%, LDO 32.0%, ENA 32.7%, ENS 13.4%, ARB 22.1%, PENDLE 25.5%, GTC 24.0%, COMP 40.7%, BAL 19.6% — none degenerate. **XAN flag explained, not excluded:** median 0.0% is an AGE artifact (token launched mid-2025; no lot CAN be >6m old before 2026-03; from 2026-03 screened HODL is a real 74-77%, coherent with its NON-CUSTODIAL XanV1 lock — locked tokens never leave holders wallets, Entry 58 — so they age in place; note ch1 and ch2 therefore partially measure the same locked stock for XAN, inherent to the lambda design, same as staked CVX/FRAX aging in place). **lambda DELTA: 6,021 -> 7,051 observed asset-months (+1,030); assets unchanged at 282** (all 43 were already lambda members — this session is DEPTH, the +1,030 months are where ch2 extends beyond each token's ch1/ch3 window). n_channels 1/2/3 = **5,331 / 1,388 / 332** (was 5,356 / 333 / 332): 2-channel asset-months +1,055. **2+ channel share 11.0% -> 24.4%**. 3-channel unchanged (none of the 43 had BOTH ch1 and ch3 — confirmed, the priority list was ch1-OR-ch3 by construction). ch2_holding standardizable in 114 monthly cross-sections, the widest of any channel. **Decision made:** accept all 43 series into lambda (assembler z-score/equal-weight logic untouched — channel inputs only). HEX not attempted (permanent deferral, Entry 66). **Rationale:** the streaming engine was validated byte-identical in session 026; every series from real Transfer events; the guard thresholds proven on the 026 build were left alone; budget rules observed with headroom on both quota days. **Downstream impact:** the remaining ch2-less lambda assets are now ONLY: coins (9, ch2 requires non-EVM archive indexers — the Entry-21 wall), non-free-chain tokens, and the ~500 >3k-holder tokens that are NOT lambda members (breadth, not depth — lower priority). The 43 new 2-channel tokens all have TVL-or-slug coverage checked in Entry 68; NV/TVL + lambda regressions can now run on a 1,388-month 2-channel cross-section.

Entry 70 — Session 027 close-out: lambda 6,021 -> 7,051 asset-months / 282 assets; TVL panel 4,999 -> 6,620 asset-months / 99 -> 130 tokens; NV/TVL now computable for all nine 3-channel assets

Date: 2026-07-04 **Spec section affected:** 3 (lambda assembly — counts/depth only; z-score/equal-weight logic untouched) + Phase-2 TVL denominator. **Session totals:** Task A (Entry 68): TVL +1,621 asset-months / +31 tokens (23 new verified slugs incl. the four missing 3-channel tokens via DL PARENT protocols, 5 reviewed other-class adds, 5 chain-level L2 series); lambda assets with TVL 49 -> 80; 3-channel asset-months

with TVL 331/332 (gap = SUSHI launch month). Task B (Entry 69): all 43 ch2-tail tokens built (+1,030 lambda asset-months, 2-channel 333 -> 1,388, 2+ share 11.0% -> 24.4%), B2 clean, 182,876 getLogs across two compliant quota days. **Decision made:** accept both panels. Depth AND denominator coverage were this session's product: the 2-channel cross-section is now 4.2x larger and the 3-channel regression set is TVL-complete. **Rationale:** all numbers from live sources (DeFiLlama keyless; Etherscan Pro under quota with headroom); every join cmc_id-only; both engines unchanged from their validated forms. **Downstream impact / next priorities (session 028+):** (a) coin staking ch1 for AVAX/BNB/NEAR/INJ/SUI/APT etc. (non-EVM, separate research effort — the deferred scope note in the kickoff); (b) the ~500 >3k-holder non-lambda breadth tokens (ch2 single-channel adds); (c) AAVE spam-excluded 2024-08 -> 2026-05 ch2 window recovery via a per-token totalSupply value cap (Entry 66 refinement, still open); (d) PQ/NVT_GL expansion (still deferred); (e) the ~140 lambda assets with neither TVL nor a defensible DL protocol (mostly Snapshot-governance/dead tokens) stay TVL-NaN — do not loosen the name-match bar.

Entry 71 — Session 028 (Task A, EVM chains): coin Channel-1 BUILT for BNB, S, GLMR, MOVR, XDC via Etherscan Pro V2 + official archive RPCs; BERA documented gap (consensus-side staking invisible to EL logs)

Date: 2026-07-04 **Spec section affected:** 3.1 (staking/locking ratio, coins); Entry-26 cross-check bar applied to coins (reconstructed series must reproduce the chain's own on-chain staked total at ~0% drift). **Asset(s)/period affected:** BNB (cmc 1839) 2024-07->2026-05 (23 mo); S/Sonic (32684) 2025-01->2026-05 (17 mo); GLMR (6836) 2022-01->2026-04 (52 mo); MOVR (9285) 2021-09->2026-05 (57 mo); XDC (2634) 2020-06->2026-05 (72 mo); BERA (24647) NOT built. New builder `phase1_channel1_pos_coins_evm.py` -> `03_data/phase1/channel1_pos_coins_evm.csv` (picked up by the `channel1_*.csv` glob). Raw event/state caches under `03_data/raw/phase1_onchain/pos_coins_evm/` (stage scripts `_s028*_fetch.py`; shared helpers `_s028_evm.py`). **What was found and built (all response-body verified live this session):** - **BNB — BUILT, replay drift +0.000%.** BSC StakeHub `0x...2002` (chainid 56) has a verified ABI; events fire. TWO kickoff premises corrected live: (1) the event data order is (`shares`, `bnbAmount`) — `bnbAmount` is the SECOND data word for both Delegated and Undelegated, not the first; (2) the correct accounting is `cum(Delegated.bnbAmount) ? cum(Undelegated.bnbAmount) + cum(RewardDistributed.reward)` — rewards accrue into the pools daily without Delegated events, and `MigrateSuccess` (BC-fusion migration, 2,070 events 2024-04-18->2024-07-14) must NOT be added because migration also emits Delegated (adding it -> +80% drift; adding rewards -> EXACT match). Cross-check: replay head = 25,717,017 BNB vs live `sum(totalPooledBNB())` over all 53 validator StakeCredit contracts = same figure to the integer (0.000%; re-run at build time +0.0005%, minutes of drift). Redelegated is pool-neutral; zero ValidatorSlashed events. WINDOW: series starts 2024-07 (first month-end after BC-fusion completed 2024-07-14); 2017->2024-06 staking lived on the retired Beacon Chain and is NOT reconstructable from BSC logs — documented gap, not interpolated. 106k events total, one quota-trivial fetch. - **S (Sonic) — BUILT, replay drift +/-0.0000% at every probed block.** SFC `0xFC00FACE...0000` (chainid 146) is a proxy; implementation ABI verified via the EIP-1967 slot. KICKOFF PREMISE CORRECTED: the naive `Delegated ? Undelegated + RestakedRewards` replay drifts +0.86% AND GROWING — block-bisection (down to block 60,010,966) proved `restakeRewards()` emits BOTH Delegated AND RestakedRewards for the same amount (live totalStake moved by the amount ONCE). Correct replay = `cum(Delegated) ? cum(Undelegated)` alone: matches archive `totalStake()` at six blocks across the full history (2M->75M) at +/-0.0000% each, and the live head exactly. Metric = SFC totalStake (includes deactivated validators' still-locked stake; ~1.2% above totalActiveStake) — flagged. 246k events. - **GLMR / MOVR — BUILT via the staking pallet's own aggregate (state reads, not reconstruction).** The ParachainStaking precompile `0x...0800` emits NO EVM logs (0 in 50k blocks on both chains) and exposes NO aggregate getter (candidateCount/round answer; totalStake() etc. do not exist) — AND Etherscan's proxy `eth_call` IGNORES the historical block tag on every chain tested (round() identical at all tags), killing the kickoff's A2d fallback as written. RESOLUTION: the OFFICIAL public RPCs (`rpc.api.moonbeam.network` / `rpc.api.moonriver.moonbeam.network`) are full archives and answer Substrate JSON-RPC, so the series is `state_getStorage(twox128("ParachainStaking")+twox128("Total"), chain_getBlockHash(month_end_block))` — the chain's own total staked (collator bonds + ALL delegations incl. bottom), one call per month, keyless. Decode cross-check: pallet Total vs `eth_call`

$\text{sum}(\text{getCandidateTotalCounted over selectedCandidates}) = 1.0537 \text{ (GLMR)} / 1.0153 \text{ (MOVR)}$ — exactly the expected superset relation (counted excludes bottom delegations + non-selected candidates). Month-end blocks resolved once via Etherscan `getBlocknobytime` (chainids 1284/1285). - **XDC** — **BUILT via Etherscan Pro balancehistory (a Pro endpoint, live-confirmed on chainid 50)**. `XDCValidator 0x...0088` holds masternode stake as NATIVE XDC. Series = the contract's native balance at each month-end block. THREE-WAY cross-check ties out exactly: live ? `getCandidateCap` (550 candidates, 265 nonzero) 2.650B <= event replay (`Propose+Vote?Unvote?Withdraw`) 2.668B (+18.0M resigned-but-unwithdrawn, still locked in the 30-day delay) <= live balance 2.701B (+32,625,000 XDC of genesis/eventless stake predating the event stream — the replay tracks `balancehistory` at this CONSTANT offset at every probed block from 40M on, 0.000% co-movement). FLAG: the balance series includes the pending-withdrawal and genesis components (~1.9% above active caps today) — kept with the flag, the RPL/xSUSHI shared-vault standard. - **BERA** — **NOT BUILT (documented gap)**. `BeaconDeposit 0x4242...4242` (chainid 80094, verified ABI) logs deposits only; withdrawals are consensus-side balance credits with NO EVM logs (ETH2 model). Cumulative deposits = 382.6M BERA — far ABOVE circulating supply (~120M), proving heavy withdraw/redeposit cycling that EL logs cannot net out. No free historical validator-balance API found (routescan 404, berascan consensus-less, hub.berachain.com API is PoL/BGT-side). Gap requires a Berachain CL (beacon-kit) archive API — none public today. **Decision made:** Accept BNB/S/GLMR/MOVR/XDC into lambda Channel 1 (5 of the 6 kickoff EVM chains); BERA logged as a gap. Every accepted series passed the Entry-26 bar at ~0% (BNB, S exact; GLMR/MOVR/XDC are state reads of the chain's own figures with decode/identity cross-checks). **Rationale:** spec ?0 verify-live discipline caught three kickoff-premise errors (StakeHub data order, StakeHub reward/migrate accounting, Sonic RestakedRewards double-emission) that would each have shipped a wrong series; the corrections are event-bisection-proven, not inferred. **Downstream impact:** the sub-channel remains `ch1_staking` (same z-score cross-section). BNB pre-2024-07 and BERA are the two documented gaps of this entry. The `_s028_evm.py` helper (keccak topics via pycryptodome, capped `getLogs` walker) and the archive-RPC pallet-storage pattern generalize to other parachains (e.g. future ASTR) — reuse before building new machinery.

Entry 72 — Session 028 (Task A2/A7): DOT/KSM still key-gated (deferred, no signup possible headless); CELO BUILT 2020-07->2026-05 via Forno archive `eth_call getTotalLockedGold()` — closes the Entry-46 gap for \$0

Date: 2026-07-04 **Spec section affected:** 3.1 (coins); Entry-46 (CELO documented gap, now closed); Entry-44 (DOT/KSM gate, unchanged). **Asset(s)/period affected:** CELO (cmc 5567) 2020-07->2026-05, 70 asset-months, BUILT into `channel1_pos_coins_evm.csv`; DOT (6636)/KSM (5034) not built. **What was found (live):** - **DOT/KSM** — **deferred**. No “subscan” key in `.api_keys.json`; the `pro.subscan.io` free-signup verification-code failure (kickoff note) stands. Per Entry 44 the build is otherwise ready (era_stat endpoint, paginate, cross-check). **MOAZZAM ACTION:** retry the free signup at `pro.subscan.io`, drop the key into `04_code/.api_keys.json` under “subscan”, and DOT/KSM become a short extension of `phase1_channel1_pos_coins_bucket2.py`. No Pro purchase. - **CELO** — **the kickoff's balancehistory route WORKS but only post-migration; the better route is Forno**. Etherscan Pro `balancehistory` on chainid 42220 answers ONLY for blocks >= ~31.06M (the 2025-03-26 L2 migration boundary; every pre-migration block returns NOTOK) and returns the RAW contract balance (includes pending withdrawals; live balance 86.0M vs `getTotalLockedGold` 78.1M = +9.2%, too dirty for the Entry-26 bar). BUT the official Forno RPC (`forno.celo.org`, keyless; mirrored by `celo.drpc.org`) serves the FULL archive INCLUDING pre-migration L1 state, so `eth_call LockedGold.getTotalLockedGold()` — the exact clean number Entry 46 identified — is readable at every historical month-end: block 10M (2021) -> 316.6M, block 20M -> 307.4M, block 32M -> 151.0M, latest -> 78.1M (Entry 46 measured 82.43M on 2026-06-26; consistent declining trend). Series built from the chain's own aggregate getter; month-end blocks via Etherscan `getBlocknobytime` (works across the migration boundary). **Decision made:** CELO accepted into lambda Channel 1 (state read of the chain's own getter; excludes pending withdrawals — cleaner than the balance-based alternative). DOT/KSM logged as deferred with the exact Moazzam action. **Rationale:** Entry 46's conclusion (“only fix is a paid balance-history endpoint or archive `eth_call` node”) was half right — the archive `eth_call` node turned out to be Celo's own free public endpoint, which nobody had probed for PRE-migration state before. Flag-don't-guess preserved: early months where circulating supply was tiny show `locking_ratio>1` (max 272%

in 2020-08); kept un-capped and flagged (SOL/AERO precedent). **Downstream impact:** Entry 46's gap classification is superseded by this entry. If Forno ever prunes pre-migration state, the raw month-end values are cached in 03_data/raw/phase1_onchain/pos_coins_evm/celo_lockedgold_history.json.

Entry 73 — Session 028 (Task A8-A10): AVAX BUILT keyless via Ava Labs' official Metrics API (resolves the Entry-42/47 pricing ambiguity: it is FREE, no key at all); NEAR and Cosmos (ATOM/INJ/SEI/KAVA) remain gaps with their gates re-verified

Date: 2026-07-04 **Spec section affected:** 3.1 (coins); Entry-42/47 gate map updates. **Asset(s)/period affected:** AVAX (cmc 5805) 2020-11->2026-05, 67 asset-months, BUILT; ATOM/INJ/SEI/KAVA/NEAR not built. **What was found (live, response-body verified):** - **AVAX — BUILT.** GET <https://metrics.avax.network/v2/networks/mainnet/metrics/validatorWeight> (and `delegatorWeight`) answers with NO key, NO signup: daily values in nAVAX back to network genesis (2,112 rows to 2020-09), paginated via `nextPageToken`. The kickoff's endpoint guesses (`glacier-api.../chains/mainnet/metrics/staking;/v2/network/staking/historic`) both 404 — the working shape is `/v2/networks/mainnet/metrics/{metric}`. SEMANTICS VERIFIED against the chain itself: P-Chain `platform.getTotalStake` (live, keyless) = 199.87M AVAX ~ `validatorWeight+delegatorWeight` (196.7M at yesterday's snapshot; stake currently moves ~2M/day) and NOT `validatorWeight` alone (159.9M) — i.e. the weights are ADDITIVE and total staked = `validatorWeight + delegatorWeight`. Month-end sample of the daily series; cross-check drift ?1.6% explained entirely by the snapshot-vs-live timing (gate set at 5%). FLAG: early months show `staking_ratio?1` vs CMC circulating (2020-09: 212M staked vs ~18M circulating — genesis staking of vesting-locked tokens CMC excludes); kept un-capped and flagged, the SOL/AERO precedent. This RESOLVES Entry-42/47's "AvaCloud pricing ambiguous, needs signup-flow check": the network staking metrics require no account whatsoever. - **NEAR — still a gap.** `NearBlocks: /v1/stats` and `/v1/validators` confirmed current-state-only (total-Stake present, no history); `api3.nearblocks.io/v1/charts` DOES serve 2,174 days of daily history back to 2020-07 but its field set (`price/supply/txns/accounts/gas`) contains NO staking figure. Pikespeak remains key-gated, pricing undisclosed. Gate unchanged from Entry 47. - **Cosmos (ATOM/INJ/SEI/KAVA) — still gaps.** `StakingRewards GraphQL (api.stakingrewards.com/public/query)` -> HTTP 401 "you should be authenticated" with no anonymous path (their key requires a commercial signup; not self-serve-free-confirmed). `Numia (data.numia.xyz)` -> HTTP 401 on every path (key required; docs route through BigQuery/Google-account territory, outside a headless session). Mintsan stays contact-sales (Entry 47). No signup attempted per the session rules. **Decision made:** AVAX accepted into lambda Channel 1 (official first-party historical series, same source-of-record treatment as ADA/Koios and XTZ/TzKT in Entry 41, PLUS a live P-Chain cross-check those never had). NEAR/Cosmos gates re-verified and left open; no purchase, no signup. **Rationale:** the Metrics API is Ava Labs' own (it powers the official explorer graphs — Entry 42's strongest-type source), now response-body verified keyless with semantics anchored to the P-Chain's own getter. **Downstream impact:** AVAX raw daily weights cached (`avax_validatorWeight.json / avax_delegatorWeight.json`). If `metrics.avax.network` ever gates, the cache preserves 2020-09->2026-07. NEAR/Cosmos remain the two structural non-EVM holdouts of the 49-coin target list, joining HBAR/SUI (engineering, Entry 45), ALGO (structural, Entry 42), EOS/ICP/APT (no source, Entry 47).

Entry 74 — Session 028 (Task B): AAVE ch2 spam window RECOVERED — the real poisoning vector was fake-value SELF-transfers; fix = engine-wide self-transfer skip (accounting identity) + per-token totalSupply value cap; all 22 nulled months restored, AAVE now 3-channel through 2026-05

Date: 2026-07-04 **Spec section affected:** 3 (Channel 2 coin-age engine); Entry 66 (the AAVE exclusion + its per-token-totalSupply refinement note, now executed); Entry 63 (denominator conventions unchanged). **Asset(s)/period affected:** AAVE (cmc 7278) ch2, all 67 months 2020-11->2026-05; engines `phase1_channel2_stream.py / _panel.py / _holding.py` (mirrored fix); no other token's shipped values change. **What was found:** the kickoff's premise (phantom lots only pollute the DENOMINATOR; swap in real totalSupply and recompute) was mechanically incomplete — phantom lots sit at EOA-class addresses and AGE past 6m, so they poison the NUMERATOR too (the first rebuild with only a tighter cap recovered the months but showed a fake HODL jump 17.8%->48.9% at exactly 2025-01 = the 2024-07 spam wave crossing the

6-month age line, and reconstructed supply still climbed 16M->62M). Root cause isolated by fetching the largest transfers in the sharpest jump month (2024-07): address 0x3d16ee6d46edb674e728b5923e2ecac4092f5920 (live AAVE balance: **0.0**) emits **Transfer events with from == to** on the REAL AAVE contract carrying fabricated values — max-uint256 (= Entry 66's 1.16e60 reading exactly), 8.0e16, 1.0e11, and sub-cap values like a fake 10,000,000 that pass ANY value cap. The FIFO engines replayed self-transfers as pop-then-append: the pop under-fills (no real lots) while the append credits the full fake value -> phantom supply at any cap. ALSO corrected: the kickoff's AAVE token address (?DDaE8) was wrong — the identity map's 0x7Fc66500c84A76Ad7e9c93437bFc5Ac33E2DDaE9 is canonical (caught because tokensupply returned 0). **The fix (two layers, in priority order):** 1. **Self-transfer skip** (if frm == to: skip) in all three engine variants. This is an ACCOUNTING IDENTITY — a self-transfer never changes any balance — not a threshold: processing one as pop+append can only (a) mint phantom supply when the value exceeds the address's live lots, and (b) wrongly refresh a real holder's lot age. Applies engine-wide going forward. 2. **PER_TOKEN_VAL_CAP = {7278: 16_000_000}** in the stream engine (Entry 66's refinement as written): an ABSOLUTE per-event cap at the token's constant on-chain totalSupply (verified live: eth_call totalSupply() == 16,000,000.0 exactly). Defense-in-depth for AAVE; the global VAL_CAP_MULT / CONTAM_MULT thresholds are UNTOUCHED, per the kickoff rule. **Verification (all passed):** - Full re-fetch + replay (5,564,724 transfers, 16.6k getLogs): reconstructed on-chain supply now 16.00M->17.97M across all 67 months (real totalSupply 16.0M constant; the residual <=+12% is ordinary sub-cap churn/Entry-49-band, far under CONTAM_MULT) — vs 1.02e9 (Entry 66) and 62M (cap-only rebuild). - **All 22 nulled months (2024-08->2026-05) recovered**; screened HODL-6m in the recovered window ranges **20.0%–31.3%** — inside the kickoff's economic-sanity gate (0.5%–80%, not 100%), continuous with the pre-window level, and the fake 2025-01 jump is gone. - Pre-spam months (<2024-04, 41 months): max |? screened HODL| = 0.000019 vs the Entry-66 stored rows — the fix is a no-op where no spam existed (the kickoff (d) cross-check). - **Panel-wide diagnostic** (_s028_selftransfer_diag.py, offline, zero network): re-replayed ALL 210 stored-event checkpoints with the fixed engine and diffed against their stored rows — **210/210 unchanged** (no month moves >1pp, no null flips). AAVE was uniquely poisoned; no panel-wide recompute is warranted. (The 50 streamed rows-only checkpoints cannot be offline-diffed; their B2 supply-vs-circ scans were already clean in sessions 026/027.) - Old checkpoint preserved as 7278_AAVE.entry66.bak. **Decision made:** accept the recovered AAVE ch2 series into lambda. AAVE is now 3-channel through 2026-05: 67/67 ch2 months, 60 three-channel asset-months; the panel's 3-channel total rises 332 -> 354 (+22 = exactly the recovered window). **Rationale:** flag-don't-guess executed both ways — the kickoff's denominator-swap reading was REJECTED on evidence (it would have shipped a numerator still carrying ~9M phantom aged AAVE), and the engine fix chosen is provably identity-preserving (210-token diff + 41-month AAVE no-op check), not a tuned threshold. **Downstream impact:** any FUTURE ch2 build/rebuild automatically gets the self-transfer skip; streamed tokens rebuilt later will silently drop this spam class. If another token ever needs a totalSupply cap, add it to PER_TOKEN_VAL_CAP with a live eth_call verification — do not loosen the rule that the global thresholds stay fixed. ORBS-class lesson recorded: fake-value SELF-transfers are a poisoning vector that value caps cannot stop — the identity skip is the structural defense.

Entry 75 — Session 028 close-out: lambda 7,051 -> 7,409 asset-months / 282 -> 289 assets; regression-ready coins (lambda?NVT_GL) 5 -> 12; AAVE 3-channel complete; 3-channel asset-months 332 -> 354

Date: 2026-07-04 **Spec section affected:** 3 (lambda assembly — counts only; z-score/equal-weight logic untouched; the assembler was re-run unmodified, new inputs picked up by the existing channel1_*.csv glob). **Session totals:** Task A (+358 coin ch1 asset-months / +7 coin assets): BNB 23, S 17, GLMR 52, MOVR 57, XDC 72, CELO 70, AVAX 67 — every series Entry-26-cross-checked at ~0% against the chain's own figure (Entries 71–73). Task B: AAVE's 22 spam-nulled ch2 months recovered via the self-transfer identity fix (Entry 74). **lambda 7,051 -> 7,409 observed asset-months / 282 -> 289 assets (coins 9 -> 16); coin lambda?NVT_GL 5 -> 12 assets / 505 overlapping coin asset-months (ADA, AVAX, BNB, CELO, ETH, GLMR, GNO, MOVR, S, SOL, TRX, XDC); 3-channel asset-months 332 -> 354; 2+ channel share 23.2%** (dip from 24.4% is compositional: +358 single-channel coin months entered the denominator). **Quota:** single Etherscan Pro day, ~45k calls total (two full AAVE re-fetches 33.1k; coin event fetches + balancehistory + cross-checks ~12k) —

under 200k/day with wide margin. Keyless sources: official Moonbeam/Moonriver RPCs (Substrate state), forno.celo.org, metrics.avax.network, api.avax.network P-Chain, rpc.soniclabs.com (diagnosis only). No signups, no purchases. **Open items for session 029+:** (a) DOT/KSM — the ONE actionable gate: free Subscan key (pro.subscan.io signup -> .api_keys.json under “subscan” -> Entry-44 era_stat build); (b) NEAR/Cosmos/BERA/HBAR/SUI/ALGO/EOS/ICP/APT gates documented in Entries 71/73 (no self-serve free path); (c) BNB pre-2024-07 Beacon-Chain era = permanent documented gap; (d) ~500 >3k-holder non-lambda breadth tokens (ch2) still deferred; (e) PQ/NVT_GL expansion still deferred; (f) coin PQ for the new lambda coins (S/GLMR/MOVR/XDC already have NVT_GL by construction of the target list — nothing to do there). Report: 03_data/SESSION028_COIN_STAKING_REPORT.md.

Entry 76 — Session 029 (Task A): Channel-2 BREADTH build — 40 of 47 non-lambda TVL-covered tokens built (36+4 across two quota days); 39 enter lambda as new assets; engine extended to BSC/Base; 7-token resume list pending

Date: 2026-07-05 **Spec section affected:** 3.2 (holding-duration channel); 3 (lambda breadth — first deliberate BREADTH session since 025). **Asset(s)/period affected:** the 47 universe tokens with confirmed TVL but ZERO lambda channels (kickoff list). **BUILT 40:** all 35 Ethereum-mainnet targets + MDX/EPS/BAKE/BTCST/MBOX (BSC). **channel2_holding.csv** 260 -> 300 tokens / 9,794 rows. **A0 multi-chain adaptation:** phase1_channel2_stream.py was ALREADY per-token multi-chain (every call passes the token’s chainid); the only constraint was the CHAIN_ID chain-name lookup in load_worklist -> extended with BSC(56)/Base(8453), nothing else touched (VAL_CAP_MULT/CONTAM_MULT=100 unchanged). **getLogs** coverage probed live: 56/137/42161/8453 all answer (heads 108.06M/89.65M/480.38M/48.20M). **Results:** 39/40 built targets entered lambda (+1,849 asset-months, all single-channel ch2). **WARP** (cmc 1166) built but EMPTY — its observed universe window (2016..2018-02) predates the mapped contract’s first Transfer entirely (every month “pre-history”) = an identity-map mismatch between the old dead cmc-1166 listing and the later-deployed 0x83e6f1E4... contract; flagged for identity review, NOT forced. Hidden giants absorbed in bounded memory (the ORBS lesson again, now at record scale): MDX 17.67M transfers (largest ch2 build ever, 135k getLogs alone), MBOX 21.2M transfers (172k getLogs), BAKE 9.0M, EPS 2.9M, FLOKI 1.67M, ATH 1.34M, AMP 1.17M. **B2 integrity scan (all 40):** 0 months above the 100x contamination threshold; worst AURORA 49.5x (legit Entry-49 heavy-lock band; treasury-locked supply vs small circ). B4: screened HODL-6m medians bounded — BMX 96.2%/MVL 84.5% are the documented illiquid-inactivity band (Entry 63 liquidity-screen advisory applies); EPIC median 0% = XAN-class age artifact. **Budget — two quota days + a QUOTA ANOMALY logged honestly:** day-1 process 186,942 getLogs (mainnet 51,109 + MDX 135,833; cap-stopped cleanly before EPS at DAILY_CAP=180k); day-2 process 281,671 (EPS 25.8k + BAKE 74.6k + BTCST 9.4k + MBOX 171.9k; cap-stopped before SFUND). MBOX alone blew through the in-process cap mid-flight — the cap only checks at TOKEN boundaries (the session-027 landmine, now demonstrated at 172k/token scale: pre-size giants before launching; holder_count does NOT predict transfer volume). The API NEVER returned a daily-limit rejection at ~292k calls on the 2026-07-05 UTC day — the assumed 200k/day hard cap did not bind (plan evidently credit-based/higher); we STOPPED anyway per the documented budget rule rather than compound the overshoot. **Pending (resume list, zero-loss from checkpoints):** SFUND(8972), MYX(36410), ADF(24796, Polygon), AVNT(38299)/KAITO(35763)/VVV(35509) (Base), RAIN(38341, Arbitrum, ~14k est — build last). Skipped by construction (non-EVM): SXP/RUNE/OSMO/SUN/CASINO; HEX permanently deferred (Entry 66). **Decision made:** accept all 40 series (39 lambda entrants + WARP flagged); resume the 7 next quota day. **Rationale:** engine unchanged from its validated form beyond the two-line chain map; B2/B4 clean; every target was pre-screened to have TVL so each new lambda asset is regression-ready the moment it lands. **Downstream impact:** lambda tokens/other with same-month TVL 80 -> 118 assets (the kickoff’s “78” figure was from the stale pre-027 coverage file). See Entry 79 for panel totals.

Entry 77 — Session 029 (Task B): all six orphaned dl_slugs REJECTED — DeFiLlama serves metadata but zero TVL data for COW/ZRX/FORTH/ENS/GTC/CHEEL

Date: 2026-07-05 **Spec section affected:** Phase 2 NV/TVL denominator (no change shipped). **What was found (live, keyless):** /protocol/{slug} for cowswap, 0x-aggregator, forth-dao, ens, gitcoin, cheelee

each answer HTTP 200 with an EMPTY top-level tvl[], EMPTY chainTvls sub-series on every listed chain, and empty currentChainTvls — DL lists the protocols but tracks no TVL. COW double-checked against the full /protocols dump + parentProtocols: CoWSwap (cmcId 19269) carries tvl null (category “DEX Aggregator” — batch-auction settlement holds no persistent liquidity), and the ONLY CoW-named TVL series (balancer-cow-amm, \$391k) belongs to parent BALANCER — not the COW token’s protocol claim. **Decision made:** REJECT all six; tvl_panel.csv and asset_onchain_identity.csv unchanged; no rebuild. GBYTE/PENGU/CYBER not re-attempted per Entry 68. **Rationale/impact:** confirms Entry 68’s “genuinely-no-TVL” classification with response-body evidence. Do NOT re-probe these six; an empty-series 200 is a verdict, not an API failure.

Entry 78 — Session 029 (Task C): coin Channel-1 BUILT for POL/MATIC, RON, KAIA(KLAY), FLR, EGLD, STRK, XRD, PEAQ — 8 series / 9 lambda assets, +322 asset-months, every one a state read of the chain’s own aggregate; 10 gaps/gates documented

Date: 2026-07-05 **Spec section affected:** 3.1 (coins); Entry-26 bar applied throughout. **Asset(s)/period affected:** MATIC(3890) 50 mo 2020-06->2024-08 + POL(28321) 21 mo 2024-09->2026-05 (listing handoff at the Polygon-2.0 boundary — one physical stake never enters lambda twice); RON(14101) 39 mo; KLAY(4256) 43 mo (full observed window); FLR(7950) 32 mo; EGLD(6892) 68 mo; STRK(22691) 19 mo; XRD(11948) 32 mo; PEAQ(14588) 18 mo. Builders: pol_series() added to phase1_channel1_pos_coins_evm.py (429 asset-months/9 assets); NEW phase1_channel1_pos_coins_native.py -> channel1_pos_coins_native.csv (251/7). Fetch stages _s029_pol_fetch.py / _s029_coin_fetch.py / _s029_coin_fetch2.py; raw caches under pos_coins_evm/ (gitignored). **Sources + kickoff corrections (all response-body verified):** - **POL/MATIC:** StakeManager 0x5e3Ef2...D908 on ETHEREUM mainnet; the aggregate is currentValidatorSetTotalStake() (3.58B live) — the kickoff’s totalStaked() is validator SELF-stake only (11.7M). Etherscan proxy ignores historical tags (Entry 71), and of the free RPCs only **eth.drpc.org serves full mainnet archive eth_call KEYLESS** (publicnode gates archive behind a personal token; llamarp/flashbots refuse). Cross-checks: drpc==etherscan at head +0.00000%; token balanceOf(StakeManager)/stake in [1.018, 1.341] every month (superset floor >=1 is the integrity condition; the 1.34 peak is the Q1-2021 delegation-churn buffer, reverting <=1.13 after). - **RON:** official api.roninchain.com RPC is PRUNED; **ronin.drpc.org is a keyless full archive**. Balance route REJECTED (+15.6% over stake at head = pending undelegations incl. revoked candidates); metric = sum(Staking.getManyStakingTotals(ValidatorSet.getValidatorCandidates(b,b)) — contract identities verified against axieinfinity/ronin-dpos-contracts deployments/ronin-mainnet. Balance/sum recorded monthly as superset check [1.000, 1.180]. The 2026 decline (235M->135M) is a genuine unstaking wave (live July figure 105.8M confirms the trend). - **KAIA/KLAY:** public-en.node.kaia.io returns “execution reverted” for historical calls, but the OFFICIAL **archive-en.node.kaia.io** answers klay_getStakingInfo(block) at any height — the node’s OWN consensus staking snapshot (units: KAIA). Cross-check: council-StakingAmounts == CnStaking contract native balances at 0.0000% (8/43 spot-checked). - **FLR:** P-chain platform.getTotalStake answers live but has no history; the build reads **PChainStakeMirror.totalSupply()** (address resolved from the FlareContractRegistry) via flare-api.flare.network archive eth_call. Cross-check: mirror vs P-chain live -0.21%. - **EGLD:** tools.multiversx.com/growth-api staking-metrics serves 2,130 DAILY points back to 2020-07-30 keyless; month-end sample; chart head vs live /economics staked -0.10%. (AVAX/Koios/TzKT source-of-record treatment.) - **STRK:** the kickoff’s contract 0x04718f... is the STRK TOKEN and 0x00ca1705... the MINTING CURVE (both ABI-verified); the real staking contract is **0x00ca1702e64c81d9a07b86bd2c540188d92a2c73cf5cc0e508d949015e7e84a7** (docs.starknet.io chain-info). blastapi retired; **rpc.starknet.java.build is a keyless archive** for starknet_call.get_total_stake() at month-end blocks (binary-searched); series from 2024-11 launch (short series kept per kickoff). STRK-only stake (v2 BTC staking power excluded). - **XRD:** the Radix Babylon Gateway (mainnet.radixdlt.com, official keyless) accepts **historical at_ledger_state timestamps** on /state/validators/list -> sum(stake_vault.balance) over all validators (single page, 245-287 items; fetch raises if it ever paginates). Babylon era only — Olympia (2021-07..2023-09) is a documented gap (old gateway retired). - **PEAQ:** peaq’s staking pallet is a **KILT fork** — storage item ParachainStaking.TotalCollatorStake {collators,delegators}; Moonbeam’s .Total is null on peaq (probed). peaq.api.onfinality.io/public is a keyless archive. Early ratio>1 (max 259%) = genesis/vesting stake, SOL/AERO/AVAX flag. - **Etherscan V2**

chainlist probed: NONE of Flare(14)/Ronin(2020)/Kaia(8217)/Cronos(25)/Chiliz(88888)/Lisk(1135) are among the 64 covered chains — all Task-C1 non-mainnet coins used native/third-party archives. **Session-028 regression fixed in passing:** the XDC cross-check compared the STATIC event-cache replay to the LIVE balance; a net -10M outflow on 2026-07-04 moved the apparent offset 32.6M->22.6M within hours and tripped the gate. Both sides now pinned at the cache's scan_to block (Pro balancehistory) — offset back to 32,625,000 EXACTLY, identity now time-invariant. **GAPS/GATES (verdict + exact action):** CRO = Cosmos-side (LCD live-only; ATOM gate, same key sourcing); **CHZ = GATE-OPEN:** rpc.chiliz.com is its own keyless archive and system contract 0x...1000 holds 2.38B CHZ (243M@5M->2.38B head, plausible total-staked trajectory) but the contract is unverified with no aggregate getter and unknown event schema — **MOAZZAM ACTION:** read total-staked off staking.chiliz.com at a known time and compare to the 0x...1000 balance; if it ties, the build is a ~30-line balance fetch (XDC pattern); LSK = no locatable staking contract (docs 404, Blockscout search empty); TON = Elector balance is a superset (stakes+credits) and live-only; FLOW = spork-bound history; DFI = ocean stats current-only (chain sunset); DASH = no free masternode-count history (stats.masternode.me dead, chainz unsupported, dashcentral current-only 2,062 MNs); WAN = explorer HTML-only; HYPE = validatorSummaries live-only (~438M staked incl. foundation); **CORE = GATE-OPEN (key):** openapi.coredao.org 401s everywhere — **MOAZZAM ACTION:** check scan.coredao.org for a free self-serve key -> .api_keys.json under “coredao” -> probe /api/stats/staking_summary. No signups attempted (session rule). **Decision made:** accept all 8 series into lambda ch1_staking. Every accepted series is the chain's own figure (state read) with an Entry-26 identity/cross-check; nothing reconstructed-and-unanchored was shipped. **Rationale:** the GLMR/MOVR/CELO state-read standard scaled to 5 more archive sources (drpc mainnet+ronin, kaia archive-en, flare-api, lava starknet) + 2 official first-party APIs (multiversx growth, radix gateway) + 1 pallet fork (peaq). **Downstream impact:** coin lambda?NVT_GL 12 -> 20 assets / 505 -> 645 coin-months (new: POL RON KLAY FLR EGLD STRK XRD PEAQ; MATIC adds lambda only, no NVT_GL series). The archive-RPC map (drpc/lava/onfinality/archive-en) generalizes — probe drpc FIRST for any future EVM-chain history need.

Entry 79 — Session 029 close-out: lambda 7,409 -> 9,580 asset-months / 289 -> 337 assets; regression-ready 92 -> 138 (coins 12->20, tokens/other 80->118); coverage CSV regenerated with a reusable builder

Date: 2026-07-05 **Spec section affected:** 3 (lambda assembly — counts only; assembler re-run unmodified, new inputs picked up by the channell_*.csv glob and the ch2 aggregate). **Session totals:** Task A (Entry 76) +1,849 asset-months / +39 assets (ch2 breadth, all TVL-covered); Task C (Entry 78) +322 asset-months / +9 coin assets. **lambda 7,409 -> 9,580 observed asset-months / 289 -> 337 assets (coins 16 -> 25); n_channels 1/2/3 = 7,860 / 1,366 / 354; 2+ share 23.2% -> 18.0%** (compositional: +2,171 single-channel months entered the denominator; the 2-ch count's 1,388->1,366 shift is the Entry-74 AAVE months having moved 2ch->3ch in session 028). **Regression-ready: coins (lambda?NVT_GL) 20 assets / 645 months + tokens/other (lambda?TVL) 118 assets / 4,194 months = 138 assets** (pre-session true baseline 92 = 12 + 80; the kickoff's “90” used the stale coverage file). **Coverage CSV:** universe_coverage_status.csv regenerated by NEW reusable 04_code/build_coverage_status.py (replaces the pre-027 inline generation; coin_staking_type carried forward as static metadata; “complete” = SAME-MONTH overlap semantics: pow_only coins on NVT alone, pos coins on lambda?NVT, tokens/other on lambda?TVL): 149 complete / 236 partial / 1,554 not_started. **Quota:** getLogs 186,942 (day-1 process) + 281,671 (day-2 process) + ~1k coin-probe calls; keyless sources did the entire Task C (drpc, lava, onfinality, archive-en.node.kaia.io, flare-api, multiversx growth-api, radix gateway). Quota anomaly documented in Entry 76. **Open items for session 030+:** (a) the 7-token Task-A resume list (Entry 76 — one clean quota day, ~30-45k calls); (b) DOT/KSM Subscan key (unchanged, Entry 72); (c) CHZ manual anchor + CORE key (Entry 78 actions); (d) WARP cmc-1166 identity-map mismatch review; (e) breadth ch2 for the ~500 >3k-holder non-lambda tokens WITHOUT TVL (lower priority than this session's TVL-covered set was); (f) PQ/NVT_GL expansion still deferred; (g) MATIC has lambda but no NVT_GL — a bitinfocharts/native-series probe could make it the 21st regression-ready coin. Report: 03_data/SESSION029_BREADTH_AND_COIN_PROBE_REPORT.md.

Entry 80 — Session 030: Task-A resume cut short at user departure — SFUND BUILT (hidden giant); MYX ABORTED mid-build (mega-giant, no checkpoint); 5 tokens untouched

Date: 2026-07-05 **Spec section affected:** 3 (lambda assembly — counts only; no engine or threshold changes). **What happened:** resume of session 029's Task A (7-token WORKLIST, Entry 76 estimated ~30-45k calls). The estimates proved wrong in the same direction as session 029's giants, but worse: - **SFUND (8972/BSC) BUILT + ACCEPTED:** tf=3,415,005 / gl=28,553 (vs ~4k estimated — a 7x hidden giant). Contract screen 20/45 top-coded; 23/23 observed months built (2023-12..2025-10). **B2 PASS** (max onchain/circ = 1.94x, nowhere near the 100x guard); **B4 PASS** (screened HODL-6m in [12.3%, 24.9%], median 20.6%, last 18.6%, no nulls). All 23 months have same-month TVL (seedify) -> regression-ready immediately. - **MYX (36410/BSC) ABORTED mid-build, NO checkpoint:** a MEGA-giant. At kill time (batch 81/128): tf=15,438,541 and climbing (would likely have passed MBOX's 21.2M record), gl ~120,440 for MYX alone. The token's Aug-2025 BSC launch region (blocks ~49M+) forced a ~2.5h silent binary-split storm between batch prints. Remaining ~47 batches at ~790s/batch put completion ~10h out — past the user's 4:30 PM departure, so the process was killed at 15:07. The streamed engine has no partial checkpoint by design (Entry 66 tradeoff), so MYX restarts from scratch next session. **Budget a dedicated session/quota day for MYX alone (~250-300k calls).** - **Untouched:** ADF (24796/Polygon), AVNT (38299/Base), KAITO (35763/Base), VVV (35509/Base), RAIN (38341/Arbitrum). **Decision made:** kill at the departure cutoff rather than leave the engine running unattended for a week (user out 7/6-7/12); accept the ~120k-call loss on MYX. SFUND accepted into ch2/lambda. **Rationale:** an unattended week-long process risks a sleep/reboot kill mid-token (same loss) plus a week of unpushed local state, violating the record-keeping rule. The quota is daily-resetting (and per Entry 76 evidently non-binding), so the redo cost is time, not budget. **Methodology lesson:** est_getlogs_calls from holder counts UNDERSHOOTS BSC tokens by 7-30x (SFUND 4k->28.5k; MYX 4k->120k+ unfinished). For BSC, treat any estimate as a lower bound and sequence SMALL CHAINS FIRST (Polygon/Base/Arbitrum), BSC last. **Downstream impact:** channel2_holding.csv 300 -> 301 tokens / 9,817 rows. **lambda 9,580 -> 9,603 asset-months / 337 -> 338 assets.** **Regression-ready 138 -> 139** (coins 20 unchanged; tokens/other lambda?TVL 118 -> 119 assets / 4,194 -> 4,217 months). Coverage regenerated: 150 complete / 235 partial / 1,554 not_started. **Quota:** ~149k getLogs this session (28,553 SFUND + ~120,440 MYX-aborted + startup), single process, no API rejection. **Open items for session 031 (Monday 2026-07-13):** (a) 6-token resume — small-first order WORKLIST=24796,35509,35763,38299,38341 then MYX=36410 alone on a fresh day (see 04_code/CLAUDE_CODE_SESSION031_TASK_A_RESUME_PROMPT.md); (b)-(g) of Entry 79 unchanged. Report: 03_data/SESSION030_TASK_A_RESUME_REPORT.md.

Entry 81 — Session 031: Task-A day-1 resume — ALL 5 SMALL TOKENS BUILT (three Base hidden giants); two OS interruptions survived; MYX still queued for its dedicated day

Date: 2026-07-24 (ran 2026-07-23 14:52 CDT -> 2026-07-24 evening; the intervening night was lost to OS sleep, see interruptions below) **Spec section affected:** 3 (lambda assembly — counts only; no engine or threshold changes; VAL_CAP_MULT/CONTAM_MULT untouched at 100). **What happened:** the 5-token day-1 WORKLIST (24796,35509,35763,38299,38341) ran to completion in small-chains-first order. Per-token: - **ADF (24796/Polygon) BUILT:** tf=367,684 / gl=1,499 (vs 1,871 est — Polygon estimate HELD, 0.8x). 9/41 months coded, 4 screened months into ch2. B2 PASS (max onchain/circ 8.2x); B4 PASS (screened HODL-6m median 3.9%, last 3.9%). - **VVV (35509/Base) BUILT — NEW ALL-TIME ch2 RECORD:** tf=27,792,859 (beats MBOX's 21.2M) / gl=82,066 (vs ~1k est — an **82x** hidden giant). 12/29 coded, 4 screened months. B2 PASS (max 2.4x); B4 PASS (median 5.7%, last 5.5%). - **KAITO (35763/Base) BUILT:** tf=8,621,771 / gl=25,870 (vs ~3k est — 8.6x). 7/54 coded, **16 screened months** (largest month contribution this session). B2 PASS (max 5.3x); B4 PASS but FLAGGED HIGH: median 64.9% / last 66.7% — consistent with KAITO's airdrop-lockup profile; inside [0,80%], kept per protocol. - **AVNT (38299/Base) BUILT:** tf=17,646,973 / gl=52,758 (vs ~4k est — 13x). 7/36 coded, 9 screened months. B2 PASS (max 7.4x); B4 PASS (median 0.0%, last 4.6% — token is ~10 months old, near-zero 6m-HODL is structural, not anomalous). - **RAIN (38341/Arbitrum) BUILT (twice):** tf=4,284,832 / gl=19,315 (vs 13,637 est — Arbitrum estimate held, 1.4x). 11/52 coded, 7 screened months. B2 PASS (max 5.0x); B4 PASS (median 0.0%, last 17.2% — young token, same pattern as AVNT). First attempt killed by an OS restart at batch 536/587 (~9,824 gl lost, no partial checkpoint by design); rebuilt from scratch same day, clean. **Interruptions**

(both OS-level, neither corrupted data): (1) **Modern Standby stall**: the machine's sleep timeout was 20 minutes; it slept on-and-off from ~18:00 on 7/23 through the 9:10 AM wake on 7/24, freezing the build mid-VVV for ~15h. Diagnosed via Kernel-Power event log + py-spy thread dump (workers healthy, blocked in-flight); the engine's 60s request timeout + 6 retries self-healed on wake with zero loss. **Sleep set to NEVER (AC+DC) via powercfg — keep it that way through MYX day.** (2) **Windows Update auto-restart at 16:40 on 7/24** killed RAIN at 90%. Sleep settings do not prevent update restarts — **pause Windows Update before the MYX run. Methodology lesson (extends Entry 80)**: holder-count estimates undershoot **Base** the same way they undershoot **BSC**: VVV 82x, KAITO 8.6x, AVNT 13x. Polygon (0.8x) and Arbitrum (1.4x) held. Treat Base like BSC in all future sequencing: estimates are lower bounds; build Base late with dedicated headroom. **Downstream impact**: channel2_holding.csv 301 -> **306 tokens / 9,857 rows** (+40 screened months). **lambda 9,603 -> 9,638 asset-months / 338 -> 341 assets. Regression-ready 139 -> 142** (coins 20 unchanged; tokens/other lambda?TVL 119 -> 122 assets / 4,217 -> 4,243 same-month rows). Accounting note: only ADF/KAITO/AVNT are net-new lambda assets — VVV and RAIN already carried ch3 lambda months; VVV's 4 ch2 months coincide exactly with its existing ch3 months (rows upgraded to multi-channel, no new rows), RAIN netted +6. TVL overlap of the new builds: ADF 4, AVNT 9, KAITO 7, RAIN 7, VVV 1 months. Coverage regenerated: **153 complete / 232 partial / 1,554 not_started. Quota**: ~191k getLogs total across the session (~28.9k on 7/23 before sleep; ~162.5k on 7/24 including the 9.8k lost RAIN first attempt). Single-day 7/24 usage ~162k under DAILY_CAP=185k; no API rejection at any point (consistent with Entry 76's credit-based-cap finding). **Open items for session 032**: (a) **MYX (36410/BSC) alone on a dedicated quota day** — budget ~250-300k calls / 12-15h wall-clock, start EARLY, sleep=never confirmed, **pause Windows Update first**; reuse the Day-2 section of 04_code/CLAUDE_CODE_SESSION031_TASK_A_RESUME_PROMPT.md. (b)-(g) of Entry 79 unchanged (DOT/KSM key, CHZ anchor, CORE key, WARP review, non-TVL breadth, MATIC NVT probe). Report: 03_data/SESSION031_TASK_A_RESUME_REPORT.md.

Entry 82 — Session 032: MYX (36410/BSC) ch2 BUILT; Task-A fully closed

Date: 2026-07-24 (overnight session; build ran ~17:58–20:37 CDT — 2.7h, not the budgeted 12–15h) **Spec section affected**: 3 (lambda assembly — counts only; no engine or threshold changes; VAL_CAP_MULT/CONTAM_MULT untouched at 100). **What happened**: MYX ch2 (HODL-wave) built clean on the dedicated quota day: **67,233 getLogs** (+40 getcode) vs the ~250–300k budget — **0.22–0.27x, the first large BSC token to come in far UNDER budget**. 10/10 months built (2025-08..2026-05, the full observed window since the Aug-2025 launch), 10 screened months. Screened HODL-6m median **3.0% / last 5.2%**. **B2 PASS** (max onchain/circ 10.5x in the launch month, decaying to 5.8x — nowhere near the 100x guard). **B4 PASS** (young token; near-zero early 6m-HODL is structural, same pattern as AVNT/RAIN). Contract screen 21/40 top-coded. **tf=22,451,143 — second-largest ch2 build ever (beats MBOX 21.2M; VVV holds the record at 27.8M) and the largest BSC build. Consistency check vs the session-030 abort**: at batch 81/128 this run printed tf=15,459,154 vs 15,438,541 at the 030 kill — near-identical, the rebuild reproduces the aborted run's data. **Anomaly (favorable, unexplained)**: session 030 logged ~120,440 gl for MYX by batch 81; this run needed only 46,346 to the same point (2.6x fewer) with identical tf. Engine unchanged and block density is fixed, so the 030 counter likely included retry storms (rate-limit churn); treat 030-derived call estimates as upper bounds. The predicted multi-hour silent stretch after batch ~61 materialized but lasted only ~1h (batch 61->66: 67s->3,659s). **Ops**: sleep=never held (no standby events). Windows Update could NOT be paused — the session shell is non-elevated and the pause keys are HKLM; verified instead that no reboot was pending and the July cumulative wave had already installed the same morning. No OS interruption occurred. For future long builds: pause WU manually from Settings before leaving, or run the session elevated. **Downstream impact**: channel2_holding.csv 306 -> **307 tokens / 9,867 rows** (+10 screened months). **lambda 9,638 -> 9,648 asset-months / 341 -> 342 assets. Regression-ready 142 -> 143** (coins 20 unchanged; tokens/other lambda?TVL 122 -> 123 assets / 4,243 -> 4,253 same-month rows). MYX has same-month TVL (myx-finance) in all 10 lambda months -> regression-ready immediately. Coverage regenerated: **154 complete / 231 partial / 1,554 not_started. Quota**: 67,273 calls total (67,233 getLogs + 40 getcode), single process, no API rejection, well under DAILY_CAP=185k — cap never approached. **Task A fully closed**: all 47 session-029 targets resolved — 41 built, 5 non-EVM skipped, 1 (WARP, cmc-1166) deferred for identity-map review

(built-but-empty, Entry 79). **Open items for session 033:** Entry-79 (b)–(g) unchanged — DOT/KSM Subscan key, CHZ manual anchor (gate-open), CORE key, WARP identity review, non-TVL breadth ch2 (~500 >3k-holder tokens), MATIC NVT_GL probe (candidate 21st regression-ready coin). Report: 03_data/SESSION032_MYX_REPORT.md.

Entry 83 — Session 033: XTZ/MATIC NVT_GL probes NEGATIVE (prompt premise wrong); JOE/SAFE gaps closed as architectural

Date: 2026-07-25 **Spec section affected:** 4.1 (NVT_GL coverage — no data changes; two NaN-marker notes refined; JOE/SAFE/DYDX gaps classified). **What happened:** Session 033's kickoff prompt directed building XTZ PQ from TzKT's /v1/statistics/daily field totalTransferred. **That field does not exist.** Verified live: the Statistics schema is supply/staking-only (totalSupply, circulatingSupply, totalFrozen, staking/delegation fields — no volume); the full 287-path TzKT swagger has NO historical volume/sum endpoint anywhere; back.tzkt.io/v1/home (browser UA+Referer required) serves only current-day volume aggregates + a 30-day price chart. The prompt's premise — “TzKT was never tried” — was correct, but trying it yields nothing: TzKT indexes everything EXCEPT aggregate historical transfer value. **XTZ (2011, 78 lambda months):** exhaustive free-source sweep, all NEGATIVE: TzStats api.tzstats.com DEAD (connection refused); TzPro unreachable; CoinMetrics community API exposes only TxCnt/TxTfrCnt for xtz (TxTfrValUSD -> 403 pro-gated; GitHub csv/xtz.csv same); bitinfocharts sentinusd-xtz EMPTY page (not the BTC default — simply no data); Messari legacy keyless API 404 (dead); CryptoCompare blockchain histo 401 key-gated. Raw TzKT operation iteration forbidden (Entry 31/32). **XTZ stays PQ=NaN;** marker refined to NaN:xtz_no_free_native_series_s033 with the full probe list. XTZ remains the largest single pq_nvtgl gap (78 lambda months) — only a paid source (Artemis/CM-pro/TzPro key) closes it. **MATIC (3890, 50 lambda months):** bitinfocharts sentinusd-matic returns the **BTC default series** (BTC-guard triggered: identical first date 2010/07/17 and last value to the BTC reference — Polygon didn't exist in 2010). CoinMetrics community rejects value metrics for 'matic'. No free keyless Polygon native-transfer-value source; lambda window ends 2024-08 at POL handoff. **MATIC stays PQ=NaN;** marker refined to NaN:polygon_native_volume_no_free_source. The “candidate 21st regression-ready coin” idea (Entry 79g) is closed negative. **JOE (11396, 6 lambda months) and SAFE (21585, 10 lambda months): PERMANENTLY CLOSED — architectural, not data-sourcing.** Both are classified asset_class=coin via consensus tags: ['staking'], but neither is a chain-native gas coin: JOE is Trader Joe's DEX governance token (Avalanche/Arbitrum, lambda from veJOE vote-escrow), SAFE is the Safe{Wallet} governance token. No native chain ? no settlement value ? no valid NVT_GL denominator exists in principle. Verified both also have ZERO rows in tvl_panel.csv ? no TVL path either. Neither has (or needs) a pq_coins.csv row. Do not re-probe. **DYDX (28324, 1 lambda month):** legitimately a coin (dYdX Chain Cosmos L1 native) but 1 lambda month (2024-03) is far too short for regression. Skip; revisit only if lambda extends via a Cosmos/Mintscan key. **Downstream impact: NONE.** pq_coins.csv row count unchanged (3,275; 2 marker notes edited). phase2_nvt_gl.py + build_coverage_status.py re-run clean and identical: NVT_GL 2,526 asset-months / 67 assets; **coins regression-ready (lambda?NVT_GL) 20 / 645 months — unchanged;** coverage 154 complete / 231 partial / 1,554 not_started. Regression-ready total stays 143. **Open items for session 034:** Entry-79 (b)–(e) unchanged — DOT/KSM Subscan key, CHZ manual anchor (gate-open), CORE key, WARP identity review, non-TVL breadth ch2 (~500 >3k-holder tokens). (f)/(g) MATIC probe now CLOSED (this entry). Report: 03_data/SESSION033_XTZ_MATIC_NVT_REPORT.md.

Entry 84 — Session 034: CHZ ch1 BUILT; Blockchair XTZ/MATIC FAILED (keyless unusable); EVM DeFi Breadth Batch 1 (101 of 102 tokens)

CHZ (4066): ch1 staking built via Chiliz Chain 2.0 public RPC (chiliz.drpc.org; eth_getBalance on 0x...1000 staking contract at month-end blocks; anchor drift -1.03% vs 2,416,757,292 CHZ confirmed 2026-07-25 on staking.chiliz.com). Window: 2023-07..2026-05 (35 months). Staking ratio range 2.35%..26.61% (real ~4x step-up 2024-06). pq_source suffix: chiliz-chain-pubRPC. CHZ already had 21 non-NaN PQ months in pq_coins.csv -> CHZ = 21st regression-ready coin.

Blockchair XTZ: FAILED. tezos/{calls,operations,transactions} all HTTP 404 (no aggregation tables exposed for Tezos); IP was then blacklisted (HTTP 430 “apply for an API key”) after ~4 anonymous requests, including

on /stats. Free anonymous tier is effectively unusable in 2026. Paid key (~\$30/mo) MIGHT unblock, but the 404s suggest Tezos aggregation may not exist at any tier - confirm with Blockchair support before paying. NOT subscribed (needs Moazzam approval). XTZ stays PQ=NaN.

Blockchair MATIC: FAILED, same blacklist; polygon/transactions also 404 before the 430s started. Same paid-key caveat. MATIC stays PQ=NaN.

EVM DeFi Breadth Batch 1: 101 of 102 tokens built (MSOL 11461 already complete -> skipped). 154,049 getLogs (est 147k), 47.56M transfers, 2,916 screened lambda months. B2 clean across the batch (no 100x contamination flags). B4 flagged-high (HODLmed>80%, kept per rule): META, TROY, SMT, YOU, BOX(3475), WHITE, HOT, BOX(2945), STRONG. Biggest builds: STRONG 6.66M tf / 20.0k gl, WSTETH 4.02M tf / 11.9k gl, XAI 2.62M tf. Survivorship targets built: CEL (Celsius, 54 scrMo), FTT (FTX, 81 scrMo), MULTI (Multichain, via prior slug).

TVL slug matching (the ABC-validation lesson applied): raw symbol match was ~40% wrong (Litentry->lighter, old-Jupiter 1503->jupiter-lend, Wrapped Solana->solana-farm, etc.). Re-matched with DL cmcId authority + name corroboration + individual verification: - 24 token-class dl_slugs written to asset_onchain_identity.csv (incl. ETHDYDX->dydx-v3, KNC 9444->kyberswap-classic, SYRUP->maple, SKY->sky-lending, MORPHO->morpho-blue). - OTHER_ADDS += MULTI/multichain (dead, 49 mo), ORC/orbit-bridge (hacked, 46 mo), MUBI/multibit-protocol, FF/falcon-finance. - CHAIN_LEVEL += METIS/chain:Metis (54 mo), XAI/chain:Xai (28 mo) per Entry-68 pattern. - REJECTED on semantics: UNFI (DL unifi cmcId=1412 collision), LOCUS (Locus Finance != Locus Chain), JUP 1503 (Solana Jupiter != 2017 Jupiter), SOL 16116 (wrapped), DAO 8420 (vesting tracker != protocol TVL), AGIX (own-token staking pool = circular), WOO woo-x (CEX reserves; used woofi-swap instead). - LST receipt tokens (wstETH/weETH/cbETH/rETH/sfrxETH/mETH/ETHx/swETH/rswETH/ezETH/rsETH/WBETH/tETH/LBTC/MSOL/aEthWETH/bUSD0) deliberately get NO protocol TVL: the receipt token's NV IS the protocol TVL (NV/TVL ~ 1 by construction, circular). - CEL and FTT have NO DeFiLlama entries (CeFi books never TVL-tracked); dead-protocol TVL defense rests on MULTI/ORC/dydx-v3/ribbon/idex-v1 etc. - ARKM, PRIME (echelon-prime): slugs valid but DL series empty (zero-TVL protocols). tvl_panel.csv rebuilt: 159 assets / 7,889 asset-months (was ~128/6,7xx).

Post-assemble: lambda 12,599 asset-months / 444 assets (was 9,648/342). Regression-ready 143 -> 173 (coins 20->21 [CHZ], tokens/other 123->152 [+29 Batch-1]). channel2_holding.csv: 408 tokens / 12,955 rows. Coverage: 184 complete / 302 partial / 1,453 not_started. Remaining EVM DeFi breadth batches: Batch 2 (session 035): 13 tokens, ~119k getLogs - WORKLIST in 034 prompt Batch 3 (session 036): stETH + MEME, ~110k getLogs Batch 4 (session 037): SHIB alone, ~128k getLogs

Entry 85 — Session 035: EVM DeFi Breadth Batch 2 (13/13 tokens); KNCL window-clip;

SNX/LEND/ANKR TVL

Batch 2 built 13/13 in one run: 74,783 getLogs (est 119k -> 0.63x), 22.9M transfers, 592 screened months. B2 clean. B4 flagged-high kept: PNT (97.9% - dead token, 5 scrMo). Long histories: FUN 105 scrMo (2017+), ANKR 85, SNX 73 (assemble window 2019-05+), SLP 58, ELON 56. Note: cmc 2691 "PNT" is Penta (dead 2018 token), NOT pNetwork - the 034 prompt's note was wrong; actual gl ~1.1k not 14.3k.

TVL decisions: - SNX (2586) -> parent slug **synthetix** (children v1+v2/v3/v4 only in /protocols; parent fetchable per Entry-68 CRV/GMX precedent). 82 mo, 73 lambda-overlap. - LEND (2239) -> **aave-v1** (AAVE 7278 keeps aave-v2; protocol-era split, no window needed). 73 mo, 4 lambda-overlap (2020-05..2020-08 - LEND died 2020-08; the low-TVL dead-token region). - ANKR (3783) -> OTHER_ADDS **ankr** (cmcId=3783 exact; other-class filter gap, same as RPL Entry 68). 67 mo, 66 overlap. - KNCL (1982) -> **kyberswap-classic** WINDOW-CLIPPED <=2021-06 via new CLIP mechanism in phase2_build_tvl_panel.py (MATIC/POL rule: KNC 9444 lambda starts 2021-07; one physical TVL never in two assets' rows same month). 9 mo 2020-10..2021-06. - MKR investigated: DL parent **maker** 400s; MakerDAO history lives in sky-lending (parent#maker) already on SKY 33038. MKR has ZERO lambda months -> no assignment, no double-count risk (SKY lambda 2025-05+ only touches post-migration months). Revisit only if MKR ever gets lambda. - SAI (2308): NO TVL by rule - liability/receipt-token circularity (SAI is the CDP stablecoin itself; NV ~ =

pegged supply, NV/TVL = inverse collateral ratio, not a valuation multiple). Same family as the LST receipt exclusion (Entry 84). SAI keeps its 25 lambda months (ch2 survivorship) but is not regression-ready. - EETH (28568): LST receipt token -> excluded (Entry 84 rule). 11 lambda months kept. - No TVL exists: FUEL, FUN, ERC20, MLK, SLP, ELON, PNT (meme/game/dead, no protocol).

tv1_panel: 163 assets / 8,120 asset-months. Post-assemble: lambda 13,191 / 457 assets. Regression-ready 173 -> 177 (coins 21, tokens/other 156). ch2 421 tokens / 13,580 rows. Coverage 188/311/1,440. Remaining breadth batches: 036 stETH+MEME (~110k gl), 037 SHIB (~128k gl).

Entry 86 - Session 036: stETH + MEME ch2 built (lambda-only; no TVL regression entry)

stETH (8085/Ethereum): 13,365 getLogs / 4,529,175 transfers / 50 screened months. B2 pass. B4 pass (HODL-6m median 13.4%, last 13.1%). TVL excluded: LST receipt circularity (NV~Lido TVL by construction, Entry 84 rule). Lambda months retained for conviction-only panel. No stETH -> lido mapping created.

MEME (28301/Ethereum): 9,079 getLogs / 2,932,933 transfers / 31 screened months. B2 pass. B4 pass (HODL-6m median 3.7%, last 15.4%). TVL: DeFiLlama HAS a memecoin protocol listing (cmcId=28301, correct contract 0xb131...cd74, category Farm) but its TVL series is EMPTY (0 data points, no currentChainTvl) -> no usable protocol TVL, lambda-only. Nuance vs prompt: cmcId match exists but carries no data; not a symbol clash (conflux MemeDex is the unrelated symbol match).

getLogs actual 22,444 total vs ~110k estimate (est was ~5x high; sparse pre-2021 blocks). Runtime ~50 min.

Post-assemble: lambda 13,272 asset-months / 459 assets. Regression-ready 177 (no change, as expected - both adds lambda-only). ch2 423 tokens / 13,661 rows. Coverage 188/313/1,438.

Remaining EVM breadth: Session 038 - SHIB (5994), ~128k getLogs est (likely high), lambda-only. (Session 037 repurposed for DOT/KSM/CORE ch1 — see Entry 87.)

Entry 87 — Keys received; Session 037 scope: DOT + KSM ch1 (Subscan) + CORE ch1 probe (CoreScan)

Date: 2026-07-27 **Keys added to .api_keys.json:** - “subscan”: 3d09e805af23487a9e1ee546338b7216 (received prior session — confirmed in place) - “coredao”: 97375a02225a40688d743659236ea82b (received this session from scan.coredao.org)

Assets unlocked: - DOT (6636) + KSM (5034): Subscan era_stat key-gated (Entry 44). With key in place, build is ready: POST polkadot.api.subscan.io/api/scan/staking/era_stat (X-API-Key header), paginate all eras, bonded_total / 1e10 (DOT) or 1e12 (KSM), map end_block_num -> timestamp via genesis_ts + block_num * 6s, bucket to months. Both have NVT_GL -> regression-ready on ch1 build. Expected: coins 21 -> 23. - CORE (23254): openapi.coredao.org 401 without key (Entry 78). Key now available. Probe: GET openapi.coredao.org/api/stats/staking_summary?apikey=. If historical series available -> build directly. If current-only -> block-level balance reads on PledgeAgent/staking contract (Core EVM chainid 1116, block time ~3s, genesis 2023-01-14). CORE has NVT_GL (40 months) -> regression-ready if ch1 confirmed. Expected: coins -> 24.

Session 037 prompt: 04_code/CLAUDE_CODE_SESSION037_DOT_KSM_CORE_PROMPT.md

Session 038 (SHIB ch2): deferred; see prior “next” note above. **Regression-ready target post-037:** 177 -> 180 (coins 21->24) if all three PASS.

Entry 88 — Session 037 RESULTS: DOT + KSM + CORE ch1 ALL BUILT (method pivot: archive-RPC state reads)

Method pivot (supersedes the Entry-87 build plan): Subscan era_stat turned out to be PER-ADDRESS (400 “address is a required field”) — Subscan exposes NO network-wide bonded history on the free plan (/api/scan/daily category Bonded returns all zeros and the key has a ~2-month history_window_exceeded cap; the subscan.io site itself was mid-upgrade, charts unreachable). Instead: raw-key state reads (twox128/twox64concat, no metadata decode) of Staking.ErasTotalStake(ActiveEra)

at month-end blocks (interpolation search on `Timestamp.Now`) from public ARCHIVE RPCs. Post-Asset-Hub-Migration months (relay staking storage cleared) read from the Asset Hub at the same timestamp: DOT relay<=2025-10 / AH from 2025-11; KSM relay<=2025-09 / AH from 2025-10. Endpoints: OnFinality public (relays); parity *-asset-hub-rpc.polkadot.io (AHs, archive verified to >=2022). Script: 04_code/session037_build_dot_ksm.py.

DOT (6636): 71 months (2020-08-31..2026-06-30) built. 1 DOT = 1e10 Planck. Latest 862,345,368 DOT. Cross-check: fresh head 881,519,184 -> drift -2.18% PASS (July growth); external anchor ~881.9M / 52.0% (Coinbase+StakingRewards 2026-07) matches fresh to 0.05%. AHM boundary continuous (831.5M -> 825.9M, -0.7%). Source: archiveRPC polkadot(+assethub):Staking.ErasTotalStake / 1e10.

KSM (5034): 76 months (2020-03-31..2026-06-30) built. 1 KSM = 1e12 Planck. DROPPED 2019-11..2020-02: pre-runtime-1050 storage only exposes `SlotStake` = MINIMUM validator backing (11k KSM), not network total — wrong metric (and pre-universe anyway). Latest 8,384,479 KSM; fresh 8,559,862 -> drift -2.05% PASS; external anchor ~8.5M / 46.0% matches. `staking_ratio` suppressed for 2020-07 (CMC circulating 2.99M < staked 5.98M — bad CMC point; 8.47M next month). Note: CMC holds KSM circulating at 8.47M through 2020-2024 -> ratios 0.77-0.93 there carry a stale denominator (universe panel remains supply authority). Source: archiveRPC kusama(+assethub):Staking.ErasTotalStake / 1e12.

CORE (23254): BUILT EXACT, 42 months (2023-01-31..2026-06-30). `staking-api.coredao.org/staking/summary/overall` is CURRENT-ONLY (round param ignored), `openapi.coredao.org/proxy/balancehistory` endpoints return empty 200s, and `rpc.coredao.org` is pruned — but Ankr (`rpc.ankr.com/core`) and dRPC (`core.drpc.org`) are ARCHIVE. Definition validated TO THE DIGIT at head: official `stakedCoreAmount` == sum over ACTIVE validators (`ValidatorSet.getValidatorOps()`, legacy fallback `currentValidatorSet(i) walk`) of `CoreAgent.candidateMap(op).amount` (0x...1011, post-StakeHub upgrade; from 2024-11) / `PledgeAgent.agentsMap(op) word0` (0x...1007, legacy; <=2024-10; word0 vs word2 differ <2%, boundary continuous with real Oct-Nov 2024 decline). Naive contract-BALANCE reads would run ~6% HIGH (335.9M vs 315.8M) — rejected. Latest 307,727,006 CORE; fresh API 315,775,339 -> drift -2.55% PASS. Ratio range 0.114-0.81 (early CMC circulating tiny). CORE-only (BTC/hashpower dual-staking excluded). Script: 04_code/session037_build_core.py. Source: core-archiveRPC eth_call active-validator stake sum.

Post-assemble: lambda 13,272 -> 13,449 asset-months / 459 -> 462 assets. Coverage 188/313/1,438 -> 189/314/1,436. Regression-ready 177 -> 178: coins 21 -> 22 (CORE enters — it already had 27 NVT_GL months; same-month lambda x NVT overlap confirmed). DOT/KSM do NOT enter: the Entry-87/prompt premise “DOT+KSM have NVT_GL” was WRONG — `nvt_gl_panel` has 68/70 rows for them but `pq_usd` ALL NULL (no PQ source) -> they move not_started -> PARTIAL (`pq_nvtgl` is now their only gap; ch1 gate CLOSED).

Bookkeeping note: the narrative “coins 21” vs coverage-file pos-coin count 20 pre-session is explained by TRX (cmc 1958): `coin_staking_type` mislabeled `pow_only` (TRON is DPoS) while having ch1 lambda (78 months) — the file counts it in the pow bucket, narrative counted it as a staking coin. Worth a one-line fix in a future universe-map pass; not touched this session.

Open items for session 038: SHIB (5994) ch2 ~128k gl est (treat as ~5x high); WARP (1166) identity review; Cosmos key -> CRO/INJ/SEI/KAVA ch1; Blockchair support email re XTZ/MATIC before paying; DOT/KSM PQ source hunt (would make them regression-ready); TRX `coin_staking_type` fix; bibliography sanity-check. Report: 03_data/SESSION037_DOT_KSM_CORE_REPORT.md

Entry 89 — Session 038: SHIB ch2 built (lambda-only; no TVL regression entry)

SHIB (5994/Ethereum): 53,683 getLogs / 17,845,462 transfers / 61 screened months (2021-05..2026-05, zero contaminated). B2 pass (no month excluded by the 100x contamination guard). B4 pass (screened HODL-6m median 78.1% <= 80%; last month 82.5%). Contract screen 13/81 candidate addresses. Actual getLogs vs 128k estimate: 0.42x (holder-count overestimate pattern of sessions 034–036 holds). Wall time ~2.1h at 8 workers.

TVL: no protocol TVL for SHIB. `shibaswap` slug in `tv1_panel` belongs to BONE (`cmc_id`=11865, ShibaSwap

governance token, 59 months already in panel) — assigning to SHIB would double-count the DEX TVL against the wrong token. SHIB -> lambda-only.

Post-assemble: lambda 13,449 -> 13,510 asset-months / 462 -> 463 assets. channel2_holding.csv 423 -> 424 tokens / 13,661 -> 13,722 rows. Coverage 189 complete / 315 partial / 1,435 not_started (SHIB not_started -> partial). Regression-ready 178 -> 178 (no change, as expected: coins 22, tokens/other 156).

EVM DeFi breadth complete (all batches 1-3b done). Next: Session 039 — DOT/KSM PQ source probe; TRX coin_staking_type fix; WARP identity review.

Entry 90 — Session 039: DOT/KSM PQ probe negative; TRX label fix; WARP closed

DOT/KSM PQ probe (Task A): NEGATIVE — no free source. Subscan /api/scan/daily (key on file): format=month not supported (day/hour/6hour only); format=day over any multi-year window -> 403 history_window_exceeded for ALL four categories (transfer/extrinsic/transaction/fee), both polkadot and kusama. Window bisection: 30-day range returns data, 90-day fails -> free window ~ 2 months, same wall session 037 hit for “Bonded”. The 30-day payload is also degenerate (2026-06-01: total=3 transfers / 5.98 DOT — not network-wide volume). Blockchair: /polkadot/stats and /kusama/stats return 200 keyless, BUT (a) the aggregation endpoint (/chain/calls?a=date(time),sum(value)&q=type(transfer)) is HTTP 404 — same no-aggregation-tables pattern as XTZ/MATIC (Entry 84), and (b) both indexes are FROZEN (DOT best block 2025-05-26, KSM 2025-05-09) — even a paid key could not cover the panel through 2026-06. Raw block iteration remains FORBIDDEN (Entry 31/32). Decision: **no free PQ source for DOT/KSM; candidates exhausted. They stay PARTIAL (gap = pq_nvtgl only). Reopens only with Subscan Pro or another paid volume series** (Blockchair ruled out — stale index).

TRX (1958) coin_staking_type fix (Task B): Label was pow_only; corrected to pos (TRON = DPoS; ch1 freezeresource series, 78 lambda months). Source of label: universe_coverage_status.csv itself — build_coverage_status.py line 33 carries coin_staking_type forward from the old file (static metadata, no live source). Edited the CSV row + re-ran the builder. TRX coverage status: complete (unchanged), but now via the pos lambda?NVT same-month path (58 overlap months) instead of the pow_only NVT-alone path. Regression-ready coins now derive cleanly to 22 WITH TRX included — the Entry-88 “coin-count +/-1” bookkeeping discrepancy is resolved; headline totals unchanged (178).

WARP (1166) identity review (Task C): PERMANENT IDENTITY MISMATCH — CLOSED. CMC id 1166 = “WARP” warppcoin.com — a 2016-02 standalone PoS coin (CMC category “coin”), inactive since 2018-05-08, own chain (chainz.cryptoid.info explorer), supply 1.1M. It NEVER deployed an ERC-20. The stored contract 0x83e6f1E41cdd28eAcEB20Cb649155049Fac3D5Aa (ch2 checkpoint: 27,257 getLogs, 0 transfers ever) is a different, later “WARP”-named token; the dl_slug polkastarter is DeFiLlama’s own wrong cmcId (Polkastarter = POLS, cmc 7208 — the Entry-52 collision). Actions: (1) phase1_build_identity_map.py gains a BAD_DL_CMCID = {“1166”} registry-drop override so rebuilds can’t resurrect the mapping; (2) identity CSV row cleared (dl_slug/address/chains blanked, dl_matched=False); (3) 54 bogus polkastarter TVL months PURGED from tvl_panel (8,120 -> 8,066 months / 163 -> 162 assets); (4) coverage now not_started. No lambda rows existed (empty checkpoint never contributed) — no lambda impact.

Post-assemble: lambda 13,510 / 463 assets (unchanged). Coverage 189 complete / 314 partial / 1,436 not_started (WARP partial -> not_started). Regression-ready 178 (coins 22, tokens/other 156) — unchanged, now internally consistent.

Open items for session 040: Cosmos key -> CRO/INJ/SEI/KAVA ch1; Blockchair support email decision (XTZ/MATIC only — DOT/KSM now ruled out); DOT/KSM PQ reopens only on Subscan Pro decision; bibliography sanity-check. Report: 03_data/SESSION039_DOTKSM_PQ_FIXES_REPORT.md

Entry 91 — Session 040: CRO/KAVA ch1 via Cosmos Archive LCD; INJ/SEI negative

Approach: free, keyless Cosmos SDK LCD cosmos/staking/v1beta1/pool with x-cosmos-block-height header at month-end blocks found via binary search on cosmos/base/tendermint/v1beta1/blocks/{height} timestamps. Candidate endpoints refreshed from the chains.cosmos.directory registry after the kickoff prompt’s CRO candidates proved DNS-dead (rest.crypto.org, notional.ventures).

Archive probe results (liveness + 365-day state-depth test): - CRO: `rest.mainnet.crypto.org` — PASS - INJ: FAIL — all 7 registry candidates pruned (no commit info found / version mismatch on immutable IAVL tree). No free archive LCD exists. - KAVA: `api.data.kava.io` (official archive) — PASS, but aggressive HTTP-420 rate limiting; required 0.6–1.5 s call pacing + 15–45 s backoff retries and two retry passes to complete. - SEI: FAIL — all 8 registry candidates pruned. **Two gateways (`sei.api.pocket.network`, `rest.cosmos.directory/sei`) are FAKE archives: they silently ignore the `x-cosmos-block-height` header and return live state.** Caught because “archive” bonded == live bonded digit-for-digit; the probe now hard-fails any node whose year-old bonded equals live bonded. This guard is load-bearing — without it SEI would have shipped a flat bogus series.

State-decode (codec) boundaries — invalid denom: on older heights: - CRO: block store is complete back to genesis (2021-03-25) but staking state decodes only from ~2025-06 onward -> 11 months built. - KAVA: current chain `kava_2222-10` restarted at height 1 on 2022-05-25 (older months have no blocks at all); state decodes only from ~2024-Q2 -> 26 months built. 2022-05..2024-03 permanently unavailable on this node. - Binary search gained a guard: if the earliest stored block postdates the target month-end, the month is skipped (previously it would silently return block 1 and mis-attribute post-restart state to earlier months).

Built (03_data/phase1/channel1_cosmos_lcd.csv, 37 rows): - CRO (3635, $\text{basecro}/10^8$): 11 months 2025-07-31..2026-05-31, ratio 0.309–0.370 - KAVA (4846, $\text{ukava}/10^6$): 26 months 2024-04-30..2026-05-31, ratio 0.091–0.127

Cross-check (latest built month vs live pool): - CRO drift: 1.81% — PASS - KAVA drift: 23.06% — WARN, investigated and explained: bonded was 99.1M at 2026-06-30 and 103.1M at ~2026-07-14 vs 127.8M live (2026-07-28) — a genuine ~25M-KAVA staking surge in the two weeks before this session, not a denom/decimal error. Series trajectory is smooth and ratio bounded.

Coverage label fixes: `coin_staking_type` `pos_possible` -> `pos` for CRO (3635) and KAVA (4846). SEI stays `pos_possible` (nothing built); INJ was already `pos`.

Environment note: user-site `pandas` was found mid-upgrade-interrupted (~`andas` remnant; `to_csv` ModuleNotFoundError, then full import failure) — reinstalled clean (`pandas` 3.0.5); `assemble` + `coverage` builders ran unmodified. The session builder writes its CSV via `stdlib csv` as defense.

Post-assemble: `lambda` 13,510 -> 13,547 asset-months / 463 -> 465 assets. Coverage 191 complete / 312 partial / 1,436 not_started. Regression-ready 178 -> 180 (coins 22 -> 24: CRO + KAVA in; tokens/other 156 unchanged).

INJ/SEI ch1 verdict: blocked on archive state access, not on method. Reopen only if a free archive LCD emerges or a dedicated indexer (paid) is approved.

Output: 03_data/phase1/channel1_cosmos_lcd.csv Builders: 04_code/session040_cosmos_lcd.py, 04_code/session040_kava_retry.py Report: 03_data/SESSION040_COSMOS_LCD_REPORT.md

Entry 92 — Session 041: HXRO moot (lambda-on-observed); SXP ch2 built; OSMO ch1 (2 mo); gap closures

Date: 2026-07-30 **Spec section affected:** 3 (lambda assembly — counts only); coverage semantics clarification.

HXRO (3748) ch2 extend — NOT RUN, task moot under existing spec: The session-041 prompt premise (“checkpoint monthly: [] -> fresh rebuild extends to 2026-05”) was a schema misread: streamed checkpoints store `rows/mblocks` (24 months, 2020-09->2022-09, intact), not `monthly`, and the stream engine skips any non-deferred checkpoint. The deeper blocker: HXRO’s panel months are `carried_forward` from 2022-10 onward (CMC top-1000 visibility lost; supply frozen at 4.285e8, subtype presumed_failed per Entry 17) while TVL starts 2023-02. `phase1_assemble_lambda.py` computes `lambda` on `status='observed'` rows ONLY (core spec rule), so no ch2 rebuild can ever create `lambda?TVL` overlap for HXRO — the observed window (->2022-09) and the TVL window (2023-02->) are disjoint. **Decision:** zero Etherscan quota spent; HXRO stays partial as a PERMANENT gap under current spec. Reopen only if Phase 3 changes the `lambda-on-observed` rule for `carried-forward` months (would affect 89,535 asset-months, not just HXRO).

SXP (4279) ch2 — BUILT (chainid 1): Coverage flag `non-EVM/etherscan_reachable:no` was wrong — caused by the unprefixed Multi-Chain address in `asset_onchain_identity.csv`. Probe confirmed `0x8ce9137d39326ad0cd6491fb5cc0cba0e089b6a9` live on Ethereum: ABI OK, supply 285,368,788.7 (18 dec), `getLogs` clean. Fixed `universe_lambda_channel_map.csv` (`chain=Ethereum, etherscan_reachable=yes, ch2_holding=Transfer-log`) and `identity` (`ethereum: prefix`). Build: `tf=569,311, getLogs=1,886, contracts screened 10/96, 77 lambda months (2019-09->2026-02), HODL-6m median 27.0% (B4 pass), no B2 contamination warnings. lambda?TVL overlap 60 months (2021-03->2026-02).`

OSMO (12220) ch1 — BUILT, 2 months (archive floor 2026-04-02): Prompt's blocks/day=15,000 (~5.75 s/block) is STALE — Osmosis now produces ~73,300 blocks/day (~1.2 s/block; epochs-module anchors). All 16 chain-registry REST endpoints + `publicnode/ecostake/quickapi` extras: pruned or fake. `osmosis.api.pocket.network` is a FAKE ARCHIVE (ignores `x-cosmos-block-height` — same landmine family as `sei.api.pocket.network`, Entry 91). Only real archive: `https://osmosis-api.noders.services`, `app-state` + block floor ~ h 58.44M = 2026-04-02 (apparent chain-wide post-upgrade state-sync point; an initial “598-day retention” estimate was an artifact of the stale 15k blocks/day constant). Built 2026-04-30 (h=60630953, ratio 0.2749) and 2026-05-31 (h=63027468, ratio 0.2630) via pool @ month-end blocks, binary-search timestamps. Drift vs live 1.77% PASS. Denom `uosmo` / 10^6 . Output: `03_data/phase1/channel1_cosmos_osmo.csv` (separate file, picked up by the assembler's `channel1_*.csv` glob). `coin_staking_type NaN -> pos. 2021-06->2026-03 unreachable keyless; reopen only on a paid indexer (Numia/ Mintscan) decision.`

Permanent gap closures (no build path): CASINO (1573): Fantom chainid 250 — not in Etherscan V2 coverage, no free alternative. RUNE (4157): THORChain — non-EVM, non-Cosmos; custom indexer required. SUN (10529): Tron — ch2 engine not adapted for Tron; TronScan API deferred.

Regression-ready semantics (recorded): regression-ready = coverage ‘complete’ AND `lambda_months > 0` (excludes the 11 `pow_only` coins complete on NVT alone); 193 complete ? 11 = 182.

Post-assemble: `lambda 13,547 -> 13,626 asset-months / 465 -> 467 assets. Coverage 193 complete / 310 partial / 1,436 not_started. Regression-ready 180 -> 182 (coins 24 unchanged; tokens/other 156 -> 158: SXP + OSMO in).`

Note: Etherscan Pro subscription lapsed after this session (2026-07-30 last day). Remaining Etherscan-dependent work: none identified (HXRO moot, not quota-blocked).

Output: `03_data/phase1/channel1_cosmos_osmo.csv; channel2_holding.csv 425 tokens / 13,799 rows`
Builders: `04_code/session041_osmo_ch1.py` Report: `03_data/SESSION041_HXRO_SXP_OSMO_REPORT.md`

Entry 93 — Cowork 2026-08-04: Growth-Levelized NV/TVL adopted for token track

Decision: The token-track valuation anchor is $NV/TVL_{GL} = MC / TVL$, replacing plain NV/TVL , so both tracks carry a growth adjustment (user: “otherwise we miss the growth metric completely”). TVL uses the identical DCF machinery and PARAMS as PQ^* (rf 4%, MRP 30%, g_{inf} 3%, $n=10$, g_{cap} [-50%,+200%], re_{floor} , β_{36} vs BTC). Because TVL is a stock (not a flow like PQ), TVL_0 = trailing-12m AVERAGE month-end TVL (≥ 6 obs), g = trailing 3y CAGR of TVL_0 with 2y/1y fallback.

Consequences: NV/TVL_{GL} panel 111 assets / 3,125 mo (2021-01..2026-05). Token regression sample 136/4,627 -> 101/2,771 (growth history requirement); combined panel 125 assets / 3,489 mo / \$451B (19.0% of universe) at 2026-05. Raw NV/TVL retained in the panel as robustness column. CAVEAT for Phase 3 robustness: 44.6% of regression asset-months hit the g_{cap} (mostly the -50% floor; median token TVL $g = -33\%/yr$, post-2021 DeFi contraction) — cap sensitivity must be checked.

Builder: `04_code/phase2_tv_l_gl.py` Output: `03_data/phase2/nv_tv_l_gl_panel.csv` Paper: Section 3.3 (TVL* equation), Table 1 funnel + Table 2 updated (commits 2b9ac26, 9308322).

Entry 94 — Cowork 2026-08-04: Phase 3 empirical design fixed

Decision: Three-element test plan (user-specified): (1) H1/H2 via pooled panel regressions with month + category FE, two-way clustered SEs, interaction + split-sample for H2; FM secondary for tokens only (coin cross-section too thin at 11-20/month). (2) H3 via Stars-minus-Avoid quadrant portfolio evaluated against self-built MONTHLY analogs of the Liu-Tsyvinski-Wu (2022 JF) three-factor model (CMKT/CSMB/CMOM from our own 1,939-asset universe; LTW original is weekly — deviation documented). (3) Horse race vs raw NVT, Metcalfe (needs BitInfoCharts active-addresses build), MVRV (needs realized- cap probe of ch2 checkpoints — NO new getLogs, Etherscan lapsed Entry 92), S2F, and technical signals (momentum family, 52wk high, MA cross); panel + spanning + sub-period designs. Full spec: 04_code/PHASE3_ANALYSIS_SPECIFICATION.md. Kickoff: 04_code/CLAUDE_CODE_PHASE3_KICKOFF_PROMPT.md (core = Tasks A-D; horse race = Phase 3b).

Conviction variable resolution: coins use ln-odds of raw ch1 staking share (SoV/MoE = $\lambda/(1-\lambda)$) requires a raw share, not the z-score index), λ_z fallback flagged; tokens use λ_z . Standardization within class-month.

Entry 95 — Session 043: Phase 3 core build decisions (Tasks A–D)

Date: 2026-08-04 **Spec section affected:** PHASE3_ANALYSIS_SPECIFICATION.md sections 1–4 (operationalizations where the spec was silent). All decisions below were fixed BEFORE any regression output was produced (honest-results clause).

Sample membership (funnel gate): coin sample = `universe_coverage_status coverage_status=='complete' AND lambda_months>0 AND asset_class=='coin'` (the 24 regression-ready coins, incl. pos_possible + POL), month has non-missing λ_z AND positive NVT_GL. A naive `coin_staking_type=='pos'` filter yields 11/396 and does NOT reproduce the paper funnel. Token sample = token rows of `nv_tvl_gl_panel` with finite positive NV/TVL_GL and non-missing λ_z . Gate verified EXACTLY (24/718, 101/2,771, medians match paper Table 2) before proceeding; `assert()` in `phase3_panel.py`.

Return machinery: monthly simple returns from `universe_panel` observed prices only, consecutive calendar month-ends (a gap in observed status = NaN link; no backfill). Cumulative `t+1..t+3 / t+1..t+6` require ALL monthly links present. Regression dep winsorized 1/99 at the monthly cross-section of the COMBINED panel (both tracks); portfolios use raw returns (spec sec. 1).

Variable operationalizations: `mom_3m = P_{t-1}/P_{t-4}-1` (months `t-3..t-1`); `mom_12_2 = P_{t-2}/P_{t-13}-1`; `r_1m` = month-`t` return. Valuation ln-ratios winsorized 1/99 POOLED WITHIN TRACK before class-month standardization. `conv_vw` = equal-weight mean of ch3 sub-channel z-scores (voting, delegation; each z-scored over all asset-months with that channel, mirroring the composite λ_z construction). Sector FE: first ‘;’-tag of `classification_table` sector consolidated to 7 groups (DEX, LendingCDP, Derivatives, Bridge, YieldStaking, Stables, Other) — 30 raw tags over 101 tokens is too granular (many singletons).

Median splits (H2 s5, H3 quadrants): `hi_conv = conv > class-month median`; `lo_val = val <= class-month median` (median asset goes to the cheap side). Split-sample difference test = pooled spec-2 + `high_val` dummy + `conv x high_val` (controls NOT interacted; conservative simple form).

Portfolio evaluation: `turnover = 0.5sum(w_t+ - w_t- / one-way per leg; net-of-cost return = gross - (TO_star + TO_avoid)cost_per_side`. Sub-period rows require ≥ 12 months (coin pre-2023 rows absent: only 20 guard-surviving months, all 2023-05 onward). Sharpe = annualized mean/sd of `(r - rf_m)`, `rf_m = 4%/12`.

Small-cluster caveat (recorded): coin track has 24 entity clusters; the split-sample above-median `t=-10.9` is not quotable — use the interacted difference test (`t=-3.99`).

RESULTS (pre-registered ladder, no tuning): H1a REJECTED at `t+1` (coin `conv t<=0.7` unconditional); H2 SUPPORTED for coins (`conv x val = -0.0169, t=-3.54`; split diff `-0.042, t=-3.99`; `+0.7%/SD` in cheap coin-months vs `-3.0%/SD` in expensive); H1b weak support (`~+0.7%/mo` per SD, `t=1.5-2.3`, strongest with sector FE / FM); H2 REJECTED for tokens (interaction `+0.003, t=0.8`, wrong sign); voting-weighted

lambda NOT better than passive (710-month subsample, all $t < 0.6$); H3 REJECTED (no SMA variant significant; coin quadrant only 20 months after breadth guard excludes 45/65 class-months). Monthly CMOM factor premium is NEGATIVE (-4.6%/mo, $t = -1.6$) — monthly analog deviation from weekly LTW documented. Full readings: 03_data/PHASE3_RESULTS_REPORT.md.

Outputs: 03_data/phase3/{regression_panel,ltw_factors_monthly,portfolio_returns}.csv, 03_data/phase3/tables/{h1h2_coin_returns}.csv
Builders: 04_code/phase3_{panel,factors,regressions,portfolios}.py

Entry 96 — Session 043: Phase 3b probes (E1 BitInfoCharts AA; E2 realized-cap)

Date: 2026-08-04 **Spec section affected:** PHASE3_ANALYSIS_SPECIFICATION.md section 5 (horse-race data builds).

E1 — BitInfoCharts active addresses: FEASIBLE but ETH-only within the coin sample.

Test coin ETH end-to-end: /comparison/activeaddresses-eth.html parsed with the existing sentinUSD Dygraph regex — 4,003 daily obs 2015-08-07..2026-08-04, monthly averaging clean (raw HTML cached at 03_data/raw/bitinfocharts/activeaddresses_eth.html). TRX/ADA/SOL return HTTP 200 STUB pages with zero data rows (checked explicitly — page-exists is not series-exists). Of the 24 sample coins only ETH has a real series; coverage is otherwise legacy PoW (BTC/LTC/DOGE/...). DECISION: Metcalfe cannot enter the cross-sectional horse race; keep only as an ETH+PoW time-series baseline if Phase 3b wants it (~10 pages, minutes of work).

E2 — ch2 checkpoint realized-cap probe: NO for the coin track (MVRV dropped). Schema survey of all 428 files in 03_data/raw/phase1_onchain/holding/: 211 events-schema checkpoints retain the FULL raw transfer list ([block, logidx, timestamp, from, to, value]) — last-move attribution IS replayable locally with zero new API calls; 215 streamed-schema checkpoints store only monthly aggregate rows (hodl_6m/hodl_12m) + mblocks — per-unit age state destroyed at stream time, unrecoverable (Etherscan lapsed, no rebuild path). BUT the 24 regression coins have ZERO ch2 checkpoints of either kind (ch2 covered EVM tokens; coin conviction is ch1 staking) — MVRV was specced as a COIN comparator (spec 5.1) and is therefore infeasible regardless of schema. Token-side overlap: 12 of 101 sample tokens events-schema, 80 streamed, 9 none. DECISION: MVRV dropped from the horse race; a 12-token MVRV side-panel remains possible but too thin as a comparator. No realized_cap.csv built.

Entry 97 — Cowork 2026-08-04: EXPLORATORY token conviction-only sorts (user-directed, post-hoc)

Context: Moazzam observed that the token conviction signal is one-dimensional (H2 interaction dead) and asked for a pure high-conviction-minus-low-conviction token sort. The session-043 comparator (median split EW) exists and fails ($\alpha + 0.33\%/mo$, $t = 0.38$).

Finding (EXPLORATORY — run after seeing session-043 results, not pre-registered): sharpening the sort strengthens the signal monotonically, as a genuine linear signal should: median EW $t = 0.38 \rightarrow$ tercile EW $t = 0.44 \rightarrow$ QUINTILE EW $\alpha + 1.71\%/mo$ $t = 2.18$ Sharpe 1.01 (50 mo); quintile VW $+ 3.11\%/mo$ $t = 1.83$. Post-2023: quintile EW $+ 1.48\%/mo$ $t = 1.79$ Sharpe 1.28 — the only portfolio variant so far that survives the sub-period. Leg breadth ~9.4 tokens; leg turnover 0.18/mo $\rightarrow$ 50bps/side cost drag ~36bps/mo (net ~+1.35%/mo EW). Median split dilutes the signal because information is in the extremes.

Sector-neutral variant INFEASIBLE at current labels: DeFiLlama compound category strings are hyper-granular (median 1 token per sector-month; only 1.1% of sector-months have ≥ 3 tokens; only 7 tokens ever qualify). Requires coarse sector remap (DEX/Lending/Yield/Derivatives/Other) before a within-sector signal can be built.

Status: exploratory, to be treated as hypothesis-generating. Confirmatory treatment queued for Phase 3b: pre-specified quintile conviction sort (EW primary), spanning vs momentum/reversal/52wk-high competitor portfolios, coarse-sector-neutralized version, cost-adjusted, sub-periods. Script: 04_code/phase3_explore_conv_quintile.py.

Entry 98 — Cowork 2026-08-04: Phase 3b design fixed (spec ?8)

Decisions (user-directed): (1) Confirmatory conviction-only token sorts: quintile EW primary, decile/tercile/VW secondary, factor alphas vs monthly LTW; exploratory origin (Entry 97) must be disclosed in the paper. (2) Coarse sector remap {DEX, Lending, Yield, Derivatives, Staking/LSD, Other} replaces raw DeFiLlama strings for token sector FE AND enables sector-level tests. (3) Single-dimension by-sector alpha tests are powered precisely BECAUSE the dead valuation dimension is dropped (double sort needs 4 joint cells and fails when conv/val correlate; single sort needs 2) — sector-neutralized full-breadth quintile primary, per-sector terciles for largest 2-3 groups as power check. (4) Spanning against reversal/momentum/52wk-high/size long-shorts is the make-or- break test given the strong token reversal (FM t=-3.4). Spec ?8; kickoff CLAUDE_CODE_PHASE3B_KICKOFF_PROMPT.md.

Entry 99 — Cowork 2026-08-04: M1–M4 mechanism framework added to paper + spec ?8.6

Decision: The coin/token H2 asymmetry (interaction strong for coins, dead for tokens) is elevated from anomaly to mechanism section. Paper Section 2.3 gains “When does conditioning bite?” — formalizes that Prop 2 requires cross-sectional variation in delta and states four discriminating mechanisms: M1 attention-dependent absorption (token premium should concentrate in small/low-turnover; conv x turnover < 0), M2 TVL denominator endogeneity (interaction should revive under sector-demeaned valuation), M3 growth-adjustment measurement (raw NV/TVL + cap-excluded), M4 seigniorage confound (coin interaction must survive staking-yield controls — a THREAT to the coin result, reported regardless of outcome). Spec ?8.6 maps each to its test; 3b kickoff Task E added. Preferred narrative if M1 holds and M2/M3 fail: single theory, two absorption regimes — coins are the partially-priced corner, tokens the unpriced corner.

Entry 100 — Session 044: Coarse sector remap rule (Task A, spec ?8.1)

Date: 2026-08-04 **Rule (fixed before any 3b estimate):** raw DeFiLlama compound category strings split on ‘;’; token assigned to the FIRST coarse group in priority order DEX > Lending > Yield > Derivatives > Staking/LSD with >=1 matching tag; no match -> Other. Tag-level case-insensitive keyword match: DEX = tag contains ‘dex’ (Dexs/DEX/DEX Aggregator); Lending = contains ‘lending’ or tag==‘CDP’; Yield = contains ‘yield’ or ‘farm’; Derivatives = contains ‘derivatives’/‘options’/‘perpetuals’; Staking/LSD = contains ‘staking’ (Liquid Staking, Staking Pool, Restaking). Literal reading of spec ?8.1 “priority-ordered”. Documented consequence: perp/derivative DEXes carrying any DEX tag (GMX, PERP, SNX, dYdX-v3-with-DEX-tag) land in DEX; only pure-derivatives strings land in Derivatives (6 tokens). Counts over 101 tokens: DEX 41, Other 21, Lending 16, Yield 15, Derivatives 6, Staking/LSD 2; median names/group-month DEX 25, Other 9, Lending 8, Yield 7, Derivatives 3, Staking/LSD 1. Largest-3 groups for per-sector tests: DEX, Other, Lending (Other is the residual grab-bag; noted in report).

Result (Task A second half): token ladder with coarse-sector FE ATTENUATES the conviction slope vs the 043 raw-7-group FE (s6_1 +0.0071 t=2.30 -> +0.0063 t=1.95; s6_2 t=1.93 -> 1.49; s6_3 t=2.01 -> 1.60; s6_4 t=1.93 -> 1.52); interaction dead under both. Part of the token conviction premium is between-coarse-sector. raw7 columns reproduce session 043 exactly (machinery check). Builders phase3b_sector_map.py + phase3b_regressions_coarse.py; outputs sector_coarse_map.csv, tables/sector_coarse_sizes.csv, tables/h1h2_sector_fe_comparison.csv.

Entry 101 — Session 044: horse-race signal operationalizations + race results (Task C, spec ?5/?8.3)

Date: 2026-08-04 **Operationalizations (fixed pre-run):** raw_val = ln(raw NVT)=ln(MC/PQ0_annual coins / ln(nv_tvl_raw) tokens, winsorized 1/99 within track (mirrors GL val treatment). S2F dual build: s2f_ln = ln(circ_supply / trailing-12m Dsupply) only where flow>0 (literal spec; 83% coin / 75% token coverage), plus supply_g12 = 12m supply growth (defined everywhere incl. deflationary months, monotone-inverse of S2F) used in the JOINT race. high52 = price / rolling-12m max (>=6 obs, George-Hwang). ma_cross = 1[MA3>MA10] monthly closes, full windows. Momentum family from the 043 panel. All standardized within class-month. Builder phase3b_signals.py -> horserace_signals.csv.

Results: (a) Panel race — COIN: no single comparator significant full-sample; the H2 interaction added to the full joint spec survives everything: $\text{conv} \times \text{val} = -0.0176$ ($t=-2.83$); sub-2024 -0.0104 ($t=-1.92$) with 52wk-high the strongest coin signal ($+0.044$, $t=3.12$) and supply growth significantly NEGATIVE (scarcity-positive). TOKEN: conv single $+0.0064$ ($t=1.97$) but attenuates to $+0.0032$ ($t=1.10$) in the joint race; only reversal survives jointly (r_1m -0.0166 $t=-2.46$; sub-2024 -0.0267 $t=-3.95$). Claim boundary recorded: token conviction is a portfolio-extremes phenomenon, not a robust linear panel slope. (b) Spanning both directions: $q5_ew$ alpha survives every single-competitor control ($+1.53\%$ to $+2.53\%$ /mo, t 1.85-3.18); the $ltw+all$ ($n=28$) and $ltw+macross$ ($n=31$) cells are underpowered ($macross_ls$ needs 10m MAs + min-3 binary legs) — reported, not treated as refutations. No competitor LS earns positive alpha on $LTW+q5$. (c) sub-2024 rows all reported. Metcalfe DESCRIPTIVE only: 7 real BitInfoCharts AA series (BTC/ETH/LTC/DOGE/BCH/DASH/ETC); ZEC = stub (0 rows); NEW LANDMINE: unknown tickers ('btg') redirect HTTP-200 to the DEFAULT btc-ltc-eth comparison chart — the Dygraph regex happily parses the BTC series; page-title guard added to `phase3b_metalcalfe.py`. BTC Metcalfe mean-reversion $t=-3.69$ (own-asset, full-sample z , look-ahead acknowledged). Outputs `tables/horserace_{panel,spanning}.csv`, `metcalfe_panel.csv`, `tables/metcalfe_summary.csv`.

Entry 102 — Session 044: confirmatory conviction-only token sorts (Task B, spec ?8.2)

Date: 2026-08-04 **Design as pre-specified** (quintile EW min-3 primary; decile/tercile/VW secondary; sector-neutral quintile via coarse-sector demeaning $n \geq 3$ else class-month; per-sector terciles for DEX/Other/Lending; coin tercile analog; NW-3 alphas vs monthly LTW; pre/post-2023; 25/50bps; turnover; spanning vs identically-built r_1m / mom_3m / 52wk-high / size quintile long-shorts). Builder `phase3b_sorts.py`.

RESULTS (Entry 97 exploratory quintile now confirmed within battery — paper must disclose origin): $q5_ew$ alpha $+1.71\%$ /mo $t=2.18$ (exact match to Entry 97), net-50bps $+1.53\%$ $t=1.95$, post-2023 $+1.48\%$ $t=1.79$, Sharpe 0.86, ~ 10.6 names/leg, turnover 0.19/0.16 per leg. SPANNING (make-or-break): alpha STRENGTHENS with reversal control ($+2.53\%$, $t=3.18$; rev_ls loading $+0.18$ — high-conviction leg tilts to recent winners, and winner-minus-loser has alpha -4.68% $t=-3.02$) and stays $+2.34\%$ ($t=2.60$) vs all four competitors. Conviction is NOT repackaged reversal. HONEST LIMITS, equal prominence: (1) decile DIES ($t=0.02$, ~ 5.8 /leg) — the “sharper sorts monotonically strengthen” narrative does NOT extend past quintiles (progression: median 0.38, tercile 0.37, quintile 2.18, decile 0.02); (2) sector-neutral quintile DIES ($t=0.39$, post-2023 negative) — premium is between coarse sectors; (3) per-sector terciles all null (DEX $t=0.89$, Other $t=-0.77$, Lending $t=-0.16$) — the by-category power test fails; (4) coin tercile analog flat ($t=0.16$); (5) $q5_vw$ bigger but noisier ($+3.11\%$, $t=1.83$). Outputs `conv_sort_returns.csv`, `tables/convsort_{alphas,stats,spanning}.csv`.

Entry 103 — Session 044: ve/plain + fee/nofee classification of the 101 tokens (Task D 2-3)

Date: 2026-08-04 **Rules:** $ve_lock='ve'$ iff governance power or the primary staking/value-accrual mechanism requires time-locking/escrow/bonding (veToken, fixed-term locked staking, bonded node/validator stakes); cooldown-only modules (stkAAVE) and unstake-anytime staking = 'plain'. $fee_share='fee'$ iff protocol fees/revenue accrue to holders/stakers via distribution, revenue-funded rewards, or systematic buyback/burn. DOMINANT-REGIME rule for mid-sample changes (BAL $\rightarrow$ veBAL 2022-03 = ve; UNI = nofee, UNification burn only from $\sim 2025-11$; AAVE = nofee, buybacks only 2025-04+; CAKE = ve via 2022-04 locked staking + 2023-04 veCAKE; 1INCH = ve via 2022-12 st1INCH Fusion locks; PENDLE = ve from 2022-11). Classified from official protocol documentation (docs domain recorded per token in the CSV); 32/101 flagged low-confidence and the split regressions re-run excluding them. Identity note: cmc 8615 EPIC = Ethernity Chain (ERN rebrand), NOT an 'Epic' protocol. Counts: ve 31 / plain 70; fee 66 / nofee 35. Output `token_gov_classification.csv` (`cmc_id`, `ve_lock`, `fee_share`, confidence, source, note); builder `phase3b_gov_classification.py`.

Entry 104 — Session 044: heterogeneity batch results (Task D, spec ?8.4 — run all, report all)

Date: 2026-08-04 **(1) Delta-lambda (conv_lz diffs, consecutive-month guard):** nothing. Token levels+changes: d1m +0.0033 t=0.88, d3m +0.0060 t=1.31 (conv level attenuates alongside); coin negative insig. Quintile sorts on changes: d1m alpha +1.33%/mo t=1.11, d3m +0.53% t=0.34. **(2) ve split REJECTED — wrong direction:** ve conv slope +0.0016 (t=0.21) vs plain +0.0047 (t=1.21); pooled conv x ve = -0.0023 (t=-0.27); excluding low-confidence -0.0120 (t=-0.94). The costly-lock prediction fails; premium sits in plain tokens if anywhere. **(3) fee split: no difference** (fee +0.0052 t=0.94 vs nofee +0.0056 t=0.87; interaction -0.0005 t=-0.05). Caveat recorded: between-token test, weak vs the model's within-token b_t comparative static. **(4) size terciles flat (t 0.32-0.51); turnover terciles TILT WRONG WAY** (lo t=0.36, mid 0.55, hi 1.41) — limits-to-arb prediction not supported. **(5) regimes:** token slope symmetric bull/bear (t=1.44/0.96); coin interaction concentrated POST-2023 (-0.0205, t=-4.32; pre-2023 flips sign on 74 obs — uninterpretable) and mildly bull-tilted (-0.0241 t=-1.83 vs -0.0088 t=-0.98). **(6) measurement:** coin interaction robust to MRP 20/40 (-0.0175/-0.0163, t=-3.55/ -3.54; PQ/TVL re-derived from emitted pq0/tvl0+g+beta, re_floor 5%) and STRONGER ex-conv-fallback (-0.0181, t=-4.67); but RAW NVT shrinks it to -0.0060 (t=-1.90) and dropping g-capped months (40% of the coin panel!) keeps magnitude (-0.0163) but kills significance (t=-0.97, n=406). Growth-levelization is load-bearing — caveat goes next to the headline t=-3.5 in the paper. Token interaction dead in every variant; token conv slope stable (+0.005/.008), significant only ex-B4 (+0.0078, t=2.08; B4 = screened HODL-6m>80% from channel2_holding, tokens only — coins have no ch2). Builder phase3b_heterogeneity.py; outputs tables/heterogeneity.csv, tables/het_portfolios.csv.

Entry 105 — Session 044: mechanism discrimination M1-M4 verdicts (Task E, spec ?8.6)

Date: 2026-08-04 **M4 staking-yield construction (new):** yield ~ = trailing-12m issuance rate / staked share = supply_g12 / logistic(conv), coin ch1_lnodds months only (logistic(conv) recovers raw lambda_ch1 exactly on those rows), winsorized 1/99 within track, class-month standardized. Approximation logged: 12m TRAILING supply growth proxies current issuance; no forward-looking or protocol-schedule data used.

VERDICTS (report table section 5): M1 attention NOT SUPPORTED — conv x turnover POSITIVE (+0.0069, t=+1.21) and D4 gradient larger in HIGH-turnover tokens (wrong sign for the prediction). M2 REJECTED — token interaction with coarse-sector-demeaned valuation stays positive (+0.0056, t=+1.27); sector-neutral-val quadrant portfolio alpha +1.50% t=1.18 n.s. M3 REJECTED (= good news for the theory reading) — token interaction ~0 under raw NV/TVL (+0.0002, t=0.05) and ex-g-cap (-0.0024, t=-0.54): the token-H2 null is NOT a measurement artifact. M4: COIN RESULT SURVIVES — conv x val = -0.0188 (t=-4.39) with staking-yield level and interaction included (conv x sy -0.0163, t=-1.58 n.s.): not a seigniorage/b_t artifact. CONSEQUENCE FOR THE PAPER: the intended “M1 supported + M2/M3 rejected” narrative loses its M1 leg; Section 2.3 must reframe (evidence pattern = between-sector, high-turnover, extremes-only -> sector- level repricing of governance value rather than individual-token attention neglect). M4 is the session's best defensive result. Outputs tables/mechanisms.csv.

Entry 106 — Cowork 2026-08-04: Full results written into paper (Sections 5, 6, abstract, intro, conclusion)

Results (5.1-5.5): coin ladder + token ladder tables; H2 asymmetry as central result; H3 power-limited null; token quintile battery with Entry-97 disclosure paragraph and full median->decile progression; spanning table (reversal-distinct); horse race summary (coin interaction survives battery at t=-2.83; token slope attenuates to t=1.10; token conviction = portfolio-extremes claim). Robustness (6.1-6.3): measurement table with BOTH coin caveats (raw-NVT attenuation t=-1.90; g-cap subsample t=-0.97) at equal prominence; M1-M4 verdict table with M1 wrong-sign reported and the between-sector reframe offered as interpretation-not-tested-mechanism; heterogeneity nulls incl. ve-prediction rejection. Abstract + intro findings rewritten to conditional-for-coins / extremes-for-tokens. Conclusion drafted. 37 pp. Moazzam to review and prune tests for the final table set.

Entry 107 — Cowork 2026-08-04: DCF misattribution fixed; P/F comparator added (spec ?8.7)

Moazzam challenged the intro's claim that DCF dominates the academic literature. He is right: pag-notta2022/biais2019 are equilibrium models, liu2021risks is factor lit. Intro para 1 recast as equilibrium (academic) vs DCF/fee-multiple capitalization (applied). Consequence: the horse race must include the implementable DCF version. ?5.1's "fee multiples are paid-data" was WRONG — DeFiLlama fees endpoints are free/keyless. Spec ?8.7 adds P/F and P/F_GL (same F* machinery) for tokens and coins, coverage-matched re-estimates, spanning vs conviction quintile, and the P/F-as-valuation interaction spec (cleanest remaining M2 test: fee denominator is not mechanically price-linked the way TVL is). To run as a short Phase 3c session. NOTE: does not conflict with Entry 30 (fees rejected as OUR PQ measure); fees here are the competitor's fundamental.

Entry 108 — Cowork 2026-08-04: Size (normalized MC) as model-free delta proxy (user-requested robustness)

Question (Moazzam): is the token H2 failure a NV/TVL measurement issue? Test with simple normalized market cap as the delta measure (big = more visible = more priced).

Tokens: conv x size = +0.0003 (t = 0.07) — nothing. Size-median split: small +0.0014 (0.32), LARGE +0.0101 (1.67) — direction WRONG for a delta story (premium sits in large tokens, echoing the M1 wrong-sign turnover result). Verdict: the token H2 null is NOT about the TVL denominator; even the model-free delta proxy shows no conditioning. Remaining measurement candidate: fee-anchored P/F denominator (spec 8.7).

Coins (new fact): conv x size = +0.0329 (t = 2.21). Conviction premium is NEGATIVE in small coins (-0.0323, t = -1.89), mildly positive in large. With BOTH interactions, conv x val survives attenuated (-0.0134, t = -1.96) alongside conv x size (+0.0314, t = 2.17); corr(val,size) = -0.10 so they are near-orthogonal conditioners. Reading: high staking in SMALL coins is a negative signal (yield-trap/dead-float candidate interpretation — small chains advertise extreme staking APRs), independent of the valuation-conditioning effect. Coin H2 keeps 10% significance in the toughest spec yet.

Script: 04_code/phase3_explore_size_delta.py. Paper: added to Section 6.1.

Entry 109 — Cowork 2026-08-04: Phase 3c scoped tokens-only; kickoff written

Moazzam directed the fee/revenue DCF comparators to TOKENS ONLY (coins dropped from ?8.7: chain fees are a toll, not a holder claim — Entry 30 logic). Two measures: P/F = MC/trailing-12m fees (practitioner multiple) and prev_gl = MC/REV* (revenue DCF, same PARAMS as PQ/TVL; revenue preferred over fees as DCF base). Core question, stated as C1 in the kickoff: does the token H2 interaction revive with a fee-anchored valuation denominator (which, unlike TVL, is not mechanically price-linked)? This is the sixth and final measurement candidate for the token conditioning null; Entry 108's five prior conditioners all failed. Coverage-matched baselines required so denominator and sample effects are separable. CLAUDE_CODE_PHASE3C_KICKOFF_PROMPT.md.

Entry 110 — Cowork 2026-08-04: Technical battery completed (spec ?8.8, 3c Task D)

Five signals added to the horse race: ma_dist (continuous MA-distance, SUPERSEDES binary MA-cross — fixes the 28-31-month underpowered spanning cells of session 044), vol12, ivol (36m vs CMKT), amihud (|r|/volume_24h snapshot proxy — caveat logged), skew36 (lottery proxy). Rationale: vol/skew are the last plausible spanners of the token conviction quintile; amihud completes the limits-to-arbitrage race; monthly RSI/MACD/Bollinger excluded as transformations of the included set; daily-native signals (MAX, true RSI) excluded for data-depth reasons — both exclusions get one justifying sentence in the paper. Runs inside session 045 (Phase 3c).

Entry 111 — Session 045: DeFiLlama fees/revenue panel built (Phase 3c Task A)

Date: 2026-08-04 **Endpoints verified live:** api.llama.fi/summary/fees/{slug}?dataType=dailyFees | [dailyRevenue](https://api.llama.fi/dailyRevenue) (daily [ts, usd] chart; HTTP 400 = no adapter exists) and [/overview/fees](https://api.llama.fi/overview/fees) (2,288 protocol +

237 chain adapters). Parent slugs accepted by /summary even when the overview lists only version children (uniswap, curve-finance, compound-finance etc. all resolve) — the overview listing is NOT authoritative for what /summary accepts; live probe per slug is.

Identity rule: `cmc_id -> dl_slug` from `tv1_panel.csv` (the map actually used in the NV/TVL build), 1:1 on all 101 sample tokens. The 4 chain-level TVL tokens (ARB, METIS, APE, BLAST) get the DL CHAIN fee adapters (arbitrum/metis/apechain/blast, verified live), flagged `chain_level_sequencer_fees` — L2 sequencer fees accrue to the DAO (a holder claim), unlike the L1 validator toll rejected in Entry 30/109; consistent with their chain-level TVL precedent (Entries 68/84). METIS and APE have no chain revenue adapter (400) — fees only.

Coverage: 55/101 tokens with fees, 51/101 with revenue; median 44 months per covered token; starts 10 pre-2021 / 13 in 2021 / 32 in 2022+. Fees-without-revenue: METIS, APE, RBN, MAV (no revenue adapter), 10 more tokens with shorter revenue than fee windows. Monthly sum by calendar month (universe month_end convention); incomplete trailing month dropped; `n_days` kept as diagnostic.

Probed-and-REJECTED slug resolutions (house conservatism — fees slug must equal the TVL identity slug or its chain mapping): **metronome** (MET, real parent match but fees start 2023-01 vs MET regression window ending 2023-06 -> zero usable DCF months); **rari** (= Rarible RARI, NOT Rari Capital RGT — collision); **nerve** (2024-09 start, wrong-protocol risk vs nerve-staking); **aurora** (chain adapter; AURORA's TVL identity is protocol-level aurora-plus; chain-toll logic applies); **bounce.tech** (2026-01 start, 0 usable regression months); **thundercore** (1 day of data). All other misses are genuine 400s (augur, loopring, hashflow, biconomy, aevo-perps, flexa, ...). **Known limitation:** DL `dydx-v3` fees cover only 2023-11..2024-10 (the wind-down), missing the 2021-22 v3 fee peak — ETHDYDX under-covered.

Raw cache `03_data/raw/phase3c/fees/` (202 files); builder `phase3c_fees_panel.py`; outputs `fees_revenue_panel.csv` (2,437 rows) + `tables/fees_coverage.csv`.

Entry 112 — Session 045: P/F and prev_gl comparator construction (Task B)

`pf` = MC / trailing-365D fee sum (≥ 6 monthly obs — PQ0 house convention). `prev_gl` = MC / `REV*` and `pf_gl` = MC / `F*` via the EXACT `phase2_nvt_gl.py` `pq_star` machinery and PARAMS (`rf` 4%, `MRP` 30%, `g_inf` 3%, `n` 10, `g_cap` [-50%,+200%] flagged, `beta36` from the regression panel, `r_e` floor 5%); `base` = trailing-12m revenue (fees); `g` = trailing 3y CAGR of the base, 2y/1y fallback, window recorded. `ln`, winsorized 1/99 pooled within track, standardized within token-month (mirrors the val treatment).

Coverage on the 2,771 token asset-months: `pf` 1,367 (51 tokens), `pf_gl` 1,178 (44), `prev_gl` 976 (35); 202 `pf_gl`-only rows (13 tokens) where fees exist but revenue does not — the flagged variant of the kickoff. `prev_gl` needs ≥ 1 y of prior base history for `g`, so its months are effectively 2023+. Sanity: median P/F 12.1, median `prev_gl` 23.0; `corr(ln P/F, raw ln NV/TVL)` = 0.48, vs GL 0.32, vs size -0.06 — distinct axis. Builder `phase3c_comparators.py`; output `fee_comparators.csv`.

Entry 113 — Session 045: C1 VERDICT — token H2 null survives the sixth (fee-anchored) measurement candidate; P/F is a real token panel signal

C1 (the session's core question): token `s4` with `val` = `ln P/F`: `conv x val` = -0.0024 (t = -0.45), n = 1,282/49 tokens. With `val` = `ln prev_gl` (primary DCF): -0.0035 (t = -0.31), n = 915/33. `pf_gl` variant: +0.0060 (t = +0.73). First time in six conditioners the SIGN goes the H2 way and splits order correctly (P/F cheap +0.0085 vs expensive +0.0023; `prev_gl` +0.0050 vs -0.0096) — but `s5_diff t` = -0.89 / -0.87: a null. Coverage-matched NV/TVL_GL baselines on the SAME subsamples equally dead (+0.70 / -0.80) -> no denominator effect; the `prev_gl`-subsample TVL interaction going negative is sample composition, not measurement. **PAPER CLAIM NOW AVAILABLE: the token conditioning null survives TVL-raw, ex-g-cap, sector-demeaning, size, turnover, AND fee/revenue anchoring — it is a fact about tokens, not measurement.**

New horse-race fact: `ln P/F` is the first valuation signal to work in the token panel — singles -0.0103 (t = -2.50) full / -0.0148 (t = -2.70) sub-2024; level term in `s4` -0.0119 (t = -3.21); survives the completed joint battery (-0.0090, t = -2.32) where raw NV/TVL never did. The DCF transform DESTROYS it (`prev_gl t` =

-0.42, pf_gl t = -0.68) and the quintile portfolio earns nothing (-0.18%/mo, t = -0.10 at ~5-6 names/leg; post-2023 +1.15%, t = 0.85) — a panel-level, not extremes, phenomenon (mirror image of conviction). Conviction's joint slope is unchanged by the fee column on the P/F subsample (+0.24 -> +0.19 t). CAVEAT logged: on the revenue-covered subsample (33 tokens, ~2023+) the conviction slope flips negative (-0.0124, t = -2.01) BEFORE any fee variable enters — composition, reported in the report headline. Neither fee comparator spans the conviction quintile (q5 alpha +1.22% t = 1.67 on pf months, +1.54% t = 1.84 on prev_gl months; direction 2 alphas null).

Entry 114 — Session 045: technical battery results (Task D) — conviction quintile SURVIVES the completed battery; 044 MA-cross cells superseded and resolved

Signals built from universe_panel only (phase3c_technicals.py): ma_dist (px/MA10-1), vol12 (>=8 obs), ivol (36m resid SD vs CMKT, >=12), amihud (ln trailing-12m mean |r|/volume_24h, >=6 obs — **CAVEAT: volume_24h is a month-end SNAPSHOT, not a monthly aggregate; noisy proxy**), skew36 (>=18 obs). Returns clipped (-90%, +300%) before moments (phase2 beta-build convention). Coverage 90-100% per track.

Token spanning verdict (the key question): q5_ew conviction alpha vs LTW + the COMPLETED 12-long-short battery = **+1.74%/mo (t = 2.45, n = 43)**; + both fee LS = +2.17% (t = 2.82, n = 35). The two underpowered session-044 MA-cross cells are SUPERSEDED by continuous ma_dist and RESOLVED in conviction's favor: +all cell +0.99% (t = 0.99, n = 28) -> +1.66% (t = 2.73, n = 43); single-MA cell +1.09% (t = 1.16, n = 31) -> +1.83% (t = 2.46, n = 50). Vol/skew (the Entry-110 candidate spanners) do not span. Direction 2: no competitor earns alpha on LTW + q5. None of the five technicals earns a significant token long-short on its own; ma_dist singles are the only significant token panel cell (-0.0120, t = -2.21 — momentum-adjacent negative, like the 044 reversal family).

Coin track: conv x val survives the completed battery: -0.0235 (t = -2.68) full, -0.0188 (t = -3.17) sub-2024 (044 eight-comparator figure was -2.83). Amihud is the strongest new coin single (-0.0223, t = -2.65; sub-2024 -3.23; illiquid-coin discount), vol12/ivol also significant sub-2024; none touches the interaction. Joint-race construction note: ma_cross REPLACED by ma_dist in the completed joint set (supersession, not addition). Exclusions sentence for the paper (monthly RSI/MACD/Bollinger redundant; daily-native MAX/RSI infeasible at free-tier depth) drafted in PHASE3C_RESULTS_REPORT.md section 6.

Entry 115 — Cowork 2026-08-04: Phase 3c results written into paper

Per PHASE3C_RESULTS_REPORT ??: (1) Section 6.1 — fee-anchored rows added to measurement table; six-conditioner definitive statement (“the conditioning failure is a fact about tokens, not about measurement”); sign-flip-but-null reported honestly; revenue-subsample conviction sign flip (-2.01) as external-validity note. (2) Section 5.5 — completed 12-comparator battery (coin interaction -2.68/-3.17); P/F as the one working token valuation comparator (panel level -2.32 joint, no portfolio translation, DCF transform kills it); “growth-levelization helps throughput flows, hurts cash flows” framing; exclusions footnote (RSI/MACD/Bollinger redundant; daily-native infeasible; amihud snapshot caveat; fee coverage 55/101). (3) Spanning table upgraded to completed battery incl. supersession note for 044's underpowered MA-cross cells (now +1.66%/t=2.73 and +1.83%/t=2.46 with continuous ma_dist); completed battery +1.74% (t=2.45); +fees +2.17% (t=2.82). (4) Intro: one sentence on P/F beating its own DCF refinement. 39 pp, clean compile. Ready for Moazzam's table-pruning review.

Entry 116 — Cowork 2026-08-04: Alternative-valuation results made visible; H3 re-run under P/F and REV* (user-directed)

Moazzam flagged that the 3c results were prose-only. Added to the paper: (1) Table horserace — full signal-by-signal race, both tracks, singles + joint rows; (2) Table altport — every alternative signal as a token quintile long-short (all null) vs the conviction quintile reference; (3) Table h2altval — the H2 interaction spec under NV/TVL_GL / ln P/F / ln MC/REV* side by side with split-samples and coverage-matched baselines. (4) NEW RUN (inline, phase3_explore_h3_altval.py): H3 quadrant re-formed with fee-anchored valuation axes. Verdict: still null under all three measures — P/F quadrant +1.67%/mo alpha, Sharpe 0.82,

but $t=1.35$ (46 mo) and DEAD post-2023 ($t=0.07$); REV* quadrant negative (-1.89%, $t=-1.24$, 31 mo, 33 excluded by breadth guard); coverage-matched NV/TVL_GL baseline $t=0.68$. Two-dimensional portfolio adds nothing over the conviction quintile under ANY valuation measure. Paper 42 pp.

Entry 117 — Cowork 2026-08-05: DEX-only conviction quintile (user-prompted exploratory)

Moazzam asked whether the within-sector null was a power issue and whether per-sector QUINTILES were ever tested. They were not (only terciles; Entry 100/session 044). DEX is the only coarse sector deep enough (39 tokens, median 25/month). Result: DEX-only conviction quintile EW = +3.48%/mo raw, +3.37%/mo LTW alpha, $t = 2.04$, 46 months, ~4.6 names/leg. DEX tercile confirms 044 ($t = 0.62$). VERDICT: the within-sector “null” was tercile dilution + thin-sector noise in the pooled sector-neutral construction, NOT absence of within-sector signal. The premium is present WITHIN the deepest sector at fine granularity; other sectors untestable at quintile granularity. The “between-sector repricing” mechanism framing (Entry 105 / paper 6.2) must be qualified accordingly. EXPLORATORY (user-prompted, post-hoc) — disclose like Entry 97. Script: 04_code/phase3_explore_dex_quintile.py.

Entry 118 — Cowork 2026-08-05: VW spanning + P/F quadrant VW (user questions)

- (1) Why EW primary: pre-registered in spec 8.2 before results; better precision (t 2.18 vs 1.83); VW 10-name legs concentrate in 1-2 large tokens. NEW: VW quintile does NOT survive the 12-battery spanning — inline identical-construction comparison: EW +1.36%/ $t=1.67$ vs VW +0.97%/ $t=0.86$ (inline battery replicates 045’s EW cell at 1.67 vs official 2.45 — construction detail differences; the EW-vs-VW comparison under identical construction is the informative one). VW’s role in the paper = microcap- robustness sentence only. (2) H3 quadrant with P/F valuation, VW variant: +2.05%/mo, $t=1.05$, Sharpe 0.64 (EW was +1.67%/ $t=1.35$, Entry 116) — best quadrant point estimates in the paper, never significant; two-dimensional construction fails under fee-anchored valuation in both weightings. Script: 04_code/phase3_explore_vw_checks.py.

Entry 119 — Cowork 2026-08-05: Pruning batch 1 executed (Moazzam’s directives)

- (1) Hypotheses UNIFIED: H1/H2/H3 stated once for both classes in new 2.4 “Testable Hypotheses”; per-class Prop 1a/1b/H1a/H1b/Prop 2 removed; 2.1/2.2 end with measurement-mapping sentences. (2) Voting-weighted refinement removed from hypotheses, results, and scorecard. (3) Token significance prose CORRECTED everywhere: significant only alone ($t=1.97$) and in FM (1.77/2.19); $t\sim 1.5$ with controls (prior “1.9-2.0 with sector FE” was stale raw7-FE numbers). (4) M1-M4 moved from theory 2.3 to 6.2 with postulate-then-test framing (mechanisms arose from results — honest learning-process narrative); 2.3 keeps only the delta-variation qualification. (5) Disclosure reworded (no “author-directed”). (6) Table restructure: horse race = joint multivariate table (all coefficients, N, assets, R2-within 0.032/0.023); singles grid -> Appendix B; Tables 7+9 MERGED (own-alpha column + q5-survival column; own alphas for rev/mom/high52/size computed inline, reversal replicates official -4.68/-3.02 exactly); Table 10 (h2altval) -> Appendix B; quadrant pooled rows dropped.
- (2) 4.2 funnel prose condensed (~1.4k chars). (8) 5.4/6.2/scorecard updated with Entry-117 DEX-quintile finding (between-sector framing qualified) and Entry-118 VW microcap sentence. 5.5 rewritten around three learnings (incrementality; coin cross-section info-free at level; P/F partial vindication with DCF twist).

Entry 120 — Cowork 2026-08-05: Proper summary statistics table (Moazzam’s request)

Old Table 2 was a coverage table, not summary stats. New Table 2: distributions (N/mean/SD/p10/p25/median/p75/p90) of main variables by track. Coins: staking share (median 38.6%), SoV/MoE (median 0.63), NVT_GL, MC, forward return, beta. Tokens: lambda_z, NV/TVL_GL, TVL, P/F (median 12.1), MC, forward return, beta. Unwinsorized; heavy-tail note motivates the log/winsorization choices; negative median returns flagged. Old coverage table -> Appendix (tab:coverage_appendix). Builder: 04_code/phase3_sumstats_table.py.

Entry 121 — Cowork 2026-08-05: Review-round edits (Moazzam’s six points)

- (1) Delta-uniformity statement rewritten to distinguish levels: between-class variation absorbed by class-separate estimation (surfaces as different avg premiums); between- sector-within-class detectable in principle; full degeneracy only if uniform across the estimation cross-section. (2) 4.2: appendix list references + survivorship paragraph (universe snapshot-based/survivorship-free; regression samples have a coverage tilt toward scale/longevity, not conditioning-on-survival). (3) 5.1: FM-vs-panel divergence explained (FM includes same controls, full cross-section monthly; equal month-weighting upweights the strong pre-2023 premium; asset clustering absorbs lambda persistence) — slope significance is estimator-sensitive because the premium is time-varying.
- (2) NEW RUN: coin quadrant guard relaxation — min3: -2.7%/mo ($t=-1.36$, 33 mo); min2: -1.5% ($t=-0.79$, 46 mo); negative at every threshold; footnote added, “mechanical” framing softened to “robust to guard, poorly powered at every setting”. (5) Tercile- dilution sentence removed from DEX passage per Moazzam (quintile already established as the instrument); passage now leads with thin-category-demeaning noise + DEX within-sector quintile; between-vs-within left explicitly unsettled. (6) NEW RUN + Table (tab:channels): token conviction by channel — HODL-6m carries essentially all signal (slope $t=1.50$, 88 tokens; $q5$ alpha +1.48%/t=1.71) vs staking/delegation/voting individually uninformative on 9-32 tokens; composite edge = governance increment; “revealed patience, not governance participation” reading; disclosed as review-stage. Script: 04_code/phase3_explore_channels_guard.py. Paper 51 pp.

Entry 122 — Cowork 2026-08-05: Figures added; working-paper ordering finalized

Two figures created (04_code/phase3_figures.py -> 05_paper/figures/): Fig 1 two-panel — (A) coin mean next-month returns by conviction-valuation cell (the H2 asymmetry visually: Star cell positive, high-conv/expensive lowest); (B) token mean returns by conviction quintile (extremes concentration). Fig 2 — cumulative log return of the token conviction quintile LS vs EW token universe (LS accumulates while universe loses value; LS series replicates official mean +2.28%/50mo exactly). Document order restructured via endfloat[tablesfirst] + processdelayedfloats after Conclusion: text (1-34) -> Tables 1-12 (36-47) -> Figures 1-2 (48-49) -> References (50-51) -> Appendices A-C with their tables (52-60). 60 pp. NEXT: introduction + abstract rewrite (Moazzam: agree the one-sentence headline claim first).

Entry 123 — Cowork 2026-08-05: Introduction rewrite, iteration 1 (Moazzam’s directives)

- (1) lambda front-loaded as THE contribution: new paragraph 2 bridges QTM with Cong-Li-Wang (2021) and Sockin-Xiong (2020) via the conviction measure — “the models supply the valuation object; lambda supplies the measurement those models have lacked.” Dual role stated upfront: velocity channel for coins (MoE present), direct network-value/membership signal for tokens (MoE absent). (2) SoV/MoE equation REMOVED from intro (stated in words, appendix ref). (3) Flow reordered: motivation -> bridge/lambda -> QTM mechanics + MoE/SoV definitions (compressed, lambda already introduced) -> why lambda informative (three traditions + token costly-signal, merged and trimmed) -> operationalization -> findings -> Contributions. (4) “Related Literature” replaced by “Contributions” paragraph: three contributions (measurement: first supply-side monetary aggregate — claim MOVED from conclusion; empirical asset pricing: first on-chain supply-side characteristic vs LTW return-based factors, conditional pricing + revealed-patience channel result; valuation practice: growth-levelized NVT formalization + horse race discipline of practitioner tools), literature comparisons woven per strand. (5) Conclusion de-duplicated: novelty claim replaced with beliefs-revealed-by-costly-actions framing; “patience not governance activity” added. 59 pp. Iteration 2 awaits Moazzam’s read.

Entry 124 — Cowork 2026-08-05: Introduction iteration 2 (Moazzam’s three directives)

- (1) Order restored with lambda as the bridge paragraph DIRECTLY after the QTM + tokenized-models framing: P1 motivation -> P2 Fisher/QTM + Cong-Li-Wang/Sockin-Xiong -> P3 lambda bridge (“its role transforms both models”; velocity channel for coins, direct network-value signal for tokens; “the models supply the valuation object; lambda supplies the measurement”) -> P4 MoE/SoV definitions + monotone result in words -> P5 why lambda informative -> P6 operationalization -> P7

findings -> P8 contributions -> roadmap. (2) Findings paragraph rebuilt around THE key finding: conditional conviction survives every measurable valuation model and technical signal jointly for coins (12-comparator regression, $t=-2.7$, nothing else predicts); token quintile alpha survives the full spanning battery ($+1.7\%/t=2.5$; reversal control RAISES it to $2.5\%/t=3.2$; no reverse-direction alpha). DCF-refinement/P-F throughput- vs-cashflow material REMOVED from findings (Moazzam: minor issues) — retained only as one clause in the methodology contribution. (3) /paragraph{Contributions} header removed; contributions now flow as a normal paragraph (“These results make three contributions...”).

Entry 125 — Cowork 2026-08-05: Long-only token quintile analysis (Moazzam’s ask)

Method (best practice, both layers reported): factor alpha (LTW, NW-3) + benchmark- relative active return / tracking error / information ratio vs EW token universe. Results (50 formation months): Q5 long-only active $+0.68\%/mo$, IR 0.41, $t=0.83$ (ns); Q1 long-only active $-1.60\%/mo$, $t=-2.25$, IR -1.10 (significant underperformance). READING: the long-short alpha is earned predominantly on the SHORT side — conviction is a loser-avoidance screen more than a winner-picker; implementable long-only version is exclusionary (drop bottom quintile). Added to 6.3 as prose. Also: intro quadrant sentence now names “Stars-minus-Avoid”. Script: 04_code/phase3_explore_longonly.py.

Entry 126 — Cowork 2026-08-05: Title, abstract finalized (pending weekend fresh-eyes review)

Title changed to “Skin in the Chain: Locked Supply and the Cross-Section of Cryptocurrency Returns” (Moazzam’s pick from fresh-read candidates). Abstract finalized per Moazzam’s own edit (lambda-first opening, theory-conditional-pricing, coin finding + joint-race survival, token conditioning failure + extremes alpha + spanning, measurement-contribution close; single-spaced). Paper state: complete 58-pp submission-shaped draft — title, abstract, intro (2 iterations), unified hypotheses, results with learnings framing, robustness + mechanisms, summary scorecard, conclusion, 12 main tables + 5 appendix tables + 2 figures in working-paper order. NEXT: Moazzam’s fresh-eyes review after the weekend (title/abstract/intro possibly revisited); then remaining polish and submission prep.

Entry 127 — Cowork 2026-08-10: AFA submission package — AI workflow documentation

Per the AFA 2027 CfP requirement (“documentation of the AI workflow, including AI conversations and human contributions, as appendices”): (1) In-paper Appendix D “AI Workflow Documentation” — two-tier workflow description (Cowork interactive / Claude Code autonomous), division of labor, record inventory, repo pointer. (2) Supplementary PDF 05_paper/AI_WORKFLOW_DOCUMENTATION.pdf (builder: 04_code/build_ai_workflow_doc.py) consolidating: human time log (66 entries), this decisions log (127 entries), all 33 verbatim kickoff prompts, all 47 session logs; TOC’d, ~1.6MB. (3) Transcript exporter 04_code/export_ai_transcripts.py — converts Claude .jsonl transcripts (Claude Code: `~/claude/projects/ C-AFA-2027-QTM-Crypto/`; Cowork: `AppData/Roaming/Claude/local-agent-mode-sessions/ **/claude/projects/`) to per-session markdown in 06_documentation/ai_transcripts/; run locally by author, then rebuild the supplementary doc.